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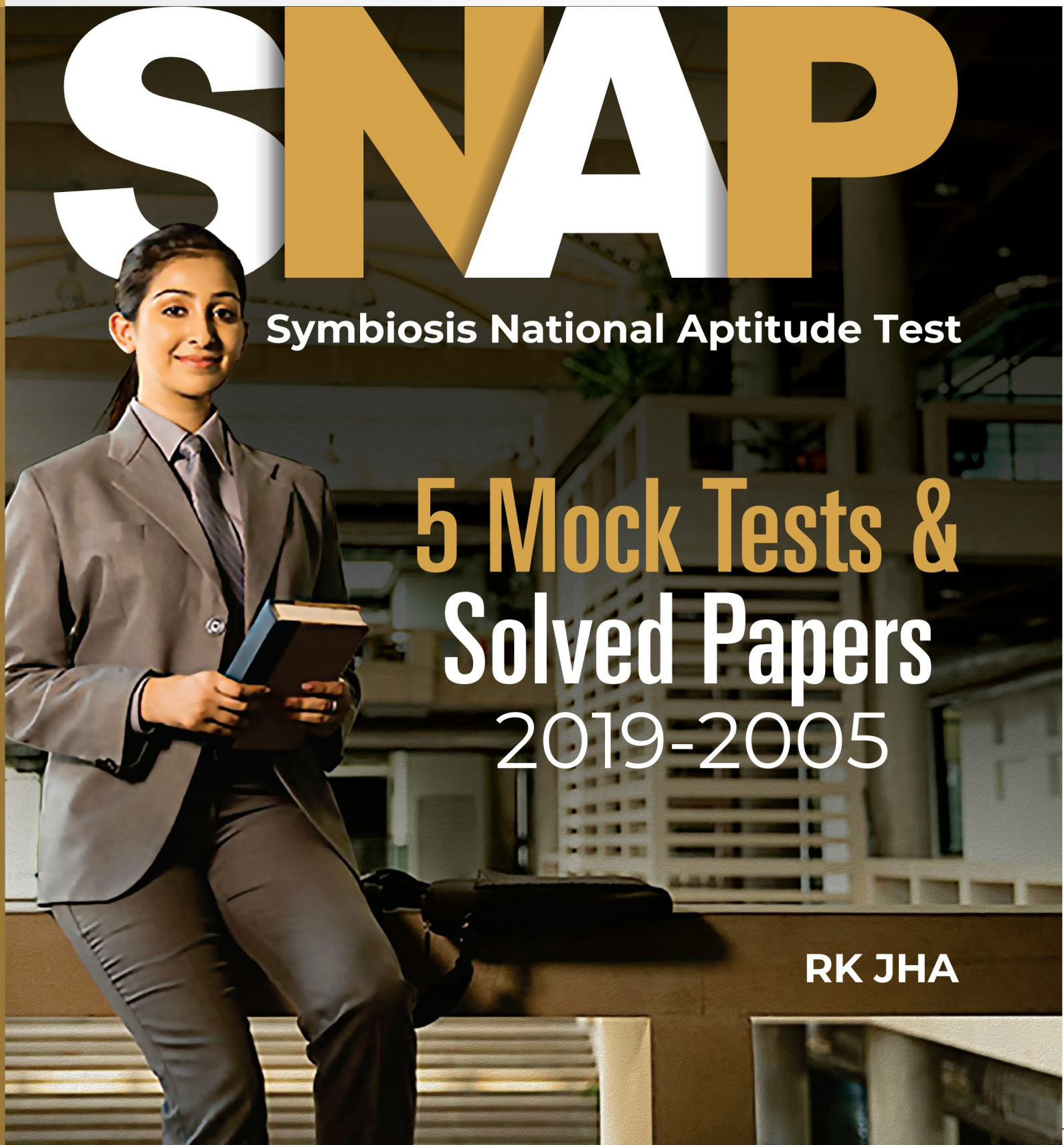

Symbiosis National Aptitude Test

5 Mock Tests & Solved Papers 2019-2005

RK JHA

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Solved Papers
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ARIHANT PUBLICATIONS (INDIA) LTD.

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Symbiosis National Aptitude Test

PREFACE

Symbiosis National Aptitude (SNAP) Test is a relatively new umbrella examination for admission to all the post-graduate institutes of Symbiosis International University (SIU). Since 2007, candidates aspiring to join MBA programs offered by institutes of Symbiosis International University have to take the common, mandatory SNAP Test.

SNAP is a mandatory written test for admission into any post-graduate programme offered by the institutes under the aegis of Symbiosis International University (SIU), established under Section 3 of the UGC Act, 1956, vide notification No.F.9.12/2001 U.3 of the Government of India. Subsequently, ten more institutes of Symbiosis have been brought under the ambit of Symbiosis International University, vide notification No.F.9. 12/2006-U.3(A)

SNAP score is valid for admission to all the post-graduate programmes of Symbiosis International University. This book has been especially designed for the aspirants of SNAP. The book contains last Thirteen years' (2018-2005) Solved Papers to make the candidates well aware of the trends and pattern of questions asked in recent years.

Furthermore Five Mock Test Papers have been provided in the book, exactly designed as per the latest pattern of the SNAP, covering every possible type of questions from each individual section. The book promises to sharpen the skills of candidates to the level of perfection and will prove to be a confidence booster for the aspirants.

I would like to convey my sincere thanks to Mr. T.K. Sinha and Mrs. Sangeeta Kedre for their assistance during the entire project. I would like to thank Mr. Shiv Kumar Kedre, Mr. Vivek Singh Meena, Mr. Sujit Kumar, Mr. N Kumar and Mr. B.B. Sinha for their cooperation. I would also like to thank Mr. Mahendra Singh Rawat & Mohd. Shahid Saifi for their kind cooperation and valuable suggestions during the project. I would also like to offer my sincerest gratitude to Mr. Deepesh Jain without whom this project would not have been possible.

Advice and suggestions for the improvement of this book will be welcomed.

WISHING ALL OF YOU BEST OF LUCK

RK Jha

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Symbiosis National Aptitude Test

OVERVIEW OF SNAP

Symbiosis National Aptitude (SNAP) Test is a relatively new umbrella examination for admission to all the post-graduate institutes of Symbiosis International University (SIU). Since 2007, candidates aspiring to join MBA programs offered by institutes of Symbiosis International University have to take the common, mandatory SNAP Test. SNAP score is valid for admission to all the post-graduate programmes of Symbiosis International University.

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ELIGIBILITY

The candidate should be a graduate in any discipline from a recognised university with a minimum score of 50%. For candidates belonging to SC/ST category, a minimum score of 45% in graduation is essential to apply for the SNAP test. However, a final year candidate may also apply for the test, provisionally. Age should not be less than 21 years.

Institute Associated with Symbiosis International University

Following are the Constituent Institutes of the Post-graduate Institutes of Symbiosis International University

1. Symbiosis Institute of Business Management (SIBM), Pune
2. Symbiosis Institute of Computer Studies & Research (SICSR), Pune
3. Symbiosis Institute of Media & Communication (SIMC), Pune
4. Symbiosis Institute of International Business (SIIB), Pune
5. Symbiosis Institute of Management Studies (SIMS), Pune
6. Symbiosis Centre for Management & Human Resource Development (SCMHRD), Pune
7. Symbiosis Institute of Telecom Management (SITM), Pune
8. Symbiosis Centre for Information Technology (SCIT), Pune

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9. Symbiosis Institute of Business Management (SIBM), Bengaluru
10. Symbiosis Institute of Operations Management (SIOM), Nasik
11. Symbiosis Institute of Health Sciences (SIHS), Pune
12. Symbiosis School of Media and Communication (SSMC), Bengaluru
13. Symbiosis School of Banking and Finance (SSBF), Pune
14. Symbiosis Institute of Business Management (SIBM), Hyderabad
15. Symbiosis School of Sports Sciences (SSSS), Pune

SNAP TEST STRUCTURE

The duration of SNAP Test is 120 min. It is an objective type test in which the candidate has to choose an appropriate response from the given options. Every wrong response accounts for 1/4 marks will be deducted.

The test is conducted in four sections, namely :

1. Analytical and Logical Reasoning
2. General English : Reading Comprehension, Verbal Ability and Verbal Reasoning
3. Quantitative Aptitude, Data Interpretation and Data Sufficiency

Number of Sections	3
Total Questions	110 (105 MCQs and 5 Non-MCQs)
Negative Marking	1/4th of Marks for the Question

SNAP PREVIOUS YEARS' ANALYSIS

Section	No. of Questions Asked										
	2019	2018	2017	2016	2015	2014	2013	2012	2011	2010	2009
General Awareness	--	25	30	30	40	40	40	40	40	40	40
Analytical and Logical Reasoning	36	35	40	40	30	30	30	30	30	30	30
General English	34	35	40	40	40	40	40	40	40	40	40
Quantitative Aptitude, Data Interpretation and Data Sufficiency	40	35	40	40	40	40	40	40	40	40	40
Total	**110**	**130**	**150**	**150**	**150**	**150**	**150**	**150**	**150**	**150**	**150**

SELECTION PROCEDURE

As per the process, the candidates securing the minimum score in the SNAP Test will be shortlisted for further rounds. The later have to appear in the Group Exercise (GE), Personal Interview (PI) and Written Ability Test (WAT).

The final merit steps will be prepared on the basis of below mentioned weightage :

Snap Score Out of 150	Group Exercise (GE)	Personal Interview (PI)	Written Ability Test (WAT)	**TOTAL**
50	10	30	10	**100**

SNAP

SYMBIOSIS NATIONAL APTITUDE TEST

SOLVED PAPER 2019*

Time : 2 hrs

INSTRUCTIONS

1. **There are 110 questions in this paper.**
2. **There are three sections viz, Section A (General English, Reading Comprehension, Verbal Reasoning, Verbal Ability) each question carry 1.5 mark, Section B (Analytical & Logical Reasoning) each question carry 1.5 mark and Section C (Quantitative Data Interpretation & Data Sufficiency) each question carry 1 mark.**
3. **In Section C also contains 5 Special (Non-MCQs) Questions carry 2 marks each.**
4. **For each incorrect attempt 1/4th of marks allotted will be deducted.**

SECTION A : GENERAL ENGLISH, READING COMPREHENSIOIN, VERBAL REASONING, VERBAL ABILITY

1. Fill in the blanks with the appropriate conjunction.

You must start at once; ..... you will be late.

(a) otherwise (b) than
(c) until (d) unless

Sol. *(a)* In the given sentence, a suggestion is given to show the result that would come out if it is not followed. Hence, otherwise will be the appropriate filler.

2. Choose the word closest in meaning to the given word.

Abnegation

(a) sacrifice (b) renunciation
(c) praise (d) indulgence

Sol. *(b)* Abnegation means an act of renouncing or rejecting something. So, renunciation is its closest meaning word.

3. Choose the word opposite in meaning to the given word.

Abjure

(a) deny (b) profess
(c) reject (d) drop

Sol. *(b)* Abjure means to reject or deny. Therefore, its antonym can be declare or profess.

4. Choose the word closest in meaning to the given word.

Lingua franca

(a) universal language (b) second language
(c) common language (d) mother tongue

Sol. *(c)* Lingua franca refers to a language that is adopted as a common language between speakers whose native languages are different. So, 'Common language' is its closest meaning word.

5. Choose the word opposite in meaning to the given word.

Petulant

(a) peevish (b) bad tempered
(c) amiable (d) impatient

Sol. *(c)* Petulant means bad-tempered or grumpy. Therefore, its antonym can be good-humoured or amiable.

6. Find the word that is the odd one out.

(a) Exacerbate (b) Alleviate
(c) Mitigate (d) Assuage

Sol. *(a)* Out of the given options, option (b) (Alleviate), (c) (Mitigate) and (d) (Assuage) means to make a problemless severe. On the other hand Exacerbate means to make a problem worse. Hence, option (a) is odd one out.

7. Change the voice of the given sentence.

I didn't realise that somebody was watching me.

(a) I didn't realise that I was being watched.
(b) I didn't realise that somebody had been watching me.
(c) I didn't realise that somebody was watching.
(d) I didn't realise that I was being watched.

Sol. *(a)* I didn't realise that I was being watched, is the correct active voice sentence of the given sentence because the sentence is in past continuous and to change it in passive voice was/were + being + V_3 should be used.

8. Complete the meaning of the given sentence.

If you behaved well, your peers …… you.

(a) will respect (b) should respect
(c) can respect (d) would respect

Sol. *(d)* The filler 'would respect' makes the sentence grammatically correct and contextually meaningful.

9. Choose the correct word from the given options.

The defence lawyer …… there was insufficient evidence to convict his client.

(a) negated (b) reiterated (c) renounce (d) exclaimed

Sol. *(b)* The correct word is 'reiterated' to fill the given blank which means repeated.

10. Choose the correct word from the given options.

The stars were …… in the sky.

(a) scintillating (b) showering
(c) invisible (d) dimming

Sol. *(a)* The correct word for the given blank is 'scintillating', which means shining.

11. Choose the word NOT having a prefix.

(a) Distemper (b) Dislike
(c) Dishonest (d) Disagree

Sol. *(a)* Out of the given option, the word distemper has no prefix. All other words have dis-as prefix to become an antonym of the root word.

12. Identify the correct figure of speech.

I must have called out to you a thousand times.

(a) simile (b) repetition
(c) hyperbole (d) oxymoron

Sol. *(c)* In the given sentence, there is an exaggeration of a thought or idea. Therefore, it is a hyperbole wherein an exaggerated statement is presented.

13. Choose the word opposite in meaning to the given word.

Servile

(a) Imperious (b) Humorous
(c) Helpful (d) Conspiratorial

Sol. *(a)* Servile means someone who is extremely willing to act to obeyed. Therefore, out of the given options, Imperious is its correct antonym, as it means giving orders and expecting to be obeyed.

14. Find a correct match of grammatical function with the usage of the word DOWN.

List I	List II
A. Noun	1. Some people can down a whole chicken in a meal.
B. Verb	2. There is a down trend in the real estate market.
C. Adjective	3. He was sent down to work in the branch office.
D. Adverb	4. In southern England the downs are so pretty.

	A	B	C	D
(a)	1	2	3	4
(c)	3	2	4	1

	A	B	C	D
(b)	4	1	2	3
(d)	2	3	1	4

Sol. *(b)* In sentence (1) the word down means to drink or to consume,. Hence, down is used as a verb.

In sentence (2), 'down' shows a characteristic quality of a trend. Hence, here it is used as an adjective.

In sentence (3), 'down' adds to the quality of the verb. Hence, herein it is used as an adverb.

In sentence (4), 'down' represents the low hills. Hence, it is used as a noun.

15. Choose the word closest in meaning to the given word.

Diaphanous

(a) hard (b) opaque (c) thick (d) transparent

Sol. *(d)* Diaphanous means sheer, light, delicate and transclucent. Hence, transparent/sheer can be its closest synonym.

16. Fill in the blanks with correct modal verb.

…… you mind if I borrowed your car?

(a) should (b) will (c) would (d) can

Sol. *(c)* As the statement is a polite request, the most appropriate modal verb to be used here is 'would'.

17. Change the voice of the given sentence.

Windowpanes are washed by cleaners.

(a) Cleaners work to wash windowpanes
(b) Cleaners washed the windowpanes
(c) Cleaners wash the windowpanes
(d) None of the above

Sol. *(c)* The given sentence is in passive voice. To convert it the subject and object would be interchanged and Ist form of verb should be used with subject.

Hence, the converted sentence would be ''Cleaners wash windowpanes''.

18. Identify the correct figure of speech.

Neeta needed new notebooks.

(a) Alliteration (b) Onomatopia
(c) Assonance (d) Paradox

Sol. *(a)* In the given sentence, there is a repetition of a consonant sound 'N'. Hence, such a repetition of sounds is a literary device called Alliteration.

19. Which part of speech is the given (bold) word?

This wood will make a good **hiding** place.

(a) Verb (b) Adverb (c) Adjective (d) Modal

Sol. *(c)* The word 'hiding' in the given sentence presents a characteristic quality of the place. So, it is adjective.

20. Choose the correct option of the following incorrect sentence.

No matter what that I do, I can't make her happy.

(a) No matter what I do, I won't make her happy.
(b) No matter what I do, I can't make her happy.
(c) No matter what I do, that I won't make her happy.
(d) No matter whatever I do, I won't ever make her happy.

Sol. *(b)* To make the given sentence grammatically correct and contextually meaningful, 'that' must be removed. Hence, option (b) is its correct answer.

21. Fill in the blank with proper conditional.

Suppose your car broke down in the middle of nowhere, what ..... do?

(a) should you (b) will you (c) can you (d) would you

Sol. *(d)* As the situation is conditional and asks what ones actions would be, 'would you' would be the most appropriate filler for the given blank.

22. Fill in the blank with appropriate option :

I ..... anything from her in a long time.

(a) did not heard (b) should not have heard
(c) haven't heard (d) None of these

Sol. *(c)* As the sentence is in present perfect form, 'haven't heard' is the most appropriate filler.

23. *Read this sentence and answer the three questions to fill the blanks therein :*

Construction : When the engineer visited the ....., the chaos of cement, steel and sand was a horrible ....., so, he had no option but to ..... that in his report.

(a) site, sight, cite (b) cite, site, sight
(c) cite, sight, site (d) sight, site, sight

Sol. *(a)* The correct order of the words to fill the blanks are site, sight, cite, to make the sentence grammatically correct and contextually meaningful. Site means a piece of land where a building was, is or will be situated. Sight means able to see in the way mentioned. Cite means to mention something as an example especially one that supports, proves or explains an idea or situation.

Directions (Q.Nos. 24-26) *Read the passage carefully and answer within the context.*

"A way to deal with frozen feelings" Every child experiences all that happens around him with total awareness. In the first seven years the child's brain is like a sponge, taking in all sensory inputs and building his idea of his surroundings. As long as the environment is safe, the child learns with incredible speed. However, when the environment is scary or stressful, the child unlearns past learning just as rapidly. In the early years of every child's life, whenever there is shock, violence, fear or pain, these intense emotions are imprinted deeply into memory. Whenever the same activity or situation is repeated, the nervous system and body subconsciously re-experience the memory of that trauma. Any emotional situation that takes us out of the present and into the past means that whenever the same kind of emotion crops up later in our life we return to the past for our reference point. If that point was at age three, we find ourselves behaving like a three-year-old. We feel childish and we behave childishly. Our feelings are the cause of this 'glitch' in our learning process. We know we should be able to make a positive change, but that doesn't change anything. The process of change need not be traumatic. We couldn't have done any better because we didn't know how to. But we should realise that was then and this is now! We can choose to choose again. It's up to us. It's our movie!

24. The "Frozen Feelings" being talked about are about

(a) negative childhood experiences
(b) childhood learning patterns
(c) inability to learn as an adult
(d) None of the above

Sol. *(a)* According to the given stanza, the frozen feeling are those experiences of shock, violence, fear or pain that has a deep impact on the child. Such a negative childhood experience is imprinted forever into the child's mind.

25. A 'glitch' is

(a) a ditch
(b) uneasy emotions
(c) sudden malfunction or breakdown
(d) learning patterns

Sol. *(c)* According to the author, a glitch refers to an unusual behaviour generally a sudden malfunction or breakdown that results due to the frozen feelings.

26. Identify the correct sentence, based on the paragraph.

(a) The process of change needs to be traumatic.
(b) We feel childish and we behave childishly.
(c) Both sentences are incorrect.
(d) Both the sentences are correct.

Sol. *(b)* As mentioned in the passage, there are glitches in ones behaviour due to frozen feelings. Due to these glitches, one feel childish and behaves childlishly.

Directions (Q.Nos. 27-31) *Read the following passage and answer the questions :*

"There are several factors that contribute to wisdom. Of these I should put first a sense of proportion; the capacity to take accounts of all the important factors in a

problem and to attach to each its due weight. This has become more difficult than it used to be owing to the extent and complexity of the specialised knowledge required of various kinds of technicians. Suppose, for example, that you are engaged in research in scientific medicine. The work is difficult and is likely to absorb the whole of your intellectual energy. You have no time to consider the effect which your discoveries or invention may have outside the field of medicine. You succeed (let us say), as modern medicine has succeeded, in enormously lowering the infant death-rate, not only in Europe and America, but also in Asia and Africa. This has the entirely unintended result of making the food supply inadequate and lowering the standard of life in the most populous parts of the world. To take an even more spectacular example, which is in everybody's mind at the present time- you study the composition of the atom from a disinterested desire for knowledge and incidentally place in the hands of powerful lunatics the means of destroying the human race. In such ways the pursuit of knowledge may become harmful unless it is combined with wisdom; and wisdom in the sense of comprehensive vision is not necessarily present in specialists in the pursuit of knowledge. Comprehensiveness alone, however, is not enough to constitute wisdom. There must be, also, certain awareness of ends of human life. This may be illustrated by the study of history. Many eminent historians have done more harm than good because they viewed facts through the distorting medium of their own passions. Hegel had a philosophy of history which did not suffer from any lack of comprehensiveness, since it started from earliest time and continued into an indefinite future. But the chief lesson of history which he sought to inculcate was that from the year A.D. 400 down to his own time, Germany had been the most important nation and the standard bearer of progress in the world. Perhaps one could stretch the comprehensiveness that constitutes wisdom to include not only intellect but also feeling. It is by no means uncommon to find men/women whose knowledge is wide but those feelings are narrow. Such men / women lack what I am calling wisdom. I think the essence of wisdom is emancipation, as far as possible, from the tyranny of the here and the now. We cannot help the egoism of our senses. Sight, sound and touch are bound up with our own bodies and cannot be made impersonal. Our emotions start similarly from ourselves. An infant feels hunger or discomfort; gradually with the years his horizon widens, and, in proportion as his thoughts and feelings become less personal and less concerned with his own physical states, he achieves growing wisdom. This is of course a matter of degree. No one can view the world with complete impartiality; however, it is possible to make a continual approach towards impartiality, on the one hand, by knowing things somewhat remote in time or space, and, on the other hand, by giving to such things their due weight in our feelings. It is this approach towards impartiality that constitutes growth in wisdom. Perhaps in this sense the wisdom can be taught. I think that this teaching should have a larger intellectual element than has been customary in what has been thought of as moral instruction. I think that the disastrous result of hatred and narrow mindedness to those who fed them can be pointed out incidentally in the course of giving knowledge. Knowledge and morals ought not to be too much separated. It is true that the kind of specialised knowledge which is required for various kinds of skills has very little to do with wisdom. But it should be supplemented in education by wider surveys calculated to put it in its place in the totality of human activities. Even the best technicians should also be good citizens, i.e. citizens of the world and not of any one nation. With every increase of knowledge and skill, wisdom becomes more necessary for every such increase augments our capacity of realising our purposes, and therefore augments our capacity for evil, if our purposes are unwise. The world needs wisdom as it has never needed it before; and if knowledge continues to increase, the world will need wisdom in the future even more than it does now.

27. According to the author what results in growth of wisdom?
(a) Widening knowledge and narrowing feelings
(b) Acquiring specialised knowledge which is required for various kinds of skills
(c) Viewing the world with complete impartiality
(d) None of the above

Sol. *(c)* According to the author, our impartiality results in the growth of wisdom.

28. According to the author the essence of wisdom is ……
(a) Deliverance from the oppression of here and now
(b) Subduing from the oppression of here and now
(c) Captivity from the oppression of here and now
(d) All of the above

Sol. *(b)* According to the author, the essence of wisdom is subduing from the oppression here and now. To get emancipated or free from ''the tyranny of the here and the now''.

29. What according to the author is the relationship between knowledge and wisdom?
(a) As human wisdom increases there is increase in knowledge created
(b) As knowledge keeps on increasing there is lesser need of wisdom
(c) As knowledge keeps on increasing there is a higher need for wisdom
(d) As growth in wisdom stops, knowledge creation stagnates.

Sol. *(c)* According to the author, as knowledge keeps on increasing there is a higher need for wisdom.

30. The example used by the author to explain the ways in which the pursuit of knowledge can be harmful, unless combined with wisdom, is

(a) the space mission
(b) medicine that lowers infant mortality across the world
(c) the progress of Germany
(d) None of the above

Sol. *(b)* The example used by the author to explain the ways in which the pursuit of knowledge can be harmful, unless combined will wisdom, is medicine that lowers infant mortality across the world.

31. What factors according to the author, contribute to wisdom?
(a) A sense of proportion, giving knowledge, study of history, emancipation
(b) A sense of proportion, dignity, knowledge, skill
(c) Comprehensiveness, a sense of proportion, awareness of the end of human life, emancipation from the tyranny of the present
(d) None of the above

Sol. *(c)* According to the author, comprehensiveness, a sense of proportion, awareness of the end of human life, emacipation from tyranny of the present, together contribute to wisdom.

Directions (Q.Nos. 32 and 33) *Read each of the components of the given sentences and mark the component with grammatical error.*

32. I. He is capable at
II. twisting any fact
III. without any suspicion
IV. at any time
(a) Only I (b) Only II (c) Only III (d) Only IV

Sol. *(a)* In statement I 'at' must be replaced by of to make the sentence grammatically correct.

33. I. My cousin brother, who lives
II. in Goa, is eager to visit us
III. in Mumbai and aspires to have
IV. a glimpse of the city
(a) Only I (b) Only II
(c) Only III (d) Only IV

Sol. *(a)* In statement I, the punctuation (,) used after my cousin brother must be removed to make the sentence grammatically correct.

34. Match the following idiomatic references to parts of the human anatomy.

List I	List II
A. Palm	1. could not tolerate the insult.
B. Foot	2. to look at with energy and desire.
C. Eye	3. to put the blame on someone else
D. Stomach	4. forced to pay the bill.

Codes

	A	B	C	D		A	B	C	D
(a)	4	2	1	3	(b)	1	3	2	4
(c)	3	4	2	1	(d)	2	4	1	3

Sol. *(c)* Parts of human anatomy and their idiomatic phrases are :
1. Palm is used when blame is put on someone else.
2. Foot is used when someone is forced to pay the bill.
3. Eye is used to look at someone with envy and desire.
4. Stomach is used to refer to the act of not tolerating the insult.

Hence , (c) is the correct option.

SECTION B : ANALYTICAL & LOGICAL REASONING

35. Choose the word that will complete the second pair in the same way as first pair
Royal Bengal Tiger : India :: Snow Leopard:
(a) Sri Lanka (b) Bangladesh
(c) Pakistan (d) Afghanistan

Sol. *(c,d)* Royal Bengal Tiger resides in India, same way Snow Leopard which resides in cold areas is found in Afghanistan as well as in Pakistan.

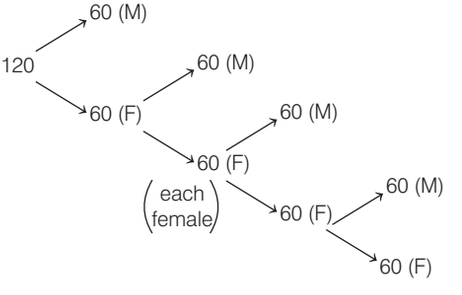
36. A new species lays exactly 120 eggs out of which 50% are male and 50% are female.. The female insect hatch and grow in a span of 20 days to lay eggs by themselves. On 1st April, 2018, an insect laid 120 eggs. Find how many eggs will be hatched, (approx) by the end of may 2018.
(a) 216000 (b) 25920000
(c) 432000 (d) None of these

Sol. *(d)* Eggs laid by insect on 1st April, 2018 are 120 eggs.

Number of males $= \frac{50}{100} \times 120 = 60$

Number of females $= \frac{50}{100} \times 120 = 60$

Now,



Hence, total number of eggs hatched
$= 60 \times 60 \times 60 \times 60 = 12960000$

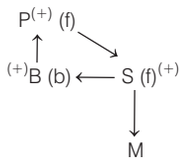
37. A+ B means A is sister of B, A/B means A is son of B, A = B means A is brother of B, A @ B means A is father of B.

Which of the following shows M is grandson of P?

(a) $P = \frac{Q}{M}$ (b) P @ B = S @ M

(c) P @ Q + M (d) P/Q/S + M

Sol. *(b)*



From the given options, option (b) only fits for the solution. Although, in this option also it is not confirmed whether M is grandson or granddaughter of P.

38. In a certain code language, if INDIA is written as 95491, then JAPAN is written as

(a) 81815 (b) 11614 (c) 11715 (d) 101614

Sol. *(c)* The alphabets from A to I are coded as numerals 1 to 9. The same trend starts from J to R and from S to Z.

A	B	C I
1	2	3 9
J	K	L ..... R
1	2	3 9
S	T	U ..... Z
1	2	3 8

Hence, from the above codes,

I	N	D	I	A
9	5	4	9	1

Similarly,

J	A	P	A	N
1	1	7	1	5

39. If 1st June, 2013 is Saturday, then 1st June, 1981 is

(a) Monday (b) Tuesday

(c) Saturday (d) Thursday

Sol. *(a)* From 1st June, 2013 to 1st June, 1981 there are 32 yr. So, there are 32 odd days.

The leap years between 1981 to 2013 are 1984, 1988, 1992, 1996, 2000, 2004, 2008, 2012

$\therefore$ Total = 8 yr

Hence, the total number of odd days

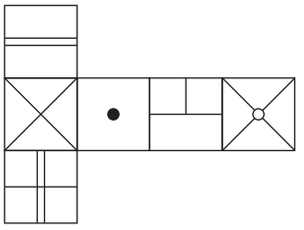
$$= (32 + 8) \div 7$$

$$= \frac{40}{7} \text{ days} = 5 \text{ odd days}$$

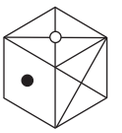
As 1st June, 2013 is Saturday.

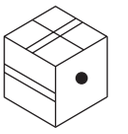
1st June, 1981 will be 5 days behind Saturday that is Monday.

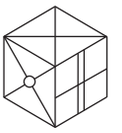
40.

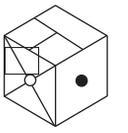


An unfolded dice is given above. Find the correct answer which can be formed by folding the unfolded dice.

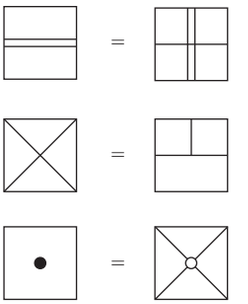
(a) 

(b) 

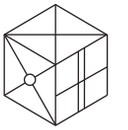
(c) 

(d) 

Sol. *(c)* The opposite faces of the dice will be



Hence, the dice formed will be



41. Fill in the missing block logically

6	2	5	4
4	1	3	2
152	7	98	?

(a) 58 (b) 48 (c) 56 (d) 72

Sol. *(c)* $6^3 - 4^3 = 216 - 64 = 152$

$2^3 - 1^3 = 8 - 1 = 7$

$5^3 - 3^3 = 125 - 27 = 98$

$4^3 - 2^3 = 64 - 8 = 56$

42. Vijay's grandfather has an old cuckoo clock. It takes 5s for the cuckoo clock to chime 5 cuckoos. How long will it take to chime 10 cuckoos?

(a) 10 s (b) 11 s (c) 10 : 25 s (d) 11 : 25 s

Sol. *(d)* It takes 5 s for the cuckoo clock to chime 5 cuckoos. i.e 1st cuckoo clock chimes at $t = 0$ s

5th cuckoo clock chimes at $t = 5$ s

Time interval $= \frac{5}{4}$ s $= 1.25$ s

For the 9th chime $= 9 \times 1.25 = 11.25$ seconds

Hence, the time taken to chime 10 cuckoos is 11 : 25 s.

43. A person wants a house such that all sides of the house face North he should build the house
(a) on the South pole
(b) on the North pole
(c) at the equator
(d) at an angle of 120° with the East direction

Sol. *(a)* The house build at South pole has all its sides facing North. And the house build at North pole has all its sides facing South.

44. In a shouting game, Rohit and Shiv responded in the proportion of Ram and Shyam's shouting.
Study the table and answer accordingly, what will come at the place of question mark(?)

	Ram	Shyam	Rohit	Shiv
9:00 am	4	4	7	7
9:15 am	3	6	2	4
9:30 am	2	6	2	?

(a) 4 (b) 8 (c) 6 (d) 2

Sol. *(c)* At 9:00 am the ratio of shouting of Ram and Shyam is 1 : 1, hence the responding of Rohit and Shiv is at a ratio of 1 : 1.
Similarly,
At 9 : 15 am the ratio is 1 : 2.
At 9 : 30 am the ratio is 1 : 3.
Hence, $\frac{1}{3}=\frac{2}{?}$
$?=6$

45. A statement is followed by two Courses of action I and II. Assume everything in the statement to be true. Decide which of the two given suggested courses of action logically follows for pursuing.

Statement Since 2018, the bulk of India's population has comprised of young working people, much more than the dependent population, children below 5 yr of age and old people above 65 yr of age. This trend will continue for the next 55 yr.

Courses of Action

I. There will be a huge increase in the GDP of the country.
II. According to a report by UNFPA, this population will be able to contribute effectively, if good health facilities, education and proper infrastructure are provided to the whole population.

(a) None follows (b) Only I follows
(c) Only II follows (d) Both I and II follow

Sol. *(c)* Only II logically follows, as if good health facilities, education and proper infrastructure are provided to the whole population, the population will be of working people for the next coming years.

46. A clock gains 10 min a day. The clock was corrected at 6:00 am. what will be the correct time when the clock shows 11:00 am, following day?
(a) 10 : 48 am (b) 11 : 00 am (c) 11 : 45 am (d) 10 : 30 am

Sol. *(a)* Normal clock - in 24 h = 1440 min
Defective clock - in 24 h = 1450 min
From today's 6 am to the following day 11:00 am.
So, it will be 29 hours.
For the defective clock the minutes will be $29\times60=1740$ min
Now, following the ratio pattern,
Normal clock – 1440 ?
Defective clock – 1450 1740
29×50 29×60
Ratio = 5 : 6
Hence, $\frac{5}{6}=\frac{1440}{?}$
$?=\frac{1440\times6}{5}=1728$ min
1728 min = 28 h 48 min which will be 10:48 am.

47. Vishal takes 3 pills with a gap of 30 min. What is the minimum time by which Vishal will get rid of his pain?
(a) 90 min (b) 60 min (c) 58 min (d) 100 min

Sol. *(b)* Let the time at which pill is taken be '*t*'. So, first pill taken at $t_1=0$ min, 2nd pill taken at $t_2=30$ min and 3rd pill taken at $t_3=60$ min
Hence, after 60 min Vishal will get rid of his pain.

48. In a mobile manufacturing company 6 staff members packed 6 mobiles in 6 min. The management wants 60 mobile to be packed in 60 min. How many staff member in total are required?
(a) 60 (b) 360 (c) 600 (d) 6

Sol. *(d)* $\frac{M_1T_1}{W_1}=\frac{M_2T_2}{W_2}$ where, M = Number of mens
T = Time taken for doing the work, W = Work done
$M_1=6$ $M_2=x$
$T_1=6$ min $T_2=60$ min
$W_1=6$ mobiles $W_2=60$ mobiles
$\therefore\ \frac{6\times6}{6}=\frac{x\times60}{60}$
$x=6$ staff members
Hence, 6 staff members in total are required to pack 60 mobiles in 60 min.

49. Choose the word that will complete the second pair in the same way as first pair.
Ornithologist : Bird :: Herpetologist:
(a) Snakes (b) Reptiles
(c) Mammals (d) None of these

Sol. *(b)* As ornithologist studies birds, similarly herpetologist studies reptiles.

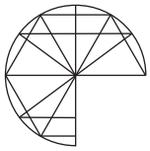
50. Amar consumed 100 laddoos from Monday to Friday. Each day, he consumed 6 more laddoos than the previous day. How many laddoos did he consume on Wednesday?
(a) 12 laddoos (b) 20 laddoos (c) 24 laddoos (d) 8 laddoos

Sol. *(b)* The total laddoos consumed in the 5 days from Monday to Friday = 100
Let laddoos consumed on Monday = a
There is an increase of 6 laddoos in consumption from the previous day.
Therefore.
$a + (a + 6) + (a + 12) + (a + 18) + (a + 24) = 100$
$5a + 60 = 100$
$a = 8$
Laddoo consumed on Monday is 8. Laddoos consumed on Wednesday = 8 + 12 = 20 laddoos

51. Find the missing term in the following series
0, 1, 2, 5, 20, 25, ?, 157,
(a) 35 (b) 150 (c) 145 (d) 40

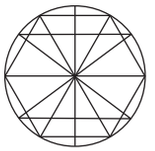
Sol. *(b)* $0 + 1 = 1$
$1 \times 2 = 2$
$2 + 3 = 5$
$5 \times 4 = 20$
$20 + 5 = 25$
$25 \times 6 = 150$
$150 + 7 = 157$
Hence, option (b) is the answer.

52. Which of the answer figures given will complete the pattern of the question figure?

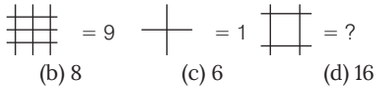


(a) (b) (c) (d)

Sol. *(a)* Clearly, option figure (a) completes the original figure which looks like the figure as shown below.

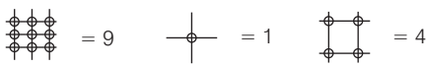


53. Find what will come at the place of question mark(?).



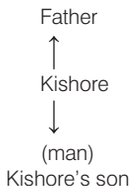
(a) 4 (b) 8 (c) 6 (d) 16

Sol. *(a)* The intersection points in the figure are taken as the answer



54. Kishore says, "that man's father is my father's son". How is that man related to Kishore?
(a) Uncle (b) Son
(c) Kishore himself (d) Cousin

Sol. *(b)*



Kishore's father's son will be either Kishore him self or his brother. So, the man will be either Kishore's son or his nephew. From the given options, the man will be Kishore's son.

55. How many times does the letter 'A' appear from 0 to 100?
(a) 2 (b) 5
(c) 1 (d) None of these

Sol. *(d)* The letter 'A' appears first time in the word 'thousand'.
So, the answer will be (zero) '0' which is not given in the options. So, option (d) will be the answer.

56. Population of Timbaktoo (2 yr ago) was 125000. Due to natural calamities people started migrating. So, population decreased at the rate of 4% per annum. How many migrated from this town in past 2 yr?
(a) 115200 (b) 105000
(c) 9800 (d) None of these

Sol. *(c)* The population 2 yr ago = 125000
Decrease in population = 4% per annum
The present population of the town
$$= \frac{96}{100} \times \frac{96}{100} \times 125000 = 115200$$
People migrated from the town in past 2 yr
$$= 125000 - 115200 = 9800$$

Directions (Q. Nos. 57-59) *Read the given information carefully and answer the questions given below.*

Five movies-Do Bigha Jameen, Sholay, 3 idiots, Chak De, Aanand screening on Monday to Friday in any order. Movie screened on Friday remains till Sunday. Screening of Do Bigha Jameen and Chak De should not be on first and last day. Chak De should be followed by Aanand. Sholay is screened immediately after Do Bigha Jameen. There is only one movie screened between Sholay and 3 idiots.

57. Which movie was screened on Friday?
(a) Aanand (b) Chak De
(c) 3 Idiots (d) Do Bigha Jameen

58. Sholay was screened on which day?
(a) Monday (b) Friday
(c) Wednesday (d) None of these

59. 3 idiots was screened on which day?
(a) Friday (b) Wednesday
(c) Tuesday (d) Monday

Solutions (Q. Nos. 57-59)

Mon - 3 idiots Tue - Do Bigha Jameen
Wed - Sholay Thurs - Chak De
Fri - Aanand

57. *(a)* The movie Aanand was screened on Friday.

58. *(c)* The movie Sholay was screened on Wednesday.

59. *(d)* The movie 3 idiots was screened on Monday.

60. Rahul asked Shyam to find N such that $N! > 10^N$ and must be the smallest integer. Shyam calculates and find N is between 10 to 15, but he asks his friend Sohan who says it is between 16 to 20. Then, Shyam asks his neighbour Suresh who says it is between 21 to 25. Then, he asks his cousin Sonal who says it is 15 between 26 to 31. Who is correct?
(a) Suresh (b) Sonal
(c) Shyam (d) None of them

Sol. *(a)* From 1, 2, 3, 4 ..... 19, 20
it is not possible i.e $N! \not> 10^N$
For $25! > 10^{25}$
It is possible.
Hence, Suresh was correct.

Directions (Q.Nos. 61-65) *Read the given information carefully and answer the questions given below.*

Eight boxes namely H to O are placed in a rack one above the other. The lowermost rack is numbered as 1 and above is 2 and so on. Each of the boxes is having different colors among Blue, Green, Yellow, Red, White, Orange, Indigo and Brown. Each of the boxes is having different serial numbers among 19, 23, 17, 31, 34, 37, 27 and 21. All the above information is not necessarily in the same order.

There are three boxes placed between the box which having serial number as 27 and the Indigo colored box. Box J's serial number is not 27. The Indigo colored box is placed adjacent to blue colored box. Box I's serial number is 37 and is not placed at the topmost rack. Box M is yellow colored and placed adjacent to the Red colored box. Box H is placed in the even numbered rack above fourth and it is placed adjacent to box which has serial number as 31. Not more than five boxes are placed above the Red colored box. Box J is placed in the seventh rack and Box H is placed adjacent to Box J. Two boxes are placed between the boxes having the serial number 21 and 34. Box J is White colored. Maximum numbers of boxes are placed between the Brown colored box and Orange colored box. Box H is not Brown colored. Box L is Green colored and it is placed adjacent to the Orange colored box. Box N's serial number is 21 and placed in the lowermost rack. The box which has serial number as 23 is placed three boxes above the Yellow colored box. Box O's serial number is 19. The box which has serial number as 17 is placed above the box which has serial number as 34.

61. What is the serial number of Box H?
(a) 23 (b) 33 (c) 17 (d) 27

62. Which among the following combination is true?
(a) Box N - Brown - 21 - 1st rack
(b) Box M - Yellow - 34 - 3rd rack
(c) Box K - Blue - 31 - 5th rack
(d) Box O - Orange - 19 - 8th rack

63. In certain way Box O is related to serial number 17, Box H is related to serial number 34 and in same way Box M is related to which of the following serial number?
(a) 21 (b) 23 (c) 37 (d) 27

64. Which of the following statement is true?
(a) Only two boxes are placed below Box M
(b) Box H is placed two boxes above the Red colored box
(c) Box L and Box K are placed adjacent to the box which has serial number 34
(d) None of the above

65. Three of the following four are alike in a certain way and hence form a group. Which of the following does not belong to the group?
(a) Box L - 34 (b) Box K - 23
(c) Box H - 19 (d) Box O - 17

Solutions (Q.Nos. 61-65)

Rack	Box	Color	Serial Numer
8	O	Brown	19
7	J	White	23
6	H	Indigo	17
5	K	Blue	31
4	M	Yellow	34
3	I	Red	37
2	L	Green	27
1	N	Orange	21

61. *(c)* The serial number of Box H is 17.

62. *(c)* Box K - Blue - 31 - 5th rack is the correct combination.

63. *(d)* As, Box O $\xrightarrow{-2}$ Serial number 17
and Box H $\xrightarrow{-2}$ Serial number 34
Similarly, Box M $\xrightarrow{-2}$ Serial number 27
Box M is related to serial number 27.

64. *(d)* None of the given option is true. Hence, the correct answer is option (d).

65. *(b)* Following the common explanations, Box K is placed in 5th rack and Box J (serial number 23) is placed in 7th rack i.e. odd number combination. Hence, option (b) does not belong to the group.

Directions (Q.Nos. 66 and 67) *In each group of question below are two conclusions followed by five set of statements. You have to choose the correct set of statements that logically satisfies given conclusions.*

66. Conclusions

Atleast some states are district.

Some towns being district is a possibility.

Statements

I. All districts are states. Some states are village. No town is district.

II. No districts is states. Some states are towns. No town is village.

III. All districts are states. Some states are towns. No town is village.

IV. No districts is states. Some states are towns. No town is village.

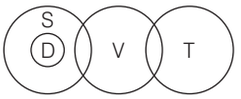
V. All districts are states. Some states are village. Some town are village.

(a) Only Statement I (b) Both Statements II and III

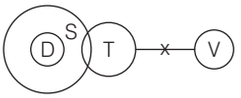
(c) Only Statement VI (d) Both Statements III and V

Sol. *(d)*

Statement V



Statement III



Hence, both Statements III and V follow.

67. Conclusions

No car is a bike.

All bikes being vans is a possibility.

Statements

I. All scooters are bikes. No bike is a moped. All cars are bikes. All mopeds are vans

II. Some scooters are bikes. No bike is a van. All cars are mopeds. All mopeds are vans.

III. Some scooters are bikes. No bike is a moped. All cars are bikes . All mopeds are vans.

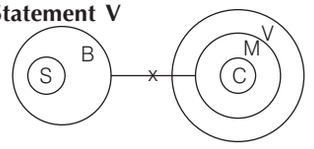
IV. All scooters are bikes. No bike is a van. All cars are mopeds . All mopeds are vans.

V. All scooters are bikes. No bike is a moped. All cars are mopeds . All mopeds are vans

(a) Only Statement V (b) Only Statement I

(c) Only Statement III (d) Both Statements III and IV

Sol. *(a)* **Statement V**



Hence, only Statement V follows.

68. In a coded language LAUGH is written as 62 7 107 37 42, then SMILE will be written as

(Note Do not include spaces in your answer.)

(a) 97 67 47 62 27 (b) 67 42 47 62 27

(c) 97 65 45 62 28 (d) 97 67 47 61 27

Sol. *(a)* LAUGH is written as 62 7 107 37 42

In the alphabetical series order.

$L = 12$ $A = 1$ $U = 21$ $G = 7$ $H = 8$

The sequence is

$(n \times 5) + 2$

$62 = (12 \times 5) + 2$

$7 = (1 \times 5) + 2$

$107 = (21 \times 5) + 2$

$37 = (7 \times 5) + 2$

$42 = (8 \times 5) + 2$

Similarly, S M I L E

$S = 19$ $M = 13$

$I = 9$ $L = 12$ $E = 5$

$(19 \times 5) + 2 = 97$

$(13 \times 5) + 2 = 67$

$(9 \times 5) + 2 = 47$

$(12 \times 5) + 2 = 62$

$(5 \times 5) + 2 = 27$

Hence, the code for SMILE will be

97 67 47 62 27

69. If '+' means '×', '–' means '÷', '÷' means '+' and '×' means '–', then what will be the value of $16 \div 64 - 4 \times 4 + 3 = ?$

(a) 20 (b) 52 (c) 15 (d) 12

Sol. *(a)* $16 \div 64 - 4 \times 4 + 3 = ?$

$16 + 64 \div 4 - 4 \times 3 = ?$

$16 + 16 - 4 \times 3 = ?$

$32 - 12 = ?$

$? = 20$

70. The question below is followed by two Arguments I and II. You have to decide which of the argument is a 'strong' argument and which is a 'weak' argument?

Statement Should young entrepreneurs be encouraged?

Arguments I. Yes, they will help in the industrial development of the country.

II. Yes, they will reduce the burden on the employment market.

(a) If only Argument I is strong

(b) If only Argument II is strong

(c) If both I and II are strong

(d) If either I or II is strong

Sol. *(c)* It is very clear that encouragement of the young entrepreneur will open up the fields for setting up of new industries. Therefore, it will help in industrial development. Consequently, more job opportunities will be created. Thus, both the arguments are strong.

SECTION C : QUANTATIVE DATA INTERPRETATION & SUFFICIENCY

71. Solve $\sqrt{\dfrac{\left(3\frac{1}{4}\right)^4 - \left(4\frac{1}{3}\right)^4}{\left(3\frac{1}{4}\right)^2 - \left(4\frac{1}{3}\right)^2}} = ?$

(a) $7\frac{7}{12}$ (b) $6\frac{5}{12}$ (c) $5\frac{5}{12}$ (d) $9\frac{7}{12}$

Sol. (c) $\sqrt{\dfrac{a^4 - b^4}{a^2 - b^2}} = \sqrt{\dfrac{(a^2 - b^2)(a^2 + b^2)}{(a^2 - b^2)}}$

$a = 3\frac{1}{4}, \quad b = 4\frac{1}{3} \quad [\because (a^2 - b^2) = (a - b)(a + b)]$

$= \sqrt{a^2 + b^2} = \sqrt{\left(3\frac{1}{4}\right)^2 + \left(4\frac{1}{3}\right)^2}$

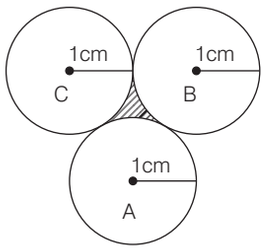
$a = 3\frac{1}{4}, \quad b = 4\frac{1}{3}$

$= \sqrt{\left(\frac{13}{4}\right)^2 + \left(\frac{13}{3}\right)^2} = 13\sqrt{\frac{1}{16} + \frac{1}{9}}$

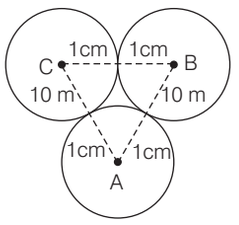
$= 13 \times \sqrt{\frac{9 + 16}{16 \times 9}} = 13 \times \sqrt{\frac{25}{144}}$

$= 13 \times \frac{5}{12} = \frac{65}{12} = 5\frac{5}{12}$

72. What is the area of the shaded portion.



Sol.



ΔABC is equilateral triangle of side 2 cm.

Then, area of $\Delta ABC = \frac{\sqrt{3}}{4} \times 2 \times 2 = \sqrt{3}$

$\left[\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} \times (\text{side})^2\right]$

Area of 3 sector of angle 60° each and radius 1 cm each.

$= 3 \times \pi \text{ (radius)}^2 \times \dfrac{\theta}{360°}$

$= 3 \times \pi (1)^2 \times \dfrac{60°}{360°} = \dfrac{\pi}{2}$

Area of shaded portion

= Area of ΔABC – Area of 3 sectors

$= \sqrt{3} - \dfrac{\pi}{2} = \dfrac{2\sqrt{3} - \pi}{2}$

73. Length of two trains is 150 m each. When they are moving in opposite direction. They cross each other in 20 s and when they are moving in same direction. Then the faster train passed slower train in 40 s. Then find the speed of faster train.

(a) 10.50 m/s (b) 11.25 m/s
(c) 30.75 m/s (d) 25 m/s

Sol. (b) Let speed of faster train = x m/s

Let speed of slower train = y m/s

Distance = Speed × Time

Relative speed (in opposite direction) = $(x + y)$ m/s

Relative speed (in same direction) = $(x - y)$ m/s

In opposite direction

$x + y = \dfrac{150 + 150}{20} \Rightarrow x + y = 15 \text{ m/s} \quad \ldots(i)$

In same direction,

$x - y = \dfrac{150 + 150}{40} = \dfrac{300}{40}$

$\Rightarrow \quad x - y = 7.5 \quad \ldots(ii)$

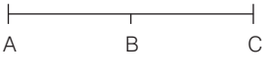
On solving Eqs. (i) and (ii), we get

$x = 11.25$ m/s

74. Brigadier Rastogi travels from A to B at 40 km/h on bike. He travels from B to C at 10 km/h on cycle. The distance from A to B is equal to B to C. Then he travels from C to A *via* B at 24 km/h by auto-rickshaw. Find his average speed.

(a) 20.5 km/h (b) 21 km/h
(c) 18 km/h (d) 19.2 km/h

Sol. (d)



Let distance from A to B LCM of 24, 10, 40 = 120 km

Time from A to B by bike $= \dfrac{120}{40} = 3$ h

Time from B to C by cycle $= \dfrac{120}{10} = 12$ h

$\left[\because \text{Time} = \dfrac{\text{Distance}}{\text{Speed}}\right]$

Time from C to A by autorickshaw $= \dfrac{240}{24} = 10$ h

$\therefore$ Average Speed $= \dfrac{\text{Total distance}}{\text{Total time}} = \dfrac{4 \times 120}{3 + 12 + 10}$

$= \dfrac{4 \times 120}{25}$

$= 19.2$ km/h

75. Rohan and Rahul are 144 km apart on point *A* and point *B* respectively. Rohan travels constantly at 8 km/h. Rahul travels 4 km in first hour, 5 km in second hour, 6 km in third hour and so on. Find the point where they well meet.

(a) 70 km from point A
(b) 74 km from point A
(c) 70 km from point B
(d) Exactly at midway between A and B

Sol. (d) Speed of Rohan = 8 km/h

Speed of Rahul in start = 4 km/h

and Rahul's speed increases 1 km/h in ever hour.

Relative speed in opposite direction AP series is formed = 12, 13, 14 ... = Cover distance per hour

Let they meet after n hour.

According to the question,

Distance between Rohan and Rahul

= Cover distance in n hour

$$144 = \frac{n}{2}[2\times 12 + (n-1)1] \quad \left[\because S_n = \frac{n}{2}[2a + (n-1)d]\right]$$

$$288 = n(24 + n - 1)$$

$$288 = n(23 + n)$$

$$n^2 + 23n + 288 = 0$$

$$n^2 - 9n + 32n + 288 = 0$$

$$(n-9)(n+32) = 0$$

$n = -32$ [it is not possible]

$n = 9$ [it is possible]

So, after 9 h Rahul and Rohan meets, then distance covered by Rohan = $9 \times 8 = 72$ km

It means, they meet exactly at midway between *A* and *B*.

76. Find the unit's place digit of

$(1!)^{1!} + (2!)^{2!} + (3!)^{3!} + \ldots + (100!)^{100!}$

(a) 2 (b) 5 (c) 7 (d) 9

Sol. (c) $5! = 5\times 4\times 3\times 2\times 1$

$= 120$

Unit digit for (5!, 6!, 7!, 8! ... ∞) is zero.

Then, $(1!)^{1!} + (2!)^{2!} + (3!)^{3!} + (4!)^{4!}$

$= (1)^1 + (2)^2 + (6)^6 + (24)^{24}$

Unit place of these numbers = 1 + 4 + 6 + 6 = 17

∴ Required unit digit = 7.

77. In how many ways can 10 books on mechanics and 8 books on quantum physics be placed in a row such that two books on quantum physics may not be together?

(a) 180 (b) 170 (c) 165 (d) 175

Sol. (c) Required ways = $^{11}C_8 = \frac{11!}{8!\times(11-8)!} = \frac{11!}{8!\times 3!}$

$$= \frac{11\times 10\times 9\times 8!}{8!\times 3\times 2\times 1} = 165$$

78. In an institute, MBA Exam is conducted and sectional cut off has been introduced into it. Any candidate appearing for the exam cannot qualify unless he clears the sectional cut off. If there are 4 sections in the paper, then what is the number of ways on applicant may fail in the exam.

Sol. There is two possibility pass (*P*) and Fail (*F*).

Then, possibility of 4 sections

$$= \begin{bmatrix} PPPF & PPFP & PFPP & FPPP \\ FFFP & FFPF & FPFF & PFFF \\ PPFF & PFFP & FFPP & FPPF \\ PFPF & FPFP & FFFF & PPPP \end{bmatrix} = 16$$

Total possibility = 16

The number of way on applicant may fail in the exam = (16 − 1) = 15

79. In how many ways letters of word POTICA can be arranged such that vowel occupy at odd positions?

(a) 36 (b) 15 (c) 42 (d) 21

Sol. (a) POTICA has three vowel.

Then, the required ways = $3!\times 3! = 6\times 6 = 36$

80. Square, Circle, Hexagon and Octagon has equal perimeter. Which has maximum area?

(a) Square (b) Circle (c) Hexagon (d) Octagon

Sol. (b) If the perimeter of figures is same then, area of figure which has maximum number of sides is maximum.

Then, circle has maximum area.

81. In how many ways can one wrap 3 Kitkat, 2 Fivestar and 3 Bar one. If atleast one Kitkat has to be there in the gift pack and the gift pack has three chocolates.

(a) 56 (b) 10 (c) 46 (d) 35

Sol. (c) Number of ways of the gift pack = $^8C_3 = 56$

Number of ways of the gift pack without Kitkat

$= {}^5C_3 = 10$

Then, number of ways of the gift pack with Kitkat

$= 56 - 10 = 46$

82. The difference between compound and simple interest for a loan is ₹114 when invested for 2 yr. The rate of interest is 6% per annum. Find the loan amount.

Sol. Difference between compound and simple interest for 2 years of a principal *P* with interest rate *r*% per annum

$$= P\left(\frac{r}{100}\right)^2$$

$$\therefore \quad 114 = P\left(\frac{6}{100}\right)^2$$

$$\Rightarrow \quad P = \frac{114\times 100\times 100}{36} \approx ₹\ 31666.66$$

83. $\log a + \log\left[\frac{a^2}{b}\right] + \log\left[\frac{a^3}{b^2}\right] + \ldots = ?$

(a) $\log\left(\frac{a^{n-1}}{b^{n+1}}\right)^{n/2}$ (b) $\log\left(\frac{b^{n-1}}{a^{n+1}}\right)^{n/2}$

(c) $\log\left(\frac{a^{n+1}}{b^{n-1}}\right)^{n/2}$ (d) $\log\left(\frac{b^{n+1}}{a^{n-1}}\right)^{n/2}$

Sol. (c) $\log a + \log\left[\frac{a^2}{b}\right] + \log\left[\frac{a^3}{b^2}\right] + \ldots$

$$= \log\left[a^1 \times \frac{a^2}{b} \times \frac{a^3}{b^2} \times \frac{a^4}{b^3} \ldots\right]$$

$[\because \log(abc\ldots) = \log a + \log b + \log c + \ldots]$

$$= \log\left[\frac{a^{\frac{n(n+1)}{2}}}{b^{\frac{n(n-1)}{2}}}\right] = \log\left[\frac{a^{n+1}}{b^{n-1}}\right]^{\frac{n}{2}}$$

84. What is the coefficient of z^3 in $-7x\,y^2\ z^3a^2b^2$.

(a) -7 (b) $7a^2b^2$ (c) $7xy^2$ (d) $-7xy^2\ a^2b^2$

Sol. (d) Given, $-7x\,y^2\ z^3a^2b^2 = (-7xy^2a^2b^2)\,z^3$

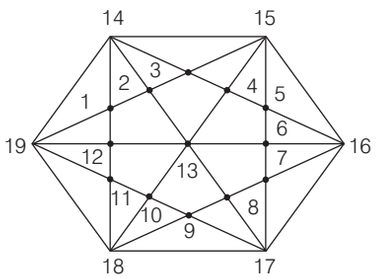
Coefficient of z^3 in $(-7xy^2a^2b^2z^3)$

$= -7xy^2a^2b^2$

85. In a regular hexagon field, ropes are tied to connect all vertices (digonal and side). What is the number of intersection points of ropes?

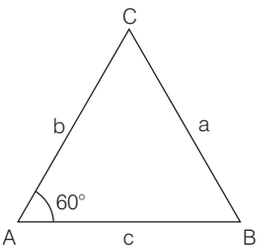
(a) 13 (b) 19 (c) 15 (d) 18

Sol. (b)



We can see that in above figure that there are 19 intersection points.

86. In the given diagram, $\angle CAB = 60°$ and $BC = a$, $AC = b$ and $AB = c$



Then which of the following is correct?

(a) $a^2 + b^2 + c^2 = abc$ (b) $a^2 = b^2 + c^2 - bc$

(c) $c^2 = a^2 + b^2 - ab$ (d) $b^2 = a^2 + c^2 - 2ac$

Sol. (b) Cosine rule

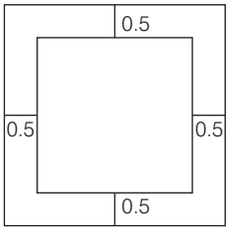
$$\cos 60° = \frac{b^2 + c^2 - a^2}{2bc}$$

$$2bc \times \frac{1}{2} = b^2 + c^2 - a^2$$

$$a^2 = b^2 + c^2 - bc$$

87. In a closed wooden box, length = 20 cm breadth = 14 cm and height = 10 cm and thickness = 5 mm. If weight of empty box is 3.462 kg, then what is the weight of 1 cm^3 of wood?

Sol. Volume of cuboid = Length × Breath × Height



Volume of wood = External volume of box with wood – Internal volume of box without wood

$= 20 \times 14 \times 10 - 19 \times 13 \times 9$

$= 2800 - 2223 = 577\ cm^3$

Total weight of wood = 3.462 kg

Weight of 577 cm^3 of wood = 3462 gm

Then, weight of 1 cm^3 of wood $cm^3 = \frac{3462}{577} = 6$ gm

88. Find the number of zeros at the end of $(5!)^{5!} + (10!)^{10!} + (50!)^{50!} + (100!)^{100!}$

(a) 120 (b) 60 (c) 240 (d) 480

Sol. (a) $(5!)^{5!}$ is add in the remaining numbers then only we have to find number of zero in $(5!)^{5!}$.

$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

Then, $(120)^{120}$ has 120 zero.

89. Ritesh is twice as good as Mitesh. Ritesh takes 30 days less than Mitesh to finish a task. How long will Ritesh and Mitesh take to complete the task together.

(a) 30 days (b) 20 days (c) 60 days (d) 15 days

Sol. (a) Let efficiency of Ritesh = $2x$

and efficiency of Mitesh = x

Mitesh takes time to complete the task = $2x$ days

and, Ritesh taken time to complete the same task

= x days.

According to the question,

$2x - x = 30$

$x = 30$ days

Mitesh's time = $2 \times 30 = 60$ days

Ritesh's time = $x = 30$ days

$\therefore$ 1 day's work Ritesh and Mitesh together

$$= \frac{1}{60} + \frac{1}{30} = \frac{2+1}{60} = \frac{1}{20}$$

$\therefore$ Ritesh and Mitesh complete the task in 20 days.

90. Mohan has a 200 L container. x L of milk is kept in the container. Mohan removes 6L milk and adds 6L water. He again replaces 6L of solution with water. Now milk and water are in the ratio 9 : 16. What is the value of x?

Sol. Add water = Initial quantity of substance

$$\times\left[1-\sqrt{\frac{\text{Final substance}}{\text{Final mixture}}}\right]$$

$$6 = x\left(1-\sqrt{\frac{9}{25}}\right) \Rightarrow 6 = x\left(1-\frac{3}{5}\right) \Rightarrow 6 = \frac{2x}{5} = 15 \text{ litre}$$

91. Indian express travels at 60 km/h and halts. For fixed time every hour. Due to halts, the average speed become 50 km/h. Find the time of halt.

(a) 10 min (b) 20 min (c) 15 min (d) 12 min

Sol. (a) Halt time

$$= \frac{\text{Speed of faster train} - \text{Speed of slower train}}{\text{Speed of faster train}} \times 60$$

$$= \frac{60-50}{60} \times 60 = 10 \text{ min}$$

Directions (Q. Nos. 92-95) *Read the given information carefully and answer the given questions.*

In 2015, 100 aspirants appeared for an exam.

They had to answer four sections Maths, DI, LR and English.

The number of students who qualified in Maths = 55.

The number of students who qualified in LR= 38.

The number of students who qualified in (Maths + English) = 30.

The number of students who qualified in (LR + English) = 15.

The number of students who qualified in (Maths + LR) = 20.

The number of students who qualified in (Maths + LR + English) = 5.

The number of students who qualified in DI = 22.

The number of students who qualified in (DI + LR) = 5.

The number of students who qualified in (DI + Maths) = 5.

The number of students who qualified in (DI + Maths + LR) = 5.

The number of students who qualified in English = 50.

Those who qualified in English, could not qualify in DI section.

92. How many qualified atleast two sections?

(a) 60 (b) 55 (c) 50 (d) 20

93. How many qualified in both Maths and LR, but not in any other section?

(a) 10 (b) 20 (c) 25 (d) 5

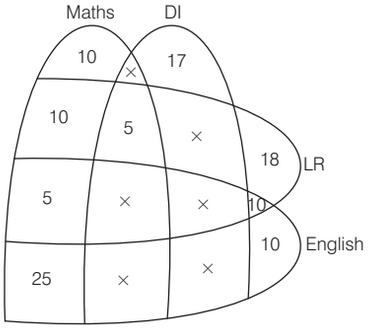
94. How many did not qualify any section?

(a) 20 (b) 10 (c) 0 (d) 5

95. How many qualified only DI section?

(a) 10 (b) 25 (c) 20 (d) 17

Solutions (Q. Nos. 92-95)



92. (b) Qualified at least two section

$= 10 + 5 + 5 + 25 + 10 = 55$ aspirants

93. (a) Qualified in both Maths and LR = 10 aspirants

94. (c) None of the student failed ie 0.

95. (d) Qualified only in DI = 17

Directions (Q. Nos. 96-98)

Year	Quantity of Rice (tonne)	% Change over previous year
20-21	134350	+ 6.25
30-31	1097172	+ 12.5
40-41	264280	+ 11.11
50-51	127890	– 09.09
60-61	201924	+ 20.00
70-71	112325	– 16.66
80-81	213465	– 25.00
90-91	169368	+ 33.33
00-01	100956	+ 50.00
10-11	23800	– 83.33

96. What is the approx production of rice in year 1949-50?

(a) 132650 tonne (b) 160479 tonne

(c) 140679 tonne (d) 120659 tonne

Sol. (c) Production of rice in year 1950-51= 127890 tonne

Production of rice in year 1950-51 is 09.09% less than that of in year 1949-50

Fraction of 09.09 $= \frac{1}{11}$

Then, production of rice in 1949-50

$$= \frac{11}{10} \times \text{production of rice in 1950-51}$$

$$= \frac{11}{10} \times 127890 = 140679 \text{ tonne}$$

97. What is the difference in the production of rice in 1969-70 and 1979-80?

(a) 149830 tonnes (b) 125600 tonnes

(c) 143980 tonnes (d) 162500 tonnes

Sol. (a) Required difference

= (Production of rice in 1979-80)
– (Production of rice in 1969-70)

$$\left[\text{Fraction } 25\% = \frac{1}{4}, 16.66\% = \frac{1}{6}\right]$$

$$= 213465 \times \frac{4}{3} - 112325 \times \frac{6}{5}$$

$$= 284620 - 134790 = 149830 \text{ tonne}$$

98. What is the the approx production in 1959-60?
(a) 168270 tonnes (b) 148070 tonnes
(c) 157270 tonnes (d) 1382890 tonnes

Sol. (a) Production of rice in 1960 -61= 201924 tonne
The production in 1959-60

$$= \frac{5}{6} \times 201924 = 168270 \text{ tonne}$$

99. What is the sum of integers from 113 to 113113, which are divisible by 7?
(a) 899235637 (b) 867265029 (c) 913952088 (d) 752643214

Sol. (c) $119 + 126 + \ldots + 113113$
This is the series of AP.
Let total number of terms be n.

$\therefore \quad l = a + (n-1)d \Rightarrow 113113 = 119 + (n-1) \times 7$

$\Rightarrow \quad 112994 = (n-1) \times 7 \Rightarrow n - 1 = 16142$

$\Rightarrow \quad n = 16143$

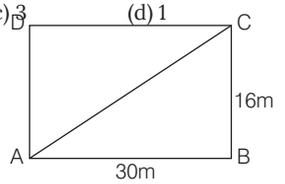
Sum of the series

$$= \frac{n}{2} \text{ (first term + last term)} = \frac{16143}{2}(119 + 113113)$$

$$= \frac{16143}{2} \times 113132 = 913952088$$

100. Robot is 4 m in length and placed at corner of 16 m × 30 m field. The robot is facing diagonally opposite corner and reaches the diagonally opposite corner in 15 s. What is the speed (in m/s) of robot?
(a) 5 (b) 2 (c) 3 (d) 1

Sol. (b) $ABCD$ is rectangle.



Then, $\angle B = 90°$

$$AC^2 = AB^2 + BC^2$$

$$AC^2 = 16^2 + 30^2$$

$$AC = \sqrt{196 + 900}$$

$$= 34\text{m}$$

Length of robot 4 m.
Then, it has to travel = 34 – 4 = 30 m

$$\text{Speed of the robot} = \frac{30}{15} = 2 \text{ m/s}$$

101. In a smart phone factory, Robot A, B, C manufacture 25%, 35% and 40% circuit board respectively. for each of the robot faulty circuit board = 5%, 4% and 2% respectively. What is the probability that the printed circuit board is defective if 1 circuit board is picked at random?
(a) 0.034 (b) 0.027 (c) 0.091 (d) 0.030

Sol. (a) Let total circuit board manufacture in factory = $100x$
Then, the circuit board manufacture by Robot

$$A = 100 \times \frac{25}{100} = 25x$$

$$\text{Robot } B = 100 \times \frac{35}{100} = 35x$$

$$\text{Robot } C = 100 \times \frac{40}{100} = 40x$$

Number of faulty circuit boards by Robot

$$A = 25x \times \frac{5}{100} = 1.25x$$

Number of faulty circuit boards by Robot

$$B = 35x \times \frac{4}{100} = 1.4x$$

Number of faulty circuit boards by Robot

$$C = 40x \times \frac{2}{100} = 0.8x$$

Total number of faulty circuit boards

$$= 1.25x + 1.4x + 0.8x = 3.45x$$

$$\therefore \text{ Required probability} = \frac{3.45x}{100x} = 0.034$$

102. Average stipend of a group of students is ₹ 50 per day. The difference between maximum and minimum stipend is ₹ 45. If these 2 students are excluded, the average decreases by ₹ 1. Minimum earning lie between ₹ 42 to ₹ 47 and the number of students is a prime number, where both the digit is also prime. What is the number of students initially?
(a) 47 (b) 43 (c) 31 (d) 37

Sol. (d) Let the minimum stipend = x
Then, the maximum stipend = $x + 45$
Let number of students = n
Then, total stipend of a group of students = $50n$ …(i)
New average = 49
Then, total stipend of a group of students

$$= 49(n - 2) \quad \ldots(ii)$$

By Eqs. (i) and (ii), we get
The stipend of a two students

$$= 50n - 49(n - 2) = n + 98$$

Then, $n + 98 = x + x + 45$

$$n + 98 = 2x + 45$$

Minimum earning lie between ₹ 42 to ₹ 47.

$$n + 98 = 2(45) + 45 \quad [x = 45]$$

$$n = 37$$

n prime possible when $x = 45$.

103. Valve A fills a bathtub in 10 h and Valve B fills the bathtub in 15 h. When A and B are opened together, later B closed after some time. Total time taken to fill the bathtub is 8 h. When was B closed?
(a) 5 h (b) 3 h (c) 8 h (d) 6 h

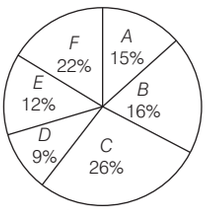
Sol. (b) Valve A fills a bathtub = 10 h
Valve B fills the bathtub = 15 h
Let B closed after x hours.

According to the question,

$$1=\frac{8}{10}+\frac{x}{15} \Rightarrow 1-\frac{4}{5}=\frac{x}{15} \Rightarrow x=3 \text{ h}$$

Directions (Q. Nos. 104 and 105) *Refer to the pie-chart and answer the given questions.*

Percentage of employees in different departments of branch 'XYZ' in the year 2014



Total number of employees = 450

104. In 2014, the number of female employees in department C was 5/13 of the total number of employees in same department. If the number of female employees in department *F* was 4 less than that in department C, what is the number of male employees department *F*?
(a) 41 (b) 42 (c) 58 (d) 54

Sol. (c) Total number of employees in department C

$$=450\times\frac{26}{100}=117$$

Female employees in department C $=117\times\frac{5}{13}=45$

Now, total number of employees in department F

$$=450\times\frac{22}{100}=99$$

Female employees in department

F = 45 – 4 = 41

∴ Male employees in department F = 99 – 41 = 58

105. The number of employees in department *E* is what per cent less than the number of employees in departments *A* and *C* together?
(a) 72 (b) 60 (c) 65 (d) 70

Sol. (d) Total number of employees in department A

$$=450\times\frac{15}{100}=\frac{135}{2}$$

Total number of employees in department C

$$=450\times\frac{26}{100}=117$$

∴ Total number of employees in department (A + C)

$$=\frac{135}{2}+117=\frac{369}{2}$$

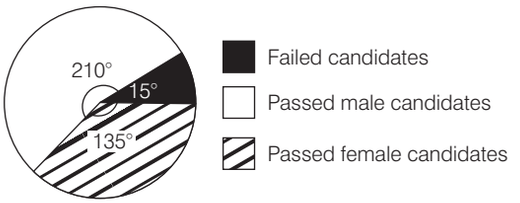
and total number of employees in department E

$$=\frac{12}{100}\times450=54$$

$$\therefore \text{Required percentage}=\frac{\frac{369}{2}-54}{\frac{369}{2}}\times100=\frac{\frac{261}{2}}{\frac{369}{2}}\times100$$

$$=\frac{261}{369}\times100=\frac{29}{41}\times100=70.73\approx70\%$$

Directions (Q. Nos. 106-109) The following pie chart shows the analysis of the result of an examination in which 5 candidates have failed. Study the chart and answer the question given below.



106. The total number of examinees was
(a) 100 (b) 120 (c) 135 (d) 150

Sol. (b) Let total number of candidates be *x*.

By pie chart, $\frac{15^\circ}{360^\circ}\times x=5$ [failed students = 5, given]

$\therefore \quad x=24\times5=120$

107. Percentage of passed female candidates with respect to total examinees is
(a) 30 (b) 37. (c) 40 (d) 45

Sol. (b) Number of passed female candidates

$$=\frac{135^\circ}{360^\circ}\times100\%=37.5\%$$

108. Percentage of passed male candidates with respect to total passed candidates is
(a) 60.8 (b) 56 (c) 71 (d) 58

Sol. (a) Percentage of passed male candidates

$$=\frac{210^\circ}{360^\circ-15^\circ}\times100\%=\frac{210^\circ}{345^\circ}\times100\%=60.8\%$$

109. Ratio of passed male candidates to the successful female candidates is
(a) 9 : 1 (b) 1 : 14
(c) 14 : 9 (d) 9 : 14

Sol. (c) Required ratio $=\frac{210^\circ}{135^\circ}=14:9$

110. Find the value of $m-n$, if

$$\frac{9^n\times3^2\times(3^{-n/2})^{-2}-(27)^n}{3^{3m}\times2^3}=\frac{1}{27}$$

(a) 1 (b) –2 (c) –1 (d) 2

Sol. (a) $\frac{9^n\times3^2\times(3^{-n/2})^{-2}-(27)^n}{3^{3m}\times2^3}=\frac{1}{27}$

$$\Rightarrow \quad \frac{3^{2n}\times3^2\times3^n-3^{3n}}{3^{3m}\times2^3}=\frac{1}{27}$$

$$\Rightarrow \quad \frac{3^{3n+2}-3^{3n}}{3^{3m}\times8}=\frac{1}{27}$$

$$\frac{3^{3n}(3^2-1)}{3^{3m}\times8}=\frac{1}{27}$$

$$\Rightarrow \quad (3^3)^{n-m}=3^{-3}$$

$$n-m=-1 \text{ or } m-n=1$$

SNAP

SYMBIOSIS NATIONAL APTITUDE TEST

SOLVED PAPER 2018*

Time : 2 hrs

INSTRUCTIONS

1. There are 130 questions in this paper.
2. There are four parts. Part A (Analytical & Logical Reasoning), Part B (General Awareness), Part C (Quantitative Aptitude) and Part D (General English).
3. Each Section contains 5 special (Non-MCQs) Questions.
4. Part A carries 40 marks, Part B carries 30 marks, Part C carries 40 marks and Part D carries 40 marks.
5. For each incorrect attempt 1/4th of marks allotted will be deducted.

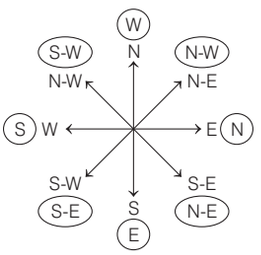
SECTION A : ANALYTICAL & LOGICAL REASONING

1. Three lamps A, B and C are covered by clothes absorbing water, petrol and oil respectively. Which of the following alternatives best represent the situation.
(a) A-off, B-glows and C-off
(b) A-off, B-glows and C-glows
(c) A-glows, B-glows and C-off
(d) All off

Sol. *(a)* A-off, B-glows and C-off.

2. A direction pole was situated on the crossing. Due to an accident the pole turned in such a manner that the pointer which was showing East, started showing South. One traveler went to wrong direction thinking to be West. In what direction actually he was travelling?
(a) East (b) South-West (c) North (d) South

Sol. *(c)* According to the given information,



Now, if a person thinks he is going towards West direction, then he is moving towards North direction actually.

3. There are three cartons, exactly one of which has money. On each box there is a statement, exactly one of which is true.
Box 1 - The money is in this box.
Box 2 - The money is not in this box.
Box 3 - The money is not in Box 1.
Which box has money?
(a) Box 1
(b) Box 2
(c) Box 3
(d) Can't be determined

Sol. *(b)* The money is in Box 2.
Box 1 - The money is in this box. (False)
Box 2 - The money is not in this box. (False)
Box 3 - The money is not in Box 1. (True)
Hence, exactly one statement is true.

4. In a row of boys facing North, A is 16th from the left and C is 16th from the right end. B, who is 4th to the right of A, is 5th to the left of C in the row. How many boys are there in the row?

* Memory Based

Sol. According to the given information,

Row of boys ⇒ 15 boys | 3 boys | 4 boys | 15 boys
A B C

∴ Total number of boys in row
$= 15 + 1 + 3 + 1 + 4 + 1 + 15 = \boxed{40}$

5. What will come in place of question mark (?) in the given series?

5, 9, 16, 29, 54, ?

(a) 200 (b) 103 (c) 106 (d) 154

Sol. *(b)* The pattern of the series is,

5 9 16 29 54 103
×2–1 ×2–2 ×2–3 ×2–4 ×2–5

6. Sunil is the son of Keshav. Simran, Keshav's sister, has a son Maruti and daughter Sita. Prem is the maternal uncle of Maruti. How is Sunil related to Maruti?

(a) Cousin (b) Uncle (c) Brother (d) Nephew

Sol. *(a)* From the given information,

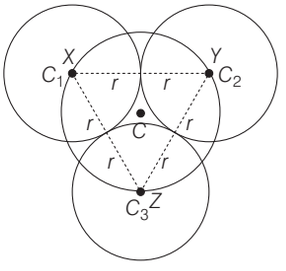
Keshav —Sister→ Simran(–) —Brother→ Prem(+)
Son ↓ Son ↓ Daughter ↘
Sunil(+) Maruti(+) Sita(–)
Sunil(+) ⋯ Cousin ⋯ Maruti(+)

Clearly, Sunil is cousin of Maruti.

7. Three identical cones with base radius '*r*' are placed on their bases so that each one is touching the other two. The radius of circle whose boundary touches the centre of these cones is

(a) smaller than r (b) equal to r
(c) greater than r
(d) depends on the height of cones.

Sol. *(c)* Let C_1, C_2 and C_3 be the base of three cones, of radius 'r'. Let C be the centre of circle formed by joining the centres of C_1, C_2 and C_3 on its boundary.



Let radius of circle with centre C be R.

Now, XYZ will be an equilateral triangle inscribed in circle C.

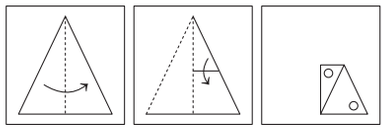
Now, side of triangle $XYZ = r + r = 2r$

We know that, radius of circumcircle $= \dfrac{\text{Side of triangle}}{\sqrt{3}}$

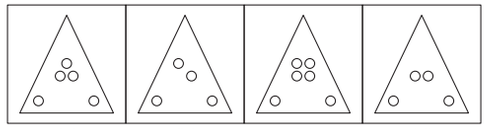
$\therefore \quad R = \dfrac{2r}{\sqrt{3}} = 1.156\, r \;\Rightarrow\; R > r$

8. A piece of paper is folded and cut as shown below in the question figures. From the given answer figures, indicate how it will appear when opened?

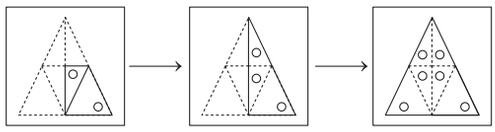
Question Figures



Answer Figures



Sol. *(c)*



9. A total of 32 players participated in a single tennis knockout tournament. What is the total number of matches played in the tournament. No match ends in a tie (knockout means if a player loses, he is out of the tournament)?

Sol. First round $= \dfrac{32}{2}$ matches $= 16$ matches

Players left $= 32 - 16 = 16$

Second round $= \dfrac{16}{2}$ matches $= 8$ matches

Players left $= 16 - 8 = 8$

Third round $= \dfrac{8}{2}$ matches $= 4$ matches

Players left $= 8 - 4 = 4$

Fourth round $= \dfrac{4}{2} = 2$ matches

Players left $= 4 - 2 = 2$

Final round $= 1$ match

∴ Total matches $= 16 + 8 + 4 + 2 + 1 = \boxed{31}$

Direction (Q. No. 10) *In the following question, two statements are given below a question. You have to decide whether the data given in the statement is sufficient to answer the question.*

Among six people A, B, C, D, E and F each having a different height who is the second tallest?

I. Only two people are shorter than D. Only two people are taller than A. E is shorter than C. D is taller than F.

II. Only three people are shorter than A. B is shorter than D but not the shortest. C is the tallest amongst them.

(a) The data in both Statements I and II together are necessary to answer the question
(b) The data in Statement II alone are sufficient to answer the question while the data in Statement I alone are not sufficient to answer the question.
(c) The data even in both Statements I and II together are not sufficient to answer the question
(d) The data either in Statement I alone or Statement II alone are sufficient to answer the question

Sol. *(a)* From Statement I,

C > E, D > F

and __ > __ > A > D > __ > __

From Statement II,

D > B, B is not the shortest.

and C > __ > A > __ > __ > __

From Statements I and II, we get

C > E > A > D > B > F

Clearly, E is second tallest.

Hence, both the statements are required.

Direction (Q. No.11) *In the following question, three statements are given below a question. You have to decide whether the data given in the statement is sufficient to answer the question.*

Among six people P, Q, R, S, T and V, each lives on a different floor of a six storey building having its six floors numbered one to six (the ground floor is numbered 1, the floor above it, number 2 and so on and the topmost floor is numbered 6). Who lives on the topmost floor?

I. There is only one floor between the floors on which R and Q live. P lives on an even numbered floor.

II. T does not live on an even numbered floor. Q lives on an even numbered floor. Q does not live on the topmost floor.

III. S lives on an odd numbered floor. There are two floors between the floors on which S and P live. T lives on a floor immediately above R's floor.

(a) Statements I and II are sufficient
(b) Statements I and III are sufficient
(c) Statements II and III are sufficient
(d) All the statements are necessary

Sol. *(c)* From Statement I,

P lives on an even numbered floor.

One floor is between floors of R and Q.

From Statement II,

T lives on an odd numbered floor and Q lives on an even numbered floor.

From Statement III,

T lives immediately above R's floor. S lives on an odd numbered floor and two floors are between P and S.

From Statements II and III

6	P
5	T
4	R
3	S
2	Q
1	V

Clearly, P lives on topmost floor.

Hence, Statements II and III are necessary.

12. Six lectures are scheduled in a week starting from Monday and ending on Sunday of the same week. Computer Science is not on Tuesday or Saturday. Finance is immediately after Operations. Statistics is not on Friday and there is one day gap between Statistics and Marketing. One day prior to the schedule of Economics there is no lecture (as that day is the off day) and Monday is not the off day. Which lecture is scheduled on Monday.

(a) Operations
(b) Statistics
(c) Computer Science
(d) Marketing

Sol. *(a)* From the given information,

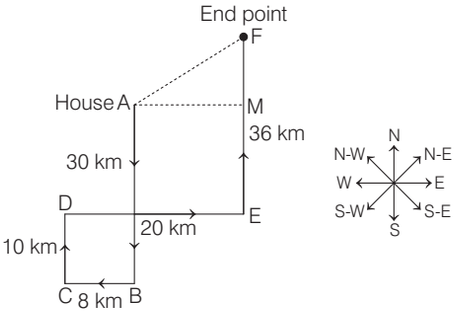
Mon	Tue	Wed	Thu	Fri	Sat	Sun
Operations	Finance	Statistics	Computer Science	Marketing	off day	Economics

Clearly, lecture on Operations is scheduled on Monday.

13. Ritika starts from her house and walks 30 km Southwards, then travels 8 km to the right and then 10 km towards the right, then again travels 20 km towards right and then 36 km towards the left. How far is she from her house and in which direction?

(a) 20 km, North-East
(b) 20 km, North
(c) 20 km, East
(d) 20 km, South-East

Sol. *(a)* From the given information,



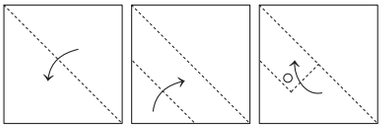
Required distance,

$$AF = \sqrt{AM^2 + MF^2}$$
$$= \sqrt{12^2 + 16^2}$$
$$= \sqrt{400} = 20 \text{ km}$$

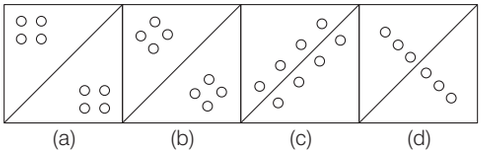
∴ She is 20 km away from her house towards North-East direction.

14. A piece of paper is folded and cut as shown below in the question figures. From the given answer figures, indicate how it will appear when opened.

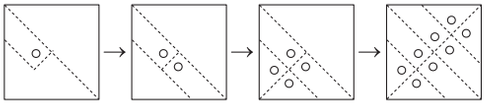
Question Figures



Answer Figures



Sol. *(c)*



Direction (Q. No. 15) *In the following question, three statements are given below a question. You have to decide whether the data given in the statement is sufficient to answer the question.*

15. Point D is in which direction with respect to point B?

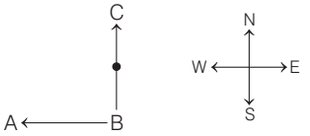
I. Point A is to the West of point B. Point C is to the North of point B. Point D is to the South of point C.

II. Point G is to the South of point D. Point G is 4 m from point B. Point D is 9 m from point B.

III. Point A is to the West of point B. Point B is exactly mid way between points A and E. Point F is to the South of point E. Point D is to the South of point F.

(a) Only I is sufficient (b) I and II are sufficient
(c) Only III is sufficient (d) All are necessary

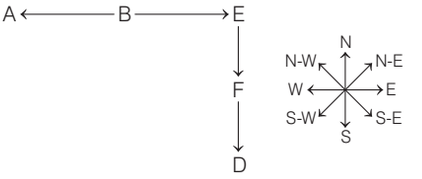
Sol. *(c)* From Statement I,



As, position of D is not clear, we cannot determine the position of D with respect to B.

No conclusion can be drawn from Statement II

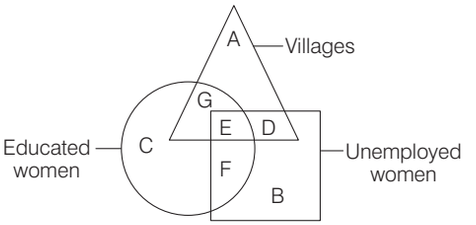
From Statement III,



Clearly, D is in South-East direction of B.

Hence, only Statement III is sufficient.

Direction (Q. No. 16) *Study the diagram given below to answer the question.*



16. Educated unemployed women in villages are represented by which letter?

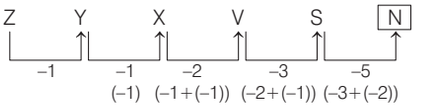
Sol. E represents educated unemployed women who live in villages.

17. What will come in place of question mark (?) in the following series?

Z, Y, X, V, S, ?

(a) O (b) N
(c) P (d) M

Sol. *(b)* The pattern of the series is



Here, the difference for the next term is obtained by adding the difference of first two terms.

18. Three pendulums of length 25cm, 36cm and 49cm are displaced from their equilibrium position. Which of them will move slowest?

(a) with length 25 cm
(b) with length 36 cm
(c) with length 49 cm
(d) All will move with same speed

Sol. *(c)* For a pendulum, $T = 2\pi\sqrt{\frac{l}{g}}$

where, T = Time period and l = length of pendulum

∵ $T \propto l$

∴ The pendulum with longest length will move slowest.

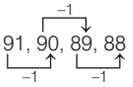
19. What will come in place of question mark (?) in the given series?

16, ?, 68, 88

(a) 06 (b) 18
(c) 63 (d) None of these

Sol. *(a)* To find the missing number, all the numbers must be rotated 180°.

∴ Series become



∴ ? = 06

20. If pig : farrow and lion : cub, then bear : ?

(a) Kit (b) Squeaker (c) Shoat (d) Cub

Sol. *(d)* As, young ones of pig are called farrow and young ones of lion are called cub. Similarly, young ones of bear are called cub.

21. What will come in place of question mark?

2	3
11	5

–5	2
–12	–2

–10	6
?	–10

(a) –70 (b) –106 (c) –15 (d) –90

Sol. *(a)* As, $2 \times 3 + 5 = 11$

$(-5) \times (2) + (-2) = -12$

Similarly, $(-10) \times (6) \times (-10) = ?$

∴ $? = -60 - 10 = -70$

22. What will come in place of question mark?

	5	
3	242	7
	9	

	12	
11	502	13
	14	

	7	
?	662	10
	21	

Sol. Here the pattern is, $5 + 7 + 9 + 3 = 24 \Rightarrow 242$

Here, the central number is obtained by placing '2' at the right side of the sum.

Similarly,

$12 + 13 + 14 + 11 = 50 \Rightarrow 502$

Now, $7 + 10 + 21 + ? = 66$

$\Rightarrow ? = 66 - 38$

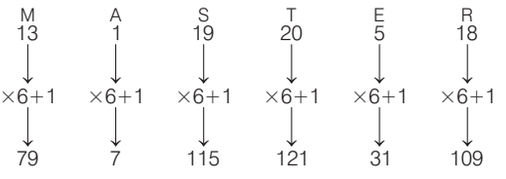
∴ $? = \boxed{28}$

23. In a certain code language 'MASTER' is coded as '79 7 115 121 31 109'. How will 'LAUGH' be coded using the same pattern.

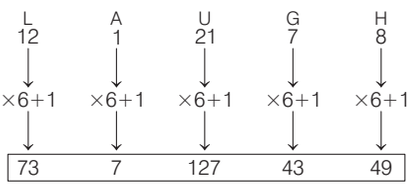
(a) 73 7 68 97 48 (b) 73 7 127 43 49

(c) 125 97 63 4 16 (d) 63 84 69 72 8

Sol. *(b)* As,



Similarly,



24. The calendar for the year 1996 is same for which year?

(a) 2024 (b) 2017 (c) 2004 (d) 1998

Sol. *(a)* Number of odd days in a leap year = 2

and number of odd days in an ordinary year = 1

Also, 1996 is a leap year

∴ The required year which has the same calendar as of 1996 must be a leap year.

Now, odd days between 1996 and 2024 = 35

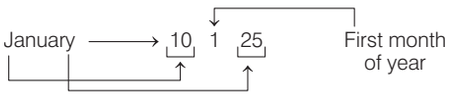
$\Rightarrow \frac{35}{7} = 0$ odd days.

∴ 2024 will have same calendar as 1996.

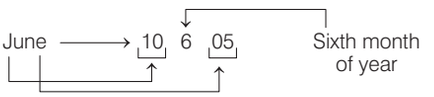
25. In a certain code language 'January' is written as '10125' and 'June' as '10605'. How will July be written in same language?

(a) 108735 (b) 101783 (c) 10673 (d) 10725

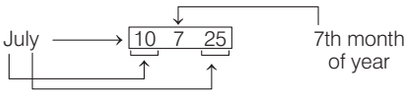
Sol. *(d)* Here, the code for each month is obtained as,



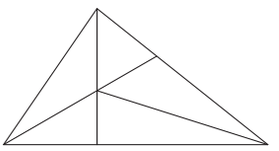
and



Similarly,

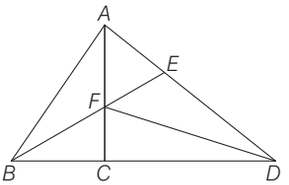


26. Count the number of triangles in the given figure.



(a) 10 (b) 12 (c) 16 (d) 18

Sol. *(b)* Naming the figure,



There are 12 triangles viz., ABF, BFC, CFD, DFE, EFA, ABE, ABC, BFD, AFD, BED, ACD, and ABD.

27. Five girls are sitting on a bench to be photographed. Seema is to the left of Rani and to the right of Bindu. Mary is to the right of Rani. Reeta is between Rani and Mary. Who is second from left in photograph?

Sol. From the given information, the arrangement is

Clearly, Seema is second from left in photograph.

Directions (Q. Nos. 28-35) *Study the given information carefully and answer the following question.*

Ten persons are sitting in twelve seats in two parallel rows containing five people each, in such a way that there is an equal distance between adjacent persons. In Row 1, A, B, C, D and E are seated and all of them are facing South, and in Row 2, P, Q, R, S and T are sitting and all of them are facing North. One seat is vacant in each row. Therefore, in the given seating arrangement each member seated in a row faces another member of the other row. All of them like a different colour i.e. red, blue, yellow, green, white, black, cream, pink, brown and violet.

A sits second to left of person who likes red colour, the one seat which is on the extreme end is adjacent to either A or the person who likes red colour. T sits at one of the extreme ends of the row. There are three persons sitting between T and S, who likes blue colour. Immediate neighbour of T faces B. One of immediate neighbour of B faces R, who likes yellow colour. There are no vacant seats adjacent to R. C likes green colour and sits second to left of vacant seat. One of immediate neighbour of R is Q and he likes white colour. One of immediate neighbour of Q faces D and he likes black colour. The one who likes cream colour sits to the immediate left of the person who likes pink colour. The one who likes brown colour sits third to left of person who likes violet colour.

28. Who among the following sits on the extreme ends of the row?
(a) E, Q (b) B, P (c) D, P (d) D, T

29. Who among the following likes violet colour?
(a) P (b) R (c) Q (d) T

30. Who among the following sits second to left of T?
(a) S (b) P (c) R (d) Q

31. Who among the following sit opposite to C?
(a) P (b) Q (c) R (d) S

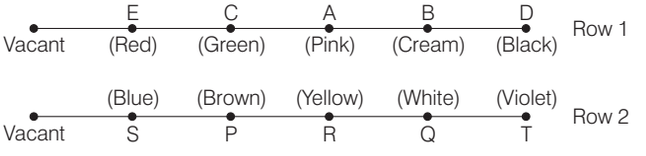
32. Who among the following likes cream colour?
(a) A (b) B (c) C (d) D

33. Who among the following is sitting between A and D?
(a) E (b) C (c) B
(d) The one who likes pink colour

34. Which of the following colour is liked by R?
(a) Yellow (b) White (c) Blue (d) Violet

35. Who among the following sits second to the left of P?
(a) Vacant seat (b) Q (c) S (d) E

Solutions (Q. Nos. 28-35) From the given information,



28. *(d)* D, T sits on the extreme ends of row.
29. *(d)* T likes violet colour.
30. *(c)* R sits second to the left of T.
31. *(a)* P sits opposite to C.
32. *(b)* B likes cream colour.
33. *(c)* B sits between A and D.
34. *(a)* R likes yellow colour.
35. *(a)* No one sits second to the left of P.

SECTION B : GENERAL AWARENESS

36. Recently who among the following has been appointed as a Chief Technology and Product Officer (CTPO) in chqbook.com?
(a) Sachin Arora (b) Parag Agrawal
(c) Dr. V Narayanan (d) Vineeth Joshi

Sol. *(a)* Chqbook.com is a financial planner in Gurugram. It especially deals in home loans. Sachin Arora is recently appointed as the Chief Technology and Product Officer (CTPO) at chqbook.com. Previously he was working for Amazon.

37. Nobel Peace Prize of 2018 was presented to
(a) Paul Romer and William Nordhaus
(b) Denis Mukwege and Nadia Murad
(c) George Smith, Frances Arnold Cregwinter
(d) James Pallison

Sol. *(b)* The Nobel prize is a set of annual international award bestowed in several categories by Swedish and Norwegian Institution. This year Noble Peace Prize was won by Denis Mukwege and Nadia Murad for their effort to end the use of sexual violence as a weapon of war and armed conflict.

38. Government of India allowed 100% FDI in which sector?

Sol. Government of India amended the 49% FDI cap in single brand retail and increased in to 100%. Now FDI in single brand retail will be permitted through automatic route and government approval will not be required. Single brand retail trading means selling different product under one brand name like Apple, Nike, Puma etc.

39. The Statue of Unity, recently inaugrated in State of Gujarat is related with which personality?
(a) Deen Dyal Upadhyay (b) Mahatma Gandhi
(c) Sardar Vallabhbhai Patel (d) Subhash Chandra Bose

Sol. *(c)* India paid tribute to Sardar Vallabhbhai Patel for his uncomparable contribution in India's development by Constructing Statue of Unity. The statue is the tallest in the world standing tall at a height of 182 m. He was the Ist Home Minister of Country and played distinct role in unification and integration of India. He unified 562 Princely State to give India its present day shape.

40. Who among the following is former and current CEO and MD of IDBI bank respectively?
(a) Mahesh Kumar Jain and B Sriram
(b) Rana Kapoor and Arjit Basu
(c) Shika Sharma and Shyamala Gopinath
(d) Amitabh Chaudhary and Chanda Kochhar

Sol. *(a)* IDBI Bank was established in 1964 by Parliament for the development of the rising Indian Industry. Recently IDBI former CEO Mahesh Kumar Jain was appointed as a Deputy Governor of RBI. On the place of Mahesh Kumar Jain, IDBI has appointed B. Sriram as Managing Director (MD) and Chief Executive Officer (CEO).

41. Under which Provision of RBI Act government can issue direction to the RBI?
(a) Section 17 (b) Section 7 (c) Section 10 (d) Section 15

Sol. *(b)* Reserve Bank of India Act, 1934 was enacted to constitute the Reserve Bank of India. Recently Section 7 of the RBI Act has come into spotlight amid the clash between Central and RBI. Over the issue of RBIs autonomy. Exercising powers under this Section government can send letter to RBI for different purpose. Government has appointed Shaktikanta Das as 25th RBI Governor.

42. Recently Assembly Election were held in which state?
(a) Chhattisgarh, Madhya Pradesh, Nagaland, Rajasthan and Telangana
(b) Chhattisgarh, Uttar Pradesh, Mizoram, Rajasthan and Andhra Pradesh
(c) Jharkhand, Madhya Pradesh, Meghalaya, Gujarat and Odisha
(d) Chhattisgarh, Madhya Pradesh, Mizoram, Rajasthan and Telangana

Sol. *(d)* Assembly Election were held in 5 state-

Chhattisgarh	-	Bhupesh Baghel
Madhya Pradesh	-	Kamal Nath
Mizoram	-	PU Zoramthanga
Rajasthan	-	Ashok Gehlot
Telangana	-	K. Chandrashekhar Rao

43. Which among the following is first Holographic phone launched recently?
(a) Red Hydrogen One
(b) Blue Hydrogen One
(c) Red Nitrogen One
(d) Black Nitrogen One

Sol. *(a)* The first Holographic phone named 'Red Hydrogen One' was launched by "AT&T" and "Verizon". The company has its roots in digital cameras for movie production. The Holographic phone can project 30 images that can be viewed without special glasses.

44. Facebook acquired which of the following video editing company?
(a) Kyro digital (b) Spark house
(c) Fayteq (d) Spin creative

Sol. *(c)* In a bid to build on its Augmented Reality (AR) ambitions, facebook has acquired German computer vision startup Fayteq. Fayteq is known of its video-editing technology that makes it possible to add and remove objects from existing videos.

45. Who won Champion of Earth award alongwith the Emmanuel Macron?

Sol. The UNEP established Champion of Earth award in 2005 as an annual award programme to recognise outstanding environmental leaders, from public and private sector. This year award was presented to Narendra Modi and Emmanuel Macron for their initiative to establish International Solar Alliance to reduce carbon emmission.

46. Name the political party established by Kamal Haasan.

Sol. Superstar Kamal Hassan announced the name of his new political party Makkal Needhi Maiam (Peoples Justice Centre). He declared that his party will follow centre approach and not toward left or right political approach. Party, is launched in Madurai, which is seen as the political capital of Tamil Nadu.

47. Mobile food delivery market place Food Panda is acquired by
(a) Uber Eats (b) Zomato
(c) Swiggy (d) Ola

Sol. *(d)* In a strategic mere to capture the food ordering market in India, online cab aggregator ola has acquired foodpande India, ola is committed for investment of $ 200 million into the food delivery business. Food panda fights bigger rivals like zomato and swiggy in the domestic market. Ola is competing with uber eats in food delivery market.

48. Biggest Health Sector Scheme launched by the GOI is
(a) Rastriya Swasth Bima Yojana
(b) Ayushman Bharat
(c) Pradhan Mantri Swasthya Suraksha Scheme
(d) National Organ Transplant Programme

Sol. *(b)* Ayushman Bharat is National Health Protection Scheme. This will cover over 10 crore poor and vulnerable families. It will providing coverage upto 5 lakh rupee per family per year for secondary and tertiary care hospitalisation. The scheme is also known as Pradhan Mantri Jan Aarogya Yojana. It is world's biggest Health Sector scheme on the basis of people covered under it.

49. More Aditya Birla Group's food and grocery retail chain will be acquired by
(a) Flipkart (b) IKEA
(c) Samara and Amazon (d) Snapdeal

Sol. *(c)* Private equity firm Samara Capital along with Amazon.cominc., the world's largest online retailer, have agreed to buy Aditya Birla group's food and grocery retail chain. Aditya Birla Groups food and grocery retail chain is known as "MORE".

50. Currently which company is the largest manufacturer of chip in the world?
(a) Qualcomm (b) Toshiba (c) Intel (d) Samsung

Sol. *(d)* Samsung, the Korean tech giants has ended reign of Intel as the world biggest seller of chipset. Chip is short for microchip, complex tiny module that store computer memory.

51. At present who is Vice-Chairman of University Grants Commission?

Sol. The UGC is a statutory body established by the UGC Act of 1956 under Ministry of HRD and is charged with co-ordination, determination and maintenance of standard of higher education. Recently Dr. Bhushan Patwardhan was appointed as the Vice-Chairman of the UGC by Government of India. Dr. Bhushan Patwardhan is a biomedical Scientist.

52. New SUV launched by Tata and Jaguar, Land Rover is
(a) Creta (b) Harrier (c) Compass (d) XUV500

Sol. *(b)* The Tata Harrier is a five seater SUV produced by Indian automaker Tata Motors. Harrier is based on a re-engineered version Jaguar, Land and Rover.

53. Which bollywood actor has signed an agreement to become Brand Ambassador of Dish TV DTH Company?
(a) Ranveer Singh (b) Salman Khan
(c) Ayushman Khurana (d) Rajkumar Rao

Sol. *(a)* India's first and largest DTH company has made Ranveer Singh as its ambassador to woo the younger audience. He is replaced by Shahrukh Khan. Ranveer will lead the Dish TV's new brand campaign 'Dish Nahi Dishkiyaon'.

54. Who won the Ramon Magsaysay Award 2018?

Sol. Ramon Magsaysay award is named after, the President of the Philippines. Prize is given to Asian individual in different field such as Public Services, Journalism, Literature etc. Bharat Vatwani is being recognised for embracing mentally offlected destitute and Sonam Wangchuk for his contribution in harnessing nature, culture and education for community progress in Ladakh.

55. Who has been appointed as a Chief Economic Advisor of India?
(a) N K Singh
(b) Krishnamurti Subramaniam
(c) Mahesh Kumar Jain (d) Amitabh Kant

Sol. *(b)* The government has appointed Krishnamurti Subramaniam as a Chief Economic Advisor for three years, Subramaniam was teacher at Indian School of Business (ISB) before taking this post. Chief Economic Advisor is responsible for determining the governments overall strategy in managing the economy and help prepare the econimical survery of India.

56. Which city of Indonesia Hosted 2018 Asian Games?
(a) Palembong and Makassar (b) Jakarta and Bondung
(c) Bondung and Makassar (d) Jakarta and Palembong

Sol. *(d)* 18th Asian Games were hosted by Indonesia in city of Jakarta and Palembong.
Indias Performance - Rank 8th
Medal - 69th
Gold - 15
Silver - 24
Bronze - 30
19th Asian Games will be Hosted by China.

57. Krishi Kalyan Abhiyaan was launched by Ministry of Agriculture and farmers welfare in .....
(a) 4th month of 2018 (b) 7th month of 2018
(c) 6th month of 2018 (d) 12th month of 2018

Sol. *(c)* With a vision of doubling farmers income by 2022 Ministry launched Krishi Kalyan Abhiyaan from 1st June, 2018 till 31st July, 2018. It will aid assist and advice farmers on how to improve farming techniques and raise their income.

58. What is India's rank in UN World Happiness Index?
(a) 133 (b) 122 (c) 110 (d) 100

Sol. *(a)* The World Happiness Report is published by the United Nation Sustainable Development Solution Network. India's ranking Reduced from 122 in 2017 to 133 in 2018. Finland was Happiest country and Burundi was last on the list. It covered 156 countries.

59. How many seats Indian National Congress won in Chhattisgarh Assembly Election?
(a) 58 (b) 61 (c) 65 (d) 68

Sol. *(d)* INC formed government in Chhattisgarh with absolute majority by capturing 68 assembly seats.
Election result in other state

	BJP	INC	
MP	109	114	
Rajasthan	73	199	
Chhattisgarh	15	68	
Mizoram	1	5	MNS - 26
Telangana	1	119	TRS - 88

60. 2018, Nobel Prize for economics was awarded to how many person?
(a) One (b) Three (c) Two (d) Four

Sol. *(c)* Nobel Economic Prize 2018 was shared by US Economist William Nordhaus and Poul Romer for integrating innovation and climate with economic growth. Nobel Economic Prize was Ist awarded in 1969 all other Nobel Prize were established in 1895. The Nobel prize is bestowed by Swedish and Norwegia Institution.

SECTION C : MATHEMATICS

61. The taxi in a city comprise of a fixed charge together with the charge for the distance covered. For a journey of 10 km, the charges paid is ₹150 and for a journey of 15 km, the charge paid is ₹220. What will a person have to pay for travelling a distance of 25 km?

(a) ₹400 (b) ₹360
(c) ₹280 (d) ₹500

Sol. *(b)* Let the fixed charge be ₹f and variable charge be ₹v per km.

According to the question,

$$f + 10v = 150 \quad \ldots(i)$$

$$f + 15v = 220 \quad \ldots(ii)$$

Solving Eqs. (i) and (ii), we get

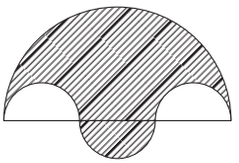
$$f = 10 \text{ and } v = 14$$

$\therefore$ Charge for 25 km $= f + 25v$

$= ₹\,(10 + 25 \times 14)$

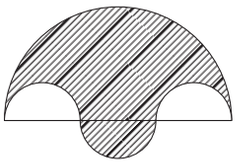
$= ₹360$

62. In the given figure, the boundary of shaded region consists of four semi-circular arcs, three smallest being equal. If the diameter of the largest is 14 cm and that of the smallest is 3.5 cm. Calculate the area of shaded region.



(a) 110.25 π cm^2 (b) 360 π cm^2
(c) 26.275 π cm^2 (d) 22.97 π cm^2

Sol. *(d)* Radius of bigger semi-circle $= \frac{14}{2} = 7$ cm



and radius of smaller semi-circle $= \frac{3.5}{2}$ cm

$\because$ Area of semi-circle $= \frac{\pi r^2}{2}$

$\therefore$ Area of shaded region

$$= \frac{\pi(7)^2}{2} + \frac{\pi}{2}\left(\frac{3.5}{2}\right)^2 - \frac{\pi}{2}\left(\frac{3.5}{2}\right)^2 - \frac{\pi}{2}\left(\frac{3.5}{2}\right)^2$$

$$= \frac{49\pi}{2} - \frac{12.25\pi}{8} = \frac{196\pi - 12.25\pi}{8}$$

$$= \frac{183.75}{8}\pi$$

$$= 22.97\ \pi \text{ cm}^2$$

63. How many six lettered words can be formed starting with M, from the word 'CUSTOM' ?

(a) 110 (b) 64 (c) 216 (d) 120

Sol. *(d)* Here, M is fixed at first position and the remaining letters can change their position.

$\therefore$ Total number of ways $= 5! = 120$

64. Output is produced according to $Q = (a - 4L)6$, where 'L' is the quantity of labour input and 'a' is the quantity of capital input. Then, which of the following alternative is correct?

(a) Output decreases as labour increases
(b) Output increases as labour increases
(c) Output decreases as capital increases
(d) None of the above

Sol. *(a)* Given, $Q = (a - 4L)6$

$\because$ Labour is combined with negative sign

$\therefore$ Output decreases as labour increases.

Hence, option (a) is correct.

65. Sonali and Meenal appear in an interview for two vacancies in the same post. The probability of Sonali's selection is 1/7 and the probability of Meenal's selection is 1/5. What is the probability that only one of them is selected?

(a) $\frac{1}{7}$ (b) $\frac{2}{7}$
(c) $\frac{3}{4}$ (d) $\frac{4}{5}$

Sol. *(b)* Let A = Event that Sonali is selected

B = Event that Meenal is selected

$\therefore \quad P(A) = \frac{1}{7}$ and $P(B) = \frac{1}{5}$

and $P(\bar{A}) = \left(1 - \frac{1}{7}\right) = \frac{6}{7}$

and $P(\bar{B}) = \left(1 - \frac{1}{5}\right) = \frac{4}{5}$

$\therefore$ Required probability

$= P(A \text{ and not } B) + P(B \text{ and not } A)$

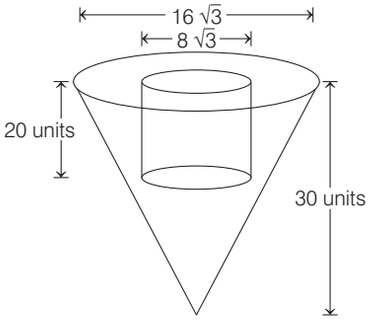
$$= \left(\frac{1}{7} \times \frac{4}{5}\right) + \left(\frac{1}{5} \times \frac{6}{7}\right)$$

$$= \frac{4}{35} + \frac{6}{35} = \frac{10}{35} = \frac{2}{7}$$

66. A cylinder of diameter $8\sqrt{3}$ units is taken out from a cone of base diameter $16\sqrt{3}$ units and height 30 units. If the height of cylinder is $\frac{2}{3}$rd of the height of cone, then find the volume of the remaining part of cone.

(a) 1920 π unit^3 (b) 1000 π unit^3
(c) 960 π unit^3 (d) 27000 π unit^3

Sol. *(c)* Radius of cylinder $= \frac{8\sqrt{3}}{2} = 4\sqrt{3}$ units



Radius of cone $= \frac{16\sqrt{3}}{2} = 8\sqrt{3}$ units

Height of cone = 30 units

Height of cylinder $= \frac{2}{3} \times 30 = 20$ units

$\because$ Volume of cone $= \frac{1}{3}\pi r^2 h$

and volume of cylinder $= \pi r^2 h$

$\therefore$ Volume of remaining part of cone

$= \frac{1}{3}\pi(8\sqrt{3})^2\,(30) - \pi\,(4\sqrt{3})^2\,(20)$

$= 64 \times 30\,\pi - 960\pi$

$= 1920\pi - 960\pi$

$= 960\pi$ unit^3

67. In how many ways can the letters of word "ASTRONAUT" be arranged?

(a) 362880 (b) 10000

(c) 90720 (d) None of these

Sol. *(c)* $\because$ There are 2A's, and 2T's, in the given word

$\therefore$ Total number of ways $= \frac{9!}{2! \times 2!}$

$= \frac{362880}{4} = 90720$

68. In a 4 km race, Anu defeats Banu by 600 m. Banu give Caira a start of 200 m and then both ends the race at the same time. How much start should Caira get from Anu to end the race at the same time?

(a) 800 m (b) 700 m (c) 500 m (d) 400 m

Sol. *(a)* Total distance = 4000 m ($\because$ 1 km = 1000 m)

$\because$ Anu defeats Banu by 600 m

$\therefore$ In same time, distance travelled by Anu = 4000 m and distance travelled by Banu

= 4000 m – 600 m = 3400 m

Now, Banu give Caira a start of 200 m

$\therefore$ In same time, distance travelled by Banu = 3400 m and distance travelled by Caira

= (3400 – 200) m = 3200 m

So, if Caira and Anu wants to finish the race together Anu should give Caira a start of 800 m.

69. *A* can complete a work in 10 days, *B* can do the same piece of work in 15 days. If they both work together for 4 days, then fraction of work left is

(a) $\frac{1}{3}$ (b) $\frac{2}{7}$ (c) $\frac{2}{3}$ (d) $\frac{5}{7}$

Sol. *(a)* *A* can complete a work in 10 days

So, 1 day work of $A = \frac{1}{10}$

Similarly, 1 day work of $B = \frac{1}{15}$

1 day work of both *A* and $B = \frac{1}{10} + \frac{1}{15}$

$= \frac{3+2}{30} = \frac{1}{6}$

4 days work of *A* and $B = \frac{4}{6} = \frac{2}{3}$

Now, fraction of work left $= 1 - \frac{2}{3} = \frac{1}{3}$

70. Simplify the given expression

$1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \ldots = ?$

(a) $\frac{2^n - 1}{2^{n-1}}$ (b) $\frac{2^{n-1} - 1}{2^{n-2}}$

(c) $2 - 2^n$ (d) $\frac{2^n - 1}{2^n}$

Sol. *(a)* Given, $1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} \ldots$

The given series is a GP,

where $a = 1$ and $r = \frac{1}{2}$

$\therefore$ Sum of the n terms of GP

$= \frac{a(1 - r^n)}{1 - r}$ ($\because r < 1$)

$= \frac{1\left(1 - \frac{1}{2^n}\right)}{1 - \frac{1}{2}} = 2\left(\frac{2^n - 1}{2^n}\right) = \frac{2^n - 1}{2^{n-1}}$

71. An elevator starts with 5 passengers and stops at 8 different floors of the building. Find out the probability of all the 5 passengers alighting at different floors.

(a) $\frac{101}{512}$ (b) $\frac{105}{512}$ (c) $\frac{107}{512}$ (d) $\frac{109}{512}$

Sol. *(b)* Favourable cases $= {}^8P_5 = \frac{8!}{(8-5)!} = 6720$

$\therefore$ Required probability $= \frac{6720}{8^5} = \frac{6720}{32768} = \frac{105}{512}$

72. If 'h', 'c' and 'v' represents height of cone, curved surface area of cone and volume of cone respectively, then the value of $3\pi v h^3$ is

(a) $c^2h^2 - 9v^2$ (b) $c^2v^2h^2$

(c) $\pi c^2 h^2 v$ (d) None of these

Sol. *(a)* Given, h = height of cone, v = volume of cone and c = curved surface area of cone.
Let r = radius of base of cone
Now, $3\pi v h^3$
$= 3\pi \times \frac{1}{3}\pi r^2 h \times h^3$ [$\because$ volume of cone $= \frac{1}{3}\pi r^2 h$]
$= \pi^2 r^2 h^4 = (\pi^2 r^2 h^2) \times h^2$ ...(i)
Now, we know that, $c = \pi r l$
$\Rightarrow$ $c = \pi r(\sqrt{r^2 + h^2})$ ($\because l^2 = r^2 + h^2$)
$\Rightarrow$ $c^2 = \pi^2 r^2 (r^2 + h^2)$
$\Rightarrow$ $c^2 = \pi^2 r^4 + \pi^2 r^2 h^2$
$\therefore$ $\pi^2 r^2 h^2 = c^2 - \pi^2 r^4$
Putting the value in Eq. (i), we get
$(c^2 - \pi^2 r^4) h^2 = c^2 h^2 - \pi^2 r^4 h^2 = c^2 h^2 - 9v^2$

73. If 20% of $x = y$, then y% of 20 is equal to
(a) 20% of x (b) 16% of x (c) 10% of x (d) 4% of x

Sol. *(d)* Given, $\frac{20}{100}x = y$
$y = \frac{x}{5}$ ...(i)
Now, $\frac{y}{100} \times 20 = \frac{x}{5} \times \frac{1}{5} = \frac{x}{25}$
$= 4\%$ of x

74. If $f(x) = \frac{x+1}{x-1}$, the ratio of x to $f(y)$, where $y = f(x)$ is
(a) $x : y$ (b) $x^2 : y^2$ (c) $1:1$ (d) $(x+1):(x-1)$

Sol. *(c)* Given, $f(x) = \frac{x+1}{x-1}$
$\therefore$ $f(y) = \frac{\frac{x+1}{x-1} + 1}{\frac{x+1}{x-1} - 1} = \frac{\frac{x+1+x-1}{x-1}}{\frac{x+1-x+1}{x-1}} = \frac{2x}{2} = x$
$\therefore$ $\frac{x}{f(y)} = \frac{x}{x} = 1:1$

75. Find the value of
$\log \tan 1° + \log \tan 2° + \ldots + \log \tan 89°$.
(a) $\frac{1}{\sqrt{3}}$ (b) 25 (c) 0 (d) 1

Sol. *(c)* Given,
$\log \tan 1° + \log \tan 2° + \ldots + \log \tan 89°$
$= \log(\tan 1° \times \tan 2° \times \ldots \times \tan 89°)$
$= \log(\tan(90-1)° \times \tan(90-2)° \ldots \tan 45° \ldots \tan 88° \cdot \tan 89°)$
$= \log(\cot 89° \cot 88° \ldots \tan 45° \ldots \tan 88° \cdot \tan 89°)$ [$\because \tan(90-\theta) = \cot\theta$]
$= \log(1 \cdot 1 \cdot 1 \cdot 1)$ [$\because \tan\theta \cdot \cot\theta = 1$ and $\tan 45° = 1$]
$= \log 1 = 0$

76. If $\log_{10} 11 = a$, then $\log_{10}\left(\frac{1}{110}\right)$ is equal to
(a) $a+1$ (b) $-(a+1)$ (c) a (d) $a-1$

Sol. *(b)* Given, $\log_{10} 11 = a \Rightarrow 10^a = 11$
Now, $\log_{10}\left[\frac{1}{110}\right] = \log_{10}\left[\frac{1}{11 \times 10}\right]$
$= \log_{10}\left[\frac{1}{10^a \times 10}\right] = \log_{10}\left[\frac{1}{10^{a+1}}\right]$
$= \log_{10} 10^{-(a+1)}$
$= -(a+1) \times \log_{10} 10$
$= -(a+1)$

77. The price of corn increases by 20% also the expenditure to bring corn from to marked increases by 5%. By how much per cent should a family decrease its consumption so as to keep their budget same?
(a) $11\frac{1}{9}\%$ (b) 13% (c) $9\frac{1}{9}\%$ (d) 25%

Sol. *(a)* Let the original price of corn and expenditure to bring it to market be ₹100 each
New price of corn $= \frac{120}{100} \times 100 = ₹120$
New expenditure $= \frac{105}{100} \times 100 = ₹105$
New total cost = ₹(120 + 105) = ₹225
Decrease in consumption = ₹(225 – 200) = ₹25
$\therefore$ Required percentage $= \frac{25}{225} \times 100 = 11\frac{1}{9}\%$

78. The cost of a TV is 150% the cost of a fridge. The cost of fridge is what per cent of the cost of TV and fridge?
(a) 45% (b) 60% (c) 25% (d) 40%

Sol. *(d)* Let the cost of fridge = ₹100
$\therefore$ Cost of TV $= \frac{150}{100} \times 100 = ₹150$
$\therefore$ Cost of fridge and TV = ₹(100 + 150) = ₹250
Required percentage $= \frac{100}{250} \times 100 = 40\%$

79. 20 students collected ₹10000 each 50% of the amount is invested in riskless government securities while the remaining in equities. The return on riskless government securities is 7.8% per annum. If the gain is equally divided among them, then value of gain per student is
(a) ₹390 (b) ₹440 (c) ₹500 (d) ₹780

Sol. *(a)* Total collection by students
= ₹ (20×10000) = ₹200000
$\therefore$ Amount invested in rickless government securities
$= \frac{50}{100} \times 200000 = ₹100000$
Return = 7.8% per annum
$\therefore$ Gain $= \frac{7.8}{100} \times 100000 = ₹7800$
$\therefore$ Gain per student $= \frac{₹7800}{20} = ₹390$

80. An umbrella is made by stitching 10 triangular pieces of cloth of two different colours, each piece measuring 20cm, 50cm and 50cm. How much cloth of each colour is required for the umbrella?

Sol. There are 10 triangles in the umbrella, 5 of each colour.

Now, area of each colour of cloth = 5 (area of 1 triangle)

Given, sides of triangle = $a = 20$ cm, $b = 50$ cm, $c = 50$ cm

Using Heron's formula,

Area of triangle $= \sqrt{s(s-a)(s-b)(s-c)}$

where, $s = \dfrac{a+b+c}{2}$

$\therefore \quad s = \dfrac{20+50+50}{2} = 60$ cm

$\therefore$ Area of triangle

$= \sqrt{60(60-50)(60-50)(60-20)}$

$= \sqrt{60\times10\times10\times40} = \sqrt{240000}$

$= 200\sqrt{6}$ cm^2

$\therefore$ Area of each colour $= 5\times200\sqrt{6} = 1000\sqrt{6}$ cm^2

81. The number of ways that 5 Marathi, 3 English and 3 Tamil books be arranged if the books of each language are to be kept together is

(a) 5820 (b) 25920
(c) 29520 (d) 27840

Sol. *(b)* Number of ways of arranging

5 Marathi books $= 5! = 120$

Number of ways of arranging 3 English books

$= 3! = 6$

and Number of ways of arranging 3 Tamil books

$= 3! = 6$

Now, these 3 sets can be arranged in 3! ways i.e. 6 ways.

$\therefore$ Total number of ways $= 120\times6\times6\times6 = 25920$

82. The difference between compound interest and simple interest for a certain sum of money for 2 yr is ₹50, the rate of interest being 5% per annum. If the same sum is invested for 3 yr at 5% per annum compound, it will become.

(a) ₹ 22475.50 (b) ₹ 23750 (c) ₹ 24000 (d) ₹ 23152.50

Sol. *(d)* Given, rate of interest = $r = 5\%$ per annum

Let Principal = ₹ P

$\because$ Difference between CI and SI for 2 yr $= P\left(\dfrac{r}{100}\right)^2$

$\therefore \quad 50 = P\left(\dfrac{5}{100}\right)^2$

$\Rightarrow \quad P = \dfrac{50\times100\times100}{25} = ₹20000$

Now, we know that, $A = P\left(1+\dfrac{r}{100}\right)^t$

$\therefore$ Amount $= 20000\left(1+\dfrac{5}{100}\right)^3$

$= 20000\left(\dfrac{21}{20}\right)^3$

$= ₹23152.50$

$\therefore$ Required sum = ₹23152.50

83. Given, 5 different green dyes, 4 different blue dyes and 3 different red dyes, the number of combination of dyes which can be chosen taking at least one green and one blue dye is

(a) 3600 (b) 3720 (c) 3800 (d) 3600

Sol. *(b)* Given, number of green dyes = 5

Number of blue dyes = 4

Number of red dyes = 3

Now, number of ways to choose at least one green and one blue dye

$= (2^5-1)(2^4-1)(2^3)$

$= (32-1)(16-1)(8) = 31\times15\times8 = 3720$

84. Sumit has to multiply a certain number by 25, but he multiplied it with 34 and the resultant is 405 more than the original result. Find the answer Sumit arrived at

(a) 1000 (b) 1125
(c) 1600 (d) 1530

Sol. *(d)* Let the number be 'a'

According to the question,

$34a - 25a = 405$

$\Rightarrow \quad 9a = 405$ or $a = 45$

$\therefore$ Result obtained by Sumit

$= 45\times34 = 1530$

85. Excluding stoppages, the speed of the bus is 50 km/h and including stoppages, it is 40 km/h. For how many minutes does the bus stop per hour?

Sol. Distance travelled by bus in 1h if stoppages are excluded = 50 km

$\because$ Original speed of bus = 50 km/h

Distance travelled by bus in 1h if stoppages are included = 40 km

$\therefore$ Difference in distance travelled $= (50-40)$ km

$= 10$ km

$\because$ Time $= \dfrac{\text{Distance}}{\text{Speed}}$

$\therefore$ Time taken to travel 10 km $= \dfrac{10}{50}$

$= \dfrac{1}{5}$ h $= 12$ min

Hence, the bus stops for 12 min every hour.

86. If taxes are 125% of prices, then average of taxes for given prices is "35, 34, 35.5, 37, 37.5 and 38".
(a) 36.17 (b) 40.20 (c) 45.20 (d) 43.17

Sol. *(c)* Average price

$$= \frac{35 + 34 + 35.5 + 37 + 37.5 + 38}{6} = \frac{217}{6}$$

Now, average taxes $= \frac{125}{100} \times$ average price

$$= \frac{125}{100} \times \frac{217}{6} = 45.20$$

87. If $R_c = m \ln\left[1 + \frac{R_m}{m}\right]$, then value of R_m is
(a) $m[e^{R_c/m} - 1]$ (b) 0
(c) 1 (d) $\frac{R_c}{m}\left[\ln\left(\frac{1}{m}\right)\right]$

Sol. *(a)* Given, $R_c = m \ln\left[1 + \frac{R_m}{m}\right]$

$$\Rightarrow \quad \ln\left[1 + \frac{R_m}{m}\right] = \frac{R_c}{m}$$

$$\Rightarrow \quad 1 + \frac{R_m}{m} = e^{R_c/m} \Rightarrow \frac{R_m}{m} = e^{R_c/m} - 1$$

$$\therefore \quad R_m = m[e^{R_c/m} - 1]$$

88. Simplify, $(a - b)^3 + (b - c)^3 + (c - a)^3$.
(a) $3(a - b)(b - c)(c - a)$ (b) $3abc$
(c) 0 (d) 1

Sol. *(a)* Let $a - b = x$, $b - c = y$ and $c - a = z$

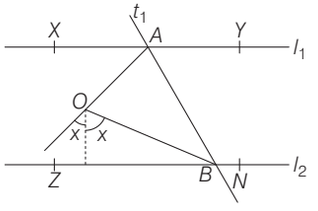
Now, $x + y + z = a - b + b - c + c - a = 0$

Now, we know that, if $x + y + z = 0$

then $x^3 + y^3 + z^3 = 3xyz$

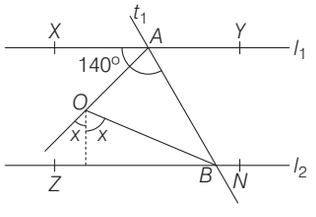
$\therefore (a - b)^3 + (b - c)^3 + (c - a)^3 = 3(a - b)(b - c)(c - a)$

89. Given, $l_1 \parallel l_2$ and $\angle XAB = 140°$ also, OA and OB are angle bisector of $\angle XAB$ and $\angle ZBA$ respectively, then find the value of x.



Sol. Given, $\angle XAB = 140°$

and OA is its bisector



$$\therefore \quad \angle OAB = \frac{140°}{2} = 70°$$

Now, $l_1 \parallel l_2$

$$\therefore \quad \angle XAB + \angle ZBA = 180°$$

or $\angle ZBA = 180° - 140° = 40°$

Also, OB is angle bisector of $\angle ZBA$

$$\therefore \quad \angle OBA = \frac{40°}{2} = 20°$$

Now, in ΔOAB

$$\angle OAB + \angle OBA + \angle AOB = 180°$$

$$\therefore \quad \angle AOB = 180° - 70° - 20° = 90°$$

Now, $x + x + \angle AOB = 180°$ (Linear Pair)

$$\therefore \quad x = \frac{180° - 90°}{2} = \boxed{45°}$$

90. The sides of a rectangular field are in the ratio 2 : 7, also the perimeter of field is 360 m, then the length and breadth of the field are
(a) 40, 140 m (b) 140, 20 m (c) 30, 210 m (d) 20, 140 m

Sol. *(a)* Let the length of rectangular field be '$2a$'m and its breadth be '$7a$'m.

$\because$ Perimeter of rectangle $= 2(l + b)$

$$\therefore \quad 2(2a + 7a) = 360$$

$$\Rightarrow \quad 9a = 180 \text{ or } a = 20 \text{ m}$$

$\therefore$ Length $= 2a = 40$m and

breadth $= 7a = 140$ m

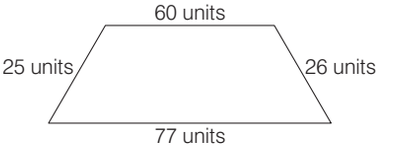
91. Simplify, $\frac{4^2 \times 2^{n+1} - 4 \times 2^n}{4^2 \times 2^{n+2} - 2 \times 2^{n+2}}$.

Sol. Given, $\frac{4^2 \times 2^{n+1} - 4 \times 2^n}{4^2 \times 2^{n+2} - 2 \times 2^{n+2}}$

$$= \frac{2^4 \times 2^{n+1} - 2^2 \times 2^n}{2^4 \times 2^{n+2} - 2 \times 2^{n+2}} = \frac{2^{n+5} - 2^{n+2}}{2^{n+6} - 2^{n+3}}$$

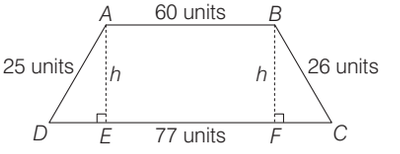
$$= \frac{2^{n+2}(2^3 - 1)}{2^{n+2}(2^4 - 2)} = \frac{8 - 1}{16 - 2} = \frac{7}{14} = \boxed{0.5}$$

92. Find the area of the given figure.



(a) 1600 unit^2 (b) 576 unit^2
(c) 1507 unit^2 (d) 1644 unit^2

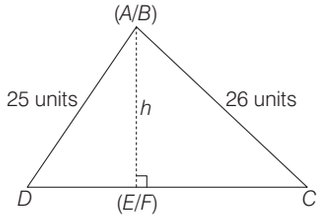
Sol. *(d)* The given figure is a trapezium.



Let AE and BF be two perpendicular drawn on CD.

$\therefore$ $AB = EF = 60$ units.

Now, $\Delta AED + \Delta BFC = \Delta ADC$



Using, Heron's formula

$\therefore$ Area of triangle $= \sqrt{s(s-a)(s-b)(s-c)}$

$= \frac{1}{2} \times \text{base} \times \text{height}$

where, $s = \frac{a+b+c}{2}$

$= \frac{25+26+17}{2} = 34$ units

$\Rightarrow \sqrt{34(34-25)(34-26)(34-17)}$

$= \frac{1}{2} \times 17 \times h$

$\Rightarrow \sqrt{34 \times 9 \times 8 \times 17} = \frac{1}{2} \times 17 \times h \Rightarrow \frac{204 \times 2}{17} = h$

$\therefore \quad h = 24$ units

Now, area of trapezium

$ABCD = \frac{1}{2}(AB + CD) \times (AE)$

$= \frac{1}{2}(60 + 77)(24) = 1644$ units^2

93. If $q = 1$, then $(p+q)^3 - (p-q)^3 - 6q(p^2 - q^2)$ is equal to

(a) 0 (b) 1 (c) 6 (d) 8

Sol. *(d)* Given,

$(p+q)^3 - (p-q)^3 - 6q(p^2 - q^2)$ and $q = 1$

$\therefore \quad (p+1)^3 - (p-1)^3 - 6(p^2 - 1)$

$\Rightarrow p^3 + 3p^2 + 3p + 1 - p^3 + 3p^2 - 3p + 1 - 6p^2 + 6$

$[\because (a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

and $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3]$

$= 6p^2 + 2 - 6p^2 + 6 = 8$

94. What will be the numerical value of MMXVIII

Sol. We know that, M = 1000, X = 10, V = 5 and I = 1

$\therefore$ MMXVIII $= 1000 + 1000 + 10 + 5 + 1 + 1 + 1 =$ 2018

95. 250 boxes of each type of dimensions $25 \times 20 \times 5$ and $15 \times 12 \times 5$ are to be covered with wood. An additional 5% of total surface area is required to cover the box with wood. If the cost of wood is ₹ 4/m^2, then total cost is

(a) ₹ 6640000 (b) ₹ 2084100 (c) ₹ 2184000 (d) ₹ 1247000

Sol. *(c)* Dimensions of 1st type of box $= 25 \times 20 \times 5$

Dimensions of 2nd type of box $= 15 \times 12 \times 5$

$\because$ Total surface area of cuboid $= 2(lb + bh + lh)$

$\therefore$ Total surface area of 1st type of box

$= 2(25 \times 20 + 20 \times 5 + 25 \times 5)$

$= 2(500 + 100 + 125) = 1450$ m^2

and total surface area of 2nd type of box

$= 2(15 \times 12 + 12 \times 5 + 15 \times 5)$

$= 2(180 + 60 + 75) = 630$ m^2

$\therefore$ Total surface area of all boxes

$= 250 \times 1450 + 250 \times 630$

$= 362500 + 157500 = 520000$ m^2

Now, area of wood required

$= \frac{105}{100} \times 520000 = 546000$ m^2

$\therefore$ Required cost $= 546000 \times 4 =$ ₹2184000

SECTION D : ENGLISH LANGUAGE

Directions (Q. Nos. 96-100) *Read the following passage and answer the questions that follows.*

Since the collective culture of an organisation is an aggregate of what is common to all of its group and individual mindsets, organisation transformation entails changing the minds of hundreds or thousands of people. Consultants Lawson and Price, listed four conditions as essential for changing mindsets, the employees must believe in the overall purpose behind the change, reporting structures, operational processes and measurement procedures must be consistent with the behaviour that people are asked to embrace, employees need to be trained to adopt to new environment and apart from top management, 'role models' at every level also need to embrace the new **dogma**. Considering the difficulties in changing mindset of existing people, most Indian organisations have found it prudent to drop them under Voluntary Retirement Schemes (VRS) and recruit new people with positive attitude to change. But this cannot be a pattern to manage change, one day or other, existing people need to be retrained to manage change. Dwelling further on our reluctance to change, Foster and Kaplan referred to 'Cultural lock-in', the inability to change the corporate culture even in the face of clear market threats. The heart of the problem is the formation of hidden sets of rules or 'mental models', that once formed are extremely difficult to change. 'Mental models' are the core concepts of the corporation, the beliefs and assumption, the cause and effect relationships, the guidelines for interpreting language and signals, the stories repeated within the corporate walls. 'Mental models' are invisible in the corporation, they are neither explicit nor examined but they are pervasive. When well crafted, 'mental models' allow management to anticipate the future and solve problems.

But once constructed, 'mental models' become self-reinforcing, self-sustaining and self-limiting.

96. According to the passage

A. For an organisation to be successful, it is essential to hire like minded people.
B. The best way to deal with staff who refuse to change their mindset is by not touting them as role models for the organisation.

Which of the statements given above is/are correct?
(a) Only A (b) Only B
(c) Both A and B (d) Neither A nor B

Sol. *(d)* According to the passage none of the statement given is correct.

97. On the basis of the passage,
A. For an organisation to redefine itself, it has to first re-evaluate its employees.
B. For an employee to become an integral part of an organisation, he has to subscribe to the views and ideology of the organisation.
C. Top management executives are the sole drivers for reinventing an organisation.
D. Now-a-days, employees can function without depending on a 'key figure' to emulate.

Which of the statements given above is / are correct?
(a) Both A and B (b) Both C and D
(c) Only B (d) Only C

Sol. *(a)* Both statements A and B are correct. An organisation needs to re-evaluate its employees. Moreover, a successful employee has to subscribe to the views and ideology of the organisation.

98. Consider the following with regards to resistance to change.
A. It is difficult for some people to think out of the box.
B. Some people resist new information and prefer to rely on past practices to conduct business.
C. Their cultural background predisposes them to shy away from risks.
D. Radical thinkers are not rewarded in today's corporate culture.
E. The Voluntary Retirement Scheme is used by some organisations to breathe new life into the organisation.

Which of the statements given above is/are correct?
(a) A, B and E
(b) A, B, C, D and E
(c) B, C and D
(d) C, D and E

Sol. *(a)* Statement A, B and E are correct with regards to resistance to change. It is difficult for some people to think out of box, some people also prefer not to take any new information and rely on past practices. VRS is used by some organisations to breathe new life into the organisation.

99. What can be a suitable title of the passage?
(a) Essentials of changing mindset
(b) Changing mindset of employees - A difficult task
(c) Importance of mental models
(d) Collective culture of an organisation

Sol. *(b)* 'Changing mind, set of employees-A difficult task' can be a suitable title.

100. What does the word 'Dogma' means as given in the passage?
(a) Conviction or belief (b) A law
(c) Self-confidence (d) Virtues

Sol. *(a)* The word 'Dogma' means a belief or set of belief's that people are expected to accept without asking questions about them.

PASSAGE II

Directions (Q. Nos. 101-106) *Read the passage given below and answer the question that follow.*

The cyber-world is ultimately ungovernable. This is alarming as well as convenient, sometimes, convenient because alarming. Some Indian politicians use this to great advantage. When there is an obvious failure in governance during a crisis they deflect attention from their own incompetence towards the ungovernable.

So, having failed to prevent nervous citizens from fleeing their cities of work by assuring them of proper protection, some national leaders are now busy trying to prove to one another and to panic-prone Indians, that a mischievous neighbour has been using the internet and social networking sites to spread dangerous rumours.

And the centre's automatic reaction is to start blocking these sites and begin elaborate and potentially endless negotiations with Google, Twitter and Facebook about access to information. If this is the official idea of prompt action at a time of crisis among communities, then Indians have more reason to fear their protectors than the nebulous mischief-makers of the cyber-world.

Wasting time gathering proof, blocking vaguely suspicious websites, hurling accusations across the border and worrying about bilateral relation; are ways of keeping busy with inessentials because one does not quite know what to do about the essentials of a difficult situation. Besides, only a fifth of the 245 websites blocked by the Centre mention the people of the North-East or the violence in Assam.

And if a few morphed images and spurious texts can unsettle an entire nation, then there is something deeply wrong with the nation and with how it is being governed. This is what its leaders should be addressing immediately, rather than making a **wrong-headed** display of their powers of censorship.

It is just as absurd and part of the same syndrome, to try to ban Twitter accounts that parody despatches from the Prime Minister's Office. To describe such forms of humour and dissent as 'misrepresenting' the

PMO-as if Twitterers would take these parodies for genuine despatches from the PMO–makes the PMO look more ridiculous than its parodists manage to using the cyber world for flexing wrong muscles is essentially not funny. It might even prove to be quite dangerously distracting.

101. According to the passage, the cyber world is
(a) beyond the imagination of people
(b) outside the purview of common people
(c) not to be governed
(d) ungovernable

Sol. *(d)* The author is of the view that the cyber world is ungovernable. (Line 1)

102. The author is of the opinion that
(a) the centre should start negotiations with Google, Twitter and Facebook
(b) the centre should help the citizens evacuate their city
(c) the centre should not block the sites
(d) the centre should arrest the guilty

Sol. *(c)* As per the passage, the author's opinion is that the Central Government should not block the sites. (Read 1st line of Para 3)

103. Which of the following is closest to the meaning of 'nebulous'?
(a) Confused (b) Vague
(c) Iridescent (d) Glowing

Sol. *(b)* Vague or hazy is closest to the meaning of nebulous. Nebulous means not developed or clear enough to describe.

104. The author's seriousness regarding the situation can best be described in the following sentences. Pick the odd one out.
(a) Our leaders should display their powers of censorship when needed
(b) If this is the official idea of prompt action at a time of crisis among communities, then Indians have more reason to fear their protectors than the nebulous mischief-maker of the cyber-world
(c) The politicians deflect attention from their own incompetence
(d) If a few morphed images and spurious texts can unsettle an entire nation, then there is something deeply wrong with the nation

Sol. *(a)* Sentence in option (a) is the odd one out as it is no where mentioned in the paragraph that our leaders should display their power of censorship.

105. Find the antonym of the word-'wrong-headed' from the options given
(a) Silly (b) Sane (c) Instable (d) Insensible

Sol. *(b)* The word 'wrong-headed' means misguided or wrong in Judgement. Its antonym is sane which means a person who acts sensibly.

106. The author warns us against
(a) not playing false with the citizens
(b) dangers inherent in the cyber-world
(c) not using the cyber-world judiciously
(d) not protecting the citizens from dangerous politicians

Sol. *(c)* The author, in this passage, warns us against not using the cyber world judiciously.

107. Choose the synonym of the word 'Inexplicable'.
(a) Unconnected (b) Chaotic
(c) Unaccountable (d) Confusing

Sol. *(c)* Unaccountable is correct synonym of inexplicable. Both words mean something that cannot be explained or accountable why it happened.

108. Choose the synonym of the word, 'Prerogative'.
(a) Privilege (b) Request (c) Desire (d) Command

Sol. *(a)* Privilege is correct synonym of the word prerogative. Both mean advantage or birth right.

Directions Q. Nos. (109 and 110) *Choose the appropriate antonyms of the given words.*

109. Camouflage
(a) Exhibit (b) Demonstrate
(c) Reveal (d) Parade

Sol. *(c)* The word camouflage means **to hide something**. Its antonym is to **reveal** which means to make something known to everyone.

110. Scurrilous
(a) Decent (b) Relevant
(c) Accurate (d) Useful

Sol. *(a)* The word **scurrilous** means derogatory or offensive. Its antonym is **decent** which means acceptable or adequate.

Directions (Q. Nos. 111 and 112) *In the given questions, choose the odd word out of the given options.*

111. (a) Bicker (b) Quarrel (c) Wrangle (d) Inveigle

Sol. *(d)* 'Inveigle' is odd one as the other three words are related to quarrel but **inveigle** means to persuade someone to do something.

112. (a) Counsel (b) Direct (c) Suggest (d) Advise

Sol. *(b)* Direct is odd one out as the other three words are associated with **giving advise** but the word **direct** means to control or be incharge of an activity, organisation etc.

Directions (Q. Nos. 113 and 114) *Fill in the blanks with the correct form of the verb given in brackets.*

113. While she ..... (stroll) in the garden, she came across her old friend.

Sol. 'Was strolling' is the appropriate answer as it denotes an action going on or happening at that time.

114. I wish I ..... (study) sincerely to clear the exam.

Sol. 'Had studied' should be used to fill the blank. 'I wish' suggests something of the past. So, we use Past Perfect Tense with it.

Direction (Q. No. 115) *Improve the sentence by using appropriate grammatical structure.*

115. Unless efforts are made and the population growth had stabilised, environmentalists predict a world wide starvation by 2020 AD.

(a) Efforts are made and the population growth is stabilised
(b) Efforts are made and the population growth stabilises
(c) Efforts is made and the population growth stabilises
(d) Efforts are made and the population growth has stabilised

Sol. *(b)* Sentence in option (b) is correct as the correct improved sentence should be **efforts are made and the population growth stabilises** as sentence is in Simple Present Tense.

Direction (Q. No. 116) *Find the correct match of the grammatical function with the usage of the word round.*

116. (i) The Earth is **round**.
(ii) Will you come **round** to our house this evening.
(iii) The child's eye **rounded** with excitement.
(iv) Indians won the first **round** of the Tennis cup.

(a) Preposition, Verb, Noun, Adjective
(b) Adjective, Preposition, Verb, Noun
(c) Noun, Adjective, Verb, Preposition
(d) Verb, Noun, Adjective, Preposition

Sol. *(b)* In first sentence **round** is an adjective, in second sentence **round** is used as **preposition**. In third sentence it is used as **verb** (past tense) and in fourth sentence it has been used as **noun**. So, option (b) is correct.

117. Rearrange the following sentence fragment (P, Q, R and S) to make meaningful sentences.

India has been a land ..... than the warrior or the administrator.

(P) but in the sense that learning has always been very highly valued.
(Q) not indeed in the sense that education has been universal.
(R) and the learned man has been held in higher esteem.
(S) of learning through the ages.

(a) PQSR (b) RQSP
(c) RSQP (d) SQRP

Sol. *(d)* Option (c) is the correct sequence of sentence for the blank is option (d) **SQRP**.

118. India has a vibrant space science programme ..... . The growth of space industry in India.

(P) the Indian space science programme has been working.
(Q) formulating an industrial participation policy towards
(R) covering astronomy, astrophysics, and space science
(S) with cooperative Indian industries and has been

(a) SQPR (b) PQRS (c) QPRS (d) RPSQ

Sol. *(d)* The correct sequence of sentences for the blank is option (d) **RPSQ**.

Directions (Q. Nos. 119 and 120) *Choose the correct sentence in Indirect speech of the sentence in Direct speech.*

119. He said, "Garima wants to take up a job."

(a) He said that Garima wants to take up a job.
(b) He told that Garima wanted to take up a job.
(c) He said that Garima wanted to take up a job.
(d) He said that Garima has wanted to take up a job.

Sol. *(c)* Option (c) is the correct sentence in indirect speech as the whole sentence will be written in Simple Past Tense.

120. He said, "Be quiet and listen to my words."

(a) He urged them to be quiet and listen to his words.
(b) He said they should be quiet and listen to his words.
(c) He requested them to be quiet and listen to my words.
(d) He urged them and said be quiet and listen to words.

Sol. *(a)* Option (a) is the correct sentence in indirect speech. Sentence is imperative type.

Directions (Q. Nos. 121 and 122) *Identify the meaning of the given idiom from the given options.*

121. To cut the crackle

(a) to humiliate
(b) to annoy someone
(c) to act in a friendly way
(d) to stop talking nonsense

Sol. *(d)* The idiom **'to cut the crackle'** means to stop talking nonsense.
e.g. The staff members were told **to cut the crackle** and get back to work.

122. A Jaundiced Eye

(a) jealousy (b) a generous view
(c) angry (d) prejudiced view

Sol. *(d)* Prejudiced or a biased view is the correct answer of this idiom.
e.g. She viewed politicians **with a jaundiced eye**.

Directions (Q. Nos. 123 and 124) *Fill in the blanks with appropriate part of speech -*

123. You must comply ..... the rules laid down by the committee.
(a) at (b) in (c) with (d) for

Sol. *(c)* 'with' is the correct word to be used here with comply. The word means to act according to an order or set of rules.

124. Walk carefully ..... you should fall.
(a) besides (b) nevertheless
(c) should (d) lest

Sol. *(d)* 'Lest' is the correct word to be used here.

Directions (Q. Nos. 125 and 126) *Write one word for the given sentence.*

125. A person who constantly thinks he is sick is a .....

Sol. hypochondriac. A hypochondriac is a person who constantly thinks he is sick.

126. An orator's speech which is high flown and pretentious is called ..... .

Sol. Bombastic is suitable to fill the blank. The word means high-sounding but with little meaning.

127. What will be the plural of Blitz.
(a) Blitxen (b) Blitzs (c) Blilzes (d) Blitze

Sol. *(c)* The plural of Blitz is made by adding 'es' to the word. So, option (c) Blitzes is the correct answer.

Direction (Q. No. 128) *Identify the correct Proverb.*

128. (a) A rolling stone collects no moss
(b) A rolling stone gathers no moisture
(c) A rolling stone gathers no moss
(d) A rolling stone creates no moss

Sol. *(c)* **'A Rolling stone gathers no moss'** is the correct proverb. It means that a person who never settles down and keeps changing his Jobs will not succeed in life.

129. Choose the correct past participle of verb **forego**.
(a) forego
(b) foregone
(c) forgo
(d) forewent

Sol. *(b)* 'Foregone' is the correct past participle of the verb **forego**.

130. Rearrange the following words to make a meaningful sentence.

is (1)/ a (2)/ tribute (3)/ children's (4)/ day (5)/ to (6)/ childhood (7)/ celebration (8).

Sol. The correct sentence is 'Children's day celebration is a tribute to childhood'.

SNAP

SYMBIOSIS NATIONAL APTITUDE TEST

SOLVED PAPER 2017*

Time : 3 hrs

INSTRUCTIONS

1. There are 150 questions in this paper.
2. There are four parts. Part A (General Awareness), Part B (Analytical & Logical Reasoning), Part C (General English) and Part D (Quantitative Aptitude).
3. Part A carries 30 marks, Part B carries 40 marks, Part C carries 40 marks and Part D carries 40 marks.
4. For each incorrect attempt 1/4th mark will be deducted.

SECTION A
General Awareness

1. The Ex-German Chancellor who was credited with Germany unification died in 2017.
(a) Gerhard Schroeder (b) Angela Merkel
(c) Helmut Schmidt (d) Helmut Kohl

Sol. *(d)* Helmut Josef Michael Kohl was a German statesman, who served as Chancellor of Germany from 1982 to 1998 (of West Germany 1982-1990 and of the reunited Germany 1990-1998) and as the Chairman of the Christian Democratic Union (CDU) from 1973 to 1998. From 1969 to 1976, Kohl was Minister President of the State Rhineland-Palatinate.

2. Which of the following was detected in the year 2016 by scientists?
(a) Magnetic wave
(b) Levitational wave
(c) Electric wave
(d) Gravitational wave

Sol. *(d)* Scientists observed gravitational waves-ripples in the fabric of space time-for the second time. The gravitational waves were detected by both of the twin Laser Interferometer Gravitational-Wave Observatory (LIGO) detectors, located in Livingston, Louisiana and Hanford, Washington, USA. Gravitational waves carry information about their origins and about the nature of gravity that cannot otherwise be obtained.

3. e-Nivaran is a new facility launched by Government of India. What is its purpose?
(a) To help the farmers to avail easy loans
(b) To help the RTI petitions
(c) To submit and track Income Tax grievances
(d) To get help for health related issues

Sol. *(c)* CBDT has launched the ambitious 'e-Nivaran' facility for online redressal of taxpayers grievances related to refunds, ITRs and PAN among others as part of its initiative to reduce instances of harassment of the public when it comes to complaints related to the IT Department.

An exclusive 'e-Nivaran' (electronic resolution) link has recently been activated on the e-filing portal of the department where taxpayers can register their complaints through their personal computer systems and receive a special PIN number on their registered mobile and email, as their unique number to keep track of the issue.

4. The technology that shares the limelight of Bitcoin by providing technological support to the said Bitcoin is called
(a) Blockcurrency (b) Blockcash
(c) Blockchain (d) Blockcipher

Sol. *(c)* Blockchain is a continuously growing list of records, called blocks, which are linked and secured using cryptography. Each block typically contains a cryptographic hash pointer as a link to a previous block, a timestamp and transaction data. By design, blockchains are inherently resistant to modification of the data.

* Memory Based

This technology came about initially as a way to verify bitcoin transactions online and to enable two parties to transact business without having to know or trust each other.

5. Who won the Hridayanath Mangeshkar award in 2016?
(a) Vishwanathan Anand (b) Amir Khan
(c) Javed Akhtar (d) Salman Khan

Sol. *(a)* Five-time world chess champion Vishwanathan Anand (46) was honoured with the Hridayanath Mangeshkar award 2016 established in 2011 by Mumbai- based socio-cultural organisation Hridayesh Art in the honour of music composer and singer Hridayanath Mangeshkar. The award felicitates and recognises successful individuals across the country from various walks of life. It carries Monetary award of ₹ 2 lakh and a memento. The previous recipients of the award include eminent personalities like Lata Mangeshkar, Asha Bhosle, Amitabh Bachchan, Hariprasad Chaurasia and AR Rahman. In 2017, this award was presented to Javed Akhtar.

6. The last person to 'walk the moon' passed away this year. He was
(a) Buzz Aldrin (b) Neil Armstrong
(c) Yuri Gagrin (d) Eugene Cernan

Sol. *(d)* Eugene Cernan was the last man to walk on the Moon. He visited the Moon's neighborhood twice, aboard Apollo 10 and Apollo 17 and also did a challenging spacewalk aboard Gemini 9. While he chose not to fly in the shuttle programme, Cernan remained involved in space as a motivational speaker and sometime television commentator, who worked on ABC broadcasts about the space shuttle.

7. In January 2017 ..... banned FM Broadcasting.
(a) France (b) Sweden (c) Denmark (d) Norway

Sol. *(d)* Norway became the first nation to start switching off its FM radio network, in a risky and unpopular leap to digital technology that will be closely watched by other countries considering whether to follow suit. The shutdown of the FM (Frequency Modulation) network, introduced in the 1950s, began in the Northern city of Bodoe on 11th January, 2017. By the end of the year, all national FM broadcasts will be closed in favour of DAB, which backers say will carry less hiss and clearer sound for listeners in the large nation cut by fjords and mountains.

8. Columbia's Juan Manuel Santos won the 2016 Nobel Prize for peace. What was his contribution that was recognised?
(a) For his efforts to bring an end to the exploitation of agrlcultural labourers
(b) For his efforts to end the country's civil war
(c) For his efforts to bring peace between latin American countries
(d) For his efforts to eradicate the use of land mines in war

Sol. *(b)* The Nobel Peace Prize was awarded to Colombia President Juan Manuel Santos, "for his resolute efforts to bring the country's more than 50-years-long civil war to an end."
Arguably the most notable accomplishment of Santos's presidency has been his success in bringing the leftist FARC (Revolutionary Armed Forces of Colombia) to the bargaining table after decades of conflict.

9. Which is India's longest tunnel?
(a) Chennai-Nashri tunnel (b) Jawahar tunnel
(c) Zoji-La tunnel (d) Banihal-Nashri tunnel

Sol. *(a)* Chennai-Nashri tunnel, India's longest highway tunnel, is on the Jammu-Srinagar national highway. The length of the tunnel is 9.28 km. The world's longest highway tunnel is 24.51 km and is in Norway. It will reduce the travel time and distance between Jammu and Srinagar by 2 hrs and 30.11 km, which will help save fuel worth ₹ 27 lakh daily.

10. How many Men's Single Wimbledon titles does Roger Federer have to his name including the 2017 title?
(a) 9 (b) 7
(c) 8 (d) 6

Sol. *(c)* Roger Federer has won 8 man's single wimbledon titles till now. Roger Federer was born on 8th August, 1981. He is a Swiss professional tennis player, who is currently ranked world No 2 in men's singles tennis by the Association of Tennis Professionals. Federer has won 19 Grand Slam singles titles, the most in history for a male tennis player and has reached a record 29 Grand Slam finals, 42 semi-final appearances and 51 quarter-final appearances.

11. Sweden based furniture maker IKEA acquired this online (and mobile) market place for finding freelance labour to carry out local on demand day-to-day tasks.
(a) Handy Man (b) Urban Clap
(c) Man Friday (d) Task Rabbit

Sol. *(d)* IKEA agreed to acquire Silicon Valley startup Task Rabbit. Task Rabbit is one of the best-known startups in the so called 'gig' economy that links freelance workers with jobs, from handy men to movers to assistants. It has about 60 employees, but over 60000 independent workers use its platform.

12. Which of the following personalities is a member of the committee of administrations of Indian Cricket during October 2017?
(a) Anjum Chopra (b) Sonika Kaliraman
(c) Diana Edulji (d) Mithaly Raj

Sol. *(c)* Former Indian women's cricket team captain Diana Edulji was named in the Supreme Court appointed four member committee of administration to the affairs of the Board of Control for Cricket in India (BCCI).

13. The Flipkart backed startup is into a social network platform exclusively for new parents.
(a) Newparents (b) Post9Months
(c) Tiny Step (d) Baby Step

Sol. *(c)* Founded by Suhail Abidi, Tiny Step is an app-first social network for parents. Tiny Step, a platform that connects parents with doctors, child service providers and other parents, has raised an undisclosed amount in its seed round of funding from Flipkart. The startup plans to use this amount to develop the product, enhance user experience and strengthen the team.

14. Who are the last two (present and the immediate past) Attorney Generals?
(a) Goolam Vahanvati and Milon Banerji
(b) K K venugopal and Goolam Vahanvati
(c) Mukul Rohatgi and Milon Banerji
(d) K K Venugopal and Mukul Rohatgi

Sol. *(d)* The Attorney General for India is the Indian Government's chief legal advisor and its primary lawyer in the Supreme Court of India. He is appointed by the President of India under Article 76(1) of the Constitution and holds office during the pleasure of the President. The 15th and current Attorney General is K K Venugopal. He replaced the Former Attorney General of India Mr. Mukul Rohatgi. He was formally appointed as with effect from 30th June, 2017.

15. Walmart acquired Men's fashion retailer ..... for $ 310 million in 2017.
(a) GAP (b) BONOBOS (c) ZARA (d) BEBE

Sol. *(b)* Walmart has agreed to acquire the men's clothing company BONOBOS for $ 310 million. BONOBOS, founded 10 years ago in New York, began by selling simple chino pants on the internet. In recent years, it has expanded to offer shirts, suits and other men's clothing.

16. Which of the following is the cyclone that hit India in December 2016?
(a) Eona (b) Nada (c) Gloria (d) Ockhi

Sol. *(b)* Cyclone Nada originated in the form of depression in the Bay of Bengal and hit Tamil Nadu and Kerala, in December 2016. There was light to moderate rainfall in both the states, with fishermen advised not to venture out into the sea because of the rough sea. With NDMA and (National Disaster Response Force) NDRF prepared to tackle the cyclone, not much damage was reported.

17. The RP-Sanjiv Goenka Group has roped in .... as the brand ambassador for its package snacking brand ..... on a bid to scale up its FMCG business.
(a) M S Dhoni, E-Vita (b) Virender Singh, Yummee
(c) P V Sindhu, Hungree (d) Virat Kohli, Too Yumm

Sol. *(d)* The RP-Sanjiv Goenka Group has roped in India cricket captain Virat Kohli as the brand ambassador for its packaged snacking brand, Too Yumm, in a bid to scale up its FMCG business which is already almost ₹ 100 crore in seven months of its launch.

18. Identify the leading business personality in the picture.

(a) Alice Vaidyan-CMD General Insurance Company
(b) Aarthi Subramanian- Exec. Director (Global Head-DEGC) TCS
(c) Kaku Nakhate President and Country Head Bank of America Merill Lynch
(d) Chitra Ramakrishna, MD & CEO National Stock Exchange

Sol. *(d)* Chitra Ramakrishna (born in 1963) was the first woman managing director and chief executive officer of the National Stock Exchange (NSE), an institution founded in the early 1990s to reform the capital market in India and now ranked as the world's largest exchange in cash market trades and as one of the top three exchanges in index and stock derivatives.

19. The Eurasia Tunnel that was opened for traffic in December 2016 connecting Europe and Asia passes underneath the ..... .
(a) Bosphorus strait (b) Dover strait
(c) Bering strait (d) Hormuz strait

Sol. *(a)* The Eurasia tunnel is a road tunnel in Istanbul, Turkey, crossing underneath the Bosphorus strait. The tunnel was officially opened on 20th December, 2016 and opened to traffic on 22nd December, 2016. It crosses the Bosphorus beneath the seabed at a maximum depth of 106 m.

20. What was the name of the mascot in the recently concluded Asian Athletic Championship 2017 held at Bhubaneshwar?
(a) Shera (b) Coby
(c) Olly (d) Vinicious

Sol. *(c)* Olly, the Olive Ridley sea turtle was the official mascot for the 22nd edition of the Asian Athletics Championships held in Bhubaneshwar from 6th to 9th July, 2017.

21. One of Olympics longest official sponsors recently ended its 41 years long sponsorship. This organisation was ..... .
(a) Pepsi (b) General Motors
(c) Adidas (d) McDonalds

Sol. *(d)* McDonalds has ended its 41-years-long Olympic Games sponsorship deal three years early. The fast-food giant had a contract running through the 2020 Tokyo Summer Olympics. McDonalds has been a sponsor since 1976 and is part of the 10Cs top sponsors programme that contributes more than $1 billion in every four-year cycle for the games.

22. On 15th February, 2017, ISRO made history by launching 104 satellites through a single rocket. What was the rocket?
(a) PSLV 3104 (b) GSLV-IRS-P3
(c) PSLV-C37 (d) GSLV-C38

Sol. *(c)* PSLV-C37 was the 39th Mission of the Indian Polar Satellite Launch Vehicle (PSLV) programme and its 16th mission in the XL configuration. PSLV-C37 successfully carried and deployed a record 104 satellites in sun-synchronous orbits. It was launched on 15th February, 2017 by the Indian Space Research Organisation (ISRO) from the Satish Dhawan Space Centre at Sriharikota, Andhra Pradesh, it broke the earlier record of launching 37 satellites by a Russian Dnepr rocket on 19th June, 2014.

23. In the world of Information Technology, SMAC has become very popular. What does the A stand for in the abbreviation?
(a) Advanced (b) Analytics
(c) Analysis (d) Artificial intelligence

Sol. *(b)* To highlight the incorporation of the aforementioned enablers of technology, the IT industry has come up with a new concept which has become the buzzword of today, 'SMAC'- an acronym for Social, Mobile, Analytics and Cloud.
It develops an ecosystem that allows businesses to enhance their operations and maximise their reach to the customers with minimal overhead.

24. Who owns the recently launched New Channel called Republic TV?
(a) Anil Ambani
(b) Subramaniam Swami
(c) Rajeev Chandrashekhar
(d) Arnab Goswami

Sol. *(d)* Republic TV is an Indian English-language news television channel. It broadcasts from its studios in Mumbai and Bengaluru. Created by former Times Now Editor-in-Chief Arnab Goswami, it was launched on 6th May, 2017. It is owned by Rajeev Chandrasekhar, an entrepreneur and an Independent politician and others including Goswami, along with his wife Samyabrata Ray.

25. In a move that will reverberate from Hollywood and Silicon Valley to TVs and smartphones around the world, the …… said Thursday (14th December, 2017) that it had reached a deal to buy most of the ..... the empire controlled by ....., in an all-stock transaction valued at approximately
(a) Walt Disney Company, 20th Century Fox, Rupert Murdoch, $70 billion
(b) Paramount Pictures, 20th Century Fox, Rupert Murdoch, $50 billion
(c) Walt Disney Company, 21st Century Fox, Rupert Murdoch, $50 billion
(d) Paramount Pictures, 21st Century Fox, Rupert Murdoch, $70 billion

Sol. *(c)* Disney on 14th December, 2017 announced the deal to acquire many parts of 21st century fox owned by Rupert Murdoch for $ 52.4 Billion in shows.

26. The language creator who constructed artificial language 'Dothraki' for the television series Game of Thrones is .....
(a) David Henchman (b) David Peterson
(c) Robert Peterson (d) Robert Henchman

Sol. *(b)* Martin's fantasy novel series A Song of Ice and Fire and its television adaptation Game of Thrones, where it is spoken by the Dothraki, nomadic inhabitants of the Dothraki Sea. The language was developed for the TV series by the linguist David J Peterson based on the Dothraki words and phrases in Martin's novels.

27. The human organ was discovered in 2017 in our digestive system.
(a) Mesentery (b) Lacteal
(c) Enterocyte (d) Ileocecal valve

Sol. *(a)* The mesentery, the new organ is found in our digestive systems and was long thought to be made up of fragmented, separate structures. But recent research has shown that it's actually one, continuous organ. However, its function is still unclear.

28. This comic book company was acquired by Netflix in 2017.
(a) Millarworld (b) DC Comics
(c) Marvel (d) Doton

Sol. *(a)* Netflix has acquired Millarworld, which is an independent comic publishing company founded by Mark Millar, a storied comic book creator who is behind a host of iconic characters and series, including Kick-Ass and Kingsman, as well as the creative force behind some of Marvel's best story arcs, including The Ultimates and Old Man Logan.

29. The country was recently boycotted by a majority of its neighbours.
(a) Oman (b) Qatar
(c) Bahrain (d) Kuwait

Sol. *(b)* Saudi Arabia and three other Arab nations have severed diplomatic ties and cut off transport links with Qatar in an extraordinary attempt to isolate the Gulf state as they claim its regional policies fuel extremism and terrorism. The unprecedented move against Doha by Riyadh, Egypt, the United Arab Emirates and Bahrain escalates a dispute over Qatar's support for political Islamist groups and perceptions that it is open to closer ties with Iran, Saudi Arabia's rival. Qatar, which shares borders with Saudi, Bahrain and the UAE, has also been expelled from a Saudi-led military coalition fighting in Yemen.

30. In January 2017, Japan launched its first ever military communications satellite. It was called
(a) Fuji-1 (b) Kirameki-2
(c) Suzuki-1 (d) Takahiro-2

Sol. *(b)* Japan launched Kirameki-2 communications satellite, which will enable it to quickly share information about ballistic missiles launched by North Korea or videos of Japanese troops deployed overseas. Japan has recently launched its first military communications satellite in a bid to provide the much needed impetus to the broadband capacity of its Self-Defense Forces. The satellite Kirameki-2 was launched from Japan Aerospace Exploration Agency's Tanegashima Space Centre, on board of H-2A rocket and successfully entered orbit.

SECTION B
Analytical & Logical Reasoning

1. What is the missing number? 2961 3870 ? 9108
(a) 1629 (b) 9018 (c) 5823 (d) 7830

Sol. *(c)* In the given series, consider each number
$2961-1692=1269 \Rightarrow 3870-783=3087$
$9108-8019=1089$
Similarly, $5823-3285=2538$
Here, each number when subtracted from its reverse, gives a number which have same digits as that of original number.

2. Four women P, Q, R, S and three men X, Y, Z play bridge, a card game for four players.
- The men and women consist of three married couples and a widow.
- The members of each married couple are never partners in a game.
- No more than one married couple ever plays in the same bridge game.

One night they played four bridge games in which partners were as follows:
Game 1 : P and X *versus* Q and Y
Game 2 : P and Z *versus* S and Y
Game 3 : Q and R *versus* Y and Z
Game 4 : R and X *versus* S and Z

Who is the widow?
(a) P (b) S (c) R (d) Q

Sol. *(c)* As per the given conditions, we have the following partners.
$Z^+ \longleftrightarrow Q^- \Rightarrow X^+ \longleftrightarrow S^-,\ Y^+ \longleftrightarrow P^-$
$\therefore$ R is the widow.

3. David pans a small tea estate in Assam, all of which is shared by him with Rahul and his friends. Rahul has a share of 3/5th of it and his friend Santosh has twice as much as John, who is also a close friend of Rahul. What fraction of the land belongs to John?
(a) 2/15 (b) 2/5
(c) 2/10 (d) Cannot be determined

Sol. *(d)* Since, nothing is given about share of David. Hence, answer cannot be determined.

4. What numbers should replace the question marks?

	7		8		12	
2	21	8	35	3	?	?
	4		11		7	

(a) 31 and 4 (b) 49 and 21
(c) 29 and 2
(d) None of the mentioned options are correct

Sol. *(b)* Here, $4-2=2$
and $8-7=1$
$\therefore$ Middle number $=21$
Also, $11-8=3$ and $8-3=5$
$\therefore$ Middle number $= 35$
Similarly, $7-3=4$ and $21-12=9$
$\therefore$ Middle number $= 49$

5. Study the rules employed in the following example and answer the question.
Input from 22 61 tea India east 55 74
Step I east from 22 61 tea lndia 55 74
Step II east 74 from 22 61 tea India 55
Step III east 74 tea from 22 61 lndia 55
Step IV east 74 tea 61 from 22 India 55
Step V east 74 tea 61 India from 22 55
Step VI east 74 tea 61 India 55 from 22

Identify the input if the result of Step II of an input is
Money 56 to 11 66 people notes 24
(a) Money 56 To 15 66 people notes 24
(b) To 15 Money 56 66 people notes 24
(c) To 15 66 Money people notes 24 56
(d) Cannot be determined

Sol. *(d)* We cannot find the input from Step II.

6. Mr. Golu, the farmer is desperate to make some money by selling his livestock. He has many animals and has a unique set of customer requirements. The customers do not want to buy animals with disabilities or strange features. Mr. Golu has a Stock of animals which include Monkeys with two tails, Sheep with six legs, cows with two udders, horses with two right hooves and chicken with two toes on each foot. Which animal will he be able to sell?
(a) Horse
(b) Cow
(c) Monkey
(d) He will not be able to sell any animal.

Sol. *(d)* Golu will not be able to sell any animal.

7. Which of the following is the greatest?

I. $\log(2+4)$ II. $(\log 2+\log 4)$

III. $\log(6-3)$ IV. $\log 6-\log 3$

(a) Only III (b) Only IV

(c) Only II (d) Both I and II

Sol. *(c)* I. $\log(2+4)=\log 6=\log(2\times 3)$

$=\log 2+\log 3$

II. $\log 2+\log 4=\log 2+2\log 2=3\log 2$

III. $\log(6-3)=\log 3$

IV. $\log 6-\log 3=\log(2\times 3)-\log 3=\log 2$

Now, $\log 2+\log 3>\log 2,\log 3$

Also, $3\log 2>\log 2+\log 3$

Hence, II is the greatest.

8. In a village located in remote part of the world 5% of the population have a birth defect and hence have only one leg, and half of the others go barefoot as they never wear shoes. What is the least number of shoes needed in the village if the total population is 10000?

(a) 5000 (b) 4750

(c) 9500 (d) 10000

Sol. *(b)* Required number of shoes

$$=\frac{95\ \%\ \text{of}\ 10000}{2}$$

$$=\frac{95\times 10000}{100\times 2}=4750$$

9. At 12:00 noon, my watch was correct. However, it then started to lose 18 minutes per hour. Four hours ago, it stopped completely and is now showing the time as 15 : 30. What is the correct time?

(a) 17:00 (b) 21:00 (c) 20:33 (d) 19:00

Sol. *(b)* Let the correct time at 15:30 be y o'clock

Then, $y-\frac{18y}{60}=\frac{7}{2}\Rightarrow y-\frac{3y}{10}=\frac{7}{2}$

$\Rightarrow \quad \frac{7y}{10}=\frac{7}{2}\Rightarrow y=5$

$\therefore$ Correct time after 4 hours = 21:00

10. 3182596 is to 65283 and 6742835 is to 53476, therefore 7496258 is to ..... ?

(a) 98765 (b) 76574

(c) 65987 (d) 85647

Sol. *(d)* As, 3182596 ends with 6 and 65283 starts with 6.

And 6742835 ends with 5 and 53476 starts with 5.

Similarly, 7496258 ends with 8 and 85647 starts with 8.

11. If 645 = 10 and 316 = 12, then 765 = ?

(a) 18 (b) 5 (c) 7 (d) 210

Sol. *(b)* As, $645=10\ \Rightarrow\ (6-4)\times 5=10$

And $316=12\ \Rightarrow\ (3-1)\times 6=12$

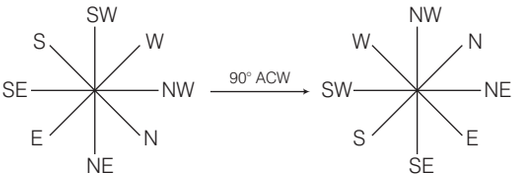
Similarly, $765=(7-6)\times 5=5$

12. A villager places a map of his farm in a gram panchayat meeting for resolving his dispute. He places the map in a way that South-East becomes North and North-East becomes West. The Sarpanch is confused. His assistant comes to help and turns the anti-clockwise by 90°. If the Sarpanch has to see the South side in the map, which side will he look?

(a) North (b) South-West

(c) South (d) West

Sol. *(b)*



If he has to see south, then he has to look in South-West direction.

13. A, B and C played some card games.

Each game had exactly one winner.

No player won 2 game in succession.

When a player was the dealer for a game he/she did not win the game.

The dealer position proceeded from A to B to C, then this order was repeated until they stopped.

The only player to win more than 2 games did not win the first game.

Who was the only player to win more than 2 games?

(a) A (b) B

(c) C (d) *Cannot be determined*

Sol. *(c)*

Dealer	A	B	C	A	B	C	A
Winner	B	C	A	C	A	B	C

$\therefore$ C won more than two games.

14. There are several aspirants who are made to sit in single row for an entrance examination. Rajesh is fifteenth from the left and Mohan is eight from the right end. If there are four aspirants between them and Mohan is to the right of Rajesh, then the total number of aspirants sitting in the row are ..... .

(a) 26

(b) 28

(c) 27

(d) None of the mentioned options are correct

Sol. *(c)*

$\therefore$ Total aspirants $=14+1+4+1+7=27$

15. Study the following information and answer the question.

In a recent performance approval done by Company X, more than 70% of the employees were found to be underperforming.

'I think that the restructuring done by the company in the previous year is responsible for the under-performance of the employees'- HR manager of Company X.

Which of the following statements weakens the statement of HR Manager of Company X?

(a) The incentives linked to performance were abolished upon restructuring creating dissatisfaction among employees.
(b) After restructuring the decision making power was taken away from employees thus delaying the work by long hours.
(c) The number of projects in Company X increased by 60% this year thereby increasing the burden on the existing employees this year.
(d) After restructuring the employees were required to report to multiple bosses leading to ambiguity at the work place.

Sol. *(c)* All the options except (c) strenghten the statement of HR manager of Company X.

Directions (Q.Nos. 16-20) *Study the following information to answer the given question.*

Twelve people are sitting in two parallel rows containing six people each in such a way that there is an equal distance between adjacent persons. In row 1- J, K, L, M, N and O are seated (but not necessarily in the same order) and all of facing South. In row 2- U, V, W, X, Y and Z are seated (but not necessarily in the same order) and all of then are facing North. Therefore, in the given seating arrangement, each member seated in a row faces another member of the other row.

M sits fourth to the left of J. The one facing J sits third to the left of Y.

Only one person sits between Y and U. U does not sit at any of the extreme ends of the line.

The one facing Z sits second to the right of K. Z does not sit at any of the extreme ends of the line.

Only two people sit between K and O.

The one facing K sits second to the left of X.

V is not an immediate neighbour of Z. L is not immediate neighbour of M.

16. Which of the following groups of people represents the people sitting at extreme ends of both the rows?

(a) M, O, X, W (b) M, K, V, W
(c) N, K, V, Y (d) J, N, U, V

17. Who amongst the following faces V?

(a) M (b) L (c) J (d) K

18. Which of the following is true with respect to the given information?

(a) K faces one of the immediate neighbours of X.
(b) V sits exactly between W and U.
(c) J faces Z.
(d) J is an immediate neighbour of K.

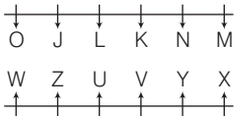
19. Which of the following is true regarding N?

(a) K sits second to right of N.
(b) V is an immediate neighbour of the person who faces N.
(c) Both L and O are immediate neighbours of N.
(d) Only one person sits between N and J.

20. Who amongst the following sits second to the right of the person who faces L?

(a) V (b) Z
(c) W (d) Y

Solutions (Q. Nos. 16-20) *As per the given information, the arrangement is as follows.*



16. *(a)* M, O, X, W are sitting at extreme ends of both the rows.

17. *(d)* K faces V.

18. *(c)* J faces Z is true.

19. *(c)* V is an immediate neighbour of Y and Y faces N.

20. *(d)* Y sits second to the right of U and U faces L.

21. If '4' is subtracted from each odd digit and '1' is added to each even digit in the number 9476582, which of the following numbers will appear twice in the new number thus formed?

(a) Only 2 (b) Both 1 and 7
(c) Both 3 and 5 (d) None

Sol. *(c)*

9	4	7	6	5	8	2
↓	↓	↓	↓	↓	↓	↓
5	5	3	7	1	9	3

∴ Both 3 and 5 appear twice.

22. Read the given information and answer the question. The government of State D was recently criticised for accepting a proposal as per which schools in the State will not conduct exams up to standard nine.

Rather, schools will promote all the students up to standard nine and from standard nine onwards the students would either be passed or failed depending upon their marks in the exams.

Which of the following may not be a reason for the criticism of the government of State D for accepting the proposal?

(a) Several parents have raised concern that students would not be willing to pay attention in class and retain the information because they do not have to write exams.

(b) Students learn to write exam papers in a stipulated period of time with practice of solving exam papers over the years and many students may not be able to grasp this directly in the standard nine.

(c) Many teachers are of the opinion that students have become very competitive and are mainly focused on acquiring marks rather than gaining knowledge.

(d) Studies suggest that students become less prone to exam anxiety and exam fear when they get used to passing exams over the years as compared to when they do not appear for an exam at all.

Sol. *(c)* Since, in Today's time students are only focussed on acquiring marks rather than on having knowledge and exposure. So, option (c) can be reason for government of State D to accept the proposal.

Directions (Q.Nos. 23-28) *Study the following information and answer the question.*

Seven friends, namely P, Q, R, S, T, U and V visit seven different countries namely Japan, Germany, China, India, Nepal, Australia and Malaysia, not necessarily in the same order, starting from Monday to Sunday (of the same week).

R visits on Thursday. Only two people visit between R and the one who visits Germany. Only four people visit between the one who visits Germany and V. The one who visits Malaysia visits immediately before V. Only two people visit between the one who visits Malaysia and P. S visits on one of the days after the one who visits Malaysia.

U visit immediately after the one who visit Japan. U does not visit Malaysia. Only three people visit between the one who visits Japan and the one who visits Nepal. The one who visits Australia visits immediately before the one who visits China. Q does not visit on Monday.

23. Which of the following countries does Q visit?
(a) China (b) Malaysia
(c) Japan (d) Nepal

24. On which of the following days does U visit a country?
(a) Friday (b) Saturday
(c) Sunday (d) Wednesday

25. Which of the following is true about T?
(a) T visits immediately before P
(b) T visits on Friday
(c) T visits China
(d) All the given options are true

26. Who amongst the following visits India?
(a) S (b) T (c) P (d) G

27. As per the given arrangement, P is related to the one who visits Japan in a certain way and V is related to the one who visits Nepal in the same way. To which of the following is R related to in the same way?
(a) The one who visits Australia
(b) The one who visits China
(c) The one who visits India
(d) The one who visits Malaysia

28. Four of the following five are alike in a certain way and thus form a group as per the given arrangement. Which of the following does not belong to that group?
(a) U-Friday (b) Q-Thursday
(c) S-Saturday (d) V-Sunday

Solutions (Q. Nos. 23-28)

Monday	T	Germany
Tuesday	P	Japan
Wednesday	U	Australia
Thursday	R	China
Friday	Q	Malaysia
Saturday	V	Nepal
Sunday	S	India

23. *(b)* Q visits Malaysia.

24. *(d)* U visits a country on Wednesday.

25. *(a)* T visits immediately before P.

26. *(a)* S visits India.

27. *(b)* R is related to the one who visits china using the given pattern.

28. *(a)* U-Friday is the odd one out.

Directions (Q.Nos. 29-32) *Study the following information carefully and answer the given question.*

When a word and number arrangement machine is given an input line of words and numbers, it arranges them following a particular rule. The following is an illustration of input and rearrangement. (All the numbers are two digit numbers)

Input 23 kinetic amount 64 nature 71 58 opium verdict 96 elderly 15

Step I opium 23 kinetic amount 64 nature 71 verdict 96 elderly 15 58

Step II elderly opium 23 kinetic amount nature 71 verdict 96 15 58 64

Step III amount elderly opium 23 kinetic nature 71 verdict 15 58 64 96

Step IV 15 amount elderly opium 23 kinetic nature 71 58 64 96 verdict

Step V 23 15 amount elderly opium kinetic 71 58 64 96 verdict nature

Step VI 71 23 15 amount elderly opium 58 64 96 verdict nature kinetic

Step VI is the last step of the above arrangement and as the intended arrangement is obtained.

As per the rules followed in the given steps, find out the appropriate steps for the given input.

Input adverb 59 36 salient 81 idioms bakery 14 launch 47 umpire 62

29. Which elements come exactly between '59' and 'bakery' in Step II of the given input?
(a) Only launch (b) Only 62
(c) Only idioms (d) Both 81 and salient

30. Which of the following combinations represents the first two and the last two element of the step VI of the given input?
(a) 81, 59, launch, bakery (b) 81, 62, salient, launch
(c) 50, 47, bakery, salient (d) 62, 14, idioms, umpire

31. If in the step III, 'idioms' interchanges its position with '81' and 'salient' also interchanges its position with '14', then which element will be third to the left of '47'?
(a) 14 (b) salient
(c) adverb (d) idioms

32. Which step are the elements 'bakery launch 47 14' found in the same order?
(a) Sixth (b) Third
(c) Fourth (d) Fifth

Solutions (Q. Nos. 29-32)

Input adverb 59 36 salient 81 idioms bakery 14 launch 47 umpire 62.

Step I adverb 59 36 salient 81 idioms bakery launch 47 umpire 62 14.

Step II idioms adverb 59 salient 81 bakery launch 47 umpire 62 14 36

Step III umpire idioms adverb 59 salient 81 bakery launch 47 14 36 62.

Step IV 47 umpire idioms adverb 59 81 bakery launch 14 36 62 salient.

Step V 59 47 umpire idioms adverb 81 bakery 14 36 62 salient launch.

Step VI 81 59 47 umpire idioms adverb 14 36 62 salient launch bakery.

29. *(d)* Both 81 and salient come exactly between 59 and Bakery.

30. *(a)* First two elements of Step VI-81, 59
Last two elements of Step VI-launch, bakery

31. *(d)* Idioms will be third to the left of 47.

32. *(b)* Step III have elements 'bakery launch 47 14' in the same order.

33. The question consists of information and two Statements numbered I and II given below it. You have to decide which of the given statements weakens or strengthens the information and decide the appropriate answer.

Information Due to increased cases of kidnapping in its vicinity, school M has made it compulsory for parents or guardians of the students to give a duly signed authority letter to the person picking up the students from school.

I. Most working parents rely on their domestic help, for picking up their children from school, who can easily coerced into forging or misusing the authority letter for monetary plans.

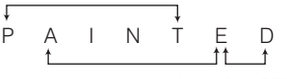
II. There is no photograph on the authority letter making it difficult to identify the person who comes to pick up the child.

(a) Both Statement I and Statement II strengthen the information.
(b) Statements I strengthens the information while Statement II weakens the information.
(c) Both Statements I and II weaken the information.
(d) Both Statements I and II are neutral are neutral Statements.

Sol. *(c)* Both Statements I and II weaken the given statements as domestic helpers can easily coerced into forging the authority letter. And no photograph on the letter can be a great drawback of the authority letter.

34. How many such pairs of letters are there in the word 'PAINTED' each of which has as many letters between them in the word in both forward and backward directions, as there are between them in the English alphabetical series?
(a) More than three (b) Two
(c) Three (d) None

Sol. *(c)*



∴ There are three such pairs PT, AE and ED.

Directions (Q.Nos. 35-39) *Study the given information carefully to answer the given question.*

In a certain code language, 'urban people prefer cars' is written as 've fm ab eg' 'profit for urban areas' is written as 'ab ep zi so' 'people demand for hike' is written as 'zi qr cd ve' 'hike in profit margin' is written as 'al jn ep cd' (All codes are two letter codes only)

35. What will be the possible code for 'urban food demand' in the given code language?
(a) qr ab zi (b) qr cr ab (c) nj qr cd (d) qr ab jn

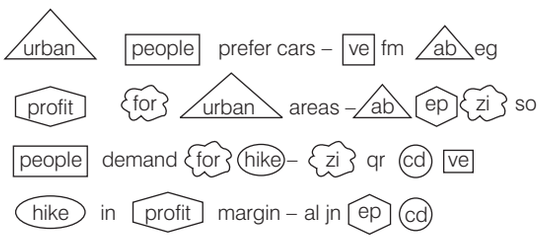
36. What is the code for areas in the given code language?
(a) ab (b) zi (c) ep (d) so

37. What does the code 'jn' stand for in the given code language?
(a) either 'hike' or 'people' (b) either 'in' or 'margin'
(c) profit (d) hike

38. In the given code language, if 'small' is coded as 'wy', then how will 'prefer small cars' be coded as?
(a) wy eg ab (b) fm ve wy
(c) eg wy fm (d) ab eg fm

39. What will be the code for 'hike' in the given code language?
(a) xl
(b) zi
(c) qr
(d) other than those given as options

Solutions (Q.Nos. 35-39)



35. *(b)* urban → ab demand → qr
∴ Possible code for 'urban food demand' is 'qr cr ab'.

36. *(d)* areas → so

37. *(b)* jn→in/margin

38. *(c)* small→wy prefer/cars→ fm/eg
∴prefer small cars → eg wy fm

39. *(d)* hike → cd

40. Read the given information and answer the question. Long term usage of antibiotics causes the disease 'Cretosis' as it decreases the secretion of hormone X. While body can endure the level of hormone X dropping to half the required number of micrograms in levels dropping to 23- micrograms needs immediate medical attention.

Which of the following can be concluded from the given statement?
(a) A patient can be said to have Cretosis only if his/her hormone 'X' levels are 23 micrograms or low.
(b) Usage of antibiotics on a short term cannot cause 'Cretosis' ever to a minor extent.
(c) All micrograms is exactly half of the amount of hormone X required daily by the body.
(d) In a patient with hormone X level of 21 microgram, if administered another 23 micrograms would bring the level to absolute normal.

Sol. *(d)* Only (d) can be deduced from the given statement.

SECTION C
General English

1. Give synonym for
Artless
(a) Pretentious (b) Innocent
(c) Devious (d) Hypocritical

Sol. *(b)* 'Artless' means 'Innocent'.

2. Choose a word from the given alternative which is opposite in meaning to the word HIATUS
(a) Sardonic (b) Pessimist
(c) Sentimental (d) Continuation

Sol. *(d)* 'Hiatus' means - a short pause or break. Hence, its opposite will be 'Continuation'.

3. Choose the right meaning of the idiom.
A pyrrhic victory
(a) A victory that is not worth winning
(b) Joyous victory
(c) Undeserving victory
(d) None of the mentioned options

Sol. *(a)* The Idiom 'A pyrrhic victory' means 'A victory that is not worth winning' because the winner has lost a lot in the process of winning.

4. Choose the correct based on the sentence.
Akhil fared poorly in the entrance test. He found it.
(a) Fathomable (b) Esoteric
(c) Manifest (d) Coherent

Sol. *(b)* Since, the performance has been poor that's why Akhil didn't well do in the exams. So, this can be said to be 'esoteric' which means 'that can be understood by only a few'.

5. Choose one word from the options given, that can fit in to all sentences.
1. You should not …… to his demands.
2. Ravi needed a loan. However, he did not …… to the loan agreement.
3. The state will …… to the union.
4. The Prince will …… to the throne.
(a) Accede (b) Precede (c) Recede (d) Secede

Sol. *(a)* In all the blanks above, 'accede' is the correct usage which means - 'to agree'.

6. Choose the correct word(s) for the underlined word(s). You cannot trust a person who plays **false and loose with** others.
(a) fastloose (b) false or loose
(c) fast or loose (d) fast and loose

Sol. *(d)* Idioms don't change their structure at all. Hence, 'false and loose' will be written as 'fast and loose'. It means to behave irresponsibly or irrmorally.

7. Choose the correct meaning of the phrase.
The team of doctors sent to the village to investigate the epidemic were exasperated. The conditions there were 'a perfect storm'.
(a) Torrential rains (b) A worst possible scenario
(c) A wonderful situation (d) A risky proposition

Sol. *(b)* The phrase 'a perfect storm' means 'a worst possible scenario'.

8. Choose the correct meaning of the phrase.
When playing the Clarinet, Jerry is 'all thumbs'.
(a) Perfunctory (b) Dancing
(c) An expert (d) Clumsy

Sol. *(d)* Phrase 'all thumbs' means 'clumsy' or awkward.

9. Identify the Compound Relative Adverb in the sentence.
I do not know the venue where the match will be played.
(a) will be (b) where (c) know (d) played

Sol. *(b)* 'where' is the relative adverb.

10. Choose one word from the options given, that can fit in to all sentences.
1. I cannot ..... this principled man.
2. These religious songs will ..... you.
3. Better suppress than ..... your desires.
4. Only delicious food can ..... my hunger.

(a) Gratify (b) Edify (c) Occupy (d) Glorify

Sol. *(a)* 'gratify' means 'to satisfy' which can be filled well in all the blanks.

11. Select the pair that best expresses the relationship similar to that expressed in this pair.
convivial : jocund
(a) lyrical : prosaic (b) umbrage : happiness
(c) timid : brave (d) folly : silliness

Sol. *(d)* The principal analogy is synonymous in relation. Hence, the answer will bear the same relation. 'Folly : Silliness' is the correct choice.

12. Rearrange the sentences to make it a readable paragraph.
1. There is a difference between foreign equity stakes and that of domestic promoters.
2. Apart from it, many other differences exist.
3. The hike in the former was important as a signal.
4. These have to be treated at different levels.
5. A similar signal is not required in respect of domestic industry.

(a) 15234 (d) 14352
(c) 23145 (d) 43215

Sol. *(b)* 14352

13. Find a correct match of grammatical function with the usage of the word REBEL.

Function	Usage
A. Noun	(i) Her rebellious moods are due to the side effects of prescription drugs.
B. Adjective	(ii) Most teenagers rebel against their elders.
C. Verb	(iii) Armed rebels moved towards the city.
D. Adverb	(iv) His body shivering with anger, Kunal's eyes fixed rebelliously.

Codes

	A	B	C	D		A	B	C	D
(a)	(i)	(iii)	(ii)	(iv)	(b)	(iv)	(ii)	(iii)	(i)
(c)	(iii)	(i)	(ii)	(iv)	(d)	(iii)	(iv)	(i)	(ii)

Sol. *(c)* Noun - Rebels
Adjective - Rebellious
Verb - Rebel
Adverb - Rebellously

14. Choose a word from the given alternatives which is opposite in meaning to the word- SABLE.
(a) Ebony
(b) Chromatic
(c) Swarthy
(d) Raven

Sol. *(b)* 'Sable' means 'black'. Hence, 'chromatic' is its correct opposite.

15. From the given alternatives, choose the one which best expresses the given sentence in Passive voice.
Who ate the last piece?
(a) The last piece was eaten by who?
(b) Who was eating the last piece?
(c) The last piece who ate?
(d) The last piece was eaten by whom?

Sol. *(d)* 'The last piece was eaten by whom?' is the correct passive voice of the given sentence.

16. Give antonym for Prevaricate (Antirignous, Deceive)
(a) Gubble
(b) Unambiguous
(c) Equivocate
(d) Tergiversate

Sol. *(b)* 'Prevari cate' means 'ambiguous'. Hence, the opposite word will be 'unambiguous'.

17. Give anytonym for Dilatory
(a) Tardy (b) Remiss
(c) Indolent (d) Diligent

Sol. *(d)* 'Dilatory' means slow and likely to cause delay. So, Diligent will be its opposite.

18. Rearrange the sentences to make it a readable paragraph.

1. Rather, it is the consequence of passive public policy.
2. It follows that the output of economy must grow.
3. High unemployment is not the inevitable result of the pace of technological change.
4. This will prevent the unemployment rate from rising.
5. We can anticipate a moderate increase in labour force, accompanied by a slow decline in the working hours.

(a) 35421 (b) 42351
(c) 24351 (d) 31524

Sol. *(d)* 31524

19. From the given alternatives, choose the one which best expresses the given sentence in Active voice.

Sugarcane is raised by some people in the Hinterland.

(a) Some people in the Hinterland raised sugarcane.
(b) Sugarcane is raised in the Hinterland by some people.
(c) In the Hinterland is raised sugarcane by some people.
(d) Some people raise sugarcane in the Hinterland.

Sol. *(d)* 'Some people raise sugarcane in the Hinterland' is the correct active voice of the given sentence.

20. Find a correct match of grammatical function with the usage of the word: SHADE

Function	Usage
A. Noun	(i) He shaded his eyes and smiles.
B. Adjective	(ii) She saw a shady garden.
C. Verb	(iii) He slunk shadily in the shadows.
D. Adverb	(iv) The shade of the tree is welcome.

Codes

A	B	C	D		A	B	C	D
(a) (iv)	(ii)	(i)	(iii)	(b)	(iv)	(ii)	(iii)	(i)
(c) (i)	(iii)	(iv)	(ii)	(d)	(i)	(ii)	(iii)	(iv)

Sol. *(a)* Noun - Shade
Adjective - Shady
Verb - Shaded
Adverb - Shadily

21. Choose the correct word based on the sentence.

Raj, normally an amicable person, gets towards the unjust.

(a) Affable (b) Benign
(c) Belligerent (d) Genial

Sol. *(c)* If someone is an amicable person and he sees something unjust towards others then it is natural that he would be angry with him. So, 'belligerent' is the correct word to be used.

22. In the following question, choose the word which best expresses the opposite meaning of the word in capital.

She was very INDUSTRIOUS by temperament.

(a) Assiduous (b) Off-the-cuff
(c) Indolent (d) Apathetic

Sol. *(c)* 'Industrious' means 'hard working'. Hence, its opposite will be 'indolent' which means 'lazy'.

23. Give anyonym for Fleece

(a) Satiate (b) Bilk
(c) Mulct (d) Rook

Sol. *(a)* 'Fleece' is the state of a person when one steals out of his greed to satisfy it. So, 'satiate' will be its opposite.

24. Give synonym for Perfunctory

(a) Precise (b) Conscientious
(c) Deliberate (d) Negligent

Sol. *(d)* 'Perfunctory' means done quickly, without taking care or interest. So, negligent is similar in meaning to it.

25. Identify the Compound Relative Pronoun in the sentence.

Whosoever diggeth a pit shall fall therein.

(a) therein (b) fall
(c) diggeth (d) whosoever

Sol. *(d)* 'Whosoever' is the compound relative pronoun.

26. Choose the correct word based on the sentence.

Rupert drained all his trust funds. He was a …… .

(a) Miser (b) Churl
(c) Scrooge (d) Wastrel

Sol. *(d)* 'Wastrel' is one who wastes his money without any planning. Hence, (d) is the correct choice.

27. Select the pair that best expresses the relationship similar to that expressed in this pair.

vacillate : be indecisive

(a) vacuous : empty (b) vindictive : vague
(c) vigilant : sleepy (d) vociferous : silent

Sol. *(a)* The principal analogy has the relation of being synonymous to each other. So, 'vacuous : empty' is the right choice.

28. Choose the correct word(s) for the underlined word(s).

If I <u>was</u> you, I would do it at once.

(a) am (b) should be
(c) would be (d) were

Sol. *(d)* For imaginary conditionals, we use the auxiliary ''were''. Hence, (d) is the suitable option.

Directions (Q.Nos. 29-40) *Read the following passages and answer the questions that follow.*

PASSAGE I

The first thing to know when talking about privacy in India is that a majority of the population does not always understand what it means. It is at times confused with shame. It's also confused with the emotion we feel when we do something that does not meet our standards or our sense of what is right. Modern Indian languages do not seem to have an exact word which captures the meaning of privacy; they are usually some variation of the words for isolation, intimacy or secret, once again hinting at a conceptual confusion. This explains the reactions of many who wonder what's the big deal about privacy because they have nothing to hide from the government anyway.

Privacy, however, is not only about hiding something or keeping it secret. It is, at its core, the right to be left alone. It does not mean that one is withdrawing from society. It is an expectation that society will not interfere in the choices made by the person so long as they do not cause harm to others. It means that one's right to eat whatever one chooses, the right to drink what one chooses, the right to love and marry whom one chooses, to wear what one chooses, among others, are rights which the state cannot interfere with.

In a society, where adults do not necessarily exercise most of these choices of their own free will (either because of family, caste or societal pressures), it is natural that the very concept of privacy seems incomprehensible. If you have grown up in a society where everything is dictated by someone else and the cost of disobedience is high to have the freedom to choose what you, will in such important matters sounds like fantasy. But it is also a common misconception that the non-well-off in India do not know or care about privacy. Millions of men and women push back daily against the oppressive hold of their families and communities and fight for the freedom to make their own choices. They may not have the right word for it, but they are creating space for themselves to exercise the right to privacy.

It is in this context that one must understand the hearings in the Supreme Court on the right to privacy. Although, the nine-judge bench has been constituted to decide whether there is a fundamental right to privacy protected under the Constitution in the specific context of the Aadhar case, privacy has many more dimensions than just data protection or surveillance by the state. A Fundamental Right to privacy, enshrined and protected in the Constitution, would mean that all persons have the right to be left alone by the state, unless such intrusion is necessitated by a just, reasonable and fair law.

The nine-judge bench was necessitated in the first place because while multiple judgements have held that there is common law right to privacy (claimed against other individuals and entities); there was doubt as to whether such a right could be claimed against government. Obviously, the Constitution does not use the word 'privacy' or we wouldn't be having these hearings. Where, then, does the right to privacy find a place in the Constitution?

To answer that it is necessary to go deep into what is meant by a Fundamental Right. At their core, such rights can be said to be the lines drawn by the Constitution delineating boundaries for the government's actions. Such boundaries necessarily imply, the petitioners' counsels have argued, that the Constitution guarantees that individuals have a right to be left alone by the state on matters of individual choice. They have argued that the earlier decisions in MP Sharma *v* Satish Chandra (1954) and Kharak Singh v State of UP (1962) were rendered relying in a narrow and pedantic interpretation of Fundamental Rights - an approach that has been discarded by the Supreme Court since the 1970s.

The Union Government has argued that it does not think that the right to privacy is a Fundamental Right protected under the Constitution. Attorney General KK Venugopal has argued that while the right to privacy may be protected as a common law right or some element of it part of a Fundamental Right, by itself, it could not per se be guaranteed as a Fundamental Right. The arguments of the Union Government and State Government supporting it have been premised on an 'originalistic' interpretation of the Constitution - that the framers never intended privacy to be a Fundamental Right available to citizens. Given the Supreme Court's recent approach where it has not been hesitant to depart from the narrow interpretation of the Constitution when the situation demands it (such as appointment of judges), perhaps this approach may not find much judicial favour. Far more worrisome is the argument that privacy is only the preserve of the well-off and the elite and protecting it through law and legal institutions may stand in the way of 'development' and poverty alleviation.

This not only misunderstands what the right to privacy means but underplays its role in allowing individuals to make free choices. It is an argument of a paternalistic and patriarchal state that knows what's good for you and will not let you make your own choices. It also clashes with the ethos of a limited government enshrined in the Constitution.

A nine-judge bench of the Supreme Court holding that the Constitution guarantees a right to privacy will, however, only settle one issue - that there is a right to privacy guaranteed against state intervention. To what extent this right can be claimed and in what circumstances the state may be allowed to intrude will have to be decided on a case by case basis. At most, the court's judgement may outline the principles on the basis of which judicial review will be carried out, but it cannot be expected to prescribe an answer for every foreseeable situation.

Whatever the final judgement', the implications will go far beyond just the Aadhar scheme and law. The law laid down by the Supreme Court on privacy could affect the course of development of the law governing reproductive rights, gay rights, beef bans, prohibition, among a host of other issues that the Indian State and society are grappling with.

29. According to the author, what is meant by a Fundamental Right?
(a) Fundamental rights can be said to be the lines drawn by the Constitution delineating boundaries for the government's actions
(b) Privacy is not a Fundamental Right
(c) All the mentioned options are correct
(d) Fundamental Right is the preserve of the well-off and the elite

Sol. *(a)* According to the author, Fundamental Right means it can be said to be the lines drawn by the constitution delineating boundaries for the government's actions.

30. The tone of this passage is
(a) social (b) philosophical (c) moral (d) reflective

Sol. *(d)* The tone of the passage is reflective, which reflects upon the idea and the concept of privacy as a Fundamental Right.

31. When the author says …… "to have the freedom to choose what you will, in such matters, sounds like fantasy", he means.
(a) To be able to do what you desire in essential matters is like a dream come true
(b) You can do what you like and not pay for it
(c) To care two hoots for somebody
(d) None of the mentioned options are correct

Sol. *(a)* The phrase : ''to have ……'' means that ''to be able to do what you desire in essential matters is like a dream come true''.

32. The rights that the state cannot interfere with are
(a) The right to drink whatever one chooses
(b) The right to wear what one chooses
(c) The right to love and marry whom one chooses
(d) All the mentioned options are correct

Sol. *(d)* All the mentioned options are correct as the state cannot interfere with any one of them.

33. The author says people usually confuse privacy with
(a) emotion (b) shame (c) hatred (d) despondency

Sol. *(d)* The author says people usually confuse privacy with despondency.

34. The most suitable title for this passage would be
(a) The Right to be Left Alone
(b) Only the Elite can Enjoy Privacy
(c) The Constitution
(d) Aadhar Card

Sol. *(a)* The most suitable title for this passage is 'The Right to be Left Alone'.

PASSAGE II

Poor Elizabeth Beckford. She was 70 years old and her crime was stealing 12lb of Gloucester cheese. For that she could have been hanged. Hundreds did in those violent, vengeful days, dancing 'the Tyburn frisk' in the words of those who crammed around the gallows to watch this favourite spectator sport of the 18th century. But the state, in its mercy, saved her life - and gave her a punishment that some would see as worse than death. She was an unwilling passenger on a fleet of 11 ships that set out from England in 1787, the first of the convoys of the criminal underclass -as the ruling elite of Georgian England saw them - sent in chains to colonise new and dangerous shores on the other side of the world.

Those 736 sad souls on that pioneering voyage would establish a new world. Though she didn't know it - and the thought would have given her no consolation as she lay crammed with others in cell-like spaces below decks - Elizabeth was a founder member of a new country, Australia.

On Thursday, more than 200 years later, those who made those dreadful voyages - 163000 in all over the years to come - are feted. 21st century Australians celebrate their convict past, taking their lead from premier John Howard, a descendant of transported folk on both sides of his family.

The shipping and court registers of the banished have long lain in the National Archive in London. Now, in the knowledge that two million of us in Britain probably have blood links with Australia's criminal forebears, they have been put online for the hundreds of thousands of amateur genealogists in this country, eager to find out more about their roots.

The history they hide may not be pleasant. Elizabeth, incredibly, was not the oldest on that first ark of despair. Dorothy Handland, a dealer in rags and old clothes, was 82. How she was expected to contribute to empire-building in a virgin land whose hardships could only be guessed at is a mystery as great as the place she was being sent to. But nonetheless she was among the waggon-loads of prisoners dragged down to the docks in Portsmouth from the sunless ship hulks at Woolwich where they had been held because the prisons were all full. They were dressed in rags, their faces pale from imprisonment, louse-ridden and thin as rakes from the slops they had been forced to live on.

Alongside the grannies were 120 other women, mostly young, like 22-years-old Elizabeth Powley. Penniless at home in Norfolk she had raided someone's kitchen for a few shillings' worth of bacon, flour and raisins and '24 ounces weight of butter valued 12d'. The death sentence on this starving girl was commuted and, as Robert Hughes, historian of the transportations notes wryly in his book, The Fatal Shore, 'she was sent to Australia, never to eat butter again'.

At least the youngest of the 'passengers', John Hudson, would never be pushed up another chimney. The nine-years old sweep was condemned to seven years' exile for theft. All on board were small-time criminals whose punishment, by the standards of later generations, in no degree fitted the crime. James Grace, 11, had taken some ribbon and a pair of silk stockings. John Wisehammer, 15, snatched some snuff from a shop counter in Gloucester.

For that, they would never see home again. The most extraordinary crime was that of William Francis, who stole a book about 'the flourishing state of the island of Tobago' from a gentleman in London. If he had had time to read it before he was caught, perhaps he had an inkling of what now lay ahead of him in a British colony far rawer than the West Indies.

There were no political prisoners, however, no rabble rousing, hay stack burning activists or trade unionists sentenced for their subversive activities, as some of today's anti-Pom Australians like to think. The women, however, were treated poorly. They arrived at the gangplank of their vessel, the Lady Penrhyn, "in a situation that stamps them with infamy", according to the officer in command of the expedition, Captain Arthur Phillip. He was appalled at their treatment by the magistrates who had sentenced them and the jailers who had held them. Whether he could guarantee them better lives at the end of their nine-months voyage was yet to be seen. What they were about to embark on was the longest journey ever attempted by such a large group of people.

Where they were going might as well have been the Moon. Crewmen, let alone convicts, believed they would never see home or their loved ones again. 'Oh my God,' wrote one officer of Marines in his journal, "all my hopes are over of seeing my beloved wife and son."

As for the country they were going to, almost nothing was known except for the promise of Captain James Cook, its discoverer, that this 'New South Wales' as he chose to call it, was now British. But, to some observers of the hang 'em tendency, the thought that the felons might be better off than if they had languished in jail provoked bitter reproach.

35. Why was Dorothy Handland sent to the 'New World'?
(a) She dealt in rags and old clothes
(b) She was a criminal
(c) She was old
(d) There is no clear reason

Sol. *(b)* Dorothy Handland was sent to "New World" for she was a criminal.

36. Who discovered Australia?
(a) John Howard (b) Arthur Phillip
(c) James Cook (d) Robert Hughes

Sol. *(c)* Australia was discovered by James Cook.

37. What was the crime of William Francis?
(a) He stole some snuff
(b) He stole ribbon and silk stockings
(c) He stole a book
(d) He stole 12 lb of butter

Sol. *(c)* William Francis stole a book.

38. Lady Penrhyn refers to
(a) a surgeon (b) a judge
(c) a criminal (d) a ship

Sol. *(d)* Lady Penrhyn refers to a ship.

39. On the basis of the passage, why was Australia developed in 1787?
(a) To send ill people to a warmer island
(b) To serve as a penal colony
(c) To employ people
(d) To serve as a military-colony

Sol. *(b)* On the basis of the passage, Australia was developed in 1787 to serve as a penal colony.

40. As per the passage, what does the word 'Pom' denote?
(a) Aboriginals (b) Australian
(c) Englishmen (d) Pomegranate

Sol. *(b)* The word 'Pom' denotes Australia.

SECTION D
Quantitative Aptitude

1. Find out next number in the series 2, 7, 17, 34, 60, ...
(a) 97 (b) 108 (c) 95 (d) 121

Sol. *(a)* 2, 7, 17, 34, 60, 97
$+(2^2+1)+(3^2+1)+(4^2+1)+(5^2+1)+(6^2+1)$

2. Kiran, Kumar and Kedar were watching cricket and eating groundnuts. They pop a groundnut in their mouth every 40 seconds, 72 seconds and 108 seconds respectively and munch it. If all of them pop a groundnut simultaneously at 8:20:00 hours, then at what time will they again pop groundnuts again in their mouth simultaneously?
(a) 8:22:20 hours (b) 8:27:12 hours
(c) 8:38:00 hours (d) 8:25:15 hours

Sol. *(c)* They pop a groundnut simultaneously at 8:20:00 hr.

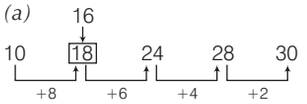
LCM of 40 sec, 72 sec, 108 sec = 1080 sec

$$= \frac{1080}{60}\text{min} = 18\text{min}$$

Hence, they pop a groundnut simultaneously after 18 min i.e., at 8:38:00.

3. Find the odd one in the series: 10, 16, 24, 28, 30
(a) 16 (b) 28
(c) 30 (d) 24

Sol. *(a)*



$\therefore$ 16 is the odd one in the series and it should be replaced by 18.

4. The probabilities that a student passes in Mathematics, Physics and Chemistry are m, p and c, respectively. Of these subjects the student has 75% chance of passing in atleast one, a 50% chance of passing in atleast two and a 40% chance of passing in exactly, two, which of the following relations are true.

(a) $p+m+c=\frac{27}{20}$ (b) $p+m+c=\frac{13}{20}$

(c) $pmc=\frac{1}{10}$ (d) Both (a) and (c)

Sol. *(a)* We have, $P(A\cup B\cup C)=\frac{3}{4}$

i.e., $P(A)+P(B)+P(C)-P(A\cap B)$

$-P(B\cap C)-P(A\cap C)+P(A\cup B\cup C)=\frac{3}{4}$

and $P(A\cap B)+P(B\cap C)+P(A\cap C)$

$-2P(A\cap B\cap C)=\frac{1}{2}$

and $P(A\cap B)+P(B\cap C)+P(A\cap C)$

$-3P(A\cap B\cap C)=\frac{2}{5}$

Solving the above equations (last two), we get

$P(A\cap B\cap C)=\frac{1}{2}-\frac{2}{5}=\frac{1}{10}$

$P(A)\,P(B)\,P(C)=\frac{1}{10}\Rightarrow pmc=\frac{1}{10}$

So, $P(A)+P(B)+P(C)-[P(A\cap B)$

$+P(B\cap C)+P(A\cap C)]+P(A\cup B\cup C)=\frac{3}{4}$

$p+m+c-\left(\frac{1}{2}+\frac{2}{10}\right)+\frac{1}{10}=\frac{3}{4}$

$p+m+c=\frac{27}{20}$

5. Mr. Murthy invested ₹ 25000 in a mutual fund three years ago. The value of the amount increased by 15% in the first year, it decreased by 15% in the second year and again increased by 10% during the third year. What is the value of the investment at the end of the three year period?

(a) 26880 (b) 28890

(c) 27500 (d) 26500

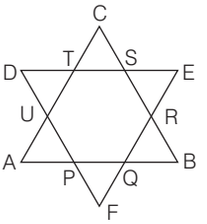
Sol. *(a)* Value of investment at the end of 3 yr

$=25000\left(1+\frac{15}{100}\right)\left(1-\frac{15}{100}\right)\left(1+\frac{10}{100}\right)$

$25000\times\frac{115}{100}\times\frac{85}{100}\times\frac{110}{100}$

= ₹ 26881.25 ≈ ₹ 26880

6. Two congruent triangles ΔABC and ΔDEF each with area 198 placed over each other as shown in the figure. AB is parallel to DE and the centres of both the triangles coincide with each other. The area of region $PQRSTU$ is .



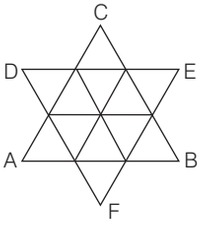
(a) 199 cm^2 (b) 132 cm^2

(c) 198 cm^2 (d) Cannot be determined

Sol. *(b)* There are total 12 similar triangles each with equal area. But a larger triangle ABC (or DEF) has only 9 smaller triangles. Out of 9 triangles, only 6 triangles are common.

$\therefore$ Area of common region

$=\frac{6}{9}\times 198=132\ cm^2$



7. An Olympic size swimming pool requires 6 hours to fill up to the brim when water is supplied simultaneously by three pipes A, B and C.

The pipes start filling at 12 o'clock in the night but if pipe C is shut off at 3 am, the swimming pool fill up to the brim at 7 am. The swimming pool is drained in the evening.

That night, if only pipe C is used to fill up the swimming pool and the filling starts at 12 o'clock in the night, what will be the time in the clock when the swimming pool will be filled up to the brim?

(a) 12 o'clock night (b) 9 am morning

(c) 6 pm evening (d) 12 o'clock noon

Sol. *(a)* Part of swimming pool filled by pipes A, B and C in 1 hr.

$\Rightarrow \quad \frac{1}{A}+\frac{1}{B}+\frac{1}{C}=\frac{1}{6}$

$\therefore$ Part of swimming pool filled by pipes A, B and C in 3 hr

$=\frac{3}{6}=\frac{1}{2}$

$\Rightarrow \quad 3\left(\frac{1}{A}+\frac{1}{B}+\frac{1}{C}\right)=\frac{1}{2}$ ...(i)

$\therefore$ Remaining part $= 1 - \frac{1}{2} = \frac{1}{2}$

Part filled by pipes A and B in 4 hr is half part of pool.

$\therefore \quad 4\left(\frac{1}{A} + \frac{1}{B}\right) = \frac{1}{2}$

$\Rightarrow \quad \frac{1}{A} + \frac{1}{B} = \frac{1}{8}$ ...(ii)

Put the value of Eq.(ii) in Eq.(i), we get

$$3\left(\frac{1}{8}\right) + \frac{3}{C} = \frac{1}{2} \Rightarrow \frac{3}{C} = \frac{1}{2} - \frac{3}{8}$$

$\Rightarrow \quad \frac{3}{C} = \frac{1}{8} \Rightarrow C = 24\,\text{hr}$

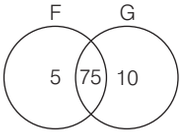
$\therefore$ Pipe C filled the pool in 24 hr. So, time in the clock when the pool is filled will be 12 o'clock night.

8. In an international school, 80% of the students know French, 85% know German and 75% know both French and German. If 40 students don't know both, then find total number of students.

(a) 400 (b) 350
(c) 460 (d) 450

Sol. *(a)* Let the total number of students be 100.

Then,



Therefore, total number of students who know either French or German or both $= 5 + 75 + 10 = 90$

Hence, total number of students don't know both $= 100 - 90 = 10$

For, $10 \rightarrow 100$

$\therefore \quad 40 \rightarrow 100 \times 4 = 400$

Hence, total number of students = 400

9. The price of an item was increased during festival season by 10%. Later the new price was decreased by 10%. If the latest price is 1000, what was the approximate original price?

(a) 1234.50 (b) 1020.50
(c) 1232.75
(d) None of the mentioned options are correct

Sol. *(d)* Let the Original Price of an item be ₹x.

New price of an item $= x \times \frac{110}{100} \times \frac{90}{100}$

According to the question,

$$1000 = x \times \frac{110}{100} \times \frac{90}{100}$$

$\Rightarrow \quad x = ₹1010.10$

$\Rightarrow \quad x \approx ₹1010$

10. John was writing an entrance test. Every question is as a multiple choice question with four probable answers and one correct answer. The probability that he knows the answer is 80%. If he correctly answered a question, then what is the probability that he was guessing?

(a) 1/5
(b) 1/17
(c) 1/27
(d) None of the mentioned options are correct

Sol. *(a)* Probability that the answer is correct $= \frac{1}{2}$

Probability that the answer is wrong $= \frac{1}{2}$

Probability that he knows the answer $= \frac{80}{100} = \frac{4}{5}$

Probability that he don't know the answer $= \frac{20}{100} = \frac{1}{5}$

Required probability $= \frac{\frac{1}{2} \times \frac{1}{5}}{\frac{1}{2} \times \frac{4}{5} + \frac{1}{2} \times \frac{1}{5}} = \frac{1}{10} \times \frac{10}{(4+1)} = \frac{1}{5}$

11. In an India-Pakistan 20 over cricket match, Pakistan played first and India had to chase a high score. Rain created havoc and the average number of overs was reduced to 18, India needed 144 runs to win. In the initial 5 overs, the average scoring rate was 6 but in next 5 overs it increased to 8. It then declined to 5 in next 5 overs. Now, what is the average needed by India in the remaining overs to win the match?

(a) 16.33 (b) 13.33 (c) 12.33 (d) 14.33

Sol. *(a)* Total runs scored in 15 overs $= 5 \times 6 + 5 \times 8 + 5 \times 5$

$= 30 + 40 + 25 = 95$

Total remaining runs to win the match $= 144 - 95 = 49$

Therefore, average required to win the match in remaining 3 overs $= \frac{49}{3} = 16.33$

12. Ravi has three sons and his age is equal to the sum of the ages of his sons. After 12 years, his age will be equal to the sum of ages of his eldest two sons. 4 years later, his age will be equal to sum of the age of the oldest and youngest. Again, 4 years after that, his age will be equal to sum of the ages of the two youngest. Find the present age of Ravi and his sons.

(a) 44, 18, 16, 12 (b) 48, 18, 16, 10
(c) 48, 20, 16, 12 (d) 48, 20, 12, 8

Sol. *(c)* Let the present ages of Ravi and his three sons be R, S_1, S_2 and S_3, respectively.

And, $S_1 < S_2 < S_3$ (Ages)

$R = S_1 + S_2 + S_3$ (given) ...(i)

After 12 years,

$$R+12=S_2+12+S_3+12$$

$$R=S_2+S_3+24-12$$

$$\Rightarrow \quad R=S_2+S_3+12 \qquad \ldots(ii)$$

After (12 + 4) years,

$$R+16=S_1+16+S_3+16$$

$$\Rightarrow \quad R=S_1+S_3+32-16$$

$$\Rightarrow \quad R=S_1+S_3+16 \qquad \ldots(iii)$$

After (12 + 4 + 4) years,

$$R+20=S_1+20+S_2+20$$

$$\Rightarrow \quad R=S_1+S_2+20 \qquad \ldots(iv)$$

On Adding Eqs. (ii), (iii) and (iv), we get

$$3R=2(S_1+S_2+S_3)+12+16+20$$

$$\Rightarrow \quad 3R=2(R)+48 \qquad [\because S_1+S_2+S_3=R, \text{ Eq. (i)}]$$

$$\Rightarrow \quad R=48 \text{ years}$$

Present age of Ravi = 48 years

And, with the option we can see that only option (c) is the correct option as in that sum of ages of son i.e., 20 + 16 + 12 = 48 years.

13. If $\log_x y=100$ and $\log_3 x=100$, then the value of y is

(a) 3^{10000} (b) 3^{100}

(c) 3^{1000} (d) 3^{10}

Sol. *(a)* $\log_x y=100$, $\log_y x=100$

Given, $\log_3 x=100 \Rightarrow x=3^{100}$

Now, $\log_x y=100 \Rightarrow y=x^{100}$

$$\Rightarrow \quad y=(3^{100})^{100} \Rightarrow y=3^{10000}$$

14. The population of a state is consistently decreasing due to migration owing lack of opportunities in the state. It has been observed that the population has decreased at a rate of 10 percent each year in the last few years. If today, the population of the state is 100 million, what will be the population of the state after 3 years?

(a) 63 million

(b) 83 million

(c) 73 million

(d) None of the mentioned options are correct

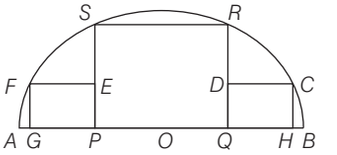
Sol. *(c)* Population of the state after 3 years

$$=100\left(1-\frac{10}{100}\right)\left(1-\frac{10}{100}\right)\left(1-\frac{10}{100}\right)$$

$$=100\times\frac{90}{100}\times\frac{90}{100}\times\frac{90}{100}=72.9 \text{ million}$$

$$\approx 73 \text{ million} \qquad \text{(approximately)}$$

15. In the figure given below, the semi-circle with centre O has a diameter $AB = 2r$. $PQRS$ is a square of maximum possible area. P and Q are lie on the diameter AB. R and S are on the arc of the semi-circle. There are two more squares of maximum possible area $EFGP$ and $CDQH$. Then $RC + FS$ = ...



(a) $\sqrt{2/5}(2r)$ (b) $2\sqrt{2}\,r/3$

(c) $\sqrt{2/5^r}$

(d) None of the mentioned options are correct

Sol. *(a)* $OR^2=OQ^2+RQ^2$

$$OR^2=5OQ^2 \qquad [\because RQ=2OQ]$$

$$\Rightarrow \text{Radius } (r)=OQ\sqrt{5},\ OQ=\frac{r}{\sqrt{5}}$$

Again, $OC^2=OH^2+HC^2$

$$r^2=(OQ+QH)^2+(QH)^2 \qquad (\because HQ=HC)$$

$$\Rightarrow \quad r^2=\left(\frac{r}{\sqrt{5}}+QH\right)^2+QH^2$$

$$\Rightarrow \quad QH=\frac{r}{\sqrt{5}}$$

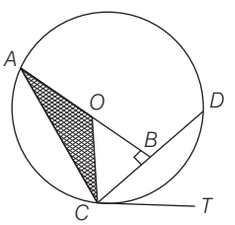
$$\therefore \quad HC=\frac{r}{\sqrt{5}}=\frac{RQ}{2}$$

$$RC=\sqrt{RO^2+OC^2}=\sqrt{\left(\frac{r}{\sqrt{5}}\right)^2+\left(\frac{r}{\sqrt{5}}\right)^2}$$

$$=r\sqrt{2/5}$$

$$\therefore \quad RC+FS=2r\sqrt{2/5}$$

16. In the given figure CD = 10 cm. CT is the tangent at C, making an angle of $\pi/4$ with CD. O is the centre of the circle. The perimeter of the shaded region is AOC is



(a) 27 cm (b) 2 cm (c) 30 cm (d) 13 cm

Sol. *(b)* $\angle OCT=90°$, $\angle DCT=45°$

$$\therefore \quad \angle OCB=45° \quad \text{and} \quad \angle COB=45°$$

$$\therefore \quad \angle AOC=180°-45°=135°$$

Now, $CD=10 \Rightarrow BC=5 \text{ cm}=OB$

$$\Rightarrow \quad OC=5\sqrt{2} \text{ cm}=OA$$

Again, $AC^2=OA^2+OC^2-2OA\cdot OC\cos 135°$

$$=2(OA)^2-2(OA)^2\cdot\cos 135°$$

$$=2(5\sqrt{2})^2-2(5\sqrt{2})^2\times\left(-\frac{1}{\sqrt{2}}\right)=100+\frac{100}{\sqrt{2}}$$

$AC^2=170.70 \Rightarrow AC\approx 13$ cm

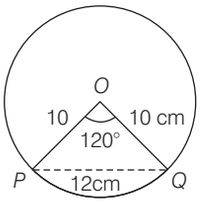
$\therefore$ Perimeter of $\Delta OAC=OA+OC+AC$

$$=5\sqrt{2}+5\sqrt{2}+13=27 \text{ cm}$$

17. In how many ways can we arrange the word CAPSTONE so that all the vowels come together?
(a) 1440 (b) 4320
(c) 2160 (d) 3480

Sol. *(b)* Required arrangements $= 6! \times 3!$
$= 6 \times 5 \times 4 \times 3 \times 2 \times 1 \times 3 \times 2 = 4320$

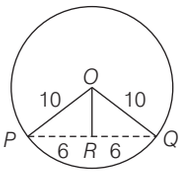
18. The area of minor segment PQ is



(a) 60.14 sq cm (b) 56.25 sq cm (c) 59.43 sq cm (d) 56.76 sq cm

Sol. *(d)* Area of sector $= OPQO = \pi r^2 \times \frac{\theta}{360°}$

$= 100\pi \times \frac{120}{360°} = \frac{100\pi}{3}$ sq cm



Now, $PR = RQ = 6$ cm

[$\because$ Perpendicular from centre to the chord bisects the chord]

$\therefore \quad OR = \sqrt{100 - 36} = \sqrt{64} = 8$ cm

Area of $\Delta POQ = \frac{1}{2} \times 12 \times 8 = 48$ sq cm

Hence, Area of minor segment $PQ = \frac{100\pi}{3} - 48$ sq cm

≈ 56.76 sq cm

19. A company employs 35 people whose average age is 45 years. The 10 most aged people have an average age that is 20% more than the average of all 35 people in the company. Three new employees of aged 32, 35 and 30 joined the company. Now, what is the difference in the average age of the same 10 most aged people and the average age of all the people in the company?
(a) 10 years (b) 9 years
(c) 12 years (d) 11 years

Sol. *(a)* Total age of 35 people $= 35 \times 45 = 1575$ years

Average age of 10 most aged people $= 45 \times \frac{120}{100} = 54$

Average age of all employees $= \frac{1575 + 32 + 35 + 30}{35 + 3}$

$= 44$ years

$\therefore$ Required difference $= 54 - 44 = 10$ years

20. On her birthday, Vrinda gets a gift box which contains 6 chocolates and 8 cup cakes. She eats 7 of them randomly. What is the probability that she ate 3 chocolates and 4 cupcakes?
(a) 0.16231 approximately (b) 0.05655 approximately
(c) 0.40792 approximately (d) 0.01165 approximately

Sol. *(c)* Required Probability $= \frac{{}^6C_3 \times {}^8C_4}{{}^{14}C_7}$

$= \frac{\frac{6!}{3!3!} \times \frac{8!}{4! \times 4!}}{\frac{14!}{7! \times 7!}}$

$= \frac{20 \times 70}{3432} = 0.40792$ approximately

21. To cross a man running at 9 km/hr in the direction of train, a train takes 40 seconds and to cross another man running at 6 km/hr in same direction it takes 30 seconds. Find the speed of the train.
(a) 20 km/hr
(b) 18 km/hr
(c) 14 km/hr
(d) Cannot be determined

Sol. *(b)* Let the speed of the train be 'x' km/hr.

According to the question,

$(x - 9) \times \frac{40}{3600} = (x - 6) \times \frac{30}{3600}$

$\Rightarrow -36x + 4x = -18 + 3x$

$\Rightarrow -4x + 3x = 18 - 36$

$\Rightarrow \quad -x = -18$

$\Rightarrow \quad x = 18$ km/hr

22. A library has 10 copies of book A, 3 copies each of books B and C, 2 copies each of books D, E, F and a single copy of books G and H. The total number of ways in which these books can be arranged on a shell is
(a) $(12!)/(10!(3!)^2(2!)^2)$
(b) $(24!)/101(3!)^2(2!)^3$
(c) $(24!)/(10!(3!)^2(2!)^2)$
(d) None of the mentioned options are correct

Sol. *(b)* Required number of ways $= \frac{24!}{10!(3!)^2(2!)^3}$

23. What is the value of $(p-1)^{q+1}$, if $p^q = 169$ and p and q are whole numbers?
(a) 1728 (b) 2187
(c) 169 (d) 144

Sol. *(a)* $p^q = 169$

since $13^2 = 169$

$\therefore \quad p = 13$ and $q = 2$

Hence, $(p-1)^{q+1} = (13-1)^{2+1} = 12^3 = 1728$

24. The population of a city increased by 5% but decreases by 0.25% each year due to emigration. Find the net increase in per cent in 3 years?
(a) 11.24%
(b) 13.23%
(c) 14.93%
(d) None of the mentioned options are correct

Sol. *(c)* Net increase each year $= 5 - 0.25 - \frac{5 \times 0.25}{100}$

$= 4.7375\% \approx 4.75\%$

Let initial population be 100.

$\therefore$ Population after 3 years $= 100 \times \left(\frac{104.75}{100}\right)^3 = 114.93$

Hence, net increase in per cent = 14.93%

25. How many factors of $2^8 \bigstar 3^6 \bigstar 5^4 \bigstar 10^5$ are multiples of 120?
(a) 594 (b) 660
(c) 540
(d) None of the mentioned options are correct

Sol. *(d)* The prime factoriation of

$2^8 \times 3^6 \times 5^4 \times 10^5$ is $2^{13} \times 3^6 \times 5^9$

120 can be prime factorised as $2^3 \times 3 \times 5$

All factors of $2^{13} \times 3^6 \times 5^9$ than can be written as multiples of 120 will be of the form $2^3 \times 3 \times 5 \times k$.

$\therefore 2^{13} \times 3^6 \times 5^9 = 2^3 \times 3 \times 5 \times k$

$\Rightarrow \quad k = 2^{10} \times 3^5 \times 5^8$

$\therefore$ Number of factors of $k = (10+1)(5+1)(8+1)$

$= 11 \times 6 \times 9 = 594$

26. The remainder obtained when 2^{97} is divided by 97 is
(a) 2
(b) 0
(c) 97
(d) None of the mentioned options are correct

Sol. *(a)* $2^{97} = 2^{96} \cdot 2$

$R(2^{96} \div 97) = 1$

[If p is a prime number and a is an integar, then a^{p-1} divided by p gives 1 as remainder]

$\therefore \quad R\left(\frac{2^{96} \cdot 2}{97}\right) = 1 \cdot 2 = 2$

27. Out of all the MBA aspirants interviewed, 50% of those who attended the first interview received the call for second interview. 90% of the people who got a call for second interview felt good about their first interview. 80% of people who did not receive a call for second interview also felt good about their first interview.

If a person felt good after his/her first interview, what is the probability that he/she will receive a call for the second interview?
(a) Approximately 73% (b) Approximately 89%
(c) Approximately 53% (d) Approximately 63%

Sol. *(c)* Required probability

$$= \frac{90\% \text{ of } 50\%}{80\% \text{ of } 50\% + 90\% \text{ of } 50\%} \times 100$$

$$= \frac{\frac{1}{2} \times \frac{9}{10}}{\frac{1}{2} \times \frac{4}{5} + \frac{1}{2} \times \frac{9}{10}} \times 100$$

$$= \frac{\frac{9}{20}}{\frac{8}{20} + \frac{9}{20}} \times 100 = \frac{9}{17} \times 100 \approx 53\%$$

28. The sum of three numbers is 138. The ratio of first to the second number is 3 : 4 and that of second to the third is 8 : 9. The first number is
(a) 48 (b) 36 (c) 54 (d) 42

Sol. *(b)* Let the three numbers be x, y and z.

$$x + y + z = 138 \quad \ldots(i)$$

$$\frac{x}{y} = \frac{3}{4} \Rightarrow 4x = 3y \quad \ldots(ii)$$

$$\frac{y}{z} = \frac{8}{9} \Rightarrow 9y = 8z \quad \ldots(iii)$$

Put the value of Eqs. (ii) and (iii) in Eq. (i), we get

$$x + y + z = 138$$

$$\Rightarrow \quad \frac{3y}{4} + y + \frac{9y}{8} = 138$$

$$\Rightarrow \quad \frac{6y + 8y + 9y}{8} = 138$$

$$\Rightarrow \quad 23y = 138 \times 8 \Rightarrow y = 48$$

$$\therefore \quad \text{First number, } x = \frac{3 \times 48}{4} = 36$$

29. If A is the set of odd integers less than 50 and B is the set of the squares of integers in A, how many elements does the intersection of A and B contain?
(a) 4 (b) 5
(c) 3 (d) Cannot be determined

Sol. *(a)* $A = \{1, 3, 5, 7, 9, 11 \ldots 49\}$, $B = \{1, 9, 25, 49, \ldots, 2401\}$

$A \cap B = \{1, 9, 25, 49\}$

$\therefore A \cap B$ contains 4 element.

30. A Bank lent a certain sum of money at a certain rate of simple interest for 2 years to David. He lent this sum at the same rate of interest but compounded annually for the same period to Pankaj, who has a bad credit history. At the end of two years, David received ₹ 2400 as compound interest but paid ₹ 2000 only as simple interest. Find the rate of interest.
(a) 20% (b) 10% (c) 30%
(d) None of the mentioned options are correct

Sol. *(d)* Let the Rate of interest be $R\%$

$$\text{CI} - \text{SI} = \text{SI}\left(\frac{R}{200}\right)$$

$$\Rightarrow 2400 - 2000 = 2000\left(\frac{R}{200}\right)$$

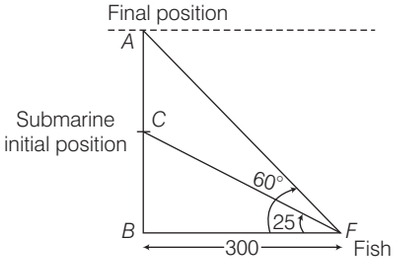
$$\Rightarrow \quad \frac{400}{10} = R \quad \Rightarrow \quad R = 40\,\%$$

31. A fish observes a submarine from the bottom of ocean, when the submarine starts moving up vertically. The angle of elevation of the submarine changes from 25 , degrees to 60 degrees in 2 minutes. The point of observation of the fish is situated 300 metres (displacement) away from the takeoff point. If the submarine is going up in straight line at a constant speed, what is the speed of the submarine? (Here, $\tan 25° = 0.466$)

(a) 4.16 m/s (b) 2.22 m/s
(c) 5.22 m/s (d) 3.16 m/s

Sol. *(d)* $\tan 25° = \frac{BC}{300}$

$$\Rightarrow \quad BC = 300 \times 0.466 = 139.8 \text{ m}$$



$$\tan 60° = \frac{AB}{300}$$

$$\Rightarrow \quad 300\sqrt{3} = AB$$

$$\Rightarrow \quad AC + BC = 300\sqrt{3}$$

$$\Rightarrow \quad AC = 519.6 - 139.8$$

$$\Rightarrow \quad AC = 379.8 \text{ m}$$

$$\therefore \text{Speed of submarine} = \frac{AC}{2\text{min}} = \frac{379.8}{2 \times 60}$$

$$= 3.16 \text{ m/s}$$

32. Gopal is planning to sell two radios. The first radio is marked ₹ 200 over and above the cost price and is sold at a discount of 20% on it. The other radio is marked 50% over and above the cost price and sold at discount of ₹ 160.

If his profit is 15% on both radios combined and 5 % on the first alone, what is the percentage of profit on the second radio?

(a) 25% (b) 10%
(c) 7% (d) Cannot be determined

Sol. *(a)* Let CP of 1st and 2nd radio be ₹x and ₹y, respectively.

$$\text{SP}_I = (200 + x)\frac{80}{100} = 160 + \frac{4}{5}x$$

$$\therefore \quad 160 + \frac{4}{5}x = \frac{105x}{100}$$

$$\Rightarrow \quad 160 = \frac{105x}{100} - \frac{4x}{5} = \frac{105x - 80x}{100}$$

$$\therefore \quad x = \frac{16000}{25} = ₹640$$

Hence, $\text{SP}_I = (200 + 640) \times \frac{80}{100} = ₹672$

Again, $\text{SP}_{II} = \frac{150y}{100} - 160$

$$\therefore 672 + \frac{150y}{100} - 160 = \frac{115}{100}(y + 640)$$

$$\Rightarrow \quad 512 + \frac{3y}{2} = \frac{115y}{100} + 736$$

$$\Rightarrow \quad 224 = \frac{35y}{100}$$

$$\Rightarrow \quad y = \frac{22400}{35} = ₹\ 640$$

$$\therefore \quad \text{SP}_{II} = ₹800$$

Hence, required profit $= \frac{800 - 640}{640} \times 100 = 25\%$

33. A factory has 900 employees, 25% of whom are men. The factory has to hire 200 additional employees to increase production. What percentage of the additional employees hired must be men in order to raise the percentage of men employees in the factory to 35%? (Assume transgender for this situation.)

(a) 70% (b) 90%
(c) 75% (d) 80%

Sol. *(d)* Total number of men employee in company before hiring

$$= 900 \times \frac{25}{100} = 225$$

Let x men are hired to raise the percentage of men to 35%.

$$\therefore \quad \left(\frac{225 + x}{900 + 200}\right) \times 100 = 35$$

$$\Rightarrow \quad \frac{(225 + x)}{1100} \times 100 = 35$$

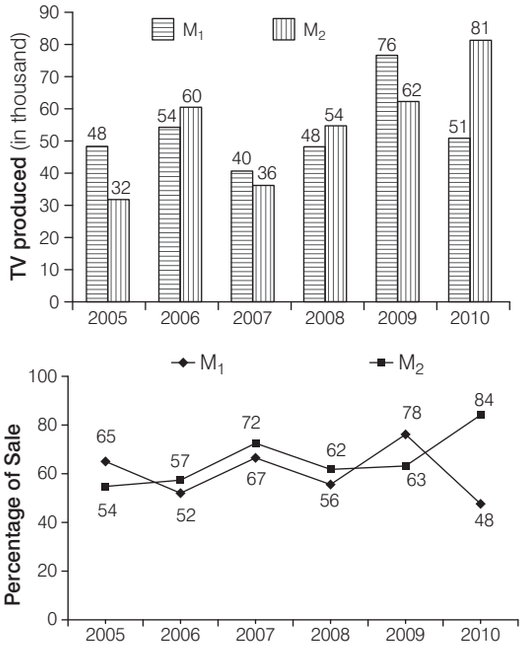
$$\Rightarrow \quad x = 385 - 225$$

$$\Rightarrow \quad x = 160$$

Thus, percentage of additional hired men employees

$$= \frac{160}{200} \times 100\% = 80\%$$

Directions (Q.Nos. 34-35) *Following bar-graph shows the number of TV models, M_1 and M_2 produced by a company in different years and the line-graph shows the percentage of sale of these models in different years.*



34. What is the total number of TV models M_1 and M_2 sold in the year 2005?
(a) 44800 (b) 48840 (c) 48480 (d) 48440

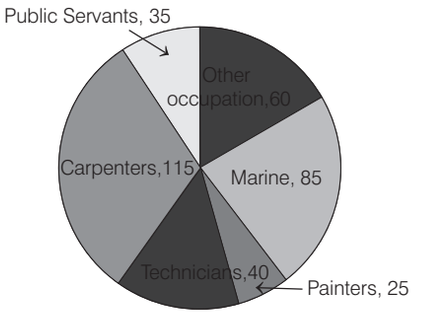
Sol. *(c)* Sale $M_1 = 48000 \times 0.65 = 31200$
Sale $M_2 = 32000 \times 0.54 = 17280$
$\therefore$ Total = 31200+17280 = 48480

35. What is the ratio of the total number of TVs of model M_2 unsold in the year 2006 to the total number of TVs of model M_2 produced in 2007?
(a) 32 : 47 (b) 41 : 60 (c) 43 : 60 (d) 47 : 60

Sol. *(d)* In 2006, unsold $M_2 = 60000 \times \dfrac{(100-57)}{100} = 25800$
In 2007, production $M_2 = 36000$
$\therefore$ Ratio $= \dfrac{25800}{36000} = \dfrac{43}{60}$

Directions (Q. Nos. 36-40)

Data in the pie chart describes the population engaged in various occupations in a German city for a year Based on the information answer questions.



36. What percent of wage earners in the city re engaged in Marine Engineering?
(a) 13.6 (b) 19.6
(c) 23.6 (d) 8.6

Sol. *(c)* Required percentage

$$= \frac{85}{85+60+35+115+40+25} \times 100$$

$$= \frac{85}{360} \times 100 = 23.6\%$$

37. If the number of persons engaged in painting is 2160, the number of persons engaged as technicians would be
(a) 1456 (b) 3456
(c) 2458 (d) 3458

Sol. *(b)* Given, $25° = 2160$

$$\Rightarrow \qquad 40° = \frac{2160}{25} \times 40 = 3456$$

38. If the total earnings from carpainters is twice that from technicians, the ratio of average wages in these occupations is
(a) 23 :8 (b) 5 : 4
(c) 16 : 23 (d) 23 : 16

Sol. *(d)* Required ratio $= \dfrac{115}{2 \times 40} = \dfrac{115}{80} = 23:16$

39. If the average earning of a wage earner from painters category was twice that from technicians, the ratio of the total earnings from these occupations is
(a) 4 : 3 (b) 5 : 4
(c) 4 : 5 (d) 3 : 4

Sol. *(b)* Required ratio $= \dfrac{25 \times 2}{40} = \dfrac{50}{40} = 5:4$

40. If the total number of wage earners increases by 10% and the number of wage earners engaged in carpentry increases by 15% in the next year, the nearest central angle of carpenters in pie chart for that year will be
(a) 120°
(b) 125°
(c) 130°
(d) None of the mentioned options are correct

Sol. *(c)* $115° = \dfrac{100}{360} \times 115\% = 31.95\%$

It is given that number of wage earners in carpentry increases by 15%.

$$\therefore \text{Increased carpenters} = \frac{31.95 \times 115}{100}$$

$$= 36.73\%$$

$$\text{Hence, required angle} = \frac{360}{100} \times 36.73$$

$$\approx 136°$$

SOLVED PAPER
2016-2005

SNAP

SYMBIOSIS NATIONAL APTITUDE TEST

SOLVED PAPER 2016*

Time : 3 hrs

INSTRUCTIONS

1. There are 150 questions in this paper.
2. There are four parts. Part A (General Awareness), Part B (Analytical & Logical Reasoning), Part C (General English) and Part D (Quantitative aptitude).
3. Part A carries 30 marks, Part B carries 40 marks, Part C carries 40 marks and Part D carries 40 marks.
4. For each incorrect attempt 1/4th mark will be deducted.

SECTION A

GENERAL AWARENESS SECTION

1. A national political party is' one which received 6% of the total votes polled in

(a) two or more states (b) the capital city
(c) four or more states (d) in all the states

Sol. *(c)* If a political party is treated as' a recognised political party in four or more states, it shall be known as a 'National Party, of India, but only so long as that political party continues to fulfill thereafter the conditions for recognition in four or more states on the results of any subsequent general election either to the House of the people or to the Legislative Assembly of any state or if it wins 2% of seats in the Lok Sabha at a General Election and these candidates are elected from three states or if it is recognised as a state party in four states.

2. The Bhakti cult spread in Maharashtra with the teaching of

(a) Saint Tukaram (b) Saint Jnaneshvara
(c) Samarth Guru Ramadas (d) Chaitanya Mahaprabhu

Sol. *(b)* Saint Jnaneshvara, introduced Varkari Movement (Vitthala Sampradaya) of Pandherpur. He was associated with Krishna Margi Sect. He was the founder of Marathi Language and literature, also wrote commentaries on Bhagwad Gita called Gyaneshwari or Bhavarthadipka. Saint Jnaneshwar was a 3rd century Maharashtrian Hindu Saint, Poet, Philosopher and Yadi of the Nath Tradition.

3. Government securities are considered liquid because they are

(a) backed by the government treasury
(b) convertible into other types of saving deposits
(c) quickly and easily marketable
(d) stable in value

Sol. *(c)* Liquid assets are those asset which can be easily convertible and with minimal impact to the price received. Assets can be easily converted without considerable price fluctuation in a liquid market and with a minimal decline in worth.

A liquid market is a type of market that posses a high level of stability and low spreads between asking and selling prices securities issued by considered risk free and their yields are often used as the benchmarks for fixed income. Securities with the same maturity.

* Memory Based

4. Bull and Bear are related to which commercial activity?

(a) Banking (b) E-commerce
(c) International trade (d) Stock market

Sol. *(d)* Both the terms are related to stock market. Investors who take a bull approach purchase securities under the assumption that they can be sold later at a higher price.

A 'bear' is considered to be the opposite of a bull. Bear investors believe that the value of a specific security or an industry is likely to decline in the future.

5. The first to come and last to leave India were

(a) The Portuguese (b) The French
(c) The English (d) The Dutch

Sol. *(a)* The Portuguese can be said to have visited India in 1498, when Vasco da Gama set his foot on calicut (Malabar Coast).

The Portuquese State of India was established in 1505. It was in 1954, that portugal cost the enclaves of Dadra and Nagar Haveli and those of Goa and Daman and Diu in December, 1961.

6. Buddha gave his first religious message at

(a) Rajagriha (b) Pataliputra
(c) Gaya (d) Sarnath

Sol. *(d)* Sarnath has the deer park where Gautama Buddha first taught the dhamma and where the Buddhist Sangha came into existence through the enlightenment of Kondanna. It is located to the North-East of Varanasi in Uttar Pradesh. Sarnath is also the site of Buddha's 'Dharmachakra parivartan' which was Buddha's first teaching after attainment of enlightenment.

7. A firm practising price discrimination will be

(a) charging different prices for different qualities of a product
(b) buying from the cheapest and selling in the dearest markets
(c) charging different prices in different markets for a product
(d) buying only from firms selling in bulk at a distance.

Sol. *(c)* Price discrimination is a pricing strategy where identical goods or services are transacted at different prices by same provider firm in different markets.

8. The practice of selling goods in a foreign country at a price below their domestic selling price is called

(a) 'diplomacy' (b) 'discrimination'
(c) 'dumping' (d) 'double pricing'

Sol. *(c)* Dumping is the policy of charging lower price in the foreign market then in the domestic market the main aim is to establish themselves in the international market.

9. In which of the following countries long staple type of cotton is mainly grown?

(a) India (b) Egypt (c) Greece (d) China

Sol. *(b)* Long-staple cotton is any cotton with fibers from 1 1/8" to 2 1/2" long.

Outstanding types of this cotton are Pima, Egyptian and Sea Island cotton. Egyptian cotton is a type of long-staple cotton, which is largely grown in the Nile river valley.

This plant contains the chemical gossypol, which reduces its susceptibility to insect and fungal damage. It is also called gossypium barbaderse.

10. Which party provided two Prime Ministers in 2 years time?

(a) BJP (b) Janata Party
(c) Janata Dal (d) Samajwadi Janata Party

Sol. *(b)* The Janata Party (People's Party) was an amalgam of Indian political parties opposed to the State of Emergency that was imposed between 1975 and 1977 by the Government of Indira Gandhi and her party, the Indian National Congress (R).

Morarji Desai became the fourth Prime Minister of India from 1977 to 1979. In 1979, Raj Narain and Charan Singh pulled out of the Janata party forcing Desai to resign from his office. Charan Singh was the fifth of the Republic of India, serving from 28th July, 1979 until 14th January, 1980.

11. The longest river in Asia is

(a) river Indus (b) river Yangtze
(e) river Hwang Ho (d) river Ganga

Sol. *(b)* The Yangtze river is known in China as the Chang Jiang or Yangtze, is the longest river in Asia and the third-longest in the world. It flows for 6300 km (3915 m) from the glaciers on the Qinghai-Tibet Plateau in Qinghai, eastward across South-West, Central and Eastern China before emptying into the East China sea at Shanghai. The river is the longest in' the world to flow entirely within one country.

12. The split between the 'Extremists' and 'Moderates' came up in the open at the Surat Congress Session in the year

(a) 1905 (b) 1906 (c) 1907 (d) 1910

Sol. *(c)* At the Surat Session in 1907, Congress leaders split into two groups moderates and extremists. In Surat, extremists tried to push the candidature of Lala Lajpat Rai for Presidentship of Congress and Moderates, Dr Rashbehari Ghosh to be the President.

The situation was saved by Lala Lajpat Rai by stepping down and Dr Rashbehari Ghosh became the President. Real trouble started when the moderates tried to repudiate the resolution on boycott, swadeshi and national education which has been adopted in 1906. This was not acceptable to the extremists. Both the factions of INC-moderates and extremists came together in Lucknow session of 1916.

13. Humayun had to run away from India after he was defeated in the battle of
(a) Panipat (b) Gogra
(e) Khanwa (d) Kannauj

Sol. *(d)* Humayun and his brother Kamran decided to fight Sher Shah but due to the differences between the brothers, Humayun fought the Battle of Kannauj in 1540 alone and lost.

Humayun became a fugitive and Sher Shah became the ruler of Agra and Delhi, thus established the Sur empire in India.

14. The highest grade and best quality coal is
(a) lignite (b) peat
(c) bituminous (d) anthracite

Sol. *(d)* Anthracite is usually considered to be the highest grade of coal and is actually considered to be metamorphic. It contains about 90% carbon with practically no volatile matter. It is largely used for heating domestically as it burns with little smoke.

15. IBRD is also known as
(a) International Bank
(b) World Bank
(c) Asian Development Bank
(d) Bank of America

Sol. *(b)* The International Bank for Reconstruction and Development (IBRD) is the first of five member institutions, which compose the World Bank Group and is headquartered in Washington DC, United States.

Together, the International Bank for Reconstruction and Development and its concessional lending arm, the International Development Association, are collectively known as the World Bank.

16. In May, 2016, Rodrigo Duterte was elected as the President of which country?
(a) Thailand (b) Switzerland
(c) Mexico (d) Philippines

Sol. *(d)* Rodrigo Duterte, also known as Digong, is a Filipino politician and jurist, who is the 16th and current President of the Philippines. He is the first Mindanaoan to hold the office, and the fourth of Visayan descent. He won the elections in May, 2016.

17. The resources which can be used continuously, year-after are called
(a) biotic
(b) abiotic
(c) non-renewable
(d) renewable

Sol. *(d)* A renewable resource is a natural resource with the ability to reproduce through biological or natural processes and replenished with the passage of time.

Renewable resources are part of our natural environment and form our ecosystem.

18. Which among the following is the fundamental quantity?
(a) Volume (b) Time (c) Velocity (d) Force

Sol. *(b)* The physical quantities which do not depend or derive from the other physical quantities are known as the fundamental physical quantities.

The fundamental quantities are mass, length, time, luminous intensity and amount of substance, electric current and temperature. All the physical quantities which are not the fundamental physical quantities but are derived from it are known as the derived physical quantities.

19. Which country launched the world's first satellite dedicated to monitoring Greenhouse gas emission in 2009?
(a) Japan (b) Brazil (c) India (d) USA

Sol. *(a)* The first satellite dedicated to monitor the greenhouse gas emissions as a part of global efforts to combat climate change was launched into space from Japan which monitors the level of greenhouse gases like carbon-dioxide and methane.

20. Who among the following was elected as the Chairman of world's biggest commercial organisation the International Chamber of Commerce in June 2016?
(a) Mukesh Ambani (b) Adi Godrej
(c) Lakshmi Narain Mittal (d) Sunil Bharti Mittal

Sol. *(d)* Airtel supremo Sunil Bharti Mittal was appointed as the Chairman of world's biggest commercial organisation, the International Chamber of Commerce in June, 2016.

21. The 'Goods and Services Tax' was proposed by a task force whose President was
(a) Vijay Kelkar (b) Montek Singh Ahluwalia
(c) Arun Jaitley (d) Narsimham

Sol. *(a)* Goods and Service Tax (GST) is a proposed system of indirect taxation. It was introduced in the Constitution as (One Hundred and First Amendment) Act, 2016. The GST is administered and governed by GST council and its chairman is Union Finance Minister of India, Arun Jaitley.

22. These waves cannot be transmitted through vaccum
(a) light (b) sound
(c) heat (d) electromagnetic

Sol. *(b)* Sound is basically a form of kinetic energy which is transmitted when a particle in a medium gets a bit of energy by being bumped by a neighbouring particle and passes same or all of this energy a long by bumping the another particle.

If there is no medium then there are no particles to bump or to be bumped and thus there would be no any sound.

23. In which of the following countries, a woman became the president for the first time in May, 2016?
(a) Taiwan (b) Myanmar
(c) Angola (d) Namibia

Sol. *(a)* Democratic Progressive Party (DPP) Chairman Tsai Ing-wen made history on May 20, 2016 as the first woman to become the President of Taiwan. The 59 year old lady elected into power on January 16, winning 56 seats, 12% of more than 12 million votes cast.

24. A passenger standing in a bus is thrown outward when the bus takes a sudden turn. This happens due to
(a) outward pull on him (b) inertia of motion
(c) change in momentum (d) change in acceleration

Sol. *(b)* Newton's first law of motion gives the concept of inertia which is of three types, inertia of rest, inertia of motion and inertia of direction.

25. USB refers to
(a) a storage device (b) a processor
(c) a port type (d) a serial bus standard

Sol. *(c)* Universal Serial Bus (USB) is designed to standardise the connection of computer peripherals (including keyboards, pointing devices, digital cameras, printers, portable media players, disk drives and network adapters) to personal computers, both to communicate and to supply Electric Power.

26. In which post was the former Police Commissioner of Delhi, BS Bassi appointed in May, 2016?
(a) The Chairman of Union Public Service Commission
(b) The Governor of Meghalaya
(c) The Director General of Border Security Force
(d) The Member of Union Public Service Commission

Sol. *(d)* BS Bassi, who was Commissioner of Police of Delhi, was appointed as the Member of Union Public Service Commission (UPSC). He was 1977 Batch IPS officer.

27. Which of the following is a better measurement of economic development?
(a) GDP (b) Disposable income
(c) NNP (d) Per capita income

Sol. *(d)* Per capita national income is the most widely used measure of the extent of a country's economic development and it indicates the actual wellbeing or financial condition of the people of that country.

28. Who, out of the following, was the first recipient of Bharat Ratna Award?
(a) BC Roy (b) S Chandrasekhar
(c) CV Raman (d) Gobind Ballabh Pant

Sol. *(c)* CV Raman : 1954; Govind Ballabh Pant: 1957; Bidhan Chandra Roy: 1961 and S Chandrashekhar: he has not won the award. CV Raman has also received Nobel Prize for Physics.

29. Who administered the oath of the Chief Minister of West Bengal to Mamata Banerjee on May 27, 2016?
(a) Ram Naik (b) Keshari Nath Tripathi
(c) TS Thakur (d) P. Sathasivam

Sol. *(b)* The oath of Chief Minister was administered by the Governor Keshari Nath Tripathi.

30. In which of the following State, Bharatiya Janata Party formed government for the first time in 2016?
(a) Assam (b) Rajasthan
(c) Kerala (d) Odisha

Sol. *(a)* The BJP led National Democratic Alliance won the majority seats in Assam making its first government in the state.

SECTION B

ANALYTICAL AND LOGICAL REASONING

1. Find the approximate value : $\sqrt{10000} + \frac{3.001}{4.987}$ of $1891.992 = ?$
(a) 2500 (b) 1230 (c) 1640 (d) 1525

Sol. *(b)* $\sqrt{10000} + \frac{3.001}{4.987}$ of $1891.992 = ?$

$\Rightarrow \quad ? \approx 100 + \frac{3}{5} \times 1900$

$= 100 + 1140 \approx 1240 \approx 1230$

2. The sum of the digits of a three-digit number is 16. If the ten's digit of the number is three times the unit's digit and the unit's digit is one-fourth of the hundredth digit, then what is the number?
(a) 446 (b) 561 (c) 682 (d) 862

Sol. *(d)* Let x, y and z be the digits at the hundredth place, ten's place and unit's place, respectively.

$\therefore \quad x + y + z = 16 \quad \ldots(i)$

$y = 3z \quad \ldots(ii)$

$z = \frac{1}{4}x \quad \ldots(iii)$

From (ii) and (iii),

$\therefore \quad y = \frac{3}{4}x \quad \ldots(iv)$

Using Eqs. (iii) and (iv) in (i), we get

$x + \frac{3}{4}x + \frac{1}{4}x = 16$

$\Rightarrow \quad x = 8$

$\therefore \quad y = 6, z = 2$

Hence the number is 862.

3. In a certain code language '123' stands for 'I am servant', '279' stands for 'servant always miserable', and '684' stands for 'poverty is curse'. Then, 'miserable' stands for which numeric?

(a) 2 (b) 7
(c) 9 (d) Cannot be determined

Sol. *(d)* 1 ② 3 ⟶ I am (servant) …(i)

② 7 9 ⟶ (servant) always miserable …(ii)

6 8 4 ⟶ poverty is curse …(iii)

From Eqs. (i) and (ii),

2 ⇒ servant

1 ⇒ I/am

3 ⇒ I/am

7 ⇒ always/miserable

9 ⇒ always/miserable

∴ miserable ⇒ [7 or 9]

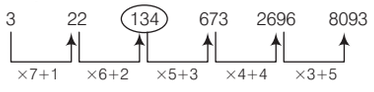
Clearly, it cannot be definitely said how will 'miserable' be written.

Direction (Q. Nos. 4-6) *Find the mising number in the series given below.*

4. 3, 22, ? 673, 2696, 8093

(a) 133 (b) 155 (c) 156 (d) 134

Sol. *(d)* The pattern is as follows



∴ ? = 134

5. 2, 8, 18, 32, 50, ?

(a) 64 (b) 72 (c) 70 (d) 68

Sol. *(b)* The pattern is as follows

$1^2 + 1^2 = 1 + 1 = 2$;

$2^2 + 2^2 = 4 + 4 = 8$;

$3^2 + 3^2 = 9 + 9 = 18$;

$4^2 + 4^2 = 16 + 16 = 32$;

$5^2 + 5^2 = 25 + 25 = 50$;

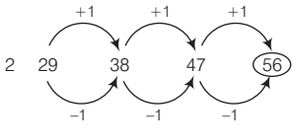
$6^2 + 6^2 = 36 + 36 = 72$

∴ ? = 72

6. 2, 29, 38, 47, ?

(a) 59 (b) 56 (c) 52 (d) 58

Sol. *(b)* The pattern is as follows



∴ ? = 56

Direction (Q. Nos. 7-9) *Study the following information to answer the given questions.*

P, Q, R, S, T, V, X and Y are seated in a straight line facing North. P sits fourth to the left of V. V sits either sixth from the left end of the line or fourth from the right end of the line. S sits second to right of R. R is not an immediate neighbour of V. T and Q are immediate neighbours of each other but neither T nor Q sits at extreme ends of the line. Only one person sits between T and X. X does sit at the extreme end of the line.

7. What is the position of Q with respect to P?

(a)Fifth to the right
(b) Immediate neighbour
(c) Second to the right
(d) Third to the left

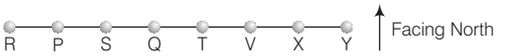
8. Which of the following represents persons seated at the two extreme ends of the line?

(a) P, V (b) Y, S (c) R, V (d) R, Y

9. How many persons are seated between R and T?

(a) One (b) Two (c) Three (d) Four

Solutions (7-9) *According to the given information, the final arrangement is as follows.*



7. *(c)* Q is second to the right of P.

8. *(d)* R and Y are sitting on the extreme ends of line.

9. *(c)* Three persons P, S and Q are seated between R and T.

10. $\left\{7\frac{1}{2} + \frac{1}{2} \div \frac{1}{2}\text{of}\ \frac{1}{4} - \frac{2}{5} \times 2\frac{1}{3} \div 1\frac{7}{8}\text{of}\left(1\frac{2}{5} - 1\frac{1}{3}\right)\right\} = ?$

(a) $3\frac{1}{5}$ (b) $2\frac{1}{24}$

(c) $4\frac{1}{30}$ (d) None of these

Sol. *(c)* Given expression,

$$= \frac{15}{2} + \frac{1}{2} \div \frac{1}{8} - \frac{2}{5} \times \frac{7}{3} \div \frac{15}{8}\ \text{of}\left(\frac{7}{5} - \frac{4}{3}\right)$$

$$= \frac{15}{2} + 4 - \frac{2}{5} \times \frac{7}{3} \div \frac{15}{8}\ \text{of}\ \frac{1}{15}$$

$$= \frac{15}{2} + 4 - \frac{2}{5} \times \frac{7}{3} \div \frac{15}{8} \times \frac{1}{15}$$

$$= \frac{15}{2} + 4 - \frac{2}{5} \times \frac{7}{3} \div \frac{1}{8}$$

$$= \frac{15}{2} + 4 - \frac{112}{15} = \frac{23}{2} - \frac{112}{15}$$

$$= \frac{121}{30} = 4\frac{1}{30}$$

11. A person travels from X to the Y at a speed of 40 kmph and returns by increasing his speed by 50%. What is his average speed for both the trips?

(a) 36 kmph
(b) 45 kmph
(c) 48 kmph
(d) 50 kmph

Sol. *(c)* Let the distance between X and Y be x km.

$\therefore$ Time taken from X to Y @ 40 km/h $= \frac{x}{40}$ h.

$\because$ Speed of person while returning from Y to X

$$= 40 \times \frac{150}{100} = 60 \text{ km/h}$$

Time taken from Y to X @ 60 km/h $= \frac{x}{60}$ h

$$\therefore \text{Average speed} = \frac{2x}{\frac{x}{40} + \frac{x}{60}} = 48 \text{ km/h}$$

12. How many M's occur in the following series such that it is preceded by W and followed by V?

X U V M R S T M W N V M W O P M W U V M W A C W M V H P N V W M W T U N

(a) 3 (b) 2
(c) 1 (d) 5

Sol. *(c)* X U V M R S T M W N V M W O P M W U V M W A C W M V H P N V W M W T U N

Clearly, there is one which is preceded by W and followed by V.

13. In a class, the number of girls is one less than the number of the boys. If the product of the number of boys and that of girls is 272, then the number of girls in the class is

(a) 15 (b) 14 (c) 16 (d) 17

Sol. *(c)* Let the number of girls and boys be x and y, respectively.

Then $x + 1 = y$

and $xy = 272$

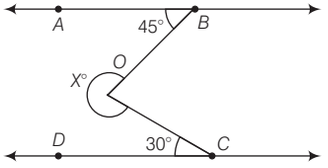
$\Rightarrow x(x+1) = 272$

$\Rightarrow x^2 + x - 272 = 0$

$\Rightarrow (x+17)(x-16) = 0$

$\therefore x = 16$

14. In the given figure, $AB \parallel CD$. Then, X is equal to



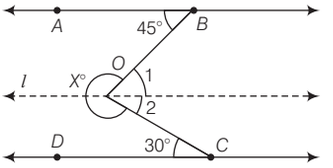
(a) 290° (b) 300°
(c) 280° (d)285°

Sol. *(d)* Through O, draw a line l parallel to both AB and CD.

Then,

$\angle 1 = 45°$ (alt. $\angle S$)

$\angle 2 = 30°$ (alt. $\angle S$)



$\Delta BOC = \angle 1 + \angle 2$

$\therefore \quad = 45° + 30° = 75°$

So, $X = 360° - \angle BOC$

$= 360° - 75° = 285°$

$\therefore \quad X = 285°$

15. If John celebrated his victory day on Tuesday, January 5, 1965, when will he celebrate his next victory day on the same day?

(a) January 5, 1970 (b) January 5, 1971
(c) Januray 5, 1973 (d) January 5, 1974

Sol. *(b)* 5th January 1965 $\Rightarrow$ Tuesday

5th January 1966 $\Rightarrow$ Wednesday

5th January 1967 $\Rightarrow$ Thursday

5th January 1968 $\Rightarrow$ Friday

5th January 1969 $\Rightarrow$ Sunday

Since, 1968 is a leap year.

5th January 1970 $\Rightarrow$ Monday

5th January 1971 $\Rightarrow$ Tuesday

Direction (Q. Nos. 16-18) *Read the following information carefully and answer the questions that follow.*

Seven friends A, B, C, D, E, F and G are sitting around a circular table facing either the centre or outside. Each one of them belongs to a different department *viz.* Finance, Marketing, Sales, HR, Corporate Finance, Investment Banking and Operations but not necessarily in the same order.

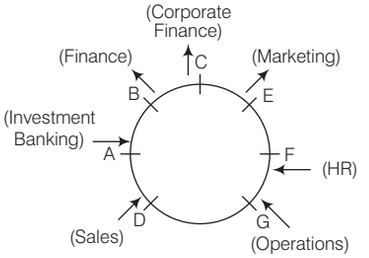
C sits third to the right of G. G faces the centre. Only one person sits between C and the person working in the HR department. Immediate neighbours of C face outside. Only one person sits between F and D. Both F and D face the centre. F works in the HR department. A works in the investment B banking department. A faces the centre. Two people sit between the persons, who work in investment, banking and marketing departments. The person, who works in corporate finance sits to the immediate left of E. C faces same direction as E. The person, who works for sales department sits to the immediate left of the person, who works for operations department.

16. For which of the following departments does B work?
(a) Finance (b) Marketing
(c) HR (d) Corporate Finance

17. What is position of B with respect to the person, who works for sales department?
(a) Immediate right
(b) Third to the left
(c) Second to the right
(d) Second to the left

18. Who sits to the immediate right of E?
(a) The person who works for marketing department
(b) C
(c) B
(d) The person who works for HR department

Solutions (16-18) *According to the given information, seating arrangement of all the seven friends is given below.*



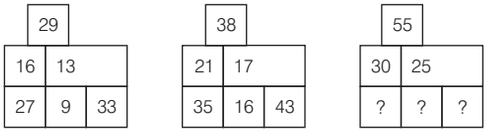
16. *(a)* B works in finance department.

17. *(d)* The person working in sales department is D and B is second to the left of D.

18. *(d)* The person who sits to the immediate right of E is F, who works in HR department.

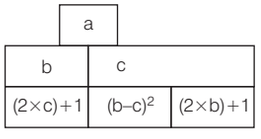
Direction (Q. Nos. 19-21) *In each of the following questions. Choose the missing terms out of the given alternatives.*

19.



(a) 65, 36, 63 (b) 53, 24, 63
(c) 49, 32, 53 (d) 51, 25, 61

Sol. *(d)* Here, the pattern is



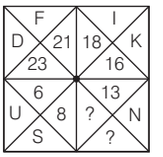
$\therefore$ First number $= (25 \times 2) + 1 = 51$

Second number $= (30 - 25)^2 = 25$

and third number $= (30 \times 2) + 1 = 61$

Hence, 51, 25 and 61 are the three missing terms.

20.

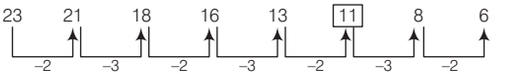


(a) 9, R (b) 11, P (c) 13, Q (d) 10, V

Sol. *(b)* The letters progress in clockwise direction starting from D and the next letter is obtained by following the below skipping pattern.

D(E)F(GH)I(J)K(LM)N(O)P(QR)S(T)U:

The numbers follows the pattern given below:



$\therefore$? = 11 and ? = *P*

21.

2	9	11	7
8	5	13	-3
7	?	10	-4
6	4	10	?

(a) 3 and (–2) (b) (–3) and (–2)
(c) 3 and 2 (d) (–3) and 2

Sol. *(a)* $2 + 9 = 11$

$9 - 2 = 7$

Also, $8 + 5 = 13$

$5 - 8 = -3$

Now, $7 + ? = 10$

$? = 10 - 7 = \boxed{3}$

Also, $4 - 6 = ?$

$? = \boxed{-2}$

22. After 9 o' clock at what time between 9 p.m. and 10 p.m. will the hour and minute hands of a clock point in opposite direction?
(a) 15 min past 9 (b) 16 min past 9
(c) $16\frac{4}{11}$ min past 9 (d) $17\frac{1}{11}$% min past 9

Sol. *(c)* At 9 o' clock, the minute hand is $9 \times 5 = 45$ mins space behind the hour hand. Hence, the minute hand will have to gain $45 - 30 = 15$ min.

Therefore, 60 min is equal to the gain of 55 min spaces.

Hence, gain of 15 min spaces equals

$$= \frac{60}{55} \times 15 = \frac{180}{11} = 16\frac{4}{11}$$

Therefore, hour and minute hands of a clock point in opposite direction after 9 o' clock at $16\frac{4}{11}$ min past 9.

23. Simplify $\dfrac{2\frac{1}{7}-2\frac{1}{2}}{2\frac{1}{4}+1\frac{1}{7}} \div \dfrac{1}{2+\dfrac{1}{2+\dfrac{1}{2-\frac{1}{2}}}}$

(a) $\frac{-1}{2}$ (b) $-\frac{1}{8}$ (c) $-\frac{1}{6}$ (d) $-\frac{1}{4}$

Sol. *(d)* Given expression,

$$= \frac{\frac{15}{7}-\frac{5}{2}}{\frac{9}{4}+\frac{8}{7}} \div \frac{1}{2+\frac{1}{2+\frac{2}{3}}}$$

$$= \frac{-5}{14} \times \frac{28}{95} \div \frac{1}{2+\frac{3}{8}} = \frac{-2}{19} \div \frac{8}{19} = \frac{-2}{19} \times \frac{19}{8} = -\frac{1}{4}$$

Direction (Q. Nos. 24-25) *Study the given information carefully and answer the given questions.*

A is the mother of B. B is the sister of C. D is the son of C. E is the brother of D. F is the mother of E. G is the granddaughter of A. H has only two children B and C.

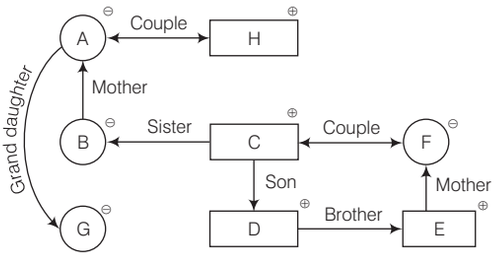
24. How is F related to H?

(a) Son-in-law (b) Daughter-in-law
(c) Father-in-law (d) Granddaughter

25. How is C related to E?

(a) Father (b) Son (c) Mother (d) Cousin

Solutions (24-25) *According to the given information, we can draw the following family tree.*



24. *(b)* F is the daughter-in-law of H.

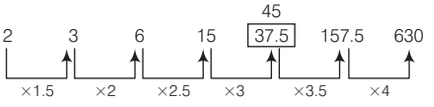
25. *(a)* C is the father of E.

26. In the given series, only one number is wrong. Find out the wrong number

2, 3, 6, 15, 37.5, 157.5, 630

(a) 3 (b) 6 (c) 15 (d) 37.5

Sol. *(d)* The pattern is as follows



Hence, number 37.5 is wrong and should be replaced by 45.

Direction (Q. Nos. 27-30) *Study the following information carefully and answer the given questions. A word and number arrangement machine when given an input line of words and numbers rearranges them following a particular rule in each step. The following is an illustration of input and rearrangement.*

Input 'sale data 18 23 for 95 then 38'

Step I data sale 18 23 for 95 then 38

Step II data 95 sale 18 23 for then 38

Step III data 95 for sale 18 23 then 38

Step IV data 95 for 38 sale 18 23 then

Step V data 95 for 38 sale 23 18 then

Step VI data 95 for 38 sale 23 then 18

and step VI is the last step of the rearrangement of the above input.

As per the rules followed in the above steps, find out in each of the following questions the appropriate step for the given input.

27. Input 'year 39 stake 47 house full 94 55'. How many steps will be required to complete the rearrangement?

(a) Five (b) Six
(c) Four (d) Seven

28. Step II of an input is 'car 83 lost ever 32 46 74 now'. How many more steps will be required to complete the rearrangement?

(a) Three (b) Four
(c) Five (d) Six

29. Step III of an input is 'and 79 code 27 18 new goal 34'. Which of the following will definitely be the input?

(a) code and 79 27 18 new goal 34
(b) code 27 18 new goal 34 and 79
(c) code 27 and 18 79 new goal 34
(d) cannot be determined

30. Input 'water full never 35 78 16 height 28' Which of the following steps will be the last?

(a) VI (b) VII (c) VIII (d) IX

Solution (27-30)

In the step I, the word coming first alphabetically comes at the 1st place from left pushing the remaining line right word; in step II, largest number comes at the 2nd position from left pushing the remaining line rightward; in Step III, the word coming second alphabetically comes at the 3rd place from left pushing the remaining line rightward; In step IV, the 2nd largest number comes at the 4th position from left pushing the remaining line right ward. Thus, word and number get arranged alternately till the words are in forward alphabetical order and the number in decreasing order.

27. *(b)* Input 'year 39 stake 47 house full 94 55'
Step I Full year 39 stake 47 house 94 55
Step II Full 94 year 39 stake 47 house 55
Step III Full 94 house year 39 stake 47 55
Step IV Full 94 house 55 year 39 stake 47
Step V Full 94 house 55 stake year 39 47
Step VI Full 94 house 55 stake 47 year 39
∴ Required steps = Six

28. *(b)* Step II Car 83 lost ever 32 46 74 now
Step III Car 83 ever lost 32 46 74 now
Step IV Car 83 ever 74 lost 32 46 now
Step V Car 83 ever 74 lost 46 32 now
Step VI Car 83 ever 74 lost 46 now 32
∴ Required more steps = Four

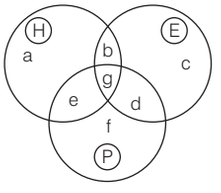
29. *(d)* In arrangement problem, previous steps cannot be determined.

30. *(a)* Input Water full never 35 78 16 height 28
Step I Full water never 35 78 16 height 28
Step II Full 78 water never 35 16 height 28
Step III Full 78 height water never 35 16 28
Step IV Full 78 height 35 water never 16 28
Step V Full 78 height 35 never water 16 28
Step VI Full 78 height 35 never 28 water 16
∴ Last step = Step VI

31. In a class of 50 students, 23 speak English, 15 speak Hindi and 18 speak Punjabi. 3 speak only English and Hindi, 6 speak only Hindi and Punjabi and 6 speak only English and Punjabi. If 9 can speak only English, then how many students speak all the three languages?

(a) 1 (b) 2
(c) 3 (d) 5

Sol. *(d)* Given $b + c + g + d = 23$ …(i)

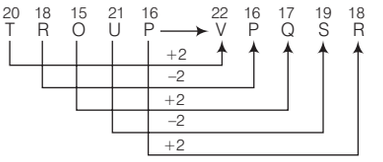


Given, $b = 3$, $d = 6$ and $c = 9$
$\therefore g = 23 - (3 + 6 + 9) = 23 - 18 = 5$

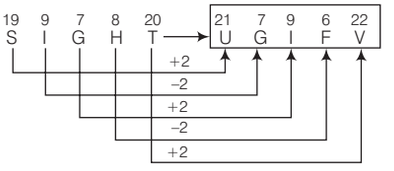
32. If in a certain code language 'TROUP' is written as 'VPQSR', then how will SIGHT be written in that

(a) UGVIF
(b) UIGVF
(c) UGIVF
(d) UGIFV

Sol. *(d)* As,



Similarly,



∴ SIGHT ⇒ UGIFV

33. **Statement** Researchers are feeling agitated as libraries are not well equipped to provide the right information to the right users at the right time in the required format. Even the users are not aware about the various services available for them.

Courses of Action

I. All the information available in the libraries should be computerised to provide faster services to the users.

II. Library staff should be trained in computer operations.

(a) None follows (b) Only II follows
(c) Both I and II follow (d) All follow

Sol. *(c)* Computerisation will definitely help the users to take best available information available in the library. Also, training to the staff in computer operations can further add to betterment of services to the users. Hence, both are feasible and therefore right courses of action.

Direction (Q. Nos. 34-36) *Read the following passage and answer the questions that follow.*

In an engineering college, five students from five different cities were elected as Secretaries by the students to perform different student activities. Each student studies in a different branch of engineering. Additionally, the following information is provided.

I. Abhishek does not stay in the Aravalli hostel, where the student from Nagpur stays.

II. The student, whose name is not Abhishek and does not study in Metallurgy, stays in Satpura hostel. He is the only student among the five to stay at Satpura hostel.

III. Hardeep neither belongs to Jodhpur nor does he study Mechanical Engineering.

IV. The student-in-charge of Cultural Activity stays in the Aravalli hostel where Civil Engineering student does not stay.

V. Sanjay and the student, who studies Metallurgy, both stay in the same hostel.

VI. The student who belong to Allahabad does not stay with the student-in-charge of the Sports activity staying at Aravalli hostel.

VII. Sanjay is not the student-in-charge of the Cultural activity.

VIII. Ravi, the student-in-charge of Mess activity, stays at Satpura hostel.

IX. The student from Patna and the student, who studies Mechanical Engineering, both stay at Aravalli hostel. They are the only two among the five students to stay at this hostel.

X. The student, who stays at Satpura hostel, studies Computer Sciences.

XI. Hemant, who does not belong to Kochi, studies Chemical Engineering. He is not the General Secretary of the Student Body.

XII. Sanjay does not belong to Allahabad.

XIII. The student from Kochi and the student-in-charge of Placement activity, both stay at the Vindhya Hostel.

34. Which of the following statements is/are incorrect?

I. The Chemical Engineering student and the student-in-charge of Cultural activity, both stay in the same hostel.

II. The student-in-charge of Placement activity is studying Metallurgy.

III. The student who belongs to Nagpur is the student-in-charge of Sport's activity.

IV. Ravi belongs to Jodhpur.

(a) Only I (b) I and II
(c) I, II and Ill (d) None of these

35. Which of the following statements is/are correct?

I. Sanjay and Abhishek stay at the same hostel.

II. The General Secretary of the student Body studies Mechanical Engineering.

III. The student who belongs to Patna is studying Metallurgy.

IV. The student who belongs to Kochi is studying Computer Science.

(a) Only I (b) Only II
(c) Only III (d) None of these

36. The student who belongs to Allahabad is studying in

(a) Metallurgy
(b) Mechanical Engineering
(c) Civil Engineering
(d) None of the above

Solutions (34-36) *The given information can be tabulise as*

Name	Activity	Hostel	Discipline	Cities
Ravi	Mess	Satpura	Computer Science	Jodhpur
Sanjay	Sports	Aravalli	Mechanical Engineering	Nagpur
Hardeep	Cultural	Aravalli	Metallurgy	Patna
Abhishek	Student Body	Vindhya	Civil Engineering	Kochi
Hemant	Placement	Vindhya	Chemical Engineering	Allahabad

34. *(b)* From the above table, we can see that the Chemical engineering student stays in Vindhya hostel while cultural activity students stay in Aravalli hostel. So, Statement I is incorrect. Statement II is also incorrect because the student in charge of Placement activity is studying Chemical Engineering.

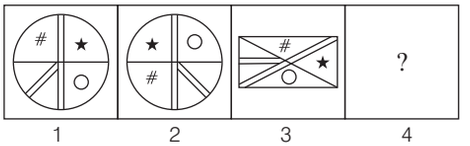
35. *(c)* From above table, it is clear that student belongs to Patna is studying Metallurgy.

36. *(d)* From above table, it is clear that the student from Allahabad is studying Chemical Engineering.

Direction (Q. Nos. 37-39) *The second figure in the first part of the problem figures bears a certain relationship to the first figure. Similarly, one of the figures in answer figures bears the same relationship to the first figure in the second part. You have to select the figure from the set of answer figures which would replace the sign of question mark (?).*

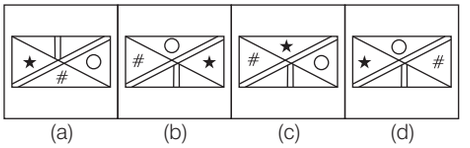
37.

Problem Figures



1 2 3 4

Answer Figures

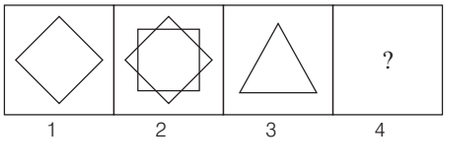


(a) (b) (c) (d)

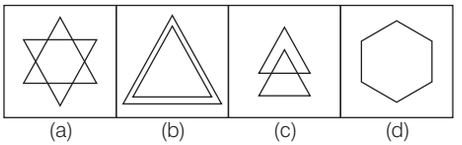
Sol. *(c)* From the first figure to the second figure, all the four smaller designs move in one sector in anti-clockwise direction.

38.

Problem Figures



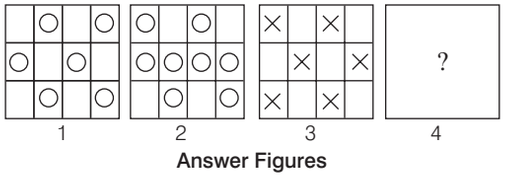
Answer Figures



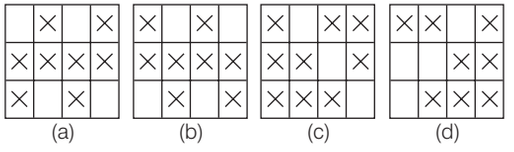
Sol. *(a)* From first figure to second figure, one similar figure is added in a set pattern.

39.

Problem Figures



Answer Figures



Sol. *(a)* From first figure to second figure, the circles of upper row move one block in anti-clockwise direction and two circles are added in the middle row. In similar way, the two crosses of upper row move one block in anti-clockwise direction and two crosses are added in the middle row.

Direction (Q. Nos. 40) *In the question below is given a statement followed by two conclusion numbered* I *and* II. *You have to assume everything in the statement to be true, then consider the two conclusions together and decide which of them logically follows beyond a reasonable doubt from the information given in the statement.*

Give answer

(a) if only conclusion I follows
(b) if only conclusion II follows
(c) if neither conclusion I nor II follows
(d) if both conclusion I and II follows

40. **Statement** From the next academic year, students will have the option of dropping Mathematics and Science for their school leaving certificate examination.

Conclusions

I. Students, who are weak in Science and Mathematics, will be benefitted.

II. Earlier students did not have the choice of continuing their education without taking these subjects.

Sol. *(d)* Both the conclusions follow. Since, the option is available from next year, it implies that this option was not available earlier. Secondly, students who are weak in Mathematics and Science will be benefitted.

SECTION C

GENERAL ENGLISH

1. "To make clean breast of " means……..

(a) To gain prominence
(b) To praise oneself
(c) To confess without being reserve
(d) To destroy before it blooms

Sol. *(c)* Idiom ''To make clean breast of'' means ''To confess without being reserve''.

2. Fate smiles ..... those who untiringly grapple with stark realities of life.

(a) with (b) over
(c) on (d) round

Sol. *(c)* The correct word to be used here in the gap is 'on' which has the correct context.

3. "Life is like a game where we are the players" is a—

(a) Simile (b) Metaphor
(c) Assonance (d) Alliteration

Sol. *(a)* Usage of 'like' shows simile.

4. "All the world's a stage and all the men and women merely players; They have their exits and their entrances;" is a—

(a) Simile (b) Metaphor
(c) Personification (d) Pun

Sol. *(b)* When the comparison is employed without the use of 'Like' then it is metaphor.

5. "Death lays its icy hands on kings" is a—

(a) Simile (b) Metaphor
(c) Personification (d) Alliteration

Sol. *(c)* When a non living object is treated as a living object then it is personification.

6. Shepherd is a word of German origin. This word can also be replaced by which of the following?

(a) Sceaphierde (b) Herder
(c) Sheepish (d) None of the above

Sol. *(a)* Shepherd can also be replaced by Sceaphierde.

7. Cotton is a word of English origin. Which of the following can replace this word?
(a) Cotone (b) Cottony
(c) Coutons (d) None of the above

Sol. *(a)* Cotton can also be replaced by Cotone.

8. Plural of 'Formula' is
(a) Formulas (b) Formulaes
(c) Formulae (d) None of the above

Sol. *(c)* The plural of 'formula' is 'formulae'.

9. The 'accommodation' in British English is known as …….. in American english.
(a) Accommodations (b) Acomodation
(c) Accommodatione (d) Acommodatione

Sol. *(a)* It is 'accommodations' in American English.

10. The 'aeroplane' in British English is known as …….. in American english.
(a) Aeroplain (b) Aerroplane
(c) Airplane (d) Aeroplanee

Sol. *(c)* It is 'airplane' in American English.

11. The 'bank holiday' in British English is known as …….. in American english.
(a) holiday (b) legal holiday
(c) vacation (d) leave

Sol. *(b)* It is 'legal holiday' in American English.

Direction (Q. Nos. 12-14) *Complete the collocation words.*

12. My grandfather was a ……. smoker, so few people were surprised when he died of oral cancer.
(a) serial (b) heavy
(c) big (d) continuous

Sol. *(b)* The correct collocation is a heavy smoker.

13. She was a/an ……. wife who loved her husband more than anything else in the whole universe.
(a) devoted (b) sincere (c) intelligent (d) cool

Sol. *(a)* The correct collocation is a devoted wife.

14. I always avoid his company because he is a crashing ……..
(a) bore (b) nuisance (c) guy (d) sound

Sol. *(a)* The correct collocation is a crashing bore.

15. Idiom "To keeps one's temper" means……
(a) To become hungry (b) To be in good mood
(c) To preserve ones energy (d) To be aloof from

Sol. *(b)* Idiom ''To keeps one's temper'' means ''to be in good mood''.

16. Idiom "To catch a tartar" means ……..
(a) To trap wanted criminal with great difficulty
(b) To catch a dangerous person
(c) To meet with disaster
(d) To deal with a person who is more than one's match

Sol. *(b)* Idiom ''To catch a tartar'' means ''to catch a dangerous person''.

17. Choose the correct option:-
(a) You don't need to come unless you want to
(b) You come only when you want to
(c) You come unless you don't want to
(d) You needn't come until you don't want to

Sol. *(a)* The correct statement is "You don't need to come unless you want to".

18. Choose the correct option:-
(a) This is one of the most important inventions of this century.
(b) This is one of the most important invention of this century.
(c) This is one of the most important invention of these century.
(d) This is one of the most important invention of centuries.

Sol. *(a)* The correct statement is ''This is one of the most important inventions of this century''.

Direction (Q. Nos. 19-21) *Read each sentence to find out whether there is any grammatical error in it. The error, if any will be in one part of the sentence. The letter of that part is the answer. If there is no error, the answer is 'D'. (Ignore the errors of punctuation, if any).*

19. (a) We discussed about the problem so thoroughly
(b) on the eve of the examination
(c) that I found it very easy to work it out
(d) No error

Sol. *(a)* We should not use 'about' with 'discuss' as it is superfluous.

20. (a) An Indian ship
(b) laden with merchandise
(c) got drowned in the Pacific Ocean
(d) No error

Sol. *(c)* It is wrong to use 'drowned'. When a person is involved then we should use ''sank''.

21. (a) I could not put up in a hotel
(b) because the boarding and lodging charges
(c) were exorbitant
(d) No error

Sol. *(a)* 'Put up' is followed by 'at' not by 'in'.

Direction (Q. Nos. 22-27) *Read the following passage carefully and answer the questions given below it. Certain words are given in bold to help you locate them while answering some of the questions.*

With Finance Commission recommendations, centre-state relations set to undergo dramatic change. Any big change requires big ideas, decisive leadership and happy coincidence of circumstances. Nothing illustrates this better than the unfolding story of cooperative federalism in India.

As Chief Minister of Gujarat, Narendra Modi had often argued that the central government implemented schemes were at odds with the state's needs and priorities. For example, schemes that provided funds for electrification were at best of limited value to Gujarat since it had already achieved near 100% electrification. This state could have spent the money provided for such a scheme more productively if allowed to use it for other purposes.

In advancing this view, Modi was joined by other Chief Ministers such as Vasundhara Raje of Rajasthan who argued that the vast numbers of central schemes further restricted their fiscal space because many of them required matching contributions by them from their otherwise untied funds. Once these matching funds were committed to access central schemes, states were left with very limited funds for even the most important expenditure items such as enforcement of law and order.

Nevertheless, this system has remained entrenched in one form or another in the last several decades on account of coincidence of three factors. First, outside of state leaders and a few economists and policy analysts, advocates of the view that true federalism means giving greater fiscal space to states and trusting them in setting their own priorities have been few and far between.

Second, the Finance Commission appointed once every five years plays a key role in the division of tax revenues between centre and states. Consistent with the first point, successive Finance Commissions held untied funds to the states at or below 30% of the divisible tax pool. Only the 13th Finance Commission exceeded this mark, setting states share at 32%.

Finally, successive central governments have chosen to transfer the bulk of the remaining funds to the states via central and centrally sponsored schemes. With nearly 8% growth over an entire decade, tax revenues have significantly expanded. Alongside, central and centrally sponsored schemes and the revenue resources they absorb have expanded dramatically as well.

Two key factors have come together to dramatically alter this equilibrium. One, India now has a prime minister who was once a chief minister and strongly feels that cooperative federalism means greater fiscal and legislative space for states. And two, by a happy coincidence, the chairman and members of the 14th Finance Commission believe in genuine federalism themselves.

Moreover, they have recognised the opportunity offered by a Prime Minister at the helm who truly believes in the power of states and their leaders. Accordingly, they have recommended that starting 2015-16, states be awarded 42% of the divisible pool of tax revenues. This is a gigantic and unprecedented 10% jump in devolution.

Predictably, the Prime Minister and his Cabinet have accepted this bold recommendation of the 14th Finance Commission. What implications does this change have? For starters, with larger transfers coming as untied funds, states also have greater responsibility in discharging their duties. It is likely that they will now have to take greater responsibility in areas that have been hitherto covered by the Centre, especially those falling on the state list. States will also have to do this in ways that are more consistent with their priorities and not according to "one size fits all" schemes. By the same token, with a lower share in the divisible pool the centre's fiscal space will shrink, requiring a rethink of central and centrally sponsored schemes.

One might ask where does Niti Aayog fit into this story? I am tempted to say that in replacing the Planning Commission by Niti Aayog, Prime Minister Narendra Modi may have anticipated the 14th Finance Commission. Under the previous regime, the centre was often seen as 'giver' and states as 'recipients' thereby making the latter feel that they were less than equal partners. In replacing the Planning Commission by Niti Aayog, the Prime Minister sought to change that equation and forge an equal relationship between the two sides. As such this change represents a step towards cooperative federalism.

We will have a more complete picture of the emerging centre-state relations, when the finance minister presents the budget. But even with what we now know, one thing is clear: in the years to come, Niti Aayog will have to play a much greater role in the knowledge space. Greater fiscal freedom combined with greater legislative freedom in the areas covered by the concurrent list of the Constitution means that states will need to play a more active role in designing their own programmes and policies.

As they do so, they will need to reach out to data, analysis and expert advice.The design of Niti Aayog as per the Cabinet Note of 1 January 2015 positions it well to provide these services. By the same token, those of us at Niti Aayog have to work hard in the months to come to build new strengths so that we do not disappoint the states that reach out to us for advice and assistance. We shall see.

22. As the Chief Minister of Gujarat, why Narendra Modi argued that the Central Government implemented schemes are at odds for the state government?

(a) Because the Central Government was not paying the equal amount of funds for all the states.
(b) Because, he being from the opposition, want to criticise the steps taken by the Central Government and leave an impression on the general public mind that what a crook central government is.
(c) Because Central Government doesn't take in the picture all the perspective while allocating the funds.
(d) Because he wanted more funds for filling his and his cabinet minister's pocket.

Sol. *(c)* As the Chief Minister of Gujarat, Narendra Modi argued that the Central Government implemented schemes are at odds for the state government because Central Government doesn't take in the picture all the perspective while allocating the funds.

23. Why Vasundhara Raje decided to join the views of Narendra Modi?

(a) Being from the same party, she had no choice but to join hand in hand with her party leader.
(b) Vasundhra Raje and Narendra Modi had a hidden plan of going against any of the good initiatives taken by the Central Government.
(c) Vasundhra Raje was also facing the atrocities of Central Government and want to raise her voice with Narendra Modi.
(d) Vasundhra Raje, being a Chief Minister of a state of India, where central schemes are restricted, the fiscal space of the state government, feels that it is injustice on the part of Central Government.

Sol. *(d)* Vasundhara Raje decided to join the views of Narendra Modi because being a Chief Minister of a state of India, where central schemes are restricted the fiscal space of the state government, feels that it is injustice on the part of Central Government.

24. The writer referred the PM's view of cooperative federalism, what do you understand by cooperative federalism?

(a) It means a more prominent role for the Centre, than states.
(b) It means the competitiveness between the Centre and the state.
(c) It means more competitiveness among states
(d) It means a sync in relation among the states and the centre.

Sol. *(d)* The writer referred the PM's view of cooperative federalism, which means a sync in relation among the states and the centre.

25. What does writer meant by "one size fits all" schemes?

(a) NITI Ayog will make schemes that would be equally beneficial for all.
(b) Previously, the schemes were more transparent and object oriented.
(c) Planning Commission were making schemes without considering the crucial scenario in the states.
(d) Planning Commission was making more precise schemes, so as to address the actual needs of the states.

Sol. *(c)* Writer meant by "one size fits all" schemes that planning Commission were making schemes without considering the crucial scenario in the states.

26. Why the Government did a 10% increase in devolution?

(a) To transfer the funds in the hands of state.
(b) Shed of the extra responsibility of the state sponsored schemes.
(c) To make state more self dependent.
(d) To increase the extent of authority on states.

Sol. *(a)* Government did a 10% increase in devolution to transfer the funds in the hands of state.

27. What is the drastic change in the policy of NITI Ayog from that was in Planning Commission?

(a) Now states will have equal equation with the centre.
(b) States will have more financial freedom
(c) Centre has given the responsibility to states to draft every scheme
(d) Both 'a' and 'b'

Sol. *(d)* The drastic change of the policy in NITI Ayog from that was in Planning Commission is that now states will have equal equation with the centre and states will have more financial freedom.

Direction (Q. Nos. 28-31) *Each question consists of two words which have a certain relationship with each other followed by four pairs of related words, Select the pair which has the same relationship.*

28. DIVA:OPERA

(a) producer:theatre (b) director:drama
(c) conductor:bus (d) thespian:play

Sol. *(d)* DIVA:OPERA is analogous to thespian:play.

29. GRAIN:SALT

(a) shard:pottery (b) shred:wood
(c) blades:grass (d) chip:glass

Sol. *(d)* GRAIN:SALT is analogous to chip:glass.

30. THRUST:SPEAR

(a) mangle:iron (b) scabbard:sword
(c) bow:arrow (d) fence:epee

Sol. *(d)* THRUST:SPEAR is analogous to fence:epee.

31. PAIN:SEDATIVE

(a) comfort:stimulant (b) grief:consolation
(c) trance:narcotic (d) ache:extraction

Sol. *(b)* PAIN:SEDATIVE is analogous to grief:consolation.

Direction (Q. Nos. 32-35) *In the following questions choose the word which best expresses the meaning of the given word.*

32. BRIEF
(a) Limited (b) Small (c) Little (d) Short

Sol. *(d)* Brief and short are synonymous.

33. EMBEZZLE
(a) Misappropriate (b) Balance
(c) Remunerate (d) Clear

Sol. *(a)* Embezzle means the misappropriation of money. Hence, misappropriate is correct.

34. VENT
(a) Opening (b) Stodge
(c) End (d) Past tense of go

Sol. *(a)* Vent is synonymous to opening.

35. AUGUST
(a) Common (b) Ridiculous
(c) Dignified (d) Petty

Sol. *(c)* August means dignified.

Direction (Q. Nos. 36-40) *In the following the questions choose the word which best expresses the opposite to the meaning of the given word.*

36. ENORMOUS
(a) Soft (b) Average (c) Tiny (d) Weak

Sol. *(c)* 'Enormous' means huge. So, its antonym will be 'tiny'.

37. COMMISSIONED
(a) Started (b) Closed
(c) Finished (d) Terminated

Sol. *(d)* 'Commissioned' means 'to continue'. Hence, 'terminated' will be its correct antonym.

38. ARTIFICIAL
(a) Red (b) Natural (c) Truthful (d) Solid

Sol. *(b)* 'Artificial' means 'man made'. So, its antonym will be 'natural'.

39. EXODUS
(a) Influx (b) Home-coming
(c) Return (d) Restoration

Sol. *(a)* 'Exodus' means 'to exclude'. So, the correct antonym will be 'influx' which means 'to include'.

40. RELINQUISH
(a) Abdicate (b) Renounce (c) Possess (d) Deny

Sol. *(c)* 'Relinquish' means 'to let go'. Hence, the correct antonym will be 'possess'.

SECTION D

QUANTITATIVE APTITUDE

1. In a 4-digit number, the sum of the first two digits is equal to that of the last two digits. The sum of the first and last digits is equal to the third digit. Finally, the sum of the second and fourth digits is twice the sum of the other two digits. What is the third digit of the number?
(a) 5 (b) 8
(c) 1 (d) 4

Sol. *(a)* Let the 4-digit number be $abcd$. Then,

$a + b = c + d$ …(i)

$b + d = 2(a + c)$ …(ii)

and $a + d = c$ …(iii)

From Eqs. (i) and (iii), $b = 2d$.

From Eqs. (i) and (ii), $3b = 4c + d$

$\Rightarrow \quad 3(2d) = 4c + d$

$\Rightarrow \quad 5d = 4c$

$\Rightarrow \quad c = \frac{5}{4} d$

Now, d can be 4 or 8. But if $d = 8$, then $c = 10$ is not possible. So, $d = 4$ which gives $c = 5$.

2. A simple mechanical device has four gears A, B, C and D such that they mesh in the order A—B—C—D. The number of teeth in these gears are 10, 21, 12, 17, respectively. If we start the mechanism from rest, how may revolutions will the largest gear have to turn before one full cycle is completed and the gears are in the position from where they started?
(a) 270
(b) 340
(c) 7140
(d) 285

Sol. *(b)* Let n be the minimum number of teeth advanced before all the wheels are in starting position again.

Hence, n must be divisible by 10, 21, 12 and 17.

Hence, n = LCM = 7140.

So, the required number of revolutions in the largest gear = 7140 / 21 = 340

3. If $a = \frac{\sqrt{5}+1}{\sqrt{5}-1}$ and $b = \frac{\sqrt{5}-1}{\sqrt{5}+1}$, then the value of $\left[\frac{a^2 + ab + b^2}{a^2 - ab + b^2}\right]$ would be

(a) 3/4 (b) 4/3
(c) 3/5 (d) 5/3

Sol. *(b)* $a = \frac{\sqrt{5}+1}{\sqrt{5}-1} = \frac{(\sqrt{5}+1)^2}{4}$

[rationalising the denominator]

$= \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2}$

$b = \frac{\sqrt{5}-1}{\sqrt{5}+1} = \frac{(\sqrt{5}-1)^2}{4}$

$= \frac{6-2\sqrt{5}}{4} = \frac{3-\sqrt{5}}{2}$

$a^2 = \left(\frac{3+\sqrt{5}}{2}\right)^2 = \frac{14+6\sqrt{5}}{4}$

$b^2 = \left(\frac{3-\sqrt{5}}{2}\right)^2 = \frac{14-6\sqrt{5}}{4}$

$ab = \left(\frac{\sqrt{5}+1}{\sqrt{5}-1}\right)\left(\frac{\sqrt{5}-1}{\sqrt{5}+1}\right) = \frac{1}{1} = 1$

$\therefore \frac{a^2 + ab + b^2}{a^2 - ab + b^2} = \frac{32}{24} = \frac{4}{3}$

4. Three shepherds *A*, *B* and *C* rented a pasture for a year. *A* grazed 22 sheeps for 4 months, *B* grazed 16 sheeps for 8 months and *C* grazed 32 sheeps for 6 months. If *C*'s share of rent is ₹ 600, the total rent for the year was

(a) ₹ 3550 (b) ₹ 200
(c) ₹ 1275 (d) ₹ 675

Sol. *(c)* *A*′ *s* monthly equivalent rent $= 22 \times 4$

B′ *s* monthly equivalent rent $= 16 \times 8$

C′ *s* monthly equivalent rent $= 32 \times 6$

$\frac{B'\text{ s monthly equivalent rent}}{C'\text{ s monthly equivalent rent}}$

$= \frac{\text{Rent paid by } B}{\text{Rent paid by } C}$

$\therefore \frac{16 \times 8}{32 \times 6} = \frac{\text{Rent paid by } B}{600}$

$\therefore$ Rent paid by $B = \frac{16 \times 8 \times 600}{32 \times 6} =$ ₹ 400

Similarly, rent paid by

$A = \frac{22 \times 4 \times 60}{32 \times 6} =$ ₹ 275

$\therefore$ Total rent $= 400 + 600 + 275 =$ ₹ 1275

5. An astronomical clock has its dial divided into 24 divisions instead of 12, and the small hand goes round in 24 h. The large hand goes round once every hour. If 24th hour is noon, then when are the hands at right angles between 24 and 1?

(a) $23\frac{2}{15}$ mins past 24 (b) $2\frac{23}{15}$ mins past 24
(c) $15\frac{15}{23}$ mins past 24 (d) $15\frac{11}{23}$ mins past 24

Sol. *(c)* Here, angle covered by the large hand in 60 min $= 360°$ and angle covered by the small hand in 60 min $= \frac{360°}{24} = 15°$. Required time when the hands are at right angles between 24 and 1.

$= \frac{60}{(360-15)} \times 90$ mins past 24

$= \frac{60}{345} \times 90$ mins past 24

$= \frac{360}{23}$ mins past 24

$= 15\frac{15}{23}$ mins past 24

6. The unit digit in the sum $(124)^{372} + (124)^{373}$ is

(a) 5 (b) 4
(c) 2 (d) 0

Sol. *(d)* $4^1 = 4;\ 4^2 = 16;\ 4^3 = 64;\ 4^4 = 256;\ 4^5 = 1024$

On dividing 372 by 4, the remainder = 0

On dividing 373 by 4, the remainder = 1

So, required unit digit

= unit's digit of the sum of $(6 + 4) = 0$

7. Find the G.C.D. of the polynomials $36x^2 - 49$ and $6x^2 - 25x + 21$

(a) $8x - 9$ (b) $9x - 5$
(c) $6x - 7$ (d) None of these

Sol. *(c)* $p(x) = 36x^2 - 49$

$= (6x)^2 - (7)^2 = (6x + 7)(6x - 7)$

$q(x) = 6x^2 - 25x + 21$

$= 6x^2 - 18x - 7x + 21$

$= 6x(x - 3) - 7(x - 3)$

$= (6x - 7)(x - 3);$

$\therefore$ GCD $= (6x - 7)$

8. Find the unit digit in the product $(4387)^{245} \times (621)^{72}$

(a) 1 (b) 2 (c) 5 (d) 7

Sol. *(d)* $7^1 = 7;\ 7^2 = 49;\ 7^3 = 343;\ 7^4 = 2401;\ 7^5 = 16809$ *i.e.*,
The digit at unit place repeats itself after power 4.
Therefore, unit's digit in the product of $(4387)^{245} \times (621)^{72}$
Units's digits in the product of $[(4387)^1 \times (621)]$

$= 7 \times 1 = 7$

9. The minute and the hour hand of a watch meet every 65 min. How much does the watch lose or gain time?

(a) 25 sec (b) 27 sec
(c) 27.16 sec (d) 30 sec

Sol. *(a)* In 60 min, the hour hand moves through an angle of 30°.

∴ In 65 min, the hour hand will move through an angle of $\frac{30}{60} \times 65 = 32.5°$

In 60 min, the minute hand moves through an angle of 360°.

∴ In 65 min, the minute hand moves through an angle of $\frac{360}{60} \times 65 = 390° = 360° + 30°$

⇒ Difference of the angles made by the hour hand and the minute hand = 2.5 (In 65 min)

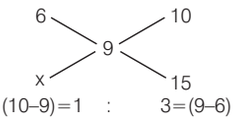
Now, an angle of 30° is made by the hour hand and the minute hand in 5 min.

∴ An angle of 2.5° is made by the minute hand in $\frac{5}{30} \times 2.5$ min, i.e. $\frac{2.5}{6}$ min, i.e. 25 sec.

10. How much Pepsi at ₹ 6 a L is added to 15 L of 'dew' at ₹ 10 a L, so that the price of the mixture be ₹ 9 a L?

(a) 5 (b) 8
(c) 10 (d) None of these

Sol. *(a)* Let x L pepsi is requied



Therefore, $\frac{x}{15} = \frac{1}{3}$

$\Rightarrow \quad x = 5$ L

11. In a village three people contested for the post of village pradhan. Due to their own interest, all the voters voted and no vote was invalid. The losing candidate got 30% votes. What could be the minimum absolute margin of votes by which the winning candidate led by the nearest rival, if each candidate got an integral per cent of votes?

(a) 4 (b) 2
(c) 1 (d) None of these

Sol. *(c)* Losing candidate = 0.3 x

∴ Other two candidates = 0.7 x

The share of winning candidate = 0.36 x

and the second ranker = 0.34 x

∴ Margin (min. possible) = 0.02 x

⇒ 2% of x

Let the minimum possible voter be 50, then

$$\frac{2 \times 50}{100} = 1$$

Hence, the minimum possible margin of votes = 1

12. The sum of first three terms of a GP is to the sum of the first six terms is 125 : 152. Find the common ratio of the GP.

(a) 3/5 (b) 3 (c) 2/5 (d) 5/8

Sol. *(a)* $$\frac{S_3}{S_6} = \frac{125}{152} = \frac{\frac{a(r^3-1)}{(r-1)}}{\frac{a(r^6-1)}{(r-1)}} = \frac{r^3-1}{r^6-1}$$

$$\Rightarrow \quad \frac{r^3-1}{(r^3)^2-1} = \frac{125}{152}$$

$$\Rightarrow \quad \frac{(r^3-1)}{(r^3+1)(r^3-1)} = \frac{125}{152}$$

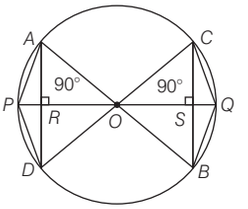
$$\Rightarrow \quad \frac{1}{r^3+1} = \frac{125}{152}$$

$$\Rightarrow \quad r^3 + 1 = \frac{152}{125}$$

$$\Rightarrow \quad r^3 = \frac{27}{125}$$

$$\therefore \quad r = \frac{3}{5}$$

13. In the adjoining figure O is the centre of the circle. $\angle AOD = 120°$. If the radius of the circle be 'r', then find the sum of the areas of quadrilaterals $AODP$ and $OBQC$.



(a) $\frac{\sqrt{3}}{2}r^2$ (b) $3\sqrt{3}r^2$
(c) $\sqrt{3}r^2$ (d) None of these

Sol. *(c)* $OQ = OB = OC = r$ (say)

[vertically opposite angles]

$\angle AOD = \angle BOC = 120°$ [By sss criterion]

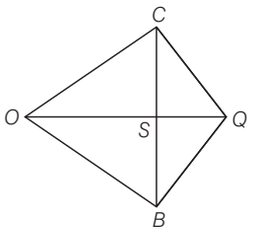
$\therefore \quad \angle BOQ = \angle COQ = 60°$

$\because \quad \frac{SB}{OB} = \sin 60° = \frac{\sqrt{3}}{2}$

$\Rightarrow \quad SB = \frac{r\sqrt{3}}{2}$

$\therefore \quad BC = 258 = r\sqrt{3}$

[a perpendicular from centre to the chord bisects the chord]



$\therefore$ Area of quadrilateral $BQCO$

$$= \frac{1}{2} \times BC \times OQ$$

$$= \frac{1}{2} \times r\sqrt{3} \times r = \frac{r^2\sqrt{3}}{2} \text{ cm}^2$$

$\therefore$ Area of both the quadrilateral $= 2\left(\frac{r^2\sqrt{3}}{2}\right) = \sqrt{3}r^2$ cm

14. An AP whose first term is unity and in which the sum of the first half of any even number of terms to that of the second half of the same number of terms is in constant ratio, the common different 'd' is given by

(a) 1 (b) 2 (c) 3 (d) 4

Sol. *(b)* Let us consider $d = 1$, then the AP is

1, 2, 3, 4, 5, 6, 7, 8

$\therefore$ $1 + 2 + 3 + 4 = 10$

and $5 + 6 + 7 + 8 = 26$

Again 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 12

$\therefore$ $1 + 2 + 3 + 4 + 5 + 6 = 21$

and $7 + 8 + 9 + 10 + 11 + 12 = 57$

$\because$ $\frac{10}{26} \neq \frac{21}{57}$

Hence $d \neq 1$

Now, consider $d = 2$, then the AP is

1, 3, 5, 7, 9, 11, 13, 15

$\therefore$ $1 + 3 + 5 + 7 = 16$

and $9 + 11 + 13 + 15 = 48$

Again 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23

$\therefore$ $1 + 3 + 5 + 7 + 9 + 11 = 36$

and $13 + 15 + 17 + 19 + 21 + 23 = 108$

$\because$ $\frac{16}{48} = \frac{36}{108} \Rightarrow \frac{1}{3} = \frac{1}{3}$

Hence, $d = 2$ is the correct answer.

15. Find the number of four digit number that are divisible by 30 and 35 but not by 140.

(a) 21 (b) 22

(c) 43 (d) 44

Sol. *(b)* If a number is divisible by 30 and 35, it is divisible by LCM (30, 35) i.e. 210.

If a number is divisible by 210 but not by 140, then it is not divisible by LCM (210, 140) i.e. 420. So, the number is divisible by 210 but not by 420. The least and the greatest four digit multiples of 210 are 1050 and 9870, respectively.

$\therefore$ The number of multiples of four-digit multiples of 210 are $\frac{9870 - 1050}{210} + 1 = \frac{8820}{210} + 1 = 43$

Similarly, the number of four-digit multiples of 420 are $\frac{9660 + 1260 + 1}{420} = 21$.

$\therefore$ The number of multiples of 210 which are not divisible by 420 are $43 - 21 = 22$

16. A covers 1/4th of his journey at 20 km/h and 1/3rd of the rest at 25 km/h and half of the rest at 30 km/h and rest at the speed of 40 km/h. What is the average speed of A?

(a) $13\frac{78}{89}$ km/h (b) 12 km/h

(c) $26\frac{86}{89}$ km/h (d) 28 km/h

Sol. *(c)* Suppose, the total distance equals to 4 times the LCM of the speeds.

$\therefore$ Total distance = 2400 km

$\therefore$ Total times $= \frac{600}{20} + \frac{600}{25} + \frac{600}{30} + \frac{600}{40}$

$= 30 + 24 + 20 + 15 = 89$ h

$\therefore$ Average speed $= \frac{2400}{89} = 26\frac{86}{89}$ km/h

17. Abhinav leaves Mumbai at 6 am and reaches Bangalore at 10 am. Praveen leaves Bangalore at 8 am and reaches Mumbai at 11 : 30 am. At what time do they cross each other?

(a) 10 am (b) 8 : 32 am (c) 8 : 56 am (d) 9 : 20 am

Sol. *(c)* Time taken by Abhinav = 4 h

Time taken by Praveen = 3.5 h

For convenience take the product of time taken by both as a distance.

Then, the distance = 14 km

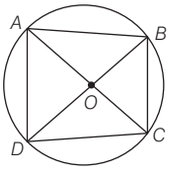
Since, Abhinav covers half of the distance in 2 hr (i.e. at 8 am)

Now, the rest half (i.e. 7 km) will be covered by both Praveen and Abhinav.

Time taken by them $= \frac{7}{7.5} = 56$ min

Thus, they will cross each other at 8 : 56 am.

18. In the given figure $ABCD$ is a cyclic quadrilateral $DO = 8$ cm and $CO = 4$ cm. AC is the angle bisector of $\angle BAD$. The length of AD is equal to the length of AB. DB intersects diagonal AC at O, then what is the length of the diagonal AC?



(a) 20 cm (b) 24 cm
(c) 16 cm (d) None of these

Sol. *(a)* $\frac{AD}{AB} = \frac{DO}{BO} = 1$ [By sss similarity]

$\Rightarrow \quad OB = OD = 8$ cm

$ABCD$ is a cyclic quadrilateral

$\therefore \quad DO \times BO = CO \times AO$

$\Rightarrow \quad AO = 16$ cm

$\therefore \quad AC = 16 + 4 = 20$ cm

19. A stamp collector likes to arrange and rearrange his collection in a symmetrical order. Sometimes he lines his stamps up in pairs, sometimes in threes, sometimes in fours, occasionally in fives, and once in a while in groups of six. After arranging them in any of these groups, he invariably has one stamp left over. When he arranges his stamps in groups of seven, he finds that there is not a single stamp to spare. How many stamps will be left over if he arranges them in groups of 8?

(a) 5 (b) 4 (c) 6 (d) 7

Sol. *(a)* Number of tickets with the collector

= LCM (3, 4, 5, 6) + 1, i.e. 60 + 1 = 61

or one more than the multiple of 60 i.e. 61, 121, 181, 241, 301 etc.

Since, it must be exactly divisible by 7, so the collector has 301 stamps and when stamps are arranged in groups of 8, he will have 5 stamps left over.

20. A can do a piece of work in 10 days, B can do it in 12 days and C can do it in 15 days. In how many days they can complete the work, working together?

(a) 6 (b) 4 (c) 5 (d) 10

Sol. *(b)* Efficiency of $A = \frac{100}{10} = 10\%$

Efficiency of $B = \frac{100}{12} = 8.33\%$

Efficiency of $C = \frac{100}{15} = 6.66\%$

Combined efficiency of A, B and $C = 10 + 8.33 + 6.66 = 25\%$

$\therefore$ Required number of days, when A, B and C worked together $= \frac{100}{25} = 4$ days

21. A balance of a trader weighs 10% less than it should be. Still the trader marks-up his goods to get the overall profit of 20%.

What is the markup on the cost price?

(a) 40% (b) 8%
(c) 25% (d) 16.66%

Sol. *(b)* Let the CP be ₹ 1 per g.

But he weighs 900 g for every 1000 g

$\therefore$ Value of good sold = 900

Now, let the markup be x%.

$\therefore \quad MP = 1000 + \frac{1000x}{100} = (1000 + 10x)$

But, MP = SP, $\therefore$ SP $= (1000 + 10x)$

Hence, profit $= \frac{(1000 + 10x) - 900}{900} \times 100 = 20$

$\Rightarrow \quad x = 8$

Thus, the markup = 8%

22. If a, b, c be respectively the sum of first p, q, r terms of an AP, then $\frac{a}{p}(q - r) + \frac{b}{q}(r - p) + \frac{c}{r}(p - q)$ equal

(a) pqr (b) $ap + bq + cr$
(c) 0 (d) $p + q + r$

Sol. *(c)* Let an AP be 2, 5, 8, 11, 14, 17, 20, 23,...

and let $p = 3$, $q = 5$, $r = 6$

and $S_p = a = (2 + 5 + 8) = 15$

and $S_q = b = (2 + 5 + 8 + 11 + 14) = 40$

and $S_r = c = (2 + 5 + \ldots + 14 + 17) = 57$

$\therefore \quad \frac{a}{p}(q - r) + \frac{b}{q}(r - p) + \frac{c}{r}(p - q)$

$= \frac{15}{3}(5 - 6) + \frac{40}{5}(6 - 3) + \frac{57}{6}(3 - 5)$

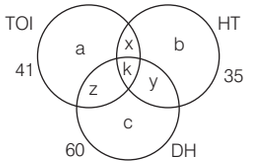
$= -5 + 24 - 19 = 0$

Direction (Q. Nos. 23-24) *A survey that* 41%, 35% *and* 60% *of the people read "Times of India", "Hindustan Times" and "Deccan Herald" respectively.* 27% *people read exactly two of the three newspapers and* 3% *read none.*

23. What percentage of people read all the three newspaper?
(a) 40% (b) 6% (c) 9% (d) 12%

24. If another survey indicates that 16% of the people read Times of India and Deccan Herald, and 14% read Hindustan Times and Deccan Herald, then what percentage of the people read only Hindustan Times (Use the data from the previous question, if necessary)
(a) 10% (b) 8% (c) 12% (d) 15%

Solutions (23-24)



23. *(b)* $x + y + z = 27\%$

$a + b + c + x + y + z + k = 97\%$

$\Rightarrow a + b + c + k = 70\%$

Now, $a + b + c + 2(x + y + z) + 3k$

$= 41 + 35 + 60 = 136\%$

$\Rightarrow a + b + c + 3k = 136 - 54 = 82\%$

And $a + b + c + k = 70\%$

$\therefore \quad 2k = 12\%$

$\Rightarrow \quad k = 6\%$

24. *(c)* $z + k = 16$

$\Rightarrow \quad z = 10 \qquad [\because k = 6]$

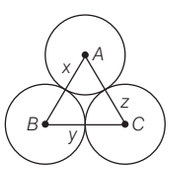
$\Rightarrow \quad y + k = 14 \Rightarrow y = 8$

$\Rightarrow \quad x = 9 \qquad [\because x + y\ z = 27]$

$\therefore \quad b = 35 - (x + k + y) = 35 - (9 + 6 + 8) = 12\%$

25. Three circles touch each other externally. The distance between their centre is 5 cm, 6 cm and 7 cm. Find the radii of the circles.
(a) 2 cm, 3 cm, 4 cm (b) 3 cm, 4 cm, 1 cm
(c) 1 cm, 2.5 cm, 3.5 cm (d) 1 cm, 2 cm, 4 cm

Sol. *(a)*



$AB = 5\text{ cm} = x + y$

$BC = 6\text{ cm} = y + z$

$AC = 7\text{ cm} = z + x$

$\therefore \quad 2(x + y + z) = 5 + 6 + 7 = 18\text{ cm}$

$\Rightarrow \quad x + y + z = 9\text{ cm}$

Now, $x = (x + y + z) - (y + z) = 9 - 6 = 3\text{ cm}$

$\Rightarrow \quad y = 5 - x = 5 - 3 = 2\text{ cm}$

$\Rightarrow \quad z = 6 - y = 6 - 2 = 4\text{ cm}$

$\therefore \quad x = 3\text{ cm},\ y = 2\text{ cm and } z = 4$

26. If a, b, c be the p th, q th and r th terms of an AP. Then, $p(b - c) + q(c - a) + r(a - b)$ equals to
(a) 0 (b) 1 (c) 3 (d) $\frac{abc}{pqr}$

Sol. *(a)* Let an AP be 3, 7, 11, 15, 19, 23 27,..... and
Let $p = 2,\ q = 3,\ r = 7$

Then, $T_p = a = 7,\ T_q = b = 11,\ T_r = c = 27$

$\therefore \quad p(b - c) + q(c - a) + r(a - b)$

$= 2(11 - 27) + 3(27 - 7) + 7(7 - 11)$

$= -32 + 60 - 28 = 0$

Hence, (a) is true.

27. A company instead of raising the mark up by 20% discounted the cost price by 20% while stiching the price tag on its project. Further the company offers a discount of 6.25% to its customer. In this process company incurs a loss of ₹ 37.5 on a single article. What is the selling price of that article?
(a) 417.5 (b) 112.5
(c) 365.5 (d) None of these

Sol. *(b)* Let CP = 100 , then

Tag price = 80

SP, after discount = 75

Total loss = 25

$\because$ ₹ 25 are being less at CP of ₹ 100.

$\therefore$ ₹ 37.5 are being less at CP of ₹ $\frac{100}{25} \times 37.5 = 150$

Hence, SP $= 150 - 37.5 = 112.5$

28. If $(5x^2 + 14x + 2)^2 - (4x^2 - 5x + 7)^2$ is divided by $(x^2 + x + 1)$ then, quotient 'q' and remainder 'r' are given by
(a) $q = (x^2 + 19x), r = 0$ (b) $q = 9(x^2 + 19x - 5), r = 0$
(c) $q = (x^2 + 19x - 5), r = 1$ (d) $q = 9(x^2 + 19x - 5), r = 1$

Sol. *(b)* $\because A^2 - B^2 = (A + B) \cdot (A - B)$

$\therefore (5x^2 + 14x + 2)^2 - (4x^2 - 5x + 7)^2$

$= [(5x^2 + 14x + 2) + (4x^2 - 5x + 7)]$

$\times [(5x^2 + 14x + 2) - (4x^2 + 5x - 7)]$

$= 9(x^2 + 19x - 5)(x^2 + x + 1)$

$\therefore \dfrac{(5x^2 + 14x + 2)^2 - (4x^2 - 5x + 7)^2}{(x^2 + x + 1)}$

$= 9(x^2 + 19x - 5)$

$\therefore \quad q = 9(x^2 + 19x - 5)$ and $r = 0$

29. If $\log_{10} x^2 - \log_{10} \sqrt{y} = 1$, find the value of y, when $x = 2$

(a) 2/9 (b) 4/25
(c) 25/4 (d) 4/9

Sol. *(b)* $\log_{10} x^2 - \log_{10} \sqrt{y} = 1$

$$\Rightarrow \log_{10}\left(\frac{x^2}{\sqrt{y}}\right) = \log_{10} 10$$

$$\Rightarrow \frac{x^2}{\sqrt{y}} = 10 \Rightarrow \frac{4}{\sqrt{y}} = 10$$

$$\Rightarrow \sqrt{y} = \frac{4}{10} = \frac{2}{5}$$

$$\Rightarrow y = \frac{4}{25}$$

30. Let a function is defined as $f(x) = \frac{a^x + a^{-x}}{2}$; $a > 0$, what is the value of $f(x + y) + f(x - y)$?

(a) $f(x) + f(y)$ (b) $f(x)\, f(y)$
(c) $2f(x)\, f(y)$ (d) $4f(x)\, f(y)$

Sol. *(c)* $f(x) = \frac{a^x + a^{-x}}{2}$, $f(y) = \frac{a^y + a^{-y}}{2}$

$$\therefore \quad f(x + y) = \frac{a^{x+y} + a^{-(x+y)}}{2} \qquad \ldots(i)$$

and $f(x - y) = \frac{a^{x-y} + a^{-x+y}}{2}$ …(ii)

Adding (i) and (ii), we get

$$f(x+y) + f(x-y) = \frac{a^{x+y} + a^{-x-y}}{2} + \frac{a^{x-y} + a^{-x+y}}{2}$$

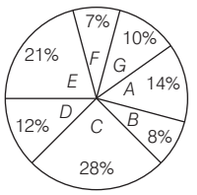
$$= \frac{1}{2}[a^x (a^y + a^{-y}) + a^{-x} (a^y + a^{-y})]$$

$$= \frac{1}{2}(a^x + a^{-x})(a^y + a^{-y})$$

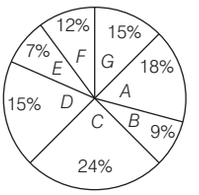
$$= 2f(x)\, f(y)$$

Direction (Q. nos. 31-33) *These questions are based on the following pie-charts' informations.*

Classification of appeared candidates in a competitive test from different institutes and qualified candidates from those institutes.



Appeared candidates = 56000



Qualified = 12000

31. What is the difference between the number of qualified candidates of institute '*C*' and those of '*E*' ?

(a) 2040 (b) 2140 (c) 2144 (d) 2120

32. What is the average number of appeared candidates from institutes *A*, *E* and *G* together?

(a) 25200 (b) 12600
(c) 8400 (d) 8600

33. What is the percentage of qualified candidates from institutes '*A*' and '*C*' taken together? (rounded to two decimal places)

(a) 21.32 (b) 22. 42 (c) 22.32 (d) 21.42

Solutions (31-33)

31. *(a)* Required difference = (24 – 7)% of 12000
= 17% of 12000
= 2040

32. *(c)* Total appeared candidates from *A, E* and *G*

$$= 56000 \times \frac{14 + 21 + 10}{100} = 560 \times 45$$

$$\therefore \text{Required average} = \frac{560 \times 45}{3} = 8400$$

33. *(d)* Required percentage

$$= \frac{12000 \times (18 + 24)\%}{56000 \times (14 + 28)\%} \times 100 = 21.42\%$$

Direction (Q. Nos. 34-36) *Study the following table carefully to answer the questions given below it.*

Terrorists killed in Poonch district in J & K over the years 2014 to 2016

Year	Location	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
2014	On LoC	6	1	0	3	26	22	18	6
	In Depth	9	21	22	21	22	52	42	14
2015	On LoC	7	9	0	6	12	4	8	10
	In Depth	28	17	16	35	21	20	12	21
2016	On LoC	5	14	2	4	8	8	0	10
	In Depth	9	6	7	34	16	16	9	7

34. What is the difference between the average number of terrorists killed on 'LoC' in 2015 and average number of terrorists killed in 'Depth' location in 2016? (only for the given months)

(a) 6 (b) 7 (c) 5 (d) 15

35. Approximately, what is the ratio between total killings in 'In depth' location in 2015 and killings on 'LoC' in that year?
(a) 7 : 2 (b) 3: 1 (c) 17 : 8 (d) 9 : 4

36. Which of the following month-and-year combination has seen maximum terrorist killings?
(a) May 2015 (b) May 2014
(c) April 2016 (d) June 2014

Solutions (34-36)

34. *(a)* Terrorists killed on 'LoC' in 2015
$= 7 + 9 + 0 + 6 + 12 + 4 + 8 + 10 = 56$
Killed in 'Depth' location in 2016.
$= 9 + 6 + 7 + 34 + 16 + 16 + 9 + 7 = 104$
$\therefore$ Required difference $= \frac{104 - 56}{8} = 6$

35. *(b)* Required ratio
$$= \frac{28 + 17 + 16 + 35 + 21 + 20 + 12 + 21}{7 + 9 + 0 + 6 + 12 + 4 + 8 + 10}$$
$$= \frac{170}{56} \approx 3 : 1$$

36. *(d)* In June 2014, total killings were $(22 + 52) = 74$, which is the highest among the given combination.

37. If $\left|\frac{30 - x}{5}\right| < 4$, then find x
(a) $< 6 < x < 6$ (b) $10 < x < 50$
(c) $x > 10$ (d) None of these

Sol. *(b)* $\left|\frac{30 - x}{5}\right| < 4$
$\Rightarrow |30 - x| < 20$
$\Rightarrow -20 < (30 - x) < 20$
$\Rightarrow -20 < 30 - x < 20$
$\Rightarrow -50 < -x < -10$
$\Rightarrow 50 > x > 10$
$\Rightarrow 10 < x < 50$

38. The least number which when divided by 12, 16 and 18 leaves 5 as remainder in each case. Find the number.
(a) 139 (b) 144
(c) 149 (d) 154

Sol. *(c)* Given, $x = 12$, $y = 16$, $z = 18$, $k = 5$
According to the formula,
Required number = (LCM of x, y and z) + k
= (LCM of 12, 16, 18) + 5
LCM of 12, 16, 18 is

2	12, 16, 18
2	6, 8, 9
3	3, 4, 9
	1, 4, 3

$\therefore$ LCM $= 2 \times 2 \times 3 \times 4 \times 3 = 144$
Required number $= 144 + 5 = 149$

39. If area of regular pentagon is $125\sqrt{3}$ cm, how long is its each side?
(a) 10 cm (b) 15 cm (c) 16 cm (d) 25 cm

Sol. *(a)* We know that,
Area of regular pentagon $= 5a^2 \frac{\sqrt{3}}{4}$ [a = side]
According to the question,
$\frac{5a^2\sqrt{3}}{4} = 125\sqrt{3} \Rightarrow a^2 = \frac{125\sqrt{3} \times 4}{5\sqrt{3}} = 100$
$\therefore$ $a = 10$ cm

40. The first 100 multiples of 10 are multiplied together. How many zeros will be there at the end of the product?
(a) 100 (b) 24
(c) 124 (d) 125

Sol. *(c)* First 100 multiples of $10 = 10, 20, 30, 40, \ldots 990, 1000$
We have to find out the number of zeros in
$10 \times 20 \times 30 \times \ldots \times 990 \times 1000$.
Which can be written as
$10^{100}(1 \times 2 \times 3 \times 4 \times \ldots \times 99 \times 100)$
$= 10^{100}\,(100!)$
Now, the number of zeros is non factorial part
$$= \frac{100}{5} + \frac{100}{25}$$
$$= 20 + 4 = 24$$
$\therefore$ Total number of zeros $= 100 + 24 = 124$

Solved Paper 2015

SNAP

Symbiosis National Aptitude Test

Instructions

1. There are total 150 questions in this paper.
2. There are four parts. Part A (General Awareness), Part B (Analytical & Logical Reasoning), Part C (General English) and Part D (Quantitative, Data Interpretation and Data Sufficiency).
3. Part A carries 40 marks, Part B carries 30 marks, Part C carries 40 marks and Part D carries 40 marks.
4. For each incorrect attempt 1/4th mark will be deducted.

Section-A General Awareness

For all questions in this section, each correct answer carries 1 mark.

1. The full form of CSIRO is
   (a) Comprehensive Scientific and Industrial Research Organisation
   (b) Cross-cultural Scientific and Industrial Research Organisation
   (c) Commonwealth Scientific and Industrial Research Organisation
   (d) Council of Scientific and Industrial Research Organisation

Sol. (c) The Commonwealth Scientific and Industrial Research Organisation (CSIRO) is the federal government agency for scientific research in Australia with its headquarters in Canberra and numerous sites around Australia. It was founded in the year 1916.

2. The full form of UNRISD is
   (a) United Nations Research Institute for Soviet Development
   (b) United Nations Research Institute for Scientific Development
   (c) United Nations Research Institute for Socio-economic Development
   (d) United Nations Research Institute for Social Development

Sol. (d) The United Nations Research Institute for Social Development (UNRISD) is an autonomous research institute within the United Nations that undertakes multi disciplinary research and policy analysis on the social dimensions of contemporary development issues. It was formed in 1963.

3. Bhaona is a presentation of the Ankia Naat of Assam. In Bhaona the cultural glimpses of ..... is/are reflected.
   (a) Assam and Orissa
   (b) Bengal
   (c) Mathura and Brindavan
   (d) All of the above

Sol. (d) Bhaona is a staging of the Ankia Naat of the state of Assam, where Ankia means one and Naat means drama. In Bhaona Cultural glimpses of Assam, Bengal, Orissa (Odisha), Mathura and Brindavan are reflected. The Sutradhar, or narrator begins the story, first in Sanskrit and then in either Brajaboli or Assamese.

4. Under which Act does the Archaeological Survey of India (ASI) protect monuments, sites and remains of national importance?
   (a) AMASR Act, 1958
   (b) AMASR Act, 1968
   (c) AMASR Act, 1978
   (d) AMASR Act, 1948

Sol. (a) The Ancient Monuments and Archaeological Sites and Remains Act (1958) provides for the preservation of ancient and historical monuments and archaeological sites and remains of national importance, for the regulation of archaeological exacavations and for protection of sculptures, carvings and other like objects.

5. In 2015, the President of India approved conferment of Padma Awards to all of the following but for _____ .
   (a) Kharag Singh Valdiya
   (b) Mohammad Yusuf Khan
   (c) Lakshmi Gopala Naidu
   (d) Saichiro Misumi

Sol. (c) The Padma Awards 2015 was not conferred upon Lakshmi Gopala Naidu.
Shri Mohammad Yusuf Khan alias Dilip Kumar received Padma Vibhushan for Arts, Dr Kharag Singh Valdiya received Padma Bhushan in the field of Science and Engineering and Shri Saichiro Misumi was conferred Padma Bhushan.

6. In Grammy Awards 2015, the best folk album was won by _____ .
   (a) Pharrell Williams - "Happy"
   (b) Old Crow Medicine Show - Remedy
   (c) Rosanne Cash - The River & The Thread
   (d) None of the above

Sol. (b) The Grammy Awards 2015 for the best folk album went to 'Old Crow Medicine Show - Remedy'. Remedy is the ninth studio album by folk, country and old time music band, old crow medicine show. The album is produced by Ted Hutt.

7. Scientists discovered a new species in the human family tree which is a small creature with a tiny brain. The new species has been named as _____ .
   (a) Homo Naledi
   (b) Dwarf Sapein
   (c) Dwarf Homo Sapein
   (d) None of the above

Sol. (a) Scientists have discovered a new species in the human family tree— Homo Naledi, a small creature with a tiny brain— after examining a huge haul of fossilised bones found in a dark chamber of a South African cave. The discovery of fossils of 15 individuals, consisting of 1550 bones represents the largest fossil hominin found on the African continent.

8. The ministry of external affairs has recently decided to change the nomenclature of Indian Based Domestic Assistance (IBDA) to _____.
   (a) Government Serving Based Domestic Assistance (GSBDA)
   (b) Swadesh Swatcha Domestic Assistance (SSDA)
   (c) Service Staff (SS)
   (d) Bharath Based Service Assistance (BBSA)

Sol. (c) The nomenclature of IBDA has been changed to 'Service Staff' which is a part of the new policy implemented in the wake of high-profile cases involving Indian diplomats and their domestic helps. The policy now addresses the rights and responsibilities of both the service staff and the diplomats.

9. The foreign exchange reserve of India consists of _____.
   (a) the foreign currency assets held by RBI and the gold holding of RBI
   (b) the gold holding of RBI and special drawing rights
   (c) the gold holding of RBI, the foreign currency assets held by RBI and special drawing rights
   (d) only the foreign currency assets held by RBI

Sol. (c) The foreign exchange reserve of India consists of foreign currency assets expressed in US dollar terms, SDRs (define and maintained by International Monetary Fund) and gold holding of RBI.

10. _____ is a charge for converting bullion into coins where free coinage is permitted. This charge is equal to the cost of bullion to coins transformation.
    (a) Bull-Coin
    (b) Brassage
    (c) Bit Coin
    (d) Coin Levy

Sol. (b) Brassage is a charge for converting bullion into coins where free coinage is permitted. This charge is equal to the cost of bullion to coins transformation.

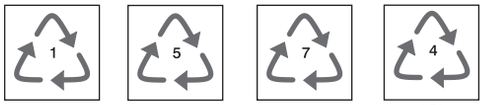
11. Microsoft introduced several new products for education customers, this includes a note-taking app called _____.
    (a) Office Note classroom
    (b) Micro Note class Notebook
    (c) Soft Note for classroom
    (d) One Note class Notebook

Sol. (d) One note class notebooks have a personal workspace for every student, a content library for handouts and a collaboration space for lessons and creative activities.

12. The National Merit Scholarship Scheme, which provided financial assistance to meritorious students from Class XI to post-graduation level in government schools/colleges/universities has been discontinued from the year _____.
    (a) 2009
    (b) 2008
    (c) 2007
    (d) None of the above

Sol. (c) The National Merit Scholarship Scheme, the objective of which was to provide financial assistance to meritorious students studying from Class XI to post-graduation level in government schools/colleges/universities has been discontinued from April 2007.

13. The symbol used for reusable microwaveable plastic ware is ____.



(a) (b) (c) (d)

Sol. (b) This is the symbol of polypropylene (pp) plastic. It is a thermoplastic polymer which is strong and tough. It has a high resistance to heat and acts as a barrier to moisture. It is used as resuable microwaveable ware, kitchen ware, yogurt container and disposable cups and plates etc.

14. Which is the national emblem of Iran, used from 1979 to 1980?

(a) (b) (c) (d)

Sol. (a) The national emblem of Iran, used from 1979 to 1980 was Lion and Sun symbol largely based on astronomical and astrological configurations : the ancient sign of the Sun in the house of Leo which itself is traced back to Babylonian astrology and near Eastern traditions.

15. Super Blood Moon, which is a rare astronomical phenomenon hasn't happened since ___ and won't happen again until ____.

(a) 1982, 2033 (b) 1980, 2035
(c) 1978, 2032 (d) 1975, 2031

Sol. (a) The Super Blood Moon, a rare astronomical phenomenon has not happened since 1982 and will not happen again until 2033.

16. The galaxy____,was initially discovered with NASA's Spitzer Space Telescope in infrared light and is believed to be at least 9 billion years old.

(a) Andromeda (b) Cosmos redshift 7
(c) Sage0536agn (d) Sunflower

Sol. (a) The Andromeda Galaxy is believed to be at least 9 billion years old and was discovered with NASA's Spitzer Space Telescope in infrared light.

17. In the year 2014, Facebook bought Whatsapp for _____ US dollars.

(a) 17 billion (b) 18 billion
(c) 19 billion (d) 20 billion

Sol. (c) Facebook acquired Whatsapp in 2014 for 19 billion US dollars making it the company's largest acquisition.

18. Shri Narendra Modi was sworn in as the Prime Minister of India on _____ at the Rashtrapati Bhavan in New Delhi.

(a) 25th May 2014 (b) 19th May 2014
(c) 26th May 2014 (d) 27th May 2014

Sol. (c) Narendra Modi sworn in as 15th Prime Minister of India on 26th May 2014. A leader of Bharatiya Janata Party, he was the Chief Minister of Gujarat from 2001 to 2014 and is the Member of Parliament from Varanasi.

19. Where is the doldrums belt located?

(a) Near the equator
(b) Near the poles
(c) Near the tropic of cancer
(d) Near the tropic of capricorn

Sol. (a) The doldrums is a colloquial expression derived from historical maritime usage, which refers to those parts of the Atlantic ocean and the Pacific ocean affected by the Intertropical Convergence Zone, a low pressure area around the equator where the prevailing winds are calm.

20. The air quality in Singapore deteriorated to a hazardous level in September 2015, forcing the city-state to shut its schools for the first time in 12 yr. This was due to _____.

(a) haze created mainly due to companies in Malaysia
(b) haze created mainly due to companies in Indonesia
(c) haze created mainly due to companies in Philippines
(d) haze created mainly due to companies in Vietnam

Sol. (b) The air quality in Singapore deteriorated to a hazardous level in September 2015 due to haze created mainly due to companies in Indonesia. Forests and brush were illegally burned in Sumatra to clear land for oil palm plantations and other farming.

21. In 1347, during the reign of Muhammed Tuglak an Afghan officer named _____ setup an independent kingdom called Bahmani Kingdom.

(a) Mahmud Gawan Bahmani
(b) Mohammad Yusuf Bahmani
(c) Hasan Gangu Bahmani
(d) Khwaja Tuglak Bahmani

Sol. (c) The Bahamani Sultanate was a Muslim state of the Deccan in South India and one of the great medieval Indian Kingdoms. It was the first independent Islamic Kingdom in South India. The founder was Alauddin Hassan Bahaman Shah also known as Hasan Gangu Bahmani.

22. The task of consolidating Mughal Kingdom was left to Akbar who was only ____ years old at the time of his accession of the throne.

(a) eleven (b) twelve
(c) thirteen (d) fourteen

Sol. (c) Akbar became the Mughal emperor at the age of thirteen years in 1556 after the death of his father Humayun.

23. Smoking in public places was prohibited nationwide from
(a) 15th August 2008 (b) 02nd October 2008
(c) 15th August 2009 (d) 02nd October 2009

Sol. (b) Smoking in public places was prohibited nationwide from 2nd October 2008 under the Prohibition of Smoking in Public Places Rules, 2008 and COTPA. The nationwide smoke– free law pertains only to public places.

24. A constitutional right can be_____, recognised and established by a sovereign state or union of states.
(a) a prerogative or a duty or a restraint of power
(b) a prerogative or a duty, a power or a restraint of power
(c) a prerogative or a duty, a power but not a restraint of power
(d) a power but not a prerogative or a duty

Sol. (b) A constitutional right can be a prerogative or a duty, a power or a restraint of power, recognised and established by a sovereign state or union of states. All constitutional rights are expressly stipulated and written in a consolidated national constitution.

25. In Nepal, the festival of lights i.e. Diwali is celebrated by some Buddhists as _____.
(a) Tihar (b) Swanti
(c) Both (a) and (b) (d) Neither (a) nor (b)

Sol. (c) Tihar or Swanti or Deepawali is a five day long Hindu festival celebrated in Nepal which comes soon after Dashain. The festival is celebrated from Trayodashi of Kartik Krishna to Kartik Shukla Dwitya every year.

26. The term "Vrajapati" used in Indian Mythology denoted _____.
(a) the Head of the Village
(b) the Head of the Family
(c) the Head of a Society
(d) the Head of Merchants

Sol. (b) The Vrajapati was in charge of commonly hold tribal pasture land and was the leader of the family. Vajrapati used to lead in the battles and later became synonymous with the Gramani.

27. One of India's most distinguished constitutional lawyers who had received brickbats for arguing in favour of Dow Chemicals in the Bhopal gas disaster case is _____.
(a) Ram Jethmalai
(b) Fali Nariman
(c) Mukul Rohatgi
(d) Pramila Nesargi

Sol. (b) Fali Sam Nariman is a distinguished Indian Constitutional jurist and senior advocate to the Supreme Court of India since 1971. He argued in favour of Dow Chemicals (the owned by Union carbide) in the infamous Bhopal gas disaster case.

28. The Nobel Peace Prize winner who gave up freedom and a life with her family in Britain, to protest against the military rule at another country, who is also the Chair of the National League for democracy is _____.
(a) Angela Merkel (b) Christine Lagarde
(c) Brito Polman (d) Aung San Suu Kyi

Sol. (d) Aung San Suu Kyi is a Burmese social democratic staliswoman, politician, President of Myanmar's National League for Democracy. She remained under house arrest in Burma from 1989 to 2010. She received Nobel Peace Prize in 1991.

29. In September 2015, it was revealed that _____ is the country which was exporting drone components worth hundreds of millions to countries that include Saudi Arabia and South Korea, to regain lost ground in a global arms race.
(a) France
(b) United Kingdom
(c) Japan
(d) United States of America

Sol. (b) The UK was exporting drone components worth hundreds of millions of pounds to countries including UAE, South Korea and others to regain lost ground in a global arms race to meet burgeoning demand for lethal UAVs.

30. In the Indian general assembly elections 2014, the BJP led NDA won ___ seats out of __ Lok Sabha seats that were announced.
(a) 335, 546 (b) 334, 545 (c) 336, 543 (d) 332, 544

Sol. (c) The BJP led NDA claimed a landslide victory in the Lok Sabha Elections 2014. The NDA looked set to win 336 seats out of 543 seats with BJP alone gaining 272 seats.

31. The BS EN 16001 solutions from BSI is applicable for _____.
(a) Energy Management Systems
(b) Environment Management Systems
(c) Energy Process Systems
(d) Environment Standard Systems

Sol. (a) BS EN 16001 is an Energy Management Systems (EnMS) enabling organisations to develop policy taking into account legal requirements and energy aspects. It applies to all types and sizes of organisations.

32. The WEEE (Waste from Electrical and Electronic Equipment) is a directive that controls _____.
(a) how electric and electronic equipment is handled and recycled
(b) how electric and electronic equipment is manufactured and handled
(c) how electric and electronic equipment is recycled
(d) how electric and electronic equipment is manufactured, handled and recycled

Sol. (c) The WEEE is a directive that controls how electric and electronic equipment is recycled.

33. The Sensex and Nifty are both indices. The base years for the BSE Sensex and Nifty are_____and _____ respectively.
(a) 1980-81 and 1990
(b) 1990-91 and 2000
(c) 1978-79 and 1995
(d) 2000-01 and 2004

Sol. (c) BSE Sensex is the benchmark index of S & P (Standard and Poor's) BSE and consists of 30 stock. Its base year is 1978-79.
Nifty is the benchmark index of NSE (National Stock Exchange) and is made up of 50 stocks. Its base year is 1995.

34. The artist who painted Irises, Sunflowers, Red Poppies, Pink Roses was _____ .
(a) Vincent van Gogh
(b) Sandro Botticelli
(c) Leonardo da Vinci
(d) Michelangelo

Sol. (a) Vincent van Gogh was a Dutch post– impressionist painter whose work had far reaching influence on 20th century art. His output includes portraits, self portraits, landscapes, still lifes, olive trees and cypresses, wheat fields and sunflowers.

35. Which Indian satellite was launched that has a fantastic timing and records 1000th of a second?
(a) GSAT-16
(b) IRNSS-1D
(c) GSAT-6
(d) Astrosat

Sol. (d) Astrosat is India's first dedicated multi-wavelength space observatory. It was launched on a PSLV–XL. It covers spectral bands from radio, optical, IR, UV and *X*-ray wavelengths.

36. W3C stands for_____.
(a) Triple Web Consortium
(b) Triple Web Consolidation Council
(c) World Wide Web Consortium
(d) World Wide Web Company

Sol. (c) World Wide Web Consortium is an international community that develops open standards to ensure the long-term growth of the web. Founded and currently led by Tim Berners— Lee, it is made up of member organisations which maintain full— time staff for the purpose of working together in the development of standards for the world wide web.

37. The country/countries that has/have resorted to Quantitative easing in the last decade is/are
(a) United States of America
(b) United Kingdom
(c) Japan
(d) All of the above

Sol. (d) Quantitative easing is a monetary policy used by central banks to stimulate the economy when standard monetary policy has become ineffective. A central bank implements quantitative easing by buying financial assets from commercial banks and other financial institutions, thus raising the price of those financial institutions, thus raising the price of those financial assets and increasing the money supply.

38. Who is the Indian badminton player who after spending an year after his shoulder injury earned a final appearance at the Korean Open in 2015?
(a) Chetan Anand (b) Ajay Jayaram
(c) Parupalli Kashyap (d) Sameer Verma

Sol. (b) Ajay Jayaram emerged as second best against top seed Chen Long of China in Seoul, in the Korean open 2015. He settled with silver medal.

39. *Tamaasha*, the traditional folk theatre form of Maharashtra has evolved from the folk forms of _____.
(a) Gondhal, Jagran and Kirtan
(b) Kirtan
(c) Gondhal, Jagran
(d) Gondhal

Sol. (a) Tamaasha is a traditional folk theatre form of Maharashtra evolved from the folk forms of Gondhal, Jagran and Kirtan. In Tamaasha, the female actress is the chief exponent of dance movements in the play.

40. The last series of wall painting in India are from _____ near Hindupur belonging to 16th century AD.
(a) Lepakshi temple
(b) Shiva temple
(c) Sri Venkateswara Swami temple
(d) Ganesh temple

Sol. (a) The last series of wall painting in India are from Lepakshi temple near Hindupur belonging to 16th century AD. The paintings are pressed within broad friezes and illustrate saivaite and secular themes. The colour scheme and the ornamentation of these figures are very pleasing and prove the highly sophisticated taste of Indian artists.

Section-B Analytical & Logical Reasoning

For all questions in this section, each correct answer carries 1 mark.

41. A bungalow has one of its room located on the first floor and there are three identical 100 watt electrical bulbs fixed in the room. Each bulb is connected to a specific switch located at the basement. There are only three switches in the basement. All the bulbs are switched off at present and you are also at the basement area. The first floor cannot be seen from the basement area. If you are allowed to use your common prudence, what is the minimum number of times that you will have to go from basement to first floor to identify which switch goes to which bulb?
(a) 03 times (b) 02 times
(c) 01 time (d) 06 times

Sol. (b) The minimum number of times one have to go from basement to first floor to identify the switches are 2.

42. Shyam is running a start-up. His initial investments are high and he is trying hard to manage and increase his cash flow. The sundry expenses that his firm incurs is negligible. He found from his accountant that the amount of prepaid expenses in the balance sheet, which were booked from the previous year to the current year was increased. Shyam also ensured that his cash funded by the venture capitalists did not reduce when compared to the previous year. The interest that he gets from his fixed deposits increases, which is in tune with his sundry expenses. The final effect on cash for this year would be _____.
(a) cash flow marginally increases
(b) cash flow exponentially increases
(c) cash flow remains the same
(d) None of the above

Sol. (a) We can say that cash flow marginally increases as prepaid expenses and interest increased and also venture capitalists fund either remained same or increased.

43. A producer of a drama theatre is creating his weekend schedule. The producer has six plays to choose from: "Made in India", "Laugh for a while", "The Life is your choice", "MBA— Maha Budhiman Aadmi", "Placements— my goal", "MBA Go Getters". The producer sets a schedule based upon the following criteria.
I. "Made in India" must be shown before "The Life is your choice" and "Laugh for a while".
II. "Laugh for a while" must be shown before "MBA— Maha Budhiman Aadmi".
III. "Placements - my goal" must be shown after "The Life is your choice" and "MBA Go Getters".
Which of the following weekend schedules are consistent with the producer's criteria?
(a) Made in India, Laugh for a while, The Life is your choice, MBA— Maha Budhiman Aadmi, Placements— my goal and MBA Go Getters
(b) MBA Go Getters, Made in India, Laugh for a while, The Life is your choice, Placements— my goal and MBA— Maha Budhiman Aadmi
(c) Made in India, MBA Go Getters, Placements— my goal, Laugh for a while, MBA— Maha Budhiman Aadmi and The Life is your choice
(d) Made in India, MBA Go Getters, MBA— Maha Budhiman Aadmi, The Life is your choice, Placements— my goal and Laugh for a while

Sol. (b) The schedules as consistent with the producer's criteria—
MBA— GO Getters, Made in India, Laugh for a while, The Life is your choice, Placements— my goal and MBA— Maha Budhiman Aadmi

44. Mr Peter gave his eldest son David a bag with 1000 gold coins. David took 230 gold coins from the bag and gave the rest to his younger brothers John, Joe and Jonathan, and advised them to distribute the balance left in the bag amongst themselves in proportion to their age which together amounted to 17.5 yr. After a lot of deliberation and discussion John, Joe and Jonathan came to a conclusion to distribute the gold coins. Their methodology was as follows: As often John took 4 gold coins, Joe took 3. As often John took 6 gold coins, Jonathan took 7. What was the age of John, Joe and Jonathan?
(a) 6 yr, 4.5 yr and 7 yr respectively
(b) 5 yr, 5.5 yr and 7 yr respectively
(c) 6 yr, 5 yr and 6.5 yr respectively
(d) 5 yr, 6.5 yr and 6 yr respectively

Sol. (a) Ratio of ages = Ratio of distribution
$\because$ Joe : John = 3 : 4
and John : Jonathan = 6 : 7
$\therefore$ Joe : John : Jonathan = 18 : 24 : 28
Let ages of Joe, John and Jonathan be $18x$, $24x$ and $28x$, respectively. [let]
$\therefore$ $18x + 24x + 28x = 17.5 \Rightarrow x = 0.25$
$\therefore$ Ages of John, Joe and Jonathan are 6 yr, 4.5 yr and 7 yr.

45. Mohan has an antique clock which strikes and makes loud gong sound every hour. It strikes the exact number of times indicating the time of the day or night. His clock takes seven seconds to strike seven O'clock, how many seconds will his clock take to strike eleven O'clock?
(a) 11.22222227 s (b) 11 s
(c) 11.66666667 s (d) None of these

Sol. (c) For 6 interval strikes = 7 s

$\therefore$ 1 interval strike $= \frac{7}{6}$ s

$\therefore$ (11 – 1) interval strike $= 10 \times \frac{7}{6} = \frac{70}{6} = 11.667$ s

46. Find the missing number.

2, 6, 20, 42, 110, _____

(a) 126 (b) 156 (c) 176 (d) 196

Sol. (b) The pattern = Prime number × Previous number

i.e. $1 \times 2 = 2 \Rightarrow 3 \times 2 = 6$

$5 \times 4 = 20 \Rightarrow 7 \times 6 = 42 \Rightarrow 11 \times 10 = 110$

Similarly, $13 \times 12 = 156$

47. Fifteen years back, Ms Kalpana had three sons Ramesh, Suresh and Rajesh. The sum of the age of Ramesh, Suresh and Rajesh was equal to half of the age of their mother. It was during the next five years when Mahesh was born. Then, the age of Ms Kalpana was equal to the sum of the ages of all her children. Time went on and years passed and Dinesh was born and age of Ramesh equaled the sum of the ages of Rajesh and Mahesh. Now, it so happened that the sum of the ages of Ramesh, Suresh, Rajesh, Mahesh and Dinesh was double the age of their mother and was also equal to the sum of the ages of Suresh and Ramesh. Also Ramesh's age was equal to sum of the ages of Mahesh and Dinesh. What is the age of Ms Kalpana?

(a) 39 yr (b) 42 yr

(c) 41 yr (d) None of these

Sol. (a) The age of Ms Kalpana is 39 yr.

48. Find the missing number.

–1, 0, 0, _ , 8

(a) 1 (b) 2 (c) 3 (d) 4

Sol. (a) The missing number is 1.

49. Ceiling Fan : Table Fan : $_{a}{}^{b}{}_{c}{}^{d}{}_{e}{}^{f}{}_{g}$: ___

a b c d e f g $_{a}{}^{b}{}_{c}{}^{d}{}_{e}{}^{f}{}_{g}$ $g^{f}{}_{e}{}^{d}{}_{c}{}^{b}{}_{a}$ None of these

(a) (b) (c) (d)

Sol. (b) One is in upper direction and other is in lower direction.

Directions (Q. Nos. 50-51) *There is a statement followed by two arguments. Choose the correct option which gives the decision on the arguments, which is derived from the statement.*

50. The new amendment of Corporate Social Responsibility (CSR) in India refers to bringing an overall positive impact on the communities, cultures, societies and environments. The fundamentals of CSR rest on the fact that not only public policy but even corporate should be responsible enough to address social issues.

Argument I Government should not enforce companies to take up CSR.

Argument II Companies moral responsibility is to take up CSR for a long run benefit.

(a) Argument II is correct but Argument I is wrong

(b) Argument I is correct but Argument II is wrong

(c) Both the arguments are false

(d) Both the arguments are correct

Sol. (a) Argument II is correct as companies moral responsibility is to take up CSR for a long run for benefit of the society. Argument I is wrong as law or rule should be implemented.

51. LPG subsidy is a stand taken by the government. Linking of Aadhar card to a bank account has been made mandatory for receiving the subsidy.

Argument I All people should give up LPG subsidies.

Argument II To fill the gap between poor and rich.

(a) Argument II is the output of Argument I

(b) Argument I is a byproduct of Argument II

(c) Argument I and Argument II are complementing to each other

(d) No logical link between Argument I and Argument II

Sol. (d) There is no logical link between the two arguments as giving up the subsidy will not fill the gap between poor and rich.

52. If CAB is coded as 723 – 5 58 in a coded language, then how will DAD be coded?

(a) 4023 – 5 4023

(b) 4090 – 5 4090

(c) 1024 – 5 1024

(d) 1246 – 5 1189

Sol. (b) As,

C	A	B
↓	↓	↓
3	1	2
↓	↓	↓
$(3^6 - 6)$	$(1^6 - 6)$	$(2^6 - 6)$
↓	↓	↓
723	– 5	58

Similarly,

D	A	D
↓	↓	↓
4	1	4
↓	↓	↓
$(4^6 - 6)$	$(1^6 - 6)$	$(4^6 - 6)$
↓	↓	↓
4090	– 5	4090

53. If MONEY is coded as 144 200 171 0 600, then DOLLAR is coded as _____.

(a) –9 200 119 119 –24 299

(b) –8 200 106 106 –20 200

(c) 122 200 102 102 10 154

(d) 120 200 101 101 08 156

Sol. (a) As,

M O N E Y

$(13^2 - 5^2)$ $(15^2 - 5^2)$ $(14^2 - 5^2)$ $(5^2 - 5^2)$ $(25^2 - 5^2)$

144 200 171 0 600

Similarly,

D O L L A R

$(4^2 - 5^2)$ $(15^2 - 5^2)$ $(12^2 - 5^2)$ $(12^2 - 5^2)$ $(1^2 - 5^2)$ $(18^2 - 5^2)$

– 9 200 119 119 – 24 299

54. There are 100 MBA aspirants in a classroom and 99% of them are engineers. How many engineers must leave the classroom in order to reduce the percentage of engineers in the classroom to 98%?

(a) 1 (b) 2 (c) 50 (d) 90

Sol. (c) Let number of engineers that must leave the classroom be 'x'.

According to the question, $\frac{99 - x}{100 - x} \times 100 = 98$

$\Rightarrow \quad 9900 - 100x = 9800 - 98x$

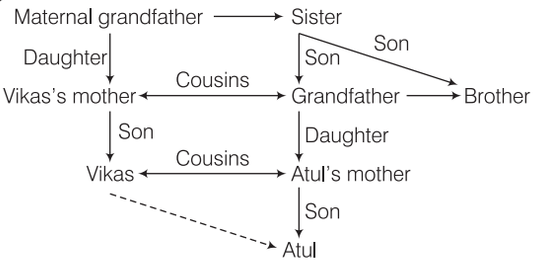
$\Rightarrow \quad 100 = 2x$

$\Rightarrow \quad x = 50$

55. Vikas was showing a photograph to his friend and pointed to a boy and told the following statement. "His name is Atul and his maternal grandfather's brother is my maternal grandfather's sister's son." How is Atul related to Vikas?

(a) They are brothers
(b) Vikas is the uncle of Atul
(c) They are distant cousins
(d) None of the above

Sol. (d)



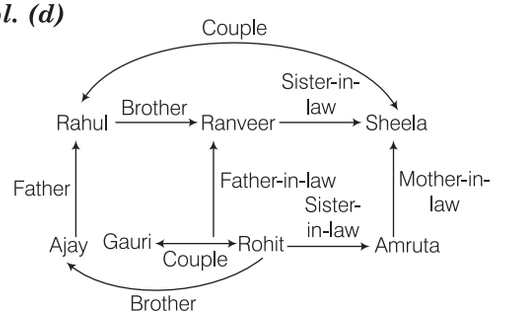
So, Vikas is maternal uncle of Atul.

56. There was a party organised and the following members attended the party : Sheela, Amruta, Rohit, Rahul, Ajay, Ranveer and Gauri.

Sheela is mother-in-law of Amruta, who is sister-in-law of Rohit. Rahul is the father of Ajay. Ajay is the brother of Rohit. Ranveer is the only brother of Rahul and the father-in-law of Gauri. Gauri was married to Rohit. How is Sheela related to Ranveer?

(a) Cousin (b) Mother
(c) Sister (d) None of these

Sol. (d)



So, Sheela is sister-in-law of Ranveer.

Directions (Q. Nos. 57-60) *Using the information given below to answer the questions.*

A chef is trying a recipe for a tasty ice cream using four ingredients. He can choose from three liquids for taste which are labelled A, B and C which are stable in nature and the choice for flavor can be from four liquids which are labelled W, X, Y and Z. For the new ice cream recipe to be tasty, there must be two liquids from the taste giving liquids. Also certain liquids cannot be mixed because of their reactions which makes it unhealthy for human consumption and the same is given below.

B cannot be mixed with W
C cannot be mixed with Y
Y cannot be mixed with Z

57. If the chef calculated that Y is the most important flavor and must be used in the recipe, which other ingredients must be part of the recipe?

(a) A, B and W (b) A, B and X
(c) A, B and Z (d) B, C and X

Sol. (b) According to the given conditions, only option (b) is possible.

58. If the chef rejected B because of its possible side effects but decided to use Z, which is a possible combination of the four ingredients in the recipe?

(a) A, C, W and Z
(b) A, X, Y and Z
(c) A,W, X and Z
(d) A, C, Y and Z

Sol. (a) As there should be two liquids, so only (a) is correct.

59. Which of the following combinations of liquids is impossible?

I. Using Y and W together

II. Using B and C together

III. Using W, X and Z together

(a) Only I (b) Only II
(c) I and III (d) I and II

Sol. (c) I and III are not possible.

60. Which of the following must always be true?

I. If C is used W must be added.

II. If Y is used B must be added.

III. If C is not used W cannot be added.

(a) I and II
(b) II and III
(c) I, II and III
(d) Only II

Sol. (c) All are true.

61. Which letter from the options given will replace the '?' in the table given below?

CAT	389376	DOG
RAT	1758276	MAT
CAB	15876	FAN
CAN	571536	?

(a) MAN (b) GIFT
(c) PAN (d) SOFT

Sol. (b) As, C A T → 3 1 20 ; D O G → 4 5 17

and $(3 + 1 + 20)^2 \times (4 + 5 + 17)^2 = 24^2 \times 26^2 = 389376$

Similarly, C A N → 3 1 14

and $(3 + 1 + 14)^2 \times ?^2 = 571536$

$\Rightarrow \quad ?^2 = 571536 \div 324$

$= 1764 = (42)^2$

$\therefore \quad ? = \text{GIFT}$

62. Which word from the given options will replace the '?' in the table given below?

WOE	1089	MISERY
VANQUISH	1308	SUBDUE
TACITURNITY	1547	SILENCE
SPLEEN	1806	?

(a) MALEVOLENCE (b) JOIN
(c) LINK (d) DRIP

Sol. (a) Here words are synonyms.

WOE → MISERY
VANQUISH → SUBDUE
TACITURNITY → SILENCE
SPLEEN → MALEVOLENCE

and 1089 (+219) 1308 (+239) 1547 (+259) 1806

63. Which number from the options given will replace the '?' in the table given below?

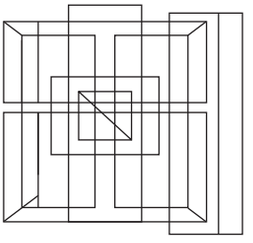
MBA	728	NCB
BCA	24	CDB
BCS	24	CDT
FCA	?	GDB

(a) 48 (b) 168
(c) 64 (d) 132

Sol. (b) As, MBA → 13 ; 728 → $(13 + 14)^2 - 1$; NCB → 14

Similarly, FCA → 6 ; 168 → $(6 + 7)^2 - 1$; GDB → 7

64. What is the least number of straight lines needed to draw the following diagram?



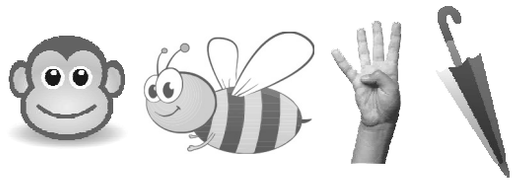
(a) 39
(b) 40
(c) 42
(d) None of the above

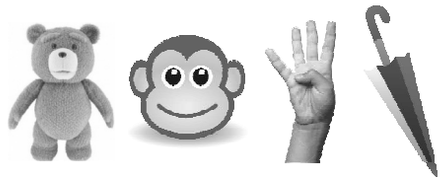
Sol. (a) Horizontal line = 15
Vertical line = 19
Digonal = 5
$\therefore$ Total line = 15 + 19 + 5 = 39

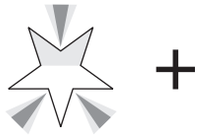
65.

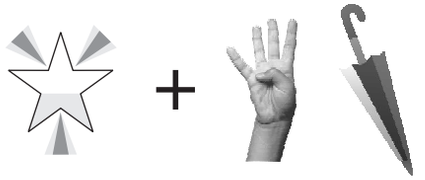
If :

Then, : ?

(a) 

(b) 

(c) 

(d) 

Sol. (d) Two new elements will be added and two will remain same.

Directions (Q. Nos. 66-70) *Read the information given below and answer the questions.*

Ten friends Matt, Sam, Pat, Tom, Sid, Alex, Kat, Jim, Jane and John are having dinner on a rectangular table. Eight facing each other along the length of the table while two facing each other along the smaller side of the table. Pat is sitting diagonally opposite to Kat, Alex is facing Jim, John is to the right of Jane, Tom is sitting between Sam and Jim, Pat is to the extreme left of Jane who is sitting on the extreme right along the length of the table. Sid is facing Tom and Matt is on the right of Sam.

66. Who do you think is sitting in front of each other along the small sides of the table?
(a) Matt and John (b) Matt and Sam
(c) Matt and Kat (d) John and Sam

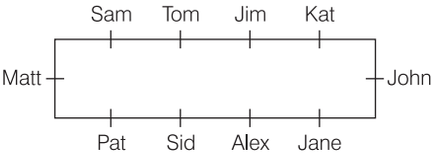
67. Sid is sitting between which of the following two?
(a) Sam and Kat (b) Pat and Alex
(c) Alex and Jane (d) Kat and Jane

68. Who are sitting diagonally opposite to each other?
(a) Sid and Tom
(b) Sam and Jane
(c) Alex and Jim
(d) Jane and Kat

69. Who is sitting to the immediate right of Jim?
(a) Sam (b) Jane
(c) Alex (d) Tom

70. Who is sitting two places left to Jane?
(a) Pat (b) Alex
(c) Sid (d) Sam

Solutions (Q. Nos. 66-70) *According to the given information, the arrangement is as follows.*



66. ***(a)*** Matt and John sit along the small sides of the table.

67. ***(b)*** Sid is sitting between Pat and Alex.

68. ***(b)*** Sam and Jane are sitting diagonally opposite to each other.

69. ***(d)*** Tom is sitting immediate right to Jim.

70. ***(c)*** Sid is sitting two places left to Jane.

Section-C General English

For all questions in this section, each correct answer carries 1 mark.

71. "To catch a tartar" means ..... .
(a) To trap wanted criminal with great difficulty
(b) To catch a dangerous person
(c) To catch a person who is more than one's match
(d) To meet with disaster

Sol. (c) The idiom "To catch a tartar" means to catch or fight with a person who is more than one's match.

72. A leopard can't change its ..... .
(a) dots (b) stripes
(c) colour (d) None of these

Sol. (d) The complete idiom is "A leopard can't change its spots".
So, the answer is (d).

73. "So sober sometimes serious Sam smiles on silly things" is a/an ..... .
(a) hyperbole (b) assonance
(c) anaphora (d) alliteration

Sol. (d) Alliteration is used in the given sentence as the consonant "S" is repeated in the sentence.

74. "The strength given by my mother is bigger than the cosmic energy in this cosmos" is ..... .
(a) rhyme (b) metaphor
(c) personification (d) hyperbole

Sol. (d) The statement is an exaggeration, so Hyperbole is the figure of speech used in the sentence.

75. The buzzing of bees is an example of ..... .
(a) Simile (b) Metonymy
(c) Onomatopoeia (d) Paradox

Sol. (c) Onomatopoeia is the figure of speech used in the phrase as "buzzing" resembles the sound that the bees make.

76. The word Cacti is of Latin origin. It can also be replaced by ..... .
(a) Cactus (b) Cats
(c) Cactuses (d) Cactusus

Sol. (c) "Cactuses" is the English plural of "Cactus" while "Cacti" is the Latin plural of Cactus. So, "Cacti" can be replaced by "Cactuses".

77. The word Trousseaux is of French origin. It can also be replaced by ..... .
(a) troussers
(b) trousseaus
(c) troussears
(d) None of the above

Sol. (b) The word "Trousseaux" is the plural of the word "Trousseau". It can be replaced by the word trousseaus.

78. The singular of the word Scarves is spelt as ..... .
(a) Scarve (b) Scarfe
(c) Scarv (d) None of these

Sol. (d) The singular form of the word "Scarves" is "Scarf". As none of the options match the word "Scarf", so the correct answer is (d).

79. The "Drawing Pins" in British English is referred to as ..... in American English.
(a) Thumb Pins (b) Board Pins
(c) Broad Pins (d) Thumbtacks

Sol. (d) "Drawing Pins" in British English is referred as "Thumbtacks" in American English.

80. "Aubergine" in Britain is referred to as ..... in United States of America.
(a) Migraine (b) Eggplant
(c) Margarine (d) Egg

Sol. (b) "Aubergine" is referred as "Eggplant" in the USA.

81. Base ball in American English is commonly referred to as ..... in British English.
(a) Run ball (b) Strike ball
(c) Rounders ball (d) Rounders

Sol. (d) "Rounders" in British English refers to "Base ball" in American English.

Directions (Q. Nos. 82-84) *Complete the collocation words.*

82. ..... Weapon
(a) Nuclear (b) Atomic
(c) Molecular (d) Electronic

Sol. (a) Only "Nuclear" fits with the word "Weapon".
So, the correct answer is (a).

83. Seminal .....
(a) News (b) River
(c) Nuance (d) Research

Sol. (d) Out of the given choices, the word "Seminal" goes with "Research". So, the correct answer is (d).

84. ..... Percentage
(a) Huge (b) Big (c) More (d) Large

Sol. (d) "Large" is the most suitable option that can be used with the word "Percentage".

85. The idiom "Against the clock" means .....
(a) Break the Rules
(b) Rushed and short on time
(c) Go back to the Past
(d) Look at the Past

Sol. (b) The idiom "Against the clock" means to rush and be short in time. So, (b) is the correct answer.

86. The idiom "Buy a lemon" means ……
(a) a superstitious way to say 'good luck'
(b) a lie which is propaganda for people to believe
(c) an unbelievable story which is told for people to believe
(d) to purchase a vehicle that constantly gives problems or stops running after you drive it away

Sol. (d) "A lemon" means a useless or faulty thing.
So, the correct answer is (d).

87. Kedar uses the following sentence to introduce himself. Choose the correct option.
(a) Myself Kedar, I belong to Mumbai
(b) Myself Kedar and I am from Mumbai
(c) Myself Kedar from Mumbai
(d) None of the above

Sol. (d) None of the given choices is a correct way of introducing oneself.

88. Ram uses the following sentence to tell the time of the day to Shyam. Which one is the correct sentence?
(a) It is 2 pm in the afternoon
(b) It is 2 pm
(c) It is 2 pm in the noon
(d) It is 2 O'clock in the afternoon

Sol. (b) The option which says "It is 2 pm" is the correct one.

Directions (Q. Nos. 89-91) *In these questions some parts have been underlined. If any rule of correct English is violated, then it could be only in the underlined part, marked as* 1, 2, 3 *or* 4. *Choose the option, which violates usage of correct English.*

89. My Parents (1) are Indian but (2) I am (3) born in Sydney (4).
(a) Only 1 (b) 1 and 3 (c) Only 3 (d) Only 4

Sol. (c) "I am" should be replaced by "I was".

90. Standing (1) on the top of the tower (2) the whole city could be seen (3).
(a) Only 1 (b) Only 2
(c) 1 and 2 (d) None of these

Sol. (a) The word "Standing" should be replaced by "If you stand" to make the sentence correct.

91. Ganesh is taller than (1) Ramesh but (2) Anoop is (3) more taller (4).
(a) Only 1 (b) Only 2
(c) Only 3 (d) Only 4

Sol. (d) The words "more taller" should be replaced by "tallest".

Directions (Q. Nos. 92-96) *Read the following passage and answer the questions.*

"There are several factors that contribute to wisdom. Of these I should put first a sense of proportion; the capacity to take account of all the important factors in a problem and to attach to each its due weight. This has become more difficult than it used to be owing to the extent and complexity of the specialised knowledge required of various kinds of technicians. Suppose, e.g. that you are engaged in research in scientific medicine. The work is difficult and is tikely to absorb the whole of your intellectual energy. You have no time to consider the effect which your discoveries or invention may have outside the field of medicine.

You succeed (let us say), as modern medicine has succeeded, in enormously lowering the infant death-rate, not only in Europe and America, but also in Asia and Africa.

This has the entirely unintended result of making the food supply inadequate and lowering the standard of life in the most populous parts of the world. To take an even more spectacular example, which is in everybody's mind at the present time — you study the composition of the atom from a disinterested desire for knowledge and incidentally place in the hands of powerful lunatics the means of destroying the human race.

In such ways the pursuit of knowledge may become harmful unless it is combined with wisdom; and wisdom in the sense of comprehensive vision is not necessarily present in specialists in the pursuit of knowledge.

Comprehensiveness alone, however, is not enough to constitute wisdom. There must be, also, certain awareness of ends of human life. This may be illustrated by the study of history.

Many eminent historians have done more harm than good because they viewed facts through the distorting medium of their own passions. Hegel had a philosophy of history which did not suffer from any lack of comprehensiveness, since it started from earliest time and continued into an indefinite future.

But the chief lesson of history which he sought to inculcate was that from the year AD 400 down to his own time, Germany had been the most important nation and the standard bearer of progress in the world. Perhaps one could stretch the comprehensiveness that constitutes wisdom to include not only intellect but also feeling.

It is by no means uncommon to find men/women whose knowledge is wide but whose feelings are narrow. Such men/women lack what I am calling wisdom.

I think the essence of wisdom is emancipation, as far as possible, from the tyranny of the here and the now. We cannot help the egoism of our senses. Sight, sound and touch are bound up with our own bodies and cannot be made impersonal.

Our emotions start similarly from overselves. An infant feels hunger or discomfort; gradually with the years his horizon widens, and, in proportion as his thoughts and feelings become less personal and less concerned with his own physical states, he achieves growing wisdom. This is of course a matter of degree.

No one can view the world with complete impartiality; however, it is possible to make a continual approach towards impartiality, on the one hand, by knowing things somewhat remote in time or space, and, on the other hand, by giving to such things their due weight in our feelings. It is this approach towards impartiality that constitutes growth in wisdom.

Perhaps in this sense the wisdom can be taught. I think that this teaching should have a larger intellectual element than has been customary in what has been thought of as moral instruction. I think that the disastrous result of hatred and narrow mindedness to those who fed them can be pointed out incidentally in the course of giving knowledge.

Knowledge and morals ought not to be too much separated. It is true that the kind of specialised knowledge which is required for various kinds of skills has very little to do with wisdom. But it should be supplemented in education by wider surveys calculated to put it in its place in the totality of human activities.

Even the best technicians should also be good citizens, i.e. citizens of the world and not of any one nation. With every increase of knowledge and skill, wisdom becomes more necessary for every such increase augments our capacity of realising our purposes, and therefore augments our capacity for evil, if our purposes are unwise.

The world needs wisdom as it has never needed it before; and if knowledge continues to increase, the world will need wisdom in the future even more than it does now.

92. According to the author, what results in growth of wisdom?
(a) Widening knowledge and narrowing feelings
(b) Acquiring specialised knowledge which is required for various kinds of skills
(c) Viewing the world with complete impartiality
(d) None of the above

Sol. (c) Viewing the world with complete impartiality results in the growth of wisdom.

93. According to the author, the essence of wisdom is
(a) deliverance from the oppression of here and now
(b) subduing from the oppression of here and now
(c) captivity from the oppression of here and now
(d) All of the above

Sol. (a) As per the author, the essence of wisdom is deliverance from the oppression of here and now.

94. What according to the author is the relationship between knowledge and wisdom?
(a) As human wisdom increases there is increase in knowledge created
(b) As knowledge keeps on increasing there is lesser need of wisdom
(c) As knowledge keeps on increasing there is a higher need for wisdom
(d) As growth in wisdom stops, knowledge creation stagnates

Sol. (c) As per the author, as knowledge keeps on increasing there is a larger need for wisdom.

95. The example used by the author to explain the ways in which the pursuit of knowledge can be harmful, unless combined with wisdom, is
(a) the space mission
(b) medicine that lowers infant mortality across the world
(c) the progress of Germany
(d) None of the above

Sol. (b) The example given by the author to elaborate the point is "medicine that lowers infant mortality rate across the world".

96. What factors according to the author, contribute to wisdom?
(a) A sense of proportion, giving knowledge, study of history, emancipation
(b) A sense of proportion, dignity, knowledge, skill
(c) Comprehensiveness, a sense of proportion, awareness of the end of human life, emancipation from the tyranny of the present
(d) None of the above

Sol. (c) The option (c) correctly lists the factors which contribute to the growth of wisdom.

Directions (Q. Nos. 97-98) *Read each of the components of the given sentences and mark the component with grammatical error.*

97. I. He is capable at
II. twisting any fact
III. without any suspicion
IV. at any time.
(a) Only I (b) Only II (c) Only III (d) Only IV

Sol. (a) "He is capable at" should be replaced by "He is capable of".

98. I. My cousin brother, who lives
II. in Goa, is eager to visit us
III. in Mumbai and aspires to have
IV. a glimpse of the city.
(a) Only I (b) II and III
(c) Only IV (d) None of these

Sol. (d) The sentence is grammatically correct. So, the answer is (d).

Directions (Q. Nos. 99-100) *Choose the correct word which best fits for the sentence to be complete and grammatically correct.*

99. It was no wonder that after the roads were closed with continuous snow fall, hotels started …… off the tourists.
(a) raking (b) taking
(c) beating (d) looting

Sol. (a) The phrase "rake off" means to take part of the money that someone earns, used especially when this seems unfair or too much. So, (a) is the correct answer.

100. When the penalty corner was saved, the players …… in toward the goal keeper to congratulate him.
(a) closed (b) went
(c) crashed (d) pooled

Sol. (a) The phrase "closed in" means "to come nearer to someone". So, answer is (a).

101. The synonym for the word "Inclement" is ..... .
(a) Stormy
(b) Intimate
(c) Advocacy
(d) Immediate

Sol. (a) The word "Inclement" means unpleasant weather especially with cold wind and rain. So, "stormy" is the synonym of "Inclement" from the given options.

102. The antonym for the word "Taciturn" is ..... .
(a) Garrulous
(b) Energetic
(c) Ephemeral
(d) Enigmatic

Sol. (a) The word "Taciturn" means a person who is reserved or uncommunicative in speech. So, the antonym of it, is "garrulous".

103. Complete with the appropriate collocation word : …… activism.
(a) Judicial
(b) Legal
(c) Prosecutional
(d) Lawful

Sol. (b) The answer of the following is Legal.

104. If Propensity : Tendency, then ..... .
(a) Prologue : Epilogue
(b) Master : Slave
(c) Audacity : Impudence
(d) Conduct : Immorality

Sol. (c) The words propensity and tendency are synonymous. Similarly, Audacity and Impudence are also synonymous. So, the answer is (c).

105. If Tepid : Hot, then …… .
(a) Jealousy : Envy (b) Hatred : Antipathy
(c) Unity : Harmony (d) Joy : Ecstasy

Sol. (b) The word "tepid" means "lukewarm". It is paired with "hot". From the given options, Hatred and Antipathy show a similar relation to tepid and hot. As, Antipathy means extreme hatred. Hence, correct answer is (b).

106. From the following words, pick the odd word out.
(a) Lampoon (b) Satire
(c) Ridicule (d) Parable

Sol. (d) "Parable" is the odd one out from the listed words as the other words are synonyms of "criticize".

107. From the following words, pick the odd word out.
(a) Euphemism (b) Maxim
(c) Aphorism (d) Dictum

Sol. (d) "Dictum" is the odd one out from the given words. The other words are synonyms of Maxim.

108. From the following words, pick the odd word out.
(a) Force (b) Intimidation
(c) Shakedown (d) Bleak

Sol. (d) "Bleak" is the odd one out from the given words as the other words are related to "Force" directly or indirectly.

Directions (Q. Nos. 109-110) *These questions have sentences which are incomplete. Pick up one phrase/clause from the options given, that will complete the sentence logically.*

109. To ensure success in a difficult task …… .
(a) one needs to be persistent
(b) persistence is needed
(c) you need a person of persistence
(d) persistence is what one needs

Sol. (a) The phrase "one needs to be persistent" completes the sentence logically.

110. The more we looked at the piece of modern art ……
(a) we liked it less
(b) better we liked it
(c) the less we liked it
(d) we liked it more and more

Sol. (c) "the less we liked it" is the only option that fits the blank properly.

Section-D Quantitative, Data Interpretation and Data Sufficiency

For all questions in this section, each correct answer carries 1 mark.

111. A batsman was having 32 runs per inning as his average after 15th inning. His average increased by 2 runs after 16th inning. Then, what was his score in the 16th inning?

(a) 64 (b) 60 (c) 46 (d) 62

Sol. (a) Total runs after 15th inning $= 15 \times 32 = 480$

Total runs after 16th inning $= 16 \times (32 + 2)$
$= 16 \times 34$
$= 544$

$\therefore$ Score in the 16th inning $= 544 - 480 = 64$

112. The least number which is a perfect square and is divisible by each of the numbers 14, 16, 18 is

(a) 6048 (b) 7056 (c) 1008 (d) 2046

Sol. (b) A number which is divisible by 14, 16, 18 is given by
LCM (14, 16, 18) = 1008
But 1008 is not a perfect square.
Now, $7 \times 1008 = 7056 = (84)^2$
$\therefore$ The least number which is a perfect square and is divisible by each of 14, 16 and 18 is 7056.

113. Four people clap after every 20 min, 30 min, 40 min and 50 min respectively. All of them clapped together at 10 : 00 am. Then, they will again clap together at

(a) 3 pm (b) 5 pm (c) 6 pm (d) 8 pm

Sol. (d) The time after which four people will clap again is given by LCM (20, 30, 40, 50) = 600 min = 10 h
$\therefore$ They will again clap together at 8 pm.

114. Three candidates A, B, C participated in an election. A gets 40% of the votes more than B. C gets 20% votes more than B. A also overtakes C by 4000 votes. If 90% voters voted and no invalid or illegal votes were cast, then what will be the number of voters in the voting list?

(a) 72000 (b) 80000 (c) 70000 (d) 78500

Sol. (b) Let the number of voters be x.

Then, votes casted $= \frac{90}{100} \times x$

According to the question,

Votes for $A = \frac{140}{100} \times$ Votes for B

Votes for $C = \frac{120}{100} \times$ Votes for B

Votes for A = Votes of C + 4000

Also, total votes for $(A + B + C) = \frac{90x}{100}$

$\Rightarrow \left(\frac{140}{100} \times \text{Votes for } B + \text{Votes for } B + \frac{120}{100} \times \text{Votes for } B\right) = \frac{90x}{100}$

$\Rightarrow \frac{360}{100} \times \text{Votes for } B = \frac{90x}{100}$

$\Rightarrow x = 4 \times (\text{Votes of } B)$

Also, $\frac{140}{100} \times \text{Votes for } B = \frac{120}{100} \times \text{Votes for } B + 4000$

$\Rightarrow \frac{20}{100} \times \text{Votes for } B = 4000$

$\Rightarrow$ Votes for $B = 20000$

$\therefore$ Total Votes casted $= x = 4 \times 20000$
$= 80000$

115. In a competitive exam, there were 5 sections. 10% of the total number of students cleared the cut off in all the sections and 5% cleared none of the sections. From the remaining candidates 30% cleared only section 1, 20% cleared only section 2, 10% cleared only section 3 and remaining 1020 candidates cleared only section 4. How many students appeared in the competitive exam?

(a) 2550 (b) 2800 (c) 3000 (d) 3200

Sol. (c) Let total number of students be x.
Students who cleared all the sections = 10%
Students who did not clear any section = 5%
So, remaining students $= 100 - (10 + 5) = 85\%$
Students who cleared section 1 = 30%
Students who cleared section 2 = 20%
Students who cleared section 3 = 10%
So, remaining students $= 100 - (30 + 20 + 10)$
$= (100 - 60)\% = 40\%$
According to the question,
40% of 85% of $x = 1020$

$\Rightarrow x = \frac{1020 \times 100 \times 100}{40 \times 85}$

$\therefore x = 3000$

116. A man sold 3/5th of his articles at a gain of 20% and the remaining at cost price. Find the gain earned in the transaction.

(a) 8 (b) 10 (c) 12 (d) 14

Sol. (c) Let CP of 1 article be ₹ 1.
Then, CP of 5 articles will be ₹ 5
SP of 3 articles $= 3 + \frac{20}{100} \times 3 =$ ₹ 3.6
and SP of remaining 2 articles = ₹ 2
$\therefore$ Profit = ₹ $[(3.6 + 2) - 5]$ = ₹ 0.6
So, profit % $= \frac{0.6}{5} \times 100$
$= 12\%$

117. A trader sells 20 articles at ₹ 54 per article after giving 10% discount and gains 50% profit. If the discount is not given, the profit gained is

(a) 56.76% (b) 66.66% (c) 62.66 % (d) 63.66%

Sol. (b) Selling price of 54 articles $= 54 \times 20 =$ ₹ 1080
Let the marked price be ₹ x.

Then, $x - \frac{10}{100} \times x = 1080$

$\Rightarrow \quad \frac{9x}{10} = 1080$

$\Rightarrow \quad x = 1200$

Now, he had gain 50%.

$\Rightarrow \quad 50 = \frac{1080 - CP}{CP} \times 100$

$\Rightarrow \quad \frac{1}{2} = \frac{1080 - CP}{CP}$

$\Rightarrow \quad CP = 2160 - 2CP$

$\Rightarrow \quad 3CP = 2160$

$\Rightarrow \quad CP = ₹\,720$

If no discount is given, then

$$\text{Profit} = \frac{1200 - 720}{720} \times 100 = \frac{480}{720} \times 100 = 66.66\%$$

118. A bottle contains 50 L of milk. From this bottle 5 L of milk was taken out and replaced by water. This process was repeated further for three times. How much milk is now contained in the bottle?

(a) 32.8 L (b) 34.4 L
(c) 36.8 L (d) 46.5 L

Sol. (a) Percentage of milk taken out $= \frac{5}{50} \times 100 = 10\%$

Milk contained in the bottle is given by

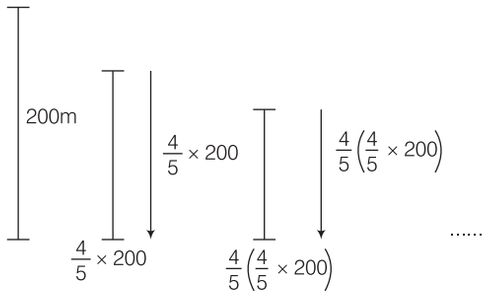
$$= 50\left(1 - \frac{10}{100}\right)^4 = 50\left(1 - \frac{1}{10}\right)^4$$

$$= 50\left(\frac{9}{10}\right)^4 = 32.8 \text{ L}$$

119. A ball is dropped from a height of 200 m. After striking the floor it re-bounces to 4/5th of the height from where it fell. The total distance it travels before coming to rest is

(a) 1200 m (b) 1600 m
(c) 1800 m (d) 1820 m

Sol. (c)



$$\text{Distance} = 200 + \frac{4}{5} \times 400 + \left(\frac{4}{5}\right)^2 \times 400 + \ldots$$

$$= 200 + 400\underbrace{\left[\frac{4}{5} + \left(\frac{4}{5}\right)^2 + \ldots\right]}_{\text{Infinite geometric series}}$$

$$= 200 + \frac{\frac{400 \times 4}{5}}{1 - \frac{4}{5}} = 200 + \frac{\frac{1600}{5}}{\frac{1}{5}} = 1800 \text{ m}$$

120. The sum of all two-digit numbers that give a remainder 2 when they are divided by 7 is

(a) 552 (b) 654 (c) 658 (d) 684

Sol. (b) Two-digit numbers which give remainder 2 when divided by 7 are 16, 23, 30, ... , 93.

$a_n = a(n-1)d$

$\Rightarrow \quad 93 = 16 + (n-1)7$

$\Rightarrow \quad 77 = (n-1)7$

$\Rightarrow \quad 11 = n - 1 \Rightarrow n = 12$

Number of such digits = 12

$\therefore$ Required sum $= \frac{n}{2}[2a + (n-1)d]$

$= 6[2 \times 16 + (12 - 1) \times 7]$

$= 6[32 + 11 \times 7]$

$= 6[32 + 77]$

$= 6 \times 109$

$= 654$

121. A man covers half of his journey by train at 90 km/h, one-third of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. The average speed during the entire journey is

(a) 22.5 km/h (b) 28.5 km/h
(c) 30.0 km/h (d) 32.5 km/h

Sol. (a) Let the total distance covered be x km.

Then, time taken by train $= \frac{\frac{x}{2}}{90} = \frac{x}{180}$ h

Time taken by bus $= \frac{\frac{1}{3}\left(\frac{x}{2}\right)}{30} = \frac{x}{180}$ h

Time taken by cycle $= \frac{\left(\frac{x}{2} - \frac{x}{6}\right)}{10} = \frac{4x}{120} = \frac{x}{30}$ h

$\therefore$ Average speed $= \frac{\text{Total distance}}{\text{Total time}} = \frac{x}{\frac{x}{180} + \frac{x}{180} + \frac{x}{30}}$

$$= \frac{x}{\frac{x}{90} + \frac{x}{30}} = \frac{x}{\frac{x + 3x}{90}}$$

$$= \frac{90}{4} = 22.5 \text{ km/h}$$

122. John's grandfather was five times older to him 5 yr ago. He would be two times of his age after 25 yr from now. What is the ratio of John's age to that of his grandfather?

(a) 7 : 11 (b) 5 : 21
(c) 3 : 11 (d) 4 : 21

Sol. (c) Let John's age be x yr and grandfather's age be y yr.

Then, $5(x - 5) = y - 5$ ...(i)

and $y + 25 = 2(x + 25)$

Solving these, we get

$x = 15$

$y = 55$

Required ratio $= \frac{15}{55} = \frac{3}{11} = 3 : 11$

123. A number when successively divided by 5 and 6 gives remainders 3 and 2 respectively. What will be the remainders, if the number is successively divided by 3 and 4?

(a) 2, 2 (b) 2, 1 (c) 1, 3 (d) 3, 4

Sol. (a) A number when successively divided by 5 and 6 leaving remainders 3 and 2 respectively can be written as

$= 6(5k + 3) + 2$

$= 30k + 20$

When $(30k + 20)$ is divided by 3; $\frac{(30k + 20)}{3} = 10k + 6$, the remainder is 2.

Now, $(10k + 6)$ is divided by 4; $\frac{10k + 6}{4} = 2k + 2$, the remainder is $2k + 2$.

If k is odd, $2k + 2$ is divisible by 4.

If k is even, the remainder when $(2k + 2)$ is divided by 4, is 2.

Hence, the remainders are 2, 2.

124. How many zeroes would be there in 1024!?

(a) 240 (b) 248 (c) 256 (d) 253

Sol. (c) The number of zeroes in $n!$ are given by

$$\left[\frac{n}{5}\right] + \left[\frac{n}{25}\right] + \left[\frac{n}{125}\right] + \ldots + \left[\frac{n}{5^k}\right]$$

Here, $n = 1024$

$\therefore$ Number of zeroes

$$= \left[\frac{1024}{5}\right] + \left[\frac{1024}{25}\right] + \left[\frac{1024}{125}\right] + \left[\frac{1024}{625}\right]$$

$= 205 + 41 + 8 + 2$

$= 256$

125. If $x = 3 + 2\sqrt{2}$, what will be the value of $x^2 + \frac{1}{x^2}$?

(a) 35 (b) 32 (c) 36 (d) 34

Sol. (d) Here, $x = 3 + 2\sqrt{2}$

$$\Rightarrow \frac{1}{x} = \frac{1}{3 + 2\sqrt{2}}$$

Rationalising the denoninator, we get

$$\frac{1}{x} = \frac{3 - 2\sqrt{2}}{(3 + 2\sqrt{2})(3 - 2\sqrt{2})} = 3 - 2\sqrt{2}$$

$$\therefore \quad \left(x + \frac{1}{x}\right) = 6$$

Also, $\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2$

$$\Rightarrow \quad x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$$

$= 6^2 - 2 = 36 - 2 = 34$

126. The unit digit in the final solution, when 13*27*63*51*98*46 is

(a) 4 (b) 8

(c) 2 (d) None of these

Sol. (a) The unit digit is given by the unit digit of

$3 \times 7 \times 3 \times 1 \times 8 \times 6 = 3024$

$\therefore$ Unit digit = 4

127. A dishonest seller sells his grocery items using a false weight and thus gains 5% for a kilogram, he uses the weight of approximately

(a) 940.251 (b) 943.123

(c) 948.238 (d) 952.381

Sol. (d) Let false weight be x.

$$\text{Then, profit\%} = \frac{\text{True weight} - \text{False Weight}}{\text{False weight}} \times 100$$

$$\Rightarrow \quad 5 = \frac{1000 - x}{x} \times 100$$

$$\Rightarrow \quad \frac{5x}{100} + x = 1000$$

$$\Rightarrow \quad \frac{105x}{100} = 1000$$

$$\Rightarrow \quad x = \frac{1000 \times 100}{105}$$

$$\therefore \quad x = ₹\ 952.381$$

128. A, B and C can do a work in 6, 8 and 12 days respectively. If they do the work together and earn ₹ 2700, what is the share of C in that amount?

(a) 600 (b) 900 (c) 1000 (d) 700

Sol. (a) A's one day work $= \frac{1}{6}$

B's one day work $= \frac{1}{8}$

C's one day work $= \frac{1}{12}$

$\therefore$ Work done together by all three in one day

$$= \frac{1}{6} + \frac{1}{8} + \frac{1}{12} = \frac{8 + 6 + 4}{48} = \frac{18}{48} = \frac{9}{24}$$

$$\therefore \text{Share of } C = \frac{\frac{1}{12}}{\frac{9}{24}} \times 2700 = ₹\ 600$$

129. How many words each of two vowels and three consonants can be formed from the letters of the word "UNIVERSAL"?

(a) 700 (b) 720 (c) 740 (d) 780

Sol. (b) Number of words of two vowels and three consonants $= {}^4C_2 \times {}^5C_3 \times 2! \times 3!$

$$= \frac{4 \times 3 \times 2!}{2! \times 2!} \times \frac{5 \times 4 \times 3!}{3! \times 2!} \times 2! \times 3!$$

$= 720$

130. Three pipes A, B and C can fill a tank in 12 h. All the pipes started working together and after 3 h, C is closed. If A and B can fill the remaining part in 10 h, then the number of hours taken by C alone to fill the tank is

(a) 100 h (b) 110 h

(c) 120 h (d) 130 h

Sol. (c) $(A + B + C)$'s 1 h work $= \frac{1}{12}$

$\Rightarrow (A + B + C)$'s 3 h work $= 3 \times \frac{1}{12} = \frac{1}{4}$

$\therefore$ Remaining work $= 1 - \frac{1}{4} = \frac{3}{4}$

$\frac{3}{4}$th work is completed by $(A + B)$ in 10 h.

$\Rightarrow$ 1 work is completed by $(A + B)$ in $\frac{10 \times 4}{3}$ h $= \frac{40}{3}$ h

$\therefore$ Work done by C alone $= \frac{1}{12} - \frac{3}{40} = \frac{10 - 9}{120} = \frac{1}{120}$

$\therefore C$ will take 120 h to complete the work alone.

131. If the numbers between 1 to 65 which will be divisible by 4 are taken and then if the number present in the units places and tens places is swapped, post which they are written in ascending order, then which of the following numbers will be at 10th place from the last?

(a) 40 (b) 24 (c) 44 (d) 25

Sol. (a) Numbers divisible by 4 are

{4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64}

$\Rightarrow$ Numbers formed by interchanging the digit places

{40, 80, 21, 61, 02, 42, 82, 23, 63, 04, 44, 84, 25, 65, 06, 46}

$\Rightarrow$ Arranging the above numbers in ascending order

{02, 04, 06, 21, 23, 25, 40, 42, 44, 46, 61, 63, 65, 80, 82, 84}

$\therefore$ The number at 10th place from last is 40.

132. Ajit, Ravi and Hari were trying to hit a target. If Ajit hits the target 5 times in 8 attempts, Ravi hits it 3 times in 5 attempts and Hari hits it 2 times in 4 attempts. What is the probability that the target is hit by at least 2 persons?

(a) 31/80 (b) 49/80 (c) 09/80 (d) 03/16

Sol. (b) Let $P(A)$ denotes the probability of target being hit by Ajit.

$$\Rightarrow \quad P(A) = \frac{5}{8}$$

Let $P(R)$ denotes the probability of target being hit by Ravi.

$$\Rightarrow \quad P(R) = \frac{3}{5}$$

Let $P(H)$ denotes the probability of target being hit by Hari.

$$\Rightarrow \quad P(H) = \frac{2}{4}$$

$\therefore$ Required probability

$= P(A)\,P(R)\,P(\overline{H}) + P(A)\,P(\overline{R})\,P(H) + P(\overline{A})\,P(R)\,P(H) + P(A)\,P(R)\,P(H)$

$$= \left(\frac{5}{8} \times \frac{3}{5} \times \frac{2}{4}\right) + \left(\frac{5}{8} \times \frac{2}{5} \times \frac{2}{4}\right) + \left(\frac{3}{8} \times \frac{3}{5} \times \frac{2}{4}\right) + \left(\frac{5}{8} \times \frac{3}{5} \times \frac{2}{4}\right)$$

$$= \frac{30 + 20 + 18 + 30}{160} = \frac{98}{160} = \frac{49}{80}$$

133. If $\frac{1}{2}\log x + \frac{1}{2}\log y + \log 2 = \log(x + y)$, then

(a) $x = -y$ (b) $x = y + 1$

(c) $x = y$ (d) $y = x + 1$

Sol. (c) $\frac{1}{2}\log x + \frac{1}{2}\log y + \log 2 = \log(x + y)$

$\Rightarrow \log x^{1/2} + \log y^{1/2} + \log 2 = \log(x + y)$ [$\because a \log b = \log_b a$]

$\Rightarrow \log(x^{1/2} \times y^{1/2} \times 2) = \log(x + y)$ [$\because \log a + \log b = \log ab$]

$\Rightarrow \quad 2\sqrt{xy} = x + y$

$\Rightarrow \quad (\sqrt{x})^2 + (\sqrt{y})^2 - 2\sqrt{x}\sqrt{y} = 0$

$\Rightarrow \quad (\sqrt{x} - \sqrt{y})^2 = 0$

$\Rightarrow \quad \sqrt{x} - \sqrt{y} = 0$

$\Rightarrow \quad \sqrt{x} = \sqrt{y} \Rightarrow x = y$

134. $\log_5 25 + \log_2(\log_3 81)$ is

(a) 1 (b) 2 (c) 3 (d) 4

Sol. (d) $\log_5 25 + \log_2(\log_3 81)$

$= \log_5 5^2 + \log_2(\log_3 3^4)$ [$\because \log_a a = 1$]

$= 2\log_5 5 + \log_2(4\log_3 3)$

$= 2 + \log_2 4$

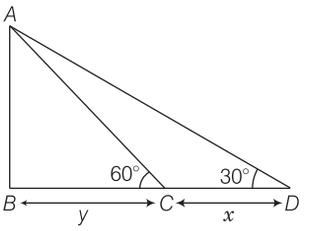
$= 2 + \log_2 2^2$

$= 2 + 2\log_2 2 = 2 + 2 = 4$

135. Peter was standing on the top of a rock cliff facing the sea. He saw a boat coming towards the shore. As he kept seeing time just flew. Ten minutes less than half of an hour, the angle of depression changed from 30 to 60. How much more time in minutes will the boat take to reach the shore?

(a) 5 (b) 10 (c) 15 (d) 20

Sol. (b)



Let $CD = x$ m and $BC = y$ m.

Let speed of boat be u m/min.

Time taken by boat to cover distance $x = 30 - 10 = 20$ min

$\therefore \quad x = u \times 20$

In ΔABD, $\frac{AB}{BD} = \tan 30° = \frac{1}{\sqrt{3}}$

$\Rightarrow \quad BD = \sqrt{3} \cdot AB$ ...(i)

In ΔABC, $\frac{AB}{BC} = \tan 60°$

$\Rightarrow \quad \frac{AB}{BC} = \sqrt{3}$

$\Rightarrow \quad BC = \frac{AB}{\sqrt{3}}$ ...(ii)

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{BD}{BC} = \frac{\sqrt{3} \cdot AB}{\frac{AB}{\sqrt{3}}} = 3$$

$\Rightarrow \quad BD = 3BC$

$\Rightarrow \quad BC + CD = 3BC$

$\Rightarrow \quad CD = 2BC$

$\Rightarrow \quad x = 2y$

$\Rightarrow \quad 20u = 2 \times y$

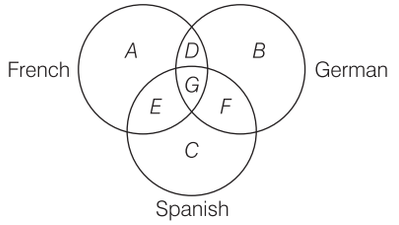
$\Rightarrow \quad y = 10u$ [time $\times$ speed]

$\therefore$ Boat will take 10 min more to reach the shore.

136. In a school where there was a compulsion to learn at least one foreign language from the choice given to them, namely German, French and Spanish. Twenty eight students took French, thirty took German and thirty two took Spanish. Six students learnt French and German, eight students learnt German and Spanish, ten students learnt French and Spanish. Fifty four students learnt only one foreign language while twenty students learnt only German. Find the number of students in the school.

(a) 60 (b) 62
(c) 70 (d) None of these

Sol. (c)



$B = 20$
and $A + B + C = 54$
$A + D + E + G = 28$
$B + D + G + F = 30$
$C + F + G + E = 32$
$D + G = 6$
$F + G = 8$
$E + G = 10$
Solving these equations, we get
$B = 20,\ F = 4,\ G = 4,\ E = 6,\ D = 2,\ A = 16,\ C = 18$
$\therefore$ Number of students $= A + B + C + D + E + F + G$
$= 70$

137. Sonali can solve 70% of the problems in a competitive exam and Nirali can solve only 60% in the same exam. What is the probability that at least one of them will solve a problem, provided selection of questions is done randomly from the same exam?

(a) 0.82 (b) 0.88
(c) 0.62 (d) 0.72

Sol. (b) Let $P(S)$ denotes the probability that the problem is solved by Sonali
and $P(N)$ denotes the probability that the problem is solved by Nirali.
Then, $P(S) = \frac{7}{10}$ and $P(N) = \frac{6}{10}$
Required probability

$$= P(S) \times P(\overline{N}) + P(N) \times P(\overline{S}) + P(S) \times P(N)$$
$$= \left(\frac{7}{10} \times \frac{4}{10}\right) + \left(\frac{6}{10} \times \frac{3}{10}\right) + \left(\frac{7}{10} \times \frac{6}{10}\right)$$
$$= \frac{28 + 18 + 42}{100} = \frac{88}{100}$$
$$= 0.88$$

138. ₹ XYZ was deposited at simple interest at a specific rate for 3 yr. Had it been deposited at 2% higher rate, it would have fetched ₹ 360 more. Find ₹ XYZ.

(a) ₹ 5500 (b) ₹ 5000
(c) ₹ 6000 (d) ₹ 4500

Sol. (c) Let rate of interest be r% and ₹ XYZ be x.
Then, according to the question,

$$\frac{x \times r \times 3}{100} + 360 = \frac{x \times (r + 2) \times 3}{100}$$
$$\Rightarrow \quad \frac{3rx + 36000}{100} = \frac{3xr + 6x}{100}$$
$$\Rightarrow \quad 6x = 36000$$
$$\Rightarrow \quad x = ₹\ 6000$$

139. A man invests certain amount at 6% per annum simple interest and another amount at 7% per annum simple interest. His income from the interest after 2 yr was ₹ 348. The ratio of first amount to second is 4 : 5. Find the total amount invested.

(a) ₹ 2600
(b) ₹ 2900
(c) ₹ 2700
(d) None of the above

Sol. (d) Let the amount invested at 6% be ₹ x and the amount invested at 7% be ₹ y.

$$\text{Then, } \frac{x \times 6 \times 2}{100} + \frac{y \times 7 \times 2}{100} = 348 \quad \text{...(i)}$$
$$\text{and} \quad \frac{x}{y} = \frac{4}{5} \quad \text{...(ii)}$$

Solving Eqs. (i) and (ii), we get

$$\frac{x \times 12}{100} + \frac{5x}{4} \times \frac{14}{100} = 348$$
$$\Rightarrow \quad \frac{12x}{100} + \frac{35x}{200} = 348$$
$$\Rightarrow \quad \frac{24x + 35x}{200} = 348$$
$$\Rightarrow \quad 59x = 348 \times 200$$
$$x = ₹\ 1179.66$$
$$\therefore \quad y = \frac{5x}{4} = ₹\ 1474.5$$
$\therefore$ Total amount invested $= x + y = 2654.2$
$\approx$ ₹ 2654

140. A person has a bag which contains 9 bulbs out of which 2 are fused and cannot be used to lighten the room. Two bulbs are selected at random. What is the probability that all the two bulbs chosen can be used to lighten the room?

(a) 5/12 (b) 7/12
(c) 9/12 (d) 10/12

Sol. (b) Number of ways of chosing two bulbs from 9 bulbs $= {}^9C_2$
Number of ways of chosing two working bulbs $= {}^7C_2$
$\therefore$ Required probability $= \frac{{}^7C_2}{{}^9C_2} = \frac{7 \times 6}{9 \times 8} = \frac{7}{12}$

141. The value of $(p-a)\times(p-b)\times(p-c)\times\ldots\times(p-z)$ is

(a) a complex polynomial which starts with p^{24}
(b) zero
(c) a complex polynomial which starts with p^{26}
(d) a complex polynomial which has several variables including p^{26} and p^{24}

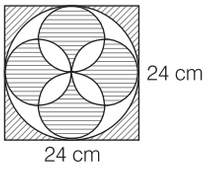
Sol. (c) $(p-a)(p-b)....(p-z) = p^{26} + a_0\ p^{25} + \ldots + a_n$

142. There are nine humans in a ship, each human has nine cages and each cage has nine huge lions and each lion has nine cubs. How many legs are there in the ship? (human have two legs, lions have four legs, cubs have four legs)

(a) 747 (b) 3258 (c) 29178 (d) 26561

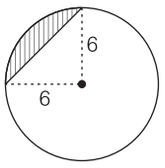
Sol. (c) Total number of legs
$= (9\times2) + (9\times9\times9\times4) + (9\times9\times9\times9\times4)$
$= 18 + 2916 + 26244$
$= 29178$

143. As shown in the figure, there is a square of 24 cm. A circle is inscribed inside the square. Inside the circle, there are four circles of equal radius which are inscribed. The total area of the shaded region in the figure given below is



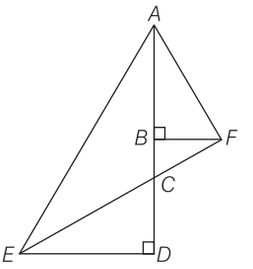
(a) $576 - 196\ \pi$
(b) $584 - 196\ \pi$
(c) $864 - 196\ \pi$
(d) None of the above

Sol. (d)



Area of shaded region $= \frac{\pi(6)^2}{4} - \frac{1}{2}\times6\times6$
$= 9\pi - 18$
In the given figure we have 8 such petals.
$\therefore$ Area of petals $= 8(9\pi - 18)$
$= 72\pi - 144$
Area of 4 small circles $= 4\times(\pi(6)^2) = 144\pi$
$\therefore$ Area of shaded region in four circles
$= 144\ \pi - (72\ \pi - 144)$
$= 72\pi + 144$
Again, Area of square – Area of bigger circle
$= (24)^2 - \pi(12)^2$
$= 576 - 144\pi$
Required area of shaded region
$= (576 - 144\pi) + (72\pi + 144) = 720 - 72\pi$

144. In the below figure, $CD = BF = 10$ units and $\angle CED = \angle BAF = 30°$. What would be the area of triangle AED?



(a) $100\times(\sqrt{2}+3)$ (b) $\frac{100}{(\sqrt{3}+4)}$
(c) $\frac{50}{(\sqrt{3}+4)}$ (d) $50\times(\sqrt{3}+4)$

Sol. (d) In right triangle CDE, $\angle CED = 30°$
Therefore, $\tan\angle CED = \tan 30° = \frac{1}{\sqrt{3}} = \frac{CD}{ED}$
Therefore, $ED = \sqrt{3}\times CD = 10\sqrt{3}$
In right triangle ABF,
$\angle BAF = 30°$
Therefore, $\tan\angle BAF = \tan 30° = \frac{1}{\sqrt{3}} = \frac{BF}{AB}$
Hence, $AB = \sqrt{3}\times BF = 10\sqrt{3}$
Triangles EDC and FBC are right triangles.
$\angle EDC = \angle FBC$. Both are right angles.
$\angle DCE = \angle BCF$. Vertically opposite angles are equal.
Because two angles of the two triangles EDC and FBC are equal, these two triangles are similar triangles.
So, $\angle BFC = \angle CED = 30°$
$\tan\angle BFC = \tan 30° = \frac{1}{\sqrt{3}} = \frac{BC}{BF}$
We know that, $BF = 10$
So, $BC = \frac{10}{\sqrt{3}}$
Now, $AD = AB + BC + CD = 10\sqrt{3} + \frac{10}{\sqrt{3}} + 10$
So, area of triangle $AED = \frac{1}{2}\ ED\times AD$
$= \frac{1}{2}\times10\sqrt{3}\times\left(10 + \frac{10}{\sqrt{3}} + 10\sqrt{10}\right)$
$= 50\times(\sqrt{3}+4)$

145. Mr Suresh went into a restaurant and had a happy lunch for ₹ 162. He paid for the same using a 500 rupee note. He was so happy with the meal and hence purchased a mini snack box for ₹ 37 and paid for the same using a 100 rupee note. Mr Suresh ensured that he collected the balance from the cashier in both cases. Although he was satisfied with the service, he did not pay any tips to the server. The next day when the cashier went and deposited the money in the bank, it was found by the banker that only the two currency notes which

were given by Mr Suresh were counterfeit notes. As per policy, the bank immediately tore both the notes using their shredding machine. What is the total loss to the restaurant in this transaction?

(a) 1200 (b) 600
(c) 799 (d) None of these

Sol. (c) Total loss to restaurant = ₹ (162 + 37 + 500 + 100)
= ₹ 799

146. If $2^x + 2^{x+1} = 48$, then the value of x^x is

(a) 04 (b) 64 (c) 256 (d) 16

Sol. (c) $2^x + 2^{x+1} = 48$

$\Rightarrow \quad 2^x (1 + 2) = 48$

$\Rightarrow \quad 2^x (3) = 48$

$\Rightarrow \quad 2^x = 16$

$\Rightarrow \quad 2^x = 2^4$

$\Rightarrow \quad x = 4$

$\therefore \quad x^x = 4^4 = 256$

Directions (Q. Nos. 147-150) *Following table shows the percentage population of six states below poverty line and proportion of male and female.*

State	% Population below poverty line	Proportion of male and female	
		Below poverty line	Above poverty line
		M : F	M : F
A	16	2 : 3	3 : 4
B	10	4 : 3	5 : 2
C	14	3 : 4	2 : 3
D	20	5 : 2	4 : 3
E	25	4 : 1	2 : 1
F	20	2 : 3	4 : 1

147. If the total population of state *A* is 5000, then what is the number of females above poverty line in state *A*?

(a) 2000 (b) 2400
(c) 2600 (d) Data inadequate

Sol. (b) Population of state *A* = 5000

$\therefore$ Population above poverty line $= \frac{84}{100} \times 5000$
$= 4200$

$\therefore$ Number of females above poverty line $= \frac{4}{7} \times 4200$
$= 4 \times 600$
$= 2400$

148. If the population of states *C* and *D* together is 20000, what is the total number of females below poverty line in the above states?

(a) 5000 (b) 6000 (c) 7200 (d) NOTA

Sol. (d) Population of states *C* and *D* together = 20000

$\therefore$ Population below poverty line of states *C* and *D* together
$= (14 + 20)\% \, 20000$
$= \frac{34}{100} \times 20000$
$= 6800$

Number of females below poverty line in state $C = \frac{4}{7}$

Number of females below poverty line in state $D = \frac{2}{7}$

$\therefore$ Total number of females below poverty line in states *C* and *D*
$= \left(\frac{4}{7} + \frac{2}{7}\right) \times 6800$
$= \frac{6}{7} \times 6800$

149. If the population of males below poverty line in state *C* is 6000 and state *E* is 10000, then what is the ratio of the total populaton of states *C* and *E* is

(a) 2 : 1 (b) 3 : 5
(c) 1 : 5 (d) NOTA

Sol. (a) Population of males below poverty line in state *C* = 6000

Population of males below poverty line in state *E* = 10000

$\therefore$ Number of females below poverty line in state
$C = \frac{4}{3} \times 6000 = 8000$

$\therefore$ Total population of state *C* below poverty line
= 8000 + 6000 = 14000

Let the total population of state *C* be x.

Then, 14% of x = 14000

$\Rightarrow \quad x = 100000$

$\therefore$ Number of females below poverty line in state
$E = \frac{1}{4} \times 10000 = 2500$

$\therefore$ Total population of *E* below poverty line = 10000 + 2500
= 12500

Let the total population of *E* be y.

Then, $\frac{25}{100} \times y = 12500$

$\Rightarrow \quad y = 50000$

$\therefore$ Required ratio $= \frac{x}{y} = \frac{100000}{50000}$
$= \frac{2}{1} = 2 : 1$

150. If in state *F* population of females below poverty line is 16000, then what is the population of males below poverty line in that state?

(a) 8000 (b) 6000
(c) 12000 (d) NOTA

Sol. (d) Population of females below poverty line = 16000

$\therefore$ Population of males below poverty line in state *F*
$= \frac{2}{3} \times 16000$
$= \frac{32000}{3}$

Solved Paper 2014

SNAP

Symbiosis National Aptitude Test

Time : 120 min **Marks** : 180

Instructions

1. There are total 150 questions in this solved paper.
2. There are four parts. Part *A* (General English), Part *B* (Quantitative, Data Interpretation and Data Sufficiency), Part *C* (General Awareness) and Part *D* (Analytical & Logical Reasoning)
3. Part *A* carries 40 marks, Part *B* carries 40 marks, Part *C* carries 40 marks and Part *D* carries 60 marks.
4. For each incorrect attempt 1/4th mark will be deducted.

Part A General English

1. Which of these is not a compound word?
   (a) Hard-wired (b) Run-of-the-mill
   (c) Bottleneck (d) Minestrone

Sol. (d) 'Minestrone' is not a compound word. It means the soup made out of the green vegetables.

2. Find the antonym of 'Native'.
   (a) Tarnished (b) Characteristic
   (c) Intrinsic (d) Exotic

Sol. (d) 'Native' means 'belonging to a place from the birth'. 'Exotic' means 'of foreign origin'. So, it is correct antonym.

3. A shortfall in supplies would knock the bottom out of the engineering industry. What does ' knock the bottom out of ' stand for?
   (a) Make it baseless
   (b) Shake its foundation
   (c) Cause it to collapse
   (d) None of the above

Sol. (c) Idiom 'knock the bottom out of something' means 'to cause a severe collapse or damage'.

4. Choose the word with correct spelling.
   (a) Demurrage (b) Demurage
   (b) Demmurrage (d) Demurragge

Sol. (a) 'Demurrage' is the correct spelling that means 'paying the late fine'.

Directions (Q. Nos. 5-9) *Fill in the blanks with proper answers.*

5. ..... the stable door after the horse has bolted.
   (a) Lock (b) Shut
   (c) Close (d) Open

Sol. (a) 'Lock' is appropriate filler. It gives the idea to close the door with a device which is operated by a key.

6. There were ..... viewers than expected.
   (a) fewer
   (b) less
   (c) many
   (d) lesser

Sol. (a) Here, comparative degree will be used. 'Less', 'lesser' are comparative, but used when amount is referred which is uncountable. 'Many' is positive degree. So, 'fewer' is appropriate.

7. ..... when, after waiting for hours, we were told the tickets were sold out.
   (a) We were at the end of our tether
   (b) We were on edge
   (c) Tempers frayed
   (d) Nerves frayed

Sol. (c) The sentence gives the idea that the people became angry when they waited for hours and didn't get the tickets. 'Tempers frayed' is used as an idiom that means the same.

8. Your proficiency in the French language depends on how hard you work. ..... if you live in the culture of your target language, you will learn even more quickly.
(a) Likewise (b) Hence
(c) Furthermore (d) In fact

Sol. (d) 'Hence' and 'In fact' are closely related, but 'in fact' is used in the sense of showing something of a practical experience.

9. The government is planning to sell PSUs ..... BHEL, SAIL and NTPC ..... first wants a regulatory agency ..... the IRDA to assess their worth.
(a) like/and/like (b) like/but/such as
(c) such as/so/like (d) such as/but/like

Sol. (b)

10. Choose the sentence which is grammatically correct.
(a) I was struck in traffic jam
(b) I struck in traffic jam
(c) I was struck up in traffic jam
(d) I was struck upon traffic jam

Sol. (a) 'I was struck in traffic jam' is the correct sentence grammatically.

11. Choose the option having correct word order in the following statements.
(a) Immigrant communities in the US have brought renewed pressure after the catastrophic terrorist attacks on the World Trade Centre in New York in 2001
(b) The aftermath of the catastrophic terrorist attacks on the World Trade Centre in New York in 2001 has brought renewed pressure on immigrant communities in the US
(c) The World Trade Centre, after the catastrophic terrorist attacks in New York in 2001, has brought renewed pressures on immigrant communities in the US
(d) In 2001, the aftermath of the catastrophic terrorist attacks on the World Trade Centre in New York has brought renewed pressures on immigrant communities in the US

Sol. (b) The sentence is depicting the consequence of the terrorist attack on World Trade Centre in New York and its aftermath on the immigrant communities. So, option (b) is the correct word order statement.

12. The football team travelled ..... train.
(a) in (b) by (c) from (d) on

Sol. (b) Preposition 'by' is appropriate usage.

13. Choose the correct synonym for 'Attenuate'.
(a) Enervate (b) Attention
(c) Awaken (d) Trigger

Sol. (a) 'Attenuate' and 'Enervate' both have similar meaning i.e. 'to weaken something'.

14. Choose the antonym for 'Cajole'.
(a) Dissuade (b) Beguile
(c) Flatter (d) Blandish

Sol. (a) 'Cajole' means 'to flatter' while 'Dissuade' means 'to counsel'.

15. Choose the correct synonym for 'Consanguinity'.
(a) Affinity (b) Corpulent
(c) Estrangement (d) Anarchy

Sol. (a) 'Consanguinity' and 'Affinity' both have similar meaning i.e. 'family relationship' and hence, they are synonyms.

Directions (Q. Nos. 16-22) *Fill in the blanks with proper answers.*

16. Autotomy is the ability of an animal to ..... a part of its body at will to save itself from predators and attackers.
(a) cast off
(b) cast away
(c) cut off
(d) cut down

Sol. (a) 'Autotomy' is the process of 'casting off' i.e. to separate a part, of one's body to escape from predation or attacks.

17. They all started ..... as the door of the dilapidated house swung open.
(a) with bated breath
(b) in beeline
(c) in the same boat
(d) a bone of contention

Sol. (a) 'Staring with bated breath' means 'to look something in anticipation or with suspense'. As the house has been in a dilapidated situation so such type of anticipation is obvious.

18. But at a time when business values are falling, stakeholders tend not to take kindly to ..... at a knock-down price.
(a) selling the family silver
(b) selling one's soul
(c) selling yourself short
(d) selling someone a pup

Sol. (b) Idiom 'to sell one's soul' means 'to do some dishonest practices for the sake of money'.

19. Although this political party has reputation for ……, its behaviour towards the public has always been ……, suggesting it may not be as insolent as has been projected by its competitors.
(a) vendibility; impertinent
(b) secrecy; uniform
(c) temerity; courteous
(d) nepotism; varied

Sol. (d) 'Nepotism' is the favouritism shown on the basis of family relationship. It is often seen in case of political parties.

20. The company believes in getting value for money and it is wrong of the employees to view its …… as …… .
(a) fore-handedness; stinginess
(b) reserve; levity
(c) insolence; shortage
(d) indolence; destitution

Sol. (b) 'Reserve' stands for the resources that a person/firm has. 'Levity' is reflection of silliness. Often we know that a firm keeps the reserve with it what others take it to be its silliness.

21. In the midst of radiantly beautiful women, this homely woman looked strikingly …… but it transpired later that she was an arbitrator and not …… .
(a) unattractive; an adjudicator
(b) incongruous; a contestant
(c) dejected; a raconteuse
(d) personable; an audience

Sol. (b) An arbitrator cannot be justified as being dejected or personable. As the sentence suggests she may be looking absurd. Hence, incongruous is an appropriate usage. Moreover, as she was a judge so she must not be a contestant.

22. This author is not outspoken on political issues and her writings are more famous for their aesthetic qualities than for their …… ones.
(a) polemical (b) commercial
(c) contemporary (d) dramatic

Sol. (a) A political issues are often gone controversial, so use of 'polemical' is appropriate.

23. Choose the word with correct spelling.
(a) Perniceous (b) Pernicious
(c) Perniciuos (d) Pernecious

Sol. (b) 'Pernicious' is the correct spelling that means 'bad or hurtful'.

Directions (Q. Nos. 24-26) *Read the given passage and answer the questions.*

Read not to contradict and confute, nor to believe and take for granted, nor to find talk and discourse, but to weigh and consider. Some books are to be tasted, others to be swallowed, and some few to be chewed and digested; that is, some books are to be read only in parts; others to be read, but not curiously; and some few to be read wholly, and with diligence and attention. Some books may also be read by deputy, and extracts made of them by others; but that would be only in the less important arguments and the meaner sort of books; else distilled books are like common distilled waters, flashy things. Reading makes a full man, conference a ready man and writing an exact man. And therefore, if a man write little, he had need have a good memory; if he confer little, he had need have a present wit; and if he read little, he had need have much cunning to seem to know that he doth not. Histories make men wise, poets witty, the mathematics subtle, natural philosophy deep, moral grave, logic and rhetoric able to contend. (Source-*Of Studies*-Francis Bacon)

24. What should be the real object of reading?
(a) To distill the contents
(b) To understand author's views and their application in day-to-day life
(c) To have a good memory
(d) To weigh and consider

Sol. (d) Real object of reading according to the passage should be to weigh and consider.

25. The word 'confute' in the passage means
(a) to argue
(b) to refuse
(c) to disapprove
(d) to doubt

Sol. (c) Here, 'confute' gives us the sense of 'disapproval'.

26. What can be done with regard to the 'meaner' sort of books?
(a) To be read but not to contradict and confute
(b) To be read curiously
(c) To be read but not curiously
(d) They can be 'read by deputy and extracts made of them by others'

Sol. (d) 'Meaner' sorts of books can be read by deputy and extracts made of them by others.

Directions (Q. Nos. 27-32) *Read the given passage and answer the questions.*

Many firms fail because when they begin exporting, they have not researched the target markets or have not developed an international marketing plan to be successful. A firm must clearly define goals objectives and potential problems. Secondly, it must develop a definitive plan to accomplish its objective regardless of problems involved. Unless the firm is fortunate enough to possess a staff with considerable expertise, it may not be able to take this crucial first step without qualified outside guidance.

Often top management is not committed enough to overcome the initial difficulties and financial requirements of exporting. It can often take more time and effort to establish a firm in a foreign market than in domestic one. Although, the early delays and costs involved in exporting may seem difficult to justify when compared to established domestic trade the exporter should take a more objective view of this process and carefully monitor international marketing efforts through these early difficulties. If a good foundation is laid for export business, the benefits derived should eventually out-weigh the investment.

Another problem area is in the selection of foreign distributor. The complications involved in overseas communications and transportation require international distributors to act with greater independence than their domestic counterparts also since a new exporter's trademarks and reputation is usually unknown in the foreign market. Foreign customers may buy on the strength of distributing agent's reputation. A firm should therefore conduct a thorough evaluation of the distributor's facilities, the personnel handling its account and managements methods employed.

Another common difficulty for new exporter is the neglect of the export market once the domestic one booms : too many companies only concentrate on exporting when there is recession. Others may refuse to modify products to meet the regulations or cultural preferences of other countries. Local safety regulations cannot be ignored by exporters. If necessary modifications are not made at the factory the distributor must make them, this is usually at a greater cost and probably not as satisfactory is it should be. It should also be noted that the resulting smaller profit margin makes the account less attractive.

If exporters expect distributing agents to actively promote their accounts, they must be trained and their performance must be continually monitored. This requires a company's marketing executive to be located permanently in the distributor's geographical areas until there is sufficient business to support the representative. The distributor should also be treated on an equal basis with domestic counterparts, *for example*, special discount offers, sales incentive programmes and special credit terms should be available.

Considering a joint venture or licensing agreement is another option for new exporters. However, many companies still dismiss international marketing as unviable. There are a number of reasons for this. There may be import restrictions in the target market. The company may lack sufficient financial resources or its product line may be too limited. Yet many products that can compete on a national basis can be successful in the majority of world markets. In general, all that is needed for success is flexibility in using the proper combinations of marketing techniques.

27. In the first paragraph, the writer suggest that firms thinking about exporting should
(a) get professional advice
(b) study international marketing
(c) identify the most profitable markets
(d) have different objectives to other exporters

Sol. (b) The firms that are thinking about the exporting business must study and research the international marketing.

28. The writer believes that if sufficient preparation is undertaken
(a) initial difficulties can be easily avoided
(b) the costs can be recovered quite quickly
(c) management will become more committed
(d) the exporter will be successful in the long term

Sol. (d) The writer believes that if sufficient preparation is undertaken then the exporter will be successful in the long term.

29. An exporter should choose a distributor who
(a) has experienced personnel
(b) has good communication skills
(c) is well established in the target market
(d) is not financially dependent on import business

Sol. (c) If the distributor of the exporter is well established in the target market, then exporter would be benefitted.

30. New exporters often make the mistake of ignoring the export market, when
(a) distribution costs are too high
(b) their product is selling well at home
(c) there is a global economic recession
(d) distributors cannot make safety modifications

Sol. (b) Exporters often ignore the exporting business, when there is a boom in domestic market.

31. For a distributor to be successful, the exporter must
    (a) focus on one particular region
    (b) finance local advertising campaigns
    (c) give the same support as to domestic agents
    (d) make sure there are sufficient marketing staff locally

Sol. (c) An exporter must give the same support as to domestic agents, so that his distributor becomes successful.

32. The writer states that some companies are reluctant to export because
    (a) there is little demand for their products
    (b) the importation of certain goods is controlled
    (c) they do not have good marketing techniques
    (d) they are not able to compete with local business

Sol. (c) Some companies are reluctant to export because they do not have good marketing techniques.

33. The kids were all coming to the pool. The pool attracted kids like
    (a) rain to the desert sands
    (b) elephants to water
    (c) a balloons to helium
    (d) flies to honey

Sol. (d) The best attraction is shown by the use of 'flies' to honey' as it is more natural to take place.

34. If thieves ..... our house, I ..... frightened to death.
    (a) will enter, would be
    (b) enter, would be
    (c) would enter, would be
    (d) enter, would have been

Sol. (b)

35. The enemy turned ..... and ran away as soon as it saw the brave warrior.
    (a) back
    (b) tail
    (c) face
    (d) feet

Sol. (b) Idiom 'turn a tail' means getting frightened and running away from the spot.

36. Vodafone ..... and initiated a fresh style in television advertising.
    (a) broke new ground (b) cut the ground
    (c) cut corners (d) prepared the ground

Sol. (a) Idiom 'breaking new ground' means 'innovating act for the very first time ever'.

37. He took to ..... his impatience indiscriminately, on whoever happened to be at hand.
    (a) giving free rein to (b) airing
    (c) venting (d) All of these

Sol. (b) 'Airing' means 'making something viral or public'. The sentence suggests that the person has made his impatience got aired.

38. He provides a ..... sketch of the political values, strategy and practices of these diverse civilisations.
    (a) thumbnail (b) thumbling
    (c) thumbprint (d) None of the above

Sol. (a) 'Thumbnail' means 'giving an idea that is concise and brief'. Subject of the sentence has given all the briefs about different attributes, hence it is the correct usage.

39. 'This desk is more square than that one' is an erroneous statement, the correct statement would be
    (a) This desk is squarer than that one
    (b) This desk is more nearly square than that one
    (c) This desk is most square than that one
    (d) This desk is much more square than that one

Sol. (a) 'More square' can well be replaced by 'squarer'.

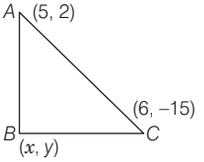
40. 'Half a loaf is better than no bread' means
    (a) God helps those who help themselves
    (b) When a bad thing happens, there is always a positive side to it
    (c) You should be happy with what you have even, if it is less than what you want
    (d) Don't be upset by trivialities

Sol. (c) This phrase has the meaning same as to 'something is better than nothing'. So, it also implies equally to 'You should be happy with what you have even, if it is less than what you want'.

Part B Quantitative, Data Interpretation and Data Sufficiency

41. The line joining the points (5, 2) and (6, –15) subtends a right angle at the point
(a) (– 1, 2) (b) (0, 7)
(c) (0, 0) (d) Insufficient data

Sol. (d) Let the point be x and y,
Since, $AB \perp BC$



$\therefore$ Slope of $AB \times$ Slope of $BC = -1$

$$\Rightarrow \quad \left(\frac{y+15}{x-6}\right)\times\left(\frac{y-2}{x-5}\right) = -1$$

Since, the data is not sufficient.
$\therefore$ We cannot find the value of points x and y.

42. Find the least positive integer, which when divided by 2 gives remainder 1, when divided by 3 gives remainder 2, when divided by 4 gives remainder 3 and when divided by 5 gives remainder 4, what would the integer be?
(a) 33 (b) 59 (c) 61 (d) 49

Sol. (b)

43. How many numbers exist between 100 and 1000, which are divisible by 7?
(a) 142 (b) 138 (c) 60 (d) 128

Sol. (d) The numbers, which are divisible by 7 between 100 to 1000 are 105, 112, 119, …, 994.
Here, we apply the formula, to find the total number,

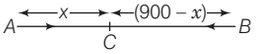
$$l = a + (n-1)d$$

$$\Rightarrow \quad 994 = 105 + (n-1)7 \quad \Rightarrow \quad 7n = 896$$

$$\therefore \quad n = 128$$

44. Two athletes A and B are running on a 900 m long track from opposite ends, their speeds are in the ratio of 4 : 5, when will they cross each other
(a) 500 m from A's end
(b) 400 m from B's end
(c) 400 m from A's end
(d) 100 m from A's end

Sol. (c) Let the speeds of A and B be $4x$ and $5x$, respectively.
Length of the tracks $AB = 900$ m
Suppose, they cross each other at point C time $= t$



Then, $\frac{4x}{5x} = \frac{x}{(900-x)} \quad \Rightarrow \quad \frac{4}{5} = \frac{x}{(900-x)}$

$\Rightarrow 3600 - 4x = 5x$

$$\Rightarrow \quad 9x = 3600 \quad \Rightarrow \quad x = 400$$

So, they cross each other 400 m from A's end.

45. Find the domain of the expression $Y = \sin x + \log x$.
(a) $(-\infty, \infty)$
(b) $(-\infty, 0)$
(c) $(0, \infty)$
(d) $(0, 1)$

Sol. (c) $\because$ Domain of $\sin x = (-\infty, \infty)$
and domain of $\log x = (0, \infty)$
$\therefore$ Domain of $(\sin x + \log x) = (-\infty, \infty) \cap (0, \infty) = (0, \infty)$

46. The expression $x^4 + 5x^2 + 36$ can be factorised as
(a) $(x^2 + x + 1)(x^2 - x + 36)$
(b) $(x^2 - x + 6)(x^2 + 4x + 6)$
(c) $(x^2 - x + 6)(x^2 + x + 6)$
(d) None of the above

Sol. (d) Here, $x^4 + 5x^2 + 36$ will remain $x^4 + 5x^2 + 36$.
So, it cannot be factorised.

47. There are 4 members in a committee. If one of them is replaced by a younger member such that the average age of the committee is as it was 6 yr ago. The new member is how many years younger than the member replaced?
(a) 42 yr
(b) 24 yr
(c) 6 yr
(d) None of the above

Sol. (b) Let the age of four members be x_1, x_2, x_3 and x_4.
Average age of four members

$$= \frac{x_1 + x_2 + x_3 + x_4}{4}$$

Let the age of new member be x_5.
Now, according to the question

$$\frac{x_1 + x_2 + x_3 + x_4 - x_4 + x_5}{4} = \frac{x_1 - 6 + x_2 - 6 + x_3 - 6 + x_4 - 6}{4} = \frac{x_1 + x_2 + x_3 + x_4 - 24}{4}$$

$$\Rightarrow \quad x_1 + x_2 + x_3 + x_4 - x_4 + x_5 = x_1 + x_2 + x_3 + x_4 - 24$$

$$\Rightarrow \quad x_5 = x_4 - 24$$

i.e. new member is 24 yr younger than the member replaced.

48. The probability that a builder will get a construction contract is 3/4 and the probability that he will not get the electrical contract is 7/9. If the probability of getting atleast one contract is 5/6, what is the probability that he will get both?
(a) $\frac{5}{36}$ (b) $\frac{5}{6}$
(c) $\frac{5}{9}$ (d) None of these

Sol. (a) Let A = Builder will get a construction contract
and B = Builder will get an electrical contract

$\therefore \quad P(A) = \frac{3}{4}$ and $P(\overline{B}) = \frac{7}{9}$

$\Rightarrow \quad P(B) = \frac{2}{9}$

Also, $P(A \cup B) = \frac{5}{6}$

$\because \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow \quad \frac{5}{6} = \frac{3}{4} + \frac{2}{9} - P(A \cap B)$

$\therefore \quad P(A \cap B) = \frac{3}{4} + \frac{2}{9} - \frac{5}{6}$

$= \frac{27 + 8 - 30}{36} = \frac{5}{36}$

49. In how many ways can 21 books on literature and 19 books on nature be placed in a row on a shelf, so that two books on nature may not be together?

(a) 4620 (b) 3672
(c) 1540 (d) None of these

Sol. (c) In order that, two books on nature and never together we must place all these books as under

$X\ L\ X\ L\ X\ \ldots\ldots\ X\ L\ X,$

where L denotes the position of literature and X that of a nature book.

Since, there are 21 books on literature, so the number of places marked X are 22.

Now, 19 places out of 22 can be chosen in ${}^{22}C_{19}$

$= {}^{22}C_3 = \frac{22 \times 21 \times 20}{3 \times 2 \times 1}$

$\therefore$ Required number of ways $= {}^{22}C_3 = 1540$

50. A sphere of radius 4 cm is dropped into a cylindrical vessel partly filled with water. The radius of the vessel is 8 cm. If the sphere is submerged completely, by how much will the surface of the water be raised?

(a) $\frac{3}{4}$ units (b) $\frac{4}{3}$ units
(c) $\frac{1}{4}$ units (d) None of these

Sol. (b) Here, radius of the sphere = 4 cm
and radius of the vessel = 8 cm

$\therefore$ Volume of the sphere $= \frac{4}{3}\pi r^3 = \frac{4}{3} \times \frac{22}{7} \times 4 \times 4 \times 4$

Increase in the volume of water = Volume of sphere

$\Rightarrow \quad \frac{22}{7} \times 8 \times 8 \times h = \frac{4}{3} \times \frac{22}{7} \times 4 \times 4 \times 4$

$\therefore \quad h = \frac{4 \times 4 \times 4 \times 4}{8 \times 8 \times 3}$

$= \frac{4}{3}$

So, the surface of water will be raised by $\frac{4}{3}$ units.

51. A solid metal cylinder of radius 5 cm and height 12 cm is melted to make solid balls of radius 1 mm. How many such balls are produced (in lakh)?

(a) 2.5 (b) 2.25
(c) 1.5 (d) None of these

Sol. (b) A solid metal cylinder of radius (r_1) = 5 cm and height (h) = 12cm.

It is melted to make a solid balls of radius r_2 = 1 mm = 0.1 cm.

Then, number of balls produced

$= \frac{\text{Volume of cylinder}}{\text{Volume of sphere (ball)}} = \frac{\pi r_1^2 h}{\frac{4}{3}\pi r_2^3}$

$= \frac{5 \times 5 \times 12 \times 3}{4 \times 0.1 \times 0.1 \times 0.1}$

$= 225000 = 2.25$ lakh

52. The value of $\log_{(-1/4)} 256$ is equal to

(a) 64 (b) 16 (c) -4 (d) 128

Sol. (c) Let $\log_{(-1/4)} 256 = x$

$\Rightarrow \quad \left(-\frac{1}{4}\right)^x = 256$

$\Rightarrow \quad \left(-\frac{1}{4}\right)^x = (4)^4$

$\Rightarrow \quad \left(-\frac{1}{4}\right)^x = (-4)^4 \qquad [\because (4)^4 = (-4)^4]$

$\Rightarrow \quad \left(-\frac{1}{4}\right)^x = \left(-\frac{1}{4}\right)^{-4}$

$\therefore \quad x = -4$

53. A salesman's commission is 6% on all sales upto ₹ 15000 and 4% on all sales exceeding this amount. He remits ₹ 30000 to his parent company after deducting his commission. Find the total sales.

(a) ₹ 31562.5
(b) ₹ 31500.5
(c) ₹ 32570.5
(d) None of the above

Sol. (a) Let total sales be x.

So, we get his commission,

$\left(\frac{6}{100} \times 15000\right) + (x - 15000)\frac{4}{100}$

$= 900 + \frac{1}{25}x - 600 = 300 + \frac{1}{25}x$

According to the question,

$x - \left(300 + \frac{x}{25}\right) = 30000$

$\Rightarrow \quad \left(x - \frac{1x}{25}\right) - 300 = 30000$

$\Rightarrow \quad \frac{24}{25}x = 30300$

$\therefore \quad x = \frac{30300 \times 25}{24} = ₹\ 31562.5$

Hence, the total sales is ₹ 31562.5.

54. If 60% of $(x - y)$ = 40% of $(x + y)$, then what per cent of x is y?

(a) 20% (b) 25% (c) 22% (d) 24%

Sol. (b) Given that, 60% of $(x - y)$ = 40% of $(x + y)$

$$\Rightarrow \quad \frac{60(x-y)}{100} = \frac{40(x+y)}{100}$$

On solving above equation, we get $(x - 4y)$

So, y is 25% of x.

55. The investments of A and B in a business is in ratio 4 : 5 and the period of their investments is in the ratio 9 : 8. Their profit will be in the ratio

(a) 9 : 10 (b) 9 : 8

(c) 8 : 11 (d) None of these

Sol. (a) Let investments of A and B be $4x$, $5x$ and period of investment is $9y$, $8y$, respectively.

Then, ratio of profit = $4x \times 9y : 5x \times 8y = 36xy : 40xy$

$\therefore$ Ratio of the profit = 9 : 10

56. A new watch was purchased and it was showing 6 O'clock. The purchaser was soon informed that the hour hand and minute hand had been interchanged and attached to the wrong axes. This resulted in the hour hand to move with a speed, which is twelve times higher than the minute hand.

Hence, the watch was taken to the mechanic for repair, at that moment the watch showed the right time! If the clock started at 6 O'clock in the correct position, then what was the first moment that the watch showed the correct time?

(a) $6\frac{5}{11}$ min past 9 (b) $5\frac{5}{11}$ min past 7

(c) $7\frac{5}{11}$ min past 7 (d) None of the above

Sol. (c) According to the question, the clock shows twelve times faster time than the original time i.e. in 1 h, this clock will cover 12 h; in 1 min, the clock will cover 12 min; in 5 min, the clock will cover 60 min or 1 h etc.

Now, look at the following table :

Original time	Clock's time
6 : 00 : 00	6 : 00 : 00
7 : 00 : 00	6 : 00 : 00
(this 6 : 00 is 12 h after the above 6 : 00)	
7 : 05 : 00	7 : 00 : 00
7 : 05 : 01	7 : 00 : 12
7 : 05 : 02	7 : 00 : 24
7 : 05 : 05	7 : 01 : 00
7 : 05 : 25	7 : 05 : 00
7 : 05 : 26	7 : 05 : 12
7 : 05 : 27	7 : 05 : 24

Now, let after x sec of this time, both time will be same

i.e. $27 + x = 24 + 12x$

$\Rightarrow \quad 11x = 3$

$\Rightarrow \quad x = \frac{3}{11}$

Hence, the required time is 7 : 05 : $27\frac{3}{11}$

i.e. $7\frac{5}{11}$ min past 7.

57. If $x + y + z = \pi$, then $\cot\frac{x}{2} + \cot\frac{y}{2} + \cot\frac{z}{2}$ will be equal to

(a) $\tan\frac{x}{2} \times \tan\frac{y}{2} \times \tan\frac{z}{2}$ (b) $\cos\frac{x}{2} \times \cos\frac{y}{2} \times \cos\frac{z}{2}$

(c) $\cot\frac{x}{2} \times \cot\frac{y}{2} \times \cot\frac{z}{2}$ (d) None of the above

Sol. (c) Given, $x + y + z = \pi$

Then, $\cot\frac{x}{2} + \cot\frac{y}{2} + \cot\frac{z}{2} = ?$

Taking two terms, we have

$$\cot\frac{x}{2} + \cot\frac{y}{2} = \frac{\cos\frac{x}{2}}{\sin\frac{x}{2}} + \frac{\cos\frac{y}{2}}{\sin\frac{y}{2}}$$

$$= \frac{\left(\cos\frac{x}{2}\cdot\sin\frac{y}{2} + \cos\frac{y}{2}\cdot\sin\frac{x}{2}\right)}{\sin\frac{x}{2}\cdot\sin\frac{y}{2}}$$

As, $x + y + z = \pi \Rightarrow x + y = \pi - z$

So, $\sin\frac{(x+y)}{2} = \cos\left(\frac{z}{2}\right)$

$$\text{LHS} = \cos\frac{z}{2} = \frac{\cos\frac{z}{2}}{\sin\frac{x}{2}\sin\frac{y}{2}} + \frac{\cos\frac{z}{2}}{\sin\frac{z}{2}}$$

$$= \cos\frac{z}{2}\cdot\left[\frac{1}{\sin\frac{x}{2}\cdot\sin\frac{y}{2}} + \frac{1}{\sin\frac{z}{2}}\right]$$

$$= \cos\frac{z}{2}\cdot\frac{\left[\sin\frac{z}{2} + \sin\frac{x}{2}\cdot\sin\frac{y}{2}\right]}{\sin\frac{x}{2}\cdot\sin\frac{y}{2}\cdot\sin\frac{z}{2}}$$

$$= \cot\frac{z}{2}\cdot\frac{\left[\sin\frac{z}{2} + \sin\frac{x}{2}\sin\frac{y}{2}\right]}{\sin\frac{x}{2}\sin\frac{y}{2}} \quad \ldots(i)$$

Now, $\sin\frac{z}{2} + \sin\frac{x}{2}\cdot\sin\frac{y}{2}$

$$= \cos\left(\frac{x}{2} + \frac{y}{2}\right) + \sin\frac{x}{2}\sin\frac{y}{2}$$

As, $\frac{x}{2} + \frac{y}{2} = \frac{\pi}{2} - \frac{z}{2}$

$$= \cos\frac{x}{2}\cos\frac{y}{2} - \sin\frac{x}{2}\sin\frac{y}{2} + \sin\frac{x}{2}\cdot\sin\frac{y}{2}$$

$$= \cos\frac{x}{2}\cos\frac{y}{2}$$

Putting the above value in Eq. (i), we get

$$\text{LHS} = \frac{\cot\frac{z}{2}\cos\frac{x}{2}\cos\frac{y}{2}}{\sin\frac{x}{2}\cdot\sin\frac{y}{2}} = \cot\frac{z}{2}\cdot\cot\frac{x}{2}\cdot\cot\frac{y}{2}$$

$$= \cot\frac{x}{2}\cdot\cot\frac{y}{2}\cdot\cot\frac{z}{2}$$

58. If $y^2 + 7ay + 40 = 0$ and $y^2 + 2ay - 60 = 0$ have a common root, then the value of a is

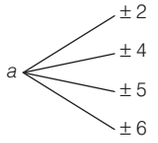
(a) ± 4 (b) ± 2
(c) ± 5 (d) ± 6

Sol. (b) Let b be the common root.

Then, $b^2 + 7ab + 40 = 0$ and $b^2 + 2ab - 60 = 0$

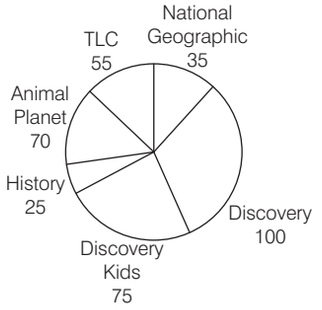
Also, $b^2 + 7ab + 40 = b^2 + 2ab - 60$

$\Rightarrow$ $5ab = -100 \Rightarrow ab = -20$



Hence, only ± 2 satisfy the above.

Directions (Q. Nos. 59-63) *These questions are based on the following pie chart, which shows the viewership of different knowledge based channels (in degrees) in the month of December, 2013. There are no overlaps in viewership of channels.*



59. During the given period for how many channels is the viewership more than 25% of the total viewership?

(a) 1 (b) 2
(c) 3 (d) None of these

Sol. (d) Given, total viewership = 360

$\therefore$ 25% of total viewership = 25% of 360° = 90°

Clearly, no channel has the viewership more than 90° more than 25% of the total viewership.

60. If 70000 people watched National Geographic on an average per day in December, 2013, then how many more people watched Discovery than History on an average per day for the same period?

(a) 360000 (b) 36000
(c) 270000 (d) None of these

Sol. (d) $\because$ Viewership of National Geographic, $35° \equiv 70000$

$\therefore$ Viewership of Discovery,

$$100° \equiv \frac{70000}{35} \times 100 = 200000$$

and viewership of History,

$$25° \equiv \frac{70000 \times 25}{35} = 50000$$

Hence, required number of people

$$= 200000 - 50000 = 150000$$

61. If the veiwership of TLC on an average was 55000 per day, then what was the viewership for all the channels on an average per day?

(a) 360000 (b) 540000
(c) 630000 (d) None of these

Sol. (a) $\because$ Viewership of TLC, $55° \equiv 55000$

$\therefore$ Viewership of all the channels 360°

$$= \frac{55000}{55} \times 360 = 360000$$

62. If the viewership of Discovery Kids channel for the first half of December is half of that of the second half of December, then what is the ratio of viewership of Discovery Kids for the second half to that of TLC for the whole month?

(a) 8 : 9 (b) 10 : 11
(c) 9 : 10 (d) None of these

Sol. (b) Let the viewership of Discovery Kids for the first and the second half of December be x and $2x$, respectively.

Then, $x + 2x \equiv 75°$

$\Rightarrow$ $3x \equiv 75°$

$\Rightarrow$ $x = 25°$

$\therefore$ Viewership of second half of December $= 2x \equiv 50°$

Also, viewership of TLC $\equiv 55°$

Hence, required ratio = 50° : 55° = 10 : 11

63. By mistake the viewership of Discovery Kids has been under quoted by 25%. If this mistake is corrected, then what is the correct share of viewership of Discovery?

(a) 45% (b) 41.55%
(c) 39.15% (d) None of these

Sol. (d) Let the correct share of viewership of Discovery be x.

Then, 75% of $x \equiv 75°$

$\therefore$ $x \equiv 75 + 25 = \frac{100}{360} \times 100\% = 27.77\%$

64. How many three digit positive integers exist that when divided by 6 leave a reminder of 4?

(a) 149 (b) 150 (c) 151 (d) 152

Sol. (a) The smallest 3-digit positive integer that when divided by 6 leaves a remainder of 4 is 106.

The largest 3-digit positive integer that when divided by 7 leaves a remainder of 4 is 994.

So, the series 106, 112, 18, ..., 994.

$\because \quad l = a + (n-1)d$

$\Rightarrow \quad 994 = 106 + (n-1)\,6$

$\Rightarrow \quad 994 = 106 + 6n - 6$

$\Rightarrow \quad 994 - 106 = 6n - 6$

$\Rightarrow \quad 888 = 6(n-1)$

$\Rightarrow \quad 148 = n - 1$

$\therefore \quad n = 149$

65. A person grows sunflowers in his square shaped farm. One sunflower takes 1 sq ft of area in his farm. This year he has increased his output by 241 sunflowers as compared to last year. The shape of the farm used for growing the sunflowers has remained the same, both the years. How many sunflowers did he in total produce this year?

(a) 12544 (b) 12996 (c) 14641 (d) 15876

Sol. (c) Let the new side of the farm be x m

$\therefore$ Area $= x^2$

$\therefore$ Number of flowers $= \dfrac{x^2}{1}$ sq ft $= x^2$

Old side of the farm be y m

$\therefore$ Area of the farm is y^2

$\therefore$ According to the question,

$x^2 - y^2 = 241$

$\Rightarrow \quad (x-y)(x+y) = 241 \times 1$

$\Rightarrow \quad x + y = 241$

And $\quad x - y = 1$

$\Rightarrow \quad 2x = 242$

$\Rightarrow \quad x = 121$

$\therefore$ Number of flowers grown $= (121)^2 = 14641$

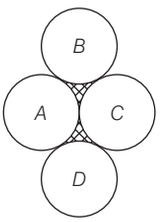
66. A box contains 11 cards with 1 to 11 consecutive numbers. If we take 5 cards in random without replacement from this box, what is the probability that the median of the numbers selected is 3?

(a) $\dfrac{13}{189}$ (b) $\dfrac{11}{128}$

(c) $\dfrac{14}{231}$ (d) None of these

Sol. (d)

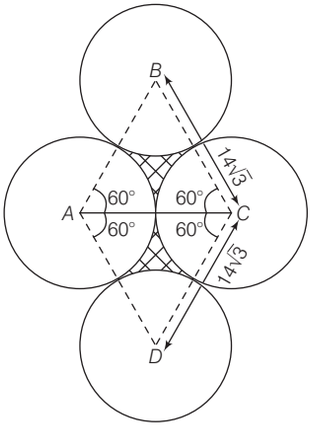
67. In the given figure, if A, B, C, D are centres of the circles and the radius of each circle is $7\sqrt{3}$ units then what is the area of the shaded region?



(a) $74\left(\sqrt{3} - \dfrac{\pi}{2}\right)$ (b) $49\left(\sqrt{3} - \dfrac{\pi}{2}\right)$

(c) $300\left(\sqrt{3} - \dfrac{\pi}{2}\right)$ (d) $294\left(\sqrt{3} - \dfrac{\pi}{2}\right)$

Sol. (d) In the figure, ΔABC and ΔADC are equilateral triangles, which radius of each circle $= 7\sqrt{3}$.



$\therefore$ Area of $\Delta ABC = \dfrac{\sqrt{3}}{4}a^2$

$= \dfrac{\sqrt{3}}{4} 14\sqrt{3} \times 14\sqrt{3}$ [where, $a = 14\sqrt{3}$ units]

$= \dfrac{3 \times \sqrt{3} \times 98}{2}$

$= \dfrac{294\sqrt{3}}{2} = 147\sqrt{3}$ sq units

Area of $ABCD = 2 \times$ Area of ΔABC

$= 2 \times 147\sqrt{3}$

$= 294\sqrt{3}$ sq units

Area of shaded part

= Area of $ABCD$ – Area of circle

$= 294\sqrt{3} - \pi(7\sqrt{3})^2$

$= 294\sqrt{3} - \pi \times 49 \times 3$

$= 294\sqrt{3} - \pi\, 147$

$= 294\left(\sqrt{3} - \dfrac{\pi}{2}\right)$ sq units

68. Instead of dividing ₹ 180 among A, B, C in the ratio $\dfrac{1}{3} : \dfrac{1}{4} : \dfrac{1}{6}$ respectively, it was divided in the ratio 3 : 4 : 6 by mistake. Who gained in this transaction?

(a) Only A
(b) Only B
(c) Only C
(d) B and C

Sol. (c) Previous ratio $= \dfrac{1}{3} : \dfrac{1}{4} : \dfrac{1}{6}$

$= 4 : 3 : 2$ [given]

So, the division of money among A, B and C is 80, 60 and 40, respectively.

$\therefore$ New ratio $= 3 : 4 : 6$

So, the division of money among A, B and C according to new ratio is 41.5, 55.38 and 83.07, respectively.

Hence, only C has gained.

69. The sum of ages of mother and daughter is 42 yr. After three years, the age of mother will be three times that of the daughter. Then, the mother's age after three years will be

(a) 28 yr (b) 33 yr
(c) 36 yr (d) 38 yr

Sol. (c) Let the age of mother = x yr
and the age of daughter = y yr
Now, according to question,

$$x + y = 42 \quad \ldots(i)$$

and $(x + 3) = 3(y + 3)$ ...(ii)

On solving Eqs. (i) and (ii), we get

$$y = 9 \text{ and } x = 33$$

$\therefore$ Mother's age after 3 yr

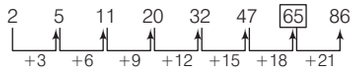
$$= 33 + 3 = 36 \text{ yr}$$

70. Find the missing number in the series

2, 5, 11, 20, 32, 47, ..., 86.

(a) 60 (b) 65
(c) 58 (d) 62

Sol. (b) The pattern of the series is as follows:



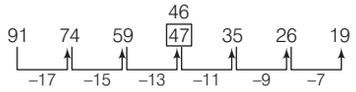
So, the missing term will be 65.

71. One number in the following series is wrong. Find out the wrong one.

91, 74, 59, 47, 35, 26, 19

(a) 47 (b) 35
(c) 26 (d) 19

Sol. (a) The pattern of the series is as follows:



Hence, 47 is wrong term, here 46 is the correct term.

72. A player can take a maximum of 4 chances to hit a bottle with a flying disc. The probability of hitting the bottle at the first, second, third and fourth shots are 0.1, 0.2, 0.35 and 0.45, respectively. What is the probability that the player hits the bottle with the flying disc?

(a) 0.6573 (b) 0.2574
(c) 0.7426 (d) None of these

Sol. (c) Probability of not hitting the bottle at first shot = P (I) = 0.9
Not hitting at second shot = P(II) = 0.8
Not hitting at third shot = P(III) = 0.65
Not hitting at fourth shot = P(IV) = 0.55
Then, probability of player hits the bottle with the flying disc

$$= (1 - 0.9 \times 0.8 \times 0.65 \times 0.55)$$
$$= (1 - 0.2574) = 0.7426$$

73. If log 2 = 0.30103 and log 3 = 0.4771, then number of digits in $(10368)^6$ will be approximately equal to

(a) 15 digits (b) 25 digits
(c) 30 digits (d) 40 digits

Sol. (b) Let $x = (10368)^6$

$\therefore$ $\log x = 6 \log 10368$
$= 6 \log (2^7 \times 3^4)$
$= 6 (7 \log 2 + 4 \log 3)$
$= 6 (7 \times 0.30103 + 4 \times 0.4771)$
$= 6 (2.10721 + 1.9084)$
$= 6 \times 4.01561 = 24.09366$

Since, the characteristic of log x is 24. It follows that x is a 25-digits number.

74. What is the measure of the angle between the hands of the clock at 8 : 24 pm?

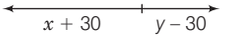
(a) 100° (b) 102°
(c) 106° (d) 108°

Sol. (d) Required angle $= 240 - 24\left(\frac{11}{2}\right)$
$= 240 - 132$
$= 108°$

75. A train after travelling for a certain distance develops a problem and decreases its speed to half its original speed and reaches its destination 45 min late. Had the problem occurred 30 km further on, it would have reached its destination 15 min earlier? What is the speed of the train?

(a) 30 km/h (b) 90 km/h
(c) 120 km/h (d) 150 km/h

Sol. (a) Let the speed of train be a km/h and $\frac{x}{a} + \frac{y}{b}$ be the time taken in condition I.
Also,



Then, time taken

$$= \frac{x + 30}{a} + \frac{y - 30}{b}$$

According to the question,

$$\frac{x}{a} + \frac{y}{b} - \frac{45}{60} = \frac{x + 30}{a} + \frac{y - 30}{b} + \frac{15}{60}$$

$\Rightarrow$ $\frac{-30}{a} + \frac{30}{b} = 1$

$\Rightarrow$ $30 (a - b) = ab$ [we know that, $a = 2b$]

$\Rightarrow$ $30 b = 2b \times b$

$\therefore$ $a = 30$ km/h

76. A manufacturer sells a good to a wholesale dealer at a profit of 18%. The wholesaler sells the same to a retailer at a profit of 20%. The retailer in turn, sells it to the customer for ₹ 30.09 thereby earning profit of 25%. What is the cost of the manufacturer?

(a) ₹ 15.30 (b) ₹ 17
(c) ₹ 20 (d) ₹ 22

Sol. (b) Let the cost of manufacturer be ₹ x.

According to the question

$$x \times \frac{118}{100} \times \frac{120}{100} \times \frac{125}{100} = 30.09$$

$$\therefore \quad x = \frac{3009 \times 100 \times 100 \times 100}{118 \times 120 \times 125 \times 100} = ₹\ 17$$

77. If ₹ 1000 is invested at an interest rate of 5% and the interest earned is added to the principal every 10 yr, then after how many years will the amount become ₹ 2000?

(a) $\frac{20}{3}$ (b) $\frac{50}{3}$

(c) $\frac{25}{4}$ (d) None of these

Sol. (b) Let $P_1 = ₹\ 1000$, $r = 5\%$, $t_1 = 10$ yr

Simple interest after 10 yr,

$$SI = \frac{P_1 \times r \times t_1}{100} = \frac{1000 \times 5 \times 10}{100} = ₹\ 500$$

Amount $= P_1 + SI$

$= 1000 + 500 = ₹\ 1500$

Let after t_2 yr, the amount become ₹ 2000.

$\therefore$ SI for t_2 yr $= 2000 - 1500 = 500$

$$= \frac{1500 \times 5 \times t_2}{100}$$

$$= \frac{20}{3}$$

$$\therefore \text{ Total time } = t_1 + t_2 = 10 + \frac{20}{3} = \frac{50}{3} \text{ yr}$$

78. Two pipes P and Q can fill a tank A in 10 h and 12 h, respectively. A third pipe R can empty a tank B in 40 h. Tank B is two times in volume compared to tank A. If P and Q are kept open to fill tank A, if R is connected to empty tank A, at the same time such that all the pipes operate simultaneously, how long will take for the tank A to be fully filled?

(a) 6 h 40 min
(b) 8 h 30 min
(c) 7 h 30 min
(d) None of the above

Sol. (c) Time taken by pipe R to empty tank $B = 40$ h

Now, volume of tank $B = 2 \times$ Volume of tank A

$\therefore$ Time taken by pipe R to empty tank $A = 20$ h

Also, time taken by pipe P to fill tank $A = 10$ h and time taken by pipe Q to fill tank $A = 12$ h

According to the question,

$$\therefore \text{ Total time taken } = \frac{1}{10} + \frac{1}{12} - \frac{1}{20}$$

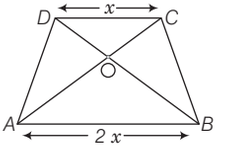
$$= \frac{6 + 5 - 3}{60} = \frac{8}{60}$$

$$= \frac{15}{2} \text{ h} = 7 \text{ h and } 30 \text{ min}$$

79. $ABCD$ is a trapezium and AB is parallel to CD. AB has a length, which is two times the length of CD. If the diagonals of the trapezium intersect at O, then the ratio of ΔAOB and ΔCOD is

(a) 1 : 4 (b) 4 : 1 (c) 1 : 2 (d) 2 : 1

Sol. (b) In trapezium $ABCD$,



$DC = x$, $AB = 2x$

Ratio of $\Delta\ AOB : \Delta\ DOC = (2x)^2 : (x)^2 = 4x^2 : x^2 = 4 : 1$

80. The MRP of a chocolate is ₹ 65. A customer purchased the chocolate for ₹ 56.16. He got two successive discounts of which one is 10%. Find the other discount of the scheme that was given by the seller.

(a) 3%
(b) 4%
(c) 5%
(d) None of the above

Sol. (b) Let the other discount be of d%.

Then, according to the question,

$$65 \times \left(\frac{100 - 10}{100}\right) \times \left(\frac{100 - d}{100}\right) = 56.16$$

$$\Rightarrow \quad 65 \times \frac{90}{100} \times \left(\frac{100 - d}{100}\right) = 56.16$$

$$\Rightarrow \quad 100 - d = \frac{5616 \times 100}{65 \times 90}$$

$$\Rightarrow \quad 100 - d = 96$$

$$\therefore \quad d = 4\%$$

Part C General Awareness

81. What is the full form of RSVP?
(a) Rigorous Session of Voice Protocol
(b) Repondez S'ilvous Plait
(c) Repondez Sans Vous Plaint
(d) Response Sought Vous Plaint

Sol. (b) RSVP stands for Repondez S'ilvous Plait. In the context of social invitations, RSVP is a request for a response from the invited person or people. It is derived from the French phrase which means 'reply if you please' or 'reply please'.

82. The Callanish Stones, which are considered to be petrified giants, as per a folklore, who did not convert to Christianity are located in which country?
(a) Norway (b) Wales (c) Scotland (d) Ireland

Sol. (c) The Callanish Stones are located in Scotland, near the village of Callanish on the West coast of Lewis. They are an arrangement of standing stones placed in a cruciform pattern with a central stone circle and are considered to be petrified giants, as per a folklore, who did not convert to Christianity.

83. Which of the following sites or monuments is not in the UNESCO list of World Heritage Sites?
(a) Humayun's Tomb at Delhi
(b) Kaziranga National Park in Asom
(c) Tirupati Temple
(d) Agra Fort

Sol. (c) Tirumala Venkateswara Temple located near Tirupati in Chittoor district of Andhra Pradesh, is a famous Vedic temple in the hill town of Tirumala. It is not in the list of UNESCO World Heritage Sites.

84. Mr. Ramon Magsaysay, in whose name the prestigious Magsaysay Award is given, belonged to
(a) Malaysia (b) Thailand
(c) Philippines (d) Indonesia

Sol. (c) Ramon del Fierro Magsaysay simply known as Mr Magsaysay was the seventh President of the Republic of the Philippines, serving from December 30, 1953 until his death in a aircraft disaster in 1957. The Magsaysay award was established in April 1957.

85. Indian musician noted for his exemplary skills at playing mandolin who died in 2014.
(a) Uppalapu Srinivas (b) Amjad Ali Khan
(c) Zarin Daruwalla (d) Ustad Usman Khan

Sol. (a) Uppalapu Srinivas was an Indian mandolin maestro belonging to the classical carnatic musical tradition of Southern India. He passed away on 19th September, 2014. Srinivas pioneered the introduction of the mandolin, a Western instrument, into classical carnatic music.

86. In the fifth meeting of SAARC Energy Ministers, which from the following has been the key area of discussion?
(a) Telecom (b) Environment
(c) Power Grid (d) Education

Sol. (c) The fifth meeting of the South Asian Association for Regional Cooperation (SAARC) Energy Ministers was held on 16th and 17th October, 2014 in New Delhi, with an aim of establishing inter-regional Power Grid Project for enhancing energy cooperation and inter-connectivity among the eight nations comprising the SAARC.

87. Malaysian Airlines Flight MH17 was a scheduled international passenger flight, presumed to have been shot down, crashed on 17th July, 2014 killing all 283 passengers and 15 crew on board was flying, from
(a) Amsterdam to Kuala Lumpur
(b) Kuala Lumpur International Airport to Beijing Capital International Airport
(c) Kuala Lumpur to Amsterdam
(d) Beijing Capital International Airport to Kuala Lumpur International Airport

Sol. (a) Malaysia Airline Flight MH17 was a scheduled international passenger flight from Amsterdam to Kuala Lumpur that crashed on 17th July, 2014, presumed to have been shot down, killing all 283 passengers and 15 crew on board.

88. Ukraine has been in news recently on account of conflict with Russia. The name of the Ukrainian President is
(a) Petro Poroshenko (b) Viktor Yushebenko
(c) Yulia Timoshenko (d) None of the above

Sol. (a) Presidential elections were held in Ukraine on 25th May, 2014, resulting in Petro Poroshenko being elected President of Ukrain.

89. Oldest brand of the Tata Group is
(a) the Taj Palace and Tower
(b) Ginger
(c) Taj Cape Town
(d) Quilon

Sol. (a) The Taj Mahal Palace and Tower, built in 1903, is the oldest brand of the Tata Group.

90. 'Gold fixing' refers to setting the price of gold. Where is the value of gold set?
(a) New York (b) London
(c) Tokyo (d) Singapore

Sol. (b) The Gold Fixing (or London Gold Fix) is the setting of the price of gold by the members of the London Gold Market Fixing Limited. The Gold Fixing also provides a published benchmark price that is widely used as a pricing medium by producers, consumers, investors and central banks.

91. Which of the following nations has declared issuance of digital currency?
(a) Ecuador
(b) Estonia
(c) United States of America
(d) United Kingdom

Sol. (a) Ecuador is the first country to create the world's first digital currency issued by the country's central bank, in an attempt to abandon the dependence on the US dollar. The new currency is expected to co-exist with the US dollar in Ecuador's economy.

92. What is the significance of EPS in the stock market parlance?
(a) Equilibrium of Private Securities
(b) Earning Per Share
(c) Elongated Period of Settlement
(d) Equity Per Share

Sol. (b) The term Earnings Per Share (EPS) represents the portion of a company's earnings, net of taxes and preferred stock dividends.

93. Which committee was appointed by RBI to provide recommendations on financial services for small businesses and low income households in the year 2013?
(a) Nachiket Mor Committee
(b) Bimal Jalan Committee
(c) Tarapore Committee
(d) Damodaran Committee

Sol. (a) Dr Nachiket Mor Committee was appointed by the RBI to provide recommendations on comprehensive financial services for small businesses and low income households in year 2013.

94. Dadabhai Naoroji, in his book 'Poverty and Un-British Rule in India' in 1867 explained about
(a) Brain Drain Theory
(b) Economic Drain Theory
(c) Wants and Means of India
(d) Benefits of British Rule

Sol. (b) Economic Drain Theory during the British colonial rule in India is the subject matter of Dadabhai Naoroji's book 'Poverty and Un-British Rule in India' published in 1867. He found out that there was always a positive balance of trade for India, the surplus being appropriated by the British for their own benefit.

95. He is a famous playwright, socialist and one of the co-founders of London School of Economics, name the person.
(a) Beatrice Webb (b) George Bernard Shaw
(c) Arnold Bennett (d) Richard Brome

Sol. (b) George Bernard Shaw was an Irish playwright, socialist and a co-founder of the London School of Economics (1895). He was also an essayist, novelist and short story writer.

96. Auto didacticism is a form of education where
(a) one automatically learns from experience of peers
(b) a set of logical deductions are taught by the teacher
(c) one teaches oneself about a subject
(d) one learns automatically from elders

Sol. (c) Auto didacticism or self-education is the act of teaching oneself about a subject or subjects in which one has had little to no formal education. Many autodidacts seek instruction and guidance from experts, friends, teachers, parents, siblings and community.

97. The author of the book 'An Inquiry into the Nature and Causes of the Wealth of Nations' is
(a) Adam Smith
(b) Karl Marx
(c) Amartya Sen
(d) Alfred Marshall

Sol. (a) 'An Inquiry into the Nature and Causes of the Wealth of Nations', generally referred to by its shortened title 'The Wealth of Nations', by Adam Smith. It was first published in 1776.

98. The universities of Bombay, Madras and Calcutta were set-up in the same year. Which was the year?
(a) 1818 (b) 1857
(c) 1947 (d) 1860

Sol. (b) The universities of Calcutta, Bombay and Madras were established in 1857. Lord Canning was Governor General at the time of establishment of these universities. These were the first three universities established by the British in the country.

99. 'Shaurya-Dridata-Karmanishtha' is the motto of
(a) Border Security Force
(b) Indian Coast Guard
(c) Indo-Tibetan Border Police
(d) National Security Guard

Sol. (c) Indo-Tibetan Border Police (ITBP) is one of the eight Central Armed Police Forces of India. The motto of ITBP is Shauray-Dridata-Karmanishtha.

100. The following is the logo of

(a) Cannes International Film Festival
(b) Berlin International Film Festival
(c) London Film Festival
(d) Amnesty International Film Festival

Sol. (a) The logo is related to Cannes International Film Festival, one of the most prestigious and published film festivals of the world. It was founded in 1946. The festival is an annual film festival held in Cannes, France.

101. The Giant Panda, world's most threatened species, is a symbol of

(a) IFAW (b) WWF (c) CEE (d) UNEP

Sol. (b) The Giant Panda has been the WWF's logo since the organisation was formed. The World Wide Fund for Nature (WWF) is an international non-governmental organisation founded on April 29, 1961. It is working on issues regarding the conservation, research and restoration of the environment.

102. Acid rain is caused mainly due to the emissions of

(a) calcium oxide and sulphur dioxide
(b) nitrogen oxide and sulphur dioxide
(c) sodium oxide and nitrogen oxide
(d) calcium oxide and carbon dioxide

Sol. (b) Acid rain is caused by a chemical reaction that begins when compounds like sulphur dioxide and nitrogen oxides are released into the air. These substances can rise very high into the atmosphere, where they mix and react with water, oxygen and other chemicals to form more acidic pollutants, known as acid rain.

103. Which of the given rivers does not flow into river Ganga?

(a) Brahmaputra (b) Gomti
(c) Gandak (d) Tapti

Sol. (d) Brahmaputra, Gomti and Gandak all are East flowing rivers and flow into river Ganga at different point. Whereas Tapti flows Westward and emptying into the Gulf of Cambay of the Arabian Sea, in the Surat district of Gujarat.

104. The international agreement for control of Green House Gas (GHG) emissions is named

(a) Adelaide Agreement
(b) Kyoto Protocol
(c) Manila Convention
(d) Kuala Lumpur Treaty

Sol. (b) The Kyoto Protocol is an international agreement setting targets for industrialised nations to cut their Green House Gas (GHG) emissions. The protocol was agreed to in 1997 in Kyoto in Japan, based on principles set out by UNFCCC.

105. The International Standard which checks the compliance of the environmental practices of an organisation against the best practices followed in the industry is

(a) ISO 9001
(b) TS 16949
(c) ISO 14001
(d) ISO 9002

Sol. (c) ISO 14001 International Standard checks the compliance of the environmental practices of an organisation. The major objective of ISO 14001 series is to promote effective and efficient environment management in an organisation.

106. Comiket, the world's largest comic festival, is held in which country?

(a) Japan (b) France
(c) Korea (d) Germany

Sol. (a) Comiket, otherwise known as the Comic Market, is the world's largest comic festival, held twice a year in Tokyo, Japan. The first Comiket was held on 21st December, 1975, with only about 32 participating circles and an estimated 600 attendees.

107. In ……, Infosys was listed in NASDAQ and in ……, Infosys became a part of NASDAQ 100.

(a) 1998; 2009 (b) 1997; 2008
(c) 1999; 2006 (d) None of these

Sol. (c) In 1999, Infosys generated revenues of $100 million and was listed in NASDAQ. Later in the year 2006, the company became part of NASDAQ-100, it was first Indian and only the company to be part of any of the major global indices.

108. The Indian team finished with 8th position in the recent Asian Games 2014 held at South Korea with a total …… tally of medals.

(a) 57 (b) 37
(c) 56 (d) 58

Sol. (a) India ended the 17th Asian Games 2014 with a total of 57 medals, including 11 gold, 9 silver, and 37 bronze, which is 8 less than the 65 medals (14 gold, 17 silver and 34 bronze). India won at the Guangzhou Games four years ago.

109. A full Moon occurs once every

(a) 14 days (b) 30 days
(c) 21 days (d) 15 days

Sol. (b) A full moon is the lunar phase that occurs when the moon is completely illuminated as seen from the Earth. A full moon occurs once every 30 days (29.53 days).

110. The river Danube empties in the

(a) Dead sea (b) Mediterranean sea
(c) Black sea (d) Red sea

Sol. (c) Stretching over 2000 mills and flowing through ten countries, the Danube is the longest river in the European Union. Starting in the German Black Forest, it empties into the Black Sea.

111. Cartography is the study of

(a) cartilages and bones (b) cartoons
(c) maps (d) carrots

Sol. (c) Cartography, the art and science of graphically representing a geographical area, usually on a flat surface such as a map or chart; it may involve the superimposition of political, cultural or other non-geographical divisions onto the representation of a geographical area.

112. Regional Rural Banks are jointly owned by
(a) Central Government, a Commercial Bank and Central bank of a country
(b) Central Government, a Commercial Bank and a State Government
(c) Central Government, Central Bank and a State Government
(d) a Commercial Bank, Central Bank of a country and a State Government

Sol. (d) Regional Rural Banks are local level banking organisations operating in different states of India. They have been created with a view to serve primarily the rural areas of India with basic banking and financial services. It is jointly owned by a Commercial Bank, Central Bank of a country and a State Government.

113. The famous Meenakshi Temple in Madurai was built by …… and beautified by the ..... .
(a) Vakatakas, Nayaks
(b) Chalukyas, Cholas
(c) Satavahanas, Cholas
(d) Pandyas, Nayaks

Sol. (d) The famous Meenakshi Temple in Madurai was built by Pandyas and later on beautified by the Nayaks. Meenakshi Amman Temple also known as Meenakshi Sundareswarar Temple is located in the city of Madurai on the Southern bank of river Vaigai.

114. Which of the given statements is not correct as per Indian Constitution?
(a) There can be a single person as the Governor for two or more states
(b) Governor is the constitutional or nominal head of the State Executive and is bound by the advice of the State Council of Ministers
(c) All executive actions of the state shall be expressed to be taken in the name of the Chief Minister
(d) The President of India consults the Governor on appointment of the High Court judges

Sol. (c) As per the Article166 of Indian Constitution, all executive action of the government of a state shall be expressed to be taken in the name of the Governor.

115. Kwan Yin has a special place in Buddhist tradition and is known
(a) to represent healing nature
(b) to bring in good luck charm
(c) as a Goddess of compassion
(d) as related to higher psychological- scientific teachings from Buddha

Sol. (c) One of the deities most frequently seen on altars in China's temple is Quan Yin, in sanskrit, her name is 'Padma-pani'. Quan Yin, along among buddhist Gods, is loved rather than feared. Regarded by the Chinese as the Goddess of compassion and mercy.

116. 'Bachpan Bachao Andolan' was initiated by
(a) Jawaharlal Nehru (b) Kailash Satyarthi
(c) Malala Yousafzai (d) Benazir Bhutto

Sol. (b) Kailash Satyarthi is an Indian Children's rights advocate and an activist againsts child labour. He founded the Bachpan Bachao Andolan in 1980, and has acted to protect the rights of more than 83000 children from 144 countries.

117. Dr Ram Manohar Lohia, the great freedom fighter and thinker, established a political party named
(a) Samajwadi Party
(b) Samyukta Socialist Party
(c) Janata Party
(d) Praja Socialist Party

Sol. (d) The Praja Socialist Party was an Indian political party in existence from 1952 to 1972. Lohia was a socialist and wanted to unite all the socialists in the world to form a potent platform. He was the founder of 'Praja Socialist Party.

118. The Nobel Prize in Physics 2014 was awarded jointly to three scientists "for the invention of efficient blue light-emitting diodes which has enabled bright and energy-saving white light sources". Who among the given four names was not in the team?
(a) Isamu Akasaki (b) Hiroshi Amano
(c) Shuji Nakamura (d) Takada Ekimano

Sol. (d) The Royal Swedish Acdemy of Sciences had decided to award the Nobel Prize in Physics for 2014 to Isamu Akasaki (Nagoya University, Japan), Hiroshi Amano (Nagoya University, Japan and Shuji Nakamura (University of California, USA).

119. Chandrashekhar Venkat (CV) Raman was awarded the Nobel Prize for investigation into 'Raman Effect', a phenomenon that deals with
(a) change in the speed of light as it passes through a transparent medium
(b) change in the atomic structure of the medium impacted by light
(c) change in the wavelength of the light as it passes through a transparent medium
(d) change of light in liquid crystals alone

Sol. (c) Raman Effect, change in the wavelength of light that occurs when a light beam is deflected by molecules as it passes through a transparent medium.

120. 'Working towards a world free of poverty' is a motto of one of these international organisations.
(a) UNESCO (b) UNICEF
(c) World Bank (d) UNCTAD

Sol. (c) 'Working towards a world free of poverty' is a motto of World Bank. The World Bank is a United Nations international financial institution that provides loans to developing countries for capital programmes.

Part D Analytical & Logical Reasoning

121. Vertex : Pyramid : ?
(a) Strand : Hair (b) Rung : Ladder
(c) Summit : Mountain (d) Frame : Picture

Sol. (c) Vertex is a highest point of pyramid. Similarly, summit is the highest point of mountain.

122. Piety : Pilgrim : Curiosity : ?
(a) Explorer (b) Pinchpenny
(c) Gamester (d) Monk

Sol. (a) Piety reflects the religiosity, which is the main quality of Pilgrim. Similarly, Curiosity demands Explorer.

Directions (Q. Nos. 123-124) *Read the paragraphs carefully and determine the main point the author is trying to make. What conclusion can be drawn from the argument? Each paragraph is followed by four statements. One statement supports the author's argument better than the others do.*

123. Now-a-days, people are so busy in their work that they do not take care of their health. A lot of diseases like diabetes, arthritis, heart attack, blood pressure are increasing due to unhealthy lifestyle. People do not find time to exercise. Stress is continuously increasing. The fast food is quick, easy and tastes better, but it is not healthy. Habits like smoking and drinking alcohol have many side effects. Polluted city environment has also negative impact on the health of citizens. To live happy and healthy life; it is important to take care of oneself.
(a) Now-a-days everybody smokes and drinks
(b) Diseases are increasing only because of pollution
(c) People are happy and stress free because their life is too busy
(d) Current city environment, life style and habits are affecting health of citizens

Sol. (d) The conclusion that can be drawn from the paragraph most appropriately and correct is option (d).

124. Diwali is a festival of lights; which signifies that good will always be better than evil. In the present day, many people have developed a lot of negativity in their minds and hearts, which has led to high corruption and higher crime rates. Value based education is the need of the day to eliminate darkness of cruelty, selfishness and greed and bring the light of wisdom, peace, kindness, truthfulness, right attitude, hard work and cooperation. Mahatma Gandhi also said that we should eradicate sins from hearts not the sinners. If a society has people with good heart, it will definitely flourish.
(a) Now-a-days nobody is good
(b) Now-a-days everybody is corrupt and criminal
(c) Value based education can save the society from crime and corruption
(d) It is not possible to fight against corruption as everybody is corrupt

Sol. (c) By reading the paragraph, we can clearly conclude that value based education can save the society. So, option (c) is best choice.

125. If X + Y means 'X is the mother of Y'; X – Y means 'X is the brother of Y'; X % Y means 'X is the father of Y' and X × Y means 'X is the sister of Y', then which of the following shows that P is the maternal uncle of Q?
(a) Q – N + M × P (b) P + S × N – Q
(c) P – M + N × Q (d) Q – S % P

Sol. (c) Let consider option (c), P – M + N × Q
From the given information
P – M ⇒ 'P is the brother of M'.
M + N ⇒ 'M is the mother of N'.
N × Q ⇒ 'N is the sister of Q'.
Therefore,

$$\underset{\text{Brother}}{P^{+}} - \underset{\text{Mother}}{M^{-}} + \underset{\text{Sister}}{N^{-}} \times Q$$

So, 'P is the maternal uncle of Q'.

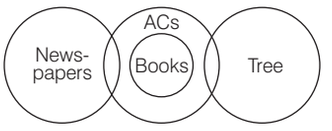
126. In the following question, three statements numbered I, II and III and four conclusions are given. Select the correct answer from the options given.

Statements
I. Some newspapers are ACs.
II. Some ACs are trees.
III. All books are ACs.

Conclusions
1. Some books are newspapers.
2. All books are trees.
3. All trees are books.
4. Some books are not newspapers.

(a) Only 1 follows
(b) 1 or 3 follows
(c) Either 1 or 4 follows
(d) 2 and 3 follow

Sol. (c)



Conclusion 1 do not follow. Conclusion 4 also do not follow, but these two conclusions forms a complementary pair. Therefore, either 1 or 4 follows.

Directions (Q. Nos. 127-128) *A statement is given followed by several assumptions. An assumption is something assumed or taken for granted. Read the statement carefully and decide which of the assumptions are implicit in the statement.*

127. **Statement** "I think that she has gone mad. Last night I saw her barking like a dog."– A tells B about Madhu.

Assumptions

I. Madhu is a friend of A.
II. B knows Madhu.
III. Some people do not bark like dogs.

(a) I and II are implicit
(b) II and III are implicit
(c) I and III are implicit
(d) All are implicit

Sol. (a) From the given Statement, it is clear that Madhu is a friend of A, because A tells B about Madhu. So, it is clear that B knows Madhu.
Therefore, Statements I and II are implicit.

128. **Statement** "The Prime Minister is expected to announce an expansion in his Ministry shortly."– A newspaper report.

Assumptions

I. The newspaper has quoted authentic sources.
II. The newspaper has reliable sources for the news.
III. The Prime Minister has the power to expand his cabinet.

(a) I and II are implicit
(b) II and III are implicit
(c) I and III are implicit
(d) All are implicit

Sol. (b) The newspaper has not quoted authentic sources, so I is not implicit; but II and III are implicit.

129. How many times in a day, do the hands of a clock coincide with each other?

(a) 12 (b) 24
(c) 22 (d) 44

Sol. (c) The hands coincide 11 times in every 12 hours. So, in a day, the hands coincide 22 times.

Directions (Q. Nos. 130-134) *These questions are based on the data given below.*

There are two circular tables in a room. Six Americans–A1, A2, A3, A4, A5 and A6 are sitting at one table and six Germans–G1, G2, G3, G4, G5 and G6 are sitting at the other table. A1 and A4 are sitting opposite to each other. A2 and A5 are also sitting opposite to each other. A3 sits to the right of A4. A2 is the only person who can translate English to German language; A3 is the only person who can translate German language to English and none of them does the *vice-versa*, unless so stated. A5 and A6 sit adjacent to each other. Similarly, G1 sits opposite to G4; G6 sits opposite to G3 and G5 sits to the right of G6. G2 and G4 sit adjacent to G3. On the table, any person can talk to another person, only as stated below. The only conversations that took place are as given below :

A1 spoke to A2; A2 to G6; G6 to A3; G6 to G5; G5 to G4; G4 to G3; G3 to G2; G2 to G1; A3 to A4; A4 to A5; A6 to A1; G1 to G6 and A5 to A6.

The conversations are one-sided i.e. A1 spoke to A2 implies that A1 is the speaker and A2 is the listener and not *vice-versa*.

130. If the person to the right of A1 wants to send a message to the person sitting to the left of G2, then the message has to pass through how many persons? (excluding the first and the last)

(a) 7 (b) 5
(c) 6 (d) None of these

131. If A3 wants to send a message to G1, then how many person must the message pass through? (excluding the first and the last)

(a) 6 (b) 4
(c) 10 (d) None of these

132. If the person sitting to the right of A2 wants to send a message to the person sitting two places to the left of A4, then what is the minimum number of people involved between them? (excluding the two people)

(a) 6 (b) 10
(c) 4 (d) None of these

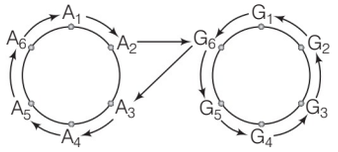
133. If every person is allowed to speak to the person who sits opposite to him directly, then what is the least number of person involved in passing a message from A3 to G1? (excluding the two people)

(a) 3 (b) 6
(c) 7 (d) None of these

134. If the order of conveying messages is reversed at both the tables i.e. 'A1 speaks to A2' (now becomes 'A2 speaks to A1' and so on) and also A2 and A3 exchange their interpreting skills, then which of the following must be true?

(a) A1 can send a message to G5 involving only 2 persons
(b) The person sitting to the right of A6 can send a message to the person sitting to the left of G2, by involving only five persons
(c) The person sitting to the right of A3, on the same table, can translate English to German language
(d) The maximum number of persons involved in the longest message in this new arrangement is more than that in the previous arrangement

Solutions (Q. Nos. 130-134) *The sitting arrangement and pattern of conversation is as follow:*



130. ***(b)*** A_6 is the person to the right of A_1 and G_3 is the person sitting to the left of G_2. So, route of send a message from A_6 to G_3 is as :

Exclude Exclude

$\boxed{A_6} \rightarrow A_1 \rightarrow A_2 \rightarrow G_6 \rightarrow G_5 \rightarrow G_4 \rightarrow \boxed{G_3}$

After excluding the first and last person i.e. A_6 and G_3, the message has to pass through 5 persons.

131. ***(c)*** Route of send a message from A_3 to G_1 is as :

Exclude

$\boxed{A_3} \rightarrow A_4 \rightarrow A_5 \rightarrow A_6 \rightarrow A_1 \rightarrow A_2 \rightarrow G_6 \rightarrow G_5 \rightarrow G_4 \rightarrow G_3 \rightarrow G_2 \rightarrow \boxed{G_1}$

Exclude

After excluding the first and last person, the message pass through 10 persons.

132. ***(b)*** A_1 is the person sitting to the right of A_2 and A_6 is the person sitting two places to the left of A_4.

So, the route of send a message from A_1 to A_6 is as :

$A_1 \rightarrow A_2 \rightarrow G_6 \rightarrow G_5 \rightarrow G_4 \rightarrow G_3 \rightarrow G_2 \rightarrow G_1 \rightarrow A_6$

As G_6 comes two times, but we count only once. According to question, after excluding the two people, there are 10 minimum persons.

133. ***(d)*** According to the question, A_3 direct speak to A_6.

Now, the route of passing a message from A_3 to G_1 is as :

$A_3 \rightarrow A_6 \rightarrow A_1 \rightarrow A_2 \rightarrow G_6 \rightarrow G_5 \rightarrow G_4 \rightarrow G_3 \rightarrow G_2 \rightarrow G_1$

After excluding the two people, there are 8 persons involved in passing a message from A_3 to G_1.

134. ***(c)*** A_2 sitting to the right of A_3 and from the question it is clear that, A_2 is the only person who can translate English to German language.

Directions (Q. Nos. 135-136) *Read the following information carefully and answer the questions that follow*

There are 3 females A, B and E and 4 males C, D, F and G standing in a straight line. No two females are together. B is to right of C, F and D are not together as A is placed between them. G is not near B or E but E and F are together. D is not to the right of B.

135. Who is in the extreme end?

(a) G and B (b) C and F
(c) B and D (d) None of these

136. Who is exactly in the middle?

(a) A (b) F
(c) E (d) None of these

Solutions (Q. Nos. 135-136) Standing pattern in a straight line is as follows :

G D A F E C B

135. ***(a)***

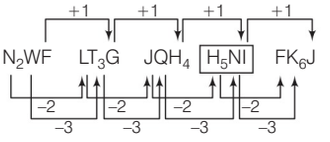
136. ***(b)***

137. Look carefully at the sequence of letters and symbols to find the pattern. Select correct pattern.

$N_2WF \quad LT_3G \quad JQH_4 \ldots FK_6J$

(a) H_5NI (b) H_5SI
(c) HR_2I_3 (d) H_5NJ

Sol. (a) The pattern is as follows



Directions (Q. Nos. 138-141) *Answer the questions based on the details given in the paragraph.*

There were 200 students who gave a competitive exam. The exam had three sections English, Mathematics and General Knowledge (GK). Ten students could not pass in any of the subjects. Number of students passed in English was 125, Mathematics was 95 and General Knowledge was 95. 55 students passed in English and Mathematics. 45 students passed in GK and Mathematics and 60 students passed in GK and English. 20 students passed in English and Mathematics but failed in GK.

138. How many students passed in GK and Mathematics, but failed in English?

(a) 30 (b) 60 (c) 45 (d) 10

139. How many students passed in English and GK, but failed in Mathematics?

(a) 60 (b) 25 (c) 20 (d) 35

140. How many students passed only in Mathematics?

(a) 30 (b) 20 (c) 25 (d) 35

141. How many students passed in all three subjects?

(a) 25 (b) 35 (c) 15 (d) 20

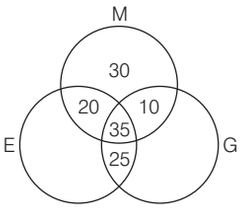
Solutions (Q. Nos. 138-141)

Given, $P(E) = 125$, $P(M) = 95$, $P(G) = 95$, $P(E \cap M) = 55$, $P(G \cap M) = 45$ and $P(G \cap E) = 35$

$P(E \cup M \cap G) = 200 - 10 = 190$

$\therefore \quad P(E \cap M \cap G) = P(E \cup M \cup G) - P(E) - P(M) - P(G) + P(E \cap M) + P(G \cap M) + P(G \cap E)$

$= 190 - 55 = 35$



138. ***(c)*** 139. ***(b)***
140. ***(a)*** 141. ***(b)***

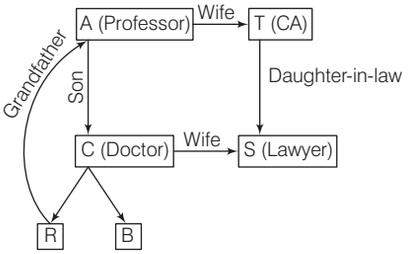
142. Answer the question based on the details given in the paragraph.

In a family of six, there are three men A, B and C and three women R, S and T. The six persons are Architect, Lawyer, CA, Professor, Doctor and Engineer by profession, but not in the same order. There are two married couples and two unmarried persons. C is not R's husband. The Doctor is married to a Lawyer. R's grandfather is a Professor. B is neither A's son, nor he is an Architect or a Professor. The Lawyer is T's daughter-in-law. C is T's son and the Engineer's father. A is married to the CA.

Which among the following is the correct pair of married couples?

(a) AS and CR
(b) AC and SR
(c) AT and CS
(d) AT and RB

Sol. (c) The family diagram can be drawn as:



Hence, the correct pair of married couples are AT and CS.

143. Five friends A, B, C, D and E have different jobs. They work with either Bank or Cinema or Pharma or Sales or Jewellery. Pick the right job in which they are employed if B is neither concerned with Bank nor with Sales. E has nothing to do with Cinema or Jewellery. C is not in Bank or Cinema. D has nothing to do in Sales or Pharma. Cinema has not given a chance to B or D nor E is a bank employee. C is not in Sales nor B is in Jewellery.

Which of the following is correct?

(a) A-Cinema; B-Sales; C-Jewellery; D-Pharma; E-Bank
(b) A-Cinema; B-Sales; C-Pharma; D-Bank; E-Jewellery
(c) A-Cinema; B-Sales; C-Jewellery; D-Bank; E-Pharma
(d) A-Cinema; B-Pharma; C-Jewellery; D-Bank; E-Sales

Sol. (d) The given information can be summarised as

Friends / Jobs	A	B	C	D	E
Bank	×	×	×	✓	×
Cinema	✓	×	×	×	×
Pharma	×	✓	×	×	×
Sales	×	×	×	×	✓
Jewellery	×	×	✓	×	×

Hence, option (d) is correct.

144. In the following question, a statement or two are followed by two conclusions.

Statement Of all the television sets manufactured in India, the 'Solar' brand as the largest sale.

Conclusions

I. The volume of sales of all the brands of television sets manufactured in India is known.
II. The production of no other television set in India is as large as that of 'Solar'.

(a) Conclusion I follows
(b) Conclusion II follows
(c) Both I and II follow
(d) Either I or II follows

Sol. (c) As given in statement that, the solar brand has the largest sale. So, it is clear that the volume of sales of all the brands of television sets manufactured in India is known. So, Conclusion I follows, because sale of solar brand is largest. So, production of no other television set in India is as large as that of 'solar'.

So, Conclusion II follows.

Directions (Q. Nos. 145-148) *Read the following information carefully and answer the questions that follow.*

Radhika, Neha, Rupa, Mahima, Sayali and Shobha joined six different renowned MBA institutes A, B, C, D, E and F and opted for six different specialisations in Management—Finance, IT, Operations, HR, Marketing and General Management. Each of them has a different educational qualification out of BCA, B.Com., CA, BBM, B.Sc. and B.Tech.

The following information is available about them

(i) Mahima joined institute B and did not take up Marketing or IT and is not a B.Tech.

(ii) The person who joined institute E opted for HR and is not a CA or a B.Sc.

(iii) The person, who is a BCA graduate, joined the institute D and took IT. Whereas, the person, who is B.Com graduate, took up General Management.

(iv) Shobha is a CA and has taken Finance.

(v) The person, who is a B.Tech. graduate has taken Operations Management and did not join institute A or C.

(vi) Radhika joined institute F, Rupa is a BCA graduate and Neha took up HR.

145. What is the educational qualification of Mahima and what is the course she opted for?
(a) B.Sc. and Finance
(b) B.Com. and General Management
(c) BBM and HR
(d) None of the above

146. Which institute did the B.Tech. Graduate join?
(a) F
(b) D
(c) A
(d) None of the above

147. The educational qualification of Sayali is
(a) B.Sc.
(b) BCA
(c) BBM
(d) CA

148. If Shobha joined the institute C, then the student who joined the institute A opted for
(a) Finance
(b) Marketing
(c) General Marketing
(d) HR

Solutions (Q. Nos. 145-148)

	A	B	C	D	E	F	Finance	IT	Operations	HR	Marketing	General Management	BCA	B.Com.	CA	BBM	B.Sc.	B.Tech.
Radhika	×	×	×	×	×	✓	×	×	✓	×	×	×	×	×	×	×	×	✓
Neha	×	×	×	×	✓	×	×	×	×	✓	×	×	×	×	×	✓	×	×
Rupa	×	×	×	✓	×	×	×	✓	×	×	×	×	✓	×	×	×	×	×
Mahima	×	✓	×	×	×	×	×	×	×	×	×	✓	×	✓	×	×	×	×
Sayali	—	×	—	×	×	×	×	×	×	×	✓	×	×	×	×	×	✓	×
Shobha	—	×	—	×	×	×	✓	×	×	×	×	×	×	×	✓	×	×	×

145. ***(b)*** The educational qualification of Mahima is B.Com. and the course is General Management.

146. ***(a)***

147. ***(a)***

148. ***(b)***

Directions (Q. No. 149-150) *Read following the information carefully and answer the given questions.*

In a village, there used live three types of tribes SIM, LIM and JIM. SIM always tells the truth, JIM always lie and LIM tell the truth and lie alternating. (They can tell truth and lie in any sequence). Three persons (of different tribes) from the village gave following statements :

NICE says "RICE is of SIM tribe; I am of JIM tribe".
MICE says "NICE is of JIM tribe; I am of SIM tribe".
RICE says "MICE is of JIM tribe; I am of LIM tribe".

149. Who is telling the truth always?
(a) NICE
(b) MICE
(c) RICE
(d) None of the above

150. Who is always speaking false?
(a) NICE
(b) RICE
(c) MICE
(d) None of the above

Solutions (Q. Nos. 149-150) From the given information, it can be concluded
MICE is of SIM tribe, therefore is always true.
NICE is of JIM tribe, therefore is always false
RICE is of LIM tribe, therefore tells truth and lies alternatively.

149. ***(b)*** 150. ***(a)***

SNAP

Symbiosis National Aptitude Test

Solved Paper 2013

Section A
(Quantitative, Data Interpretation & Data Sufficiency)

For all questions in this section, each correct answer carries 1 mark.

1. Dinesh can do a piece of work in 15 days and Harish can do it in 12 days. If Dinesh, Harish and Subodh together can do it in 4 days, in how many days can Subodh alone do this work?

(a) 12 (b) 15 (c) 10 (d) 8

➥ *(c)* Dinesh can do a piece of work in 1 day $= \frac{1}{15}$ work

Harish can do a piece of work in 1 day $= \frac{1}{12}$ work

Now, according to the question,

Dinesh, Harish and Subodh together can do in 1 day

$= \frac{1}{4}$ work

So, $\frac{1}{15} + \frac{1}{12} + \frac{1}{\text{Subodh}} = \frac{1}{4}$

$\Rightarrow \frac{1}{\text{Subodh}} = \frac{1}{4} - \left(\frac{1}{12} + \frac{1}{15}\right) = \frac{1}{4} - \left(\frac{5+4}{60}\right) = \frac{1}{4} - \frac{9}{60}$

$\Rightarrow \frac{1}{\text{Subodh}} = \frac{15-9}{60} = \frac{6}{60} = \frac{1}{10}$ work

So, Subodh can alone do this work in 10 days.

2. A village has 2000 people, out of which 1100 are male and 900 are female. The overall literacy rate is 95%, while only 3% male are not literate. What percentage of the female population is literate?

(a) 92.55% (b) 93.65%
(c) 94% (d) None of these

➥ *(a)* In village, total number of people = 2000

Number of males = 1100

Number of females = 900

$\because$ Number of male illiterate $= \frac{1100 \times 3}{100} = 33$

$\therefore$ Literate $= 1100 - 33 = 1067$

Overall literacy rate = 95 %

Let women literacy rate = x

Now, according to the question,

$\Rightarrow 2000 \times \frac{95}{100} = 1067 + \frac{900 \times x}{100}$

$\Rightarrow 20 \times 95 = 1067 + 9x$

$\Rightarrow 9x = 1900 - 1067 \Rightarrow 9x = 833$

$\because x = \frac{833}{9} = 92.55\%$

3. A sum of money is sufficient to pay Ravi's wages for 18 days or Ajit's wages for 12 days. The same money is sufficient to pay the wages of both for how many days?

(a) 9 days (b) 8.5 days
(c) 7.2 days (d) 5.5 days

➥ *(c)* Let Ravi's wages be ₹ x.

Now, Ravi's wages ₹ x for 1 work in 18 days.

So, Ravi's wages ₹ x for 1 day $= \frac{1}{18}$ work

Similarly, Ajit's wages ₹ x for 1 day $= \frac{1}{12}$ work

Now, taken together Ravi's and Ajit's wages ₹ x for 1 day

$= \frac{1}{18} + \frac{1}{12} = \frac{5}{36}$ work

$\because$ Ravi's and Ajit's wages ₹ x for 1 work

$= \frac{36}{5}$ days $= 7.2$ days

4. What is the number to be added to 1/4th of 30% of 120, so that it is 50% more than 25% of 40?

(a) 6 (b) 7 (c) 5 (d) 4

➥ *(a)* Let the number to be added = x

Now, according to the question,

$$x + \frac{1}{4} \times \frac{30\% \times 120}{100} = \frac{50\% \times 25\% \times 40}{100 \times 100} + \frac{25 \times 40}{100}$$

$$\Rightarrow \quad x + 9 = 5 + 10 \quad \Rightarrow \quad x = 15 - 9$$

$$\therefore \quad x = 6$$

5. 10 yr ago the average age of all 25 teachers of a school was 45 yr and 4 yr, ago 1 teacher retired from her post at the age of 58. After 1 yr a new teacher whose age was 54 yr was recruited from outside. The present average age of all the teachers is

(a) $55\frac{17}{25}$ yr (b) $55\frac{4}{5}$ yr (c) $54\frac{18}{25}$ yr (d) $49\frac{1}{2}$ yr

➥ *(c)* 10 yr ago, average age of 25 teachers

$$\Rightarrow \quad \frac{(x_1 - 10) + (x_2 - 10) + \cdots + (x_{25} - 10)}{25} = 45$$

$$\Rightarrow \quad \frac{x_1 + x_2 + \cdots + x_{25}}{25} - \frac{250}{25} = 45$$

$$\Rightarrow \quad \frac{x_1 + x_2 + \cdots + x_{25}}{25} = 45 + 10 = 55$$

According to the question, age of one teacher

$$x - 4 = 58 \quad \therefore \quad x = 58 + 4 = 62$$

$$\frac{x_1 + \cdots + x_{24} + 62}{25} = 55$$

$$\Rightarrow \quad \frac{x_1 + \cdots + x_{24}}{24} = 55 - 2.58$$

$$\Rightarrow \quad \frac{x_1 + \cdots + x_{24}}{24} = 52.42$$

Again, according to the question, new teacher's age

$$x - 4 + 1 = 54$$

$$\therefore \quad x = 57$$

So, $$\frac{x_1 + \cdots + x_{24} + 57}{25} = 52.42 + \frac{57}{25}$$

$$\Rightarrow \quad \frac{x_1 + \cdots + x_{24} + 57}{25} = 54.7$$

6. X can complete a task in 15 days and Y can complete the same task in 20 days. X started the task alone and after 5 days Y joined X. After working together for 3 days, X had to leave the work. How long did it take for Y to complete the remaining work?

(a) 4.5 days (b) 5 .3 days
(c) 2.5 days (d) 6.3 days

➥ *(d)* $X - 15 \quad 4$

$Y - 20 \quad 3 \quad 60$

X can do in 15 days and Y can do in 20 days.

LCM of 15 and 20 = 60 days.

Now, total number of work = 60

X works for 5 days.

So, his work = $5 \times 4 = 20$

Remaining work = $60 - 20 = 40$

X and Y together work for three days = $(4 + 3) \times 3 = 21$

Again, remaining work = $40 - 21 = 19$

X leave work means remaining work is completed by Y.

So, Y can complete remaining work = $\frac{19}{3} = 6.3$ days

7. Find out the value of

$$\frac{[(2.093 + 0.089)^2 - (2.093 - 0.089)^2]}{[(2.093)(0.089)]}.$$

(a) 4 (b) 3.79 (c) 4.12 (d) 2.09

➥ *(a)* $\frac{[(2.093 + 0.089)^2 - (2.093 - 0.089)^2]}{[(2.093)(0.089)]}$

As we know that, $a^2 - b^2 = (a + b)(a - b)$

$$= \frac{(2.093 + 0.089 - 2.093 + 0.089)(2.093 + 0.089 + 2.093 - 0.089)}{2.093 \times 0.089}$$

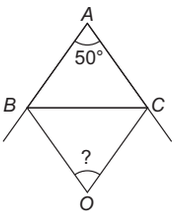
$$= \frac{(0.089 + 0.089) \times (2.093 + 2.093)}{2.093 \times 0.089}$$

$$= \frac{2 \times (0.089) \times 2 \times (2.093)}{2.093 \times 0.089} = \frac{4 \times 0.089 \times 2.093}{2.093 \times 0.089} = 4$$

8. In a ΔABC, the measure of $\angle A$ is 50° and the external bisectors of $\angle B$ and $\angle C$ meet at O. The measure of $\angle BOC$ is

(a) 140° (b) 40°
(c) 105° (d) 65°

➥ *(d)*



Here, $$\angle CBO = \frac{180 - \angle ABC}{2} \quad \text{... (i)}$$

Similarly, $$\angle BCO = \frac{180 - \angle ACB}{2} \quad \text{... (ii)}$$

According to the question,

$\angle CBO$ and $\angle BCO$ are angle bisectors of external angles

Now, adding Eqs. (i) and (ii)

$$\angle CBO + \angle BCO = \frac{360° - \angle ABC - \angle ACB}{2} \quad \text{...(iii)}$$

Here, ABC is triangle in which $\angle A = 50°$.

Bisector of angle B and C meet at O (external bisector).

As we know that,

$$\angle BAC + \angle ABC + \angle ACB = 180°$$

$$\therefore \quad \angle ABC + \angle ACB = 180 - \angle A \quad \text{...(iv)}$$

Again, in ΔBOC

$$\angle CBO + \angle BOC + \angle OCB = 180°$$

Now, from Eq. (iii),

$$\frac{360° - \angle ACB - \angle ABC}{2} + \angle BOC = 180°$$

$$\Rightarrow \quad 360° - (\angle ACB + \angle ABC) + 2\angle BOC = 360°$$

$$\Rightarrow \quad \angle ACB + \angle ABC = 2\angle BOC$$

Again, from Eq. (iv),

$$180° - \angle A = 2\angle BOC$$

$$\therefore \quad \angle BOC = \frac{180 - \angle A}{2} = 90 - \frac{\angle A}{2}$$

$$= 90 - \frac{50}{2} = 90 - 25 = 65^{\circ}$$

9. The average salary of all the 60 employees in an office is ₹12000 per month. If the number of executives is twice the number of non-executive employees, then the average salary of all the non-executive employees is

(a) ₹ 9000 (b) ₹ 8000
(c) ₹ 6000 (d) Cannot be determined

➥ *(d)* Let the number of employees is 60.

So, average salary of employees

$$\frac{x_1 + x_2 + x_3 + \cdots + x_{60}}{12} = 12000$$

Now, according to the question

Number of non-executives = x

Number of executives = $2x$

So, 20 employees are non-executive and 40 employees are executive but for finding the average salary of non-executive employees it must be given the salary of only one class otherwise it will not be possible to find average salary.

10. Two pots A and B contain syrup and water in the ratio 3 : 7 and 5 : 2, respectively. Find the ratio in which these mixtures could be mixed to get a new mixture containing syrup and water in the ratio of 2 : 3.

(a) 22 : 7 (b) 4 : 3
(c) 16 : 9 (d) None of these

➥ *(a)* Let them be mixed in ratio $x : y$.

In 1st mixture Syrup $= \frac{3x}{10}$

Water $= \frac{7x}{10}$

In 2nd mixture Syrup $= \frac{5y}{7}$

Water $= \frac{2y}{7}$

Now, according to the question,

$$\frac{3x}{10} + \frac{5y}{7} : \frac{7x}{10} + \frac{2y}{7} = \frac{2}{3}$$

$$\Rightarrow \quad \frac{21x + 50y}{70} : \frac{49x + 20y}{70} = \frac{2}{3}$$

$$\frac{21x + 50y}{49x + 20y} = \frac{2}{3}$$

$$\Rightarrow \quad 63x + 150y = 98x + 40y$$

$$110y = 35x$$

$$\frac{x}{y} = \frac{110}{35} = \frac{22}{7}$$

11. A truck carrying goods has travelled for 3 days. The 1st day it travelled for 12h at an average speed of 40 km/h, the 2nd day it travelled for 10 h at an average speed of 50 km/h, the third day it completed 360 km travelling for 8 h. The average speed of the truck (in km/h) for all the three days is

(a) 45.22 (b) 49.58 (c) 43.73 (d) 44.66

➥ *(d)* 1st day 12 h with average speed = 40 km/h

$\because$ 1st day covers total distance = $40 \times 12 = 480$ km

Similarly, 2nd day covers total distance

$= 10 \times 50 = 500$ km

$\because$ 3rd day covers total distance = 360 km

$$\text{Average speed} = \frac{\text{Total distance of three days}}{\text{Total time in three days}}$$

$$= \frac{480 + 500 + 360}{12 + 10 + 8}$$

$$= \frac{1340}{30} = \frac{134}{3} = 44.66 \text{ km/h}$$

12. The HCF of two numbers is 8 and their LCM is 96. If one of the numbers is 32, then the other is

(a) 24 (b) 42
(c) 16 (d) None of these

➥ *(a)* According to the formula,

HCF × LCM = First number × Second number

Given, HCF = 8 and LCM = 96

One of the numbers = 32

$$\Rightarrow \quad 8 \times 96 = 32 \times \text{Second number}$$

$$\therefore \text{Second number} = \frac{8 \times 96}{32} = 24$$

13. The ratio between present ages of A and B is 3 : 4. The present age of B is 10 yr more than the age of A 5 yr ago. Find out the present age of B.

(a) 12 yr (b) 17yr
(c) 20 yr (d) None of these

➥ *(c)* Ratio of present ages of A and $B = \frac{3}{4}$

Let present age of $A = x$

5 yr before, A's age $= x - 5$

According to the question,

B's present age $= x - 5 + 10 = x + 5$

Now, $\frac{3}{4} = \frac{x}{x + 5}$

$$\Rightarrow \quad 3x + 15 = 4x \quad \Rightarrow \quad x = 15 \text{ yr}$$

$\therefore$ Present age of $B = 5 + 15 = 20$yr

14. The circumference of the front wheels of a vehicle is 3.5 m and that of the back wheels is 3.0 m. If the vehicle is moving at a speed of 54 km/h, the shortest time in which both the wheels will make a whole number of turns is

(a) 6.4 s (b) 4 s (c) 2.1 s (d) 1.4 s

➥ *(d)* The circumference of front wheels and back wheels are 3.5 m and 3.0 m.

$$\text{Front wheels} = 3.5\text{m} = \frac{35}{10}\text{m}.$$

$$\text{Back wheels} = 3.0\,\text{m} = \frac{30}{10}\text{m}.$$

Now, LCM of front wheels and back wheels circumference is $= \frac{210}{10} = 21\,\text{m}$.

So, $\text{Speed} = 54\,\text{km/h} = 54 \times \frac{5}{18}\text{m/sec}.$

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\text{Time} = \frac{21\,\text{m}}{54 \times \frac{5}{16}\,\text{m/sec}} = \frac{21 \times 18}{54 \times 5} = \frac{7}{5} = 1.4\,\text{sec}$$

15. If p is 81% of q, then what percentage of p is q?

(a) 81% (b) 101.9% (c) 112.3% (d) 123.45%

➥ *(d)* $p \quad q$

81 100

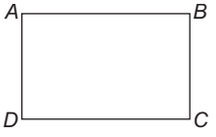
$$\therefore \text{Percentage of } p \text{ is } q\% = \frac{100}{81} \times 100$$

$$= \frac{100 \times 100}{81} = 123.45\%$$

16. The length of a rectangular plot is 15m more than its breadth. If the cost of fencing the plot is ₹ 2700 at the rate ₹ 30 per metre, what is the length (in metre) of the plot?

(a) 25 (b) 35
(c) 30 (d) None of these

➥ *(c)*



Let the breadth of a rectangular plot be x m.

Length = $(x + 15)$ m

Perimeter of rectangular plot = $2(l + b)$

So, $2(x + x + 15) \times 30 = 2700 \Rightarrow 2(2x + 15) \times 30 = 2700$

$\Rightarrow \quad 4x + 30 = 90 \quad \Rightarrow \quad 4x = 60 \quad \Rightarrow \quad x = 15\text{m}$

So, length of plot = 15 + 15 = 30 m

17. In a class, 70% of the students opted for History, 45% of the students opted for Geography and 20% opted for neither History nor Geography. What percentage of the class opted for both the subjects?

(a) 35% (a) 25% (c) 30% (d) 20%

➥ *(c)* Number of students selected History $= 80 \times \frac{70}{100} = 56$

Number of students selected Geography

$= (80 \times 45)/100 = 36$

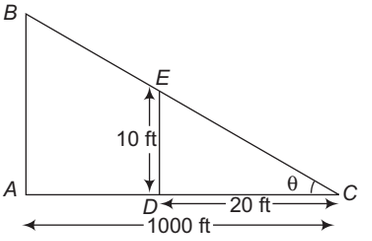
Number of students who selected History and Geography both = 24

Percentage of students who selected History and Geography both subjects $= \frac{24 \times 100}{80} = 30\,\%$

18. A boy lying on the ground 1000 ft away from a multistored building, can see the top of the building exactly in line with the top of a 10 ft tall flag post. If the flag post is 20 ft away from the boy, what is the height of the building?

(a) 500 ft (b) 600 ft (c) 800 ft (d) 1000 ft

➥ *(a)*



In ΔABC, $\tan\theta = \frac{AB}{AC} = \frac{AB}{1000}$ ...(i)

Again, in ΔDEC $\tan\theta = \frac{DE}{DC} \Rightarrow \tan\theta = \frac{10}{20} = \frac{1}{2}$ ...(ii)

From Eqs. (i) and (ii), $\frac{AB}{1000} = \frac{1}{2}$

$\therefore \quad AB = 500\,\text{ft}$

19. 21 boys turn out for an NCC parade. Out of these, 19 are wearing NCC caps and 11 are wearing NCC shoes. There are no boys without one or the other. How many boys are wearing the full uniform?

(a) 2 (b) 6 (c) 9 (d) 10

➥ *(c)* Total number of boys = 21

Number of boys wearing Wear NCC caps = 19

Number of boys wearing NCC shoes = 11

$\therefore$ Full uniform boys = (19 + 11) – 21 = 9

20. A number is increased by 20% and then is reduced by 20%. The number

(a) does not change (b) decreases by 4%
(c) increases by 4% (d) increases by 8%

➥ *(b)* By formula , $x + y + \frac{xy}{100}$

Here, $x = +20$ and $y = -20$

$$= +20 - 20 + \frac{20 \times (-20)}{100} = -4\%$$

So the number is decreased by 4%.

21. In a class of 42, there are 22 boys and 20 girls. The class teacher has to appoint a girl monitor and a boy monitor. In how many ways can the selection be made?

(a) 22 (b) 420 (c) 42 (d) 440.

➥ *(c)* 42 < 22 Boys, 20 Girls

Selection of monitor = ${}^{22}C_1 + {}^{20}C_1 = 22 + 20 = 42$

22. Arvind and Mohan are 20 km apart. If they travel in opposite directions, they meet each other after half an hour. If they travel in the same direction, they meet after 2 h. If Arvind travels faster than Mohan, his speed is

(a) 15 km/h (b) 20 km/h
(c) 25 km/h (d) 30 km/h

➥ *(a)*

Arvind Mohan
20 km

Both travel in opposite direction

A B
20 km

Relative velocity = $V_1 + V_2$

Here, $V_1 = 10$

Velocity of B, $V_1 + V_2 = \frac{x}{t} = \frac{20}{1/2} \Rightarrow V_1 + V_2 = 40$ ...(i)

Again, they both travel in same direction and meet each other after 2 h

So, $V_1 - V_2 = \frac{20}{2} = 10$ ...(ii)

On solving Eqs. (i) and (ii),

$\Rightarrow V_1 = \frac{50}{2} = 25$ km/h

Now, put the value of V_1 in Eq. (i)

$25 + V_2 = 40 \Rightarrow V_2 = 40 - 25 = 15$ km/h

Arvind's speed = 15 km/h

23. Let $A = \{1, 2, 3, 4, 5, 6\}$

How many subsets of A can be formed with just two elements, one even and one odd?

(a) 6 (b) 8 (c) 9 (d) 10

➥ *(c)* $A = \{1, 2, 3, 4, 5, 6\}$

Set of even and odd numbers

= (2, 1) (2, 3) (2, 5) (4, 1) (4, 5) (4, 3) (6, 1) (6, 3) (6, 5)

= Total number of subsets 9.

24. A grocer sells a soap marked ₹ 30 at 15% discount and gives a shampoo sachet costing ₹ 1.5 free with each soap. He, then makes 20% profit. His cost price per soap is

(a) ₹ 15 (b) ₹ 16 (c) ₹ 18 (d) ₹ 20

➥ *(d)* A soap's marked price = ₹ 30

Let the cost price per soap be ₹ x.

According to the question,

$$30 \times \frac{(100 - 15)}{100} - 1.5 = x \times \frac{120}{100}$$

$\Rightarrow 30 \times \frac{85}{100} - 1.5 = x \times \frac{12}{10} \Rightarrow 30 \times \frac{17}{20} - 1.5 = x \times \frac{6}{5}$

$\Rightarrow \frac{51}{2} - 1.5 = \frac{6x}{5} \Rightarrow 25.5 - 1.5 = \frac{6x}{5}$

$\Rightarrow 24 = \frac{6x}{5}$

$\therefore x = 20$

25. Which one of the following is correct?

(a) $\log(m + n) = \log m + \log n$
(b) $\log(m - n) = \log m - \log n$
(c) $\log(m \times n) = \log m + \log n$
(d) $\log m^n = (\log m)^n$

➥ *(c)* $\log(m \times n) = \log m + \log n$

26. The sum of two numbers is 15, their product is 30. The sum of their reciprocals is

(a) 2 (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{4}{15}$

➥ *(c)* Let α and β be the two numbers.

So that, their reciprocals are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.

According to the question,

$\alpha + \beta = 15$ ...(i)

and $\alpha\beta = 30$ ...(ii)

Then, $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{15}{30} = \frac{1}{2}$

27. Three persons Amar, Akbar and Anthony agree to pay their hotel bills in the ratio of 3 : 4 : 5. Amar pays the first day's bill which amounts to ₹ 26.85. Akbar pays the second day's bill which amounts to ₹ 42.75 and Anthony pays the third day's bill which amounts to ₹ 53. When they settle their accounts, which of the following happens?

(a) Amar gives ₹ 3 to Anthony
(b) Akbar gives ₹ 2 to Amar
(c) Amar gives Akbar ₹ 1.95 and ₹ 2 to Anthony
(d) None of the above

➥ *(c)* Amar : Akbar : Anthony

3 : 4 : 5

Total amount paid by these three persons

$= 26.85 + 42.75 + 53 = 122.6$

Akbar paid amount = $\frac{4}{12} \times 122.6 =$ ₹ 40.86

Amar paid amount = $\frac{3}{12} \times 122.6 =$ ₹ 30.65

Anthony paid amount = $\frac{5}{12} \times 122.6 =$ ₹ 51.08

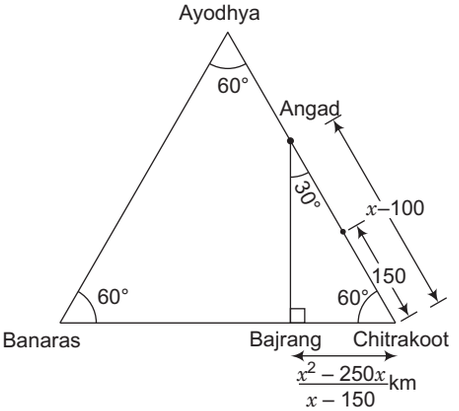
So, the option (c) is correct.

28. Inspired by the 'Golden quadrilateral project', the UP Government recently accomplished a diamond triangular project. Under this project, the State Government laid down 6 lane roads connecting three cities Ayodhya, Banaras and Chitrakoot, which are equally separated from each other *i.e.,* in terms of geometry they form an equilateral triangle. Angad and Bajrang start simultaneously from Ayodhya and Banaras respectively, towards Chitrakoot. When Angad covers 100 km, Bajrang covers such a distance that the distance between Angad and Bajrang makes 90° angle with the road joining Banaras to Chitrakoot. When Bajrang reaches Chitrakoot, Angad is still 150 km away from Chitrakoot.

What is the distance between Ayodhya and Chitrakoot?

(a) 250 km (b) 450 km
(c) 300 km (d) None of these

➥ *(c)* According to the question,



Ayodhya, Banaras and Chitrakoot made equilateral triangle and distance between them is equal. Let the distance be x km.

Now,

When Bajrang reaches Chitrakoot, then Angad is 150 km away from Chitrakoot.

distance travelled by Bajrang when Angad travells 100km $= \frac{100x}{x-150}$

Rest distance of Bajrang $= x - \frac{100x}{x-150} = \frac{x^2-250x}{x-150}$

In ΔDOC

$$\cos 60° = \frac{DC}{OC} = \frac{b}{h} \Rightarrow \frac{1}{2} = \frac{\frac{x^2-250x}{x-150}}{(x-100)}$$

$$\Rightarrow \frac{1}{2} = \frac{x^2-250x}{(x-150)(x-100)}$$

$$\Rightarrow x^2-150x-100x+15000 = 2x^2-500x$$

$$\Rightarrow x^2-250x+15000 = 2x^2-500x$$

$$\Rightarrow x^2-250x-15000 = 0$$

$$\Rightarrow x^2-300x+50x-15000 = 0$$

$$\Rightarrow x^2(x-300)+50(x-300) = 0$$

$$\Rightarrow (x-300)(x+50) = 0$$

$$\Rightarrow x = 300, x \neq -50$$

So, distance between Ayodhya and Chitrakoot is 300 km.

29. If $(1-1/2)\ (1-1/3)\ (1-1/4)\ (1-1/5)...(1-1/40) = \frac{x}{40}$, then what is the value of x ?

(a) 13 (b) 39 (c) 21 (d) 1

➥ *(d)* $\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{5}\right)...\left(1-\frac{1}{40}\right) = \frac{x}{40}$

$$\Rightarrow \frac{1}{2}\times\frac{2}{3}\times\frac{3}{4}\times\frac{4}{5}\times\frac{5}{6}\times\cdots\times\frac{39}{40} = \frac{x}{40} \Rightarrow \frac{1}{40} = \frac{x}{40}$$

$$\therefore x = 1$$

30. The smallest number which when divided by 12, 16, 18 gives 7 as remainder in each case, is

(a) 103 (b) 135 (c) 151 (d) 187

➥ *(c)* 151 is the smallest number which when divided by 12, 16 and 18, gives remainder 7.

31. If $\left(\frac{a}{b}\right)^{2x-3} = \left(\frac{b}{a}\right)^{x-9}$, then x is equal to

(a) – 6 (b) – 2 (c) 12 (d) 4

➥ *(d)* $\left(\frac{a}{b}\right)^{2x-3} = \left(\frac{b}{a}\right)^{x-9}$

As we know that, $\left(\frac{a}{b}\right)^{x} = \left(\frac{1}{b/a}\right)^{-x}$

$$\Rightarrow \left(\frac{a}{b}\right)^{2x-3} = \left(\frac{1}{a/b}\right)^{x-9} \Rightarrow \left(\frac{a}{b}\right)^{2x-3} = \left(\frac{a}{b}\right)^{-(x-9)}$$

Both having same base, so

$(2x-3) = -(x-9) \Rightarrow 2x-3 = -x+9 \Rightarrow 3x = 12$

$$\therefore x = 4$$

32. The population of vultures in a locality decreases by a certain rate (compounded annually). If the current population of vultures is 29160 and the ratio of the decrease in population for the second year and the third year be 10: 9, then what was the population of vultures 3 yr ago?

(a) 30000 (b) 35000 (c) 40000 (d) 50000

➥ *(c)* Given Ratios

$$\Rightarrow \frac{\text{Decrease in population in 2nd yr}}{\text{Decrease in population in 3 rd yr}} = \frac{10}{9}$$

So, $\frac{100}{100-r} = \frac{10}{9} \Rightarrow 900 = 1000-10r$

$$\Rightarrow 10r = 1000-900 \Rightarrow 10r = 100$$

$$\because r = 10\%$$

Now, Formula for 3 yr when compounded annually decrease

$\Rightarrow$ $29160 = \text{Present population}\left(1-\frac{10}{100}\right)^3$

$\Rightarrow$ $29160 = \text{Present population}\left(\frac{90}{100}\right)^3$

$\because$ Present population $= \frac{29160\times100\times100\times100}{90\times90\times90} = 40000$

So, 3 yr ago population of vultures was 40000

33. Anna sold his car to Boney at a profit of 20% and Boney sold it to Chakori at a profit of 10 %. Chakori sold it to a mechanic at a loss of 9.09%. The mechanic spent 10% of his purchase price on repairs and then sold it at a profit of 8.33% to Anna once again.

(a) Anna's loss is 23% (b) Anna's loss is 29%
(c) Anna's gain is 5% (d) Anna's gain is 4%

➥ *(a)* Anna $\xrightarrow{+20\%}$ Boney $\xrightarrow{+10\%}$ Chakori $\xrightarrow{-9.09\%}$ Mechanic $\xrightarrow[+8.33\%]{-10\%}$ Anna (again)

Suppose, Anna's cost of his car = 100

Then, Anna takes profit of 20% = $100\times\frac{120}{100}=120$

Boney takes profit of 10% = $120\times\frac{110}{100}=132$

Chakori takes loss of 9.09%

$$=\frac{132\times(100-9.09)}{100}=120.0012$$

Now,mechanic spents 10% on car's maintenance.

$\therefore$Car's price for mechanic = $120.0012\times\frac{110}{100}=132.00132$

Again, mechanic takes profit of 8.33%

$$=132.00132\times\frac{108.33}{100}=₹142.9978$$

Anna's loss = 142.99 – 120 =22.99

Since, Anna sells the car at ₹120.50.

Anna having money from this car is ₹ 120

$$=22.99=23\%\,(\text{loss})$$

34. A circle and a square have the same perimeter. Then,

(a) the area of the circle is greater
(b) the area of the square is $1/\pi$ times that of the circle
(c) the area of the square is greater than that of a circle
(d) the areas are equal

➥ *(a)* Suppose a circle having radius = r and square having sides a.

From the question, both circle and square have equal perimeter, then

$$2\pi r=4a \Rightarrow r=\frac{4a}{2\pi} \quad \ldots(i)$$

$\therefore$ $$a=\frac{2\pi r}{4}=\frac{\pi r}{2} \quad \ldots(ii)$$

Area of circle = $\pi r^2=\pi\left(\frac{4a}{2\pi}\right)^2$

Put the value of $r=\frac{4a}{2\pi}=\pi\frac{16a^2}{4\pi^2}=\frac{16a^2}{4\pi}=\frac{4a^2}{\pi}=\frac{28}{22}a^2$

Area of square = a^2

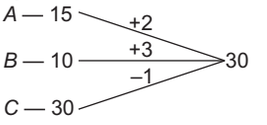
On comparing, area of square with area of circle, we find

The area of circle is greater than the area of square.

35. Two taps A and B can fill a cistern in 15 h and 10 h, respectively. Tap C can empty full cistern in 30 h. All the three taps were opened for 2 h, when it was remembered that the emptying tap had been left open, it was then closed. How many more hours would it take for the cistern to be filled?

(a) 30 min (b) 1.2 h (c) 24 min (d) 35 min

➥ *(b)* A and B fill a cistern in 15 h and 10 h respectively but C can empty it in 30 h. So, LCM of A, B and C is 30.



Now, A, B and C all these three taps are open for

2 h = (2 + 2) + (3 + 3) + (– 1 – 1) = 4 + 6 – 2 = 8 L

Remaining to fill up = 30 – 8 = 22 L

Now, tap C is closed, then taps A and B are open to fill the tap cistern 1 h = 2 + 3 = 5 L

Time taken by taps A and $B=\frac{22}{5}=4.4$ h

36. At Atharva Prakashan every book goes through 3 phases-typing, composing and binding. There are 16 typists, 10 composers and 15 binders. A typist can type 8 books in each hour, a composer can compose 12 books in each hour and a binder can bind 12 books in each hour. All the people at Atharva Prakashan work for 10 h per day and each person is trained to do only one type of job. How many books can be prepared in each day?

(a) 1500 (b) 1200 (c) 1440 (d) 1380

➥ *(b)* At Atharva Prakashan

Every employee work for 10 h and each employee is trained to do one type of job.

So, 16 typists type books in 1 h = 16 × 8

16 typists type books in 10 h = 16 × 8 × 10 = 1280

Composer

10 composers compose books in 1 h = 10 × 12

10 composers compose books in 10 h = 10 × 12 × 10 = 1200

Similarly, **Binding**

15 binders bind books in 1 h = 15 × 12

15 binders bind books in 10 h = 15 × 12 × 10 = 1800

In one day a book passes through three processes so, only 1200 books are prepared.

37. A contractor employed a certain number of workers to finish constructing a road in a certain scheduled time. Some time later, when a part of work had been completed, he realised that the work delayed by three-fourth of the scheduled time. So, he at once doubled the number of workers and thus he managed to finish the road on the scheduled time. How much work had been completed before increasing the number of workers?

(a) $\frac{77}{9}\%$ (b) $\frac{100}{7}\%$
(c) 20% (d) None of these

➥ *(d)* $\frac{m_1 d_1 t_1}{w_1} = \frac{m_2 d_2 t_2}{w_2} \Rightarrow \frac{x \times dt}{w_1} = \frac{m_2 \times d_2 (t + \frac{3}{4}t)}{w_2}$

$\therefore$ Total work increase $= \frac{3/4t}{t} \times 100 = 75\%$

Directions (Q. Nos. 38-40) *Read the information carefully to answer the questions that follow.*

A train enters into a tunnel *AB* at A and exits at *B*. A jackal is sitting at *O* in another by-pass tunnel *AOB*, which is connected to *AB* at *A* and *B*, where *OA* is perpendicular to *OB*. A cat is sitting at *P* inside the tunnel *AB* making the shortest possible distance between *O* and *P*, such that $AO : PB = 30 : 32$. A train before entering into the tunnel AB blows the whistle (or siren) somewhere before *A*. On hearing the whistle, if the jackal and the cat run towards *A*, they would meet with an accident with the train exactly at *A*, the entrance to the tunnel. Further, if the jackal moves towards *B* instead of *A* (on hearing the whistle) it would again meet with an accident exactly at *B*, the exit of the tunnel, with the same train coming from the same direction.

38. What is the ratio of speeds of the jackal and the cat?

(a) 4 : 3 (b) 5 : 3
(c) 1 : 1 (d) None of these

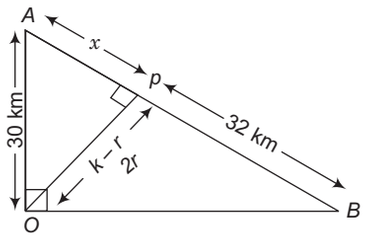
39. The ratio of speed of the jackal to the speed of train is

(a) 5 : 1 (b) 3 : 5
(c) 1 : 5 (d) None of these

40. If the tunnels were to be connected from *O* to *P* and the jackal moves from *O* to *A* through *P*, it will meet with train at M_1. Then, AM_1 is

(a) 20 km (b) 16 km
(c) 10 km (d) None of these

Hint (Q. Nos. 38–40) *AB* is tunnel, let $AP = 10x$, then



ΔAPO and ΔPOB are similar, then

$$AP = \frac{AO^2}{AB} \Rightarrow x = \frac{(30)^2}{32 + x} \Rightarrow (32 + x)\, x = 900$$

$$\Rightarrow 32x + x^2 = 900 \Rightarrow x^2 + 32x - 900 = 0$$

$$\therefore x = \frac{-32 \pm \sqrt{1024 - 4 \times (-900)}}{2} = \frac{-32 \pm \sqrt{1024 + 3600}}{2}$$

$$= \frac{-32 \pm \sqrt{4624}}{2} = \frac{-32 \pm 68}{2} = \frac{-32 + 68}{2} \text{ and } \frac{-32 - 68}{2}$$

$(\because x \neq \frac{-100}{2}$, it is negative)

$= 36/2 = 18$

38. *(a)* $\frac{\text{Speed of jackal}}{\text{Speed of cat}}$

$$= \frac{\text{Distance covered by jackal time}}{\text{Distance covered by cat time}}$$

$$= \frac{\frac{30}{t}}{\frac{18}{t}} = \frac{30}{18} = \frac{5}{3}$$

$\therefore$ Ratio = 5:3

39. *(d)* $\frac{\text{Speed of jackal}}{\text{Speed of cat}} = Po^2 + (18)^2 = (30)^2$

$$Pr^2 = 900 - 324$$

$$\Rightarrow Pr^2 = 576$$

$$\Rightarrow Po = 24$$

$$\frac{\text{Speed of jackal}}{\text{Speed of train}} = \frac{\text{Distance covered by jackal time}}{\text{Distance covered by train time}}$$

$$= \frac{\frac{40}{t}}{\frac{150 + 50}{t}} = \frac{40}{200} = \frac{1}{5}$$

$$\Rightarrow \frac{\text{Speed of jackal}}{\text{Speed of train}} = \frac{1}{5} = 1:5$$

40. *(c)* Since, when the train arrives at *A*, the jackal can move 30km. So, at the time when train is at *A* the jackal will cover 6km from *P* on *PA* in addition to 24 km at *OP*. Now, the rest distance at *AP* is 12 km, this remaining distance will be covered by train and jackal accending to their respective speeds.

Here, speeds of jackal to train = 1:5

So, distance covered by train $= 12 \times \frac{5}{6} = 10$ km. and

distance covered by jackal $= 12 \times \frac{1}{6} = 2$ km.

So, jackal will meet train at M_1 whcih is 10km away from *A*.

Section B
(General English)

For all questions in this section, each correct answer carries 1 mark.

Directions (Q. Nos. 41–44) *Choose the correct synonym of the word in capital letters from the four options given below.*

41. ENTRENCHED

(a) Filled up (b) Fortified
(c) Followed by (d) Kept down

➥ ***(b)*** 'Entrenched' means 'to put in a position of strength'. So, its correct synonym would be 'fortified' which also means 'to protect by strength'.

42. RECTITUDE

(a) renew (b) resort
(c) integrity (d) liberate

➥ ***(c)*** 'Rectitude' means 'having a strong moral virtue'. So, the correct synonym would be 'integrity' which too means the 'unity with a moral principle'.

43. INVIDIOUS

(a) unacceptable (b) enticing
(c) breathtaking (d) inverted

➥ ***(a)*** 'Invidious' means 'being hateful'. So, the correct synonym would be 'unacceptable' which also means 'not being loved'.

44. AFICIONADO

(a) affection (b) good friend
(c) tour guide (d) ardent fan

➥ ***(d)*** 'Aficionado' means 'someone who is deeply devoted to someone/something.' Correct synonym, therefore would be 'ardent fan' which has similar meaning.

Direction (Q. No. 45) *Choose the correct alternative for the following sentence.*

45. Despite immense development in almost all the fields, human still cannot control nature and ..... to do so.

(a) probably not be able
(b) probably will never able
(c) probably will never be able
(d) perhaps be never having ability

➥ ***(a)*** Among all the given option, (a) is the most appropriate choice.

Direction (Q. No. 46) *The sentence given below is jumbled and each part is numbered. You are required to arrange it in the most logical and grammatically acceptable sequence.*

46. a path/(a) put up a notice/(b) to another/(c) from one place/(d) the public/(e) Halton Borough Council/(f) about its/(g) plans to move/(h) to tell/(i)

(a) e,b,i ,f ,g,h ,a ,d ,c (b) f,h ,a ,d ,c ,b ,i ,g ,e
(c) e,h ,a ,d ,c ,b ,i ,g ,f (d) f,b ,i ,e ,g ,h ,a ,d ,c

➥ ***(d)*** The correct sentence order is
'Subject + Verb + Infinitive + Object + Complement.'

Directions (Q. Nos. 47-48) *Choose the correct spelling*

47. Disgraphia is a deficiency in the ability to write, spell or put down thoughts on paper. The underlined word is spelled incorrectly. Which of the following is the correct spelling?

(a) Disgrephia (b) Disgruphia
(c) Dysgraphya (d) Dysgraphia

➥ ***(d)*** Deficiency in the ability to write, spell or putting on thoughts is called 'Dysgraphia'. So, it is correct spelling.

48. PANGERYRIC means 'to pay tribute to'. The underlined word is spelt incorrectly. Mark the correct spelling.

(a) Panegyric (b) Penagyric
(c) Penagiryc (d) None of these

➥ ***(a)*** 'Paying tribute' is panegyric. So, it is correct spelling.

Directions (Q. Nos. 49-50) *Identify two most appropriate options from those given below that can complete the sentence without changing the meaning*

49. I ..... scorpions, they are diabolic.

1. hate 2. scare 3. loath
4. wary 5. cagey

(a) 1, 5 (b) 5, 3 (c) 1, 3 (d) 2, 4

➥ ***(d)*** Since, scorpions are dangerous, so people are scared of them.

50. To fully understand (A) ..... work such as George Orwell's Animal Farm, one must be able to differentiate (B) ..... events of the plot from the abundant extended metaphors.

Blank 'A'	Blank 'B'
1. an allegorical	4. literal
2. a tragic	5. superficial
3. a parabolic	6. figurative

(a) 2, 4 (b) 1, 4 (c) 3, 4 (d) 1, 5

➥ ***(d)*** 'Animal farm' is an allegorical writing. The characters in it are all superficial, so option (d) gives the right combination.

Direction (Q. No. 51) *Find out the two words that are nearly the same or opposite in meaning and indicate the correct combination provided in the four options*

51. 1. Attenuate 2. Repent 3. Make thin 4. Fore

(a) 1 and 3 (b) 3 and 2
(c) 2 and 4 (d) 1 and 4

➥ ***(a)*** 'Attenuate' means 'to make something weak'. On the other hand, 'Making thin' also has similarity with 'weakening'. So, '1and 3' is the right combination.

Direction (Q. No. 52) *Choose the word nearest in the meaning to the word underlined.*

52. A rapier is a

(a) baton (b) rascal
(c) quick sharp hit (d) long thin light sword

➥ ***(d)*** 'Rapier' is a long thin light sword.

Directions (Q. Nos. 53-54) *Choose the correct alternatives for the following idioms.*

53. Raise cain

(a) re-incarnate (b) cause trouble
(c) bring up (d) look after

➥ ***(b)*** Idiom 'raise cain' means 'to create or to cause a trouble'. Look at the sentence: Terrorists must stop raising cains in our country.

54. Go to seed

(a) harvesting (b) planting
(c) become unfit and down (d) financial ruin

➥ ***(c)*** Idiom 'going to seed' means becoming 'unfit and run down'. It has a meaning closer to 'get or become uncaring'.

e.g., All his labour and hard work has gone to seed as he couldn't pass the exam.

Directions (Q. Nos. 55-58) *Read the following passage and answer the questions that follow.*

The next ingredient is a very remarkable one: Good Temper. 'Love is not easily provoked'. Nothing could be more striking than to find this here. We are inclined to look upon bad temper as a very harmless weakness. We speak of it as a mere infirmity of nature, a family failing, a matter of temperament, not a thing to take into very serious account in estimating a man's character and yet here, right in the heart of this analysis of love, it finds a place and the Bible again and again returns to condemn it as one of the most destructive elements in human nature. The peculiarity of ill temper is that it is the vice of the virtuous. It is often the one blot on an otherwise noble character. You know men who are all but perfect and women who would be entirely perfect but for an easily ruffled quick-temper or 'touchy' disposition. This compatibility of ill temper with high moral character is one of the strongest and saddest problems of ethics. The truth is there are two great classes of sins—sins of the Body, and sins of disposition. The prodigal sin may be taken as a type of the first, the Elder Brother of the second. Now, society has no doubt whatever as to which of these is the worse. Its brand falls, without a challenge, upon the Prodigal. But are we right? We have no balance to weigh one another's sins and coarser and finer are but human words; but faults in the higher nature may be less venial than those in the lower, and to the eye of him who is love, a sin against love may seem a hundred times more base. No form of vice, not worldliness, not greed of gold, not drunkenness itself does more to un-christianise society than evil temper. For embittering life, for breaking up communities, for destroying the most sacred relationship, for devastating homes, for withering up men and women, for taking the bloom off childhood,in short for sheer gratuitous misery-producing power, this influence stands alone. Jealousy, anger, pride, uncharity, cruelty, self-righteousness, touchiness, doggedness, sullenness-in varying properties these are the ingredients of all ill temper. Judge, if such sins of disposition are not worse to live in, and for others to live with than sins of the body. There is really no place in Heaven for a disposition like this. A man with such a mood could only make Heaven miserable for all the people in it. Excerpts from passages by Henry Drummond.

55. What is the popular notion about bad temper?

(a) It is a serious malady
(b) It is a harmless weakness
(c) It is a peculiarity
(d) It shows strength of character

➥ ***(b)*** 'Bad temper is a harmless weakness'. The answer can be inferred better from the 2nd and 3rd lines of the passage.

56. What does the word 'venial' mean?

(a) forgivable sin or mistake
(b) deer meat
(c) wicked
(d) insect

➥ ***(a)*** 'Venial' has a similar meaning to that of 'forgivable sin or mistake'.

e.g., Murder is not a venial act.

57. The Bible condemns which of the following as one of the most destructive elements in human nature?
(a) Anger (a) Laziness
(c) Procrastination (d) Splurging

➥ ***(a)*** Bible condemns 'anger' as the most destructive elements in human nature.

58. Find the exact word in the passage which means easily or quickly offended.
(a) Uncharity
(b) Self righteousness
(c) Touchiness
(d) Pride

➥ ***(c)*** Passage has the word 'touchiness' which means easily or quickly offended.

Direction (Q. No. 59) *Fill in the blank with the correct option.*

59. 'If you are morose' you feel __.
(a) excited (b) sleepy
(c) peaceful (d) gloomy

➥ ***(d)*** When someone is 'morose' he is in a sad, dejected and gloomy state.

Directions (Q. Nos. 60-61) *Name the part of speech of the underlined word in the following sentences.*

60. We scored as many goals as <u>them</u>.
(a) Conjunction (b) Pronoun
(c) Adverb (d) Preposition

➥ ***(b)*** 'Them' is personal pronoun of third person and it is in objective case.
e.g., We love <u>them</u> very much.

61. There is more evidence <u>yet</u> to be offered.
(a) Verb (b) Adverb
(c) Preposition (d) None of these

➥ ***(b)*** 'Yet' is an adverb meaning 'until now'. Look at the sentence. We have, gone too far <u>yet</u> a lot has to be covered.

Direction (Q. No. 62) *Identify a synonym to replace the underlined word(s).*

62. The baby is in the <u>carrying basket</u>.
(a) bassinet (b) tub
(c) baby bag (d) preambulator

➥ ***(d)*** Bassinet — A bed for baby.
Tub — An object in which one takes bath.
Baby bag — A bag that a mother has over her back or front in which she carries her baby.
Preambulator — It is the carriage basket in which baby is taken forward. It has wheels.

Direction (Q. No.63) *Which of the following does not denote the underlined word?*

63. The festival of <u>Christmas.</u>
(a) Noel (b) Yule (c) Crimbo (d) Halloween

➥ ***(d)*** Christmas is known as Noel, Yule, Crimbo but it is not known as to be 'Halloween'.

64. Choose the correct option to fill in the blank with the correct grammatical usage.
He has his finger ..... the pulse of the nation.
(a) in (b) below (c) on (d) over

➥ ***(c)*** 'Having a finger on the pulse' means being aware of the recent development.
e.g., The manager of the company <u>has a finger on the pulse of</u> the issues related to its growth and development.

65. Identify the error in the following sentence by choosing from the given alternatives
'Where do you live' asked the stranger?
(a) an exclamation should be used instead of a question mark
(b) the question mark should be used inside the quotation marks
(c) the question mark should be used outside the quotation marks
(d) None of the above

➥ ***(b)*** 'Where do you live?' asked the stranger. This is the correct way to explain it.

Directions (Q. Nos. 66-67) *In each of the following questions the part of the sentence which is in capitals and has been underlined may have an error. Correct the error.*

66. Unless he <u>DOES NOT GIVE UP</u> smoking, we cannot be sure of any improvement in his health.
(a) has not given up (b) gives up
(c) did not give up (d) no improvement

➥ ***(b)*** The correct syntax is
[Unless/until + Simple Present + Subject + Verb + Complement]
e.g., Unless <u>he comes</u> we have to wait here.

67. A recently carried out meta-analysis of two decades of published research <u>DOES NOT SUGGEST THAT SHOULD BE AN ASSOCIATION BETWEEN COFFEE DRINKING AND CORONARY AILMENTS.</u>
(a) suggests that there is no association between coffee drinking and coronary ailments
(b) any association between and coronary ailments is not suggested
(c) shows that coffee drinking should not be suggested to have an association with coronary ailments
(d) suggests the association between coffee drinking and coronary ailments on a macro level

➥ ***(a)*** Sentence that has 'does not' in a negative sense is a confusing one when there is a complex sentence. Moreover use of 'should' is also inappropriate.

68. In the following sentences, the order of the sentences has been jumbled up. Make the best choice to find out the correct sequence so as to form a meaningful paragraph.

A. The following represents a condensed statement of what we think we now know about the relationship between education and employment

B. Although the linkages between education and employment are complex and in the past, often analysed with simplistic notions of casualty, recent research results have yielded new insights about the nature of these linkages.

C. In the interests of brevity, the argument is put forward as a series of major propositions and derivatives strategies relating to the education employment nexus.

D. Intensive research efforts are currently being supported through the developing world both by national governments and international donor agencies in the hope of improving, understanding of the nature and causes of rising unemployment.

E. The massive problems of widespread and chronic unemployment and underemployment in less developed nations will remain among the most serious challenges to the development policy during the next several decades.

(a) ACBDE (b) BACDE (c) CABDE (d) DBEAC

➥ *(a)* ACBDE is the correct order.

69. Given below are four sentences, each of which may or may not have an error. Following them are four options. Select the option indicating the error free statement (s).

A. Fruits from Madhavs' garden were stolen.

B. Fruit's are sold here.

C. All the students of the students' council agreed to raise funds.

D. The dress's texture was coarse.

(a) A and B (b) A and C (c) Only C (d) B and D

➥ *(c)* Madhavs' → Error of apostrophe.
Fruit's → It should be 'fruits.'
Dress's → No 's' is needed.
So, only option (c) is correct.

70. Choose the odd word out.

(a) Esoteric (b) Exigent (c) Occult (d) Mystical

➥ *(b)* Exigent is the odd word. Esoteric, occult, mystical all have similar meaning in the question as 'mysterious' but 'exigent' means 'urgent.

71. Choose the word or phrase that best completes the sentence.

The rebels sought to overcome the ..... strength of the police forces by engaging in ..... tactics.

(a) lack of ..... diversionary
(b) augmented ..... peaceful
(c) preponderant ..... guerilla
(d) power ..... foolish

➥ *(c)* 'Preponderant' means 'dominating'. The 'guerilla tactics' reflects the idea of a warfare where fight takes the unconventional way. Here, rebels (the opponents) have demanded to overcome the preponderant (dominant) strength that police have by the (guerilla) tactics.

Directions (Q. Nos. 72–74) *Read the following passage and answer the questions that follow.*

My new mistress proved to be all she appeared when I first met her at the door, a woman of the kindest heart and feelings. She had never had a slave under her control previously and prior to her marriage she had been dependent upon her own industry for a living. She was by trade a weaver and by constant application to her business, she had been in a good degree preserved from the lighting and dehumanising effects of slavery. I was utterly astonished at her goodness. I scarcely knew how to behave towards her. My early instruction was all out of place. The crouching servility, usually so acceptable a quality in a slave, did not answer when manifested toward her. Her favor was not gained by it; she seemed to be disturbed by it. She did not deem it to be impudent or unmannerly for a slave to look in the face. The meanest slave was put fully at ease in her presence and none left without feeling better for having seen her. But alas! This kind heart had but a short time to remain such. The fatal poison of irresponsible power was already in her hands and soon commenced its internal work.

Very soon I went to live with Mr and Mrs Auld; she very kindly commenced to teach me the A, B, C. After I had learnt this, she assisted me in learning to spell words of three and four letters. Just at this point of my progress, Mr Auld found out what was going on and at once forbade Mrs. Auld to instruct me further, telling her that it was unlawful, as well as unsafe, to teach a slave to read. Further he said, 'If you give a slave an inch, he will take a mile. A slave should know nothing but to obey his master-to do as he is told to do. Learning would spoil the best slave in this world 'Now,' said he ''If you teach that boy (speaking of myself) how to read, there would be no keeping him. It would forever unfit him to be a slave. He would at once become unmanageable, and of no value to his master. As to him, it could do him no good but a great deal of harm. It would make him discontented and unhappy. 'These words sank deep into my heart, stirred up sentiments within that lay slumbering and called into existence an entirely new train of thought. I now understood what had been to me a most perplexing difficulty-the white man's power to enslave the black man. From that moment, I understood that pathway from slavery to freedom. Though conscious of the difficulty of learning without a teacher, I set out with high hope, and a fixed purpose, at whatever cost of trouble, to learn how to read. The very decided manner with which my master spoke, and strove to impress his wife with the evil consequences of giving me instruction, served to convince me that he was deeply sensible of the truths he was uttering. It gave me the best assurance that I

might rely with the utmost confidence on the results which, he said, would flow from teaching me to read. What he most dreaded that I most desired. What he most loved, that I most hated. That which to him was great evil, to be carefully shunned, was to me a great good, to be diligently sought; and the argument which he so warmly urged, against my learning to read, only served to inspire me with a desire and determination to learn. In learning to read, I owe almost as much to the bitter opposition of my master, as to the kindly aid of my mistress. I acknowledge the benefit of both.

72. The author's main purpose in this passage is to
(a) describe a disagreement between a woman and her husband
(b) analyse the reason for the education of slaves
(c) describe a slave's literacy as a means of freedom
(d) dramatise a slave's change in attitude towards his mistress

➥ ***(c)*** The passage describes the ways how a slave can use literacy as a means of freedom for himself.

73. For which of the following reasons does Mr Auld forbid his wife to educate the slave?
1. Providing slaves with an education violates the law.
2. He believes slaves lack the capacity for education.
3. He fears education would leave the slave less submissive.

(a) Only 1 (b) Only 3
(c) 1 and 2 (d) 1 and 3

➥ ***(d)*** Mr Auld forbade his wife to teach the slave as it was unlawful and the slave would be unmanageable. So,option (d) is correct.

74. The tone of author in acknowledging his debt to his master can be best described as
(a) sentimental and nostalgic
(b) cutting and ironic
(c) petulant and self-righteous
(d) resigned but wistful

➥ ***(a)*** The author has acclaimed that he had got the benefit from the statements of his master. He remembers it after a very long time. So, he is sentimental as well as nostalgic.

75. Replace the underlined phrase with the one which is closest to it in meaning. I DROPPED A CLANGER when I mentioned her ex- husband.
(a) Inadvertently caused embarrassment
(b) Let go of a clanger
(c) Toppled a lampshade
(d) None of the above

➥ ***(a)*** 'Dropping a clanger' means 'causing inadvertent embarrassment.' Look at the sentence.

Directions (Q. Nos. 76-78) *Fill in the blanks in the sentences with the appropriate choices.*

76. I should not have ..... to talk in such a ..... strain especially when I had not studied the man to whom I was talking.
(a) daring ..... commanding
(b) tried ..... bold
(c) ventured ..... peremptory
(d) emboldened ..... reckless

➥ ***(b)*** Daring is not properly formed. It should be 'dared.' Ventured means to take risk.When ventured is combined with peremptory, it gives the idea that there has been a very commanding way of telling something and since the person is not very much familiar with the man he is talking to, he has risked it. So, option (b) is correct.

77. The ..... of Agatha Christie's argument is that human nature remains the same, wherever you are.
(a) acne (b) pinnacle (c) nadir (d) crux

➥ ***(d)*** 'Crux' means 'the most vital or important part.

78. The ..... play caused me to squirm in my seat, but she began to ..... her eyes in a way that irritated me.
(a) delirious ..... spot (b) grin ..... wet
(c) disastrous ..... damp (d) maudlin ..... daub

➥ ***(a)*** 'Delirious' means something that is 'in an imbalanced state' mentally and 'spot' means 'to mark.'

Directions (Q. Nos. 79-80) *Select the choice that best replaces the underlined part of the sentence given in the question*

79. There they stood on the top shelf, all 20 volumes, their red binding and gold lettering undoubtedly making them the most attractive between all the books' on the bookshelf
(a) their red binding and gold lettering undoubtedly making them the most attractive between all the books on
(b) their red binding and gold lettering undoubtedly making them the most attractive among all the books on
(c) their red binding and gold lettering undoubtedly made them the most attractive among the books in
(d) their red binding and gold lettering undoubtedly making them seem the most attractive among all the books on

➥ ***(b)*** Option (a) has used 'between' which is wrong as there are a lot of books. Option (c) has used 'made' which is inappropriate. Option (d) has used 'seem' which does not suit the frame of sentence. But option (b) is the most appropriate one.

80. Everything in the world decays, it comes to an end and also dies inevitably.
(a) decays, it comes to an end and also dies inevitably
(b) decays, comes to an end and dies inevitably
(c) decays and as it comes to an end it also dies inevitably
(d) decay, they an end,' and then they die inevitably

➥ ***(b)*** When verbs of the same form are being used consecutively, then 'it' should not be used; 'also' should be avoided. So, the correct sequence should be [decays, comes to an end and dies].

Section C
(Analytical and Logical Reasoning)

For all questions in this section, each correct answer carries 2 marks.

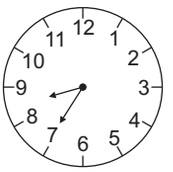
81. What will be the measurement of the angle made by the hands of a clock when the time is 8 : 35?

(a) 45° (b) 32.5° (c) 47.5° (d) 30°

➥ *(a)* 35 min 8 past $= 8 + \frac{35}{60} = \frac{7}{2}$

Angle traced by hour hand (points to the 8) in $\frac{17}{2}$h

$$\frac{360}{120} \times \frac{17}{2} = 255$$



Angle traced by minute hand (points to the 7) in 35 min

$$= \frac{360}{60} \times 35 = 210$$

∴ Required angle = 255 – 210 = 45°

82. Find the odd man out.

(a) DIRAMLOG (b) ERENIGEN
(c) UIHCSSBI (d) AFOLDFID

➥ *(b)* According to the given information,

(a) DIRAMLOG - MARIGOLD

(b) ERENIGEN - ERENIGEN

(c) UIHCSSBI - HIBISCUS

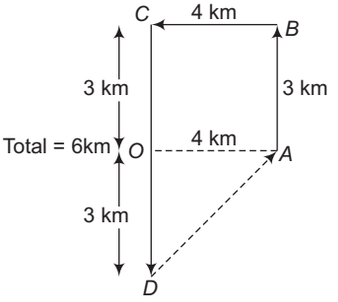
(d) AFOLDFID - DAFFODIL

All the terms or optionals except 'ERENIGEN' are related to the 'flowers'.

83. Raghu is at point *A*. He walks 3 km to the North and then turns to his left. He walks 4 km in this direction. He turns left again and walks 6 km. If he wishes to reach point A again, in which direction should he be walking and what distance will he have to cover?

(a) South-East,5 km (b) South-East,4 km
(c) North-East,5 km (d) North-East,4 km

➥ *(c)* Raghu's starting point is *A*. He walked 3 km to North and turned his left in the direction of West and walked 4 km. He turned left again in the direction of South and walked 6 km and finally he turned in the direction of North-East to reach point *A*. The distance from point *D* to point *A* is 5 km.



Note that, *CB* = *OA* is 4 km.

$$OA = 4 \Rightarrow OD = 3$$

So, $(AD)^2 = (OA)^2 + (OD)^2 = (4)^2 + (3)^2 = 16 + 9 = 25$

∴ $AD = 5$ km

Directions (Q.Nos.84-85) *Read the following information carefully and answer the questions.*

Six students A, B, C, D, E and F participated in a dancing competition wherein they won prizes 12000, 10000, 8000, 6000, 4000, 2000 according to the position secured. The following information is known to us:

(a) A won less money than B

(b) The difference between the winning of C and F was ₹ 2000

(c) The difference between the winning of D and F was at least ₹ 4000

(d) E won the ₹ 8000 prize

84. Which of the following could be the ranking from first place to sixth place of students?

(a) A, D, E, B, F, C (b) B, A, E, C , F, D
(c) F, B ,E ,A ,C ,D (d) B, A, E,D,C, F

➥ *(d)* On the basis of given information in the question, the ranking from first place to sixth place of students could be

B – 12000
A – 10000 (*A* won less money than *B*)
E – 8000
D – 6000 (*E* won ₹ 80000 Prize)

C – 4000 and the difference between winning of *C* and *F* was ₹ 2000.

F – 2000 and the difference between winning of *D* and *F* was at least ₹ 4000.

85. If *A* won ₹ 4000, how much in total did *C* and *F* win?

(a) ₹ 6000 (b) ₹ 10000
(c) ₹ 22000 (d) ₹ 18000

➥ *(c)* If *A* won ₹ 4000, then total of *C* and *F* win.

According to the given information,

C – 12000
F – 10000 (Difference between was at least ₹ 4000)

E – 8000
B – 6000
A – 4000 (*A* won less money than *B*)

D – 2000

The total of C and F = 12000 + 10000 = ₹ 22000

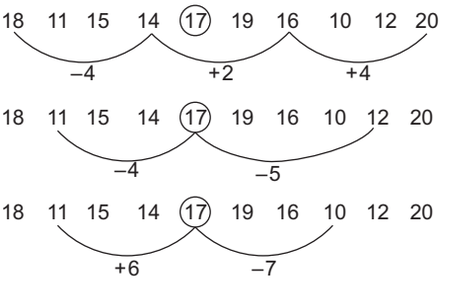
86. Where should the number 17 be placed to fit into the sequence?

18,11,15,14,19,16,10,12,20

(a) Between 14 and 19 (b) Between 18 and 11
(c) Between 19 and 16 (d) Between 12 and 20

➥ *(a)* Given, 18 11 15 14 19 16 10 12 20

We put 17 in between 14 and 19 and the following pattern as below



87. There are three facts and you are given three more statements. You have to determine which of these, if any is also a fact.

Fact 1. All kids like to play.

Fact 2. Some kids like ice-cream.

Fact 3. Some kids look like their mothers

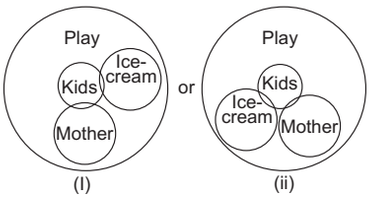
Statement 1. All kids who like ice-cream look like their mothers.

Statement 2. Kids who like ice-cream also like to play.

Statement 3. Kids who like to play do not look like their mothers,

(a) Only 1 (b) Only 2
(c) 2 and 3 (d) None can be a fact

➥ *(b)*



The above two figures show the facts which are given in the information. Statement 1 'All kids who like ice-cream look like their mothers' is true from figure (II) but not valid with figure (I).

Statement 2 'Kids who like ice-cream also like to play,' is valid from both the figures.

Statement 3 'Kids who like to play do not look like their mothers' is valid from figure (I) but not with figure (II).

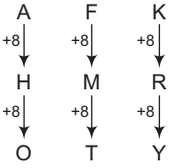
So, option (b) is correct.

88. Insert the missing letter

A	F	K
H	M	R
	T	Y

(a) O (b) P (c) Q (d) R

➥ *(a)* The letters in each column follow this pattern,



Direction (Q. No. 89) *Six statements are followed by options consisting of three statements put together in a specific order. Choose the option which indicates a valid argument, that is, the third argument is a conclusion drawn from the earlier two statements.*

A. Amar is unhappy
B. Amar is honest
C. Some magicians are honest
D. No honest man is a magician
E. No magician is happy
F. Amar is not a magician

(a) FED (b) BFA (c) AEF (d) BDF

➥ *(d)* On the basis of given information, the arguments as below

From option (a),

F → Amar is not a magician.
E → No magician is happy.
D → No honest man is a magician.

From option (b),

B → Amar is honest.
F → Amar is not a magician.
A → Amar is unhappy.

From option (c),

A → Amar is unhappy.
E → No magician is happy.
F → Amar is not a magician.

From option (d),

B → Amar is honest.
D → No honest man is a magician.
F → Amar is not a magician.

It is clear from the above options that BDF is the correct or valid argument.

Directions (Q. Nos. 90-91) *The table below presents data on Gross National Product (GNP)and Net National Product (NNP) for India for six years, Answer questions based on these data.*

Gross National Product and Net National Product

	GNP (in ₹ crore)	NNP (in ₹ crore)
1988-89	348210	309289.
1989-90	402931	357285
1990-91	470252	418057
1991-92	541593	479001
1992-93	615831	543566
1993-94	695342	615273

90. The year during which both GNP and NNP have shown almost equal rate of growth-

(a) 1989-90 (b) 1990-91
(c) 1991-92 (d) 1992-93

➥ *(a)* The year during both GNP and NNP have shown almost equal rate of growth is 1989-90. This is because of the less difference between them.

Year	GNP	NNP	Difference
1989-90	402931	357285	45646
1990-91	470252	418057	52195
1991-92	541593	479001	62592
1992-93	615831	543566	72265

The year 1989-90 having less difference.

91. Which of the following statements could be stated to be true?

(a) GNP has been growing during the period under review
(b) NNP has been growing during the period under review
(c) The growth pattern has been different for both indicators during period under review
(d) None of the above

➥ *(c)* It is clear from the given table that the growth pattern has been different for both indicators during period under review. Reason behind this-The GNP and NNP both have been growing during the period under review but in a different pattern.

92. If PEACE is coded as RGCEG, then how is MICKY coded in that code? '

(a) MOUSE (b) OKEMA (c) LHBJX (d) JDLJ

➥ *(b)*

Basic Word	P	E	A	C	E
	+2	+2	+2	+2	+2
Coded Word	R	G	C	E	G

Similarly,

Basic Word	M	I	C	K	Y
	+2	+2	+2	+2	+2
Coded Word	O	K	E	M	A

93. A set of symbols is given indicating the terms for which they will be used in the questions following them.

Θ = equal to, φ = not equal to, Δ = greater than, x= less than, + = not greater than, $ = not less than A Δ B Δ C does not imply which of the following options?

(a) A $ B Δ C (b) C + B x A (c) A x B × C (d) C + B φ A

➥ *(d)* After interchanging the symbols of the given information, the options are as follows

(a) A $ B D C→ A> B> C

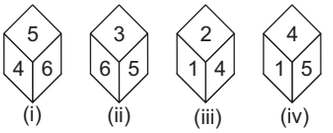
(b) C+ B× A→ C< B< A

(c) A× B H C→ A< B= C

(d) C+ B Q C→ C< B≠ A

A D B D C→ A> B> C, already given in the question. According to this statement, only option (d) does not imply the following options.

94. Given below are 4 pictures of a cube



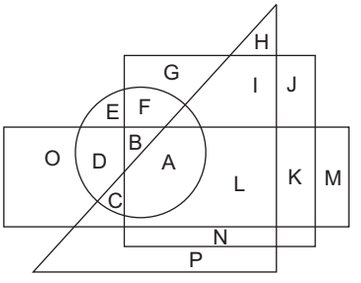
Which number is on the face opposite to 3?

(a) 1 (a) 2 (c) 4 (d) 5

➥ *(c)* From figures (i),(ii) and (iii),we conclude that 5, 6, 1 and 2 lie adjacent to 4. Hence, 3 must lie opposite 4 and *vice-versa.*

From figures (i) and (iv), we find that 1 is opposite to 6, 2 is opposite to 5 and 4 is opposite to 3.

Directions (Q. Nos. 95–96)



Rectangle indicates Faculty members, Square indicates Research scholars, Circle indicates Indians and Triangle indicates Americans.

95. In the given figure show the area which indicates faculty members who are research scholars but are neither Americans nor Indians.

(a) L (b) N
(c) K (d) G

➥ *(c)* The area which indicates faculty members who are research scholars but are neither American nor Indians is 'K'.

96. In the given figure show the area which indicates Indian Americans who are faculty members but not research scholars.

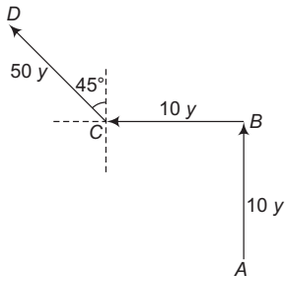
(a) A (b) B (c) C (d) F

➥ *(c)* The area 'C' indicates Indian-Americans who are faculty members but not research scholars.

97. While facing East you turn to your left and walk 10 yards. Then, turn to your left and walk 10 yards and now turn 45° to your right and go straight to cover 50 yards. Now, in what direction are you with respect to the starting point?

(a) North-East (b) North
(c) South-East (d) North-West

➥ *(d)* Let *A* is the initial point of a man. He faces East and turned his left in the direction of North and walked 10 yards.

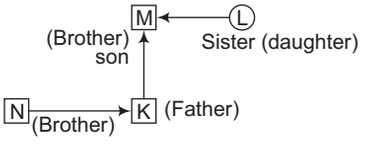


Then, turned his left in the direction of West and walked 10 yards. Now, he turned 45° to his right and walked 50 yards straight in the same direction. Now, the direction of a man with respect to his starting point is North-West.

98. If P– Q means Q is son of P, P × Q means P is brother of Q, P ÷ Q means Q is sister of P and P + Q means P is mother of Q, which of the following is definitely true about N × K – M ÷ L?

(a) K is father of L and M
(b) L is daughter of K and niece of uncle N
(c) K is father of L and M-his son and daughter respectively
(d) M is uncle of K's brother N

➥ *(a)* N × K– M ÷ L, according to the given information,



From the figure above, it is clear that K is the father of L and M.

Directions (Q. Nos. 99-100) *Five girls Rama, Sudha, Tara, Uma and Veena share an apartment and have distributed the task as one girl making breakfast per day, Monday through Friday, one of the five dishes upama, dosa, idli, uttapa and paratha.*

- Veena does not make uttapa and does not cook on Tuesday.
- Sudha makes parathas but not on Monday or Friday.
- Upama is made on Thursday.
- Tara makes her dish which is not uttapa on Wednesday.
- Idli is made on Friday but not by Uma.
- Rama cooks on Monday

99. What does Tara cook on Wednesday?

(a) Upama (b) Paratha
(c) Idli (d) Dosa

100. What day does Uma prepare her dish?

(a) Monday (b) Tuesday
(c) Thursday (d) Friday

Solutions (Q. Nos. 99-100) *On the basis of given information, the table as below*

						Days
Girls	Upma	Dosa	Idli	Uttapa	Paratha	
Rama	×	×	×	✓	×	Monday
Sudha	×	×	×	×	✓	Tuesday
Tara	×	✓	×	×	×	Wednesday
Uma	✓	×	×	×	×	Thursday
Veena	×	✓	×	×	×	Friday

➥ *(d)* Tara cooks dosa on Wednesday.

➥ *(c)* Uma prepares dish on Thursday.

Direction (Q. No. 101) *Answer the following question on the basis of the following data*

Kunal, Rahul, Mohit and Charan are constables on duty in Delhi, and they are trying to recollect who amongst them has been working the most number of days without leave. Here, are the facts : Kunal has put in more days of consecutive duty than has Rahul, who has got in five days of consecutive duty. Mohit has gone fifteen days straight without leave, more than Kunal and Rahul combined) Charan has put in eight consecutive days of duty, which is less than what Kunal has done.

101. How many days has Kunal gone without leave?

(a) 6 (b) 9 (c) 12 (d) 4

➥ *(b)* On the basis of given information,

Rahul's consecutive duty—5 days

Mohit's gone straight without leave—15 days

Charan's consecutive duty—8 days

According to the given information,

Kunal has put more than Rahul

and Charan has put less days duty than Kunal.

It is clear from the option (b) *i.e.,* Kunal's number of days = 9 and Rahul's number of days = 5

Rahul + Kunal = 5 + 9 = 14

Kunal's number of days is more than Rahul and less than Charan's number of days and total of Rahul and Kunal is less than Mohit's number of days.

Directions (Q. Nos. 102-104) *Five hundred school children were asked about cricketers they liked. 80 said they like Sachin Tendulkar, 60 said they liked Sourav Ganguly,*

90 said they liked Mahendra Singh Dhoni.
10 said they liked all them 40 said they liked
Sachin and Sourav, 50 liked Sachin and Dhoni, while 30 said they liked Sourav and Dhoni.

102. How many, out of the 500, didn't choose Sachin, Sourav or Dhoni?

(a) 280 (b) 380
(c) 180 (d) 480

103. How many like Sourav but not Tendulkar?

(a) 21 (b) 22
(c) 23 (d) None of these

104. How many like fewer than two cricketers?

(a) 410 (b) 400
(c) 390 (d) 395

Solutions (Q. Nos. 102-104)

Sachin = $b + g + d = 30 + 10 + 40 = 80$

Sourav = $b + g + f = 30 + 10 + 20 = 60$

Dhoni = $d + g + f + e = 40 + 10 + 20 + + 20 = 90$

All cricketers = 10 Sachin + Sourav

$= b + g = 40\ (30 + 10)$

Sachin + Dhoni = $g + d = 10 + 40 = 50$

Sourav + Dhoni = $g + e = 10 + 20 = 30$

➥ *(b)* Total = 500

Who didn't choose Sachin, Sourav or Dhoni

= (10 + 40 + 20 + 30 + 20 + 0 + 0 = 120)

out of 500 is 500 – 120 = 380

➥ *(d)* None of the options is true according to the question.

➥ *(b)* 400 like fewer than two cricketers. It is clearly shown by

380 + 20 = 400

105. If 7 is coded as CBRT343, then 9 is

(a) CBRT27 (b) SQRT81
(c) CBRT729 (d) CBRT6561

➥ *(c)* 7 is coded as C B R T 343

343 is the cube of 7 *i.e.,* ($7^3 = 343$)

Similarly, 9 is coded as C B R T 729

729 is the cube of 9 *i.e.,* ($9^3 = 729$)

106. If HELLO is coded as 15 | 12 | 12 | 5 | 8, then WHERE is coaded as

(a) 16|13|I3|6|9 (b) 5|18|5|8|23
(c) 5|5|18|23|8 (d) 5|5|18|8|23

➥ *(b)* **Basic word** H E L L O

Coded word 15 12 12 5 8

'H' has 8th position in the alphabetical order and 'O' has 15th position as in the alphabetical order. Both the positions of numeric form are interchanged as shown in the above figure. Similarly. 'L' has 12th position which is interchanged with 'E''s position *i.e.*, '5' and so on.

In the same way,

Basic word W H E R E

Coded word 5 18 5 8 23

107. Complete the following series 3 10 59.

(a) 402 (b) 403 (c) 404 (d) 405

➥ *(a)* 3, 10, 59.

This series is based on the following pattern

$10 - 3 = 7 = 7$

$59 - 10 = 49 = 7 \times 7 \Rightarrow 7^2$

$402 - 59 = 343 = 7 \times 7 \times 7 \Rightarrow 7^3$

108. If WOOD is coded as 23 | 225 | 4, then MEET is coded as

(a) |3|5|5|20 (b) 13|10|20
(c) 13|25|20 (d) None of these

➥ *(c)* **Basic Word** W O O D ($(15)^2$)

Coded Word 23 225 4

In this 'W' has 23rd position in the alphabetical order, 'D' has '4' position, 'O' has 15th position in the alphabetical order, letter 'O' has 15th position and 225 is the square of 15th number.

Similarly,

Basic word M E E T ($(5)^2$)

Coded word 13 25 20

'M' has 13th position, 'T' has 20th position and 'E' has 5th position and 25 is the square of number 5.

Directions (Q. Nos. 109-110) *Answer the questions based on the following information, which gives data about certain coffee producers in India.*

S. No.	Brands	Production ('000 tonnes)	Capacity utilisation (%)	Sales ('000 tonnes)	Total sales value (₹ in crore)
1.	Brooke Bond	2.97	76.50	2.55	31.15
2.	Nestle	2.48	71.20	2.03	26.75
3.	Lipton	1.64	64.80	1.26	15.25
4.	MAC	1.54	59.35	1.47	17.45
	Total (including others)	11.60	61.30	10.67	132.80

109. What is the maximum production capacity (in '000 tonnes) of Liption for coffee?

(a) 2.53 (b) 2.85
(c) 2.24 (d) 2.07

➥ *(a)* The production capacity for lipton = 64.80% for 1.64 thousand tonne
maximum capacity is 100%
For 100% it would be $= \left(\frac{100}{64.8} \times 1.64\right)$
$= 2.53$ thousand tonne

110. What percent of the total market shares (by sales value) is controlled by others ?

(a) 60% (b) 32%
(c) 67% (d) Data insufficient

➥ *(b)* Sales of coffee (other)
$= 132.80(31.15 + 26.75 + 15.25 + 17.45)$
$= 132.80 - 90.6 = 42.2$
$\therefore$ Required per cent $= \frac{42.2}{13.2} \times 100 = 32\%$ approximately

Section-D
(General Awareness)

For all questions in this section, each correct answer carries 1 mark.

111. The economically diverse region of MENA nations refers to the countries in

(a) Middle East and North African Countries
(b) Middle East and North American Countries
(c) Mediterranean Europe and North Atlantic Countries
(d) None of the above

➥ *(a)* A region encompassing approximately 22 countries in the Middle East and North Africa (MENA). The MENA region accounts for approximately 6% of the world's population, 60% of the world's oil reserves and 45% of the world's natural gas reserves. Due to the region's substantial petroleum and natural gas reserves, MENA is an important source of global economic stability.

112. Open market operations refer to the sale and purchase by the RBI of

(a) Foreign currency (b) Gold
(c) Government bonds (d) All of these

➥ *(c)* An Open Market Operation (also known as OMO) is an activity by a central bank to buy or sale Government bonds in the open market. A central bank uses them as the primary means of implementing monetary policy.

113. The agency that estimates national income in India is

(a) RBI
(b) Planning Commission
(c) Central Statistics Organisation
(d) None of the above

➥ *(c)* The Central Statistical Organisation (CSO) was set up in the cabinet secretariat on 2nd May, 1951. CSO is responsible for coordination of statistical activities in India and evolving and maintaining statistical standards. It is located in Delhi.

114. Panchatantra was written by

(a) Brahmbhatt
(b) Vishnu Sharma
(c) Jaya Deva
(d) None of these

➥ *(b)* The Panchatantra is an ancient Indian inter-related collection of animal fables in verse and prose, in a frame story format. It was written by Vishnu Sharma.

115. IRDA is the Indian regulator for the
(a) Banking sector (b) Mutual funds
(c) Rural Development (d) None of these

➥ *(d)* Insurance Regulatory and Development Authority (IRDA) is an autonomous apex statutory body which regulates and develops the insurance industry in India. It was constituted by a Parliament of India Act called Insurance Regulatory and Development Authority Act, 1999 and duly passed by the Government of India.

116. The Speaker of the Lok Sabha is elected by the
(a) Members of Lok Sabha
(b) President of India
(c) Leader of the majority party in the Lok Sabha
(d) None of the above

➥ *(c)* In the Lok Sabha, the lower house of the Indian Parliament, both presiding officers- the Speaker and the Deputy Speaker are elected from among its members by a simple majority of members present and voting in the house. As such, no specific qualifications are prescribed for being elected the Speaker. The Constitution only requires that Speaker should be a member of the house.

117. The Finance Minister's Budget speech presented to the Indian parliament every year usually has two parts. What does the Part A relate to ?
(a) Taxation Proposals (b) General Economic Survey
(c) Both (a) and (b) (d) None of the above

➥ *(b)* The budget speech of the finance minister is usually in two parts. Part A deals with general economic survey of the country while part B relates to taxation proposals.

118. Which two institutions are popularly referred to as the 'Bretton Woods Twins'?
(a) International Monetary Fund and World Bank
(b) World Bank and Asian Development Bank
(c) Asian Development Bank and International Monetary Fund
(d) United Nations and World Trade Organisation

➥ *(a)* The Bretton Woods Twins refers to the two multilateral organisations created at the Bretton Woods conference in 1944. They are the World Bank and the International Monetary Fund.

119. Based on the statements given below, Special Drawing Rights is best described to include which of the following statements?
1. It is a basket of currencies reviewed every 5 yr.
2. It was created to reduce the dependencies on gold and US dollar.
3. It is a supplementary foreign exchange reserve assets maintained by IMF.
(a) 1 and 2 (b) 2 and 3 (c) 1, 2 and 3 (d) Only 3

➥ *(c)* Special Drawing Rights (SDRs) are supplementary foreign exchange reserve assets defined and maintained by the International Monetary Fund (IMF).

120. Which NATO country's parliament was the first to vote in favour of the move to make military service compulsory for women?
(a) Italy (b) Germany
(c) Norway (d) None of these

➥ *(c)* Norway's parliament, in June 2013, has voted overwhelmingly to conscript women into its armed forces, becoming the first European and first NATO country to make military service compulsory for both genders.

121. The market for government securities is also referred to as
(a) Bullion market (b) Gilt-edge market
(c) Secondary market (d) Money market

➥ *(b)* Gilt-edged securities are bonds issued by certain national governments. The term is of British origin and originally referred to the debt securities issued by the bank of England, which had a gilt (or gilded) edge. Hence, known as gilt-edged securities or gilts for short.

122. From April 2010 the savings bank interest is calculated
(a) at 4% per annum
(b) on daily balance method based on the closing balance maintained every day
(c) lowest balance available between 10th and the final day of the month
(d) None of the above

➥ *(b)* From 1st April, 2010, interest on all saving bank account deposits is being calculated on a daily basis, thereby earning account holders higher interest income. This is due to the fact that the RBI has instructed banks to change the mechanism of interest income calculation. The calculation is done on the 'daily balance method', with the rate of interest remaining the same at 3.5% per annum.

123. Financial Action Task Force (FATF) is an Inter-governmental Body established in 1989. It is concerned with
(a) Money laundering
(b) Stock markets
(c) Foreign Direct Investment
(d) Foreign Institutional Investors

➥ *(a)* The Financial Action Task Force (FATF) is an intergovernmental organisation founded in 1989 on the initiative of the G7. The purpose of the FATF is to develop policies to combat money laundering and terrorism financing.

124. Which of the following sectors in India was not nationalised by Government at any point of time?
(a) Banking (a) Insurance
(c) Coal (d) None of these

➥ *(d)*

125. Which of the following is not a Palindrome?
(a) A car, a man, a maraca
(b) A man, a plan, a canal, Panama
(c) As you sow, so you reap
(d) None of the above

➥ *(c)* A palindrome is a word, phrase, number or other sequence of symbols or elements, whose meaning may be interpreted the same way in either forward or reverse direction. Famous examples include 'Amor Roma', 'A man, a plan, a canal, Panama'.

126. The movie 'Life of Pi' is based on an award winning novel written by
(a) Peter Carey (b) Alan Hollinghurst
(c) Yann Martel (d) None of these

➥ *(c)* 'Life of Pi is a 2012 American 3D live-action computer-animated adventure drama film based on Yann Martel's 2001 novel of the same name.

127. LIDL, Walgreen, ASDA and Kroger are
(a) Automobile companies
(b) Retailers
(c) Electronic product manufacturers
(d) None of the above

➥ *(b)*

128. Which bank was the first to introduce the Automated Teller Machine (ATM)in India?
(a) Hongkong and Shanghai Banking Corporation (HSBC)
(b) Citibank
(c) Standard Chartered Bank
(d) HDFC Bank

➥ *(a)* Hongkong and Shanghai Banking Corporation (HSBC) introduced ATM concept in India, first time in 1987.

129. The Three Gorges Dam, is the world's largest hydropower project and is located in
(a) China (a) Brazil
(c) Russia (d) None of these

➥ *(a)* The three Gorges Dam is a hydroelectric dam that spans the Yangtze River by the town of Sandouping, located in Yiling district, Yichang, Hubei Province, China. The Three Gorges Dam is the world's largest power station in terms of installed capacity (22500 MW).

130. India signed the Kyoto Protocol in the year
(a) 2000 (b) 1998
(c) 2002 (d) 1995

➥ *(c)* The Kyoto Protocol to the United Nations Framework Convention on Climate Change (UNFCC) is an international treaty that sets binding obligations on industrialised countries to reduce emissions of greenhouse gases. India signed the Kyoto Protocol in the year, 2002.

131. Given below are the creators of some well-known cartoon characters. Identify the creator cartoon pair that does not match.
(a) Elzie Crisler Segar-Popeye
(b) Dan Yaccarino-Oswald the Octopus
(c) William Hanna-Richie Rich
(d) Walt Disney-Donald Duck

➥ *(c)* William Hanna was an American animator, director, producer, voice actor and cartoon artist. Richie Rich is a fictional character in the Harvery comics universe. It was created by Alfred Harvey and Warren Kremer.

132. The goals of monetary policy do not include
(a) full employment (b) price stability
(c) maximising tax revenue (d) None of these

➥ *(d)* Monetary policy is the process by which monetary authority of a country, generally a central bank (or RBI) controls the supply of money in the economy by exercising its control over interest rates in order to maintain price stability and achieve high economic growth.

133. The Corporate Head Quarters of Tata Group is
(a) Bombay House (b) Tata Towers
(c) JRD Centre (d) None of these

➥ *(a)* Bombay House is a historic privately owned building in Mumbai that serves as the head office of the Tata Group.

134. Dronacharya Award for Coaching for the year 2013 was received by Narinder Singh Saini for ' which discipline?
(a) Archery (b) Boxing
(c) Hockey (d) None of these

➥ *(c)*

135. This company signed an agreement with the Government of India for the Passport Seva Project in 2008.
(a) Infosys Technologies
(b) Mahindra Satyam
(c) Tata Consultancy Services
(d) Wipro

➥ *(c)*

136. Recommendations of which committee led to the reforms in the banking sector in India in the nineties?
(a) Malhotra Committee
(b) Narasimham Committee
(c) Mayaram Committee
(d) Damodaran Committee

➥ *(b)* Narasimham Committee-I was formed in 1991 and Narasimham Committee-II was formed in 1998 and both were related to banking sector reforms.

137. Which country is the largest producer of newsprint?
(a) Indonesia (b) Finland
(c) Canada (d) None of these

➥ *(c)*

138. India's Direct Cash Transfer Scheme draws similarities from models of which country?
(a) Brazil (b) Germany
(c) Australia (d) Sri Lanka

➥ *(a)* Direct Benefit Transfer or DBT is an anti-poverty programme launched by the Government of India on 1st January, 2013. This programme aims to transfer subsidies directly to the people living below poverty line.

139. Who put forth the theory of 'Zero Defects' ?
(a) W Edward Deming
(b) Philip Crosby
(c) Joseph Juran
(d) Martin Hall

➥ *(b)* Zero Defects was a management-led programne to eliminate defects in industrial production that enjoyed brief popularity in American industry in the late 1960s and early 1970s. It was given by Philip Crosby.

140. Logic of induction is very close to
(a) the logic of observation
(b) the logic of sampling
(c) the logic of the controlled variables
(d) None of the above

➥ *(b)*

141. How many pairs of chromosomes exist in a human cell?
(a) 12 (b) 23
(c) 34 (d) 45

➥ *(b)* In humans, each cell normally contains 23 pairs of chromosomes, for a total of 46.

142. International Date Line roughly follows the longitude of
(a) 180 degrees (b) 90 degrees
(c) 360 degrees (d) None of these

➥ *(a)* International Date Line (IDL) passes through the middle of the Pacific ocean, roughly following the 180° longitude but it deviates to pass around some territories and Island groups.

143. Daylight Saving Time (DST) implies adjusting the clock by 1h
(a) forward near the start of spring and backward in autumn
(b) forward near the start of autumn and backward in spring
(c) forward near the start of spring and forward in autumn
(d) backward near the start of autumn and backward spring

➥ *(a)* Daylight Saving Time (DST) is the practice of advancing clocks during the lighter months so that evenings have more daylight and mornings have less. Typically clocks are adjusted forward one hour near the start of spring and are adjusted backward in the autumn.

144. Who among the following sports persons are twins?
(a) Serena Williams and Venus Williams
(b) Mark Waugh and Steve Waugh
(c) Irfan Pathan and Yusuf Pathan
(d) None of the above

➥ *(b)* Mark Waugh and Steve Waugh, a set of twins who played test and ODI cricket for Australia from the 1980s to the 2000s.

145. Individuals with this blood type are universal red cell donors
(a) A^- (b) B^+ (c) AB (d) O^-

➥ *(d)* The universal red cell donor has type O^- blood type. Blood group AB^+ is called universal receptor, which can receive any blood type.

146. 'Relationship beyond banking' is the tag line of
(a) State Bank of India (b) Central Bank of India
(c) Bank of India (d) Bank of Maharashtra

➥ *(c)*

147. Find the odd man out.
(a) Sarod (a) Sitar (c) Shehnai (d) Santoor

➥ *(c)* Sarod, Sitar and Santoor are played with hand but Shehnai is played with mouth.

148. Which instrument measures Blood Pressure?
(a) Sphygmomanometer (a) Hygrometer
(c) Saccharimeter (d) Spherometer

➥ *(a)* A Sphygmomanometer or blood pressure meter is a device used to measure blood pressure, composed of an inflatable cuff to restrict blood flow and a mercury or mechanical manometer to measure the pressure.

149. Find the odd person out.
(a) Shikha Sharma (b) Chanda Kochhar
(c) Shubhalakshmi Panse (d) Kiran Mazumdar Shaw

➥ *(d)* Shikha Sharma – MD and CEO of Axis Bank
Chanda Kochhar – MD and CEO of ICICI Bank
Shubhalakshmi Panse – MD of Allahabad Bank
Kiran Mazumdar Shaw – MD of Biocon Limited

150. Extradition is a process whereby
(a) one country transfers a suspected or convicted criminal to another country
(b) olden traditions are rejuvenated as a part of cultural events
(c) additional diction efforts are taken for language improvisation
(d) extraction of the tar is carried out from the mines

➥ *(a)* Extradition is the official process whereby one country transfers a suspected or convicted criminal to another country. Between countries, extradition is normally regulated by treaties, where extradition is compelled by laws, such as among subnational jurisdiction, the concept may be known more generally as rendition.

SNAP

Symbiosis National Aptitude Test

Solved Paper 2012

Max. Marks : 180 Time Alloted : 120 Min.

Section I *Quantitative and Data Interpretation and Sufficiency*

1. A dice is rolled three times and sum of three numbers appearing on the uppermost face is 15. The chance that the first roll was a four is
(a) 2/5 (b) 1/5
(c) 1/6 (d) None of these

2. A boat covers a distance of 30 km downstream in 2 h while it takes 6 h to cover the same distance upstream. What is the speed of the boat (in km/h)?
(a) 5 (b) 7.5
(c) 13 (d) None of these

3. A five-digit numbers is formed by using the digits 1, 2, 3, 4 and 5 without repetitions. What is the probability that the number is divisible by 4?
(a) 1/5 (b) 5/6
(c) 4/5 (d) None of these

4. If the algebraic sum of deviations of 20 observations measured from 23 is 70, mean of these observations would be
(a) 24 (b) 25
(c) 26 (d) None of these

5. An alloy of gold and silver weighs 50 g. It contains 80% gold. How much gold should be added to the alloy, so that percentage of gold is increased to 90?
(a) 50 g (b) 60 g (c) 30 g (d) 40 g

6. Weekly incomes of two persons are in the ratio of 7 : 3 and their weekly expenses are in the ratio of 5 : 2. If each of them saves ₹ 300 per week, then the weekly income of the first person is
(a) ₹ 7500 (b) ₹ 4500 (c) ₹ 6300 (d) ₹ 5400

7. Wheat is now being sold at ₹ 27 per kg. During last month its cost was ₹ 24 per kg. Find by how much per cent a family reduces its consumption, so as to keep the expenditure fixed?
(a) 10.2% (b) 12.1% (c) 12.3% (d) 11.1%

8. There are 10 stations on a railway line. The number of different journey tickets that are required by the authorities is
(a) 92 (b) 90
(c) 91 (d) None of these

9. The radius of a circle is so increased that its circumference increased by 5%. The area of the circle then increases by
(a) 12.5% (b) 10.25% (c) 10.5% (d) 11.25%

10. In how many ways can the letters of the word 'ABACUS' be rearranged such that the vowels always appear together?
(a) 6! / 2! (b) 3! · 3!
(c) (3! · 3!)/2! (d) (4! · 3!)/2!

11. In 4 yr, ₹ 6000 amounts to ₹ 8000. In what time at the same rate will ₹ 525 amount to ₹ 700?
(a) 2 yr (b) 3 yr
(c) 4 yr (d) 5 yr

Directions (Q. Nos. 12-14) *Study the following information to answer the given questions.*

At the start of a game of cards, J and B together had four times as much money as T, while T and B together had three times as much as J. At the end of the evening, J and B together had three times as much money as T, while T and B together had twice as much as J. B lost ₹ 200.

12. What fraction of the total money did T have at the beginning of the game?
(a) 1/3 (b) 1/8 (c) 2/9 (d) 1/5

13. What fraction of the total money did J win/lose?
(a) Won 1/12 (b) Lost 1/6
(c) Lost 1/3 (d) Won 1/5

14. What amount did B start with?
(a) ₹ 575 (b) ₹ 375 (c) ₹ 825 (d) ₹ 275

15. If a and b are negative and c is positive, which of the following statement/s is/are true?
I. $a - b < a - c$
II. If $a < b$, then $a/c < b/c$
III. $1/b < 1/c$
Select the correct answer using the codes given below
(a) Only I (b) Only II
(c) Only III (d) II and III

16. The diagonal of a square is $4\sqrt{2}$ cm. The diagonal of another square whose area is double that of the first square is
(a) 8 cm (b) $8\sqrt{2}$ cm
(c) $4\sqrt{2}$ cm (d) 16 cm

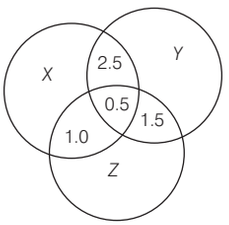
17. The maximum length of a pencil that can be kept in a rectangular box of dimensions 8 cm × 6 cm × 2 cm, is
(a) $2\sqrt{13}$ cm (b) $2\sqrt{14}$ cm
(c) $2\sqrt{26}$ cm (d) $10\sqrt{2}$ cm

Directions (Q. Nos. 18-19) *These question are based on the information given below*

The Venn diagram given below shows the estimated readership of 3 daily newspapers (*X*, *Y* and *Z*) in a city. The total readership and advertising cost for each of these papers is as below.

Newspaper	Readership (in lakh)	Advertising cost (₹ per sq cm)
X	8.7	6000
Y	9.1	6500
Z	5.6	5000

The total population of the city is estimated to be 14 million. The common readership (in lakh) is indicated in the given Venn diagram.



18. The number of people (in lakh) who read atleast one newspaper is
(a) 4.7 (b) 11.9 (c) 17.4 (d) 23.4

19. The number of people (in lakh) who read only one newspaper is
(a) 4.7 (b) 11.9 (c) 17.4 (d) 23.4

20. Sonali invests 15% of her monthly salary in insurance policies. She spends 55% of her monthly salary in shopping and on household expenses. She saves the remaining amount at ₹ 12750. What is Sonali's monthly income?
(a) ₹ 42500 (b) ₹ 38800 (c) ₹ 40000 (d) ₹ 35500

21. How many kg of tea worth ₹ 25 per kg must be blended with 30 kg of tea worth ₹ 30 per kg, so that by selling the blended variety at ₹ 30 per kg, there should be a gain of 10%?
(a) 36 kg (b) 40 kg (c) 32 kg (d) 42 kg

22. In an examination, out of 480 students 85% of the girls and 70% of the boys passed. How many boys appeared in the examination, if total pass percentage was 75%?
(a) 370 (b) 340
(c) 320 (d) 360

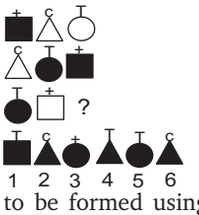
23. 300 g of salt solution has 40% salt in it. How much salt should be added to make it 50% in the solution?
(a) 40 g (b) 60 g
(c) 70 g (d) 80 g

24. What number should replace the question mark in the image below?

6	12	?
48	96	192

(a) 18 (b) 20
(c) 22 (d) 24

25. What image from bottom row should replace the question mark?
(a) 4
(b) 6
(c) 1
(d) None of the above



26. A five-digit numbers divisible by 3 is to be formed using numerical 0, 1, 2, 3, 4 and 5 without repetition. The total number of ways this can be done is
(a) 122 (b) 210 (c) 216 (d) 217

27. A contract is to be completed in 50 days and 105 men were set to work, each working 8 h a day. After 25 days, 2/5th of the work is finished. How many additional men be employed, so that the work may be completed on time, each man now working 9 h per day?
(a) 34 (b) 36 (c) 35 (d) 37

28. *A* can build up a structure in 8 days and *B* can break it in 3 days. *A* has worked for 4 days and then *B* joined to work with *A* for another 2 days only. In how many days will *A* alone build up the remaining part of the structure?
(a) 10 days (b) 9 days
(c) 12 days (d) None of these

29. The first two terms of a geometric progression add up to 12. The sum of the third and the fourth terms is 48. If the terms of the geometric progression are alternately positive and negative, then the first term is
(a) – 2 (b) – 4 (c) – 12 (d) 8

30. The mean of the numbers a, b, 8, 5, 10 is 6 and the variance is 6.80. Then, which one of the following gives possible values of a and b?
(a) $a = 0, b = 7$ (b) $a = 5, b = 2$
(c) $a = 3, b = 4$ (d) $a = 2, b = 4$

31. An agent sells goods of value of ₹ 15000. The commission which he receives at the ratio of $12\frac{1}{2}$ % is
(a) ₹ 1875 (b) ₹ 2125 (c) ₹ 2000 (d) ₹ 2700

32. $\sqrt{110.25} \times \sqrt{0.01} \div \sqrt{0.0025} - \sqrt{420.25}$ is equal to
(a) 0.75 (b) 0.50 (c) 0.64 (d) 0.73

Directions (Q. Nos. 33-35) *Study the following information to answer the given questions.*

The table below shows the number of people who responded to a survey about their favourite style of music. Use this information to answer the following questions to the nearest whole percentage.

Style of Music \ Age	15-20	21-30	31 +
Classical	6	4	17
Pop	7	5	5
Rock	6	12	14
Jazz	1	4	11
Blues	2	3	15
Hip-Hop	9	3	4
Ambient	2	2	2
	33	33	68

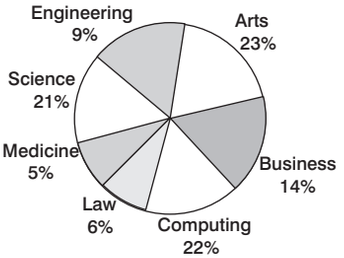
33. What percentage of respondents under 31 indicated that Blues is their favourite style of music?
(a) 7.1% (b) 7.6% (c) 8.3% (d)14.1%

34. What percentage of respondents aged 21-30 indicated a favourite style other than Rock music?
(a) 64% (b) 60% (c) 75% (d) 36%

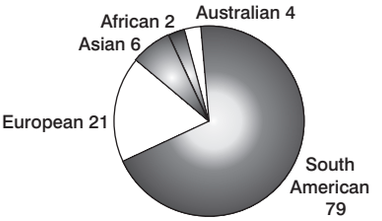
35. What percentage of the total sample indicated that Jazz is their favourite style of music?
(a) 6% (b) 8%
(c) 22% (d) 12%

Directions (Q. Nos. 36-40) *Study the following information to answer the given questions.*

The pie charts below show the percentage of students in each faculty at North-West University and the number of non-US students in the Arts faculty. These percentages have been rounded to the nearest whole number. There are a total of 1049 students in Arts faculty. Use this information to answer the following questions



Students by Faculty



Arts Students (non-US)

36. What percentage of students in the Arts faculty are non-US students?
(a) 14% (b) 9%
(c) 30% (d) 11%

37. How many students are there in the Engineering faculty?
(a) 420 (b) 410
(c) 390 (d) 440

38. How many students are there at the university?
(a) 4650 (b) 4560
(c) 4640 (d) 4450

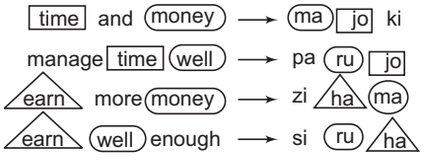
39. If 6% of Science students are Asian, how many Asian students are studying Science?
(a) 48 (b) 66
(c) 120 (d) 57

40. There are 34 European medical students. What percentage of the faculty does this represent?
(a) 14% (b) 18%
(c) 16% (d) 15%

Section II *Analytical and Logical Reasoning*

Directions (Q. Nos. 41-43) *Study the following information to answer the given questions.*

'time and money' is written as 'ma jo ki',
'manage time well' is written as 'pa ru jo',
'earn more money' is written as 'zi ha ma' and
'earn well enough' is written as 'si ru ha'.



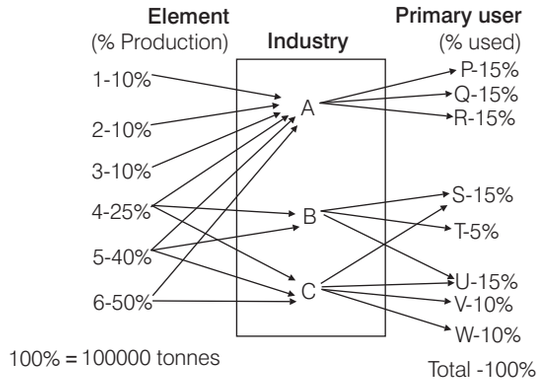
41. What is the code for 'earn'?
(a) si (b) ru
(c) ha (d) Cannot determined

42. Which of the following represents 'more time'?
(1) pa jo (2) zi ki (3) ma ki (d) jo zi

43. What is the code for 'manage'?
(a) ru (b) pa
(c) jo (c) ha

44. Lou observes that, if flight 409 is cancelled, then the manager could not possibly arrive in time for the meeting. But the flight was not cancelled. Therefore, Lou concludes, the manager will certainly be on time. Evelyn replies that even, if Lou's promises are true, his argument is fallaciou. and therefore she adds, the manager will not arrive on time after all.
Which of the following is the strongest thing that we can properly say about this discussion?
(a) Evelyn is mistaken in thinking Lou's argument to be fallacious and so her own conclusion is unwarranted.
(b) Evelyn is right about Lou's argument but nevertheless her own conclusion is unwarranted.
(c) Since, Evelyn is right about Lou's argument, her own conclusion is well supported.
(d) Since, Evelyn is mistaken about Lou's argument, her own conclusion must be false.

45. Cars are safer than planes. 50% of plane accidents result in death, while only one per cent of car accidents result in death.
Which of the following, if true, would most seriously weaken the argument above?
(a) Planes are inspected more often than cars.
(b) The number of car accidents is several hundred thousand times higher than the number of plane accidents.
(c) Pilots never fly under the influence of alcohol, while car drivers often do.
(d) Plane accidents are usually the fault of air traffic controllers not pilots.



Directions (Q. Nos. 46-48) *Refer to the chart below showing annual production and answer the questions that follow.*

46. Which industry/industries contribute/ to company S?
(a) Industries A and B (b) B and C
(c) A and C (d) Only B

47. Industry B processes what percentage of the total production of listed elements?
(a) 25% (b) 65%
(c) 40% (d) Cannot be determined

48. Of the listed elements processed by the industry A, how many tonnes are produced annually?
(a) 30000 (b) 50000
(c) 100000 (d) None of these

Directions (Q. Nos. 49-52) *In each of the following questions there are two blanks marked I and II.*

The words to fill in these blanks are given against I as (A), (B), (C), (D) and II as (P), (Q), (R), (S) respectively. The right words to fill in these blanks are given as four alternatives. The words on either side of the sign (::) have a similar relationship. That alternative which signifies this relationship is your answer.

49. I. : Increase :: Descend : II
I. (A) Grow (B) Ascend (C) Rise (D) Price
II. (P) Reduce (Q) Down (R) Decrease (S) Mountain
(a) AR (b) RB
(c) CP (d) DQ

50. Modern : I :: II : Old
I. (A) Ancient (B) Death (C) Famous (D) Civilization
II. (P) Industrialisation (Q) Young (R) Fashion (S) Western
(a) AQ (b) AS (c) BP (d) CR

51. Part : I :: Class : II
I. (A) Section (B) Whole (C) School (D) Students
II. (P) Student (Q) School (R) Teachers (S) Rooms
(a) AR (b) BQ (c) CP (d) DS

52. Summit : Apex :: I : II
I. (A) beautiful (B) Picture (C) Attractive (D) Enchanting
II. (P) Comfortable (Q) Pretty (R) Healthy (S) Brave
(a) AQ (b) BP
(c) DS (d) CR

53. If in a certain code, 'BEAUTIFUL' is coded as '573041208', 'BUTTER' as '504479', how is 'FUTURE' coded in that code?
(a) 201497 (b) 204097
(c) 704092 (d) 204079

54. How many Mondays are there in a particular month of a particular year, if the month ends on Wednesday?
(a) 4 (b) 5
(c) 3 (d) Cannot be determined

Directions (Q. Nos. 55-56) *There are two rows of numbers in each question. The upper row is complete and in the lower row one number is missing. Find a suitable number to fill the blank space, so that the symmetry with the upper row is maintained.*

55. 17 102 12
15 ? 10
(a) 211 (b) 75 (c) 125 (d) 117

56. 12 336 14
15 ? 16
(a) 220 (b) 480 (c) 125 (d) 450

Directions (Q. Nos. 57-58) *Study the following informa- tion to answer the given questions.*

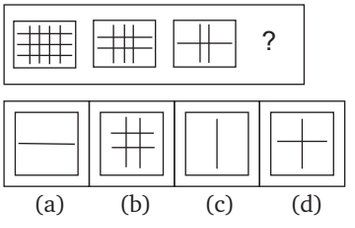
Seven People A, B, C, D, E, F and G are planning to enjoy boating.There are only two boats and the following conditions are to be kept in mind.
I. A will go in the same boat in which E is to go.
II. F cannot go in the boat in which C is, unless D is also accompanying.
III. Neither B nor C can be given the boat in which G is.
IV. The maximum number of persons in one boat can be four only.

57. If F and B are in one boat, which of the following statements is true?
(a) G is in the other boat
(b) D is in the other boat
(c) C is in the other boat
(d) E is with F and B in one boat

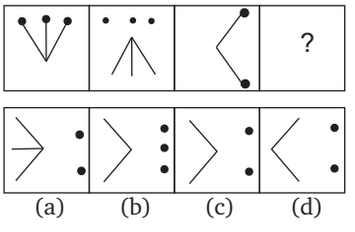
58. If E gets the boat with F, which of the following is the complete and accurate list of the people, who must be sitting in other boat?
(a) F and E (b) G and A
(c) D and A (d) C, D and B

Directions (Q. Nos. 59-60) *A series of figures has been shown on the left. Find the figure in the place of '?' from the figures on the right.*

59.



60.



61. Complete the following series.
GMSY, IOUA, KQWC, ?
(a) MSYE (b) NSYE (c) MYTE (d) MSYF

Directions (Q. Nos. 62-65) *Following is a new plan called 'Airtel One', Introduced by Airtel (GSM Service Provider) for its post-paid customers.*

Pulse Rate	60 s		
Price of Plan	₹ 99		
Free Airtime on Plan	Nil		
Incoming Calls(₹)	Free		
Outgoing Calls ₹/minute	Airtel	GSM / CDMA	Landline / WLL
Local rates	₹ 1	₹ 1	₹ 2
STD rates			
50-200 km	₹ 1.50	₹ 1.50	₹ 2
200-500 km	₹ 1.50	₹ 2.50	₹ 2.50
500 + km	₹ 1.50	₹ 3.00	₹ 3.50
ISD			
USA, Canada, Europe (Fixed Line)		₹ 7	
Gulf, Europe (Mobile) SAARC		₹ 10	
Rest of the world		₹ 40	
SMS			
Local		₹ 1.50	
National		₹ 2.00	
International		₹ 5.00	
Value Added Service (₹)	₹ 3.00 and ₹ 6.00 Depending on the service		
Monthly bill = Call Charges + SMS Charges + Price of Plan			

62. If Mohan made STD calls (within 100 km) for 30 min and spent 100 min on local calls with 30% on landline, 40% on GSM and 30% on Airtel and Rohan spent 18 min on STD within 150 km and spent 120 min on local calls with 30%, 40% and 30% on GSM, Iandline and Airtel respectively, then who spent more?
(a) Mohan
(b) Rohan
(c) Both spent the same amount
(d) Cannot be determined

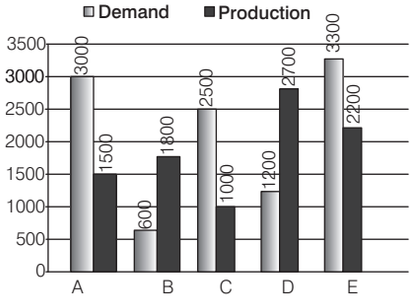
63. A new SMS scheme was introduced at 60 paise/local SMS, with an additional monthly charge of ₹ 35. Who will not benefit from the scheme? A person sending
(a) 38 local SMS a month
(b) 40 local SMS a month
(c) 60 local SMS a month
(d) 59 local SMS a month

64. A bill of ₹ 199 per month is definitely not possible when you make only
(a) 30 calls to Airtel on STD (500 + km) and 55 local calls in a month
(b) 16 calls to Airtel on STD (15 km and 76 local calls in a month)
(c) 10 STD calls (250 km) each to Airtel, GSM and landline respectively and 30 locals in a month
(d) 8,4 and 7 calls to Airtel, GSM and landline respectively on STD (500 + Kms) and 55 local calls in a month

65. A person makes 12 min of ISD calls in a month. If 80% of his ISD bill came from calling the rest of the world, then the ISD calls for the minimum duration were made to which of the following group of countries?
(a) USA, Canada, Europe (Fixed Line)
(b) Gulf, Europe (Mobile), SAARC
(c) Rest of the world
(d) Cannot be determined

Directions (Q. Nos. 66-67) *These questions are based on the graph which shows the demand and production statistics of 5 TV companies.*

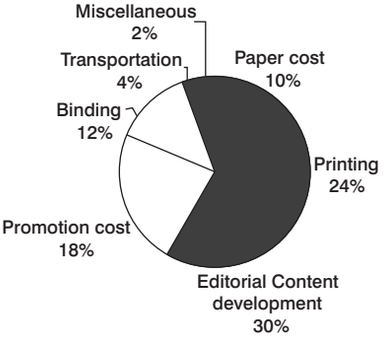
66. What is the difference between the average demand and



the average production of the five companies taken together?
(a) 1400 (b) 400 (c) 280 (d) 138

67. The production of the company D is how many times of the production of the company A?
(a) 1.8 (b) 1.5 (c) 2.5 (d) 1.11

Directions (Q. Nos. 68-69) *The following questions are based on the pie chart given below which gives the expenditure incurred in printing a magazine.*



68. What is the angle for the sector representing paper cost?
(a) 10° (b) 36° (c) 23.5° (d) 45°

69. For a given issue of the magazine, the 2% of the cost is ₹ 2000 and the print-run is 12500 copies. What should be the sale price if the publisher desires a profit of 5%?
(a) ₹ 5 (b) ₹ 7.5 (c) ₹ 8 (d) ₹ 8.40

70. If 'water' is called 'food', 'food' is called 'tree', 'tree' is called 'sky', 'sky' is called 'wall', on which of the following does a 'fruit' grow?
(a) Water (b) Food (c) Tree (d) Sky

Section III *General English*

Direction (Q.Nos. 71-72) *Choose the correct option to fill the blank for correct grammatical use.*

71. She is a close friend of mine.
(a) my (b) I (c) me (d) mine

72. He took care of her before she was restored to health.
(a) when (b) before (c) after (d) till

73. Choose the correct option.
The man ..... his appearance completely since then.
(a) is changing (b) changed
(c) has changed (d) is changed

74. From the options below, choose the word with the incorrect spelling.
(a) Diarrhea (b) Diaper (c) Dichotomy (d) Dias

75. Choose the word spelt correctly
(a) superintentent (b) seive
(c) alloted (d) dissipatc

Direction (Q. Nos. 76-79) *Fill in the blanks with the correct pair of words.*

76. He lives ..... Bengaluru ..... 115, Richmond Road.
(a) at, in (b) in, at (c) at, on (d) in, on

77. 'Aurally challenged' is a ..... for the .....
(a) metaphor, blind (b) euphemism, deaf
(c) smile, disabled (d) synonym, dumb

78. If error is a ....., fault is .....
(a) defect, a mistake (b) mistake, a defect
(c) slip, an inaccuracy (d) blunder, to blame

79. The man has ..... the rules of ethical conduct; he is ..... a beast.
(a) flaunted, literally (b) ignored, basically
(c) broken, as (d) flouted, virtually

Directions (Q. Nos. 80-81) *Which punctuation mark is missing in the following sentences?*
(a) inverted commas (b) semicolon
(c) comma (d) hyphen

80. Part of Australia is known to the natives as The Outback.

81. I know, that you want to learn to drive Rima but you are too young.

Directions (Q. Nos. 82-83) *Select as option the word closest in meaning to the given word.*

82. Veracious
(a) False (b) Varied (c) Image (d) Truthful

83. Perturb
(a) Stipulate (b) Turn around
(c) Disturb greatly (d) Compatible

Direction (Q. Nos. 84-85) *Choose the option that is the closest in meaning to the phrases in bold.*

84. He was a king, who ruled his subject with a high hand.
(a) sympathetically (b) oppressively
(c) democratically (d) generously

85. I racked my brains to solve this difficult problem.
(a) I read a large number of books
(b) I consulted several people
(c) I used my commonsense
(d) I subject my mind to hard thinking

Directions (Q. Nos. 86-87) *Select the most appropriate meaning of the underlined idiomatic phrase.*

86. I was supposed to give a speech to my English class, but I got cold feet and I didn't go.
(a) My English class-room makes my feet feel too cold
(b) My feet got cold so I didn't go
(c) I got too nervous and I didn't go
(d) It was winter time and I had no socks for my feet

87. Take care of what you say! You will have to eat your words!
(a) You have no food to eat
(b) You will have to take back what you have said
(c) You are not good with your language
(d) None of the above

88. Which two sentences convey the same idea?
1. Shouldn't they have checked your tickets?
2. I wonder, if they should have checked your tickets.
3. I want to know, if they checked your tickets.
4. They should have checked your tickets.
(a) 2 and 3 (b) 1 and 2 (c) 1 and 4 (d) 2 and 4

Directions (Q. Nos. 89-90) *Choose a single word, which is opposed to the meaning contained in the sentence.*

89. He was magnanimous and his benevolence made him give to charity.
(a) mean (b) cruel (c) snobbish (d) tyrannical

90. Biannual is
(a) once is two years (b) every year
(c) twice a year (d) after every two years

91. Choose the odd one out
(a) temporal (b) ephemeral
(c) transient (d) eternal

92. Choose the correct option If 'inter' mean; between as in interstate, 'intra' as in intravenous means
(a) into (b) onto
(c) with in (d) without

93. Match the List-I (parts of speach) to List-I (thier usage)

List I (Parts of speach)	List II (Their usage)
A. Adjective	1. The fire engine come rushing down the hill.
B. Preposition	2. He was seen the ups and downs of life.
C. Noun	3. Down with the tyrant.
C. Verb	4. The porter was hit by the down train

Codes

	A	B	C	D
(a)	4	1	2	3
(b)	2	1	4	3
(c)	1	2	4	3
(d)	1	3	2	4

Directions (Q. Nos. 94-95) *Choose the segment with the error. If there is no error, choose 'd'.*

94. Match the items in List I with those in List II

List I	List II
A. As deaf as	1. gall
B. As bitter	2. an eel
C. As unpredictable as	3. a post
D. As slippery as	4. The weather

Codes

	A	B	C	D		A	B	C	D
(a)	3	2	4	1	(b)	3	1	4	2
(c)	4	2	1	3	(d)	1	3	2	4

95. Match the List I (Parts of spech) to List II (their usage).

List I (Parts of speech)	List II (Their usage)
A. Adjective	1. Rain comes from above
B. Preposition	2. His conduct is above suspicion.
C. Noun	3. Look above the mantel piece.
D. Verb	4. The above information is for public

Codes

	A	B	C	D		A	B	C	D
(a)	4	3	1	2	(b)	3	2	1	4
(c)	4	1	2	3	(d)	2	4	1	3

Directions (Q. Nos. 96-97) *Choose the segment with the error. If there is no error, choose 'd'.*

96. Idli and Sambar (a) make the breakfast (b) in the South. (c) No error (d)

97. Your account should (a) have been credited (b) with three months interest. (c) No error (d)

98. Choose the sentence where the underlined word is used appropriately
(a) Who's car are you planning to borrow?
(b) Whose planning to have a birthday party?
(c) He is the person who you met at the junction.
(d) Does he know whom to call about the hoted reservations?

Directions (Q. Nos. 99-100) *Choose the segment with the error. If there is no error, choose 'd'.*
(a) if the first sentence is correct.
(b) if the second sentence is correct.
(c) if both sentences are correct.
(d) if both sentences are wrong.

99. 1. Can you imagine his forgetting his own birthday?
2. Can you imagine him forgetting his own birthday?

100. 1. Recently, I read about a unique wedding that took place in the newspaper.
2. Recently, I read in the newspaper about a unique wedding that took place.

101. Complete the given sentence by choosing the correct phrase
You cannot succeed unless
(a) you do not work hard (b) you shall not work hard
(c) You will work hard (d) you work hard

102. Re-arrange the scrambled segments in logical order to make a complete sentence
1. to places, where more opportunities are available
2. and so there is a great demand for English
3. for professional and economic growth
4. because it takes one outside one's own community
5. English is the language of opportunities
(a) 1, 2, 5, 3, 4 (b) 5, 4, 1, 3, 2
(c) 3, 1, 4, 2, 5 (d) 2, 1, 3, 5, 4

103. Choose the best construction
(a) If you would have taken care you wouldn't have got typhoid.
(b) If you took care you wouldn't have got typhoid.
(c) If you take care you wouldn't have got typhoid.
(d) If you had taken care you wouldn't have got typhoid.

Directions (Q. Nos. 104-105) *Choose the correct option.*

104. At times, we are all ..... to be mistaken.
(a) apt (b) likely
(c) considered (d) able

105. He hardly cares, ...?
(a) does he (b) doesn't he (c) will he (d) won't he

Directions (Q. Nos. 106-107) *Identify the figures of speech in the following sentences.*

106. As proud as a peacock.
(a) Metaphor (b) Simile
(c) Apostrophe (d) Epigram

107. Death lays his icy hand on kings.
(a) Personification (b) Exclamation
(c) Simile (d) Anticlimax

Direction (Q. Nos. 108-110) *Read the passage carefully and answer within the context.*

A Way to Deal with Frozen Feelings

Every child experiences all that happens around him with total awareness. In the first seven years, the child's brain is like a sponge, taking in all sensory inputs and building his idea of his surroundings. As long as the environment is safe, the child learns with incredible speed. However, when the environment is scary or stressful, the unlearns past learning just as rapidly. In the early years of every child's life, whenever there is shock, violence, fear or pain, these intense emotions are imprinted deeply into memory. Whenever the same activity or situation is repeated, the nervous system and body subconsciously re-experience the memory of that trauma.

An emotional situation that takes us out of the present and into the past means that whenever the same kind of emotion crops up later in our life, we return to the past for our reference point. If that point was at age three, we find ourselves behaving like a three-years-old. We feel childish and we behave childishly. Our feelings are the cause of this 'glitch' in our learning process. We know we should be able to make a positive change, but that doesn't change anything.

The process of change need not be traumatic. We couldn't have done any better because we didn't know to. But we should realise that was then and this is now! We can choose to choose again. It's up to us. It's our movie!

108. The "Frozen Feelings" being talked about are about.
(a) negative childhood experiences
(b) Childhood learning patterns
(c) Inability to learn as an adult
(d) none of the above

109. A 'glitch' is
(a) a ditch
(b) uneasy emotions
(c) sudden malfunction or breakdown
(d) learning patterns

110. Identify the correct statement, based on the paragraph.
(a) The process of change needs to be traumatic
(b) We feel childish and we behave childishly
(c) Both sentences are incorrect
(d) Both the sentences are correct

Section IV *General Awareness*

111. Dendrochronology is
(a) The method of scientific dating based on the analysis of tree-ring growth patterns.
(b) a core sample from the accumulation of snow and ice over many years that have re-crystallised and have trapped air bubbles from previous time periods.
(c) is the study of glaciers, or more generally ice and natural phenomena that involve ice.
(d) the scientific study of landforms and the processes that shape them.

112. Which company launched the first mobile phone operation in India?
(a) Bharti Airtel (b) Essar
(c) Max Touch (d) Modi Telstra

113. Which is the longest running English language TV cartoon in history, first launched by Warner Brothers in 1969 exclusively for television and is now a popular fare on the Cartoon Network?
(a) Scooby-Doo (b) Tom and Jerry
(c) Popeye (d) Johnny Bravo

114. The expansion for BIFR, in the context of the Indian Industry is
(a) Board for Industrial and Financial Reconstruction
(b) Bureau for Industrial and Financial Reconstruction
(c) Board for Investment and Financial Reconstruction
(d) Bureau for Investment and Financial Reconstruction

115. What is Ekistics?
(a) The water sports related study
(b) The Science of body mass
(c) The art of card tricks
(d) The Science of human settlements

116. What is Red Herring in an IPO?
(a) Prospectus
(b) Submission of Form
(c) Funds Generated during IPO
(d) Minimum Offer Per Share

117. Eight O'clock coffee, a US based coffee marketer, is a group company of
(a) Tata (b) ITC (c) Britannia (d) Unilever

118. Bio-diesel is extracted from the following plant
(a) Hibiscus (b) Aloe Vera
(c) Jatropha (d) Chamomile

119. Which woman tennis player won the maximum number of Grand Slam singles titles in the history of tennis?
(a) Steffi Graf (b) Martina Navratilova
(c) Billi Jean King (d) Margaret Court

120. What is the nickname of New York City?
(a) Big Sweet Lime (b) Big Apple
(c) Big Grape (d) None of these

121. Which was the first Indian company to be listed on NASDAQ?
(a) TCS (b) Wipro
(c) Infosys (d) Cognizant

122. 'TAREGNA' a village in Bihar was in the news because of
(a) Rahul Gandhi's election campaign
(b) Viewing of the solar eclipse
(c) A visit by Hillary Clinton
(d) The Naxal massacre of tribals

123. The driver for Formula One's Force India Team is
(a) Mark Webber (b) Adrian Sutil
(c) Felipe Massa (d) Jensen Button

124. (MDRT)—Million Dollar Round Table consists of
(a) World's richest industrialists
(b) World's most powerful golfers
(c) Global Association of Life Insurance professionals
(d) World's richest sportpersons

125. Who declares the 'Credit Policy' of India?
(a) The Finance Ministry
(b) Reserve Bank of India
(c) The Commerce Ministry
(d) The Indian Banks' Association (IBA)

126. Name of the body which regulates the Stock Exchanges in India is
(a) CBI (b) CID
(c) SHCIL (d) SEBI

127. Rafflesia arnoldii is its botanical name and it holds a world record in the plant world. What's the record?
(a) Tallest cacti
(b) Largest bloom
(c) Longest surviving tree
(d) Smallest plant

128. Temujin was a famous historical character of the world, but we know him by a different name. Which one?
(a) Babur (b) Genghis Khan
(c) Kublai Khan (d) Emperor Nero

129. What is Short Selling in stock market lingo?
(a) Selling the shares which you do not own.
(b) Selling a share after owning it for a short period of time.
(c) Selling all the shares in your booty, which are not productive.
(d) Selling of shares-which are quoted for a short period of time.

130. Mahatma Gandhi was nominated for Nobel Peace prize for the first time in the year 1937 and last time in the year 1948. How many times in between these two years was he nominated?
(a) 1 (b) 3
(c) 2 (d) 0

131. Who among the following has been awarded with Rajiv Gandhi Khel Ratna Award, 2012?
(a) Vijay kumar (b) Yogeshwar Dutt
(c) Both (a) and (b) (d) Sushil Kumar

132. Who has won Women's Single Title US Open, 2012?
(a) Victoria Azarneka (b) Serena Williams
(c) Sara Errani (d) Roberta Vinci

133. Which bank has launched Click-2-Card Service on August 28, 2012?
(a) PNB (b) SBI
(c) ICICI (d) HDFC

134. Who has been elected as the President of Somalia by the country's Parliament on September 11, 2012?
(a) Mohammed Osman Jawari
(b) Muse Hussan Abdulle
(c) Sharif Sheikh Ahmed
(d) Hassan Sheikh Mahmood

135. Who has been appointed as the chairman of National Book Trust (NBT) on September 6, 2012?
(a) Sukumar Azhikode (b) A Sethumadhavan
(c) KN Tilak Kumar (d) None of these

136. Who has Written the book 'Still Counting The Dead'?
(a) Frances Harrison (b) Faisal Devji
(c) JK Rowling (d) None of these

137. How many scientists has been selected for Shanti Swarup Bhatnagar Prize 2012?
(a) 11 (b) 10
(c) 12 (d) 13

138. Who were the first twins to play test cricket?
(a) Merv and Kim Hughes
(b) Ian and Greg Chappel
(c) Surinder and Mohinder Amarnath
(d) Steve and Mark Waughs

139. Which country has the largest rail network in the world?
(a) India (b) China
(c) USA (d) UK

140. Who won the 2009 French Open tennis ladies title?
(a) Svetlana Kuznetsova (b) Kristina Mladenovic
(c) Maria Sharapova (d) Venus Williams

141. Who invented the microchip?
(a) Robert Noyce and Jack Kilby
(b) Larry Page and Sergey Brin
(c) Sabeer Bhatia
(d) Gordon Moore and Robert Noyce

142. Which is the largest landlocked country in the world?
(a) Switzerland (b) China
(c) Kazakhstan (d) Mongolia

143. Which country makes Panama Huts?
(a) Ecuador (b) Panama (c) Cuba (d) Bolivia

144. Who was the Prime Minister of India when the 42nd Amendment Bill bringing in the changes in the Constitution was passed?
(a) Rajiv Gandhi (b) VP Singh
(c) Indira Gandhi (d) Morarji Desai

145. Which one of the following planets lost its planet status recently?
(a) Moon (b) Neptune (c) Pluto (d) Saturn

146. One barrel of oil is approximately equivalent to how many litres?
(a) 200 (b) 150 (c) 120 (d) 160

147. Five Year Plans in India are finally approved by the
(a) Union Cabinet
(b) President on the Advice of Prime Minister
(c) National Development Council
(d) Planning Commission

148. Many a times we read in the newspapers about Southern Cone of South America; which includes Argentina, Chile, Paraguay, Uruguay and Peru. Which is the most spoken language of this geographical area?
(a) Portugese (b) Spanish
(c) French (d) English

149. Which one of the following is not a member of the Organisation of Petroleum Exporting Countries (OPEC)?
(a) Algeria (b) Brazil (c) Ecuador (d) Nigeria

150. Who invented 'computer laptop'?
(a) Arthur Wynna - USA (b) Q Daimler - Germany
(c) Sinclair - Britain (d) None of these

Answer *with* Explanations

1. (*d*) Total number of favourable outcomes, $n(S) = 6^3 = 216$

Combinations of outcomes for getting sum of 15 on uppermost face = (4,5,6), (5,4,6), (6,5,4), (5,6,4), (4,6,5), (6,4,5), (5,5,5), (6,6,3), (6,3,6), (3,6,6)

Now, outcomes on which first roll was a four, $n(E)$

$= (4,5,6), (4,6,5)$

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{2}{216} = \frac{1}{108}$

2. (*d*) Speed of boat downstream $= \frac{\text{Distance covered}}{\text{Time taken}}$

$= \frac{30}{2} = 15 \text{ km/h}$

Now, speed of boat downstream $= \frac{\text{Distance covered}}{\text{Time taken}}$

$= \frac{30}{6} = 5 \text{ km/h}$

$\therefore$ Speed of boat $= \frac{\text{Speed downstream} + \text{Speed upstream}}{2}$

$= \frac{15+5}{2} = \frac{20}{2} = 10 \text{ km/h}$

3. (*d*) Number of five-digit numbers which can be formed with 1, 2, 3, 4 and 5 without repetition,

$n(S) = 5 \times 4 \times 3 \times 2 \times 1 = 120$

Number of five-digit numbers which are formed with 1, 2, 3, 4 and 5 are divisible by 4

$= 2(3 \times 2)$

$n(E) = 12$

As last two digit must be 12 or 24.

$P(E) = \frac{n(E)}{n(S)} = \frac{12}{120} = \frac{1}{10}$

4. (*d*) Mean $= \frac{\text{Sum of observation}}{\text{Number of observation}}$

Sum of observation is given as $= 70$

Number of observation $= 20$

$\text{Mean} = \frac{70}{20} = 3.5$

But here observation starts from 23.

So, mean is $23 + 3.5 = 26.5$

5. (*a*) Gold in 50 g of alloy $= \frac{80 \times 50}{100} = 40 \text{ g}$

Now, according to question,

$\Rightarrow \quad \frac{40+x}{50+x} = \frac{90}{100}$

$\Rightarrow \quad 100(40+x) = 90(50+x)$

$\Rightarrow \quad 10(40+x) = 9(50+x)$

$\Rightarrow \quad 400 + 10x = 450 + 9x$

$x = 450 - 400 = 50 \text{ g}$

Thus, 50 g of gold must be added to make it 90%.

6. (*c*) Ratio of income $= 7 : 3$

and ratio of expenses $= 5 : 2$

So, $\quad a = 7, b = 3, \ c = 5, \ d = 2, \ x = ₹\ 300$

$\therefore$ Income of 1st person $= \frac{x(d-c)}{(ad-bc)} \cdot a$

$= \frac{300(2-5)}{14-15} \times 7 = \frac{300 \times -3 \times 7}{-1} = ₹\ 6300$

7. (*d*) Old price of rice = ₹ 24 per kg

New price of rice = ₹ 27 per kg

$\therefore$ Increase in price of rice

= New price − Old price

$= 27 - 24 = ₹\ 3$ per kg

$\therefore$ Increased percentage in price $= \frac{3}{24} \times 100 = 12\frac{1}{2}\%$

Now, reduction in consumption, so as to keep the expenditure fixed.

According to the formula,

$= \left(\frac{a}{100+a} \times 100\right)\% = \left(\frac{12.5}{100+12.5} \times 100\right)\% = 11.1\%$

8. (*b*) There are 10 stations on railway line.

So, the number of different journey tickets between two stations from given 10 stations from one side,

$^{10}C_2 = \frac{10 \times 9}{2} = 45$

Similarly, number of different journey tickets from other side

$= 45$

$\therefore$ Total number of tickets to be generated by authorities

$= 45 + 45 = 90$

9. (*a*) Increase in circumference of circle = 5%

$\therefore$ Increase in radius is also 5%.

Now, increase in area of circle is given by the following formula

$= \left(2a + \frac{a^2}{100}\right)\%$

where, a = Increase in radius $= \left(2 \times 5 + \frac{5 \times 5}{100}\right)\% = 12.5\%$

10. (*d*) There are 6 letters in 'ABACUS'

$\therefore$ Number of vowels = 3 (A, A, U)

As these 3 vowels has to be together, then we have to arrange 4 letters.

$\therefore$ Ways of arranging 4 letters $= 4!$

Ways of arranging 3 vowels $= \frac{3!}{2!}$

(since, there are 2 A, so it is divided by 2)

Now, possible number of combinations are $4! \times \frac{3!}{2!}$.

11. (*c*) Amount = ₹ 8000, Time (T) = 4 yr, Principal (P) = ₹ 6000, Rate (R%) = ?

Simple Interest (SI) $= (A - P)$

$= \text{Amount} - \text{Principle} = 8000 - 6000 = ₹\ 2000$

According to the formula of simple interest,

$\text{SI} = \frac{P \times R \times T}{100}$

$2000 = \frac{6000 \times R \times 4}{100}$

$R = \frac{2000 \times 100}{6000 \times 4} = \frac{25}{3}\%$

Now, again
Amount (A) = ₹ 700
Principal (P) = ₹ 525, Time (T) = ?

$$\text{Rate } (R) = \frac{25}{3}\%$$

Simple Interest = $A - P$ = ₹ (700 – 525) = ₹ 175
Using formula,

$$\text{SI} = \frac{P \times R \times T}{100} \Rightarrow 175 = \frac{525 \times \frac{25}{3} \times T}{100}$$

$$\therefore \quad T = \frac{175 \times 100 \times 3}{525 \times 25} = 4 \text{ yr}$$

12. (d) Let the money with J = x
The money with B = y
The money with T = z
According to the question,

$$x + y = 4z \quad \text{...(i)}$$

and $$z + y = 3x \quad \text{...(ii)}$$

On solving Eqs. (i) and (ii) we get

$$\begin{array}{r} x + y = 4z \\ z + y = 3x \\ (-) \quad - \quad - \\ \hline x - z = 4z - 3x \end{array}$$

$$\Rightarrow \quad x + 3x = 4z + z \Rightarrow 4x = 5z$$

$$\therefore \quad x = \frac{5}{4}z$$

On putting value of x in Eq. (i), we get

$$\frac{5}{4}z + y = 4z$$

$$y = 4z - \frac{5}{4}z = \frac{16z - 5z}{4} = \frac{11}{4}z$$

Now, ratio of money with J, B and T, respectively J : B : T

$$\frac{5}{4}z : \frac{11}{4}z : z \Rightarrow \frac{5}{4} : \frac{11}{4} : 1 \Rightarrow 5 : 11 : 4$$

Now, T's share in total money $= \frac{4}{5+11+4} = \frac{4}{20} = \frac{1}{5}$

13. (a) Using 2nd condition,
Let the money with J in end = x_1
The money with B in end = y_1
The money with T in end = z_1
According to the condition given in question,

$$x_1 + y_1 = 3z_1 \quad \text{...(iii)}$$

and $$z_1 + y_1 = 2x_1 \quad \text{...(iv)}$$

On subtracting Eq. (iv) from Eq. (iii), we get

$$\begin{array}{r} x_1 + y_1 = 3z_1 \\ z_1 + y_1 = 2x_1 \\ (-) \quad - \quad - \\ \hline x_1 - z_1 = 3z_1 - 2x_1 \end{array}$$

$$\Rightarrow \quad 3x_1 = 4z_1$$

$$\therefore \quad x_1 = \frac{4}{3}z_1$$

On putting value of x_1 in Eq. (iii), we get

$$\frac{4}{3}z_1 + y_1 = 3z_1$$

$$y_1 = 3z_1 - \frac{4}{3}z_1 = \frac{9z_1 - 4z_1}{3} = \frac{5}{3}z_1$$

Now, their ratio of money at the end of the day

$$= \frac{4}{3}z_1 : \frac{5}{3}z_1 : z_1 = 4 : 5 : 3$$

Now, J's share in the evening was $\frac{4}{12} = \frac{1}{3}$
J's share in the morning was $\frac{5}{20} = \frac{1}{4}$
So, J's share increases, hence he win.
Now, winning share $= \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$

14. (c) Now, let total money is a.
Then, by condition given in question for B,

$$\frac{11}{20}a - \frac{5}{12}a = 200$$

As B lost ₹ 200.

$$\frac{66a - 50a}{120} = 200$$

$$a = 1500$$

Now, B's amount in starting $= \frac{11}{20} \times 1500 =$ ₹ 825

15. (c) By given condition
a and b = – ve
c = + ve
Then, $\frac{1}{b}$ = – ve

$$\frac{1}{c} = + \text{ ve}$$

So, $$\frac{1}{b} < \frac{1}{c}$$

16. (a) Diagonal of square = $\sqrt{2}\,a$
where a = Side of square $4\sqrt{2} = \sqrt{2}\,a$,

$$a = 4 \text{ cm}$$

Now, area of square = $a^2 = (4)^2 = 16$
Side of a square whose area is 2×16.

$$\therefore \quad a_1^2 = 32; \; a_1 = \sqrt{32}; \; a_1 = 4\sqrt{2}$$

Now, diagonal of new square = $\sqrt{2}\,a = \sqrt{2} \times 4\sqrt{2} = 8$ cm

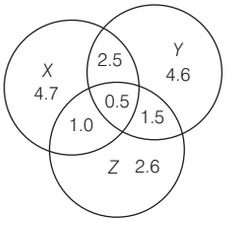
17. (c) Length of largest pencil that can be kept in a box

$$= \sqrt{l^2 + b^2 + h^2}$$

where, $l = 8$ cm, $b = 6$ cm, $h = 2$ cm

$$= \sqrt{64 + 36 + 4} = \sqrt{104} = 2\sqrt{26} \text{ cm}$$

18. (c) Venn diagram can be rearranged as



So, total number of reader who read atleast one newspaper
= 4.7 + 4.6 + 2.6 + 0.5 + 1.0 + 1.5 + 2.5 = 17.4 (in lakh)

19. (b) Number of reader who read only one newspaper
= 4.7 + 4.6 + 2.6 = 11.9 lakh

20. (*a*) Total salary spent on insurance, shopping and household expenses $= 15 + 55 = 70\%$

$\therefore$ Saving $= 100 - 70 = 30\%$

Let the total salary be x.

$\therefore$ 30% of $x = ₹\ 12750$

$\Rightarrow \quad \frac{30 \times x}{100} = 12750 \Rightarrow x = \frac{12750 \times 100}{30} = ₹\ 42500$

21. (*a*) Let the quantity of tea worth ₹ 25 $= x$

Then, by condition given in question,

$$(25x + 30 \times 30) \times \frac{110}{100} = 30\,(30 + x)$$

$\Rightarrow \quad (25x + 900)11 = 10\,(900 + 30x)$

$\Rightarrow \quad 275x + 9900 = 9000 + 300x$

$\Rightarrow \quad 300x - 275x = 900$

$$x = \frac{900}{25} = 36\,\text{kg}$$

22. (*c*) Total number of students = 480

Percentage of total student passed

$$= 75\% \text{ of total student} = \frac{75}{100} \times 480 = 360 \text{ students}$$

Now, using the condition from the question, let the number of boys be x.

Then,

70% of $x + 85\%$ of $(480 - x) = 360$

$$\Rightarrow \quad \frac{70 \times x}{100} + \frac{85 \times (480 - x)}{100} = 360$$

$\Rightarrow \quad 70x - 85x + 40800 = 36000$

$\Rightarrow \quad 40800 - 36000 = 85x - 70x \Rightarrow 4800 = 15x$

$$\therefore \quad x = \frac{4800}{15} = 320$$

$\therefore$ There are 320 boys who appeared for the examination.

$\therefore \quad (120 + x)\,2 = 300 + x$

$\therefore \quad 240 + 2x = 300 + x$

$$x = 60\,\text{g}$$

23. (*b*) 40% is salt in 300 g of salt solution.

The quantity of salt $= \frac{40 \times 300}{100} = 120\,\text{g}$

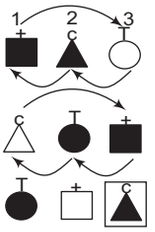
Now, by the condition in the question, $\frac{120 + x}{300 + x} = \frac{50}{100}$

24. (*d*) Series is given as

$\underline{6} \times 2 \rightarrow \underline{12} \times 2 \rightarrow 24$

$\underline{48} \times 2 \rightarrow \underline{96} \times 2 \rightarrow \underline{192}$

25. (*b*) In the given arrangement transition takes place as follows



Also, the unshaded figure follow the pattern of change from

$$3 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 1$$

26. (*c*) A five digit which divisible by 3 is formed when sum of digits is also divisible by 3.

So, combination formed using six digits, which are divisible by 3

$$= 5 + 4 + 3 + 2 + 1 = 15$$

$$= 5 + 4 + 2 + 1 + 0 = 12$$

So, set of numbers are (5, 4, 3, 2, 1) and (5, 4, 2, 1, 0).

Number formed by using 1st set $= 5 \times 4 \times 3 \times 2 \times 1 = 120$

Similarly, using 2nd set $= \underline{4} \times 4 \times 3 \times 2 \times 1 = 96$

Hence, using 2nd set underlined place cannot be filled by 0 otherwise it will become a four digit numbers.

$\therefore$ Total numbers $= 120 + 96 = 216$

27. (*c*) Using formula,

$$\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$$

Here, $M_1 = 105, D_1 = 25, H_1 = 8, W_1 = \frac{2}{5}$

Now, let the additional men be x.

Then, $M_2 = 105 + x,\ D_2 = 25,\ H_2 = 9, W_2 = 1 - \frac{2}{5} = \frac{3}{5}$

On putting these values in the above formula, we get

$$\frac{105 \times 25 \times 8}{2/5} = \frac{(105 + x) \times 25 \times 9}{3/5}$$

$$\Rightarrow \quad \frac{105 \times 8}{2} = \frac{(105 + x) \times 9}{3}$$

$\Rightarrow \quad 105 \times 4 = (105 + x) \times 3$

$\Rightarrow \quad 105 \times 4 = 105 \times 3 + 3x \Rightarrow 3x = 105$

$$\therefore \quad x = 35 \text{ men}$$

28. (*d*) A's one day work $= \frac{1}{8}$

B's one day work in breaking the building $= \frac{1}{3}$

Now, according to the question, A's four days work

$$= 4 \times \frac{1}{8} = \frac{1}{2}$$

Now, A's and B's two days work $= 2\left(\frac{1}{8} - \frac{1}{3}\right) = 2 \times \frac{-5}{24} = \frac{-10}{24}$

Total work done in 6 days $= \frac{1}{2} + \left(\frac{-10}{24}\right) = \frac{12 - 10}{24} = \frac{2}{24} = \frac{1}{12}$

$\therefore$ Remaining work $= 1 - \frac{1}{12} = \frac{11}{12}$

Now, A has to complete the work, so A can complete in x days.

$$\therefore \quad \frac{1}{8} \times x = \frac{11}{12}$$

$$x = \frac{11 \times 8}{12} = \frac{11 \times 2}{3} = 7\frac{1}{3} \text{ days}$$

29. (*c*) Let the first term of GP $= a$ and common ratio $= r$

So, series is as follows a, ar, ar^2 and ar^3.

Now, conditions given in question are

$$a + ar = 12 \quad \ldots(i)$$

$$ar^2 + ar^3 = 48 \quad \ldots(ii)$$

From Eqs. (i) and (ii),

$$r^2\,(a + ar) = 48$$

$$r^2\,(12) = 48$$

On putting value of $(a+ar)$ from Eq, (i), we get

$$r^2=\frac{48}{12}$$
$$r^2=4$$
$$r=-2,2$$

As given in the question terms alternate as positive and negative, so taking $r=-2$ and putting value in Eq. (i),

$$a+a(-2)=12$$
$$\Rightarrow \quad a-2a=12 \Rightarrow -a=12$$
$$\therefore \quad a=-12$$

30. (*c*) $\text{Mean}=\frac{\text{Sum of observations}}{\text{Number of observations}}$

$$6=\frac{a+b+8+5+10}{5}$$
$$30=a+b+23$$
$$a+b=7 \quad \ldots(\text{i})$$

Now, $a=7-b$

$$\text{Variance}=\frac{\left[\begin{array}{l}(a-6)^2+(b-6)^2+(8-6)^2\\ +(5-6)^2+(10-6)^2\end{array}\right]}{5}$$

$$=6.80\times 5=(a-6)^2+(b-6)^2+(2)^2+(-1)^2+(4)^2$$

On putting value of a from Eq. (i), we get

$$34=(7-b-6)^2+b^2+36-12b+4+1+16$$
$$\Rightarrow \quad 34=(1-b)^2+b^2-12b+57$$
$$\Rightarrow \quad 34=b^2-2b+1+b^2-12b+57$$
$$2b^2-14b+24=0 \quad \ldots(\text{ii})$$

On solving above equations, we get

$$b=3,4 \text{ and } a=4,3$$

Now, only option (c) match the above with $a=3$ and $b=4$.

31. (*a*) Agent receives commission in the ratio $=12\frac{1}{2}\%$

Total value of goods sold = ₹ 15000

$\therefore$ Agents commission $=12\frac{1}{2}\%$ of 15000

$$=\frac{25}{2}\times\frac{15000}{100}=₹\,1875$$

32. (*b*) $\sqrt{110.25}\times\sqrt{0.01}\div\sqrt{0.0025}-\sqrt{420.25}$

$$\Rightarrow \sqrt{10.5\times10.5}\times\sqrt{0.1\times0.1}\div\sqrt{0.05\times0.05}-\sqrt{20.5\times20.5}$$
$$\Rightarrow \quad 10.5\times0.1\div0.05-20.5=10.5\times2-20.5$$
$$=21.0-20.5=0.50$$

33. (*b*) Number of respondents under 31 who's favourite style of music is blues

$$=2+3 \quad \begin{cases}2 \text{ from age group of } 15\text{-}20\\ 3 \text{ from age group of } 21\text{-}30\end{cases}$$
$$=5$$

Total respondents below 31 = 33 + 33 = 66

$\therefore$ Percentage of respondents $=\frac{5}{66}\times100=7.57\%\approx 7.6\%$

34. (*a*) Total respondents between 21-30 who's favourite style is other than rock $=(4+5+4+3+3+2)=21$ and total respondents $=33$

$\therefore$ Per cent of respondents who's favourite style is other than rock

$$=\frac{21}{33}\times100$$
$$\approx 64\%$$

35. (*d*) Total respondent who's favourite style of music is Jazz $=(1+4+11)=16$

Total respondent $=(33+33+68)=134$

$\therefore$ Percentage of respondent $=\frac{16}{134}\times100\approx12\%$

36. (*c*) Total non-US arts faculty $=21+6+2+4=33$

Total arts faculty $=21+6+2+4+79=112$

$\therefore$ Percentage of non-US arts faculty

$$=\frac{33}{112}\times100\approx30\%$$

37. (*b*) As their are 1049 students in arts which constitutes 23% of total students by faculty

So, $\frac{23\times x}{100}=1049$

$\therefore \quad x=\frac{1049\times100}{23}=4560$ students

Total students $=4560$

Percentage of Engineering student $=9\%$

Total number of Engineering student $=\frac{9\times4560}{100}=410$ students

38. (*b*) Total number of students $=4560$

Calculated in question number 37.

39. (*d*) Total number of Science students

$$=21\% \text{ of total students}=\frac{21\times4560}{100}\approx 958 \text{ students}$$

Now, there are 6% Asian students

$\therefore$ Total Asian students $=6\%$ of total Science students

$$=\frac{6\times958}{100}\approx 57 \text{ students}$$

40. (*d*) Total number of Medical students $=5\%$ of total students

$$=\frac{5\times4560}{100}\approx 228 \text{ students}$$

Percentage of medical student who are European 34% of 228

$$=\frac{34\times228}{100}\approx15\%$$

41. (*c*) Code for 'earn' is 'ha'.

42. (*d*) more time → jo zi

43. (*b*) Code for 'manage' → pa

44. (*b*) As Lou's arguement are not fallacious. He has reasons to support his arguement but Evelyn statement does not have any reason to support his arguement.

45. (*b*) In the statement, it is concluded that cars are safer than plane and 50% plane accident results in death but number of car accident are several hundred thousand times more, so this result in more deaths in car accidents. So, this statement seriously weaken the above statement.

46. (*b*) Company's is contributed by B and C.

47. (*d*) Industry B's production cannot be determined as it is not clear from the figure.

48. (*d*) A's contribution to primary users = Usage of $(P+Q+R)$ together

Now, $(P+Q+R)$'s consumption = (15 + 15 + 15) = 45%

Now, here 100% of manufacture accounts to 100000 tonnes.

So, production of A will be 45% of total production.

As we cannot calculate percentage of elements given to Industry A for production from different elements.

∴ A's annual production is 45% of 100000

$$= \frac{45\times100000}{100} = 45000 \text{ tonnes}$$

49. (*b*) Decrease is opposite of Increase similarly opposite of descend is ascend.

50. (*a*) Opposite to modern is ancient and in the same way young is opposite of old.

51. (*b*) By combining parts, it become complete or as a whole and in the same way combining classes it become a school.

52. (*a*) Summit and apex are same in meaning *i.e.*, on the top similarly beautiful and pretty are also save meaning.

53. (*b*)

B	E	A	U	T	I	F	U	L
↓	↓	↓	↓	↓	↓	↓	↓	↓
5	7	3	0	4	1	2	0	8

B	U	T	T	E	R
↓	↓	↓	↓	↓	↓
5	0	4	4	7	9

So, code of FUTURE is

F	U	T	U	R	E
↓	↓	↓	↓	↓	↓
2	0	4	0	9	7

54. (*d*) There are months of 30, 31 and 28 days and last day of month is Wednesday.

So, using 30 days, there are 4 Mondays.

Using 31 days, there are 5 Mondays.

So, it cannot be defermined.

55. (*b*) For 1st condition

$$17\times\frac{12}{2}=102$$

Similarly, $15\times\frac{10}{2}=?$

$$?=75$$

56. (*b*) For 1st condition,

$$\left(\frac{12}{3}\times4\right)\times\left(\frac{14}{2}\times3\right)=336$$

$$\left(\frac{15}{3}\times4\right)\times\left(\frac{16}{2}\times3\right)=?$$

$$?=480$$

57. (*a*) Using the conditions given in question.

Data can be arranged as

Boat –1⇒ (A, E, G) (C, D, F, B) ⇐ Boat –2

So, when F and B are on same boat, then G is in the other boat.

58. (*d*) By rearranging above data using condition from the question. Data is arranged as

Boat 1⇒ (C, B, D) (F, E, G, A) ⇐ Boat 2

∴ It is clear from figure C, B and D are on same boat.

59. (*c*) In the given arrangement, one line from vertical and one from horizontal is removed to form next figure.

So, next figure will be given in option (c).

60. (*c*) In the arrangement given, the change from figure 1 is that black dots remain at the same place and rest is rotated 180°. Similarly, for answer figure dots will remain at their places and rest is rotated by 180°.

So, answer figure is given in option (c).

61. (*a*)

G	M	S	Y
+2↓	+2↓	+2↓	+2↓
I	O	U	A
+2↓	+2↓	+2↓	+2↓
K	Q	W	C
+2↓	+2↓	+2↓	+2↓
M	S	Y	E

62. (*b*) Spending of Mohan is calculated below 30 min STD (within 100 km)

$$=30\times1.5=₹\,45$$

30% of 100 min in local landline $=\left(\frac{30\times100}{100}\right)\times2=₹\,60$

30% of 100 min on Airtel local $=\left(\frac{30\times100}{100}\right)\times1=₹\,30$

40% of 100 min on GSM local $=\left(\frac{40\times100}{100}\right)\times1=₹\,40$

Total spending $=(45+60+40+30)=₹\,175$

Total spending of Rohan is

18 min of STD within 150 km $=18\times1.50=₹\,27$

30% of 120 min of GSM local $=\left(\frac{30\times120}{100}\right)\times1=₹\,36$

30% of 120 min of Airtel local $=\left(\frac{30\times120}{100}\right)\times1=₹\,36$

40% of 120 min of landline local $=\left(\frac{40\times120}{100}\right)\times2=₹\,96$

Total spendings $=(27+36+36+96)=₹\,195$

∴ Rohan spends more than Mohan.

63. (*a*) Person sending 38 local SMS a month is not benefitted as Using old scheme cost of 38 local scheme

$$=38\times1.5=₹\,57$$

Using new scheme total cost is

$$=38\times0.6+35=22.8\times35=₹\,57.8$$

So, using new scheme he had a loss of 80 paise in all.

64. (*a*) By making 30 calls to Airtel on STD (500 + km) amounts to

$$=30\times1.5=₹\,45$$

55 local calls, let these all local calls on landline.

Then, this amounts to $=55\times2=₹\,\frac{110}{155}$

So, bill of ₹ 199 is not possible in condition (a).

65. (*d*) As it is not cleard from the statement that how much was the bill, so we cannot determine the duration of minimum ISD calls made to any countries.

66. (*c*) Total demand of all the company

$$=3000+600+2500+1200+3300=10600$$

$$\text{Average}=\frac{\text{Total demand}}{\text{Number of company}}=\frac{10600}{5}=2120$$

Similarly, total production of all companies

$$=(1500+1800+1000+2700+2200)=9200$$

$$\text{Average}=\frac{\text{Total production}}{\text{Number of company}}=\frac{9200}{5}=1840$$

∴ Difference in average demand and average production is
$$= 2120 - 1840 = 280$$

67. (*a*) Production of company D = 2700

and production of company A = 1500

$$\therefore \text{Ratio of production} = \frac{\text{Production of company D}}{\text{Production of company A}} = \frac{2700}{1500} = 1.8$$

So, Production of company D is higher by 1.8 tonnes than production of company A.

68. (*b*) 10% of is the paper cost of total.

So, It is equal to $\frac{10}{100}$ and this must be equal to $\left(\frac{x}{360}\right)^{\circ}$.

$$\frac{10}{100} = \frac{x^{\circ}}{360^{\circ}}$$

$$\Rightarrow \quad x = 36^{\circ}$$

69. (*d*) When 2% of cost is ₹ 2000.

Then, 100% of cost is ₹ 100000.

The publisher wants 5% profit on whole.

So, 5% of total cost $= \frac{5}{100} \times 100000 =$ ₹ 5000

Total amount or SP of magazine
$$= 100000 + 5000 =$$ ₹ 105000

Total prints published = ₹ 12500

∴ Cost of one magazine
$$= \frac{\text{SP of Magazine}}{\text{Total Magazine}} = \frac{105000}{12500} =$$ ₹ 8.40

70. (*d*) Fruit grows on tree and here tree is called as sky.

So, Fruit grow on sky.

71. (*d*) mine

72. (*b*) 'before', she restored to health restored means to ring back to its original condition. (Before is used in reference to two event).

73. (*c*) 'has changed' (Since is often used with present perfect or past perfect time).

74. (*d*) Dias — 'dais'

75. (*d*) dissipate

76. (*b*) He lives in Bengaluru at 115, Richmond. ('in' is used for big place, town city or country 'At' is used for a point).

77. (*b*) euphemism, deaf (hearing disorder).

78. (*b*) If error is a mistake fault is a defect.

79. (*d*) The man has flouted the rules of ethical conduct, he is virtually a beast flouted. (Means to show contempt for or to mock). (Ethical is related to morals)

80. (*a*) Pat of Australia is known to the natives, as 'The outback' (Specific name of a place).

81. (*c*) I know that you want to learn to drive Rima but, you are to young.

82. (*d*) Veracious means truthful (one, who observes truth).

83. (*c*) Perturb means to disturb greatly (to agitate, to throw into confusion).

84. (*b*) With a high hand means oppressively or violently.

85. (*d*) I subjected my mind to hard thinking. (To think very hard usually in order to remember or to find the solution to a problem).

86. (*c*) I got too nervous and I didn't go. (Suddenly become too frightened to do something one had planned to do).

87. (*b*) You will have to take back what you have said. (Force to admit that he said before was wrong).

88. (*c*) 1, 4

89. (*b*) magnamious means greatness of mind benevolence means generous of charitable antonym of it is cruel, which means hard hearted painful or sagafe.

90. (*c*) twice a year

91. (*d*) (a) temporal means penetrating to time
(b) ephemeral means short lived.
(c) Transient means short duration
(d) eternal-everlasteng.
In all the option, wer found they are bounded except (*d*) eternal so its the odd one.

92. (*c*) 'intra' means 'within'

93. (*b*) **94.** (*b*)

95. (*d*) His conduct is above suspicious (Adjective)
The above the mantel piece (Preposition)
The above information is for the public. (adverb)
Rain comes from above (noun).

96. (*c*) Replace in with 'of' (of is used for the location).

97. (c) credited 'by' three months interst. (By is usually implies before time or date).

98. (*d*) 'Whose' car are you planning to borrow?
who is planning to have a birthday party? He is the peson whom you met at the Junction.

99. (*b*) **100.** (*b*)

101. (*c*) You will work hard (will is used with uneles)

102. (*b*) 5, 4, 1, 3, 2

103. (*a*)

104. (*b*) Likely (Probability was there to be mistaken).

105. (*b*) Doesn't he?

106. (*b*) Simile (Comparision is made).

107. (*a*)personification (death is represanted as a person).

108. (*a*) The 'frozen feelings' being talked about the negative childhood experiences". The passage is talking about the negative experiences of the life, which a child faced in his past, which has been deep frozen or settled in his mind, which effected his whole life. He never been able to forget the trauma, he had passed through in his childhood. 'In the early years ..... deeply into the memory."

109. (*c*) Glitch means sudden malfunction or breakdown or same irregularity.

110. (*a*) According to passage, "The process of change needs to be traumatic" this is a correct statement, if one try to overcome from his weakness and fear, he can do it, one should try to understand that which had happrmrf in his childhood, it was his past, now he should try to live in his present and we must struggle to save our future its totally in our hands.

Although option (b) is also confirmed in the second last paragraph of a passage, but it shows the malfunctioning of our behaviour.

111. (*a*) Dendrochronology or tree-ring dating is the scientific method of dating based on the analysis of patterns of tree rings, also known as growth rings. Dendrochronology can date the time at which tree rings were formed in many types of wood to the exact calendar year.

112. (*d*) Modi Telstra company (on July 31, 1995) launched the first mobile phone operation in India. The company was a joint venture between India's Modi Group and Australian Telecom Gaint Telstra. At present, the company has been acquired by BHARTI Group and is part of the AIRTEL brand.

113. (*a*) Scooby-Doo, the longest-running TV cartoon in the past. It was launched in 1969. The series was created for Hanna-Barbera productions which was later absorbed by warner Bros.

114. (*a*) The Board for Industrial and Financial Reconstruction (BIFR) is an agency of the Government of India, part of the Department of Financial Services of the Ministry of Finance. Its objective is to determine of industrial companies and to assist in reviving those that may be viable and shutting down the others. It was established in 1982.

115. (*d*) The term Ekistics (coined by Konstontions Apostolos Doxiadis in 1942) applies to the Science of human settlements. It includes regional, city community planning and dwelling design.

116. (*a*) A red herring prospectus, as a first or preliminary prospectus, is a document submitted by a company (issuer) as part of a public offering of securities (either stocks or bonds).

117. (*a*) Eight O'Clock coffee is the brand name of the light roast of coffee introduced by the American Supermarket Chain A and P in 1859.

118. (*c*) Jatropha is a genus of flowering plants in the spurge family, Euphorbiaceae. It contains approximately 170 species of succulent plants, shrubs and trees.

119. (*d*) Margaret Court also known as Margaret Smith Court, is a retired world number 1 tennis player and Christian minister from Australia. She won 66 Grand Slams in her life.

120. (*b*) 'The Big Apple' is a nickname for New York City. It was first popularised in the 1920s by John J Fitz Gerald, a sports writer for the New York morning Telegraph.

121. (*c*) Infosys, India's second largest outsourcer, become the first Indian company to be listed on the Nasdaq in 1999.

122. (*b*) TAREGNA in Bihar has been chosen as the best place in India to watch a total solar eclipse on 22 July.

123. (*b*) Adrian Sutil is a German racing driver, who has raced in Formula one. He continued to race with the team under their new guise Force India in 2008, where he remained until 2011.

124. (*c*) The Million Dollar Round Table (MDRT) described itself as the premier Association of Financial Professionals. It was formed in 1927 to help insurance salespeople.

125. (*b*) RBI declares the 'Credit Policy' of India.

126. (*d*) Securities and Exchange Board of India (SEBI) is a board (autonomous body) created by the Government of India in 1988 and given statutory form in 1992 with the SEBI Act 1992, with its head office at Mumbai. It is the regulator of securities markets in India.

127. (*b*) The flower with the world's largest bloom as the Rafflesia arnoldii. This rare flower is found in the rainforests of Indonesia.

128. (*b*)

129. (*a*) Short selling is the selling of a stock that the seller doesn't own. More, a short sale is the sale of a security that isn't owned by the seller, but that is promised to be delivered.

130. (*b*) Mahatma Gandhi was nominated five times for the Nobel peace prize in 1937, 1938, 1939, 1947 and 1948.

131. (*c*) Vijay Kumar (shooting) and Yogeshwar Dutt (wrestling) have been awarded with Rajiv Gandhi Khel Ratna Award, 2012 on August 29, 2012 by President of India.

132. (*b*) Serena Williams has won women's Single Title US Open 2012 by defeating Victoria Azarneka.

133. (*b*) SBI has launched Click-2-Card service which is a new online application service, aimed at netting about 3000 customers in the next four months.

134. (*d*) Hassan Sheikh Mahmood replaced the Former President of the Transational Federal Government, Sheikh Sharif Sheikh Mohammed.

135. (*b*) A Sethumadhavan, Veteran Malayalam frictional writer, has been appointed as the Chairman of NBT. It is the first time a Malayalam writer is given the post after late Sukumar Azhikode.

136. (*a*) Still counting the dead describes the decades of civil war between the Tamil Tiger Guerrillas and the Government reached its bloody climax.

137. (*a*) 11 scientists have been selected for Shanti Swarup Bhatnagar prize 2012, India's premier awards in the field of Science and Technology.

138. (*d*) Mark Waughs and Steve Waughs are the first twin to play International Cricket.

139. (*c*) The US has the largest railway system in the world-about 240000 km connecting 48 continental states. That is enough track to circle the earth five times.

140. (*a*) Svetlana Kuznetsova won the 2009 French Open tennis ladies title.

141. (*a*) The integrated chip circuit also known as 'The chip' or microchip was invented by both Jack kilby and Robert Noyce.

142. (*c*) A landlocked country is a country that has no access to the ocean at its borders. Kazakhstan is the world's ninth largest country and is also the world's largest landlocked country.

143. (*b*) **144.** (*c*)

145. (*c*) Pluto orbits beyond the orbit of Neptune. It is much smaller than any of the official planets and now classified as a 'dwarf planet'.

146. (*d*) One barrel = 42 US gallons

= 158.987 L = 160 L

147. (*d*) The economy of India is based in part on planning through its Five Year Plans which are developed, executed and monitored by the Planning Commission.

148. (*b*)

149. (*b*) The Organisation of the Petroleum Exporting Countries (OPEC) is an inter-governmental organisation of twelve oil producing made up of Algeria, Angola, Ecuador, Iran, Iraq, Kuwait, Libya, Nigeria, Qatar, Saudi Arabia, the United Arab Emirates and Venezuela.

150. (*d*) The computer laptop was invented by Adam Osborne in 1981.

SNAP

SYMBIOSIS NATIONAL APTITUDE TEST

Solved Paper 2011

Time : 120 min **Marks :** 180

Section I **General Awareness**

For all questions in this section, correct answers carry 1 *mark each.*

1. 'Swayamsiddha' is a scheme launched by the Government of India to help
(1) school children only
(2) health workers only
(3) senior citizens without any regular income
(4) women only

2. What is 'Share swap'?
I. A business takeover in which acquiring company uses its own stock to pay for the acquired company.
II. When a company uses its own share to get some short term loan for working capital requirement.
III. When companies are require to float a new issue to earn capital for their expansion programmes, each shareholder gets some additional preferential share. The process is known as 'share swap'.
(1) I only (2) I and II
(3) III only (4) None of these

3. Regarding "carbon credits" which of the following statements is not correct?
(1) The carbon credit system was ratified in conjunction with Kyoto Protocol.
(2) Carbon credits are awarded to those countries or groups who have reduced the green house gases below their emission quota.
(3) The goal of carbon credit system is to reduce carbon dioxide emission.
(4) Carbon credits are traded at a price fixed from time to time by the United Environment Programme.

4. Both Foreign Direct Investment (FDI) and Foreign Institutional Investor (FII) are related to investment in a country. Which one of the following statements best represents an important difference between the two?
(1) FII helps bring better management skills and technology while FDI only bring capital.
(2) FII helps in increasing capital availability in general, while FDI only targets specific sectors.
(3) FDI flows only into secondary markets while FII targets primary markets.
(4) FII is considered to be more stable than FDI.

5. In context of global oil prices, "Brent crude oil" is frequently referred to in the news.
What does this term imply?
I. It is a major classification of crude oil.
II. It is sourced from North sea.
III. It does not contain sulphur.
Which of the statement(s) given above is /are correct?
(1) II only (2) I and II
(3) I and III (4) I, II and III

6. With reference to "Look East Policy" of India, consider the following statements :
I. India wants to establish itself as an important regional player in the East Asian affairs.
II. India wants to plug the vacuum created by the termination of Cold War.
III. India wants to restore the historical and cultural ties with its neighbours in South-East and East Asia.
Which of the statement(s) given above is/are correct?
(1) I only (2) I and III
(3) III only (4) I, II and III

7. Recently 'oilzapper' was in news. What is it?
(1) It is eco-friendly technology for the remediation of oily sludge and oil spills.
(2) It is the latest technology developed under sea oil exploration.
(3) It is genetically engineered high biofuel-yielding maize variety.
(4) It is the latest technology to control accidentally caused flames from oil wells.

8. Why is the offering of 'teaser loans' by commercial banks a cause of economic concern?
I. The teaser loans are considered to be an aspect of sub-prime lending and banks may be exposed to the risk of defaulters in future.
II. The teaser loans are given to inexperienced entrepreneurs to set up manufacturing or export units.
Which of the statement(s) given above is/are correct?
(1) I only (2) II only
(3) Both I and II (4) Neither I nor II

9. Why is the Government of India disinvesting its equity in the Central Public Sector Enterprises (CPSEs)?
I. The government intends to use the revenue earned from disinvestment mainly to pay the external debt.
II. The government no longer intends to retain the management control of CPSEs.
Which of the statement(s) given above is/are correct?
(1) I only (2) II only
(3) Both I and II (4) Neither I nor II

10. The book 'Unto this Last' which influenced Gandhiji, was authored by
(1) Boris Yeltsin (2) Pushkin
(3) Ruskin Bond (4) John Ruskin

11. Besides the USA, India has signed, with which of the following countries, the agreement being named as "Cooperation Agreement on the Development of Peaceful Uses of Nuclear Energy"?
(1) Italy (2) Germany (3) France (4) Australia

12. Which of the following country is not a member of the Nuclear Suppliers Group (NSG)?
(1) Russia (2) USA (3) Italy (4) Iran

13. Many times we read in the newspapers that several companies are adopting the FCCBs route to raise capital. What is the full form of FCCB?
(1) Foreign Currency Convertible Bonds
(2) Foreign Convertible Credit Bonds
(3) Financial Consortium and Credit Bureau
(4) None of the above

14. As per the reports published in various newspapers, RBI has asked banks to make plan to provide banking services in villages having population of 2000. This directive issued by the RBI will fall in which of the following categories?
(1) Plan for financial inclusion.
(2) Efforts to meet priority sector lending.
(3) Extension of Internet and branchless banking.
(4) None of the above

15. Which of the following is/are treated as artificial currency?
(1) ADR (2) GDR
(3) Both (1) and (3) (4) SDR

16. Which of the following terms indicates a mechanism used by commercial banks to provide credit to the government?
(1) Cash Credit Ratio
(2) Debit Service Obligation
(3) Liquidity Adjustment Facility
(4) Statutory Liquidity Ratio

17. Who among the following was not part of the drafting committee of the Lokpal Bill?
(1) Salman Khurshid (2) Anna Hazare
(3) Arvind Kejriwal (4) Kiran Bedi

18. Criminal procedure is a subject of which of the following lists?
(1) State list (2) Concurrent list
(3) Union list (4) Both (1) and (3)

19. One of the world's biggest uranium resources was recently found in
(1) Andhra Pradesh (2) Maharashtra
(3) Jharkhand (4) Uttarakhand

20. Which of the following was the first venture of Kishore Biyani?
(1) Pantaloons (2) Big Bazaar
(3) Central (4) Future Capital Holdings

21. Who among the following is often referred to as "Father of India's Green Revolution"?
(1) Dr. Verghese Kurien
(2) Dr. Amrita Patel
(3) MS Swaminathan
(4) Gurudev Khush

22. Who among the following directed the movie 'Peepli Live'?
(1) Ashutosh Gowarikar (2) Kiran Rao
(3) Anusha Rizvi (4) Seema Chisti

23. The point at which solid, liquid and gaseous forms of a substance co-exist is called
(1) sublimation (2) distillation point
(3) triple point (4) melting point

24. The limit beyond which the stars suffer internal collapse is called the
(1) Raman Effect (2) Chandrashekhar Limit
(3) Aurora Borealis (4) Quasan Zone

25. UNDP reports publishes every year the Human Development Index. Which of the following is not a criteria used to measure the Human Development Index?
(1) Health (2) Education
(3) Living Standards (4) Human Rights

26. The software company I-flex Solutions was originally a division of which famous financial services company?
(1) Citicorp (2) ICICI
(3) HSBC (4) ABN Amro Bank

27. Which of the following is not a member of SAARC?
(1) Bhutan (2) Bangladesh
(3) Burma (4) Maldives

28. Which article of the Indian Constitution recognizes Hindi in Devanagari Script as the official language of India?
(1) Article 345 (2) Article 343
(3) Article 348 (4) Article 334

29. India has finally woken up to the needs of the country's elderly. With the number of people in the 60-plus age group in India expected to increase to 100 million in 2013 and to 198 million in 2030, the Health Ministry is all set to roll out the
(1) National Programme for Health Care of the Elderly
(2) National Programme for Senior Citizens
(3) National Programme for Old Aged
(4) Rashtriya Vriddha Swasthya Yojana

30. Which of the following statement(s) is/are correct?
I. Non-resident Indians (NRIs) can now cast votes in their home constituencies in India.
II. The NRI can cast his vote by postal balloting.
(1) I only (2) II only
(3) Both I and II (4) Neither I nor II

31. In a landmark move, which of these State Governments has set up a Savarna Aayog, a commission to identify the deprived and underprivileged families among upper castes?
(1) Uttar Pradesh (2) Bihar
(3) Madhya Pradesh (4) Rajasthan

32. Which of these countries has announced one billion US dollar aid for the reconstruction of Nalanda University?
(1) Japan (2) Singapore (3) Malaysia (4) China

33. What is the name of the in-house magazine to be published by the Indian Railways, to be distributed to the passengers on high end trains?
(1) Rail Vishwa (2) Rail Bandhu
(3) Rail Mail (4) Rail Patra

34. The government, in February 2011, set up a task force to create a way to provide direct subsidies to the ultimate beneficiaries on petroleum working gas and fertilizers. The task force will be headed by
(1) Nandan Nilekani (2) Apoorva Sinha
(3) Sangam Chitra (4) All of these

35. Which of these countries has been added in the 'BRIC' (Brazil-Russia-India-China) group in of emerging economies?
(1) South Korea (2) Venezuela
(3) South Africa (4) Malaysia

36. Which of the following countries was the host of First Asian Yoga Championship held recently?
(1) Thailand (2) Cambodia (3) Vietnam (4) Laos

37. Who among the following has been defeated by Saina Nehwal, who has won the Swiss Open Grand Prix Gold Title recently, becoming the first Indian to do so?
(1) Ji Hyun Sung (2) Wang Lin
(3) Kamilla Rytter Juhl (4) Shinta Mulia Sari

38. On March 6, 2011, the Supreme Court held that has ample powers to enact legislation with respect to extra-territorial aspects for the security of India?
(1) President (2) Prime Minister
(3) Supreme Court (4) Parliament

39. Which country has appointed Brigadier General Ravinder Singh as its next Army chief, the first Sikh in nearly 30 years to be given the force's batton?
(1) Thailand (2) Singapore (3) Malaysia (4) Indonesia

40. Anant Pai, better known as 'Uncle Pai', who recently died, was the creator of which among the following comic series?
(1) Chacha Chaudhary (2) Chandamama
(3) Indrajaal (4) Amar Chitra Katha

Section II Analytical & Logical Reasoning

For all questions in this section, correct answers carry 2 mark each.

Directions (Q. Nos. 41-42) *Each question consists of a set of statements. Assume that each one of these statements is individually true. Each of the four choices consists of a subset of these statements. Choose the subset as your answer where the statements therein are logically consistent among themselves:*

41. A. Only if the water level in the coastal areas rises, then the people change their lifestyle.
B. People change their lifestyle only if they are rewarded.
C. If people are rewarded, then they will not change their lifestyle.
D. If the temperature rises, then the water level in the coastal areas rises.
E. Whenever the water level in the coastal area rises, then the temperature rises.
F. Unless the people change their lifestyle, temperature rises.
G. People are rewarded.
H. Water level in the coastal areas does not rise.
(1) C, D, F, G and H (2) G, F, D, B and H
(3) E, F, G, H and B (4) None of these

42. A. If Kumar sings, then the audiences sleep.
B. If Kumar sings, then the audiences dance.
C. Unless audience do not dance, the concert will be successful.
D. Only if the audience dance, the concert will be successful.
E. If Vina dances, then Kumar sings.
F. Kumar sings only if Vina dances.
G. Vina dances
H. The concert is successful.
(1) C, F, G, B and H (2) A, C, F, G and H
(3) E, C, G, B and H (4) Both (2) and (3)

Directions (Q. Nos. 43-45) *These questions are based on the data given below.*

There are only four members of a family *viz.*, A, B, C and D and there is only one couple among them. When asked about their relationships, following were their replies :
I. A → B is my son. D is my mother.
II. B → C is my wife. D is my father.

III. C → D is my mother-in-law. A is my daughter.
IV. D → A is my grand-daughter. B is my daughter-in-law.

43. Who always speaks the truth?
(1) A (2) B (3) C (4) D

44. How is B related to C?
(1) Father (2) Mother
(3) Wife (4) Husband

45. Which of the following statements must be true?
(1) A's grandmother alternates between the truth and lie.
(2) C's wife always speaks the truth.
(3) A's grandfather always speaks the truth.
(4) B's daughter always tells lie.

Directions (Q. Nos. 46-48) *These questions are based on the following information.*

A cube of 7cm × 7cm × 7cm is kept in the corner of a room and painted in three different colours, each face in one colour. The cube is cut into 343 smaller but identical cubes.

46. How many smaller cubes do not have any face painted?
(1) 125 (2) 180 (3) 144 (4) 216

47. How many smaller cubes have exactly one colour on them?
(1) 108 (2) 72 (3) 36 (4) 24

48. How many smaller cubes have at the most two faces painted?
(1) 343 (2) 342 (3) 256 (4) 282

Directions (Q. Nos. 49-50) *These questions are based on the following information.*

Amit was driving in New Town, where all roads either North-South or East-West forming a grid. Roads were at a distance of 1 km from each other in parallel.

49. Amit started at the intersection of streets no. 7 and 8. He drove 3 km North, 3 km West and 4 km South. Which further route could bring him back to his starting point?
I. 3 km East, then 2 km South
II. 1 km North, then 3 km East
III. 1 km North, then 2 km West
(1) I only (2) II only
(3) I and II (4) II and III

50. After driving as stated in question no. 49 above, Amit did not return to his starting point, but instead drove 4 km East and 1 km North. How far is he from his starting point?
(1) 5 km (2) 4 km (3) 1 km (4) 7 km

Directions (Q. Nos. 51-55) *Refer to the following data and answer the questions that follow:*

A numerical machine accepts two values X and Y. Then, it updates these values as $X = XY$ and $Y = Y + 1$ in every step. The machine stops at $X \geq N$.

51. For $X = 3$, $Y = 2$ and $N = 100$, how many steps are performed before the machine stops?
(1) 2 (2) 3 (3) 4 (4) 5

52. In the above question (51), what is the final value of X ?
(1) 6 (2) 20 (3) 72 (4) 360

53. In the above question (51), what is the final value of Y ?
(1) 4 (2) 5 (3) 6 (4) 20

54. If the value of N is changed to 500, what would be the final value of X ?
(1) 360 (2) 500 (3) 560 (4) 2160

55. If $X = 2$ and $Y = 3$, what should be the minimum value of N such that final value of Y is 7?
(1) 300 (2) 360 (3) 720 (4) 860

Directions (Q. Nos. 56-60) *Refer to the following statements and answer the questions:*

Seven students Priya, Ankit, Raman, Sunil, Tony, Deepak and Vicky take a series of tests. No two students get similar marks. Vicky always scores more than Priya. Priya always scores more than Ankit. Each time either Raman scores the highest and Tony gets the least, or alternatively Sunil scores the highest and Deepak or Ankit scores the least.

56. If Sunil is ranked sixth and Ankit is ranked fifth, which of the following can be true?
(1) Vicky is ranked first or fourth
(2) Raman is ranked second or third
(3) Tony is ranked fourth or fifth
(4) Deepak is ranked third or fourth

57. If Raman gets the highest, Vicky should be ranked not lower than
(1) second (2) third (3) fourth (4) fifth

58. If Raman is ranked second and Ankit is ranked first, which of the following must be true?
(1) Sunil is ranked third (2) Tony is ranked third
(3) Priya is ranked sixth (4) None of these

59. If Sunil is ranked second, which of the following can be true?
(1) Deepak gets more than Vicky
(2) Vicky gets more than Sunil
(3) Priya gets more than Raman
(4) Priya gets more than Vicky

60. If Vicky is ranked fifth, which of the following must be true?
(1) Sunil scores the highest (2) Raman is ranked second
(3) Tony is ranked third (4) Ankit is ranked second

61. In 2002, according to a news poll, 36% of the voters had leaning towards party "Y". In 2004, this figure rose to 46%. But in another survey the percentage was down to 40%. Therefore, the party "Z" is likely to win the next election. Which of the following, if true, would seriously weaken the above conclusion?
(1) People tend to switch their votes at the last minute
(2) It has been showed that 85% of the voters belonging to the party "Y" vote in an election as compared to 80% of the voters belonging to party "Z"
(3) 35% of people favour party "Z"
(4) No one can predict how people will vote

62. Inflation rose by 5% over the second quarter, by 4% during the first quarter and higher than 3% recorded during the same time last year. However, the higher price index did not seem to alarm National Stock Index as stock prices remain steady.
Which of the following, if true, could explain the reaction of National Stock Index?

(1) RBI announced that it will take necessary corrective measures.
(2) Stock prices were steady because of a fear that inflation would continue.
(3) Economists warned that inflation would continue.
(4) Much of the quarterly increase in the price level was due to a summer drought effect on food price.

Direction (Q. No. 63) *Pick up the appropriate analogy.*

63. Birth : Dirge
(1) Sunset : Sunrise
(2) Security Check : Arrival
(3) Marriage : Alimony
(4) Welcome Address : Vote of Thanks

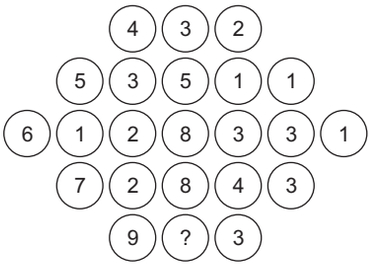
64. Beautiful beaches attract people, no doubt about it. Just look at this city's beautiful beaches, which are among the most overcrowded beaches in the state.
Which of the following exhibits a pattern of reasoning most similar to the one exhibited in the argument above?
(1) Moose and bear usually appear at the same drinking hole at the same time of day. Therefore, moose and bear must grow thirsty at about the same time.
(2) Children who are scolded severely tend to misbehave more often than other children. Hence, if a child is not scolded severely that child is less likely to misbehave.
(3) This software programme helps increase the work efficiency of its users. As a result, these users have more free time for other activities.
(4) During weather my dog suffers from fleas more than during cooler weather. Therefore, fleas must thrive in a warm environment.

65. No national productivity measures are available for underground industries that may exist but remain unreported. On the other hand, at least some industries that are run entirely by self-employed industrialists are included in national productivity measures.
From the information given above, it can be validly concluded that
(1) there are at least some industries run entirely by self employed industrialists that are underground industries
(2) no industries that are run entirely by self employed industrialists operate underground
(3) there are at least some industries other than those run entirely by self-employed industrialists that are underground industries
(4) there are at least some industries run entirely by self employed industrialists that are not underground industries

66. Nilu has never received a violation from the Federal Aviation Administration during her 16 yr flying career. Nilu must be a great pilot.
Which of the following can be said about the reasoning above?
(1) The definitions of the terms create ambiguity.
(2) The argument uses circular reasoning.
(3) The argument is built upon hidden assumptions.
(4) The argument works by analogy.

67. Many people argue that the death penalty deters murder. However, the notorious killer Tom Hanks deliberately moved to a state that imposes the death penalty just before embarking on a series of ferocious murders. Thus, it seems clear that the existence of the death penalty does not serve as a deterrent to murder.
The argument above may best be characterized as
(1) an appeal to emotion
(2) a flawed analogy
(3) a general conclusion based on a specific example
(4) circular reasoning

68. What number should replace the question mark?



(1) 1 (2) 4 (3) 12 (4) 6

69. The fewer restrictions there are on the advertising of legal services, the more lawyers there are who advertise their services, and the lawyers who advertise a specific service usually charge less for that service than lawyers who do not advertise. Therefore, if the state removes any of its current restrictions, such as the one against advertisements that do not specify fee arrangements, overall consumer legal costs will be lower than, if the state retains its current restrictions.
If the statements above are true, which of the following must be true?
(1) Some lawyers who now advertise will charge more for specific services if they do not have to specify fee arrangements in the advertisements.
(2) More consumers will use legal services if there are fewer restrictions on the advertising of legal services.
(3) If the restrictions against advertisements that do not specify fee arrangements is removed, more lawyers will advertise their services.
(4) If more lawyers advertise lower prices for specific services, some lawyers who do not advertise will also charge less than they currently charge for those services.

70. Which of the following, if true, would most seriously weaken the argument concerning overall consumer legal costs?
(1) The state is unlikely to remove all of the restrictions that apply solely to the advertising of legal services.
(2) Lawyers who do not advertise generally provide legal services of the same quality as those provided by lawyers who do advertise.
(3) Most lawyers who now specify fee arrangements in their advertisements would continue to do so even in the specification were not required.
(4) Most lawyers who advertise specific services do not lower their fees for those services when they begin to advertise.

Section III Quantitative Aptitude

For all questions in this section, correct answers carry 1 *mark each.*

71. A train travelling at 36 km/h crosses a platform in 20 s and a man standing on the platform in 10 s. What is the length of the platform in metres?
(1) 240 m (2) 100 m
(3) 200 m (4) 300 m

72. By walking at 4/5th of his usual speed, a man reaches office 10 min later than usual. What is his usual time?
(1) 20 min (2) 40 min
(3) 30 min (4) 50 min

73. A man and a woman 81 miles apart from each other, start travelling towards each other at the same time. If the man covers 5 miles/h to the women's 4 mile/h, how far will the woman have travelled when they meet?
(1) 27 (2) 36
(3) 45 (4) None of these

74. Two people were walking in opposite directions. Both of them walked 6 miles forward, then took right and walked 8 miles. How far is each from starting positions?
(1) 14 miles and 14 miles (2) 10 miles and 10 miles
(3) 6 miles and 6 miles (4) 12 miles and 12 miles

75. Four men and three women can do a job in 6 days. When 5 men and 6 women work on the same job, the work gets completed in 4 days. How long will 2 women and 3 men take to do the job?
(1) 18 (2) 10 (3) 8.3 (4) 12

76. Ram completes 60% of a task in 15 days and then takes the help of Rahim and Rachel. Rahim is 50% as efficient as Ram is and Rachel is 50% as efficient as Rahim is. In how many more days will they complete the work?
(1) $\frac{121}{3}$ (2) $\frac{51}{7}$ (3) $\frac{40}{7}$ (4) $\frac{65}{7}$

77. *A* and *B* can do a piece of work in 21 and 24 days, respectively. They start the work together and after some days *A* leaves the work and *B* completes the remaining work in 9 days. After how many days did *A* leave?
(1) 5 (2) 7 (3) 8 (4) 6

78. A trader makes a profit equal to the selling price of 75 articles when he sold 100 of the articles. What % profit did he make in the transaction?
(1) 33.33% (2) 75% (3) 300% (4) 150%

79. In a 100 m race, if *A* gives *B* a start of 20 m, then *A* wins the race by 5 s. Alternatively, if *A* gives *B* a start of 40 m the race ends in a dead heat. How long does *A* take to run 200 m?
(1) 10 s (2) 20 s (3) 30 s (4) 40 s

80. A 4 cm cube is cut into 1cm cubes. What is the percentage increase in the surface area after such cutting?
(1) 4% (2) 300% (3) 75% (4) 400%

81. A number G236G0 can be divided by 36, if G is
(1) 8
(2) 6
(3) 1
(4) More than one values are possible.

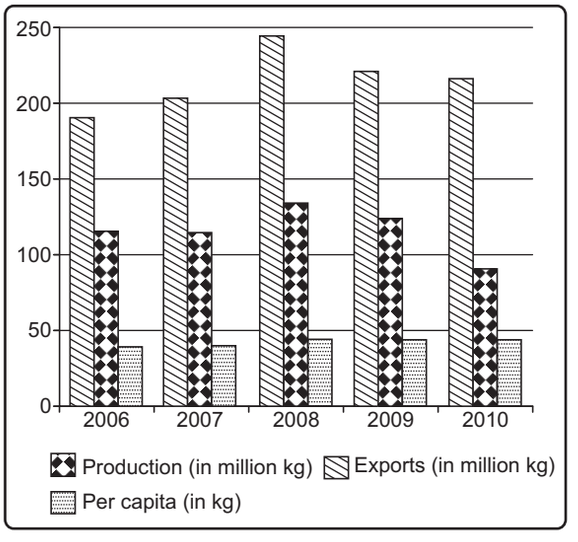
82. Amit can do a work in 12 days and Sagar in 15 days. If they work on it together for 4 days, then the fraction of the work that is left, is
(1) 3/20 (2) 3/5
(3) 2/5 (4) 2/20

83. A rectangular park 60 m long and 40 m wide has two concrete crossroads running in the middle of the park and rest of the park has been used as a lawn. If the area of the lawn is 2109 sq m, then what is the width of the road?
(1) 2.91 m (2) 3 m
(3) 5.82 m (4) None of these

84. A bag contains 5 white and 3 black balls; another bag contains 4 white and 5 black balls. From any one of these bags a single draw of two balls is made. Find the probability that one of them would be white and other black ball.
(1) 275/504 (2) 5/18
(3) 5/9 (4) None of these

Directions (Q. Nos. 85-88) *Answer the questions on the basis of the information given below.*

The following bar graph gives the production, exports and per capita consumption of rice in country A, for the five years from 2006 to 2010.



Consumption = Production – Exports
Per Capita Consumption = (Consumption) ÷ (Population)

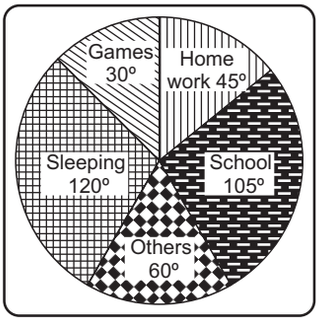
85. In which year was the percentage increase in the consumption of rice over the previous year, the highest?
(1) 2007 (2) 2008 (3) 2009 (4) 2010

86. What is the population of country A in the year 2008 (in million)?
(1) 2.64 million (2) 2.72 million
(3) 2.79 million (4) 2.85 million

87. The ratio of exports to consumption in the given period was the highest in the year
(1) 2006 (2) 2007 (3) 2008 (4) 2009

88. In which of the given years was the population of country A, the highest?
(1) 2007 (2) 2008 (3) 2009 (4) 2010

Directions (Q. Nos. 89-93) *The following pie chart shows the hourly distribution (in degrees) of all the major activities of a student.*



89. The percentage of time, which he spends in school, is
(1) 38% (2) 30%
(3) 40% (4) 25%

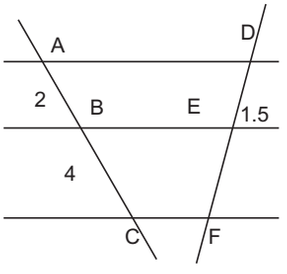
90. How much time (in percent) does he spend in games in comparison to sleeping?
(1) 30% (2) 40%
(3) 25% (4) None of these

91. If he spends the time in games equal to the homework and remains constant in other activities, then the percentage decrease in time of sleeping is
(1) 15% (2) 12.5%
(3) 20% (4) None of these

92. What is the difference in time (in hours) spent in school and in homework?
(1) 2 (2) 3 (3) 4 (4) 8

93. If he spends 1/3rd time of homework in Mathematics, then the number of hours he spends in rest of the subjects in homework is
(1) 1 (2) 2 (3) 3 (4) 4

94. Three parallel lines are cut by two transversals as shown in the given figure. If $AB = 2$ cm, $BC = 4$ cm and $DE = 1.5$ cm, then the length of EF is



(1) 2 cm (2) 3 cm
(3) 3.5 cm (4) 4 cm

95. $\log_{10} 10 + \log_{10} 10^2 + \ldots + \log_{10} 10^n$
(1) $n^2 + 1$ (2) $n^2 - 1$
(3) $\frac{(n^2+n)}{2} \cdot \frac{n(n+1)}{3}$ (4) $\frac{n^2+n}{2}$

96. The sum of a number and its reciprocal is thrice the difference of the number and its reciprocal. The number is
(1) $\pm\sqrt{2}$ (2) $\pm\frac{1}{\sqrt{2}}$ (3) $\pm\frac{1}{\sqrt{3}}$ (4) $\pm\sqrt{3}$

97. The total number of natural numbers that lie between 10 and 300 and are divisible by 9 is
(1) 32 (2) 30 (3) 33 (4) 34

98. If ${}^nC_x = 56$ and ${}^nP_x = 336$, find n and x.
(1) 7, 3 (2) 8, 4 (3) 8, 3 (4) 9, 6

99. One side of an equilateral triangle is 24 cm. The mid-points of its sides are joined to form another triangle whose mid-points are in turn joined to form still another triangle. This process continues indefinitely. Find the sum of the perimeters of all the triangles.
(1) 144 cm (2) 72 cm (3) 536 cm (4) 676 cm

100. The probability that a leap year selected at random contains either 53 Sundays or 53 Mondays, is
(1) 17/53 (2) 1/53
(3) 3/7 (4) None of these

101. Find the intercepts made by the line $3x + 4y - 12 = 0$ on the axes
(1) 2 and 3 (2) 4 and 3
(3) 3 and 5 (4) None of these

102. The average of 4 distinct prime numbers a, b, c, d is 35, where $a < b < c < d$. a and d are equidistant from 36 and b and c are equidistant from 34 and a, b are equidistant from 30 and c and d are equidistant from 40. The difference between a and d is
(1) 30 (2) 14
(3) 21 (4) Cannot be determined

103. Ramsukh Bhai sells rasgulla (a favourite Indian sweets) at ₹ 15 per kg. A rasgulla is made up of flour and sugar in the ratio 5 : 3. The ratio of price of sugar and flour is 7 : 3 (per kg). Thus, he earns 66 2/3 profit. What is the cost price of sugar?
(1) ₹ 10/kg (2) ₹ 9/kg (3) ₹ 18/kg (4) ₹ 14/kg

104. A reduction of 20% in the price of sugar enables a person to purchase 6 kg more for ₹ 240. What is the original price per kg of sugar?
(1) ₹ 10/kg (2) ₹ 8/kg (3) ₹ 6/kg (4) ₹ 5/kg

105. A solid sphere is melted and recast into a right circular cone with a base radius equal to the radius of the sphere. What is the ratio of the height and radius of the cone so formed?
(1) 4 : 3 (2) 2 : 3
(3) 3 : 4 (4) None of these

106. The speed of scooter, car and train are in the ratio of 1 : 4 : 16. If all of them cover equal distance, then the ratio of time taken/velocity for each of the vehicle, is
(1) 256 : 16 : 1 (2) 1 : 4 : 16
(3) 16 : 4 : 1 (4) 16 : 1 : 4

107. B is twice efficient as A and A can do a piece of work in 15 days. A started the work and after a few days B joined him. They completed the work in 11 days, from the starting. For how many days did they work together?
(1) 1 day (2) 2 days
(3) 6 days (4) 5 days

108. *A*, *B*, *C* and *D* purchased a restaurant for ₹ 56 lakhs. The contribution of *B*, *C* and *D* together is 460% that of *A*, alone. The contribution of *A*, *C* and *D* together is 366.66% that of B's contribution and the contribution of *C* is 40% that of *A*, *B* and *D* together. The amount contributed by *D* is
(1) 10 lakhs (2) 12 lakhs (3) 16 lakhs (4) 18 lakhs

109. The salary of Raju and Ram is 20% and 30% less than the salary of Saroj, respectively. By what percent is the salary of Raju more than the salary of Ram?
(1) 33.33% (2) 50% (3) 15.18% (4) 14.28%

110. The radius of a wire is decreased to one-third and its volume remains the same. The new length is how many times the original length?
(1) 2 times (2) 4 times (3) 5 times (4) 9 times

Section IV General English

For all questions in this section, correct answers carry 1 *mark each.*

Directions (Q. Nos. 111-116) *Read the following passage and answer the questions that follow.*

DH Lawrence-1885-1930 : The Supreme Triumph

For man, the vast marvel is to be alive. For man as for flower and beast and bird, the supreme triumph is to be most vividly, most perfectly alive. Whatever the unborn and the dead may know, they cannot know the beauty, the marvel of being alive in the flesh. The dead may look after the afterwards. But the magnificent here and now of life in the flesh is ours, and ours alone, and ours only for a time. We ought to dance with rapture, that we should be alive and in the flesh, and part of the living, incarnate cosmos. I am part of the sun as my eye is the part of me. That I am part of the Earth my feet below know the perfectly, and my blood is part of the sea. My soul knows that I am a part of the human race, my soul is an inorganic part of the great human soul, as my spirit is a part of my nation. In my own very self, I am part of my family. There is nothing of me that is alone and absolute except my mind, and we shall find that the mind has no existence by itself, it is only the glitter of the sun on the surface of the waters-Apocalypse, 1931.

111. By 'triumph' the author means
(1) sin (2) loss (3) sorrow (4) victory

112. When the dead look after the afterwards the living should look at life
(1) forever (2) for some months
(3) for only a short while (4) in the past

113. By 'rapture' the author means an emotion involving great
(1) trepidation (2) thrill (3) fear (4) joy

114. By the last line "It .....waters", the author means that the mind is only
(1) a mirage (2) an illusion (3) magic (4) a reflection

115. The tone of this passage is
(1) social (2) moral
(3) reflective (4) philosophical

116. The most suitable title for this passage would be
(1) The Surface of the Waters
(2) My Mind
(3) The Human Race
(4) Alive and Kicking

Directions (Q. Nos. 117-119) *Choose the correct option.*

117. The repetition of messages or the use of superfluous expressions is called …… .
(1) redundancy (2) hyperbole (3) alliteration (4) allegory

118. Ink : Pen : Paper
(1) Watch : Dial : Strap (2) Book : Paper : Words
(3) Farmer : Plough : Field (4) Colour : Brush : Canvas

119. Realia
(1) Theoretical constructs (2) Fabricated examples
(3) Objects from real life (4) Based on reality

Directions (Q. Nos. 120-125) *Choose the grammati- cally correct option from the following.*

120. (1) 'Are these gloves belonging to you?' she asked.
(2) 'Does this gloves belong to you?' she asked.
(3) 'Do these gloves belongs to you?' she asked.
(4) 'Do these gloves belong to you?' she asked.

121. (1) I live in a house in a street in the countryside. The street is called "Bear Street" and the house is old—more than 100 years old!
(2) I live in the house in the street countryside. The street is called "Bear Street" and the house is old—more than 100 years old!
(3) I live in a house in the street in the countryside. The street is called "Bear Street" and the house is old—more than 100 years old!
(4) I live in a house in a street in the countryside. The street is called "Bear Street" and a house is old—more than 100 years old!

122. (1) The teachers will be able to visit our schools and compare our teaching methods to their own.
(2) The teachers will be able to pay a visit to our schools and compare teaching methods for their own.
(3) The teachers will be able to visit our schools and compare our teaching methods with their own.
(4) The teachers will be able to visit our school and compare their teaching method with their own.

123. (1) Could you give me the amount that you filled out in the check which was sent?
(2) Could you give me the amount what you filled out in the check you sent?
(3) Could you give me the amount for which you filled out in the check you sent?
(4) Could you give me the amount wherein you filled out in the check you sent?

124. (1) I have completed the work yesterday.
(2) I did completed the work yesterday.
(3) I have had completed the work yesterday.
(4) I completed the work yesterday.

125. (1) The train couldn't stop in time and crashed with the truck.
(2) The train couldn't stop in time and crashed into the truck.
(3) The train couldn't stop in time and crashed against the truck.
(4) The train couldn't stop in time and crashed before the truck.

Directions (Q. Nos. 126-130) *Choose the correct synonymous word or description for each italicized word.*

The Jan Lokpal Bill, also *referred to* **(126)** as the citizens' ombudsman bill, is a proposed *independent* **(127)** anti-corruption law in India. Anti-corruption social activists proposed it as a more effective improvement on the original Lokpal Bill, which is currently being proposed by the Government of India. The Jan Lokpal Bill aims to effectively *deter* **(128)** corruption, redress *grievances* **(129)** of citizens, and protect whistle-blowers. If made into law, the bill would create an independent *ombudsman* **(130)** body called the Lokpal. It would be empowered to register and investigate complaints of corruption against politicians and bureaucrats without prior government approval.

126. (1) described as (2) included in
(3) supported for (4) reformed as

127. (1) self-centered (2) impartial
(3) self-seeking (4) self-possessed

128. (1) swell (2) prevent
(3) propel (4) lucubrate

129. (1) complaints (2) dishonesty
(3) committees (4) opinions

130. (1) a government appointee who investigates complaints by private persons against bureaucrats and/or politicians.
(2) a government appointee who investigates complaints by government against common citizens.
(3) a government appointee who investigates complaints by citizens against citizens.
(4) a government appointee who investigates complaints by government against government officials.

Directions (Q. Nos. 131-134) *In the following sentences, one word or a phrase is used wrongly. Choose the word that must be changed or modified or deleted to make the sentence correct. There are sentences without any errors too.*

131. The committee **came to** a decision **to discuss** in detail **about** assorted problems that people have been facing for **a long time**.
(1) came to (2) to discuss
(3) about (4) a long time

132. I know you must not **see eye to eye** with the philosophy of Ramkrishna but you must admit that he **had had** tremendous **influence over** a great many followers.
(1) see eye to eye (2) had had
(3) influence over (4) No error

133. The Principal **inquired with** the students, if they would like their teacher **to repeat** the lesson **again**.
(1) inquired with (2) to repeat
(3) again (4) No error

134. One of the **security guards** rushed **forward**; unlocked the gate and asked whether I had anything **objectionable**.
(1) security guards (2) forward
(3) objectionable (4) No error

Directions (Q. Nos. 135-136) *Rearrange the following sentence fragments* (P, Q, R *and* S) *to make meaningful sentences.*

135. I always told them ..... and consideration for me.
P. How for me she
Q. Always exuded warmth
R. Was like a family's
S. Elder and how she has
(1) PRSQ (2) PQSR (3) PRQS (4) SQPR

136. A year or ..... foreign languages
P. I picked up a liking for learning
Q. the largest urban area and primary
R. so after leaving Bangkok
S. city of Thailand and a place of unique beauty
(1) RPQS (2) RQSP (3) RSPQ (4) SPQR

Directions (Q. Nos. 137-139) *Choose the correct option.*

137. This book ..... five sections.
(1) comprises of (2) comprises
(3) consists (4) comprises to

138. My grandfather left most of his money to an NGO; the rest went directly to my daughter and ..... .
(1) I (2) me (3) myself (4) myself too

139. Parts of a country behind the coast or a river's banks.
(1) Isthumus (2) Archipelago
(3) Hinterland (4) Swamp

Directions (Q. Nos. 140-141) *In the questions below, the sentences have been given in active/passive voice. From the given alternatives, choose the one which best expresses the given sentence in passive/active voice.*

140. Who is creating this mess?
(1) Who has been created this mess?
(2) By whom has this mess been created?
(3) By whom this mess is being created?
(4) By whom is this mess being created?

141. You should open the wine about three hours before you use it.
(1) Wine should be opened about three hours before use.
(2) Wine should be opened by you three hours before use.
(3) Wine should be opened about three hours before you use it.
(4) Wine should be opened about three hours before it is used.

Direction (Q. Nos. 142) *In the question below, the passage consist of six sentences. The first and the sixth sentences are given.* (S1, S6) *The middle four sentences in each have been removed and jumbled up. These are labeled as* P, Q, R *and* S. *Find out the proper order for the four sentences.*

142. **S1** Metals are today being replaced by polymers in many applications.
P. Above all, they are cheaper and easier to process making them a viable alternative to metals.
Q. Polymers are essentially a long chains of hydrocarbons molecules.
R. Today polymers as strong as metals have been developed.
S. These have replaced the traditional chromium plated metallic bumpers in cars.
S6 Many Indian Institutes of Science and Technology run special programmes on polymer science.
The proper sequence should be
(1) QRSP (2) RSQP (3) RQSP (4) QRPS

Directions (Q. Nos. 143-144) *Choose the word which best expresses the meaning of the given word.*

143. FRUGALITY
(1) Foolishness (2) Extremity (3) Enthusiasm (4) Economy

144. HARBINGER
(1) Massenger (2) Steward (3) Forerunner (4) Pilot

Directions (Q. Nos. 145-146) *Choose the word which is the exact opposite of the given words.*

145. EXODUS
(1) Influx (2) Home-coming (3) Return (4) Restoration

146. EQUANIMITY
(1) Resentment (2) Dubiousness
(3) Duplicity (4) Excitement

Direction (Q. Nos. 147) *Find the odd one out from the group of words.*

147. (1) Bludgeon (2) Dragon
(3) Black Jack (4) Order

Direction (Q. Nos. 148) *Each pair of capitalized words given is followed by four pair of words. Select the pair that does not express a relationship similar to that expressed by the capitalized pair.*

148.

KERNAL	:	SHELL
(1) Caterpillar	:	Pupa
(2) Larva	:	Cacoon
(3) Lassitude	:	Syncope
(4) Passenger	:	Car

149. Both of them ..... since their childhood.
(1) are working here
(2) work here
(3) have been working
(4) are liking to work

150. Although initial investigations pointed towards him ..... .
(1) the preceding events corroborated his involvement in the crime
(2) the additional information confirmed his guilt
(3) the subsequent events established that he was guilty
(4) the subsequent events proved that he was innocent

Answers with Solutions

1.	(4)	**2.**	(1)	**3.**	(4)	**4.**	(2)	**5.**	(2)	**6.**	(2)	**7.**	(1)	**8.**	(1)	**9.**	(1)	**10.**	(4)
11.	(3)	**12.**	(4)	**13.**	(1)	**14.**	(1)	**15.**	(4)	**16.**	(4)	**17.**	(4)	**18.**	(2)	**19.**	(1)	**20.**	(1)
21.	(3)	**22.**	(3)	**23.**	(3)	**24.**	(2)	**25.**	(4)	**26.**	(1)	**27.**	(3)	**28.**	(2)	**29.**	(1)	**30.**	(1)
31.	(2)	**32.**	(4)	**33.**	(2)	**34.**	(1)	**35.**	(3)	**36.**	(1)	**37.**	(1)	**38.**	(4)	**39.**	(2)	**40.**	(4)

41. (3) We can choose the true subset by taking each statement.
G: People are rewarded.
C: If people are rewarded, then they will not change their lifestyle.
F: Unless people change their lifestyle, temperature rises.
This can also be put in the following manner.
If people do not change their lifestyle, then temperature rises.
Thus, if the temperature doesn't rise, then people change their lifestyle.
Here, according to G and C, people do not change their lifestyle and so temperature rises.
D: If temperature rises, then water level in the coastal area rises.
H: Water level in the coastal area doesn't rise.
Here, D is in accordance with G, C and F but H is not.
So, option 1 is incorrect.
B: People change their lifestyle only if they are rewarded.
This means that if people have changed their lifestyle, then they have been rewarded.
Note that if people are rewarded, they may or may not have changed their lifestyle.
G: People are rewarded.
From B and G, we don't know whether people have changed their lifestyle or not.
F: Unless the people change their lifestyle, temperature rises.
This can also be put as – If the people don't change their lifestyle, then temperature rises.
But as we don't know whether people have changed their lifestyle or not, we cannot conclude whether temperature has risen or not.
D: If temperature rises, then water level in the coastal area rises.

H: Water level in the coastal area doesn't rise.

As we don't know whether temperature has risen or not, H cannot be concluded using D.

Hence, option 2 is also incorrect.

E: Whenever the water level in the coastal areas rises, then the temperature rises.

H: Water level in the coastal area doesn't rise.

Thus, from these two statements the temperature may or may not rise.

While discussing the previous option, we have proved that from statements B, F and G we also come to the same conclusion that the temperature may or may not rise.

Hence, statements E, F,G, H and B are logically consistent among themselves.

42. (3) From option 2, statement A says – If Kumar sings, then the audiences sleep.

Out of the other 4 statements C, F, G and H, none of the statements mention anything about sleep and hence option 2 is incorrect.

If option 2 is incorrect, option 4 is also incorrect as it mentions both (2) and (3).

C: Unless audience does not dance, the concert will be successful.

This can also be put as – If the audience dance, the concert will be successful.

F: Kumar sings, only if Vina dances.

This means that – If Kumar has sung, then Vina has danced.

But if Vina has danced, Kumar may or may not have sung.

B: If Kumar sings, then the audience dances.

We don't know whether Kumar has sung or not and so we don't know whether the audience have danced or not.

Hence, we cannot conclude that the concert is successful as given in statement H.

So, option 1 is also logically incorrect.

E: If Vina dances, then Kumar sings.

G: Vina dances.

Thus, from these two statements we know that Kumar sings.

B: If Kumar sings, then audiences dance.

Thus, audiences have danced.

C: Unless audience does not dance, the concert will be successful.

This can also be put as – If the audience dance, the concert will be successful.

Thus, the concert is successful as mentioned in statement H.

Hence, statements E, C, G, B and H are logically consistent among themselves.

43. (4) The person with both statements true will be the one whose statements help to establish the relationships among all 4 members and especially between B and C. This is because one of the questions explicitly asks for the relationship between B and C and there is no answer option such as "None of the above" or "Data Insufficient".

Observing A B is A's son and D is A's mother. However, there is no information about C. Therefore, the relationship between B and C cannot be established. Therefore, both of A's statements cannot be true.

Observing B B is C's husband and D is B's father. However, there is no information on A. A could be B and C's child or B's sibling. Therefore, both of B's statements cannot be true. Consider C D is C's mother-in-law and A is C's daughter. Since, the family has only four members, B has to be C's spouse. Thus, the overall relationship between the four members is known. It is also known that B and C are a couple. However, it cannot be said whether B is the husband or wife of C. Therefore, both of C's statements also cannot be considered true.

Observing D A is D's grand-daughter and B is D's daughter-in-law. Again, since there are only 4 members in the family, C should be B's husband, D's son and A's father.

Thus, D is the grand-parent, B and C are the couple where C is the husband and A is the grand-daughter. Thus, D's statements give the overall relationships as well as the specific relationship between B and C. Therefore, D is the person who always speaks the truth.

44. (3) Observing the solution of Q. 42, we can conclude that B is C's wife.

45. (4) Observing each option.

D is either A's grandfather or grandmother. Since, D always speaks the truth, option 3 may or may not be true (depending on the gender of D) while option 1 is definitely false.

B is C's wife. Based on the fact that both of D's statements are true, both of B's statements turn out to be false. Therefore, option 2 is also false.

A is B's daughter. Based on the fact that both of D's statements are true, both of A's statements turn out to be false.

Thus, B's daughter always tells lies.

46. (4) Observing the information given in question that the cube is kept in the corner of the room. Assume that opposite faces are painted red, blue and green. Since the cube is in the corner, one red, blue and green face each will be adjacent to the two walls and floor and will not be painted. So, there are only three faces that are painted. Since, the cube is cut into 343 identical cubes, it means that each face is cut into 7 equal layers along each dimension. Thus, there are 7 similar layers along the length, breadth as well as height. Since only one face is painted along each dimension, if we remove one layer each along the length, breadth as well as height, the remaining 6 layers in each dimension will be unpainted.

Hence, number of cubes that have no face painted

$$6^3 = 216$$

47. (1) Consider the 3 faces that are painted. Let us first consider the number of cubes that have at least one colour painted on them.

Consider the top face. Assume it is painted blue. It will have $7^2 = 49$ cubes with at least one colour on them.

Now, consider the front face. Assume it is painted red. It will also have 49 cubes with at least one colour. However, the top 7 cubes on this face are also common to the top face and have been counted once.

$\therefore$ Number of cubes to be considered from front face

$$= 7 \times 6 = 42$$

Finally, consider the side face which will be painted green and which will also have 49 cubes with at least one colour.

However, it also has 7 cubes common to the top face, 7 cubes common with the front face and 1 cube common to both the top and front face.

∴ Number of cubes to be considered from side face
$= 49 - (7 + 7 - 1) = 36$

∴ Number of cubes with at least one colour on them
$= 49 + 42 + 36 = 127$

Now, there are 7 cubes common to the top and front face. Similarly, there are 7 cubes common to the front and side face as well as 7 cubes common to the top and side face. Here, the cube at the intersection of all three faces will have red, blue as well as green colour, while the remaining 6 faces in each case will have a red-blue or red-green or blue-green combination.

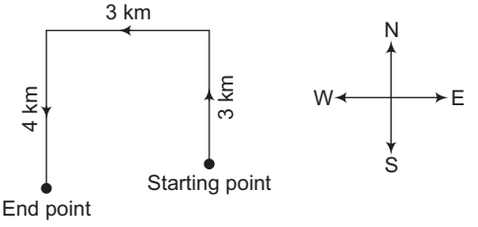
Thus, there is only 1 cube that is painted in 3 colours while there are 6 × 3 *i.e.*, 18 cubes that are painted in exactly 2 colours.

Hence, number of cubes that have exactly one colour on them
$= 127 - (1 + 18) = 108$

48. (2) Number of cubes that have at the most 2 faces painted
= Total number of cubes
– Number of cubes that have 3 faces painted.

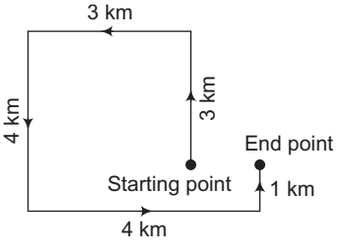
From the solution of Q. 47, number of cubes that have at the most 2 faces painted = 343 – 1 = 342

49. (2) After driving 3 km North, 3 km West and 4 km South, Amit is 1 km to the South and 3 km to the West of his starting point. So, he will have to move 1 km North and 3 km East to reach the point from where he started.



Hence, Amit will have to move 1 km North and 3 km East to reach the starting point.

50. (3) Driving as stated in the solution of Q. 49, Amit will reach the starting point if he moves 1 km North and 3 km East. If he moves 1 km North and 4 km East, he will be 1 km West from his starting point.



51. (3) $X = XY$ and $Y = Y + 1$ is the process that takes place in every step and the machine stops at $X \geq N$.

Input : $X = 3$, $Y = 2$ and $N = 100$.

Step 1 : $X = 3 \times 2 = 6$, $Y = 3$

Step 2 : $X = 6 \times 3 = 18$, $Y = 4$

Step 3 : $X = 18 \times 4 = 72$, $Y = 5$

Step 4 : $X = 72 \times 5 = 360$, $Y = 6$

Hence, 4 steps are performed before the machine stops.

52. (4) In Q. 51, the final value of $X = 360$.

53. (3) In Q. 51, the final value of $Y = 6$.

54. (4) If $N = 500$, then by using the solution of question 51, we can further calculate step 5 as follows :

Step 5 : $X = 360 \times 6 = 2160$, $Y = 7$.

Hence, the value of $X = 2160$.

55. (1) **Input :** $X = 2$, $Y = 3$

Step 1 : $X = 2 \times 3 = 6$, $Y = 4$

Step 2 : $X = 6 \times 4 = 24$, $Y = 5$

Step 3 : $X = 24 \times 5 = 120$, $Y = 6$.

Thus, for any value of N greater than 120, the value of Y will become 7.

The correct answer should be 121.

Hence, nearest value of 121 here is 300.

Solutions (Q. Nos. 56-60)

Vicky always scores more than Priya and Priya always scores more than Ankit. Thus, Vicky's rank is always numerically lower than Priya's and Priya's rank is always numerically lower than Ankit's.

Either of the following two situations take place.

If Raman scores the highest, then Tony scores the least or if Sunil scores the highest, then Deepak or Ankit scores the least. Thus, if Raman is ranked 1st, then Tony is ranked 7th and if Sunil is ranked 1st, then either Deepak of Ankit is ranked 7th.

56. (4) The question states that Sunil is ranked 6th and Ankit is ranked 5th.

Thus, Raman is ranked 1st and Tony is ranked 7th.

So, the following situations are possible for ranks 2, 3 and 4.

	2 – Vicky, 3 – Priya, 4 – Deepak
or	2 – Vicky, 3 – Deepak, 4 – Priya
or	2 – Deepak, 3 – Vicky, 4 – Priya.

Vicky can never be 1st, Raman will never be 2nd or 3rd, Tony is ranked 7th and hence cannot be 4th or 5th.

Deepak can be 3rd or 4th according to the above obtained conditions.

57. (3) If Raman gets the highest, he will be ranked 1st. Accordingly, Tony will be ranked 7th.

Now, Priya and Ankit are always ranked numerically higher than Vicky.

Hence, Vicky can never be ranked lower than 4th as if he is ranked 5th, then there will not be enough positions to accommodate both Ankit and Priya.

58. (4) The question states that Ankit's rank is 1st.

This is not possible because he always scores less than Vicky and Priya.

59. (1) If Sunil's rank is 2nd, then Raman's rank is 1st and Tony's rank is 7th.

Vicky's lowest numerical rank will be 3rd and so he will always get less than Sunil.

Raman is ranked 1st and so Priya will always get less than Raman.

Vicky scores more than Priya is given in the question data.

Hence, by eliminating the options, we can conclude that it is possible for Deepak to get more than Vicky.

60. (1) If Vicky's rank is 5th, then Priya's rank is 6th and Ankit's rank is 7th. Hence, Sunil's rank is highest or Sunil scores the highest.

61. (3) The conclusion is that Party Z is likely to win the next election. A weakening statement would have to show party Y having a favourable advantage or party Z having a clear disadvantage.

Option 1 is loose and does not provide anything concrete.

Option 2 shows the percentage of people voting from each party but fails to bring out the vantage point of a single party.

Option 4 does not weaken the conclusion.

Option 3 clearly states a percentage of people in favour of party Z, which is less than 50%.

62. (4) The information given to us is about inflation rising in quarters of a year.

Option 1 talks about the RBI taking "necessary measures" which is ambiguous.

Option 2 and 3 do not provide a strong explanation for the statement.

Option 4 is the right explanation. It talks about the quarterly increase in inflation and is in line with the information given.

63. (3) Birth - The act or process of bearing young;
Dirge - A funeral hymn or lament.
Similarly : Marriage - is a social union or legal contract; alimony - allowance made to a wife out of her husband's estate or income for her support, upon her divorce.

64. (4) The reasoning in the main statement is that, 'because beaches are beautiful, they are crowded'.

Option 1 only states that Moose and bear grow thirsty at the same time, it does not give a reason for this.

Option 2 shows an 'If not A then B' type of relationship.

Option 3 does not show a relationship between two variables.

Option 4, shows a similar relation to the one described in the main statement.

65. (4) Option 1 is false as there is no available measure for underground industries.

Option 2 assumes that all self-employed industries are not underground, which is erroneous.

Option 3 states that industries 'except for self-employed ones' may be running underground. This is a wrong conclusion.

Option 4 is correct. The latter statement states that some self-employed industries are included in national productivity measures and only non-underground industries are included in national productivity measures.

66. (3) This statement has an underlying assumption that 'receiving a violation from the FAA is a hindrance to an excellent performance record'.

67. (3) The example of Tom Hanks is specific. It is just one case upon which the broad conclusion of 'death penalty not serving as a deterrent to murder' has been made.

68. (4) Starting from upper row (the number at the center of each row is the sum of the other numbers divided by 2)

1st row $\rightarrow (4+2)\div 2=3$

2nd row $\rightarrow (5+3+1+1)\div 2=5$

3rd row $\rightarrow (6+1+2+3+3+1)\div 2=8$

4th row $\rightarrow (7+2+4+3)\div 2=8$

5th row $\rightarrow (9+3)\div 2=\boxed{6}$

69. (3) The latter part of the information states that removal of restrictions such as the ones including fee arrangements for advertisements will cause reduction in overall consumer legal costs and the former part of the information states that fewer restrictions on advertising will lead to an increase in lawyers advertising for their services.
Hence, option (3) is true if the given statements information is true.

70. (4) The information clearly mentions that those lawyers who advertise for their services charge lesser than those who do not advertise. Hence, option (4) seriously weakens the argument concerning consumer legal costs.

71. (2) Speed of train = 36 km/h = $36\times 5/18$ m/s = 10 m/s
Let the length of the platform be P and length of the train be T.
Now, the train crosses the platform in 20 s and the man in 10s .

Hence, the train travelled, $(P+T)$ in 20 s and the distance T in 10 s.

Hence, it travelled P distance in 20 – 10 = 10 s.

Hence, P = 10 × 10 = 100 m.

72. (2) Let t be the usual time taken by the man.
Now, if he travels at 4/5th of his usual speed ,then time taken by him will be $5/4\times t$.

Hence, $5/4\times t=t+10$

Hence, $t=40$ min

73. (2) Relative speed of man and woman is $(4+5)=9$ miles/h
Hence, time required to cover 81 miles $=81/9=9$ h
In 9 h, the woman will travel $9\times 4=36$ miles.

74. (2) Assume that one of them walked 6 miles towards East and then 8 miles South.
Hence, he is 6 miles East and 8 miles South of the original position.

Hence, he is $\sqrt{(6^2+8^2)}=10$ miles from his original position.

Similarly, we can prove that the other person is also 10 miles from his original position.

75. (3) Let a man takes m days and a woman takes w days to complete the work.
Therefore, according to the question,

$$\frac{4}{m}+\frac{3}{w}=\frac{1}{6} \quad \text{...(i)}$$

$$\frac{5}{m}+\frac{6}{w}=\frac{1}{4} \quad \text{...(ii)}$$

Solving Eqs. (i) and (ii), we get,

$m=36$ and $w=54$

Hence, (2 women and 3 men)'s 1 day work

$2/54 + 3/36 = 13/108$

Hence, they will complete the work in $108/13 = 8.3$ days

Note: *In actual paper, all the options were incorrect. Hence, we have changed one of the options to include the correct answer.*

76. (3) Since, Ram completes 60% of the work in 15 days.
Therefore, he will complete 100% of the work in

$$\frac{15}{60}\times 100=25 \text{ days}$$

Now, since Rahim is 50% as efficient as Ram.
Therefore, time taken by Rahim to complete the work =50
And since Rachel is 50% as efficient as Rahim,
Therefore, time taken by Rachel to complete the work =100
Hence, Ram, Rahim and Rachel together will complete

$\frac{1}{25}+\frac{1}{50}+\frac{1}{100}=\frac{7}{100}$ of the work in one day.

Hence, they will complete the work in 100/7 days.
Hence, they can complete the remaining 40% work in

$0.4\times\frac{100}{7}=\frac{40}{7}$ days

Note: *In actual paper, all the options were incorrect. Hence, we have changed one of the options to include the correct answer.*

77. (2) In 9 days, B will do $9/24 = 3/8$ of the work.
Hence, A and B together did $1 - 3/8 = 5/8$ of the work together.
Now, A and B can complete $\frac{1}{21}+\frac{1}{24}=\frac{5}{56}$ of the work in one day
Hence, they will complete 5/8 of the work in $\frac{5}{8}\times\frac{56}{5}=7$ days

78. (3) Let selling price of each article be ₹ 1.
∴ Selling price of 100 article = ₹ 100
∴ Profit = ₹ 75
∴ Cost price = 100 – 75 = 25
Hence, profit percentage = 75/25 × 100% = 300%

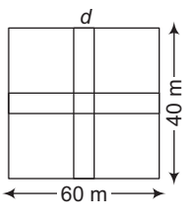
79. (3) Let V_a and V_b be the speeds of A and B, respectively.
∴ $$\frac{V_a}{100}+5=\frac{80}{V_b} \quad ...(i)$$
$$\frac{V_a}{100}=\frac{60}{V_b} \quad ...(ii)$$
Solving Eqs. (i) and (ii), we get
$$V_b = 4, V_a = 100/15$$
Hence, A will takes $\frac{200}{\frac{100}{15}}=30$ s

80. (2) Surface area of the original cube
$$= 2(4\times4+4\times4+4\times4)$$
$$=2\times3\times16 = 6 \times 16 \text{ cm}^2$$
Now, there will 64 new cubes of side 1 cm.
∵ Surface area of a cube
$$=2(1\times1+1\times1+1\times1)$$
$$=2\times3=6 \text{ cm}^2$$
∴ surface area of these 64 cubes = 64 × 6 cm^2
Hence, percentage increase in surface area is
$$=\frac{64\times6-16\times6}{16\times6}\times100\%=300\%$$

81. (1) G236G0 is divisible by 36 *i.e.*, by 9 and 4
Therefore, G + 2 + 3 + 6 + G + 0 = 2G + 11 is divisible by 9. Also, G should be an ever number.
Only G = 8 satisfies the given condition.

82. (3) In one day, Amit and Sagar can do $\frac{1}{12}+\frac{1}{15}=\frac{3}{20}$ of the work.
∴In 4 days, they will complete $\frac{3}{20}\times4=3/5$ of the work.
Hence, fraction of the work remaining after 4 days $=1-\frac{3}{5}=\frac{2}{5}$

83. (2) Let the width of the road be d.
Hence, Area of lawn = Area of park – Area of road
$= 2400 - (40d + 60d - d^2)$
$\Rightarrow 2109 = 2400 - 100d + d^2$
$\Rightarrow d^2 - 100d + 297 = 0$
$\Rightarrow d = 97$ or $d = 3$.



84. (1) Probability of choosing either bag is 1/2.
Now, from the first bag, we can choose one white and one black ball in 5 × 3 = 15 ways.
While, two balls can be selected in ${}^8C_2 = 28$ ways
Hence, probability that the two balls chosen are of different colour = 15/28
Similarly, we can prove that the probability that two balls chosen from the second bag are of different colour = 20/36
Hence, the required probability
$$=\frac{1}{2}\times\frac{15}{28}+\frac{1}{2}\times\frac{20}{36}=\frac{275}{504}$$

Solutions (Q. Nos. 85-88)
Consumption = Production – Exports
Per Capita Consumption = Consumption/Population
∴ Population = Consumption/Per Capita Consumption
Thus, the consumption and population for each year can be calculated as shown in the table below:

Year	Production	Exports	Consumption	Per Capita Consumption	Population	% Increase in Consumption	Export/ Consumption
2006	186.5	114	72.5	36.25	2.00		1.57
2007	202	114	88	35.2	2.50	21.38	1.30
2008	238	130	108	38.7	2.79	22.73	1.20
2009	221	116	105	40.5	2.59	– 2.78	1.10
2010	215	88	127	42	3.02	20.95	0.69

85. (2) The percentage increase in the consumption of rice was the highest in 2008.

86. (3) From the table, the population of country A in 2008 was 2.79 million.

87. (1) From the table, the ratio of consumption to exports in the given period was the highest in 2006 (1.57).

88. (4) From the table, the population of country A was the highest in the year 2010 (3.02 million).

89. (2) The percentage of time spent in school
$$=\frac{105}{360}\times100\%=29.17\%\approx30\%$$

90. (3) Since, the angle subtended by "games" is 30° while that subtended by "sleeping" is 120° , therefore the required percentage is $(30/120)\times100\%$ *i.e.,* 25%.

91. (2) Since, the student spends an equal time in games and home work, he can either increase the time spent on games or decrease the time spent on home work.
It is also given that the time spent on other activities remains constant while the decrease in time spent on sleeping is to be found.
This can be happen only if the time spent on games is increased.
So, the angle subtended for games should increase from 30° to 45° . If this happens, the angle subtended for sleeping will decrease from 120° to 105°.
∴ Percentage decrease $=\frac{(120-105)}{120}\times100\%=12.5\%$

92. (3) Difference in time spent in school and home work is

$$\frac{(105-45)}{360}\times 24 = 4 \text{ h}$$

93. (2) Time spent on home work $= \frac{45}{360}\times 24 = 3$ h

Out of this, one-third *i.e.*, one hour is spent on Mathematics.

So, the remaining two hours are spent on the rest of the subjects.

94. (2) Since, the given three lines are paralled and cut by two transversals. Therefore,

$$\frac{AB}{BC} = \frac{DE}{EF}$$

Substituting value of AB, BC and DE, we get

$$EF = 2 \times 1.5 = 3$$

95. (4) $\log 10 + \log 10^2 + \ldots + \log 10^n$

$\log_{10} 10 + 2\log_{10} 10 + \ldots + n\log_{10} 10$

$= 1 + 2 + 3 + \ldots + n$

$= n(n+1)/2$

96. (1) Let a be the required number.

Therefore, $\left(a + \frac{1}{a}\right) = 3\left(a - \frac{1}{a}\right)$

$$a + \frac{1}{a} = 3a - \frac{3}{a}$$

$\Rightarrow \quad a^2 = 2 \quad \Rightarrow \quad a = \pm\sqrt{2}$

97. (1) Smallest and largest numbers, in the given region, which are divisible by 9 are 18 and 297, respectively.

Hence, the required number

$= (297 - 18)/9 + 1$

$= 31 + 1 = 32$

98. (3) ${}^nC_x = \frac{n!}{x!(n-x)!} = 56$ ...(i)

${}^nP_x = \frac{n!}{(n-x)!} = 336$ ...(ii)

Dividing Eq (ii) by Eq. (i), we get

$x! = 6$

$x = 3$

Then, from Eq (ii),

$$\frac{n!}{(n-3)!} = 336$$

$\Rightarrow n(n-1)(n-2) = 6 \times 7 \times 8 \Rightarrow n = 8$

99. (1) Perimeter of the largest equilateral triangle $= 3 \times 24 = 72$ cm

Now, triangle formed by joining the mid-points will have perimeter half of that of original triangle.

Hence, the required sum $= 72 + 36 + 18 + \ldots$ upto infinite terms

This is nothing but an infinite GP, with first term (a) $= 72$ and common ratio (r) $= \frac{1}{2}$

Hence, the required sum $= \frac{a}{1-r} = \left(\frac{72}{1-\frac{1}{2}}\right) = 144$ cm

100. (3) A leap year contains 52 weeks and 2 extra days.

These two days can be either of

(Sun, Mon), (Mon, Tue), (Tue, Wed), (Wed, Thur), (Thur, Fri), (Fri, Sat) or (Sat, Sun)

Probability that the leap year has 53 Sundays or 53 Mondays

$$P(S \cup M) = P(S) + P(M) - P(S \cap M)$$

$$= \frac{2}{7} + \frac{2}{7} - \frac{1}{7} = \frac{3}{7}$$

101. (2) We have, $3x + 4y = 12$

$\Rightarrow \quad x/4 + y/3 = 12$

This is nothing but the intercept form of the equation of the given line.

Hence, intercepts on x and y axes are 4 and 3, respectively.

102. (2) Given, a and d are equidistant from 36.

This implies that average of a and d is 36.

Therefore, $a + d = 72$

Similarly, we have,

$b + c = 68$, $a + b = 60$

and $c + d = 80$

Now, $a < b$ and $a + b = 60$

Hence, $a < 30$ and $b > 30$

Similarly, $b < 34$ and $c > 34$

And $c < 40$ and $d > 40$

Now, only primes which satisfy these conditions are

$a = 29$, $b = 31$, $c = 37$ and $d = 43$

Hence, $d - a = 43 - 29 = 14$

103. (4) After selling at ₹ 15 per kg, Ramsukh earns a profit of 66.66%.

Therefore, cost price of rasgulla is ₹ 9/kg.

Now, ratio of flour and sugar is 5 : 3.

Therefore, 1 kg of rasgulla is made up of 5/8 kg of flour and 3/8 kg of sugar.

Let price of 1 kg of flour $= 3k$

and price of 1 kg of sugar $= 7k$

Hence, price of 1 kg of rasgulla

$$\frac{3}{8}\times 7k + \frac{5}{8}\times 3k = 9 \quad \Rightarrow \quad k = 2$$

Hence, cost price of sugar $= 7k = 7 \times 2 =$ ₹ 14/kg

104. (1) Let cost price of sugar be ₹ x per kg.

Therefore, reduced cost price = $0.8x$

According to the question,

$240/x + 6 = 240/0.8x$

$\Rightarrow \quad 300/x = 240/x + 6$

$\Rightarrow \quad 60/x = 6$

$\Rightarrow \quad x =$ ₹ 10 per kg

105. (4) Volume of a solid sphere is given as $\frac{4\pi r^3}{3}$, where r is the radius of the sphere.

Volume of a right circular cone is $\frac{\pi r^2 h}{3}$, where r is the radius of the base of the cone and h is the height of the cone. The solid sphere is recast into a right circular cone with base radius equal to the radius of the sphere.

$\therefore \quad \frac{4\pi r^3}{3} = \frac{\pi r^2 h}{3} \quad \Rightarrow \quad h = 4r$

$\Rightarrow \quad \frac{h}{r} = \frac{4}{1}$

Thus, the ratio of the height to the radius is 4 : 1.

106. (1) Assume that the scooter, the car and the train travel 16 km.
Let their speeds be 1 km/h, 4 km/h and 16 km/h, respectively.
Thus, time taken by the scooter, the car and the train are 16 h, 4 h and 1 h, respectively.
Ratio of time taken to the velocity of each vehicle will be
$(16/1) : (4/4) : (1/16) = 16 : 1 : (1/16) = 256 : 16 : 1$

107. (2) A can do a piece of work in 15 days.
B is twice as efficient as A.
So, B can do the same piece of work in 7.5 days.
Assume that there is 15 units of work to be done.
A does 1 unit per day and B does 2 units per day.
For the first x days, A worked alone and hence work done is $1 \times x = x$ units.
For the next $(11 - x)$ days, A and B worked together. In 1 day, they do $(1 + 2) = 3$ units.
So, in $11 - x$ days, they do $33 - 3x$ units.
$\therefore \quad 33 - 3x + x = 15$
$\Rightarrow \quad 2x = 18$ and $x = 9$.
Thus, A worked alone for 9 days the A and B worked together for 2 days.

108. (4) Let the contributions of A, B, C and D in purchasing a restaurant be a, b, c and d, respectively.
Then, according to the question,

$$a + b + c + d = 5600000 \quad \text{...(i)}$$
$$b + c + d = 4.6a \quad \text{...(ii)}$$
$$a + c + d = 3.666\, b \quad \text{...(iii)}$$
$$0.4\,(a + b + d) = c$$
$$a + b + d = 2.5c \quad \text{...(iv)}$$

Using Eqs. (i) and (ii), we get
$a = 5600000/5.6 = 1000000$ ...(v)
Using Eqs. (i) and (iii), we get

$$b = 5600000/4.66$$
$$= 1201716.73 \quad \text{...(vi)}$$

Using Eqs. (i) and (iv), we get

$$c = 5600000/35 = 1600000 \quad \text{...(vii)}$$

Using Eqs. (i), (v), (vi) and (vii), we get $d \approx 1800000$

109. (4) Let the salary of Saroj be ₹ 100.
Given that salary of Raju and Ram is 20% and 30% less than Saroj's salary.
Hence, Raju's salary $= 80\%$ of 100
$= ₹\,80$
and Ram's salary $= 70\%$ of $100 = ₹\,70$
Thus, required percentage $= \frac{80 - 70}{70} \times 100\%$
$= 14.28\%$

110. (4) Let the radius of the wire be r and its length be l.
A wire is cylindrical in nature where the length is considered to be its height.
Therefore, volume of this wire is $\pi r^2 l$.
Let new radius and new length of the wire be r' and l', respectively
$\therefore \quad r' = r/3$
Now, as the volume remains constant,

$$\pi r^2 l = \pi \left(\frac{r}{3}\right)^2 l'$$

$\Rightarrow \quad l' = 9l$
Thus, new length is 9 times the original length.

111. (4) The synonym of 'triumph' here in this passage, is 'victory'.

112. (3) Refer to the fourth sentence of the passage.

113. (4) 'Rapture' means 'a feeling of great joy'.

114. (4) Here in this phrase, 'glitter' means 'reflection'.

115. (4) The passage is from the beginning to the end is talking about the philosophy of life.

116. (4) The first sentence in itself indicates the theory so it must be the title of the passage.

117. (1) Repetition of anything is termed 'redundancy'.

118. (4) In this analogy, the relationship is in between material and object. So, this is the right choice.

119. (3) 'Realia' clearly expresses 'the things or objects from the real life'.

120. (4) No other sentence is grammatically correct.

121. (1) No other sentence is structurally proper.

122. (3) All other sentences have error of prepositional use and the placement of pronoun.

123. (1) All other sentences are grammatically unacceptable.

124. (4) All other sentences have error in the usage of tense.

125. (2) The correct phrase is 'crashed into'.

126. (1) **127.** (2) **128.** (2) **129.** (1) **130.** (1)

131. (3) This sentence doesn't need 'about'.

132. (2) It should be ''It had ..... '' in place of ''It had had ..... ''.

133. (3) After the use of 'repeat', the use of 'again' is uncalled for.

134. (4) **135.** (1) **136.** (2) **137.** (2) **138.** (2)

139. (1) **140.** (4) **141.** (4) **142.** (1) **143.** (4)

144. (3) **145.** (1) **146.** (4) **147.** (4) **148.** (4)

149. (3) The word 'since' indicates that it is a sentence of Present Perfect Continuous.

150. (4) The word 'Although' is the indicator of the right option for the answer.

SNAP

Symbiosis National Aptitude Test

Solved Paper 2010

Section I
(Quantitative and Data Interpretation & Sufficiency)

For all questions in this section, correct answer carries 1 mark each.

1. A dice is rolled three times and sum of three numbers appearing on the uppermost face is 15. The chance that the first roll was a four is

(a) $\frac{2}{5}$ (b) $\frac{1}{5}$ (c) $\frac{1}{6}$ (d) None of these

2. A boat covers a distance of 30 km downstream in 2 h while it takes 6 h to cover the same distance upstream. What is the speed of the boat in km/h?

(a) 5 (b) 7.5 (c) 13 (d) 18

3. A five-digit number is formed by using the digits 1, 2, 3, 4 and 5 without repetitions. What is the probability that the number is divisible by 4?

(a) $\frac{1}{5}$ (b) $\frac{5}{6}$ (c) $\frac{4}{5}$ (d) None of these

4. If the algebraic sum of deviations of 20 observations measured from 23 is 70, mean of these observations would be

(a) 24 (b) 25 (c) 26 (d) None of these

5. An alloy of gold and silver weighs 50 g. It contains 80% gold. How much gold should be added to the alloy so that percentage of gold is increased to 90?

(a) 50 g (b) 60 g (c) 30 g (d) 40 g

6. Weekly incomes of two persons are in the ratio of 7 : 3 and their weekly expenses are in the ratio of 5 : 2. If each of them saves ₹ 300 per week, then the weekly income of the first person is

(a) ₹ 7500 (b) ₹ 4500 (c) ₹ 6300 (d) ₹ 5400

7. Wheat is now being sold at ₹ 27 per kg. During last month its cost was ₹ 24 per kg. Find by how much per cent a family reduces its consumption so as to keep the expenditure fixed.

(a) 10.2% (b) 12.1% (c) 12.3% (d) 11.1%

8. There are 10 stations on a railway line. The number of different journey tickets that are required by the authorities is

(a) 92 (b) 90 (c) 91 (d) None of these

9. The radius of a circle is so increased that its circumference increased by 5%. The area of the circle then increases by

(a) 12.5% (b) 10.25% (c) 10.5% (d) 11.25%

10. In how many ways can the letters of the word *ABACUS* be rearranged such that the vowels always appear together?

(a) $\frac{6!}{2!}$ (b) $3! \times 3!$ (c) $\frac{3! \times 3!}{2!}$ (d) $\frac{4! \times 3!}{2!}$

11. In 4 yr, ₹ 6000 amounts to ₹ 8000. In what time at the same rate will ₹ 525 amount to ₹ 700?

(a) 2 yr (b) 3 yr (c) 4 yr (d) 5 yr

Directions (Q. Nos. 12 to 14) *Based on the information given below.*

At the start of a game of cards, *J* and *B* together had four times as much money as *T*, while *T* and *B* together had three times as much as *J*.

At the end of the evening, *J* and *B* together had three times as much money as *T*, while *T* and *B* together had twice as much as *J*. *B* lost ₹ 200.

12. What fraction of the total money did *T* have at the beginning of the game?

(a) $\frac{1}{3}$ (b) $\frac{1}{8}$ (c) $\frac{2}{9}$ (d) $\frac{1}{5}$

13. What fraction of the total money did *J* win/lose?

(a) Won $\frac{1}{12}$ (b) Lost $\frac{1}{6}$ (c) Lost $\frac{1}{3}$ (d) Won $\frac{1}{5}$

14. What amount did *B* start with?

(a) ₹ 575 (b) ₹ 375 (c) ₹ 825 (d) ₹ 275

15. If *a* and *b* are negative, and *c* is positive, which of the following statements is/are true?

(I) $a - b < a - c$ (II) If $a < b$, then $a/c < b/c$ (III) $1/b < 1/c$

(a) I only (b) II only (c) III only (d) II and III

16. The diagonal of a square is $4\sqrt{2}$ cm. The diagonal of another square whose area is double that of the first square is

(a) 8 cm (b) $8\sqrt{2}$ cm (c) $4\sqrt{2}$ cm (d) 16 cm

17. The maximum length of a pencil that can be kept in a rectangular box of dimensions 8 cm × 6 cm × 2 cm is

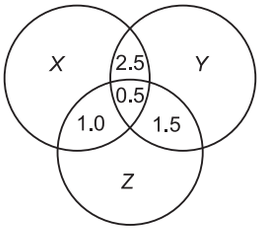
(a) $2\sqrt{13}$ cm (b) $2\sqrt{14}$ cm (c) $2\sqrt{26}$ cm (d) $10\sqrt{2}$ cm

Directions (Q. Nos. 18 and 19) *Based on the information given below.*

The Venn diagram given below shows the estimated readership of 3 daily newspapers (*X, Y* and *Z*) in a city. The total readership and advertising cost for each of these papers is as below.

Newspaper	Readership (lakh)	Advertising cost (₹ per cm^2)
X	8.7	6000
Y	9.1	6500
Z	5.6	5000

The total population of the city is estimated to be 14 million. The common readership (in lakh) is indicated in the given Venn diagram.



18. The number of people (in lakh) who read at least one newspaper is

(a) 4.7 (b) 11.9 (c) 17.4 (d) 23.4

19. The number of people (in lakh) who read only one newspaper is

(a) 4.7 (b) 11.9 (c) 17.4 (d) 23.4

20. Sonali invests 15% of her monthly salary in insurance policies. She spends 55% monthly salary in shopping and on household expenses. She saves the remaining amount ₹ 12750. What is Sonali's monthly income?

(a) ₹ 42500 (b) ₹ 38800 (c) ₹ 40000 (d) ₹ 35500

21. How many kilograms of tea worth ₹ 25 per kg must be blended with 30 kg of tea worth ₹ 30 per kg so that by selling the blended variety at ₹ 30 per kg there should be a gain of 10%?

(a) 36 kg (b) 40 kg (c) 32 kg (d) 42 kg

22. In an examination, out of 80 students 85% of the girls and 70% of the boys passed. How many boys appeared in the examination if total pass percentage was 75%?

(a) 370 (b) 340 (c) 320 (d) 360

23. 300 g of salt solution has 40% salt in it. How much salt should be added to make it 50% in the solution?

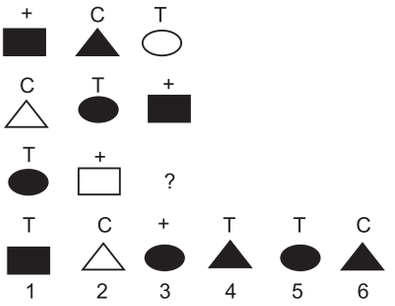
(a) 40 g (b) 60 g (c) 70 g (d) 80 g

24. What number should replace the question mark in the image below?

6	12	?
48	96	192

(a) 18 (b) 20 (c) 22 (d) 24

25. What image from bottom row should replace the question mark?



(a) 4 (b) 6 (c) 1 (d) None of these

26. A five-digit number divisible by 3 is to be formed using numerical 0, 1, 2, 3, 4 and 5 without repetition. The total number of ways this can be done is

(a) 122 (b) 210 (c) 216 (d) 217

27. A contract is to be completed in 50 days and 105 men were set to work, each working 8 h a day. After 25 days, 2/5th of the work is finished. How many additional men be employed so that the work may be completed on time, each man now working 9 h a day?

(a) 34 (b) 36 (c) 35 (d) 37

28. *A* can build up a structure in 8 days and *B* can break it in 3 days. *A* has worked for 4 days and then *B* joined to work with *A* for another 2 days only. In how many days will *A* alone build up the remaining part of the structure?

(a) 10 days (b) 9 days (c) 12 days (d) None of these

29. The first two terms of a geometric progression add up to 12. The sum of the third and the fourth terms is 48. If the terms of the geometric progression are alternately positive and negative, then the first term is

(a) – 2 (b) – 4 (c) – 12 (d) 8

30. The mean of the numbers a, b, 8, 5, 10 is 6 and the variance is 6.80. Then which one of the following gives possible values of a and b?

(a) $a = 0,\ b = 7$ (b) $a = 5, b = 2$ (c) $a = 3, b = 4$ (d) $a = 2,\ b = 4$

31. An agent sells goods of value of ₹ 15000. The commission which he receives at the ratio of $12\frac{1}{2}\%$ is

(a) ₹ 1875 (b) ₹ 2125 (c) ₹ 2000 (d) ₹ 2700

32. $\sqrt{110.25} \times \sqrt{0.01} \div \sqrt{0.0025} - \sqrt{420.25}$ equals to

(a) 0.75 (b) 0.50 (c) 0.64 (d) 0.73

Directions (Q. Nos. 33 to 35) *Based on the information given below.*

The table below shows the number of people who responded to a survey about their favourite style of music. Use this information to answer the following questions to the nearest whole percentage.

Age	15-20	21-30	31+
Classical	6	4	17
Pop	7	5	5
Rock	6	12	14
Jazz	1	4	11
Blues	2	3	15
Hip-Hop	9	3	4
Ambient	2	2	2
	33	33	68

33. What percentage of respondents under 31 indicated that Blues is their favourite style of music?

(a) 7.1 (b) 7.6 (c) 8.3 (d) 14.1

34. What percentage of respondents aged 21-30 indicated a favourite style other than Rock music?

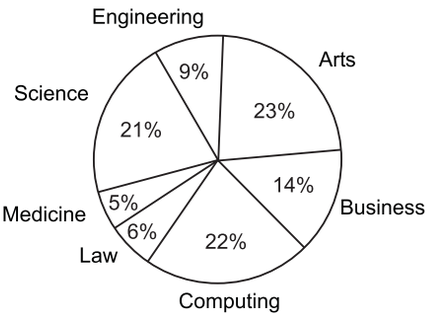
(a) 64% (b) 60% (c) 75% (d) 36%

35. What percentage of the total sample indicated that Jazz is their favourite style of music?

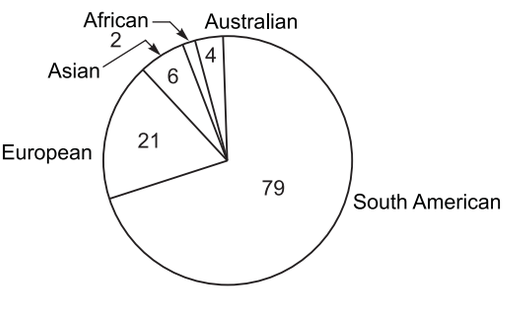
(a) 6% (b) 8% (c) 22% (d) 12%

Directions (Q. Nos. 36 to 40) *Based on the information given below.*

The pie charts below show the percentage of students in each faculty at North-West University and the number of non-US students in the Arts faculty. These percentages have been rounded to the nearest whole number. There are a total of 1049 students in Arts faculty. Use this information to answer the following questions.



Students by faculty



Arts students (non-US)

36. What percentage of students in the Arts faculty are non-US students?
(a) 14% (b) 9% (c) 30% (d) 11%

37. How many students are there in the Engineering faculty?
(a) 420 (b) 410 (c) 390 (d) 440

38. How many students are there at the University?
(a) 4650 (b) 4560 (c) 4640 (d) 4450

39. If 6% of Science students are Asian, how many Asian students are studying Science?
(a) 48 (b) 66 (c) 120 (d) 57

40. There are 34 European medical students. What percentage of the faculty does this represent?
(a) 14% (b) 18% (c) 16% (d) 15%

Section II
(Analytical and Logical Reasoning)

For all questions in this section, correct answer carries 1 mark each.

41. There are 3 societies *A, B, C* having some tractors each.
A gives *B* and *C* as many tractors as they already have.
After some days *B* gives *A* and *C* as many tractors as they have.
After some days *B* gives *A* and *C* as many tractors as they have.
Finally each has 24 tractors. What is the original number of tractors each had in the beginning?
(a) *A*-29, *B*-21, *C*-12 (b) *A*-39, *B*-21, *C*-12 (c) *A*-21, *B*-12, *C*-29 (d) *A*-21, *B*-12, *C*-39

42. Although most of the fastest growing jobs in today's economy will require a college degree, many of the new jobs being created from home health aide to desktop publisher require knowledge other than gained from earning a degree. For workers in those jobs, good basic skills in reading, communication, and mathematics play an important role in getting a job and developing a career.
From the information given above it can be validly concluded that, in today's economy
(a) skills in reading, communication and mathematics play an important role in developing a career as a desktop publisher
(b) the majority of the new jobs being created requires knowledge other than that gained from earning a college degree
(c) a job as a home health aide will rely more on communication skills than on a basic skills in reading and mathematics
(d) if a job is one of the fastest growing jobs, it will require a college degree

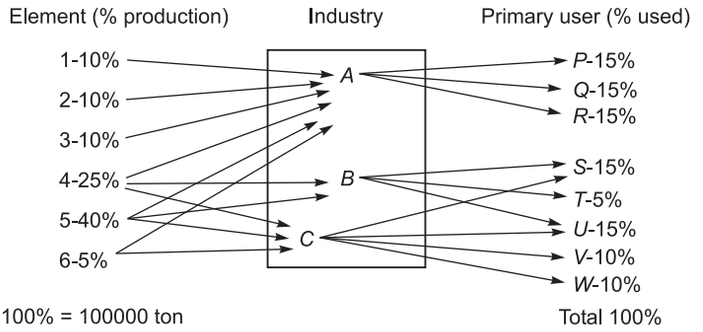
43. According to the National Agricultural Aviation Society (NAAS), without the use of crop protection products to control insects, weeds, and diseases, crop yields per acre will drop by more than 50%. The first aerial application of insecticide occurred in 1921, and it was a huge success. By contrast, in today's economy all aircraft that are classified as aerial applicators do more than just apply insecticide; today, they also spread sees and apply fertilizer.
From the information given above it cannot be validly concluded that
(a) according to the NAAS, if crop yields per acre never drop by more than 50%, then crop protection products have been used to control insects, weeds, and diseases
(b) in today's economy, any aircraft that cannot be used to apply fertilizer cannot be classified as an aerial applicator
(c) in today's economy, if an aerial applicator is used, then it will be able to spread seed and to apply fertilizer
(d) according to NAAS, if crop yields per acre drop by more than 50%, then crop protection products have not been used to control insects, weeds, and diseases

44. Lou observes that if flight 409 is cancelled, then the manager could not possibly arrive in time for the meeting. But the flight was not cancelled. Therefore, Lou concludes, the manager will certainly be on time. Evelyn replies that even if Lou's premises are true, his argument is fallacious. And therefore, she adds, the manager will not arrive on time after all.

Which of the following is the strongest thing that we can properly say about this discussion?
(a) Evelyn is mistaken in thinking Lou's argument to be fallacious, and so her own conclusion is unwarranted
(b) Evelyn is right about Lou's argument, but nevertheless her own conclusion is unwarranted
(c) Since Evelyn is right about Lou's argument, her own conclusion is well supported
(d) Since Evelyn is mistaken about Lou's argument, her own conclusion must be false

45. Cars are safer than planes. 50% of plane accidents result in death, while only 1% of car accidents result in death.
Which of the following, if true, would most seriously weaken the argument above?
(a) Planes are inspected more often than cars
(b) The number of car accidents is several hundred thousand times higher than the number of plane accidents
(c) Pilots never fly under the influence of alcohol, while car drivers often do
(d) Plane accidents are usually the fault of air traffic controllers, not pilots

Directions (Q. Nos. 46 to 48) *Refer to the chart below showing annual production and answer the questions that follow.*



46. Which industry/industries contributes/contribute to Company *S*?
(a) Industries *A* and *B* (b) *B* and *C* (c) *A* and *C* (d) *B* only

47. Industry *B* processes what percentage of the total production of listed elements?
(a) 25% (b) 65% (c) 40% (d) Cannot be determined

48. Of the listed elements processed by the Industry *A*, how many tons are produced annually?
(a) 30000 (b) 50000 (c) 100000 (d) 55000

Directions (Q. Nos. 49 to 52) *In each of the following questions there are two blanks marked I and II. The words to fill in these blanks are given against I as (A, B, C, D) and II as (P, Q, R, S) respectively. The right words to fill in these blanks are given as four alternatives. The words on either side of the sign (::) have a similar relationship. That alternative which signifies this relationship is your answer.*

49. I : Increase :: Descend : II

I. (A) Grow	(B) Ascend	(C) Rise	(D) Price
II. (P) Reduce	(Q) Down	(R) Decrease	(S) Mountain
(a) AR	(b) RB	(c) CP	(d) DQ

50. Modern I :: II : Old

I. (A) Ancient	(B) Death	(C) Famous	(D) Civilization
II. (P) Industrialisation	(Q) Young	(R) Fashion	(S) Western
(a) AQ	(b) AS	(c) BP	(d) CR

51. Part : I :: Class : II

I. (A) Section	(B) Whole	(C) School	(D) Students
II. (P) Student	(Q) School	(R) Teachers	(S) Rooms
(a) AR	(b) BQ	(c) CP	(d) DS

52. Summit : Apex :: I : II

I. (A) Beautiful	(B) Picture	(C) Attractive	(D) Enchanting
II. (P) Comfortable	(Q) Pretty	(R) Healthy	(S) Brave
(a) AQ	(b) BP	(c) DS	(d) CR

53. If in a certain code, BEAUTIFUL is coded as 573041208, BUTTER as 504479, how is FUTURE coded in that code?

(a) 201497 (b) 204097 (c) 704092 (d) 204079

54. How many Mondays are there in a particular month of a particular year, if the month ends on Wednesday?

(a) 4 (b) 5 (c) 3 (d) Cannot be specified

Directions (Q. Nos. 55 and 56) *Based on the information given below.*

There are two rows of numbers in each questions. The upper row is complete and in the lower row one number is missing. Find a suitable number to fill the blank space so that the symmetry with the upper row is maintained.

55.

17	102	12
15	?	10

(a) 211 (b) 75 (c) 125 (d) 117

56.

12	336	14
15	?	16

(a) 220 (b) 480 (c) 125 (d) 450

Directions (Q. Nos. 57 and 58) *Based on the information given below.*

Seven people *A, B, C, D, E, F, G* are planning to enjoy boating. There are only two boats, and the following conditions are to be kept in mind.

(I) *A* will go in the same boat in which *E* is to go.

(II) *F* cannot go in the boat in which *C* is, unless *D* is also accompanying.

(III) Neither *B* nor *C* can be given the boat in which *G* is.

(IV) The maximum number of persons in one boat can be four only.

57. If *F* and *B* are in one boat, which of the following statements is true?

(a) *G* is in the other boat (b) *D* is in the other boat
(c) *C* is in the other boat (d) *E* is with *F* and *B* in one boat

58. If *E* gets boat with *F*, which of the following is the complete and accurate list of the people who must be sitting in other boat?

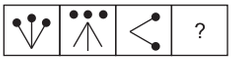
(a) *F* and *E* (b) *G* and *A* (c) *D* and *A* (d) *C, D* and *B*

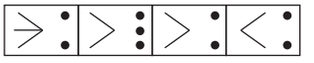
Directions (Q. Nos. 59 and 60) *A series of figures has been shown on the left. Find the figure in the place of ? (question mark) from the figures on the right.*

59.

(a) (b) (c) (d)

60.





(a) (b) (c) (d)

61. Complete the following series.

(*GMSY, IOVA, KQWC,* ?)

(a) *MSYE* (b) *NSYE* (c) *MYTE* (d) *MSYF*

Directions (Q. Nos. 62 to 65) *Based on the information given below.*

Following is a new plan called 'Airtel One', introduced by Airtel (GSM service provider) for its post-paid customers.

Pulse Rate	60 sec		
Price of Plan	₹ 99		
Free Airtime on Plan	Nil		
Incoming Calls (₹)	Free		
Outgoing Calls (₹/min)	Airtel	GSM/CDMA (10 Digit)	Landline/WLL
Local Rates	₹ 1	₹ 1	₹ 2
STD rates			
50-200 km	₹ 1.50	₹ 1.50	₹ 2
200-500 km	₹ 1.50	₹ 2.50	₹ 2.50
500 + km	₹ 1.50	₹ 3.00	₹ 3.50
ISD			
USA, Canada, Europe (Fixed Line)	₹ 7		
Gulf, Europe (Mobile) SAARC	₹ 10		
Rest of the World	₹ 40		
SMS			
Local	₹ 1.50		
National	₹ 2.00		
International	₹ 5.00		
Value Added Service (₹)	₹ 3.00 and ₹ 6.00 Depending on the service		
Monthly bill = Call charges + SMS charges + Price of plan			

62. If Mohan made STD calls (within 100 km) for 30 min and spent 100 min on local calls with 30% on landline, 40% on GSM and 30% on Airtel, and Rohan spent 18 min on STD within 150 km and spent 120 min on local calls with 30%, 40% and 30% on GSM, landline and Airtel respectively, then who spent more?

(a) Mohan
(b) Rohan
(c) Both spent the same amount
(d) Cannot be determined

63. A new SMS scheme was introduced @60p/local SMS, with an additional monthly charge of ₹ 35. Who will not benefit from the scheme? A person sending

(a) 38 local SMS a month
(b) 40 local SMS a month
(c) 60 local SMS a month
(d) 59 local SMS a month

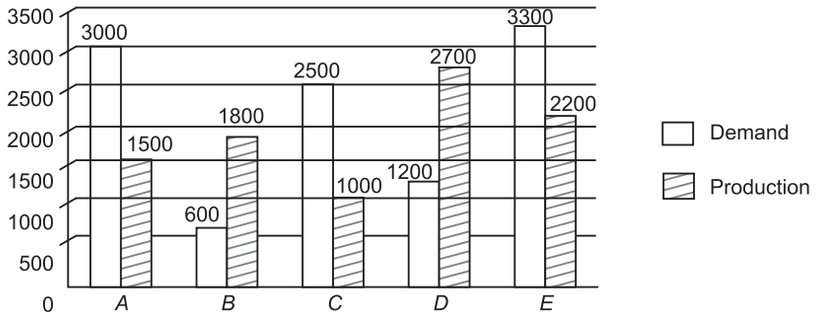
64. A bill of ₹ 199/Month is definitely not possible when you make only

(a) 30 calls to Airtel on STD (500+km) and 55 local calls in a month
(b) 16 calls to Airtel on STD (15 km) and 76 local calls in a month
(c) 10 STD calls (250 km) each to Airtel, GSM and landline respectively and 30 local calls in a month
(d) 8, 4 and 7 calls to Airtel, GSM and landline respectively on STD (500 + km) and 55 local calls in a month

65. A person makes 12 min of ISD calls in a month. If 80% of his ISD bill came from calling the rest of the World, then the ISD calls for the minimum duration were made to which of the following groups of countries?

(a) USA, Canada, Europe (Fixed line)
(b) Gulf, Europe (Mobile), SAARC
(c) Rest of the World
(d) Cannot be determined

Directions (Q. Nos. 66 and 67) *These questions are based on the graph which shows the demand and production statistics of* 5 *TV companies.*



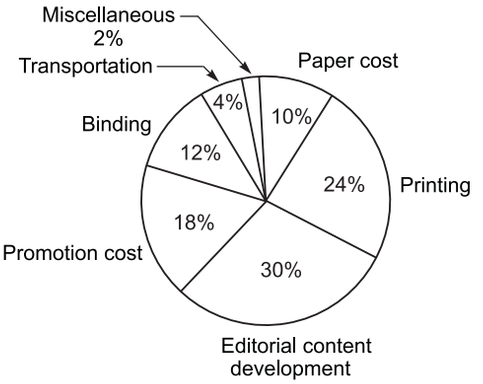
66. What is the difference between the average demand and the average production of the five companies taken together?

(a) 1400 (b) 400 (c) 280 (d) 138

67. The production of the Company *D* is how many times of the production of the Company *A*?

(a) 1.8 (b) 1.5 (c) 2.5 (d) 1.11

Directions (Q. Nos. 68 and 69) *The following questions are based on the pie-chart given below which gives the expenditure incurred in printing a magazine.*



68. What is the angle for the sector representing paper cost?

(a) 10° (b) 36° (c) 23.5° (d) 45°

69. For a given issue of the magazine, the 2% of the cost is ₹ 2000 and the print-run is 12500 copies. What should be the sale price, if the publisher desires a profit of 5%?

(a) ₹ 5 (b) ₹ 7.5 (c) ₹ 8 (d) ₹ 8.40

70. If 'water' is called 'food', 'food' is called 'tree', 'tree' is called 'sky', 'sky' is called 'wall', on which of the following does a 'fruit' grow?

(a) Water (b) Food (c) Tree (d) Sky

Section III (General English)

For all questions in this section, correct answer carries 1 mark each.

Directions (Q. Nos. 71 and 72) *Choose the correct option to fill the blank for correct grammatical use.*

71. She is a close friend of …… .
(a) my (b) I (c) me (d) mine

72. He took care of her …… she was restored to health.
(a) when (b) before (c) after (d) till

73. Choose the correct option : The man …… his appearance completely since then.
(a) is changing (b) changed (c) has changed (d) is changed

74. From the options below, choose the word with the incorrect spelling.
(a) Diarrhea (b) Diaper (c) Dichotomy (d) Dias

75. Choose the word spelt correctly.
(a) Superintendant (b) Seive (c) Alloted (d) Dissipate

Directions (Q. Nos. 76 to 79) *Fill in the blanks with the correct pair of words.*

76. He lives …… Bengaluru …… 115, Richmond Road.
(a) at, in (b) in, at (c) at, on (d) in, on

77. 'Aurally challenged' is a …… for the …… .
(a) metaphor, blind (b) euphemism, deaf (c) simile, disabled (d) synonym, dumb

78. If error is a ……, fault is …… .
(a) defect, a mistake (b) mistake a defect (c) slip, an inaccuracy (d) blunder, to blame

79. The man has …… the rules of ethical conduct; he is …… a beast.
(a) flaunted, literally (b) ignored, basically (c) broken, as (d) flouted, virtually

Directions (Q. Nos. 80 and 81) *Which punctuation mark is missing in the following sentences?*
(a) Inverted commas (b) Semicolon (c) Comma (d) Hyphen

80. Part of Australia is known to the natives as The Outback.

81. I know that you want to learn to drive Rima but you are too young.

Directions (Q. Nos. 82 and 83) *Select as option the word closest in meaning to the given word.*

82. Veracious
(a) False (b) Varied (c) Image (d) Truthful

83. Perturb
(a) Stipulate (b) Turn around (c) Disturb greatly (d) Compatible

Directions (Q. Nos. 84 and 85) *Choose the option that is the closest in meaning to the phrases in bold.*

84. He was a king who ruled his subjects **with a high hand.**
(a) sympathetically (b) oppressively (c) democratically (d) generously

85. **I racked my brains** to solve this difficult problem.

(a) I read a large number of books
(b) I consulted several people
(c) I used my commonsense
(d) I subjected my mind to hard thinking

Directions (Q. Nos. 86 and 87) *Select the most appropriate meaning of the underlined idiomatic phrase.*

86. I was supposed to give a speech to my English class, but I got cold feet and I didn't go.

(a) My English class-room makes my feet feel too cold
(b) My feet got cold so I didn't go
(c) I got too nervous and I didn't go
(d) It was winter time and I had no socks for my feet

87. Take care of what you say! You will have to eat your words!

(a) You have no food to eat
(b) You will have to take back what you have said
(c) You are not good with your language
(d) None of these

88. Which two sentences convey the same idea?

1. Shouldn't they have checked your tickets?
2. I wonder if they should have checked your tickets.
3. I want to know if they checked your tickets.
4. They should have checked your tickets.

(a) 2, 3 (b) 1, 2 (c) 1, 4 (d) 2, 4

Directions (Q. Nos. 89 and 90) *Choose a single word which is opposed to the meaning contained in the sentence.*

89. He was magnanimous and his benevolence made him give to charity.

(a) Mean (b) Cruel (c) Snobbish (d) Tyrannical

90. Biannual is

(a) once in two years (b) every year (c) twice a year (d) after every two years

91. Choose the odd one out.

(a) Temporal (b) Ephemeral (c) Transient (d) Eternal

92. Choose the correct option.

If 'inter' mean; between as in interstate, 'intra' as in intravenous means …… .

(a) into (b) onto (c) within (d) without

93. Match the parts of speech (in bold) to their usage (down).

1. Adjective
2. Preposition
3. Noun
4. Verb

5. The fire engine came rushing **down** the hill.
6. He has seen the ups and **downs** of life.
7. **Down** with the tyrant!
8. The porter was hit by the **down** train.

(a) 1-8, 2-5, 3-6, 4-7 (b) 1-6, 2-5, 3-8, 4-7 (c) 1-5, 2-6, 3-7, 4-8 (d) 1-5, 2-7, 3-6, 4-8

Directions (Q. No. 94) *Match the items in Column I with those in Column II. Choose the correct answer combination given below.*

94. Match the following.

	Column I		Column II
1.	As deaf as	5.	gall
2.	As bitter as	6.	an eel
3.	As unpredictable as	7.	a post
4.	As slippery as	8.	the weather

(a) 1-7, 2-6, 3-8, 4-5 (b) 1-7, 2-5, 3-8, 4-6 (c) 1-8, 2-6, 3-5, 4-7 (d) 1-5, 2-7, 3-6, 4-8

Directions (Q. No. 95) *Match the parts of speech (in bold) to their usage.*

95. Match of following.

1. Adjective	5. Rain comes from **above**.
2. Adverb	6. His conduct is **above** suspicion.
3. Noun	7. Look **above** the mantel piece.
4. Preposition	8. The **above** information is for the public.

(a) 1-8, 2-7, 3-5, 4-6 (b) 1-7, 2-6, 3-5, 4-8
(c) 1-8, 2-5, 3-6, 4-7 (d) 1-6, 2-8, 3-5, 4-7

Directions (Q. Nos. 96 and 97) *Choose the segment with the error. If there is no error, choose* (d).

96. Idli and Sambar (a) make the breakfast (b) in the South. (c) (d)

97. Your account should (a) have been credited (b) with three months' interest. (c) (d)

98. Choose the sentence where the underlined word is used appropriately.
(a) Who's car are you planning to borrow?
(b) Whose planning to have a birthday party?
(c) He is the person who you met at the junction.
(d) Does he know whom to call about the hotel reservations?

Directions (Q. Nos. 99 and 100) *Choose:*

(a) if the first sentence is correct (b) if the second sentence is correct
(c) if both sentences are correct (d) if both sentences are wrong

99. 1. Can you imagine his forgetting his own birthday?
2. Can you imagine him forgetting his own birthday?

100. 1. Recently I read about a unique wedding that took place in the newspaper.
2. Recently I read in the newspaper about a unique wedding that took place.

101. Complete the given sentence by choosing the correct phrase.
You cannot succeed unless
(a) you do not work hard (b) you shall not work hard
(c) you will work hard (d) you work hard

102. Rearrange the scrambled segments in logical order to make a complete sentence.
1. to places where more opportunities are available
2. and so there is a great demand for English
3. for professional and economic growth
4. because it takes one outside one's own community
5. English is the language of opportunities

(a) 1, 2, 5, 3, 4 (b) 5, 4, 1, 3, 2
(c) 3, 1, 4, 2, 5 (d) 2, 1, 3, 5, 4

103. Choose the best construction.
(a) If you would have taken care you wouldn't have got typhoid.
(b) If you took care you wouldn't have got typhoid.
(c) If you take care you wouldn't have got typhoid.
(d) If you had taken care you wouldn't have got typhoid.

Directions (Q. Nos. 104 and 105) *Choose the correct option.*

104. At times, we are all …… to be mistaken.

(a) apt
(b) likely
(c) considered
(d) able

105. He hardly cares, …… ?

(a) does he
(b) doesn't he
(c) will he
(d) won't he

Directions (Q. Nos. 106 and 107) *Identify the figures of speech in the following sentences.*

106. As proud as a peacock.

(a) Metaphor
(b) Simile
(c) Apostrophe
(d) Epigram

107. Death lays his icy hand on kings.

(a) Personification
(b) Exclamation
(c) Simile
(d) Anticlimax

Directions (Q. Nos. 108 to 110) *Read the passage carefully and answer within the context.*

"A way to deal with Frozen Feelings"

Every child experiences all that happens around him with total awareness. In the first seven years the child's brain is like a sponge, taking in all sensory inputs and building his idea of his surroundings. As long as the environment is safe, the child learns with incredible speed. However, when the environment is scary or stressful, the child unlearns past learning just as rapidly.

In the early years of every child's life, whenever there is shock, violence, fear or pain, these intense emotions are imprinted deeply into memory. Whenever the same activity or situation is repeated, the nervous system and body subconsciously re-experience the memory of the trauma.

Any emotional situation that takes us out of the present and into the past means that whenever the same kind of emotion crops up later in our life we return to the past for our reference point. If that point was at age three, we find ourselves behaving like a three year old. We feel childish and we behave childishly. Our feelings are the cause of this 'glitch' in our learning process. We know we should be able to make a positive change but that doesn't change anything.

The process of change needs not be traumatic. We couldn't have done any better because we didn't know how to. But we should realize that was then and this is now! we can choose to choose again. It's up to us. It's our movie!

108. The "Frozen Feelings' being talked about are about

(a) negative childhood experiences
(b) childhood learning patterns
(c) inability to learn as an adult
(d) None of these

109. A 'glitch' is

(a) a ditch
(b) uneasy emotions
(c) sudden malfunction or breakdown
(d) learning patterns

110. Identify the correct statement, based on the paragraph.

(a) The process of change needs to be traumatic.
(b) We feel childish and we behave childishly.
(c) Both sentences are incorrect.
(d) Both sentences are correct.

Section IV (General Awareness)

For all questions in this section, correct answer carries 1 mark each.

111. Dendrochronology is
(a) the method of scientific dating based on the analysis of tree-ring growth patterns
(b) a core sample from the accumulation of snow and ice over many years that have recrystallized and have trapped air bubbles from previous time periods
(c) the study of glaciers or more generally ice and natural phenomena that involves ice
(d) the scientific study of landforms and the processes that shape them

112. Which company launched the first mobile phone operate on in India?
(a) Bharti Airtel (b) Essar (c) Max Touch (d) Modi Telstra

113. Which is the longest running English language TV cartoon in history, first launched by Warner Brothers in 1969 exclusively for television, and is now a popular fare on the Cartoon Network?
(a) Scooby Doo (b) Tom and Jerry (c) Popeye (d) Johnny Bravo

114. The expansion for BIFR, in the context of the indian Industry is
(a) Board for Industrial and Financial Reconstruction (b) Bureau for Industrial and Financial Reconstruction
(c) Board for Investment and Financial Reconstruction (d) Bureau for Investment and Financial Reconstruction

115. What is Ekistics?
(a) The water sports related study (b) The science of body mass
(c) The art of card tricks (d) The science of human settlements

116. What is Red Herring in an IPO?
(a) Prospectus (b) Submission of form
(c) Funds generated during IPO (d) Minimum offer per share

117. Eight O'clock Coffee, a US based coffee marketer, is a group company of
(a) Tata (b) ITC (c) Britannia (d) Unilever

118. Bio-diesel is extracted from the following plant
(a) Hibiscus (b) Aloe Vera (c) Jatropha (d) Chamomile

119. Which woman tennis player won the maximum number of Grand Slam singles titles in the history of tennis?
(a) Steffi Graf (b) Martina Navratilova (c) Billi Jean King (d) Margaret Court

120. What is the nickname of New York city?
(a) Big Sweet Lime (b) Big Apple (c) Big Grape (d) None of these

121. Which was the first Indian company to be listed on NASDAQ?
(a) TCS (b) Wipro (c) Infosys (d) Cognizant

122. "TAREGNA"—a village in Bihar was in the news because of
(a) Rahul Gandhi's election campaign (b) viewing of the solar eclipse
(c) a visit by Hillary Clinton (d) the Naxal massacre of tribals

123. The driver for Formula One's Force India Team is
(a) Mark Webber (b) Adrian Sutil (c) Felipe Massa (d) Jensen Button

124. (MDRT)–Million Dollar Round Table consists of
(a) World's richest industrialists (b) World's most powerful golfers
(c) Global Association of Life Insurance professionals (d) World's richest sports persons

125. Who declares the "Credit Policy" of India?
(a) The Finance Ministry
(b) Reserve Bank of India
(c) The Commerce Ministry
(d) The Indian Banks' Association (IBA)

126. Name of the body which regulates the Stock Exchanges in India is
(a) CBI (b) CID (c) SHCIL (d) SEBI

127. Rafflesia arnoldii is its botanical name and it holds a World record in the plant World. What's the record?
(a) Tallest cacti
(b) Largest bloom
(c) Longest surviving tree
(d) Smallest plant

128. Temujin was a famous historical character of the World but we know him by a different name, which one?
(a) Babur (b) Genghis Khan (c) Kublai Khan (d) Emperor Nero

129. What is *Short Selling* in stock market lingo?
(a) Selling the shares which you do not own
(b) Selling a share after owning it for a short period of time
(c) Selling all the shares in your booty which are not productive
(d) Selling of shares which are quoted for a short period of time

130. Mahatma Gandhi was nominated for Nobel Peace prize for the first time in the year 1937 and last time in the year 1948. How many times in between these two years was he nominated?
(a) 1 (b) 3 (c) 2 (d) 0

131. Who Owns the ill-fated Bhopal plant of Union Carbaid, now?
(a) Daewoo (b) Du Pont (c) Dow Chemical (d) None of these

132. Harpo is a magazine published by
(a) Oprah Winfrey (b) Michelle Obama (c) Sudha Murthy (d) None of these

133. "We have learned that if we provide people with an occasion and an excuse to shop, they will come," are the golden words of whom?
(a) Mukesh Ambani
(b) Kishore Biyani
(c) Sam Walton of Wal-Mart
(d) Kumar Mangalam Birla

134. Which Indian biscuit brand is the single largest selling biscuit in the world in its segment?
(a) Parle G (b) Priya Gold (c) Tiger (d) Hide and Seek

135. Which book, written by Rashmi Bansal, profiles the story of 25 IIM-Ahmedabad graduates?
(a) Stay Hungry Stay Foolish
(b) The Elephant, the Tiger and the Cell Phone
(c) The Five Dollar Smile
(d) None of these

136. Who has been named as the voice and Data telecom person of the year?
(a) Manoj Kohli (b) Sunil Bharti Mittal (c) SP Shukla (d) Vikram Mehmi

137. How 'the mighty fall' and 'why some companies never give in' has been authored by
(a) Brian Tracy (b) Ram Charan (c) Jim Collins (d) CK Prahalad

138. Which economist came up with the idea of the 'Invisible Hand'
(a) Adam Smith (b) Keynes (c) Amartya Sen (d) Ricardo

139. Find the odd one out channel owned by
(a) TV 18- Raghav Behl
(b) NDTV- Prannoy Roy
(c) Sun TV- Rajnikant
(d) Zee TV- Subhash Chandra

140. LAB 126 is a subsidiary of a well known dot com. It designs and market a gadget, a rage in US. Name the gadget.
(a) Kindle ebook reader (b) iphone (c) iPOD (d) Zozo

141. 'Let's begin' is the new tagline of
(a) Genpact (b) Corus (c) Essar (d) Hockey India

142. Name the real estate of Dubai World which is behind Dubai's sovereign default scare.
(a) Khalifa (b) Nabeel (c) Nakheel (d) Telenor

143. Which car showed the fastest growth in sales during Q3-09 in India?
(a) Santro (b) Alto (c) Wagon R (d) Ambassador

144. Aegis is the BPO arm of
(a) AV Birla group (b) Essar (c) Bharti Enterprises (d) Infosys

145. Name the company that has selected Infosys, TCS, Wipro and IBM for its application development and application maintenance contracts.
(a) British Petroleum (b) Shell (c) BMW (d) Daimler Benz

146. In relation with the RBI, NDS is an electronic platform for facilitating dealing in Government Securities and Money Market Instruments. NDS Stands for
(a) National Depository System (b) Negotiated Dealing System
(c) National Dealing System (d) Negotiable Deals System

147. In the context of Mutual Funds, ELSS means
(a) Endowment Linked Savings Scheme (b) Equity Linked Savings Scheme
(c) Established Line Savings Scheme (d) Entrepreneur Line Savings Scheme

148. Which is the famous product manufactured by the American company Anheuser Busch?
(a) Coca Cola (b) Budweiser beer (c) Cadburys chocolates (d) None of these

149. Which company is the world's largest, based on its composite score of sales, profits, assets and market value?
(a) General Motors (b) General Electric (c) Microsoft (d) Wal-Mart

150. Charagh Din is a famous brand of
(a) Electricity lamps (b) Shirts (c) Sunscreens (d) biscuits

Answers with Explanations

1. (*b*) The sum of numbers can be 15 in the following three ways.

***Case* (i)** $15 = 3 + 6 + 6$

The first, second and third throws can be (3, 6, 6), (6, 3, 6) and (6, 6, 3) respectively.

$\therefore$ Total number of ways in which 3, 6 and 6 can be obtained is 3.

***Case* (ii)** $15 = 4 + 5 + 6$

The first, second and third throws can be either of 4, 5 and 6.

$\therefore$ Total number of ways in which 4, 5 and 6 can be obtained is 6.

***Case* (iii)** $15 = 5 + 5 + 5$

The first, second and third throws can be 5, 5 and 5.

$\therefore$ Total number of ways in which 5, 5 and 5 can be obtained is 1.

$\therefore$ The total number of ways in which the sum of throws can be 15 is

$$3 + 6 + 1 = 10$$

The total number of ways in which the first roll will be a 4 is 2.

$$\therefore \quad \text{Required chance} = \frac{2}{10} = \frac{1}{5}$$

Hence, option (b).

2. (*) Let b and s be the speeds of the boat and stream respectively.

As per the given conditions,

$$\frac{30}{b+s} = 2 \qquad \ldots(i)$$

and $$\frac{30}{b-s} = 6 \qquad \ldots(ii)$$

Dividing Eqs. (i) and (ii),

$$\frac{b-s}{b+s} = \frac{1}{3}$$

$$\therefore \quad b = 25$$

Substituting in Eq. (i),

$$\frac{30}{2s-s} = 2$$

$$\therefore \quad s = 5$$

$$\therefore \quad b = 10 \text{ km/h}$$

The correct answer is not there in the options.

3. (*a*) A number divisible by 4 formed using the digits 1, 2, 3, 4 and 5 has to have the last two digits 12 or 24 or 32 or 52.

In each of these cases, the five-digit number can be formed using the remaining 3 digits in

$$3 \times 2 \times 1 = 6 \text{ ways}$$

$\therefore$ A number divisible by 4 can be formed in

$$6 \times 4 = 24 \text{ ways}$$

Total numbers that can be formed using the digits 1, 2, 3, 4 and 5 without repetitions $= 5! = 120$

$$\therefore \text{ Required probability} = \frac{24}{120} = \frac{1}{5}$$

Hence, option (a).

4. (*d*) Let a, b, c, d, e and f be six numbers, such that

$$a = e + m$$
$$b = e - k$$
$$c = e - l$$
$$d = e - j$$
$$f = e + p$$

Then, the sum of deviations of a, b, c, d, e and f from e is $(m - k + l - j + p)$

The mean of a, b, c, d, e and f is given by

$$\frac{a+b+c+d+e+f}{6}$$

$$= \frac{e+m+e-k+e-l+e-j+e+e+p}{6}$$

$$= \frac{6e + (m-k-l-j+p)}{6}$$

$$= \frac{6e + \text{Sum of deviations}}{6}$$

Using the similar logic, the required mean

$$= \frac{20 \times 23 + 70}{20} = 26.5$$

Hence, option (d).

5. (*a*) Let x gram of gold be added to the alloy to increase the percentage of gold in it to 90%.

$$\therefore \quad \frac{80\% \text{ of } (50+x)}{50+x} = \frac{90}{100}$$

$$\therefore \quad \frac{40+x}{50+x} = \frac{90}{100}$$

$$\therefore \quad x = 50 \text{ g}$$

Hence, option (a).

6. (*c*) Let the incomes be $7x$ and $3x$ and let the expenses be $5y$ and $2y$.

$$\therefore \quad 7x - 5y = 300 \qquad \ldots(i)$$

and $$3x - 2y = 300 \qquad \ldots(ii)$$

Solving Eqs. (i) and (ii) simultaneously, we get

$$x = 900 \quad \text{and} \quad y = 1200$$

$\therefore$ Income of first person $= 7x = 7 \times 900 =$ ₹ 6300

Hence, option (c).

7. (*d*) Assume the family consumes 1 kg usually.

To keep expenditure at ₹ 24, its new consumption should be $\frac{24}{27} = \frac{8}{9}$ kg

$\therefore$ Percentage decrease in consumption

$$= \frac{\left(1 - \frac{8}{9}\right)}{1} \times 100 = 11.1\%$$

Hence, option (d).

8. (*b*) From a certain station, there will be a ticket for each of the other 9 stations and there are 10 stations on the railway line.

$\therefore$ The number of different journey tickets

$= 9 \times 10 = 90$

Hence, option (b).

9. (*b*) Circumference of a circle $= 2\pi r$, where r is the radius.

Circumference $\propto r$

As the circumference increases by 5%, the radius also increases by 5%.

$\therefore$ New radius $= 1.05\, r$

$\therefore$ As area $\propto$ (radius)^2

$\therefore$ New area $= (1.05)^2 \times$ Old area

$= 1.025 \times$ Old area

$\therefore$ Percentage increase in area $= 10.25\%$

Hence, option (b).

10. (*d*) The three vowels in *ABACUS* are *A*, *A* and *U*. These three can be arranged among themselves in $\frac{3!}{2!} = 3$ ways.

As the three vowels are to appear together, we consider them as one entity. Thus we have four letters; (*AAU*), *B*, *C* and *S* to be arranged.

This can be done in 4! ways.

$\therefore$ Required number of ways $= \frac{4! \times 3!}{2}$

Hence, option (d).

11. (*c*) We can see that $\frac{8000}{6000} = \frac{700}{525} = \frac{4}{3}$

$\therefore$ The time that 525 requires to grow to 700 will be the same as the time that 6000 requires to grow to 8000.

$\therefore$ Required number of years = 4

Hence, option (c).

12. (*d*) At the start of the game *J* and *B* together had 4 times the money that *T* had.

Let the total money be *m*.

Then, $j + b + t = 4t + t = 5t = m$

$\therefore$ The total money is divisible by 5.

Similarly, from the other three statements in the data, we find that the total money is divisible by 4 and 3 as well.

$\therefore$ The total money is divisible by

$4 \times 3 \times 5 = 60$

Let the total money be $60y$.

At the start of the game,

$j + b = 4t$

$\therefore \quad j + b + t = 5t = 60y$

$\therefore \quad t = 12y$

$\therefore \quad j + b = 48y \quad \ldots(i)$

Also, $t + b = 3j$

$\therefore \quad 4j = 60y$

$\therefore \quad j = 15y$

Substituting in Eq. (i),

$15y + b = 48y$

$\therefore \quad b = 33y$

$\therefore$ The amounts with *J*, *B* and *T* at the start were $15y$, $33y$ and $12y$ respectively.

Now, at the end of the game,

$j + b = 3t$

$\therefore \quad 4t = 60y$

$\therefore \quad t = 15y$

$\therefore \quad j + b = 45y \quad \ldots(ii)$

Also, $t + b = 2j$

$\therefore \quad 3j = 60y$

$\therefore \quad j = 20y$

Substituting in Eq. (ii), we get

$b = 25y$

$\therefore$ The amounts with *J*, *B* and *T* at the end were $20y$, $25y$ and $15y$ respectively.

$\therefore$ The fraction of total money that *T* had at the beginning was $\frac{12y}{60y} = \frac{1}{5}$

Hence, option (d).

13. (*a*) *J* won $20y - 15y = 5y$

$\therefore$ Required fraction $= \frac{5y}{60} = \frac{1}{12}$

Hence, option (a).

14. (*c*) As *B* lost ₹ 200,

$\therefore \quad 33y - 25y = 200$

$\therefore \quad 8y = 200$

$\therefore \quad y = 25$

$\therefore \quad 33y = 825$

$\therefore$ *B* started with ₹ 825.

Hence, option (c).

15. (*d*) Here *a* and *b* are negative and *c* is positive.

Let $a = -4$, $b = -3$ and $c = 2$

(I) $a - b = -4 + 3 = -1$

and $a - c = -4 - 2 = -6$

So, $a - b > a - c$

So, I is not true.

(II) Taking the same values of *a*, *b* and *c*, we have

$\frac{a}{c} = -\frac{4}{2} = -2$

$\frac{b}{c} = -\frac{3}{2} = -1.5$

Since $-2, -1.5, \frac{a}{c} < \frac{b}{c}$

So, II is true.

(III) For any values of *b* and *c*,

$\frac{1}{b} < \frac{1}{c}$ since *b* is negative and *c* is positive.

So, III is true.

So, statements II and III are true.

Hence, option (d).

16. (*a*) Let the length of the side of the original square be *x*.

Then , the length of the diagonal

$= \sqrt{x^2 + x^2} = 4\sqrt{2}$

$\therefore \quad 2x^2 = 32$

$\therefore \quad x^2 = 16$

$\therefore \quad x = 4$

Now, the area of the other square is twice that of the first square.

So, the area of the second square $= 2x^2 = 32$

If the length of the second square is y, then

$$y^2 = 32$$

$\therefore$ Length of the diagonal of this square

$$= \sqrt{2y^2} = \sqrt{64} = 8 \text{ cm}$$

Hence, option (a).

17. (c) Let the length, breadth and height of the rectangular box be l, b and h respectively. Then the maximum length of the pencil that can be accommodated in the box will be equal to the length of its greatest diagonal, which is the body diagonal.

This length is given as $\sqrt{l^2 + b^2 + h^2}$.

Dimensions of the box are given as $8 \times 6 \times 2$.

So, the maximum length of the pencil that can be kept in the box $= \sqrt{8^2 + 6^2 + 2^2} = \sqrt{104} = 2\sqrt{26}$ cm

Hence, option (c).

18. (c) Total readership of X is 8.7 lakh.

$\therefore$ $X + 2.5 + 0.5 + 1 = 8.7$

$\therefore$ $X = 4.7$

Also, total readership of Y is 9.1 lakh.

$\therefore$ $2.5 + 0.5 + 1.5 + Y = 9.1$

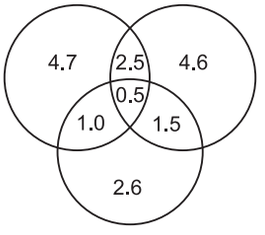
$\therefore$ $Y = 4.6$

Also, total readership of Z is 5.6 lakh.

$\therefore$ $Z + 1 + 0.5 + 1.5 = 5.6$

$\therefore$ $Z = 2.6$

Therefore, we have



$\therefore$ Number of people who read at least one paper

$= 4.7 + 4.6 + 2.6 + 1 + 1.5 + 0.5 + 2.5 = 17.4$

Hence, option (c).

19. (b) From the solution to the previous question,

Number of people who read only one newspaper

$$= 4.6 + 4.7 + 2.6 = 11.9$$

Hence, option (b).

20. (a) Percentage of Sonali's salary invested in insurance policies = 15%

Percentage of salary spent in shopping and household expenses = 55%

$\therefore$ Percentage of salary saved

$$= 100 - (15 + 55)$$

$$= 30\%$$

Let Sonali's monthly salary be x.

$\therefore$ 30% of $x = 12750$

$$x = \frac{12750 \times 100}{30}$$

$$= ₹\ 42500$$

Hence, option (a).

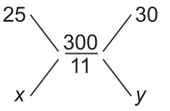
21. (a) There are two varieties of tea, one worth ₹ 25 per kg and the other worth ₹ 30 per kg.

Now, by selling the blended variety at ₹ 30 per kg, profit should be 10%.

So, the cost price of the blended tea should be

$$\frac{30}{1.1} = ₹\ \frac{300}{11} \text{ kg}$$

This problem can be depicted through alligation as below



We get $\frac{x}{y} = \frac{30}{25}$

Amount of tea of ₹ 30 per kg to be used is 30 kg. If the amount of tea worth ₹ 25 per kg is a.

$\therefore$ $\frac{30}{25} = \frac{a}{30}$

$\therefore$ $a = \frac{30 \times 30}{25} = 36$ kg

Hence, option (a).

22. (*) Of the total 80 students, let there be b boys and g girls.

$\therefore$ $b + g = 80$ ...(i)

Total pass percentage was 75%.

So, number of students who passed

$$= 0.75 \times 80 = 60$$

Now, it is given that 85% of girls and 70% of boys passed.

$\therefore$ $0.7b + 0.85g = 60$ ...(ii)

Solving Eqs. (i) and (ii), we get

$$g = 26.67 \text{ and } b = 53.33$$

None of the options has the correct answer.

23. (b) 300 g solution has 40% salt in it.

So, the amount of salt in it

$$= \frac{40 \times 300}{100} = 120 \text{ g}$$

For the salt percentage to be 50%,

$$\frac{120 + x}{300 + x} = \frac{1}{2}$$

$\therefore$ $240 + 2x = 300 + x$

$\therefore$ $x = 60$ g

Hence, option (b).

24. (d) Here, each number in the second row is obtained by multiplying the corresponding number of the first row by 8.

$\therefore$ The number that should replace the question mark

$$= \frac{192}{8} = 24$$

Hence, option (d).

25. (b) The rectangle, triangle and the circle each moves one step to the left in every subsequent row. Thus, the rightmost figure in the third row is the triangle. We can also see that all the three colours, white, grey and black are present in each row. So, the third row should have a black triangle. In the first and second rows, the rectangle is

labelled "+", the triangle "C" and the circle "T". Thus, the black triangle in row three should also be labelled "C".

Hence, option (b).

26. (*c*) Using the digits 0, 1, 2, 3, 4 and 5, five-digit numbers divisible by 3, can be formed using the following combinations.

***Case* (i)** 1, 2, 3, 4, 5

Total number of numbers formed using these digits

$$= 5! = 120$$

***Case* (ii)** 0, 1, 2, 4, 5

Total number of numbers formed using these digits

$$= 4 \times 4 \times 3 \times 2 = 96$$

Thus, total numbers = 120 + 96 = 216

Hence, option (c).

27. (*c*) Since $\frac{2}{5}$ th of the work is completed in the 25 days, remaining $\frac{3}{5}$ th of the work is to be completed in 25 days.

Let x men work for 25 days to complete $\frac{3}{5}$ th of the work.

$$\frac{M_1D_1H_1}{W_1} = \frac{M_2D_2H_2}{W_2}$$

$$\therefore \quad \frac{25 \times 105 \times 8 \times 5}{2} = \frac{x \times 25 \times 9 \times 5}{3}$$

$$\therefore \quad x = \frac{105 \times 8}{2 \times 3} = 140$$

$\therefore$ Additional men employed = 140 – 105 = 35

Hence, option (c).

28. (*d*) A can build the structure in 8 days.

$\therefore$ Fraction of structure built in a day by $A = \frac{1}{8}$

Similarly, fraction of structure broken by B in a day $= \frac{1}{3}$

Amount of work done by A in 4 days $= \frac{4}{8} = \frac{1}{2}$

Now, both A and B work together for 2 days.

So, the fraction of structure built in 2 days

$$= 2\left(\frac{1}{8} - \frac{1}{3}\right) = \frac{2 \times (-5)}{24} = \frac{-5}{12}$$

Fraction of structure still to be built

$$= \frac{1}{2} + \frac{5}{12} = \frac{11}{12}$$

If A takes x days to build up the remaining structure, then

$$\frac{x}{8} = \frac{11}{12}$$

$$\therefore \quad x = \frac{22}{3}$$

Hence, option (d).

29. (*c*) Let the first term and the ratio of the geometric progression be a and r respectively.

$$\therefore \quad a + ar = 12$$

$$\therefore \quad a(1 + r) = 12 \qquad \ldots(i)$$

Also, $\quad ar^2 + ar^3 = 48$

$$\therefore \quad ar^2(1 + r) = 48 \qquad \ldots(ii)$$

Dividing Eq. (ii) by Eq. (i),

$$\frac{ar^2(1 + r)}{a(1 + r)} = \frac{48}{12} = 4$$

$$\therefore \quad r^2 = 4$$

$$\therefore \quad r = \pm\sqrt{4} = \pm 2$$

Since, the terms of the geometric progression are alternately positive and negative, taking $r = -2$

$\therefore$ From Eq. (i), $\quad a(1 - 2) = 12$

$$\therefore \quad a = -12$$

Hence, option (c).

30. (*c*) Mean of the given numbers $= \frac{a + b + 23}{5}$

Mean is given as 6.

$$\therefore \quad \frac{a + b + 23}{5} = 6$$

$$\therefore \quad a + b = 30 - 23 = 7$$

So, option (d) can be eliminated.

Now, variance $= \frac{1}{5}[(a - 6)^2 + (b - 6)^2 + (5 - 6)^2 + (10 - 6)^2 + (8 - 6)^2]$

$$\therefore \quad \frac{1}{5}[(a - 6)^2 + (b - 6)^2 = 21] = 6.8$$

$$\therefore \quad (a - 6)^2 + (b - 6)^2 + 21 = 34$$

$$\therefore \quad (a - 6)^2 + (b - 6)^2 = 13$$

Only option (c) fits into the above equation.

Hence, option (c).

31. (*a*) Value of goods = ₹ 15000

Commission received $= 12\frac{1}{2}\%$

$$\therefore \quad \text{Commission received} = \frac{12.5 \times 15000}{100} = 1875$$

Hence, option (a).

32. (*b*) $\sqrt{110.25} = 10.5$

$\sqrt{0.01} = 0.1$

$\sqrt{0.0025} = 0.05$

$\sqrt{420.25} = 20.5$

$$\therefore \quad \sqrt{110.25} \times \sqrt{0.01} \div \sqrt{0.0025} \times \sqrt{420.25}$$

$$= 10.5 \times \frac{0.1}{0.05} - 20.5$$

$$= \frac{1.05}{0.05} - 20.5$$

$$= 21 - 20.5 = 0.5$$

Hence, option (b).

33. (*b*) The total number of respondents under the age of

$$31 = 33 + 33 = 66$$

The number of respondents under the age of 31 who indicated that Blues is their favourite style of music

$$= 2 + 3 = 5$$

$\therefore$ The required percentage $= \frac{5}{66} \times 100 \approx \frac{15}{2} \approx 7.5\%$

Hence, option (b).

34. (*a*) There were 33 respondents in all in the 21-30 age group.

There were 12 respondents in this group who preferred Rock music.

$\therefore$ There were 33 – 12 = 21 students who favoured a style other than Rock music.

$\therefore$ The required percentage $= \frac{21}{33} \times 100 \approx 64\%$

Hence, option (a).

35. (*d*) The total number of respondents = 33 + 33 + 68 = 134

The number of respondents who indicated that Jazz is their favourite style of music

$= 1 + 4 + 11 = 16$

$\therefore$ The required percentage $= \frac{16}{134} \times 100 \approx 12\%$

Hence, option (d).

36. (*d*) The total number of students in the Arts faculty = 1049

The number of non-US students in the Arts faculty

$= 79 + 21 + 6 + 2 + 4 = 112$

$\therefore$ The required percentage

$= \frac{112}{1049} \times 100 \approx 10.67\%$

The closest option is 11%.

Hence, option (d).

37. (*b*) Let the number of total students in the University be x.

There are 1049 students in the Arts faculty, which represents 23% of the total students.

$\therefore \quad 1049 = 0.23x$

The number of students in the Engineering faculty is 9% of the total.

$\therefore$ The total number of Engineering students

$= \frac{9}{100} \times \frac{1049}{0.23} \approx 410$ students

Hence, option (b).

38. (*b*) 1049 students in Arts represent 23% of the total students.

Let the total number of students in the University be x.

$\therefore \quad 1049 = 0.23x$

$\therefore \quad x = \frac{1049 \times 100}{23} = 4560$

$\therefore$ The total number of students is 4560.

Hence, option (b).

39. (*d*) The Science students are 21% of the total number of students.

6% out of these are Asians.

The total number of students in the University = 4560

$\therefore$ The number of Asian students studying Science

$= \frac{6}{100} \times \frac{21}{100} \times 4560 \approx 57$

Hence, option (d).

40. (*d*) The total number of medical students is 5% of the total students.

$\therefore$ The number of medical students

$= \frac{5}{100} \times 4560 = 228$

Out of these, 34 are from Europe.

This represents $\frac{34}{228} \times 100 = 14.9\% \approx 15\%$ of the faculty.

Hence, option (d).

41. (*b*) *A* gives *B* as many tractors that they already have.

$\therefore$ Number of tractors with *A* > Number of tractors with *B* and *C*.

Hence, option (b).

42. (*a*) Option (b) can be eliminated as "many" is not the same as 'majority'.

The passage states that "…good basic skills in reading, communication, and mathematics play an important role…" and not reliance on communication skills over basic skills in reading and mathematics. Eliminate option (c).

The passage states 'most' of the fastest growing jobs. Option (d) implies 'all' the fastest growing jobs. Eliminate option (d).

The passage mentions that good basic skills in reading, communication and mathematics are important prerequisites for jobs that do not require a college degree-jobs such as that of a desktop published, as mentioned in option (a).

Hence, option (a).

43. (*d*) Options (a), (b) and (c) can be concluded from the data given in the passage.

With the help of binary logic, we arrive at the following to be valid conclusions :

If *X*, then *Y* and If not *Y*, then not *X*

The following conclusions may or may not be valid.

If *Y*, then *X* and If not *X*, then not *Y*

The passage states that "If there are no crop protection products (*X*), then crop yield per acre will drop by more than 50% (*Y*)".

Option (d) states that "If crop yield per acre drops by more than 50% (*Y*), then crop protection products have not been used (*X*)".

Hence, the conclusion in option (d) may or may not be valid.

Hence, option (d).

44. (*b*) This is a case of "If *X*, then *Y*", where *X* is a sufficient but not a necessary condition for *Y* to occur. If flight 409 is cancelled (*X*), then the manager cannot arrive on time (*Y*).

The other valid conclusion would be Not *Y*, therefore Not *X*-The manager can arrive on time, therefore flight 409 is not cancelled.

There are other two conclusions that may or may not be valid-they may or may not be sufficient conditions :

If *Y*, then *X*-The manager cannot arrive on time, therefore flight 409 is cancelled.

If not *X*, then not *Y*-Flight 409 is not cancelled, therefore the manager can arrive on time. Lou's conclusion is based on Not *X*, therefore not *Y*, which may or may not be true-hence, Lou's argument is fallacious.

Evelyn states that the manager will not arrive at time-there is no possibility that the manager can arrive on time. But, based on data, if flight 409 is not cancelled (not *X*), then the manager can arrive on time (not *Y*) may or may not happen. Hence, Evelyn's conclusion that the manager will not arrive on time is unwarranted.

Hence, option (b).

45. (*b*) The passage assumes that a higher percentage of accident related deaths in planes compared to cars implies cars are safer.

We need an argument that attacks the premise that percentage is the only way to determine safety; or we need an argument that brings in another elements that shows cars are not as safe as percentages make them to be.

Options (a), (b) and (d) touch upon other aspects which merely add data to the main statement. If the number of car accidents is several hundred thousand times higher than the number of plane accidents, it nullifies the premise of percentages being the deciding factor as well as presents data that supports that cars are not as safe as they are made out to be vis-a-vis planes.

Hence, option (b).

46. (*b*) From the figure, it is clear that only Industries *B* and *C* contribute to Company *S*.

Hence, option (b).

47. (*d*) Observe that Industry *B* processes elements 4 and 5.

But elements 4 and 5 are also processed at Industries *A* and *C*.

Hence, the proportion of element 4 that gets processed in Industry *B* is unknown.

Similarly, the proportion of element 5 that gets processed in Industry *B* is unknown.

∴ The percentage of total production of listed elements processed by Industry *B* cannot be determined.

Hence, option (d).

48. (*c*) Industry *A* processes all the listed elements, *ie,* elements *A*, *B*, *C*, *D* and *E*.

The total annual production of all these elements is 100000 ton.

Hence, option (c).

49. (*b*) RB shows the correct analogy with exact antonyms in both the pairs-Ascend : Increase :: Descend : Decrease.

Hence, option (b).

50. (*a*) AS introduces 'Western' which does not logically fit in the analogy.

AQ gives the perfect antonymous analogous relationship - Modern : Ancient :: Young : Old.

Hence, option (a).

51. (*b*) BQ gives the correct relationship-Part : whole :: Class : School.

Hence, option (b).

52. (*a*) Summit and Apex are synonyms. The only option which comes close to being a synonym is AQ-Beautiful : Pretty.

Hence, option (a).

53. (*b*) The code for *B* is 5, code for *E* is 7 and so on.

Based on this coding system, the code for future is 204097.

Hence, option (b).

54. (*d*) If the month that ends on Wednesday have 31 days, there will be 5 Mondays otherwise will have 4 Mondays. Therefore, cannot be specified.

Hence, option (d).

55. (*b*) From the first row, $\frac{(17 \times 12)}{2} = 102$

∴ From the second row, $\frac{(15 \times 10)}{2} = 75$

Hence, option (b).

56. (*b*) From the first row, $12 \times 14 \times 2 = 336$

∴ From the second row, $15 \times 15 \times 2 = 480$

Hence, option (b).

57. (*a*) From condition (III), *B* and *G* are in different boats.

∴ If *F* and *B* are in one boat, *G* is in the other boat.

Hence, option (a).

58. (*d*) From condition (IV), the maximum number of persons in one boat can be four only.

∴ There must be at least three people in the complete list of people who must be sitting in either boat.

Hence, option (d).

59. (*c*) In the first figure there are 4 vertical and 3 horizontal lines.

In the following figures, both the vertical and horizontal lines reduce by 1.

∴ In the fourth figure there will be only one vertical line.

Hence, option (c).

60. (*c*) In the first figure there are 3 shaded balls on 3 lines.

In the second figure the shaded balls remain in the same places and the lines are turned in the opposite direction.

In second figure there are 2 shaded balls on two sticks.

∴ In the fourth figure the lines are turned in the opposite directions.

Hence, option (c).

61. (*a*) In the alphabetic order *I* is second to the right of *G* and *K* is second to the right of *I*, similarly *M* is second to the right of *K*.

Therefore, the first alphabet of the next element in the series will be *M*.

Therefore, we can eliminate option (b).

Now, *O* is second to the right of *M*, *Q* is second to the right of *O* and similarly *S* is second to the right of *Q*.

∴ We can eliminate option (c).

In options (a) and (d) only the last letter is different.

Consider the last letters. *A* is second to the right of *Y*, *C* is second to the right of *A* and similarly *E* is second to the right of *C*.

Hence, option (a).

[**Note :** There is a misprint in the second word. The third letter should be *U* instead of *V*.]

62. (*b*) Mohan spent 30% on landline, 40% on GSM and 30% on Airtel and spent 30 min and 100 min on STD and landline calls respectively.

This breakup can also be shown in a tabular manner as follows :

Mohan	Airtel	GSM	Landline
STD (minute)	9	12	9
Local (minute)	30	30	30

Mohan falls in the category of 50-200 km.

Hence, Mohan's STD cost is

$(9 \times 1.5) + (12 \times 1.5) + (9 \times 2)$

$= 13.5 + 18 + 18 = ₹ 49.5$

His cost of local calls is $(30 \times 1) + (40 \times 1) + (30 \times 2)$

$= ₹ 130$

Price of plan = ₹ 99

Hence, Mohan's total cost

$= 99 + 49.5 + 130 = ₹ 278.5$

Rohan spent 40% on landline, 30% on GSM and 30% on Airtel and spent 18 min and 120 min on STD and landline calls respectively.

This breakup can also be shown in a tabular manner as follows:

Rohan	Airtel	GSM	Landline
STD (minute)	5.4	5.4	7.2
Local (minute)	36	36	48

Rohan falls in the category of 50–200 km.

Hence, Rohan's STD cost is

$(5.4 \times 1.5) + (5.4 \times 1.5) + (7.2 \times 2)$

$= 8.1 + 8.1 + 14.4 = ₹ 30.6$

His cost of local calls is

$(36 \times 1) + (36 \times 1) + (48 \times 2) = ₹ 168$

Price of plan = ₹ 99

Hence, Rohan's total cost

$= 99 + 30.6 + 168 = ₹ 297.6$

Hence, Rohan spent more.

Hence, option (b).

63. (*a*) Solve this problem by comparing the charges incurred while sending 38, 40, 60 and 59 local SMSes respectively under the old scheme and the new scheme.

Old scheme			New scheme			
Number	Rate	Total cost	Monthly charge	Number of SMS	Rate	Total cost
38	1.5	57	35	38	0.6	57.8
40	1.5	60	35	40	0.6	59
60	1.5	90	35	60	0.6	71
59	1.5	88.5	35	59	0.6	70.4

Thus, a person sending 38 local SMS per month actually pays 80 paise more. Hence, option (a).

64. (*d*) Note that this question actually asks for a plan which is not possible at all.

The other three plans may be possible in some cases.

The fixed price of the plan is ₹ 99.

Hence, the call charges should be exactly ₹ 100 in three out of four cases.

Consider option (a).

30 calls to Airtel on STD (500 + km) cost the user ₹ 1.5 per call.

Hence, the total STD cost = $30 \times 1.5 = ₹ 45$

If the person makes all his local calls to Airtel numbers or to GSM phones, he will be charged ₹ 1 per call.

Hence, his total local call cost

$= 55 \times 1 = ₹ 55$

Hence, the total calling cost

$= 45 + 55 = ₹ 100$

Hence, a bill of 99 + 100

= ₹ 199 is possible.

Consider option (b).

16 calls to Airtel on STD (15 km) cost the user ₹ 1.5 per call.

Hence, the total STD cost

$= 16 \times 1.5 = ₹ 24$

If the person makes all his local calls to Airtel numbers or to GSM phones, he will be charged ₹ 1 per call.

Hence, his total local call cost

$= 76 \times 1 = ₹ 76$

Hence, the total calling cost

$= 24 + 76 = ₹ 100$

Hence, a bill of 99 + 100 = ₹ 199 is possible.

Consider option (c).

10 STD calls (250 km) each to Airtel, GSM and landline respectively.

Hence, the total STD cost

$= 10 \times (1.5 + 2.5 + 2.5) = ₹ 65$

He makes 30 local calls in all.

Hence, if the person makes 25 calls to Airtel and 5 calls to a landline, he gets charged as ₹ 1 and ₹ 2 respectively, *ie,*

$(25 \times 1) + (5 \times 2) = ₹ 35$

Hence, the total calling cost

$= 65 + 35 = ₹ 100$

Hence, a bill of 99 + 100 = ₹ 199 is possible.

Consider option (d).

The person makes 8, 4 and 7 calls to Airtel, GSM and landline respectively on STD (500 + km).

Hence, his total STD cost

$= (8 \times 1.5) + (4 \times 3) + (7 \times 3.5) = ₹ 49.5$

He makes 55 local calls a month.

Even if he makes all the calls to Airtel, he gets charged at ₹ 1.

Hence, his minimum local call charges are ₹ 55.

Hence, his minimum total call charges are

$49.5 + 55 = ₹ 104.5$

Hence, a total of ₹ 199/month cannot be obtained for this breakup.

Hence, option (d).

65. (*d*) 80% of the ISD bill came from calling the rest of the World.

Hence, 20% of the bill came from the other regions.

The total time spent on ISD calling was 12 min.

Since, the ISD rates for each region vary, the 12 min cannot be proportionately divided based on the bill.

Consequently, it is not possible to find out the calling duration for the other two regions.

Hence, the answer cannot be determined.

Hence, option (d).

66. (*c*) Total demand

$= 3000 + 600 + 2500 + 1200 + 3300$

$= 10600$ unit

$$\text{Average demand} = \frac{\text{Total demand}}{\text{Number of TV companies}}$$

$$= \frac{10600}{5} = 2120 \text{ unit}$$

Total production $= 1500 + 1800 + 1000 + 2700 + 2200$

$= 9200$ unit

$$\text{Average production} = \frac{\text{Total production}}{\text{Number of TV companies}}$$

$$= \frac{9200}{5} = 1840 \text{ unit}$$

Difference between the average demand and the average production $= 2120 - 1840 = 280$

Hence, option (c).

67. (*a*) Production of Company $D = 2700$

Production of Company $A = 1500$

$$\frac{\text{Production of Company } D}{\text{Production of Company } A} = \frac{2700}{1500} = 1.8$$

ie, Production of Company *D* is 1.8 times the production of Company *A*.

Hence, option (a).

68. (*b*) Paper cost is 10% of the total cost.

Angle for sector representing paper cost

$$= \left(\frac{10}{100}\right) \times 360°$$

$= 36°$

Hence, option (b).

69. (*d*) 2% of the cost is ₹ 2000 for 12500 copies.

∴ 1% of the cost is ₹ 1000.

Using this information,

The total expenditure = ₹ 100000

$$\therefore \quad \text{Cost for 1 copy} = \left(\frac{100000}{12500}\right) = ₹ 8$$

Sale price for a profit of 5% $= 8 \times 1.05 =$ ₹ 8.40

Hence, option (d).

70. (*d*) 'Tree' is called 'sky'. Hence, logically 'fruit' grows on 'sky'.

Hence, option (d).

71. (*d*) We need a possessive pronoun in the blank.

72. (*d*) 'Till' is used to denote a certain time period.

73. (*c*) The blank needs to have the Past Perfect Tense.

74. (*) All four words are spelled correctly.

75. (*a*) The correct spellings are superintendent.

76. (*d*) 'In/On' are the correct sequence of preposition.

77. (*b*) 'Aural' us related to 'the ear'.

78. (*a*) Error refers to a minor flow while fault to mistake.

79. (*d*) 'Flonted' means to 'to defy'.

80. (*a*) 'The Outback'

81. (*c*) I know…… Rima, but…..

82. (*d*) Veracious means truthful.

83. (*c*) to perturb someone is to disturb them.

84. (*b*) With a high land means to rule in a dogmatic and oppressive manner.

85. (*d*) Rack the brains means to search thoroughly.

86. (*c*) The idiom means to be nervous.

87. (*b*) The idiom means to gulp one's pride and to take back what they said.

88. (*c*) Option 1 and 4 conveying, the idea that 'They should have checked your tickets'.

89. (*a*) Magnanimous means large hearted/generous.

90. (*c*) Biannual means 'twice in a single year'.

91. (*d*) Eternal means forever whereas the other three options refer to something that is short lived.

92. (*a*) 'Intravenous' means into/through the veins.

93. (*a*) Options b,c,d get eliminated as 1-8 is the correct pair.

94. (*b*) Options c and d get eliminated as 3-8 is the correct pair. 'As bitter as gall' is the correct simile.

95. (*a*) Options b and d get eliminated as 1-8 form the correct pair between a and c. a is the correct combination as 3-5 is the correct pair.

96. (*b*) b should be 'makes' as idli & sambhar are complementary items.

97. (*c*) by three months's interest.

98. (*d*) Options a,b,c should have been whose, who is, whom respectively.

99. (*b*) Error of pronoun.

100. (*b*) Modifier error.

101. (*d*) The idea presented is that one has to work hard to succeed.

102. (*b*) Sentence 5 has to be the starting of the sentence and only option b has 5 at the starting.

103. (*d*) Only one conditional is required. Hence, option a is incorrect and option d is correct. Options b and c are incorrect tense wise.

104. (*b*) Refer to the word 'at times'.

105. (*a*) Notice that it is a question statement.

106. (*b*) 'as-as' constructions are called similes.

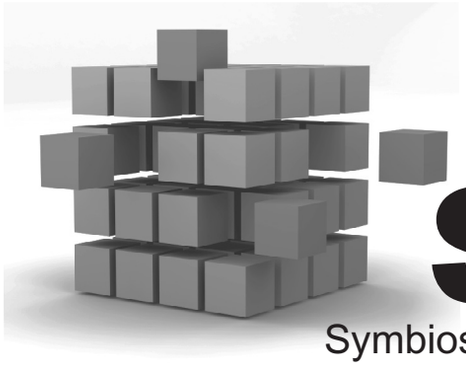
107. (*a*) Death has been personified as a being with hands.

108. (*d*) Frozen feelings refer to the reaction of a person a negative emotional situation that occurred in his/her childhood.

109. (*c*) Refer to para 3, the anchor talks about behaving like a 3 year old child while going through certain negative emotions.

110. (*b*) Refer to para 3 line 4.

111. (*a*)	**112.** (*d*)	**113.** (*b*)	**114.** (*a*)	**115.** (*d*)
116. (*d*)	**117.** (*a*)	**118.** (*c*)	**119.** (*d*)	**120.** (*b*)
121. (*c*)	**122.** (*b*)	**123.** (*b*)	**124.** (*c*)	**125.** (*b*)
126. (*d*)	**127.** (*b*)	**128.** (*b*)	**129.** (*a*)	**130.** (*b*)
131. (*c*)	**132.** (*a*)	**133.** (*b*)	**134.** (*a*)	**135.** (*a*)
136. (*a*)	**137.** (*c*)	**138.** (*a*)	**139.** (*c*)	**140.** (*a*)
141. (*c*)	**142.** (*c*)	**143.** (*d*)	**144.** (*b*)	**145.** (*a*)
146. (*b*)	**147.** (*b*)	**148.** (*b*)	**149.** (*b*)	**150.** (*b*)



SNAP 2009

Symbiosis National Aptitude Test

Solved Paper

Quantitative and Data Interpretation & Data Sufficiency Section

For all questions in this section, correct answers carry 1 mark each.

1. A dice is rolled three times and sum of three numbers appearing on the uppermost face is 15. The chance that the first roll was a four is
(1) 2/5 (2) 1/5
(3) 1/6 (4) None of these

2. A boat covers a distance of 30 km downstream in 2 h while it takes 6 h to cover the same distance upstream. What is the speed of the boat in kms per hour?
(1) 5 (2) 7.5
(3) 13 (4) 18

3. A five digit number is formed by using the digits 1,2,3,4 and 5 with without repetitions. What is the probability that the number is divisible by 4?
(1) 1/5 (2) 5/6
(3) 4/5 (4) None of these

4. If the algebraic sum of deviations of 20 observations measured from 23 is 70,mean of these observations would be
(1) 24 (2) 25
(3) 26 (4) None of these

5. An alloy of gold and silver weighs 50 g. It contains 80% gold. How much gold should be added to the alloy so that percentage of gold is increased to 90?
(1) 50 g (2) 60 g
(3) 30 g (4) 40 g

6. Weekly incomes of two persons are in the ratio of 7 : 3 and their weekly expenses are in the ratio of 5 : 2. If each of them saves Rs 300 per week, then the weekly income of the first person is
(1) Rs 7500 (2) Rs 4500
(3) Rs 6300 (4) Rs 5400

7. Wheat is now being sold at Rs 27 per kg. During last month its cost was Rs 24 per kg. Find by how much per cent a family reduces its consumption so as to keep the expenditure fixed.
(1) 10.2% (2) 12.1%
(3) 12.3% (4) 11.1%

8. There are 10 stations on a railway line. The number of different journey tickets that are required by the authorities is
(1) 92 (2) 90
(3) 91 (4) None of these

9. The radius of circle is so increased that its circumference increased by 5%. The area of the circle then increases by
(1) 12.5% (2) 10.25%
(3) 10.5% (4) 11.25%

10 In how many ways can the letters of the word ABACUS be rearranged such that the vowels always appear together?
(1) 6!/2! (2) 3!×3!
(3) (3!×3!)/2! (4) (4!×3!)/2!

11. In 4 yr, Rs 6000 amounts to Rs 8000. In what time at the same rate will Rs 525 amount to Rs 700?
(1) 2 yr (2) 3 yr
(3) 4 yr (4) 5 yr

> **Directions (Q. 12-14):** *At the start of a game of cards, J and B together had four times as much money as T, while T and B together had three times as much as J.*
> *At the end of the evening, J and B together had three times as much money as T, while T and B together had twice as much as J. B lost Rs* 200.

12. What fraction of the total money did *T* have at the beginning of the game?
(1) 1/3 (2) 1/8
(3) 2/9 (4) 1/5

13. What fraction of the total money did *J* win/lose?
(1) Won 1/12 (2) Lost 1/6
(3) Lost 1/3 (4) Won 1/5

14. What amount did *B* start with?
(1) Rs 575 (2) Rs 375
(3) Rs 825 (4) Rs 275

15. If a and b are negative, and c is positive, which of the following statement/s is /are true?
(I) $a - b < a - c$ (II) if $a < b$, then $a/c < b/c$
(III) $a/b < a/c$
(1) I only (2) II only
(3) III only (4) II and III only

16. The diagonal of a square is $4\sqrt{2}$ cm. The diagonal of another square whose area is double that of the first square is
(1) 8 cm (2) $8\sqrt{2}$ cm
(3) $4\sqrt{2}$ cm (4) 16 cm

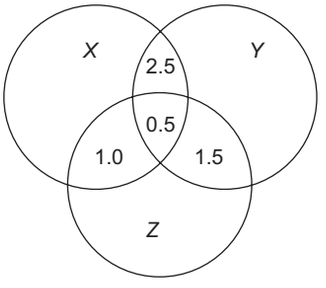
17. The maximum length of a pencil that can be kept in a rectangular box of dimensions 8 cm × 6 cm × 2 cm, is
(1) $2\sqrt{13}$ cm (2) $2\sqrt{14}$ cm
(3) $2\sqrt{26}$ cm (4) $10\sqrt{2}$ cm

Directions (Q. 18-19): *Questions are based on the information given below.*

The Venn diagram given below shows the estimated readership of 3 daily newspapers (*X*,*Y* & *Z*) in a city. The total readership and advertising cost for each of these papers is as below.

Newspapers	Readership (lakhs)	Advertising cost (Rs per sq cm)
X	8.7	6000
Y	9.1	6500
Z	5.6	5000

The total population of the city is estimated to be 14 million. The common readership (in lakhs) is indicated in the given Venn diagram.



18. The number of people (in lakhs) who read at least one newspaper is
(1) 4.7 (2) 11.9
(3) 17.4 (4) 23.4

19. The number of people (in lakhs) who read only one newspaper is
(1) 4.7 (2) 11.9
(3) 17.4 (4) 23.4

20 Sonali invests 15% of her monthly salary in insurance policies. She spends 55% of her monthly salary in shopping and on household expenses. She saves the remaining amount of Rs 12750. What is Sonali's monthly income?
(1) Rs 42500 (2) Rs 38800
(3) Rs 40000 (4) Rs 35500

21. How many kg of tea worth Rs 25 per kg must be blended with 30 kg of tea worth Rs 30 per kg so that by selling the blended variety at Rs 30 kg there should be a gain of 10%?
(1) 36 kg (2) 40 kg
(3) 32 kg (4) 42 kg

22. In an examination, out of 480 students 85% of the girls and 70% of the boys passed. How many boys appeared in the examination if total pass percentage was 75%?
(1) 370 (2) 340
(3) 320 (4) 360

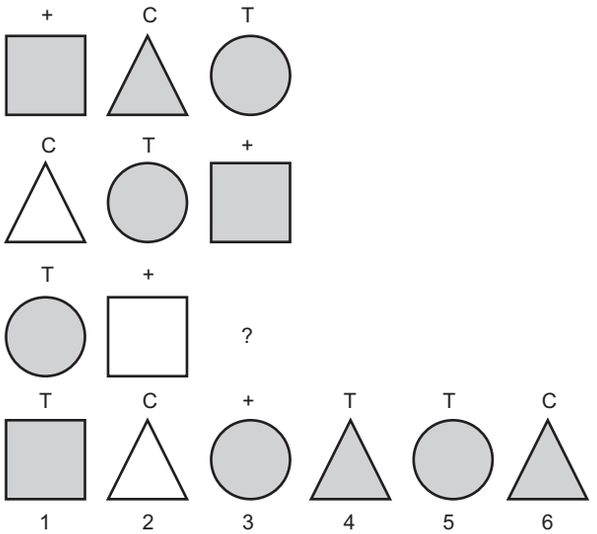
23. 300 g of salt solution has 40% salt in it. How much salt should be added to make it 50% in the solution?
(1) 40 kg (2) 60 kg
(3) 70 kg (4) 80 kg

24. What number should replace the question mark in the image below?

6	12	?
48	96	192

(1) 18 (2) 20
(3) 22 (4) 24

25. What image from bottom row should replace the question mark?



(1) 4 (2) 6
(3) 1 (4) None of these

26. A five-digit number divisible by 3 is to be formed using digits 0,1,2,3,4 and 5 without repetition. The total number of ways this can be done is
(1) 122 (2) 210
(3) 216 (4) 217

27. A contract is to be completed in 50 days and 105 men were set to work, each working 8 h a day. After 25 days, 2/5 of the work is finished. How many additional men be employed so that the work may be completed on time, each man now working 9 h a day?
(1) 34 (2) 36
(3) 35 (4) 37

28. *A* can built up a structure in 8 days and *B* can break it in 3 days. *A* has worked for 4 days and then *B* joined to work with *A* for another 2 days only. In how many days will A alone built up the remaining part of the structure?
(1) 10 days (2) 9 days
(3) 12 days (4) None of these

29. The first two terms of a geometric progression add up to 12. The sum of the third and the fourth terms is 48.If the terms of the geometric progression are alternately positive and negative, then the first term is
(1) – 2
(2) – 4
(3) – 12
(4) 8

30. The mean of the numbers a, b, 8, 5, 10 is 6 and the variance is 6.80. Then which one of the following gives possible values of a and b ?
(1) $a = 0, b = 7$
(2) $a = 5, b = 2$
(3) $a = 3, b = 4$
(4) $a = 2, b = 4$

31. An agent sells goods of value of Rs 15000. The commission which he receives at the ratio of $12\frac{1}{2}\%$ is
(1) Rs 1875
(2) Rs 2125
(3) Rs 2000
(4) Rs 2700

32. $\sqrt{110.25} \times \sqrt{0.01} \div \sqrt{0.00025} - \sqrt{420.25}$ equal to
(1) 0.75 (2) 0.50
(3) 0.64 (4) 0.73

Directions (Q. 33-35): *The table below shows the number of people who responded to a survey about their favourite style of music. Use this information to answer the following questions to the nearest whole percentage.*

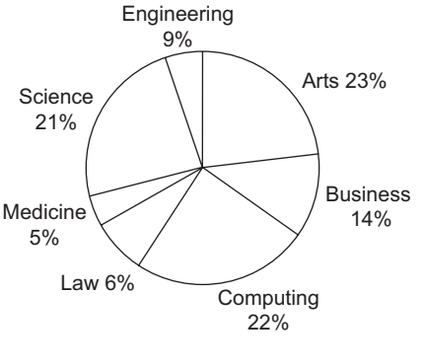
Age	15-20	21-30	31+
Classical	6	4	17
Pop	7	5	5
Rock	6	12	14
Jazz	1	4	11
Blues	2	3	15
Hip – Hop	9	3	4
Ambient	2	2	2
	33	33	68

33. What percentage of respondents under 31 indicated that Blues is their favourite style of music?
(1) 7.1 (2) 7.6
(3) 8.3 (4) 14.1

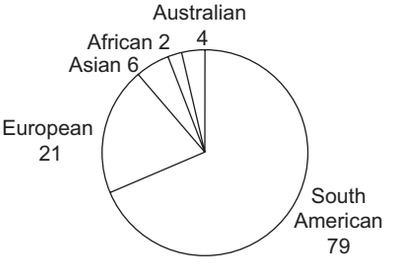
34. What percentage of respondents aged 21-30 indicated a favourite style other than Rock music?
(1) 64% (2) 60%
(3) 75% (4) 36%

35. What percentage of the total sample indicated that Jazz is their favourite style of music?
(1) 6% (2) 8%
(3) 22% (4) 12%

Directions (Q. 36-40): *The pie charts below show the percentage of students in each faculty at North West University and the number of non-US students in the Arts faculty. These percentages have been rounded to the nearest whole number. There are a total of* 1049 *students in Arts faculty. Use this information to answer the following questions.*



Students by Faculty



Arts Students (Non-US)

36. What percentage of students in the Arts faculty are non-US students?
(1) 14% (2) 9%
(3) 30% (4) 11%

37. How many students are there in the Engineering faculty?
(1) 420 (2) 410
(3) 390 (4) 440

38. How many students are there at the University?
(1) 4650 (2) 4560
(3) 4640 (4) 4450

39. If six per cent of Science students are Asian. How many Asian students are studying Science?
(1) 48% (2) 66%
(3) 120% (4) 57%

40. There are 34 European medical students. What percentage of the faculty does this represent?
(1) 14% (2) 18%
(3) 16% (4) 15%

■ Analytical & Logical Reasoning Section

For all questions in this section, correct answers carry 2 marks each.

41. There are 3 societies A, B, C having some tractors each. A gives B and C as many tractors as they already have.
After some days B gives A and C as many tractors as they have.
After some days C gives A and B as many tractors as they have.
Finally each has 24 tractors. What is the original number of tractors each had in the beginning?
(1) A-29, B-21, C-12 (2) A-39, B-21, C-12
(3) A-29, B-12, C-29 (4) A-21, B-12, C-39

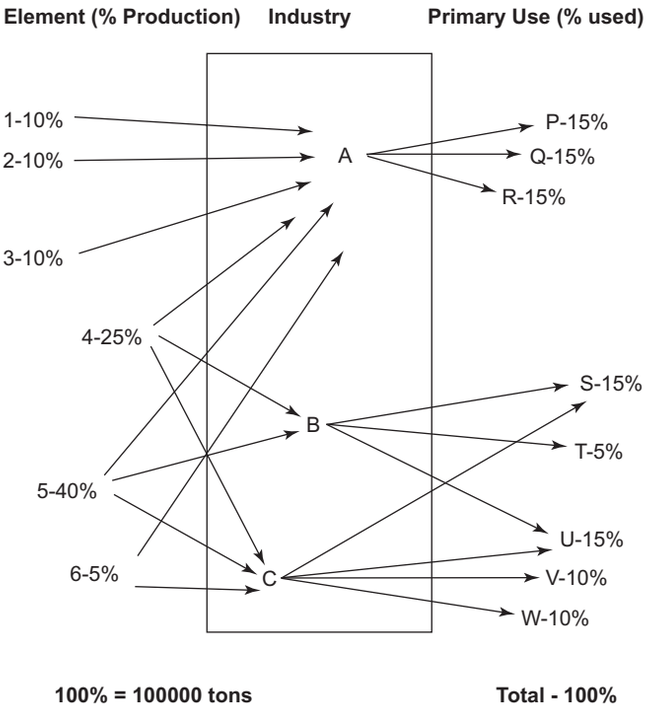
42. Although most of the fastest growing jobs in today's economy will require a college degree, many of the new jobs being created from home health aide to desktop publisher require knowledge other than gained from earning a degree. For workers in those jobs, good basic skills in reading, communication and mathematics play an important role in getting a job and developing a career.
From the information given above it can be validly concluded that, in today's economy
(1) skills in reading, communication and mathematics play an important role in developing a career as a desktop publisher.
(2) the majority of the new jobs being created require knowledge other than that gained from earning a college degree
(3) a job as a home health aide will rely more on communications skills than on a basic skills in reading and mathematics.
(4) if a job is one of the fastest growing jobs, it will require a college degree

43. According to the National Agricultural Aviation Society (NAAS), without the use of crop protection products to control insects, weeds, and diseases, crop yields per acre will drop by more than 50% . The first aerial application of insecticide occurred in 1921 and it was huge success. By contrast, in today's economy all aircraft that are classified as aerial applicators do more than just apply insecticide; today, they also spread seeds and apply fertilizer.
From the information given above it CANNOT be validly conclude that
(1) according to the NAAS, if crop yields per acre never drop by more than 50%, then crop protection products have been used to control insects, weeds, and diseases.
(2) in today's economy any aircraft that cannot be used to apply fertilizer cannot be classified as an aerial applicator
(3) in today's economy, if an aerial applicator is used, then it will be able to spread seed and to apply fertilizer
(4) according to NAAS, if crop yields per acre drop by more than 50%, then crop protection products have not been used to control insects, weeds, and diseases.

44. Lou observes that if flight 409 is cancelled, then the manager could not possibly arrive in time for the meeting. But the flight was not cancelled. Therefore, Lou concludes, the manager will certainly be on time.Evelyn replies that even if Lou's premises true, his argument is fallacious. And therefore, she adds, the manager will not arrive on time after all.
Which of the following is the strongest thing that we can properly say about this discussion?
(1) Evelyn is mistaken in thinking Lou's argument to be fallacious, and so her own conclusion is unwarranted.
(2) Evelyn is right about Lou's argument, but nevertheless her own conclusion is unwarranted.
(3) Since Evelyn is right about Lou's argument, her own conclusion is well supported.
(4) Since Evelyn is mistaken about Lou's argument, her own conclusion must be false.

45. Cars are safer than planes. Fifty per cent of plane accidents result in death, while only one per cent of car accidents result in death.
Which of the following, if true, would most seriously weaken the argument above?
(1) Planes are inspected more often than cars.
(2) The number of car accidents is several hundred thousand times higher than the number of plane accidents.
(3) Pilots never fly under the influence of alcohol, while car drivers often do.
(4) Plane accidents are usually the fault of air traffic controllers, not pilots.

Directions (Q. 46-48): *Refer to the chart below showing annual production and answer the questions that follow.*



46. Which industry/industries contribute/contributes to company S?
(1) industry A and B only (2) B and C only
(3) A and C only (4) Smallest plant

47. Industry B processes what percentage of the total production of listed elements?
(1) 25% (2) 65%
(3) 40% (4) Cannot be determined

48. Of the listed elements processed by the Industry A, how many tons are produced annually?
(1) 30000 (2) 50000
(3) 100000 (4) 55000

Directions (Q. 49-52): *In each of the following questions there are two blanks marked I & II. The words to fill in these blanks are given against I as (A, B, C, D) and II as (P, Q, R, S) RESPECTIVELY. The right words to fill in these blanks are given as four alternatives. The words on either side of the sign (: :) have a similar relationship. That alternative which signifies this relationship is your answer.*

49. I : Increase : : Descend : II

I. (A) Grow (B) Ascend
(C) Rise (D) Price
II. (P) Reduce (Q) Down
(R) Decrease (S) Mountain

(1) AR (2) RB
(3) CP (4) DQ

50. Modern : I : : II : Old

I. (A) Ancient (B) Death
(C) Famous (D) Civilization
II. (P) Industrialisation (Q) Young
(R) Fashion (S) Western

(1) AQ (2) AS
(3) BP (4) CR

51. Part : I : : Class : II

I. (A) Section (B) Whole
(C) School (D) Students
II. (P) Student (Q) Shchool
(R) Teachers (S) Room

(1) AR (2) BQ
(3) CP (4) DS

52. Summit : Apex : I : II

I. (A) Beautiful (B) Picture
(C) Attractive (D) Enhancing
II. (P) Comfortable (B) Pretty
(C) Healthy (D) Brave

(1) AQ (2) BP
(3) DS (4) CR

53. If in a certain code BEAUTIFUL is coded as 573041208, BUTTER as 504479, how is FUTURE coded in that code?

(1) 201497 (2) 204097
(3) 704092 (4) 204079

54. How many Mondays are there in particular month of a particular year if the month ends on Wednesday ?

(1) 4 (2) 5
(3) 3 (4) Cannot be specified

Directions (Q. 55-56): *There are two rows of numbers in each question. The upper row is complete and in the lower one number is missing. Find a suitable number to fill the blank space so that the symmetry with the upper row is maintained.*

55.

17	102	12
15	?	10

(1) 211 (2) 75
(3) 125 (4) 117

56.

12	336	14
15	?	16

(1) 220 (2) 480
(3) 125 (4) 450

Directions (Q. 57-58): *Seven People A, B, C, D, E, F, G are planning to enjoy boating. There are only two boats, and the following conditions are to be kept in mind.*

(i) A will go in the same boat in which E is to go.
(ii) F cannot go in the same boat in which C is, unless D is also accompanying.
(iii) Neither B nor C can be given the boat in which G is.
(iv) The maximum number of persons in one boat can be four only.

57. If F and B are in one boat, which of the following statements is true ?

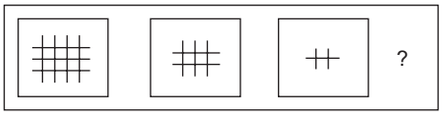
(1) G is in the other boat
(2) D is in the other boat
(3) C is in the other boat
(4) E is with F and B in one boat

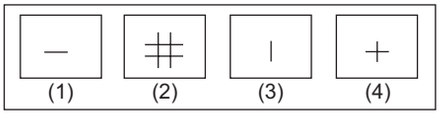
58. If E gets the boat with F, which of the following is the complete and accurate list of the people who must be sitting in one boat ?

(1) F and E (2) G and A
(3) D and A (4) C, D and B

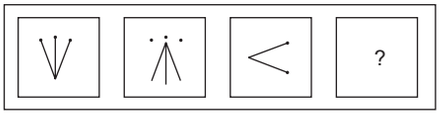
Directions (Q. 59-60): *A series of figures has been shown. Find the figure in the place of ? from the figures given below.*

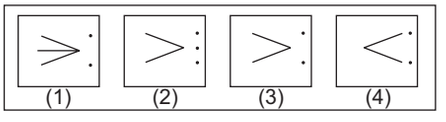
59.





60.





61. Complete the following series.
(GMSY, IOUA, KQWC, ?)

(1) MSYE (2) NSYE
(3) MYTE (4) MSYF

Directions (Q. 62-65): *Following is a new plan called 'Airtel one' introduced by Airtel (GSM service Provider) for its post-paid customers.*

Pulse Rate	60 Secs		
Price of Plan	Rs 99		
Free Airtime on Plan	Nill		
Incoming Calls (Rs)	Free		
Outgoing Calls (Rs/min.)	Airtel	GSM/CDMA	Landline/ WLL
		(10 Digit)	
LOCAL RATES	Re 1	Re 1	Rs 2
STD RATES			
50-200 km	Rs 1.50	Rs 1.50	Rs 2
200-500 km	Rs 1.50	Rs 2.50	Rs 2.50
500 + km	Rs 1.50	Rs 3.00	Rs 3.50
ISD			
USA, Canada, Europe (Fixed Line)	Rs 7		
Gulf, Europe (Mobile) SAARC	Rs 10		
Rest of the world	Rs 40		
SMS			
Local	Rs 1.50		
National	Rs 2.00		
International	Rs 5.00		
Value Added Service (Rs)	Rs 3.00 and Rs 6.00 Depending on the service		
Monthly bill = Call charges + SMS charges + Price of Plan			

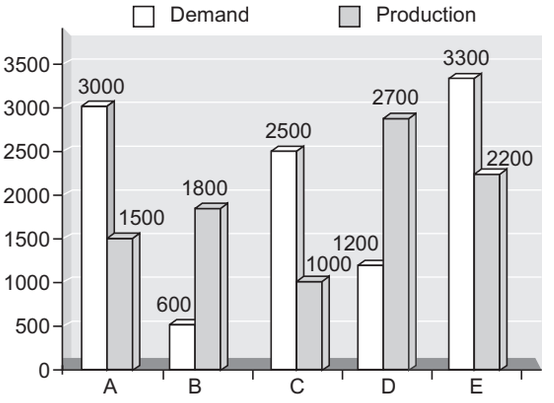
62. If Mohan made STD calls (within 100 km) for 30 min and spent 100 min on local calls with 30% on landline. 40% in GSM and 30% on Airtel, and Rohan spent 18 min on STD within 150 km and spent 120 min on local calls with 30%, 40% and 30% on GSM, landline and Airtel respectively, then who spent more?
(1) Mohan
(2) Rohan
(3) Both spent the same amount
(4) Cannot be determined

63. A new SMS scheme was introduced @60p/local SMS, with an additional monthly charge of Rs 35. Who will not benefit from the scheme ? A person sending
(1) 38 local SMS a month
(2) 40 local SMS a month
(3) 60 local SMS a month
(4) 59 local SMS a month

64. A bill of Rs 199/Month is definitely not possible when you make only
(1) 30 calls to Airtel on STD (500 + km) and 55 local calls in a month
(2) 16 calls to Airtel on STD (150 km) and 76 local calls in a month
(3) 10 STD calls (250 km) each to Airtel, GSM and landline respectively and 30 local calls in a month
(4) 8, 4 and 7 calls to Airtel, GSM and landline respectively on STD (500 + km) and 55 local calls in a month

65. A person makes 12 min of ISD calls in a month. If 80% of his ISD bill from calling the rest of the world, then the ISD calls for the minimum duration were made to which of the following group of countries?
(1) USA, Canada, Europe (Fixed Line)
(2) Gulf, Europe (Mobile), SAARC
(3) Rest of the World
(4) Cannot be determined

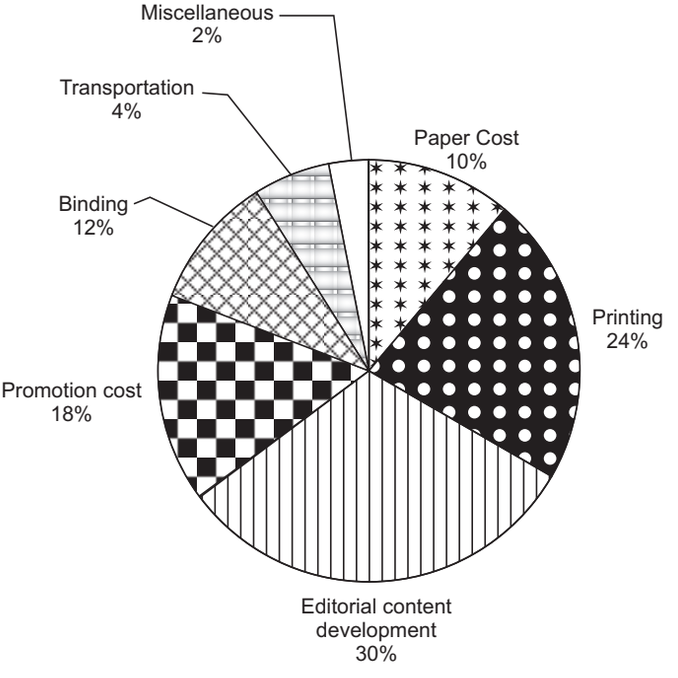
Directions (Q. 66-67): *These questions are based on the graph which shows the demand and production statistics of 5 TV companies.*



66. What is the difference between the average demand and the average production of the five companies taken together?
(1) 1400 (2) 400
(3) 280 (4) 138

67. The production of the company D is how many times of the production of the company A?
(1) 1.8 (2) 1.5
(3) 2.5 (4) 1.11

Directions (Q. 68-69): *The following questions are based on the pie-chart given below which gives the expenditure incurred in printing a magazine.*



68. What is the angle for the sector representing paper cost?
(1) 10 degrees (2) 36 degrees
(3) 23.5 degrees (4) 45 degrees

69. For a given issue of the magazine, the 2% of the cost is Rs 2000 and the print-run is 12500 copies. What should be the sale price if the publisher desires a profit of 5%?
(1) Rs 5 (2) Rs 7.5
(3) Rs 8 (4) Rs 8.40

70. If 'water' is called 'food' , 'food' is called 'tree', 'tree' is called 'sky', 'sky' is called 'wall', on which of the following does a 'fruit' grow ?
(1) Water (2) Food
(3) Tree (4) Sky

■ General English Section

For all questions in this section, correct answers carry 1 mark each.

Directions (Q. 71-72): *Choose the correct option to fill the blank for correct grammatical use.*

71. She is a close friend of _____.
(1) my (2) I
(3) me (4) mine

72. He took care of her _____ she was restored to health.
(1) when (2) before
(3) after (4) till

73. Choose the correct option: **The man _____ his appearance completely since then.**
(1) is changing (2) changed
(3) has changed (4) is changed

74. From the options below, choose the word with the incorrect spelling
(1) Diarrhea (2) Diaper
(3) Dichotomy (4) Dias

75. Choose the word spelt correctly
(1) superintendant (2) sieve
(3) allotted (4) dissipate

Directions (Q. 76-79): *Fill in the blanks with the correct pair of words.*

76. He lives _____ Bangaluru _____ 115, Richmond Road.
(1) at in (2) in at
(3) at on (4) in on

77. 'Aurally challenged' is a _____ for the _____.
(1) metaphor blind
(2) euphemism deaf
(3) simile disable
(4) synonym dump

78. If error is a _____ fault is _____.
(1) defect – a mistake (2) mistake – a defect
(3) slip – an inaccuracy (4) blunder – to blame

79. The man has _____ the rules of ethical conduct; he is _____ a beast.
(1) flaunted – literally (2) ignored - basically
(3) broken - as (4) flouted – virtually

Directions (Q. 80-81): *Which punctuation mark is missing in the following sentences?*

(1) inverted commas (2) semicolon
(3) comma (4) hypen

80. Part of Australia is known to the natives as The Outback.

81. I know that you want to learn to drive Rima but you are too young.

Directions (Q. 82-83): *Select as option the word closest in meaning to the given word.*

82. Veracious
(1) False (2) Varied
(3) Image (4) Truthful

83. Perturb
(1) Stipulate (2) Turn around
(3) Disturb greatly (4) Compatible

Directions (Q. 84-85): *Choose the option that is the closest in meaning to the phrases in bold.*

84. He was a king who ruled his subjects **with a high hand.**
(1) sympathetically (2) oppressively
(3) democratically (4) generously

85. **I racked my brains** to solve this difficult problem.
(1) I read a large number of books.
(2) I consulted several people.
(3) I used my commonsense.
(4) I subjected my mind to hard thinking.

Directions (Q. 86-87): *Select the most appropriate meaning of the underlined idiomatic phrase.*

86. I was supposed to give a speech to my English class, but I got cold feet and I didn't go.
(1) My English class-room makes my feet feel too cold.
(2) My feet got cold so I didn't go.
(3) I got too nervous and I didn't go.
(4) It was winter time and I had no snacks for my feet.

87. Take care of what u say! You will have to eat your words!
(1) You have no food to eat.
(2) You will have to take back what you have said.
(3) You are not good with your language.
(4) None of the above

88. Which two sentences convey the same idea?
1. Shouldn't they have checked your tickets?
2. I wonder if they should have checked your tickets.
3. I want to know if they checked your tickets.
4. They should have checked your tickets.
(1) 2, 3 (2) 1, 2
(3) 1, 4 (4) 2, 4

Directions (Q. 89-90): *Choose a single word which is opposed to the meaning contained in the sentence.*

89. He was magnanimous and his benevolence made him give to charity.
(1) mean (2) cruel
(3) snobbish (4) tyrannical

90. Biannual is
(1) once in two years (2) every year
(3) twice a year (4) after every two years

91. Choose the odd one out
(1) temporal (2) ephemeral
(3) transient (4) eternal

92. Choose the correct option
If 'inter' means between as in interstate, 'intra' as in intravenous means _____.
(1) Into (2) onto
(3) within (4) without

93. Match the part of speech (in bold) to their usage (**down**)

1. Adjective	5. The fire engine came rushing **down** the hill.
2. Preposition	6. He has seen the ups and **downs** of life.
3. Noun	7. **Down** with the tyrant!
4. Verb	8. The porter was hit by the **down** train.

(1) 1-8, 2-5, 3-6, 4-7 (2) 1-6, 2-5, 3-8, 4-7
(3) 1-5, 2-6, 3-7, 4-8 (4) 1-5, 2-7, 3-6, 4-8

Directions (Q. 94): *Match the items in the column I with those in column II. Choose the correct answer combination given below.*

94.

I	II
1. As deaf as	5. gall
2. As bitter as	6. an eel
3. As unpredictable as	7. a post
4. As slippery as	8. the weather

(1)	(2)	(3)	(4)
1-7	1-7	1-8	1-5
2-6	2-8	2-5	2-7
3-8	3-8	3-5	3-6
4-5	4-6	4-7	4-8

Directions (Q. 95): *Match the part of speech* (*in bold*) *to their usage.*

95. ABOVE

1. Adjective	5. Rain comes from **above**.
2. Adverb	6. His conduct is **above** suspicious.
3. Noun	7. Look **above** the mantel piece.
4. Preposition	8. The **above** information is for the public.

(1)	(2)	(3)	(4)
1-8	1-7	1-8	1-6
2-7	2-6	2-5	2-8
3-5	3-5	3-6	3-5
4-6	4-8	4-7	4-7

Directions (Q. 96-97): *Choose the segment with error. If there is no error, choose '4'.*

96. Idli and Sambar make the breakfast in the south.
(1) (2) (3) (4)

97. Your account should have been credited with three month's interest
(1) (2) (3) (4)

98. Choose the sentence where the underlined word is used appropriately.
(1) Who's car are you planning to borrow?
(2) Whose planning to have a birthday party?
(3) He is the person who you met at the junction.
(4) Does the know whom to about the hotel reservations?

Directions (Q. 99-100): *Choose*
(1) *if the first sentence is correct*
(2) *if the second sentence is correct*
(3) *if both sentences are correct*
(4) *if both sentences are wrong*

99. 1. Can you imagine his forgetting his own birthday?
2. Can you imagine him forgetting his own birthday?
(1) (2) (3) (4)

100. 1. Recently I read about a unique wedding that took place in the newspaper.
2. Recently I read in the newspaper about a unique wedding that took place.
(1) (2) (3) (4)

101. Complete the given sentence by choosing the correct phrase.
You cannot succeed unless
(1) You do not work hard.
(2) You shall not work hard.
(3) You will work hard.
(4) You work hard.

102. Re-arrange the scrambled segments in logical order to make a complete sentence.
1. to place where more opportunities are available
2. and so there is a great demand for English
3. for professional and economic growth
4. because it takes one outside one's own community
5. English is language of opportunities
(1) 1, 2, 5, 3, 4 (2) 5, 4, 1, 3, 2
(3) 3, 1, 4, 2, 5 (4) 2, 1, 3, 5, 4

103. Choose the best construction
(1) If you would have taken care you wouldn't have got typhoid.
(2) If you took care you wouldn't have got typhoid.
(3) If you take care you wouldn't have got typhoid.
(4) If you had taken care you wouldn't have got typhoid.

Directions (Q. 104-105): *Choose the correct option.*

104. At times, we are all ____ to be mistaken.
(1) apt (2) likely
(3) considered (4) able

105. He hardly cares, _____ ?
(1) does he (2) doesn't he
(3) will he (4) won't he

Directions (Q. 106-107): *Identify the figures of speech in the following sentences.*

106. As proud as a peacock.
(1) Metaphor (2) Simile
(3) Apostrophe (4) Epigram

107. Death lays his icy hand on kings.
(1) Personification (2) Exclamation
(3) Simile (4) Anticlimax

Directions (Q. 108-110): *Read the passage carefully and answer within the context.*

"A way to deal with frozen feelings"

Every child experiences all that happens around him with total awareness. In the first seven years the child's brain is like a sponge, taking in all sensory inputs and building his idea of his surroundings. As long as the environment is safe, the child learns with incredible speed. However, when the environment is scary or stressful, the child unlearns past learning just as rapidly.

In the early years of every child's life, whenever there is shock, violence, fear or pain, these intense emotions are imprinted deeply into memory. Whenever the same activity or situation is repeated, the nervous system and body subconsciously re-experience the memory of that trauma.

Any emotional situation that takes us out of the present and into the past means that whenever the same kind of emotion crops up later in our life we return to the past for our reference point. If that point was at age three, we find ourselves behaving like a three-year-old. We feel childish and we behave childishly. Our feelings are the cause of this 'glitch' in our learning process. We know we should be able to make a positive change, but that doesn't change anything.

The process of change need not be traumatic. We couldn't have done any better because we didn't know how to. But we should realize that was then and this is now! We can choose to choose again. It's up to us. It's our movie!

108. The "Frozen Feelings" being talked about are about
(1) negative childhood experiences
(2) childhood learning patterns
(3) inability to learn as an adult
(4) None of the above

109. A 'glitch' is
(1) a ditch (2) uneasy emotions
(3) sudden malfunction or breakdown
(4) learning patterns

110. Identify the correct sentence, based on the paragraph
(1) The process of change needs to be traumatic.
(2) We feel childish and we behave childishly.
(3) Both sentences are incorrect.
(4) Both the sentences are correct.

■ General Awareness : General Knowledge, Current Affairs, Business Scenario

For all questions in this section, correct answers carry 1 mark each.

111. Dendrochronology is
(1) The method of scientific dating based on the analysis of tree-ring growth patterns.
(2) A core sample from the accumulation of snow and ice over many years that have re-crystallized and have trapped air bubbles from previous time periods.
(3) is the study of glaciers or more generally ice and natural phenomena that involve ice.
(4) The scientific study of landforms and the processes that shape them.

112. Which company launched the first mobile phone operation in India?
(1) Bharti Airtel (2) Essar
(3) Max touch (4) Modi Telstra

113. Which is the longest running English language TV cartoon in history, first launched by Warner Brothers in 1969 exclusively for television, and in now popular fare on the cartoon Network?
(1) Scooby Doo (2) Tom and Jerry
(3) Popeye (4) Johnny Bravo

114. The expansion for BIFR, in the context of the Indian Industry is
(1) Board for Industrial and Financial Reconstruction
(2) Bureau for Industrial and Financial Reconstruction
(3) Board for Investment and Financial Reconstruction
(4) Bureau for Investment and Financial Reconstruction

115. What is Ekistics?
(1) The water sports related study
(2) The science of Body Mass
(3) The art of card tricks
(4) The science of human settlements

116. What is Red Herring in an IPO?
(1) Prospectus
(2) Submission of Form
(3) Funds Generated during IPO
(4) Minimum offer per share

117. Eight O'clock coffee, a US based coffee marketer, is a group company of
(1) Tata (2) ITC
(3) Britannia (4) Unilever

118. Bio-diesel is extracted from the following plant
(1) Hibiscus (2) Aloe vera
(3) Jatropha (4) Chamomile

119. Which woman tennis player won the maximum number of Grand Slam singles titles in the history of tennis?
(1) Steffi Graf (2) Martina Navratilova
(3) Billi Jean King (4) Margaret court

120. What is the nickname of New York city?
(1) Big Sweet Lime (2) Big Apple
(3) Big Grape (4) None of these

121. Which was the first Indian company to be listed in NASDAQ?
(1) TCS (2) Wipro
(3) Infosys (4) Cognizant

122. 'TAREGNA'- a village in Bihar was in the news because of
(1) Rahul Gandhi's election campaign
(2) Viewing of the solar eclipse
(3) A visit by Hillary Clinton
(4) The Naxal massacre of tribals

123. The driver for Formula One's Force India Team is
(1) Mark Webber (2) Adrian Sutil
(3) Felipe Massa (4) Jensen Button

124. (MDRT)- Million Dollar Round Table consists of
(1) World's richest industrialists
(2) World's most powerful golfers
(3) Global Association of Life Insurance professionals
(4) World's richest sports persons

125. Who declares the 'Credit Policy' of India?
(1) The Finance Ministry
(2) Reserve Bank of India
(3) The Commerce Ministry
(4) The Indian Bank's Association

126. Name of the body which regulates the stock Exchanges in India is
(1) CBI (2) CID
(3) SHCIL (4) SEBI

127. Rafflesia arnoldii is its botanical name and it holds a world record in the plant world. What's the record?
(1) Tallest cacti (2) Largest bloom
(3) Longest surviving tree (4) Smallest plant

128. Temujin was a famous historical character of the world, but we know him by a different name, which one?
(1) Babur (2) Genghis Khan
(3) Kublai Khan (4) Empereor Nero

129. What is short selling in stock market lingo?
(1) Selling the shares which you do not own.
(2) Selling a share after owning it for a short period of time.
(3) Selling all the shares in your booty which are not productive.
(4) Selling of shares which are quoted for a short period of time.

130. Mahatma Gandhi was nominated for Nobel Peace prize for the first time in the year 1937 and last time in the year 1948. How many times in between these two years was he nominated?
(1) 1 (2) 3
(3) 2 (4) 0

131. Which organization instituted the Nobel Prize in Economics?
(1) Svenska Handelsbanken
(2) International Monetary Fund
(3) Economics Department of Oxford University
(4) Sveriges Riksbank

132. Film stars have been endorsing Lux beauty soap since ages. Who was the first actress of Bollywood to do so?
(1) Madhubala (2) Leela Naidu
(3) Suraiya (4) Leela Chitnis

133. India's interstate cricket tournament is named after Kumar Shri Ranjitsinghji Vibhaji Jadeja or in short Ranji, the great cricket legend. He played his cricket in England and also played for English national team. Which country he used to play for in the league cricket of England?
(1) Sussex (2) Essex
(3) Somerset (4) Surrey

134. As a practice, all major states capitals have offices of Reserve Bank of India with only one exception. Which city of India has an office of RBI without being a state capital?
(1) Pune (2) Allahabad
(3) Baroda (4) Nagpur

135. This distinguished Indian was born on 28 December 1937 in Surat. He has completed the Advanced Management Program at Harvard Business School, USA. He turned down a job offer from IBM to join his family business and today he is one the top business leaders of India. Name the tycoon.
(1) Ratan Tata (2) Anil Ambani
(3) Azim Premji (4) Anand Mahindra

136. It's known to be used in Egypt in 3000 BC, but its name had been derived from old French expression. It is made from the fermentation of ethanol in a process that yields its key ingredient, ethanoic acid. What is it?
(1) Tobasco Sauce (2) Phenol
(3) Vinegar (4) Soya Sauce

137. The name of this Japanese company literally means 'three diamonds'. Which one?
(1) Sanyo (2) Mutsuhito
(3) Mitsubishi (4) Suzuki

138. Who were the first twins to play test cricket?
(1) Merv and Kim Hughes
(2) Ian and Greg Chappel
(3) Surinder and Mohinder Amarnath
(4) Steve and Mark Waugh

139. Which country has the largest rail network in the world?
(1) India (2) China
(3) USA (4) UK

140. Who won the 2009 French open tennis ladies tittle?
(1) Svetlana Kuznetsova (2) Kristina Mladenovic
(3) Maria Sharapova (4) Venus Williams

141. Who invented the microchip?
(1) Robert Noyce and Jack Kilby
(2) Larry Page and Sergey Brin
(3) Saber Bhatia
(4) Gordon Moore & Robert Noyce

142. Which is the largest landlocked country in the world?
(1) Switzerland (2) China
(3) Kazakhstan (4) Mongolia

143. Which country makes Panama hats?
(1) Ecuador (2) Panama
(3) Cuba (4) Bolivia

144. Who was the Prime Minister of India when the 42nd Amendment Bill bringing in the changes in the constitution was passed?
(1) Rajiv Gandhi (2) VP Singh
(3) Indira Gandhi (4) Morarji Desai

145. Which one of the following planets lost its planet status recently?
(1) Moon (2) Neptune
(3) Pluto (4) Saturn

146. One barrel of oil is approximately equivalent to how many litres?
(1) 200 (2) 150
(3) 120 (4) 160

147. Five year plans in India are finally approved by the
(1) Union Cabinet
(2) President on the advice of prime minister
(3) National Development Council
(4) Planning commission

148. Many a time we read in the newspapers about Southern Cone of South America which includes Argentina, Chile, Paraguay, Uruguay and Peru. Which is the most spoken language of this geographical area?
(1) Portuguese (2) Spanish
(3) French (4) English

149. Which one of the following is not a member of the Organization of Petroleum Exporting Countries (OPEC)?
(1) Algeria (2) Brazil
(3) Ecuador (4) Nigeria

150. Who invented 'computer laptop'?
(1) Arthur Wynna-USA
(2) Q. Daimler-Germany
(3) Sinclair-Britain
(4) None of these

Answers

1. (2)	**2.** (*)	**3.** (1)	**4.** (4)	**5.** (1)	**6.** (3)	**7.** (4)	**8.** (2)	**9.** (2)	**10.** (4)
11. (3)	**12.** (4)	**13.** (1)	**14.** (3)	**15.** (4)	**16.** (1)	**17.** (3)	**18.** (3)	**19.** (2)	**20.** (1)
21. (1)	**22.** (3)	**23.** (2)	**24.** (4)	**25.** (2)	**26.** (3)	**27.** (3)	**28.** (4)	**29.** (3)	**30.** (3)
31. (1)	**32.** (2)	**33.** (2)	**34.** (1)	**35.** (4)	**36.** (4)	**37.** (2)	**38.** (2)	**39.** (4)	**40.** (4)
41. (2)	**42.** (1)	**43.** (4)	**44.** (2)	**45.** (2)	**46.** (2)	**47.** (4)	**48.** (3)	**49.** (2)	**50.** (1)
51. (2)	**52.** (1)	**53.** (2)	**54.** (4)	**55.** (2)	**56.** (2)	**57.** (1)	**58.** (4)	**59.** (3)	**60.** (3)
61. (1)	**62.** (2)	**63.** (1)	**64.** (4)	**65.** (4)	**66.** (3)	**67.** (1)	**68.** (2)	**69.** (4)	**70.** (4)
71. (4)	**72.** (4)	**73.** (3)	**74.** (4)	**75.** (4)	**76.** (2)	**77.** (2)	**78.** (2)	**79.** (4)	**80.** (1)
81. (3)	**82.** (4)	**83.** (3)	**84.** (2)	**85.** (4)	**86.** (3)	**87.** (2)	**88.** (3)	**89.** (1)	**90.** (3)
91. (4)	**92.** (3)	**93.** (1)	**94.** (2)	**95.** (1)	**96.** (2)	**97.** (4)	**98.** (4)	**99.** (2)	**100.** (2)
101. (4)	**102.** (2)	**103.** (4)	**104.** (2)	**105.** (1)	**106.** (2)	**107.** (1)	**108.** (1)	**109.** (3)	**110.** (2)
111. (1)	**112.** (4)	**113.** (*)	**114.** (1)	**115.** (4)	**116.** (1)	**117.** (1)	**118.** (3)	**119.** (4)	**120.** (2)
121. (3)	**122.** (2)	**123.** (2)	**124.** (3)	**125.** (2)	**126.** (4)	**127.** (2)	**128.** (2)	**129.** (1)	**130.** (2)
131. (4)	**132.** (4)	**133.** (1)	**134.** (4)	**135.** (1)	**136.** (3)	**137.** (3)	**138.** (4)	**139.** (3)	**140.** (1)
141. (1)	**142.** (3)	**143.** (1)	**144.** (3)	**145.** (3)	**146.** (4)	**147.** (4)	**148.** (2)	**149.** (2)	**150.** (4)

Note: '*' No option is correct

Answers with Explanations

1. (2) The sum of three numbers can be 15 in the following three ways.
(i) $15 = 3 + 6 + 6$
The first, second and third throws can be (3, 6, 6), (6, 3, 6) and (6, 6, 3) respectively.
$\therefore$ Total number of ways in which 3, 6 and 6 can be obtained is 3.
(ii) $15 = 4 + 5 + 6$
The first, second and third throws can be either of 4, 5 and 6.
$\therefore$ Total number of ways in which 4, 5 and 6 can be obtained is 6.
(iii) $15 = 5 + 5 + 5$
The first, second and third throws can be 5, 5 and 5.
$\therefore$ Total number of ways in which 5, 5 and 5 can be obtained is 1.
$\therefore$ The total number of ways in which the sum of throws can be 15 is $3 + 6 + 1 = 10$
The total number of ways in which the first roll will be a 4 is 2
$\therefore$ Required chance $= \frac{2}{10} = \frac{1}{5}$

2. (*) Let x and y be the speed of boat and stream respectively.
When boat goes downstream
$$\frac{30}{x+y} = 2 \quad \ldots(i)$$
and when boat goes upstream
$$\frac{30}{x-y} = 6 \quad \ldots(ii)$$
Dividing Eq.(i) by Eq. (ii),
$$\frac{x-y}{x+y} = \frac{1}{3}$$
$\therefore \quad x = 2y$
Substituting the value of x in Eq. (i)
$$\frac{30}{2y+y} = 2$$
$\therefore \quad y = 5$ km/h
$\therefore \quad x = 10$ km/h
The correct answer is not there in the options.

3. (1) A number divisible by 4 formed using the digits 1, 2, 3, 4 and 5 should have the last two digits 12, 24, 32 or 52.

In each of these cases, the five digit number can be formed using the remaining 3 digits in $3 \times 2 \times 1 = 6$ ways.

$\therefore$ A number divisible by 4 can be formed in $6 \times 4 = 24$ ways.

Total numbers that can be formed using the digits 1, 2, 3, 4 and 5 without repetitions $= 5! = 120$

$\therefore$ Required probability $= \frac{24}{120} = \frac{1}{5}$

4.(4) $\text{Mean} = \frac{\text{No. of observations} \times \text{deviation from Fixed value} + \text{Sum of deviation}}{\text{No. of observations}}$

$= \frac{20 \times 23 + 70}{20} = 26.5$

5.(1) Let x g of gold should be added to the alloy to increase the percentage of gold in it to 90%.

$\therefore \quad \frac{80\% \text{ of } 50 + x}{50 + x} = \frac{90}{100}$

$\Rightarrow \quad \frac{40 + x}{50 + x} = \frac{9}{10}$

$\Rightarrow \quad 400 + 10x = 450 + 9x$

$\therefore \quad x = 50 \text{ g}$

6. (3) Let their incomes be Rs $7x$ and Rs $3x$ and let the expenses be Rs $5y$ and Rs $2y$.

$\therefore \quad 7x - 5y = 300 \quad \ldots(i)$

and $\quad 3x - 2y = 300 \quad \ldots(ii)$

From Eqs. (i) and (ii)

$x = 900$ and $y = 1200$

Income of first person $= 7x = 7 \times 900$

$= \text{Rs } 6300$

7. (4) Last month suppose the family consumption was 1 kg. To keep expenditure at Rs 24, its new consumption should be $\frac{24}{27} = \frac{8}{9}$ kg.

$\therefore$ Percentage decrease in consumption

$= \frac{\left(1 - \frac{8}{9}\right)}{1} \times 100$

$= 11.1\%$

8. (2) From a certain station, there will be a ticket for other 9 stations and there are 10 stations on this railway line.

$\therefore$ The number of different journey tickets $= 9 \times 10 = 90$

9.(2) Circumference of a circle $= 2\pi r$, where r is the radius,of the circle.

When circumference increases by 5%, the radius also increases by 5%.

$\therefore$ New radius $= 1.05r$

$\therefore$ As area $\propto (\text{radius})^2$

$\therefore$ New area $= (1.05)^2 \times$ old area

$= 1.1025 \times$ old area

$\therefore$ Increase in area $= 0.1025$ of old area

$\therefore$ Percentage increase in area $= 10.25\%$

10. (4) The three vowels in ABACUS are A, A and U. These three can be arranged among themselves in 3!/2! ways.

As the three vowels are to appear together, we consider them as one letter. Thus, we have four letters; (AAU), B, C and S.

This can be done in 4! ways.

$\therefore$ Required number of ways $= \frac{4! \times 3!}{2!}$

11. (3) We can see that $\frac{8000}{6000} = \frac{700}{525} = \frac{4}{3}$

$\therefore$ The time that 525 requires to grow to 700 will be the same as the time that 6000 requires to grow to 8000.

$\therefore$ Required time $= 4$ yr

For (Q. 12-14) At the start of the game J and B together had 4 times as much money as T had.

Let the money that J, B and T had be j, b and t respectively.

Then, $\quad j + b + t = 4t + t = 5t$

$\therefore$ The total money is divisible by 5.

Similarly, from the other three statements in the data, we find that the total money is divisible by 4 and 3 as well.

$\therefore$ The total money is divisible by LCM of 3,4 and 5=60.

Let the total money be $60x$.

At the start of the game,

$j + b = 4t$

$\therefore \quad j + b + t = 5t = 60x$

$\therefore \quad t = 12x$

$\therefore \quad j + b = 48x \quad \ldots(i)$

$t + b = 3j$

$\therefore \quad t + b + r = 60x$

$\Rightarrow \quad 4j = 60x$

$\therefore \quad j = 15x \quad \ldots(ii)$

From Eqs. (i) and (ii)

$15x + b = 48x$

$\therefore \quad b = 33x$

$\therefore$ The amounts with J, B and T at the start of the game were $15x$, $33x$ and $12x$ respectively.

Now, at the end of the game,

$j + b = 3t$

$\therefore \quad 4t = 60x$

$\therefore \quad t = 15x$

$\therefore \quad j + b = 45x \quad \ldots(iii)$

Also, $\quad t + b = 2j$

$\therefore \quad 3j = 60x$

$\therefore \quad j = 20x \quad \ldots(iv)$

From Eqs. (iii) and (iv)

$b = 25x$

$\therefore$ The amounts with J, B and T at the end were $20x$, $25x$ and $15x$ respectively.

12. (4) The fraction of total money that T had at the beginning was

$\frac{12x}{60x} = \frac{1}{5}$

13. (1) J won $20x - 15x = 5x$

$\therefore$ Required fraction $= \frac{5x}{60x} = \frac{1}{12}$

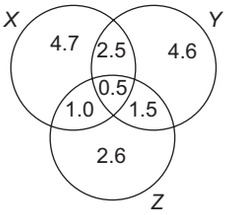
14. (3) As B lost Rs 200,
$\therefore \quad 33x - 25x = 200$
$\therefore \quad 8x = 200$
$\therefore \quad x = 25$
$\therefore \quad 33x = 825$
$\therefore$ B started with Rs 825.

15. (4) Here a and b are negative and c is positive.
Let $a = -6, b = -4$ and $c = 2$
(I) $a - b = -6 + 4 = -2$
and $a - c = -6 - 2 = -8$
So, $a - b > a - c$
So, I is not true.
(II) Taking the same values of a, b and c,
we have $a/c = -6/2 = -3$
$b/c = -4/2 = -2$
Since, $-3 < -2$, $a/c < b/c$
So, II is true.
(III) For any value of b and c,
$1/b < 1/c$, since b is negative and c is positive.
So, III is true.
So, statements II and III are true.

16. (1) Let the length of the side of first square be x.
Then, the length of the diagonal $= \sqrt{x^2 + x^2} = 4\sqrt{2}$
$\therefore \quad 2x^2 = 32$
$\therefore \quad x^2 = 16$
$\therefore \quad x = 4$
Now, the area of the an other square is twice that of the first square.
So, the area of the second square $= 2x^2 = 32$
If the length of the second square is y, then $y^2 = 32$
$\therefore$ Length of the diagonal of this square $= \sqrt{2y^2}$
$= \sqrt{64}$
$= 8$ cm

17. (3) **Formula** If the length, breadth and height of the rectangular box are l, b and h respectively, then the maximum length of the pencil that can be kept in the box will be equal to the length of its greatest diagonal,
This length is given as $\sqrt{l^2 + b^2 + h^2}$
Dimensions of the box are given as $8 \times 6 \times 2$.
So, the maximum length of the pencil that can be kept in the box
$= \sqrt{8^2 + 6^2 + 2^2} = \sqrt{104} = 2\sqrt{26}$ cm

For (Q. 18-19)
Total readership of X is 8.7 lakhs.
$\therefore \quad X + 2.5 + 0.5 + 1 = 8.7$
$\therefore$ People who read only newspaper $X = 4.7$ lakhs.
Total readership of Y is 9.1 lakhs
$\therefore \quad 2.5 + 0.5 + 1.5 + Y = 9.1$
$\therefore$ People who read only newspaper $Y = 4.6$ lakhs.
Also, total readership of Z is 5.6 lakhs
$\therefore \quad Z + 1 + 0.5 + 1.5 = 5.6$
People who read only newspaper $Z = 2.6$ lakhs
Therefore,we have

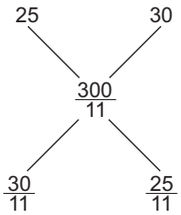


18. (3) Number of people who read at least one paper $= 4.7 + 4.6 + 2.6 + 1 + 1.5 + 2.5 + 0.5 = 17.4$

19. (2) From the above venn diagram
Number of people who read only one newspaper $= 4.6 + 4.7 + 2.6 = 11.9$

20. (1) Percentage of Sonali's salary invested in insurance policies = 15%
Percentage of salary spent in shopping and household expenses = 55%
$\therefore$ Percentage of her salary saved $= 100 - (15 + 55)$
$= 30\%$
Let Sonali's monthly salary be Rs x.
$\therefore \quad 30\%$ of $x = 12750$
$$x = \frac{12750 \times 100}{30} = \text{Rs } 42500$$

21. (1) There are two varieties of tea, one worth Rs 25 per kg and the other worth Rs 30 per kg.
Now, by selling the blended variety at Rs 30 kg, profit should be 10%.
So, the cost price of the blended tea should be
$$\frac{30}{1.1} = \text{Rs}\frac{300}{11}\text{per kg}$$
This problem can be depicted through alligation as below



$\therefore$ Ratio in which the tea worth Rs 25 per kg must be blended with tea worth Rs 30 per kg $= \frac{30}{25}$
$\therefore$ When tea worth Rs 30 per kg is 30 kg. Then, quantity of tea worth Rs 25 per kg $= \frac{30}{25} \times 30 = 36$ kg

22. (3) Of the 480 students, let there be x boys and y girls.
$\therefore \quad x + y = 480 \quad \ldots(i)$
Total pass percentage was 75%.
So, number of students who passed
$= 0.75 \times 480 = 360$
Now, it is given that 85% of the girls and 70% of the boys passed.
$\therefore \quad 0.7x + 0.85y = 360 \quad \ldots(ii)$
Solving Eqs. (i) and (ii), we get
$x = 320$ and $y = 160$

23. (2) 300 g solution has 40% salt in it.

So, the amount of salt in it $=\frac{40\times300}{100}=120\,g$

For the salt percentage to be 50%,let x g of salt should be added

$$\frac{120+x}{300+x}=\frac{50}{100}=\frac{1}{2}$$

$\therefore \quad 240 + 2x = 300 + x$

$\therefore \quad x = 60$ g

24. (4) In the second row each number is obtained by multiplying the corresponding number in the first row by 8.

$\therefore$ The number that should replace the question mark $= 192/8 = 24$

25. (2) The rectangle, triangle and the circle each move one step to the left in every subsequent row and the colour of two figures is changed. Thus, the rightmost figure in the third row is the triangle. We can also see that both the colours, white and grey are present in each row. So, the third row should have a grey triangle. In the first and second rows, the rectangle is labeled '+', the triangle 'C' and the circle 'T'. Thus, the grey triangle in row three should also be labeled 'C'.

26. (3) Using the digits 0, 1, 2, 3, 4 and 5, five digit numbers divisible by 3, can be formed when the sum of all the digits is divisible by 3.

There are two possible combinations.

(i) 1, 2, 3, 4, 5
Total number of numbers formed using these digits = 5! = 120

(ii) 0, 1, 2, 4, 5
Total number of numbers formed using these digits = 4 × 4 × 3 × 2 = 96

Thus, total numbers = 120 + 96
= 216

27. (3) Since, $\frac{2}{5}$th of the work is completed in the 25 days, remaining $\frac{3}{5}$th of the work is to be completed in 25 days.

Let x men should work for 25 days to complete $\frac{3}{5}$th of the work.

$$\frac{M_1 D_1 H_1}{W_1}=\frac{M_2 D_2 H_2}{W_2}$$

$$\therefore \quad \frac{105\times25\times8\times5}{2}=\frac{x\times25\times9\times5}{3}$$

$$\therefore \quad x=\frac{105\times25\times8\times5\times3}{25\times9\times5\times2}=140$$

$\therefore$ Additional men be employed = 140 − 105
= 35

28. (4) A can build the structure in 8 days.

In 1 day A can built $=\frac{1}{8}$ part of structure

In 1 day B can break $=\frac{1}{3}$ part of structure

Amount of work done by A in 4 days = 4/8 = 1/2

Work done by A and B working togethor in 2 days.

$$=2\left(\frac{1}{8}-\frac{1}{3}\right)=\frac{2\times(-5)}{25}=\frac{-5}{12}$$

Fraction of structure still to be built

$$=\frac{1}{2}+\frac{5}{12}=\frac{11}{12}$$

Let A will build up the remaining structure, in x days. Then,

$$\frac{x}{8}=\frac{11}{12}$$

$$\therefore \quad x=\frac{22}{3}$$

29. (3) Let the first term and the common ratio of the Geometric Progression be a and r respectively.

$\therefore \quad a + ar = 12$

$\therefore \quad a(1 + r) = 12 \quad \ldots$ (i)

and $\quad ar^2 + ar^3 = 48$

$\therefore \quad ar^2(1 + r) = 48 \quad \ldots$(ii)

Dividing Eq. (ii) by Eq. (i),

$$\frac{ar^2(1+r)}{a(1+r)}=\frac{48}{12}=4$$

$\therefore \quad r^2 = 4$

$\therefore \quad r=\pm\sqrt{4}=\pm2$

Since, the terms of the Geometric Progression are alternately positive and negative, so $r = -2$

$\therefore$ From Eq. (i), $\quad a(1 - 2) = 12$

$\therefore \quad a = -12$

30. (3) Mean of the given numbers $=\frac{a+b+8+5+10}{5}=6$

$\therefore \quad a + b = 30 - 23 = 7$

So, option 4 can be eliminated.

Now, variance

$$\therefore \quad =\frac{1}{5}[(a-6)^2+(b-6)^2+(5-6)^2+(10-6)^2+(8-6)^2]$$

$$\therefore \quad \frac{1}{5}[(a-6)^2+(b-6)^2+21]=6.8$$

$\therefore (a - 6)^2 + (b - 6)^2 + 21 = 34$

$\therefore \quad (a - 6)^2 + (b - 6)^2 = 13$

Only option 3 fits into the above equation.

31. (1) Value of goods = Rs 15000

Commission he receives $=12\frac{1}{2}\%$

$\therefore$ Total commission received $=\frac{12.5\times15000}{100}$

$=$Rs 1875

32. (2) $\sqrt{110.25}\times\sqrt{0.01}\div\sqrt{0.0025}-\sqrt{420.25}$

$=10.5 \times 0.1 \div 0.05 - 20.5$

$=10.5\times\frac{0.1}{0.05}-20.5$

$= 10.5 \times 2 - 20.5$

$= 21 - 20.5 = 0.5$

33. (2) The total number of respondents under 31 = 33 + 33 = 66
The number of respondents under 31 who said that Blues is their favourite style of music = 2 + 3 = 5
$\therefore$ The required percentage $= \frac{5}{66} \times 100 \approx 7.5\%$

34. (1) There were 33 respondents in all in the 21-30 age group.
There were 12 respondents in this group whose favourite style was Rock music.
$\therefore$ There were 33 – 12 = 21 respondents whose favourite style was other than Rock music.
$\therefore$ The required percentage $= \frac{21}{33} \times 100 \approx 64\%$

35. (4) The total number of respondents = 33 + 33 + 68 = 134
The number of respondents who said that Jazz is their favourite style of music = 1 + 4 + 11 = 16
$\therefore$ The required percentage $= \frac{16}{134} \times 100 \approx 12\%$

36. (4) The total number of students in the Arts faculty = 1049
The total number of non-US students in the Arts faculty = 79 + 21 + 6 + 2 + 4 = 112
$\therefore$ The required percentage $= \frac{112}{1049} \times 100 \approx 10.67\%$
$\approx 11\%$

37. (2) Let the number of total students in the University be x.
23% of the total students = 1049
$\Rightarrow$ $0.23x = 1049$
The number of students in the Engineering faculty is 9% of the total students.
$\therefore$ The total number of Engineering students
$$= \frac{9}{100} \times \frac{1049}{0.23} \approx 410 \text{ students}$$

38. (2) From the previous solution, if there are total x students.
Then, $0.23x = 1049$
$$x = \frac{1049}{0.23} = 4560$$

39. (4) The Science students are 21% of the total number of students.
6% of Science students are Asians.
The total number of students in the University = 4560
$\therefore$ The number of Asian students studying Science
$$= \frac{6}{100} \times \frac{21}{100} \times 4560 \approx 57$$

40. (4) Medical students are 5% of the total students.
$\therefore$ The total number of medical students
$$= \frac{5}{100} \times 4560 = 228$$
Out of these, 34 are European.
This represents $\frac{34}{228} \times 100 = 14.9\% \approx 15\%$ of this faculty.

41. (2) From (i) statement A gives B and C as many tractors that they already have.
$\therefore$ Number of tractors with A > Number of tractors with B and C together.

42. (1) Option 2 can be eliminated as 'many' and 'majority' are not same.
The passage states that "... good basic skills in reading, communication and mathematics play an important role …" and not reliance on communication skills over basic skills in reading and mathematics. Option 3 can be eliminated.
The passage states 'most' of the fastest growing jobs. Option 4 implies 'all' the fastest growing jobs. Option 4 can be eliminated.
The passage mentions that good basic skills in reading, communication and mathematics are important prerequisites for jobs that do not require a college degree-jobs such as that of a desktop publisher, as mentioned in option 1.

43. (4) Options 1, 2 and 3 can be concluded from the data given in the passage.
The passage states that "If there are no crop protection products, then crop yield per acre will drop by more than 50%."
Option 4 states if crop yield per acre drops by more than 50%, then crop protection products have not been used.
Hence, the conclusion in option 4 cannot be concluded because drop can be due to other reasons.

44. (2) If flight 409 is cancelled, then the manager cannot arrive on time.
The other valid conclusion would be the manager can arrive on time, therefore flight 409 is not cancelled.
There are other two conclusions that may or may not be valid- they may or may not be sufficient conditions.
The manager cannot arrive on time, therefore flight 409 is cancelled.
Flight 409 is not cancelled, therefore the manager can arrive on time.
Evelyn states that the manager will not arrive at time- there is no possibility that the manager can arrive on time. But, based on data, if flight 409 is not cancelled, then the manager can arrive on time may or may not happen. Hence, Evelyn's conclusion that the manager will not arrive on time is unwarranted.

45. (2) The passage assumes that a higher percentage of accident related deaths in planes compared to cars implies cars are safer.
We need an argument that attacks the premise that percentage is the only way to determine safety; or we need an argument that brings in another element that shows cars are not as safe as percentages make them to be.
Options 1, 3 and 4 are not related to the main statement.
If the number of car accidents is several hundred thousand times higher than the number of plane

accidents, it nullifies the premise of percentages being the deciding factor as well as presents data that supports that cars are not as safe as they are made out to be vis-a-vis planes.

46. (2) From the chart, it is clear that only Industry B and C contribute to company S.

47. (4) Observe that Industry B processes element 4 and 5.
But elements 4 and 5 are also processed by Industry A and C.
Hence, the proportion of element 4 that gets processed in Industry B is unknown.
Similarly, the proportion of element 5 that gets processed in Industry B is unknown.
So, the percentage of total production of listed elements processed by Industry B cannot be determined.

48. (3) Industry A processes all the listed elements, *ie*, Element 1,2,3,4 and 5.
The total annual production of all these elements is 100000 tons.

49. (2). RB shows the same relationship as opposite of decrease is increase and opposite of descend is ascend.
$\therefore$ Decrease : Increase :: Descend : Ascend

50. (1) Opposite of modern is ancient and opposite of young is old.
So, Modern : Ancient :: Young : Old
$\therefore$ AQ will be the correct pair.

51. (2) As superset of part is whole.
Similarly, superset of class is school.
So, Part : Whole :: Class : School
$\therefore$ BQ will be the correct pair.

52. (1) Summit and Apex are synonyms. The only option which is close of being a synonym is AQ.
Beautiful : Pretty.

53. (2)

B	E	A	U	T	I	F	U	L		B	U	T	T	E	R
↓	↓	↓	↓	↓	↓	↓	↓	↓	and	↓	↓	↓	↓	↓	↓
5	7	3	0	4	1	2	0	8		5	0	4	4	7	9

Based on this coding system, the code for FUTURE is 204097.

54. (4) If the month that ends on Wednesday have 31 days, there will be 5 Mondays otherwise will have 4 Mondays. So, cannot be specified.

55. (2) From the first row, $(17 \times 12)/2 = 102$
Similarly, from the second row, $(15 \times 10)/2 = 75$

56. (2) From the first row, $12 \times 14 \times 2 = 336$
Similarly, from the second row, $15 \times 16 \times 2 = 480$

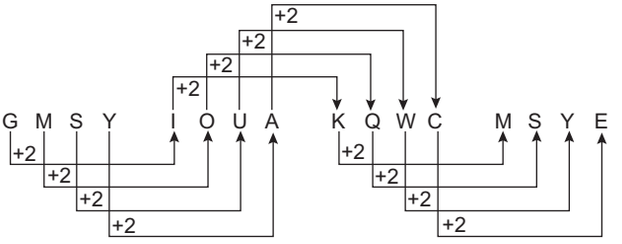
57. (1) From statement (III), B and G must be on different boats.
$\therefore$ If F and B are in one boat, G is in the other boat.

58. (4) From statement (IV), the maximum number of persons in one boat can be four only.
$\therefore$ There must be at least three people in the complete list of people who must be sitting in either boat. Only option (4) has 3 persons.

59. (3) In the first figure there are 4 vertical and 3 horizontal lines.
In the following figures both the vertical and horizontal lines reduce by 1.
$\therefore$ In the fourth figure there will be only one vertical line.

60. (3) In the first figure there are 3 black circles on 3 lines.
In the second figure the black circles remain in the same places and the lines are turned in the opposite direction.
In second figure there are 2 black circles on two lines.
$\therefore$ In the fourth figure the lines are turned in the opposite direction and circles remain in the same places.

61. (1)



62. (2) Mohan spent 30% on landline, 40% on GSM and 30% on Airtel and spent 30 min and 100 min on STD calls and landline calls respectively.
This data can be arranged in a tabular manner.

Mohan	Airtel	GSM	Landline
STD (minutes)	9	12	9
Local (minutes)	30	40	30

Mohan made STD calls in the range of 50-200 km.
Hence, Mohan's STD calls cost is $(9 \times 1.5) + (12 \times 1.5) + (9 \times 2) = 13.5 + 18 + 18 =$ Rs 49.50
His local calls cost is $(30 \times 1) + (40 \times 1) + (30 \times 2) =$ Rs 130
Price of plan = Rs 99
Hence, Mohan's total cost = (49.50 + 130) + 99
= Rs 278.50
Rohan spent 40% on landline, 30% on GSM and 30% on Airtel and spent 18 min and 120 min on STD calls and landline calls respectively.
This data can be arranged in a tabular manner.

Rohan	Airtel	GSM	Landline
STD (minutes)	5.4	5.4	7.2
Local (minutes)	36	36	48

Rohan made STD calls in the range of 50-200 km.
Hence, Rohan's STD calls cost is $(5.4 \times 1.5) + (5.4 \times 1.5) + (7.2 \times 2) = 8.1 + 8.1 + 14.4 =$ Rs 30.60
His local calls cost is $(36 \times 1) + (36 \times 1) + (48 \times 2) =$ Rs 168
Price of plan = Rs 99
Hence, Rohan's total cost = (30.60 + 168) + 99
= Rs 297.60
Hence, Rohan spent more.

63. (1) Solve this problem by comparing the charges incurred while sending 38, 40, 60 and 59 local SMS respectively under the old scheme and the new scheme.

Number of SMS	Old Scheme		New Scheme		
	Rate	Total Cost	Monthly Charge	Rate	Total Cost
38	1.5	57	35	0.6	57.8
40	1.5	60	35	0.6	59
60	1.5	90	35	0.6	71
59	1.5	88.5	35	0.6	70.4

Thus, a person sending 38 local SMS per month will not benefit from the scheme.

64. (4) To answer this question we have to consider all the options.

(1) Cost of 30 STD (500 + km) calls on Airtel = 30 × 1.5= Rs 45
Cost of 55 local calls on Airtel or GSM/CDMA = Rs 55
Total bill = 45+ 55+ 99 = 199

(2) Cost of 16 STD (150 km) calls on Airtel = 16× 1.5 = Rs 24
Cost of 76 local calls on Airtel or GSM/CDMA = Rs 76
Total bill = 24 + 76 + 99 = Rs 199

(3) Cost of 10 STD (250 km) call each = 10× (1.5 + 2.5 + 2.5)=Rs 65
Out of 30 local calls if 25 are on Airtel and 5 on landline.
The cost = 25 × 1 + 5 × 2 = Rs 35
Total bill = 65 + 35 + 99 = Rs 199

(4) Cost of STD (500 + km)
= 8 × 1.5 + 4 × 3 + 7 × 3.5 = Rs 48.50
Cost of 55 local calls = Rs 55
Total bill = 48.50 + 55 + 99 = Rs 202.50

65. (4) 80% of the ISD bill came from calling the rest of the world.
Hence, 20% of the bill came from the other regions.
The total time spent on ISD calling was 12 min.
Since, the ISD rates for each region vary, the 12 min cannot be proportionately divided based on the bill.
So, it is not possible to find out the calling duration for the other two regions.
Hence, the answer cannot be determined.

66. (3) Total demand = 3000 + 600 + 2500 + 1200 + 3300
= 10600 unit

$$\text{Average demand} = \frac{\text{Total demand}}{\text{Number of TV companies}}$$

$$= \frac{10600}{5} = 2120 \text{ unit}$$

Total production = 1500 + 1800 + 1000 + 2700 + 2200 = 9200 unit.
Average production

$$= \frac{\text{Total production}}{\text{Number of TV companies}} = \frac{9200}{5} = 1840 \text{ unit}$$

Difference between the average demand and the average production
= 2120 − 1840
= 280

67. (1) Production of company D = 2700
Production of company A = 1500

$$\frac{\text{Production of company D}}{\text{Production of company A}} = \frac{2700}{1500} = 1.8$$

∴ Production of the company D is 1.8 times of the production of the company A.

68. (2) Paper cost is 10% of the total cost.
∴ Angle for the sector representing paper cost = (10/100) × 360 degrees = 36 degrees.

69. (4) 2% of the cost is Rs 2000 for 12500 copies.

$$\text{The total cost} = 2000 \times \frac{100}{2} = \text{Rs } 100000$$

∴ Cost for 1 copy = (100000/12500) = Rs 8
For a profit of 5% SP of a copy
= 8 × 1.05 = Rs 8.40.

70. (4) we know that 'Fruit' grows on 'Tree' and 'Tree' is called 'Sky'. So, 'Fruit grow on 'Sky'.

71. (4) The blank requires a nominative possessive pronoun in the first person. Therefore, the correct word is 'mine'. When the possessive comes before the noun, 'my' is used (*eg*, *My* close friend). When the possessive comes after the noun, 'mine' is used (*eg*, A close friend of *mine*).

72. (4) Logically, options 1 and 3 can be eliminated, since he would not take care of her 'when' or 'after' she was restored to health. Although 'before' may fit in the context of the sentence, 'till' is the best word to bring out the meaning that he took care of her till she was restored to health.

73. (3) The words 'since then' in the given sentence necessitate the use of the present perfect tense.
Therefore, the verb should be 'has changed'.

74. (4) The word 'diarrhoea' may seem to be incorrectly spelt for users of British English; however, it is correctly spelt as 'diarrhea' according to US English.
'Diaper' and 'dichotomy' have been spelt correctly.
The word 'dais', meaning a raised platform usually at the front of a room for the purpose of a lectern or a seat of honour, has been incorrectly spelt as 'dias'.

75. (4) The word 'dissipate' has been spelt correctly. The correct spelling of the other words is 'superintendent', 'sieve' and 'allotted'.

76. (2) The correct preposition in the first blank should be 'in', since one stays 'in' a city. This eliminates options 1 and 3. In the second blank, the correct preposition is 'at', since one stays 'at' an address.

77. (2) 'Aural' means 'of or pertaining to the ear or the sense of hearing'.
'Aurally challenged' is a term used to refer to people who are deaf or hearing impaired.
A 'euphemism' is a figure of speech involving 'the substitution of a mild, indirect, or vague expression for one thought to be offensive, harsh or blunt'.
Therefore, the use of the term 'aurally challenged' is a 'euphemism', since it is a mild term substituted for the harsher word 'deaf'.

78. (2) An error could be a mistake, slip or blunder, but not a defect. Therefore, option 1 can be eliminated.
'Blunder' is 'a gross, stupid, or careless mistake' and is a higher degree than the others; therefore, option 4 can also be eliminated.
A 'fault' is defined as 'a defect or imperfection'.

79. (4) 'Flaunt' means 'to display conspicuously'. This logically contradicts the sentence, since a man who 'flaunts' the rules of ethical conduct would not be considered a beast. This eliminates option 1.
Similarly, a man who 'ignores' the rules of ethical conduct would also not be considered a beast. This eliminates option 2.
A beast would be one who 'breaks' or 'flouts' the rules. 'Flout' means 'to treat with disdain, scorn or contempt' and fits the context perfectly.
In the second blank, the word 'as' does not fit the sentence grammatically. This eliminates option 3.

80. (1) Inverted commas, also known as quotation marks, are used either to enclose the exact words of a speaker or to denote words as titles or when a special meaning is conferred upon them.
The correct sentence should be: Part of Australia is known to the natives as 'The Outback'.

81. (3) Commas are used to mark off words used in addressing people and also before certain coordinative conjunctions.
The correct sentence should be: I know that you want to learn to drive, Rima, but you are too young.

82. (4) 'Veracious' means 'truthful or honest'.

83. (3) 'Perturb' means 'to disturb or disquiet greatly in mind'.

84. (2) The phrase 'with a high hand' means 'in an arrogant or dictatorial manner'.
The words sympathetically, democratically and generously do not fit in this context.
The word closest in meaning is "oppressively", which means 'unjustly harsh or tyrannical'.

85. (4) To "rack one's brains" means 'to try very hard to think or to remember something'.

86. (3) The noun phrase 'cold feet' refers to 'a loss of confidence, or a sudden onset of uncertainty, fear or nervousness'. It is in no way related to the literal meaning of the temperature of one's feet.

87. (2) "To eat one's words" is an idiom meaning 'to retract one's statement, especially with humility'.
The most appropriate meaning from the given options is 'to take back what you have said'.

88. (3) The first sentence is in the form of a question, "Shouldn't they have checked your tickets?" expressing the speaker's opinion that 'I think they should have checked your tickets'.
The second sentence starts with "I wonder if they should have..." The word 'wonder' brings in an element of doubt or speculation.
The third sentence starts with "I want to know if they checked..." This indicates that the speaker does not know whether the tickets were checked or not, but is asking about it. There is no opinion expressed here.
The fourth sentence is a definite opinion that "They should have checked..." This sentence conveys the same idea as expressed in the first sentence.

89. (1) 'Magnanimous' means 'generous' and 'benevolence' means 'charitableness'. At first glance, none of the options seem to be the exact opposite of the meaning given in the sentence.
'Cruel' means 'wilfully causing pain to others'; 'snobbish' means 'condescending or overbearing to others'; 'tyrannical' means 'unjustly cruel, harsh or severe'. Therefore, options 2, 3 and 4 can be eliminated.
Although the common meaning of being 'mean' is being 'nasty or unkind', when used with reference to money, being mean refers to being 'stingy or miserly'.

90. (3) Biannual has two meanings. Firstly, it means occurring twice a year, which is also referred to as semiannual. Secondly, it also means occurring every two years, which is also called 'biennial'. Since, the word 'biennial' is specifically used for an event that occurs every two years or even something that lasts for two years, biannual is more commonly used to refer to something that occurs twice a year.

91. (4) 'Temporal', 'ephemeral' and 'transient' mean 'lasting a very short time; short-lived; transitory'. This is the exact opposite of 'eternal'.

92. (3) 'Intravenous' means 'within a vein'.

93. (1) Sentence 8 has used 'down' as an adjective as it tells something more about the train (noun).
This eliminates options 2, 3 and 4.

94. (2) The third fragment 'as unpredictable as' is used only in reference to the 'weather'. This establishes the 3-8 link.
The fourth fragment 'as slippery as' is used only in reference to 'an eel'. This establishes the 4-6 link.
This eliminates option 1.

95. (1) The word 'above' in sentence 8 is an adjective as it tells us more about 'information' (noun).
This establishes the 1-8 link, eliminating options 2 and 4.
The word 'above' in sentence 5 is a noun. This eliminates option 3.

96. (2) There is an error in fragment 2 of the sentence. The article 'the' has been incorrectly placed.

97. (4) There is no error in this sentence.

98. (4) Option 1 should read, 'whose car'.
Option 2 should read, 'who is'.
Option 3 should read, 'whom'.

99. (2) The word 'him' is the object of the word 'imagine' and 'forgetting his own birthday' is a participial phrase describing 'him'.
If we use 'his forgetting', then the gerund 'forgetting' links to the word 'his'. In this case the whole phrase 'his forgetting his own birthday' becomes the object, which is not appropriate.

100. (2) Sentence 1 is incorrect as it gives the impression that the wedding was held in a newspaper. It has a modifying error.
Sentence 2 has no grammatical errors.

101. (4) Options 1 and 2 are incorrect as two negations make the sentence incorrect. The given sentence requires a positive conclusion.
The sentence also needs to be in the present tense and not the future tense. This eliminates option 3.

102. (2) The logical start of the sentence is fragment 5 as it introduces the subject under consideration – English.
This eliminates options 1, 3 and 4.

103. (4) Option 1 is incorrect as it has a tense error. The words 'would have' cannot be repeated twice in a single sentence.
Options 2 and 3 are incorrect as they are both tense errors.
Sentence 4 has no such errors.

104. (2) Option 1 is incorrect because 'apt' means 'inclined, disposed'. People cannot be inclined to be mistaken.
Options 3 and 4 are also incorrect as a person cannot be 'considered' to be mistaken or be 'able' to be mistaken.
In the given context 'likely' fits in the best.

105. (1) When using question tags, if the first part of the sentence has a negation, then the question tag would be positive. Here 'hardly' signifies negation, hence the tag needs to positive. This eliminates options 2 and 4.
Option 3 is incorrect as 'will he' signifies future while the action is taking place in the present.
Therefore, only a question tag in the present tense will fit in correctly.

106. (2) A metaphor is an implied comparison.
A simile is a stated comparison between two fundamentally dissimilar things that have at least one point in common. A simile is usually introduced by 'like', 'as' or 'so'.
Apostrophe is a direct address to the dead, to the absent, or to a personified object or idea. This figure of speech is a special form of personification. Although in apostrophe, certain human qualities such as understanding are implied when the direct address is made as if addressing a person present.
An epigram is a statement or any brief saying in prose or poetry, in which there is an apparent contradiction.
Here the sentence provided is a simile as there is a comparison between someone and a peacock.

107. (1) A 'personification' is a figure of speech in which an inanimate object or abstraction is endowed with human qualities or abilities.
An 'exclamation' is a figure of speech in which the exclamatory form is used to draw greater attention to a point than a mere statement. Exclamation is little more than a cry, a sudden expression for a range of emotions such as surprise, pleasure, anger, disgust, or pain.
A 'simile' is a stated comparison between two fundamentally dissimilar things that have at least one point in common. A simile is usually introduced by 'like', 'as' or 'so'.
'Anticlimax' is the opposite of climax, a sudden descent from higher to lower. It is chiefly used for the purpose of satire or ridicule.
Here the sentence provided is a 'personification' as death, an inanimate object, is endowed with the human quality of laying hands.

108. (1) The passage is titled, "A way to deal with frozen feelings." The passage highlights the effects of shock, violence, fear or pain that every child experiences in the early years. The emotions mentioned in the passage are negative. No positive emotions have been mentioned in the passage. Therefore, we can come to the conclusion that 'Frozen feelings' are about negative childhood experiences.
Options 2 and 3 are incorrect as they are an outcome of frozen feelings.

109. (3) A 'glitch' is 'a defect or malfunction that occurs in a machine or plan'. The passage mentions, "Our feelings are the cause of this 'glitch' in our learning process." What is this "glitch" in the context of the passage? The passage mentions that "whenever the same kind of emotion crops up later in our life we return to the past for our reference point." This means that there occurs some kind of malfunction or breakdown which interferes with the learning process.

110. (2) Option 1 is incorrect as the passage states "The process of change need not be traumatic".
Option 2 is correct and is mentioned in the following extract from the passage, "We feel childish and we behave childishly."

111. (1) Dendrochronology is the method of scientific dating based on the analysis of tree-ring growth patterns.

112. (4) Modi Telstra launched the first mobile phone operation in India.

113. (*) The correct answer for this question is not given in the options.

114. (1) The expansion of BIFR, in the context of the Indian Industry is Board for Industrial and Financial Reconstruction.

115. (4) Ekistics is the science of human settlements.

116. (1) Red Herring is a Prospectus in an IPO.

117. (1) Eight O'clock Coffee, a US based coffee marketer, is a group company of Tata.

118. (3) Bio-diesel is extracted from Jatropha.

119. (4) Margaret Court is the woman tennis player who won the maximum number of Grand Slam singles titles in the history of tennis.

120. (2) 'Big Apple' is the nickname of New York city.

121. (3) Infosys was the first Indian company to be listed on NASDAQ.

122. (2) 'TARENGA' - a village in Bihar was in the news because of viewing of the solar eclipse.

123. (2) The driver for Formula One's Force India Team is Adrian Sutil.

124. (3) (MDRT) - Million Dollar Round Table consists of Global Association of Life Insurance professionals.

125. (2) Reserve Bank of India declares the 'Credit Policy' of India.

126. (4) SEBI is the body which regulates the Stock Exchanges in India.

127. (2) Rafflesia arnoldii holds a world record in the plant world for being the largest bloom.

128. (2) Temujin, a famous historical character of the world was Genghis Khan.

129. (1) Short selling in the stock market lingo is selling the shares which you do not own.

130. (2) Mahatma Gandhi was nominated for Nobel Peace prize for the first time in the year 1937 and last time in the year 1948. He was nominated 3 times (1938, 1939 and 1947) in between these two years.

131. (4) Sveriges Riksbank instituted the Nobel Prize in Economics.

132. (4) The first actress of Bollywood to endorse Lux beauty soap was Leela Chitnis.

133. (1) Kumar Shri Ranjitsinghji Vibhaji Jadeja played for Sussex in the league cricket of England.

134. (4) Nagpur is the city which has an office of RBI without being a state capital.

135. (1) The tycoon described in the question is Ratan Tata.

136. (3) Vinegar is the substance which derived its name from old French expression and made from the fermentation of ethanol in a process that yields its key ingredient, ethanoic acid.

137. (3) The name of the Japanese company which literally means 'three diamonds' is Mitsubishi.

138. (4) Steve and Mark Waugh were the first twins to play test cricket.

139. (3) USA has the largest rail network in the world.

140. (1) Svetlana Kuznetsova won the 2009 French open tennis ladies title.

141. (1) Robert Noyce and Jack Kilby invented the microchip.

142. (3) Kazakhstan is the largest landlocked country in the world.

143. (1) Ecuador makes Panama hats.

144. (3) Indira Gandhi was the Prime Minister of India when the 42nd Amendment Bill bringing the changes in the Constitution was passed.

145. (3) Pluto lost its planet status recently.

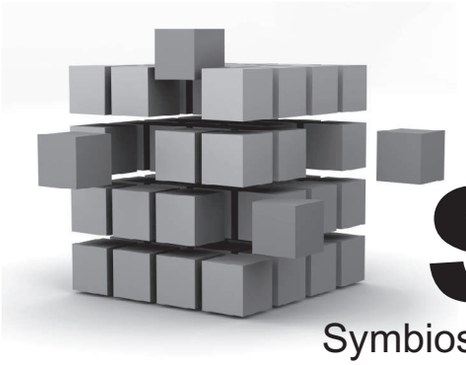
146. (4) One barrel of oil is approximately equivalent to 160 litres.

147. (4) Five year plans in India are finally approved by the Planning commission.

148. (2) The most spoken language in the Southern Cone of South America which includes Argentina, Chile, Paraguay, Uruguay and Peru is Spanish.

149. (2) Brazil is not a member of the organization of petroleum exporting countries (OPEC).

150. (4) Adam Osborne invented 'computer laptop'. This answer is not mentioned in the options.



SNAP 2008

Symbiosis National Aptitude Test

Solved Paper

General English

For the following 5 questions in this section, correct answers carry 2 marks each.

1. Match the following idiomatic references to parts of the human anatomy

(1) palm (2) foot
(3) eye (4) stomach
(5) could not tolerate the insult
(6) to look at with envy and desire
(7) to put the blame on someone else
(8) forced to pay the bill

(1) 1-8, 2-6, 3-5, 4-7 (2) 1-5, 2-7, 3-6, 4-8
(3) 1-7, 2-8, 3-6, 4-5 (4) 1-6, 2-8, 3-5, 4-7

2. Find the maximum number of times any one of the given words fits the set of sentences

RAISE ARISE AROSE RISE

(i) Opportunities will …… and you must grab them.
(ii) A hot wind …… from the desert.
(iii) I …… at dawn on most days.
(iv) A mood of optimism …… among the people.

(1) in all four sentences (2) in 3 sentences
(3) in 2 sentences (4) in 1 sentence

3. Which two sentences in the following convey the same idea ? Choose from the combinations listed below :

(1) He is in a fool's paradise.
(2) He can't see the wood for the trees.
(3) He can't distinguish between reality and fancy.
(4) He is unable to separate unimportant details from the really important ones.

(1) 2, 3 (2) 2, 4
(3) 1, 4 (4) 1, 3

4. Find the correct match of grammatical function with usage for the word 'THEN'.

	Function		Usage
(1)	Noun	(5)	He was the then King of Nepal.
(2)	Adjective	(6)	I have not heard about him since then.
(3)	Adverb	(7)	He is not feeling well then how can he possibly go for the picnic.
(4)	Conjunction	(8)	He was not a graduate then.

(1) 1-8, 2-5, 3-7, 4-7 (2) 1-6, 2-5, 3-8, 4-7
(3) 1-7, 2-5, 3-6, 4-8 (4) 1-8, 2-7, 3-6, 4-5

5. We can never make our beliefs regarding the world certain. Even scientific theory of a most rigorous and well-confirmed nature is likely to change over a decade or even tomorrow. If we refuse to even try to understand, then it is like resigning from the human race. Undoubtedly life of an unexamined kind is worth living in other respects—as it is no mean thing to be a vegetable or an animal. It is also true that a man wishes to see this speculative domain beyond his next dinner.

From the above passage it is clear that the author believes that

(1) men would do well not to speculate
(2) progress in the scientific field is impossible
(3) one should live life with the dictum 'what will be will be'
(4) men are different from animals as far as their reasoning abilities are concerned

For the following questions in this section, correct answers carry 1 mark each.

Directions (Q. 6-11) *Read the passage and answer within its context.*

Rajendra K. Pachauri, head of the Intergovernmental Panel on Climate Change, is getting nightmares because of the Nano, Tata's soon-to-be-launched Rs One lakh car. Sunita Narain of the Centre for Science and Environment (CSE) says that it is not the Nano by itself but cars overall that give her nightmares. The villains in my nightmares are neither the Nano nor cars overall, but stupid government policies that subsidize and encourage pollution, adulteration and congestion.

Sanctimonious greens call the Nano disastrous because of its affordability-millions more will now clog roads and consume more fossil fuel. This is elitism parading as virtue. Elite greens own cars, but cannot stand the poorer masses becoming mobile, since the consequent congestion will eat into the time of the elite!

More logical would be a protest against big cars that use more space and fuel, or highly polluting old cars. Instead, green hypocrites aim at a new car with the lowest cost, best mileage and

least emissions. The Nano will not burden us with too many cars. India has very few cars per person by world standards. London and New York have ultra-high car densities, yet have clearer air than Delhi. Our problem is too many bad policies, not too many cars.

We subsidize vehicles on a gargantuan scale invisible to lay folk. Roads and flyovers cost crores to build and maintain, yet road use is free (save on a few toll roads). Traffic police and lights are costly, yet are provided free. These invisible subsidies starve cities of funds to expand roads and public transport.

Land in cities now costs lakhs per square metre. Yet parking is free in the suburbs, and often costs just Rs. 10 per day in city centres. A single parking space of 23 square meters occupies land worth Rs. 40 lakhs. A car occupies more space than an office desk, yet the desk space pays full commercial rent while parking space costs just about Rs. 10 per day.

Daily parking charges range from $30 (Rs 630) in Washington to $60 (Rs 1260) in New York. CSE launched a sensible campaign to raise parking fees in Delhi to Rs 120 per day, but was foiled. So, parking space now exceeds green space, a scathing comment on priorities.

The world price of crude oil has risen 13 fold since 1998 to over $ 130 per barrel, but Indian petrol prices have barely doubled. Left Front politicians, who once wanted to soak the rich, now want to subsidize them. Under-recoveries of oil companies' total may be Rs. 2,00,000 crore, even after a recent price hike. This is far more than the cost of Sarva Shiksha Abhiyan (education for all) and the Employment Guarantee Scheme put together.

We sanctimoniously lecture rich countries to reduce their green house emissions, yet subsidize our own. Diesel is subsidized to be cheaper than petrol. So, Indian car makers produce the highest proportion of diesel cars in the world. Diesel fumes contain suspended particles that are highly toxic. This subsidy kills.

So does kerosene provided at throwaway prices, ostensibly to benefit poor villagers. One third of all kerosene is used to adulterate petrol and diesel. That causes horrendous pollution even in the greenest of cars.

What's the way forward ? We must abolish subsidies and raise taxes on vehicles and fuels to reflect their full social cost. The biggest but least visible subsidy is for parking, and we should start there.

Many car owners in the West take public transport to work since parking space downtown is costly and scarce. We should levy parking fees on an hourly, not daily, basis. Rs. 10 per hour could be a starting point in the metros.

In parts of Tokyo, you cannot own a car unless you own a private parking space. This is too extreme for India, but indicates the future path. If we charge owners the full social cost of parking, people will buy smaller and perhaps fewer vehicles, and fewer still will take them to work. That will slash congestion and pollution.

Cities should levy stiff annual taxes on vehicles, not a one-time tax, and use the revenue to constantly expand public transport and roads. This will create economic synergy : Private transport will finance public transport. London and New York have high density public transport as well as high car density.

Apart from underground rail, cities need elevated roads to ease congestion and pollution. Lata Mangeshkar helped kill a proposal for an elevated road near her Mumbai flat; perhaps she felt her throat and singing would be affected. She did not care that the throats of poor people living on the pavements were far worse affected by fumes, and might get relief if some fumes were diverted to a higher level. What elitism!

Next, some medicine that will be really bitter, politically. The excise duty on all automotive vehicles should be raised to reflect their social costs. Fuel subsidies should be abolished. Price differentials between petrol, diesel and kerosene should be removed, ending incentives for adulteration. Diesel cars should bear a heavy additional cess to finance improved healthcare for those affected by their emission of harmful particulate matter.

That is a long, politically difficult agenda. Only part of it will ever be achieved. Yet that is the way to go, rather than agitate the Nano.

6. By 'Sanctimonious greens' the writer refers to
(1) aristocratic environmentalists
(2) the rich
(3) environmentalists with a 'holier than thou' attitude
(4) those who decry deforestation

7. The elite are
(1) jealous of Nano owners
(2) afraid of traffic jams and depletion of fossil fuel
(3) afraid of reaching their destinations late
(4) full of disdain that the poor can afford cars

8. The paradox of the situation is that
(1) bigger cars mean more fuel, more space and more pollution
(2) though India has fewer cars the Nano will bring more pollution
(3) London and New York have more cars and less pollution
(4) though India is smaller than the US its cars cause more pollution

9. In saying 23 square meters of parking space costs 40 lakhs, the writer is
(1) caustic (2) exaggerating
(3) sarcastic (4) ironical

10. The writer blames India for
(1) subsidizing kerosene whereby greenhouse emissions are indirectly subsidized
(2) subsidizing diesel
(3) for increasing the cost of parking by the hour
(4) for not making it mandatory for car owners to own parking space

11. The most suitable title for this passage is
(1) Polluting Politics
(2) No No Nano
(3) Submerge Subsidies
(4) More Cars, Less Pollution

12. The plural of Virus is
(1) Viruses (2) Virae
(3) Virii (4) Virus

13. If the following segments of a sentence are to be rearranged in logical order as A, B, C, D where would '3' be placed
(1) to see that students do not altogether forget to write especially during exam time
(2) the education groups are now asking for hand writing classes
(3) thanks to mobile testing and computer literacy
(4) writing in long hand is becoming a vanishing art

(1) A (2) B
(c) C (4) D

14. If leaf is to leaves and knife is to knives, then belief is to ……
(1) beliefs (2) believes
(3) belief (4) believing

15. Choose the sentence where the underlined word is used correctly
(1) This latest novel is a pedestrian story about spies.
(2) The exam paper is not pedestrian but difficult.
(3) This is the pedestrian highway.
(4) Every week we are forced to listen to a pedestrian lecture.

16. When the fire alarm rang …… left the building immediately.
(1) all (2) everyone
(3) all the people (4) every person

17. In the following sentence choose the erroneous segment(s).
He is one of those people (A)/who thinks (B)/he owns the world. (C)
(1) error in segment A
(2) error in segment B
(3) error in segment A and C
(4) error in segment B and C

18. Choose the correct meaning for the word cynic—
(1) the person who is selfish.
(2) the person who is concerned about others.
(3) the person who isn't misanthropic.
(4) the person who believes that people always act from selfish motives.

19. Choose the word with correct spelling
(1) catagories (2) diarrhoea
(3) ommission (4) inaugarate

20. Pick out the right sentences.
(1) I will go with you.
(2) There was nobody I could go with.
(3) I have a glass with painting on it.
(4) The curtains do not match with the furniture.
(1) 1 and 2 (2) 2 and 3
(3) 1 and 4 (4) All

21. About the following pair of phrases, choose the correct option.
(i) A two days' visit
(ii) A two day's visit
(1) The first phrase is erroneous
(2) The second phrase is erroneous
(3) Both phrases are erroneous
(4) Both phrases are correct

Directions (Q. 22-24) *Read the following passage and answer within its context.*

Nearly two thousand years have passed since a census decreed by Caesar Augustus became part of the greatest story ever told. Many things have changed in the intervening years. The hotel industry worries more about overbuilding than overcrowding, and if they had to meet an unexpected influx, few inns would have managed to accommodate the weary guests. Now it is the census taker that does the traveling in the fond hope that a highly mobile population will stay put long enough to get a good sampling. Methods of gathering, recording and evaluating information have presumably been improved a great deal. And where then it was the modest purpose of Rome to obtain a simple head count as an adequate basis for levying taxes, now batteries of complicated statistical series furnished by governmental agencies and private organizations are eagerly scanned and interpreted by sages and seers to get a clue for future events.

The Bible does not tell us how the Roman census takers made out, and as regards our more immediate concern, the reliability of present day economic forecasting, there are considerable differences of opinion. They were aired at the celebration of the 125th anniversary of the American Statistical Association. There was the thought that business forecasting might well be on its way from an art to a science, and some speakers talked about new-fangled computers and high-faulting mathematical systems in terms of excitement and endearment, which we, at least in our younger years when these things mattered, would have associated more readily with the description of a fair maiden.

But others pointed to a deplorable record of highly esteemed forecasts and forecasters with a batting average below that of the Mets and the President-elect of the Association cautioned that "high-powered statistical methods are usually in order where the facts are crude and inadequate, statisticians assume."

We left his birthday party somewhere between hope and despair and with the conviction, not really newly acquired, that proper statistical methods applied to ascertainable facts have their merits in economic forecasting as long as neither forecaster nor public is deluded into mistaking the delineation of probabilities and trends for a prediction of certainties of mathematical exactitude.

22. According to the passage, taxation in Roman times was based on
(1) mobility (2) wealth
(3) population (4) census takers

23. The author refers to the Mets primarily in order to
(1) show that sports do not depend on statistics
(2) contrast verifiable and unverifiable methods of record keeping
(3) indicate the changes in attitudes from Roman days to the present
(4) illustrate the failure of statistical predictions

24. The author's tone can best be described as
(1) jocular (2) scornful
(3) pessimistic (4) humanistic

25. Disinterested is closest in meaning to
(1) bored (2) unbiased
(3) not interested (4) indifferent

26. Choose the option which is the nearly opposite in meaning to BERATE
(1) grant (2) praise
(3) refer (4) purchase

27. Arrange the following in the right order to make a complete sentence
(i) with interconnected vibrating balls and springs

(ii) in a naive sense, a field in physics may be envisioned as if space were filled
(iii) as the displacement of a ball from its rest position
(iv) and the strength of the field can be visualized
(1) (ii), (i), (iv), (iii) (2) (i),(ii), (iii), (iv)
(3) (iv), (iii), (ii), (i) (4) (iii), (iv), (i), (ii)

28. Find the odd one out
(1) latent (2) natural
(3) inborn (4) inherent

29. He told the teacher that …… .
(1) he was liked by the whole class
(2) you are liked by the whole class
(3) he is liked by the whole class
(4) you were liked by the whole class

30. Match the several meanings of the word COMPLEX with their appropriate usages.

	Meaning		Usage
(1)	complicated	(5)	A new sports complex is coming up for the Common Wealth Games.
(2)	abnormal state of mind	(6)	Culture is a complex whole of many things.
(3)	group of structures	(7)	She has a complex about being overweight.
(4)	mixture	(8)	His motives in carrying out the crime were complex.

(1) 1-6, 2-8, 3-7, 4-5 (2) 1-8, 2-7, 3-5, 4-6
(3) 1-5, 2-7, 3-6, 4-8 (4) 1-8, 2-5, 3-6, 4-7

31. Which does not make a sensible word/phrase when added to the word
Honey
(1) pot (2) suckle
(3) comb (4) taste

Directions (Q. 32-35) *Read the passage carefully and answer within the context.*

In September of 1929, traders experienced a lack of confidence in the stock market's ability to continue its phenomenal rise. Prices fell. For many inexperienced investors, the drop produced a panic. They had all their money tied up in the market, and they were pressed to sell before the prices fell even lower. Sell orders were coming in so fast that the ticker tape at the New York Stock Exchange could not accommodate all the transactions.

To try to reestablish confidence in the market, a powerful group of New York bankers agreed to pool their funds and purchase stock above current market values. Although the buy orders were minimal, they were counting on their reputations to restore confidence on the part of the smaller investors, thereby affecting the number of sell orders. On Thursday, October 24, Richard Whitney, the Vice President of the New York Stock Exchange and a broker for the J.P. Morgan Company, made the effort on their behalf. Initially it appeared to have been successful, then, on the following Tuesday, the crash began again and accelerated. By 1932, stocks were worth only twenty percent of their value at the 1929 high. The results of the crash had extended into every aspect of the economy, causing a long and painful depression, referred to in American history as the Great Depression.

32. The New York bankers conted on
(1) Current market values
(2) The number of sell orders
(3) Confidence
(4) Their reputation

33. The cause of downfall of share market was
(1) Inexperienced investors
(2) Phenomenal decrease
(3) Lack of confidence in stock market's ability
(4) Panic amongst investors

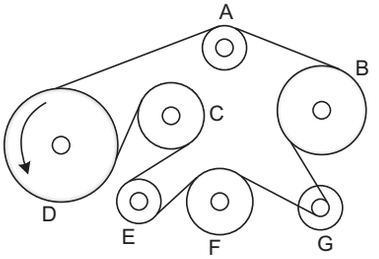
34. Choose the word in the passage that is an antonym of "minimal"
(1) negligible (2) minimum
(3) maximal (4) significant

35. Identify the correct sentence
(1) The office is opposite to the bank.
(2) The office is opposite the bank.
(3) The office is opposite from the bank.
(4) The office is opposite of the bank.

▪ Analytical and Logical Reasoning

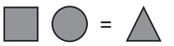
For all questions in this section, correct answers carry 2 marks each.

36. A band passes around all the wheels so that they can all be turned by the driving wheel. When the driving wheel turns in the direction shown. Which way will the wheel B turn ?

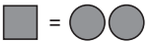


(1) Clockwise (2) Cannot move
(3) Anti-clockwise (4) Either way

37. In a certain language, (A) 'Sun shines brightly' is written as 'ba lo sul'; (B) 'Houses are brightly lit' as 'kado udo ari ba'; and (C) 'Light comes from sun' as 'dapi kup lo nro'. What words will be written for 'sun' and 'brightly' ?
(1) lo, ba (2) ba, lo
(3) snl, lo (4) ba, sul

38. Given are the following three equations
(1) 
(2)
(3)

How many circles are equivalent to a square, as per the ratios in the three given equations ?

(1) (2) (3) (4) 

39. Each child in a family has at least 4 brothers and 3 sisters. What is the smallest number of children the family might have ?
(1) 7 (2) 8
(3) 9 (4) 10

40. In the following question two statements are followed by two conclusions numbered I and II.
Assume the two statements are true even if they are at variance with commonly known facts. Then pick the correct answer from the choices given below.
A. Only conclusion I follows
B. Only conclusion II follows
C. Both conclusions I and II follow
D. Neither conclusion I nor conclusion II follows
Statements Some doctors are fools. Joshi is a doctor.
Conclusions I. Joshi is a fool.
II. Some fools are doctors.
(1) A (2) B
(3) C (4) D

41. Debu walks towards East then towards North and turning 45° right walks for a while and lastly turns towards left. In which direction is he walking now ?
(1) North (2) East
(3) South-East (4) North-West

Directions (Q. 42-44) *Read the following instructions and answer.*

I. There is a rectangular wooden block of length 4 cm, height 3 cm and breadth 3 cm.
II. The two opposite surfaces of 4 cm × 3 cm are painted yellow on the outside.
III. The other two opposite surfaces of 4 cm × 3 cm are painted red on the outside.
IV. The remaining two surfaces of 3 cm × 3 cm are painted green on the outside.
V. Now, the block is cut in such a way that cubes of 1 cm × 1 cm × 1 cm are created.

42. How many cubes will have only one colour ?
(1) 10 (2) 12
(3) 14 (4) 18

43. How many cubes will have no colour ?
(1) 1 (2) 2
(3) 4 (4) 8

44. How many cubes will have any two colours ?
(1) 32 (2) 24
(3) 16 (4) 12

Direction (Q. 45) *Read the following about the grid given below and answer.*

The cells in this grid contain the digits 1 to 9 in random order.
Column A contains no odd digits.
Cell C3 minus Cell C2 equals 4.
The sum of three digits in Row 1 is 17.
Number 7 is in column B; its left hand neighbor is not 4.
The digits of column C add upto 14.
2 is not in the same horizontal row as 8; and 9 is not immediately below 3.

45. Which cell holds the number 9 ?

	A	B	C
1			
2			
3			

(1) B1 (2) B3
(3) C2 (4) C1

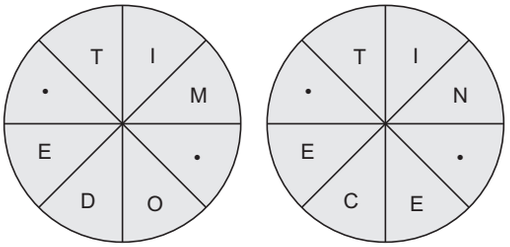
46. Replace the questions mark with the right option.
4, 32, 288, ?, 31680
(1) 25600 (2) 2880
(3) 7420 (4) 10000

47. In the Sunday bazzar, Jamuna sells her lemons at Rs 0.50 for two. Her neighbor Seema has a little smaller lemons; she sells hers at Rs 0.50 for three. After a while, when both ladies have the same number of lemons left, Seema is called away. She asks her neighbour to take care of her goods. To make things simple, Jamuna puts all lemons in one big pile, and starts sellings five lemons per one rupee. When Seema returns, at the end of the day, all lemons have been sold. But when they start dividing the money, there appears to be a shortage of Rs 3.50. Supposing they divide the money equally, how much does Jamuna lose with this deal ?
(1) Rs 10.50 (2) Rs 11.50
(3) Rs 42.00 (4) Rs 52.50

48. There are two cups, one containing orange juice and one containing an equal amount of lemonade. One teaspoon of the orange juice is taken and mixed with the lemonade.Then a teaspoon of this mixture is mixed back into the orange juice. Is there more lemonade in the orange juice or more orange juice in the lemonade ?
(1) More orange juice in the lemonade
(2) More lemonade in the orange juice
(3) Equal amount of each juice between the two cups
(4) None of the above

49. Consider the statement and decide which of the assumptions are implicit
"In the present period of economic hardships, education and small family norm may lead the nation to progress and prosperity".
Assumptions
A. Education and small family norms are directly related to nations progress.
B. Big families find it difficult to bear the cost of education.
(1) Only A is implicit
(2) Only B is implicit
(3) Both A and B are implicit
(4) Neither A nor B is implicit

50. Fill in the blanks to find two words that are synonyms



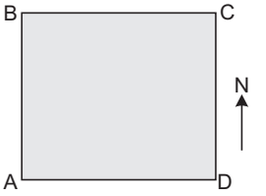
(1) KS, ST (2) MS, NT
(3) ST, DN (4) MS, DN

51. Beautiful beaches attract people, no doubt about that. Just look at the city's most beautiful beaches, which are amongst the most overcrowded places in the state.
Which of the following exhibits a pattern of reasoning similar to the one exhibited in the argument above ?
(1) Moose and bear usually appear at the same drinking hole at the same time of day. Therefore, moose and bear must be feeling thirsty at about the same time.
(2) Children who are scolded severely tend to misbehave more often than other children. Hence if a child is not scolded severely, that child is less likely to misbehave
(3) During warm weather my dog suffers more fleas than during cool weather. Therefore, fleas must thrive in a warm environment
(4) Tally accounting software helps increase the work efficiency of its users. As a result, these users have more time for other activities

52. Abdul, Mala, and Chetan went bird watching. Each of them saw one bird that none of the others did. Each pair saw one bird that the third did not and one bird was seen by all three. Of the birds Abdul saw, two were yellow. Of the birds Mala saw, three were yellow. Of the birds Chetan saw, four were yellow. How many yellow birds were seen in all ? How many non-yellow birds were seen in all ?
(1) 7 yellow birds and 3 non yellow birds
(2) 5 yellow birds and 2 non yellow birds
(3) 4 yellow birds and 2 non yellow birds
(4) 3 yellow birds and 2 non yellow birds

53. In each of the following two sets I and II, find the word or pair of words different from the other three words or pair of words
I. J. Lake K. Brook L. Stream M. River
II. J. Weighty-Heavy K. Broad-Wide
L. Big-Large M. Tiny-Small
(1) I-J, II-J (2) I-K, II-M
(3) I-K, II-J (4) I-J, II-K

54. A, B, C and D are standing on the four corners of a square field as shown in the figure. From the positions shown in the figure. A walks to the North position and B walks to the East position while C decides to walk two sides in anti-clockwise direction. B walks to North and then changes his mind to take the previous position. Identify the choice with correct positions.



(1) A and B occupy the same position
(2) C and D occupy the same position
(3) D and B are in their original positions
(4) B and C are in diagonally opposite positions

55. A gambler bet on a horse race, but the bookee would not tell him the results of the race. The bookee gave clues as to how the five horses finished which may have included some tie and would not pay the gambler off unless the gambler could determine how the five horses finished based on the following clues
- Penn Fe finished before Night Marvel and after Wish Bones.
- Wish Bones tied with Penn Fe if and only if Hallelujah did not tie with Sundae.
- Penn Fe finished as many places after Sundae as Sundae finished after Wish Bones if and only if Wish Bones finished before Night Marvel.

The gambler thought for a moment, then answered correctly. How did the five horses finish the race ?
(1) Sundae came in first. Wish Bones and Hallelujah tied for second place. Penn Fe came in fourth. Night Marvel came in fifth.
(2) Wish Bones came in first. Sundae and Penn Fe tied for second place. Hallelujah came in fourth. Night Marvel came in fifth.
(3) Wish Bones came in first. Sundae and Hallelujah tied for second place. Penn Fe came in fourth. Night Marvel came in fifth.
(4) Penn Fe came in first. Night Marvel and Hallelujah tied for second place. Wish Bones came in fourth. Sundae came in fifth.

56. In a school drill, a number of children are asked to stand in a circle. They are evenly spaced and the 6th child is diametrically opposite the 16th child. How many children are made to stand in the circle ?
(1) 16 (2) 20
(3) 22 (4) None of these

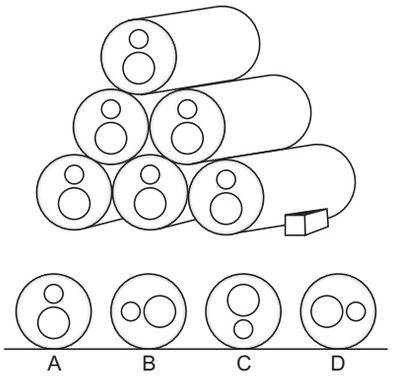
57. In this question insert the missing number at the sign of interrogation.

8	4	9	5
5	7	3	4
3	4	5	8
39	44	60	?

(1) 62 (2) 72
(3) 60 (4) 70

58. Steel cylinders are made so that each one has a large and small hole through the middle. In the drawing six cylinders have been stacked on top of each other. To stop the cylinders from rolling on the smooth floor they are wedged by heavy blocks at each side of bottom row. If the heavy

blocks are removed what would be the position of cylinders when they stopped rolling ?



(1) A (2) B
(3) C (4) D

Directions (Q. 59-61) *Use the information given below to answer.*

(i) There is a group of 5 persons A, B, C, D and E.
(ii) In the group there is one badminton player, one chess player and one tennis player.
(iii) A and D are unmarried ladies and do not play any games.
(iv) No lady is a chess player or a badmintor player.
(v) There is a married couple in the group of which E is the husband.
(vi) B is the brother of C and is neither a chess player nor a tennis player.

59. Which of the groups has only ladies ?
(1) ABC (2) BCD
(3) CDE (4) None of these

60. Who is the tennis player ?
(1) B (2) C
(3) D (4) E

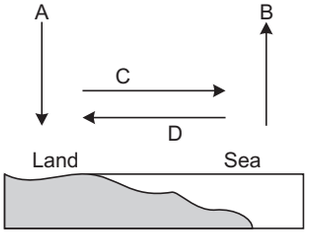
61. Who is the wife of E ?
(1) A (2) B
(3) D (4) None of these

Direction (Q. 62) *Consider the following statements and answer the question.*

M, N, O and P are all different individuals.
M is the daughter of N.
N is the son of O.
O is the father of P.

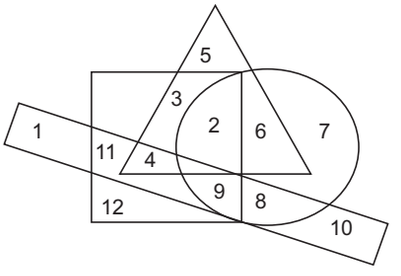
62. Which among the following statements is contradictory to the above premises ?
(1) P is the father of M
(2) O has three children
(3) M has one brother
(4) M is the granddaughter of O

63. The drawing shows a cross section where the land meets the sea. The section covered is 5 km. On a hot day, in which direction, indicated by four arrows, is the wind most likely to blow ?



(1) A (2) B
(3) C (4) D

Directions (Q. 64-65) *In the diagram below, the circle stands for 'educated', square stands for 'hard working', triangle for 'urban people' and rectangle for 'honest'. The different regions of the diagram are numbered from 1 to 12. Study the diagram carefully and answer the questions.*



64. Uneducated urban, hard working and honest people are indicated by
(1) 3 (2) 11
(3) 9 (4) 4

65. Non-urban educated people who are neither hard working nor honest are indicated by
(1) 5 (2) 7
(3) 10 (4) 11

■ General Awareness Section

Directions (Q. 66-70) *For the following 5 questions in this section, correct answers carry 2 marks each.*

66. A. M. Touring award is considered as the Nobel Prize in the field of computers, given annually by Association for Computing Machinery co-sponsored by Intel and Google. Who was A M Touring in whose memory the award was instituted ?
(1) First Chairman of Intel
(2) A British Mathematician
(3) The legendary Dean of Stanford who revolutionized the computer education in the US
(4) An US cognitive scientist in the field of artificial intelligence (AI) and co-founder of Massachusetts Institute of Technology's AI laboratory

67. Find the most accurate description of 'Bt cotton'.
(1) A variety of cotton that grows fast
(2) A genetically modified crop of cotton whose DNA has been altered by introducing an extra gene

(3) A variety of cotton that is immune to diseases
(4) A genetically modified crop of cotton whose DNA has been designed in the lab

68. Las Vegas, US-based tour operator AMX Company has filed a trademark patent for which of the following Tagline ?
(1) "God's Own Country"
(2) "Heaven on Earth'
(3) "Discover Incredible India"
(4) "Come Alive with India"

69. Which out of the following holds the highest number of shares of ICICI Bank ?
(1) Banks Financial Institutes and Insurance
(2) FIIs
(3) General public
(4) Private Corporate Bodies

70. Indian Standard Time is based on the longitude of 82.5 degrees passing through which of the following places ?
(1) Mirzapur, U.P.
(2) Jabalpur, M.P.
(3) Nagpur, Maharashtra
(4) Jaipur, Rajasthan

Directions (Q. 71-100) : *For the following questions in this section, correct answers carry 1 mark each.*

71. The Black Box of an aircraft is an important part as all the conversations and data are recorded therein. Although it is named black box, but its colour is not black. What is the colour of the black box in a commercial airplane ?
(1) Brown (2) Yellow
(3) Red (4) Orange

72. Find the person who plays the odd sport out of the following
(1) Navaratilova (2) Michael Phelps
(3) David Beckham (4) Prakash Padukone

73. Sariska and Ranthambore are the reserves for which of the following animals ?
(1) Lion (2) Deer
(3) Tiger (4) Bear

74. March, 2008 witnessed a turning point in the history of which of the following Himalayan Kingdoms when democracy was ushered in replacing monarchy ?
(1) Bhutan (2) Nepal
(3) Sikkim (4) None of these

75. Neil Armstrong brought back a rock from the moon. On earth
(1) the mass of the rock will change
(2) the weight of the rock will change
(3) mass and weight will change
(4) none of the above

76. Milk, Cheese and Eggs are the source of
(1) Vitamin C and A (2) Vitamin A and D
(3) Vitamin C and D (4) Vitamin B and C

77. In August 2008, India's longest runway for passenger aircraft was commissioned in
(1) Mumbai (2) New Delhi
(3) Hyderabad (4) Bengaluru

78. The first ever public hearing in India, almost like a referendum, on the fate of SEZ was held during the month of September 2008 in/at
(1) Verna IT/ITeS SEZ in Goa
(2) Haryana for Reliance Haryana SEZ
(3) Pen, Maharashtra
(4) POSCO India, Steel SEZ Phase II, Orissa

79. A1 Ahram is
(1) a university in Damascus
(2) a newspaper in Cairo
(3) a mosque in Riyadh
(4) a small mountain range in Jordan

80. India's largest and first multi-national pharmaceutical giant Ranbaxy is being bought over by
(1) Matrix Pharma (2) GVK Bio-sciences
(3) Merck (4) None of these

81. Find the odd product out of the following
(1) Pamper (2) Dove
(3) Tide (4) Pantene

82. In the internet sphere, 'Opera' is the name of a
(1) web conference site
(2) web advertising firm
(3) software for webinars
(4) web browser

83. Which of the following pair is not correct ?
(1) Shahid Parvez : Sitar
(2) Rashid Khan : Vocal Music
(3) Ayaan Ali Khan : Sarod
(4) Romu Muzumdar : Guitar

84. At what frequency SENSEX calculation is carried out ?
(1) Every half an hour (2) Every hour
(3) Real time (4) 6 times a day

85. The second largest manufacturer of CD's, DVD's and other optical media in the world is
(1) Sony (2) Intel
(3) Philips (4) Moser Baer

86. Which of the following is a legal right and not a fundamental right ?
(1) Right to liberty of thought and expression
(2) Right against exploitation
(3) Right to property
(4) Right to practice religion

87. The term Net Shot is associated with
(1) Badminton (2) Tennis
(3) Table Tennis (4) Volley Ball

88. What is Dry Ice ?
(1) Solidified Carbon Dioxide
(2) Transparent plastic decoy which looks like Ice
(3) Sulfur dioxide at –57° Celsius
(4) Chemically treated Ice to make it dry

89. The rail-based mass rapid transit system in Mumbai has been awarded to a consortium of companies led by
(1) Reliance Infrastructure
(2) Reliance Industries
(3) GMR Infrastructure
(4) Lanco Infrastructure

90. If bilirubin is high in a human body, which organ is most affected ?
(1) Pancreas (2) Liver
(3) Kidney (4) Large intestine

91. Who of the following is one of the most celebrated Photo Journalists in India ?
(1) Gautam Rajadhyaksha
(2) Raghu Rai
(3) Anjii Reddy
(4) Sudhir Dar

92. Inflation implies
(1) rise in budget deficit
(2) rise in general price index
(3) rise in price of consumer goods
(4) rise inmoney supply

93. The India-US Nuclear Deal is called 123 Agreement. What does 123 denote ?
(1) The deal is governed by the clause number 123 of International Atomic Energy Agency, in the context of peaceful use of atomic energy
(2) This number denotes a cluster of clauses of Hyde Act of US Government
(3) This is the simple number related to chain reaction for creation atomic energy
(4) Section 123 of the US Atomic Energy Act

94. In the United States of America, the President is elected
(1) By the electorate
(2) By the electoral college
(3) By the two national political parties
(4) By the US Congress

95. The term 'Uruguay Round' is associated with an important world organization. Which one ?
(1) UNDP (2) GATT
(3) WTO (4) World Bank

96. Siebel is a software firm that is now taken over by
(1) IBM (2) Accenture
(3) Gartner (4) Oracle

97. First Indian motion picture insured by a General Insurance company is
(1) Hum Apke Hain Kaun
(2) Taal
(3) Dil se
(4) Ashoka

98. India's first coalition government in New Delhi was formed under the leadership of
(1) V.P. Singh
(2) P.V. Narasimharao
(3) Morarji Desai
(4) Atal Behari Vajpayee

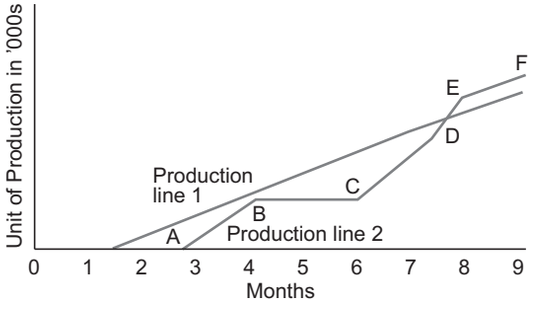
99. WiMax stands for
(1) Wireless internet for microwave access
(2) Wireless inter-operability for microwave access
(3) Worldwide internet for microwave access
(4) Worldwide inter-operability for microwave access

100. El Nino is
(1) A sea storm
(2) A war ocean current
(3) A tropical disturbance
(4) Another name of typhoon

■ Quantitative and Data Interpretation and Sufficiency Section

Directions (Q. 101-105) *For the following 5 questions in this section, correct answers carry 2 marks each.*

101. A factory is to commission two production lines. Production line 1 is to use existing technology. Production line 2 is to use the latest innovations in technology and, while promising to achieve considerable advances in productivity, it will take longer to start and is likely to experience teething problems. Graph indicates the productive record of each product line. Refer to graph to answer the following.



A. Can the duration of reported breakdown be established ?
B. Can the loss of production be quatified ?
(1) Both questions A and B can be answered
(2) Only A question can be answered
(3) Only B question can be answered
(4) Neither question can be answered

102. From a book, a number of consecutive pages are missing. The sum of the page numbers of these pages is 9808. Which pages are missing ?
(1) The page 9808 is missing.
(2) The pages 291 up to 322 are missing.
(3) The pages 291 up to and including 322 are missing.
(4) Either 1 or 3.

103. In the following series, find the one number that is wrong
2, 3, 13, 37, 86, 167, 288.
(1) 3 (2) 13
(3) 37 (4) 86

104. Two sea trawlers left a sea port simultaneously in two mutually perpendicular directions. Half an hour later, the shortest distance between them was 17 km and another 15 min later one sea trawler was 10.5 km farther from the origin than the other. Find the speed of each sea trawler.
(1) 16 km/h, 30 km/h
(2) 18 km/h, 24 km/h
(3) 20 km/h ,22 km/h
(4) 18 km/h, 36 km/h

105. The image below indicates the number of residents at 5 hotels on 1 Feb and 1 July 1998. Which hotel had the greatest increase in the total number of adult residents on 1 July 1998 compared with 1 Feb of that year ?

01-Feb-98				
Hotel	Female Adults	Male Adults	Female Children	Male Children
V	32	64	8	11
W	11	28	3	2
X	47	84	14	7
Y	22	42	5	6
Z	63	104	18	23

01-July-98				
Hotel	Female Adults	Male Adults	Female Children	Male Children
V	41	73	16	14
W	18	37	7	6
X	47	92	12	17
Y	37	59	19	13
Z	71	112	26	20

(1) V (2) W
(3) Y (4) Z

For the following questions in this section, correct answers carry 1 mark each.

106. This question consists of a question and two statements numbered I and II. Decide whether the data given in the statements are sufficient to answer the question.
What is the 57th number in the series of numbers.
I. Each number in the series is three more than the preceding number.
II. The ten number in the series is 29.
(1) The data in statement I alone is sufficient to answer the question while the data in Statement II alone is not sufficient to answer the question.
(2) The data in statement II alone is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question.
(3) If the data either in statements I and II alone are sufficient to answer the question.
(4) If the data in both statements I and II together are necessary to answer the question.

107. The cost of levelling and turfing a square field at Rs 160 per hectare is Rs 2624.46. The cost of surrounding it with a railing costing 25 paise per metre is
(1) Rs 350 (2) Rs 375
(3) Rs 400 (4) Rs 405

108. When the Sun rays inclination increases from 30° to 60°, the length of the shadow of a tower decreases by 60 m. Find the height of the tower.
(1) 50.9 m (2) 51.96 m
(3) 48.8 m (4) None of these

Directions (Q. 109-112) *A survey of movie goers from five cities A, B, C, D and E is summarized below. The first column gives the percentage of viewers in each city who watch less than one movie in a week. The second column gives the total number of viewers who view one or more movies per week.*

City	I	II
A	60	2400
B	20	3000
C	85	2400
D	55	2700
E	75	8000

109. How many viewers in city C watch less one movie a week ?
(1) 2040 (2) 13600
(3) 16000 (4) 3600

110. Which city has the highest number of viewers who watch less than one movie a week ?
(1) City E (2) City D
(3) City B (4) City C

111. A city with the second lowest number of movie watchers is
(1) City E (2) City D
(3) City B (4) City C

112. The total number of all movie goers in the five cities who watch less than one movie per week is
(1) 50000 (2) 36000
(3) 18500 (4) 45250

113. The 288th term of the sequence a, b, b, c, c, c, d, d, d, d, ... is
(1) u (2) v
(3) w (4) x

114. The inequalilty of $p^2 + 5 < 5p + 14$ can be satisfied if
(1) $p \le 6, p = -1$ (2) $p = 6, p = -2$
(3) $p \le 6, p \le 1$ (4) $p \ge 6, p = 1$

115. Answer the question based on the following table.

Relative Sweeteners of Different Substances			
Lactose	0.16	Maltose	0.32
Glucose	0.74	Sucrose	1.00
Saccharin	675.00		

What is the ratio of glucose to lactose in a mixture as sweet as maltose ?
(1) 8 : 21 (2) 1 : 3
(3) 3 : 2 (4) 16 : 9

116. A colourless cube is painted blue and then cut parallel to sides to form two rectangles solids of equal volume. What percentage of surface area of each of new solids is not painted blue ?
(1) 25 (2) 16
(3) 20 (4) 18

117. There are 10 stations on a railway line.The number of different journey tickets that are required by the authorities is
(1) 10! (2) 90
(3) 81 (4) 10

118. A and B throw one dice for a stake of Rs 11, which is to be won by the player who first throws a six. The game ends when the stake is won by A or B. If A has the first throw, what are their respective expectations ?
(1) 5 and 6 (2) 6 and 5
(3) 11 and 0 (4) 10 and 1

119. Which investment gives a better return, assuming the face value of shares to be Rs 10 ?
A. 5% stock at 75, subject to 30% income tax.
B. 4% stock at 90, tax free.
(1) B (2) A
(3) Both A and B (4) None of these

120. Four stacks containing equal number of chips are to be made from 11 orange, 9 white, 13 black and 7 yellow chips. If all these chips are used and each stack contains at least one chip of each colour, what is the maximum number of white chips in any one stack ?
(1) 3 (2) 4
(3) 5 (4) 6

121. A 14.4 kg gas cylinger runs for 104 h when the smaller burner on the gas stove is fully opened while it runs for 80 h when the larger burner on the gas stove is fully opened. Which of these values are the closest to the percentage difference in the usage of gas per hour, of the smaller burner over the larger burner ?
(1) 26.23% (2) 30%
(3) 32.23% (4) 23.07%

122. Study the question and the statements given below. Decide whether any information provided in the statement(s) is redundant and/or can be dispensed with, to answer it.
It 7 is added to numerator and denominator each of fraction a/b, will the new fraction be less than the original one ?
Statement I a = 73, b = 103
Statement II The average of a and b is less than b
Statement III a – 5 is greater than b – 5
(1) II and either I or III
(2) Only I or II
(3) Any two of them
(4) Any one of them

123. Coefficient of variation is useful for studying
(1) Risk (2) Disparity
(3) Consistency (4) All of these

124. A cyclist drove 1 km, with the wind in his back, is 3 min and drove the same way back, against the wind in 4 min. If we assume that the cyclist always puts constant force on the pedals, how much time would it take him to drive 1 km without wind ?
(1) $2\frac{1}{3}$ min (2) $3\frac{3}{7}$ min
(3) $2\frac{3}{7}$ min (4) $3\frac{7}{12}$ min

125. A, B and C started a business by investing 1/2, 1/3 and 1/6 of the capital respectively. After 1/3 of the total time, A withdrew his capital completely and after 1/4 of the total time B withdrew his capital. C kept his capital for the full period. The ratio in which total profit is to be divided amongst the partners is
(1) 1 : 2 : 1 (2) 4 : 1 : 4
(3) 2 : 1 : 2 (4) 1 : 2 : 2

126. A number lock consists of 3 rings each marked with 10 different numbers. In how many cases the lock cannot be opened ?
(1) 3^{10} (2) 10^3
(3) 30 (4) 999

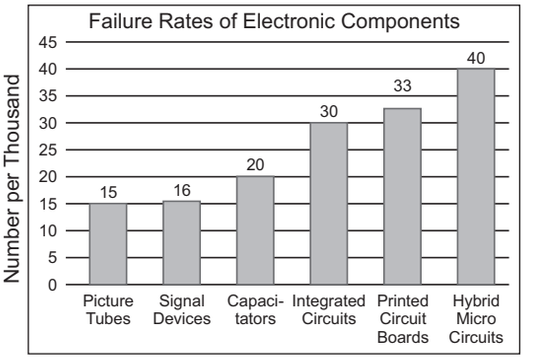
127. A person buys 18 local tickets for Rs 110. Each first class ticket costs Rs 10 and each second class ticket costs Rs 4. What will another lot of 18 tickets in which the number of first class and second class tickets are interchanged cost ?
(1) 112 (2) 118
(3) 121 (4) 124

128. A clock loses 12 min every 24 h. It is set right at 7 : 25 pm on Monday. What will be the time when the clock shows 1 : 45 pm the following day ?
(1) 1 : 20 : 35 pm (2) 1 : 35 : 50 pm
(3) 1 : 25 : 35 pm (4) None of these

129. In a row at a bus stop, A is 7 from the left and B is 9 from the right. They both interchange their pesitions. A becomes 11 from the left. How many peoples are there in the row ?
(1) 18 (2) 19
(3) 20 (4) 21

130. A merchant wants to make profit by selling food grains. Which of the following would maximize his profit ?
I. Sell product at 30% profit
II. Increase the price by 15% over the cost price and reduce weight by 15%
III. Use 700 g of weight instead of 1 kg.
IV. Mix 30% impurities in grains and sell it at cost price
(1) III
(2) II and I
(3) II
(4) All give the same profit

Directions (Q. 131-134) *Study the following chart carefully and answer.*



131. From the failure data of electronics components presented here, which statement is true ?
(1) Integrated circuits and Printed circuit boards are more reliable.
(2) Capacitors are more reliable than Picture tubes.
(3) Higher number of component failures is likely to be due to Picture tubes.
(4) The least reliable component is Hybrid micro circuits.

132. Which of the following components has a failure rate 25% more than that of signal devices ?
(1) Picture tubes
(2) Capacitors
(3) Integrated circuits
(4) Printed circuit boards

133. Lowest priority for investing in any changes or additions to the component manufacturing units, in the company's investment plans, may be given to the following

(1) Printed circuit boards and Hybrid micro circuits
(2) Capacitors and Integrated circuits
(3) Picture tubes and signal devices
(4) Signal devices and Capacitors

134. For the equipments using Integrated circuit boards : 400, Capacitors : 240 and Printed circuit boards : 120 to run with minimum downtime, how many spares should be kept in the store respectively ?
(1) 12, 4, 3 (2) 12, 5, 4
(3) 5, 12, 4 (4) 12, 4, 4

135. The water from a roof, 9 sq m in area, flows down to a cylindrical container of 900 cm^2 base. To what height will the water rise in cylinder if there is a rainfall of 0.1 mm ?
(1) 0.1 cm (2) 0.1 m
(3) 0.11 cm (4) 1 cm

Answers

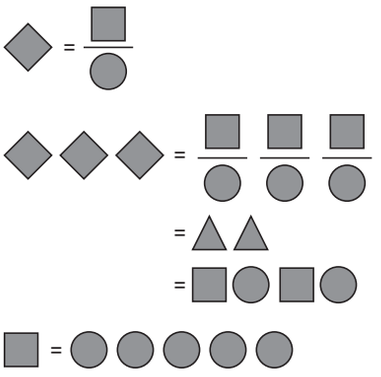
1. (3)	**2.** (3)	**3.** (4)	**4.** (2)	**5.** (1)	**6.** (1)	**7.** (4)	**8.** (3)	**9.** (3)	**10.** (2)
11. (1)	**12.** (1)	**13.** (2)	**14.** (1)	**15.** (3)	**16.** (3)	**17.** (4)	**18.** (4)	**19.** (2)	**20.** (1)
21. (2)	**22.** (3)	**23.** (1)	**24.** (1)	**25.** (4)	**26.** (2)	**27.** (1)	**28.** (1)	**29.** (1)	**30.** (2)
31. (2)	**32.** (4)	**33.** (4)	**34.** (3)	**35.** (2)	**36.** (4)	**37.** (1)	**38.** (1)	**39.** (3)	**40.** (2)
41. (4)	**42.** (1)	**43.** (2)	**44.** (3)	**45.** (2)	**46.** (2)	**47.** (1)	**48.** (1)	**49.** (1)	**50.** (4)
51. (3)	**52.** (2)	**53.** (1)	**54.** (4)	**55.** (3)	**56.** (2)	**57.** (2)	**58.** (3)	**59.** (4)	**60.** (2)
61. (4)	**62.** (4)	**63.** (4)	**64.** (4)	**65.** (2)	**66.** (3)	**67.** (2)	**68.** (2)	**69.** (1)	**70.** (1)
71. (4)	**72.** (4)	**73.** (3)	**74.** (1)	**75.** (2)	**76.** (2)	**77.** (4)	**78.** (1)	**79.** (3)	**80.** (4)
81. (1)	**82.** (1)	**83.** (3)	**84.** (3)	**85.** (4)	**86.** (3)	**87.** (2)	**88.** (1)	**89.** (4)	**90.** (2)
91. (2)	**92.** (4)	**93.** (4)	**94.** (2)	**95.** (3)	**96.** (4)	**97.** (2)	**98.** (3)	**99.** (2)	**100.** (2)
101. (1)	**102.** (2)	**103.** (2)	**104.** (1)	**105.** (3)	**106.** (3)	**107.** (4)	**108.** (2)	**109.** (2)	**110.** (1)
111. (2)	**112.** (4)	**113.** (4)	**114.** (1)	**115.** (1)	**116.** (3)	**117.** (2)	**118.** (2)	**119.** (1)	**120.** (4)
121. (2)	**122.** (3)	**123.** (4)	**124.** (2)	**125.** (3)	**126.** (4)	**127.** (4)	**128.** (2)	**129.** (2)	**130.** (1)
131. (4)	**132.** (2)	**133.** (3)	**134.** (2)	**135.** (4)					

Answers with Explanations

36. (4) In first rotation it will move anti-clockwise while in alternate rotation it will move in clockwise direction.

37. (1) From (A) and (C) sun is written as 'lo' and from (A) and (B) brightly is written as 'ba'.

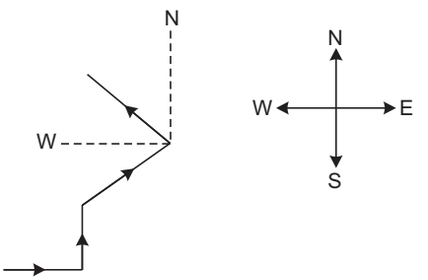
38. (1) From 2.



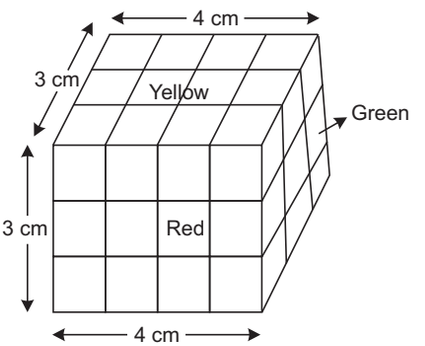
39. (3) For having at least 4 brothers, the number of male child should be 5 at least and for having 3 sisters at least, the number of female child should be 4 at least. So, the smallest number of children in the family might be 5 + 4 = 9.

40. (2) Conversion of statement 'I' to 'I'.

41. (4) North-West



42. (1) Red = 2 × 2 = 4,
Yellow = 2 × 2 = 4
Green = 1 × 2 = 2,
Total = 10



43. (2) 2 (two)

44. (3) Red + Yellow = $2 \times 4 = 8$
Red + Green = $1 \times 4 = 4$
Yellow + Green = $1 \times 4 = 4$
Total = 16

45. (2) According to the given information the grid willl be as B3 will hold the number 9.

S. No.	A	B	C
1	6	3	8
2	2	7	1
3	4	9	5

46. (2) $4 \times 8 = 32$
$32 \times 9 = 288$
$288 \times 10 = 2880$
$2880 \times 11 = 31680$

47. (1) According to the question,
When Seema is called away then both of them have 13 lemon left (*ie*, LCM of 2, 3, 5).
If both of them sell these lemons individually then they will get $\frac{30}{4}$ = Rs 7.50 and $\frac{30}{6}$ = Rs 5 *ie*, a total of Rs 7.50 + Rs 5 = Rs 12.50.
But Jamuna gets = $\frac{65}{5}$ = Rs 12 *ie*, she looses 50 p. But in the question it has been given that she looses Rs 3.50. Hence, number of lemons

$$\therefore \quad \text{Loss of Jamuna} = \frac{210}{4} - \frac{1}{2} \times \frac{420}{5}$$
$$= 52.50 - 42 = 10.50$$

48. (1) Obviously more orange juice in the lemonade.

49. (1) Only A is Implicit.

50. (4) IMMODEST — INDECENT

51. (3) Exhibits the similar pattern of reasoning.

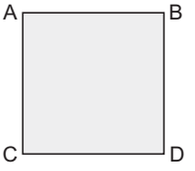
52. (2)

Abdul	Mala	Chetan	AM	MC	CA	AMC
1	1	1 Y	1 Y	1 Y	1 Y	1 Y

Abdul → 2 yellow = AM + AMC
Mala → 3 yellow = AM + MC + AMC
Chetan → 4 yellow = MC + CA + AMC + C
∴ 5 yellow birds and 2 non-yellow birds.

53. (2) I-K, II-M

54. (4) According to the information given obviously from the figure B and C are in diagonally opposite position.



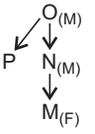
55. (3) According to the information given
1 Wish Bones
2/3 Tied Sundae and Halleujah tied for second place
4 Penn Fe
5 Night Marvel.

56. (2) The total number of students in the circle
$= 6 + 16 - 2 = 20$

57. (2) $(8 + 5) \times 3 = 39$
$(4 + 7) \times 4 = 44$
$(9 + 3) \times 5 = 60$
$\therefore \quad (5 + 4) \times 8 = 72$

For (Q. 59-61) $A_{(F)}$ — Do not play any game.
$B_{(M)}$ — Badminton player
$C_{(F)}$ — Tennis player
$D_{(F)}$ — Do not play any game.
$E_{(M)}$ — Chess player

62. (4)



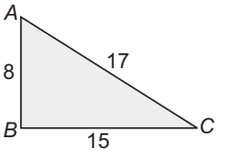
Obvious M is the granddaughter of 'O'.

101. (1) Loss of production cannot be quantified from the given graph as selling price of the unit and the cost of production has not been given.

102. (2) Sum of the page numbers from 291 to 322
$= (1 + 2 + 3 + \ldots + 322) - (1 + 2 + 3 + \ldots + 290)$
$= \frac{322 \times 323}{2} - \frac{290 \times 291}{2}$
$= 52003 - 42195 = 9808$

103. (2) $2 \underset{1}{\leftrightarrow} 3 \underset{10}{\leftrightarrow} 13 \underset{24}{\leftrightarrow} 37 \underset{49}{\leftrightarrow} 86 \underset{81}{\leftrightarrow} 167 \underset{121}{\leftrightarrow} 288$
12 should come in place of 13 as the difference between two numbers is the square of odd numbers in the increasing order.
ie, $(1)^2 = 1$, $(3)^2 = 9$, $(5)^2 = 25$ and so on.

104. (1) $17^2 = 8^2 + 15^2$
We can get the speed of each sea trawler, by using the first part of the information.



105. (3) Total number of adult residents is equal to the sum of female adults and male adults.

	1st Feb, 1998	1 July, 1998	Increase
V	32 + 64 = 96	41 + 73 = 114	18
W	11 + 28 = 39	18 + 37 = 55	16
X	47 + 84 = 131	47 + 92 = 139	8
Y	22 + 42 = 64	37 + 59 = 96	32
Z	63 + 104 = 167	81 + 112 = 193	26

106. (3) Since, number is preceded by 3
$\therefore$ Common difference = 3
Again, $t_{10} = 29$
$\therefore \quad a + 9d = 29$
$a = 2$
∴ Both statements are together required.

107. (4) Area of field = $\frac{2624.46}{160}$ = 1604025 hectare
= 40.53102 Acare
= 164150.64 m^2

Side $= \sqrt{164150.64} = 405.15$

Cost $= 405.15 \times 0.25 =$ Rs $405.15 \approx$ Rs 405

108. (2) Let height of town $= x$

$$\tan 60° = \frac{x}{y}$$

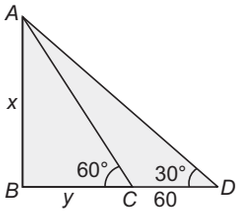
$$y = \frac{x}{\sqrt{3}}$$

$$\tan 30° = \frac{x}{y + 60}$$

$$\sqrt{3}x = y + 60$$

$$\Rightarrow \quad \sqrt{3}x - \frac{x}{\sqrt{3}} = 60°$$

$$x = 30\sqrt{3} = 51.96 \text{ m}$$



109. (2) In City C the number of viewers who watch more than one movie a week is 2400 which is 15% of the total number of veiwers.

$\because$ $15\% = 2400$

$\therefore$ $85\% = 13600$

110. (1) Those city have the highest number of viewers who have maximum percentage of viewers who watch less thanone movie in a week and also has the maximum number of viewers who view one or more movies per week.

For city E the required number of viewers

$25\% = 8000$

$\therefore$ $75\% = 24000$

111. (2)

City	I	II
A	60% (3600)	2400 (40%)
B	20% (750)	3000 (80%)
C	85% (13600)	2400 (15%)
D	55% (3300)	2700 (45%)
E	75% (24000)	8000 (25%)

112. (4) Total number of all movie goers in the five cities who watch less than one movie per week

$= 3600 + 750 + 13600 + 3300 + 24000$

(Data from solution of Q. no. 111)

$= 45250$

113. (4) Here, given sequence are in AP whose $a = 1$, $d = 1$ $S_n = 288$

$\therefore$ 24th term of alphabat will be the answer

i.e., $= x$

115. (1) In the given question, we have to get a mixture of the ratio of glucose to lactose which will be as sweet as maltose. Now, after solving the given ratio.

(1) $\underline{0.74} \times 8 + \underline{0.16} \times 21 = 0.32 \times 29\,(8 + 21)$

$5.92 + 3.36 = 9.28$

$9.28 = 9.28$

(2) $0.74 \times 1 + 0.16 \times 3 = 0.32\,(1 + 3)$

$0.74 + 0.48 = 0.32 \times 4$

$1.22 = 1.28$

(3) $0.74 \times 3 + 0.16 \times 2 = 0.32\,(3 + 2)$

$2.22 + 0.32 = 0.32 \times 5$

$2.54 = 1.60$

(4) $0.74 \times 16 + 0.16 \times 9 = 0.32\,(16 + 9)$

$11.84 + 1.44 = 0.32 \times 25$

$13.28 = 8.00$

So, in option (1) we see that the ratio of glucose and lactose given results in a mixture as sweet as maltose.

116. (3) Area of painted portion is $10x^2$. If side of cube is $2x$.

Area which is not painted $= 4x^2$

$\therefore$Required percentage $= \frac{2x^2 \times 100}{10x^2} = 20$

117. (2) Total number of tickets from one station $= {}^9C_1 = 9$

But it can be reapet by 10 station.

$\therefore$ Total number of tickets $= 10 \times 9 = 90$

119. (1) Return on investment A $= 75 \times \frac{5}{100} = 3.75$

Tax given (@ 30%) on investment A

$$= 3.75 \times \frac{30}{100} = 1.125$$

Net return on investment A $= 3.75 - 1.125 = 2.625$

Return on investment B $= 90 \times \frac{4}{100} = 3.6$

(Investment B is tax free)

$\therefore$ Investment B gives a better return.

120. (4) Each colour is compulsory in each stack. We have 9 white chips. These chips have to be distributed in all the 4 stacks. Considering maximum number of white chips in any one stack, the chips may be distributed as 6, 1, 1, 1 in each stack. Hence, maximum number of white chips in any one stack can be 6.

121. (2) Uses by smaller burner $= \frac{14.4}{104} = 0.1384$

Uses by larger burrer $= \frac{14.4}{80} = 0.18$

$\therefore$ Difference $= 0.18 - 0.1384 = 0.0416$

Percentage $= \frac{0.0416 \times 100}{0.1384} = 30.05 \approx 30\%$

124. (2) Let his speed is x and y is speed of wind.

$$\therefore \quad \frac{1}{x + y} = \frac{3}{60} \quad \Rightarrow \quad x + y = 20$$

$$\frac{1}{x - y} = \frac{4}{60} \quad \Rightarrow \quad x - y = 15$$

$$x = \frac{35}{2} \text{ km/L}$$

$$\therefore \quad \text{Time} = \frac{1}{35/2} = \frac{2}{35} \text{ h}$$

$$= \frac{2}{35} \times 60 = 3\frac{3}{7} \text{ min}$$

125. (3) Required ratio $= \frac{1}{2} \times \frac{1}{3} : \frac{1}{3} \times \frac{1}{4} : \frac{1}{6}$

$= \frac{1}{6} : \frac{1}{12} : \frac{1}{6} = 2 : 1 : 2$

126. (4) Total number of ways of lock $= 10^3 = 1000$

$\therefore$ Since, one case it will open.

$\therefore$ No. of case when it will not open $= 1000 - 1 = 999$

127. (4) Let number of first class ticket $= x$

$\therefore$ Number second class tickets $= 18 - x$

$10x + 3(18 - x) = 110$

$\therefore \quad x = 8$

$\therefore$ Second class tickets $= 10$

$\therefore$ Now, cost $= 10 \times 10 + 3 \times 8 = 124$

128. (2) After 18 h 20 min it will lose more than 9 min.

$\therefore$ Correct time is 1 : 35 : 50 pm

129. (2) Total number of people is $= 7 + 4 + 8 = 19$

130. (1) He will gain maximum profit when he use 700*g* weight.

ie, $= \frac{300}{700} \times 100 = 42.85\%$

132. (2) 25% more of signal devices are $= 16 \times \frac{125}{100}$

$= 20$ Capacitors

134. (2) Number of Integrated circuit Boards $= \frac{30}{1000} \times 400 = 12$

Number of capacitor $= \frac{20}{1000} \times 240 = 4.8 \approx 5$

Number of printed circuit board

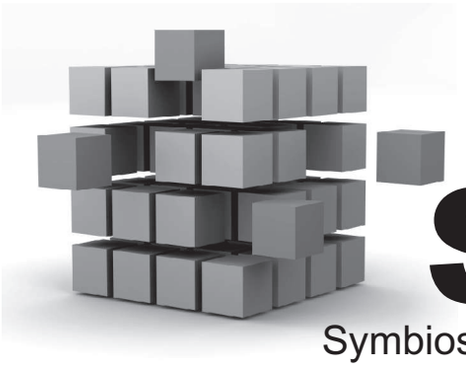
$= \frac{33}{1000} \times 120 = 3.96 \approx 4$

135. (4) Volume of rain = Volume of container

$90000 \times \frac{0.1}{10} = 900 \times h$

$900 = 900h$

$h = 1$ cm



SNAP 2007

Symbiosis National Aptitude Test

Solved Paper

Quantitative and Data Interpretation and Sufficiency

1. Thirty days are in September, April, June and November. Some months are of thirty one days. A month is chosen at random. Then its probability of having exactly three days less than maximum of 31 is
(1) 15/16 (2) 1
(3) 3/48 (4) None of these

2. In 4 yr, the simple interest on a certain sum of money is 7/25 of the principal. The annual rate of interest is
(1) 4% (2) 4.5%
(3) 7% (4) 9%

3. Inside a triangular park, there is a flower bed forming a similar triangle. Around the flower bed runs a uniform path of such a width that the sides of the park are exactly double the corresponding sides of the flower bed. The ratio of areas of the path to the flower bed is
(1) 1 : 1 (2) 1 : 2
(3) 1 : 3 (4) 3 : 1

4. The intersection of two cubes cannot be
(1) cube
(2) triangle
(3) a rectangle
(4) None of the above

5. The table below shows the temperatures in degrees Fahrenheit in a city over one week.

Noontime Temperatures						
Mon	Tue	Wed	Thu	Fri	Sat	Sun
66	78	75	69	78	77	70

If m represents the median temperature, f represents the temperature that occurs most often, and a represents the average (arithmetic mean) of the seven temperatures, which of the following is the correct order of m, f and a ?
(1) $a < m < f$ (2) $a < f < m$
(3) $m < a < f$ (4) $m < f < a$

Directions (Q. 6-8) *Use the table about the speed of a train over a 3 h period. The time count does not begin from starting of train.*

Time (minutes)	Speed at given time (in km/h)
0	40
30	45
45	47.5
60	50
90	55
120	60
150	65
180	70

6. How fast was the train travelling $2\frac{1}{2}$ h after the beginning of the time period ?
(1) 50 km/h (2) 55 km/h
(3) 60 km/h (4) 65 km/h

7. During the three-hour period shown in the table, the speed of the train increased by
(1) 25% (2) 100%
(3) 75% (4) 125%

8. At time t, measured in minutes after the beginning of the time period, which of the following gives the speed of the train in accordance with the table ?
(1) $t/6$ (2) $6t$
(3) $40 + t$ (4) $40 + t/6$

9. A ladder is lying against a wall which is 5 m high. If the ladder slips 2 m away from the wall, the top of the ladder touches the foot of the wall. The length of the ladder is
(1) 5 m (2) 5.25 m
(3) 7.25 m (4) 4 m

10. The profits of Biscuits India Ltd. soared by 32% in the year 2006–07 as compared to year 2005–06. By what per cent did Biscuits India's Sales increase in 2006–07 compared to the previous year ? (Assume : Profit = Sales – Expenses)
Decide whether the information given in the two statements is sufficient to solve the problem.

Statement (A) Expenses in 2006–07 were Rs 1400 crores, as compared to Rs 1220 crore in 2005–06.
Statement (B) Sales in 2006–07 were Rs 4300 crore.
(1) Any one of the two statements (A) or (B) taken alone, is sufficient to answer the question
(2) Each of the statement (A) or (B) taken alone, is sufficient to answer the question
(3) Both statements, taken together, are sufficient to answer the question, but neither of them alone is sufficient
(4) Both the statements together are insufficient to answer the question

11. Two identical trains *A* and *B* running in opposite directions at same speed take 2 min to cross each other completely. The number of bogies of *A* are increased from 12 to 16. How much more time would they now require to cross each other ?
(1) 40 s (2) 50 s
(3) 60 s (4) 20 s

12. The average of nine numbers is *M* and the average of three of these is *P*. If the average of remaining numbers is *N*, then
(1) $M = N + P$ (2) $2M = N + P$
(3) $3M = 2N + P$ (4) $3M = 2P + N$

13. A special lottery is to be held to select a student who will live in the only deluxe room in a hostel. There are 100 Year-III, 150 Year-II, and 200 Year-I students who applied. Each Year-III's name is placed in the lottery 3 times; each Year-II's name, 2 times; and each Year-I's name,1 time. What is the probability that a Year-III's name will be chosen ?
(1) 1/8 (2) 2/9
(3) 2/7 (4) 3/8

14. How many arrangements can be formed out of the letters of the word EXAMINATION so that vowels always occupy odd places ?
(1) 72000 (2) 86400
(3) 51600 (4) 64000

15. Area of a square natural lake is 50 sq km. A diver wishing to cross the lake diagonally, will have to swim a distance of
(1) 10 miles (2) 12 miles
(3) 15 miles (4) None of these

16. In a school drill, a number of children are asked to stand in a circle. They are evenly spaced and the 6th child is diametrically opposite the 16th child. How many children are made to stand in the circle ?
(1) 16 (2) 20
(3) 22 (4) None of these

17. A man purchased 40 fruits; apples and oranges for Rs 17. Had he purchased as many oranges as apples and as many apples as oranges, he would have paid Rs 15. Find the cost of one pair of an apple and an orange.
(1) 70 paise (2) 60 paise
(3) 80 paise (4) 1 rupee

18. Consider the following piecewise definition of the function *f*.

$$f(x) = \begin{cases} 3 - x, & \text{if } x < 0 \\ x^2 + 2, & \text{if } x \geq 0 \end{cases}$$

Evaluate $f(-3)$.
(1) 6 (2) 0
(3) 11 (4) –7

19. A man earns 6% SI on his deposits in Bank *A* while he earns 8% simple interest on his deposits in Bank *B*. If the total interest he earns is Rs 1800 in three years on an investment of Rs 9000, what is the amount invested at 6% ?
(1) 3000 (2) 6000
(3) 4000 (4) 4500

20. Bags I, II and III together have ten balls. If each bag contains at least one ball, how many balls does each bag have ? Decide whether the data given in the statements are sufficient to answer the question.
Statement (A) Bag I contains five balls more than bag III.
Statement (B) Bag II contains half as many balls as bag I.
(1) Statement (A) alone is sufficient, but statement (B) alone is not sufficient to answer the question asked
(2) Statement (B) Alone is sufficient, but statement (A) alone is not sufficient to answer the question asked
(3) Both statements (A) and (B) together are sufficient to answer the question asked, but neither statement alone is sufficient to answer the question asked
(4) Each statement alone is sufficient to answer the question asked

21. The number 311311311311311311311 is
(1) divisible by 3 but not by 11
(2) divisible by 11 but not by 3
(3) divisible by both 3 and 11
(4) neither divisible by 3 nor by 11

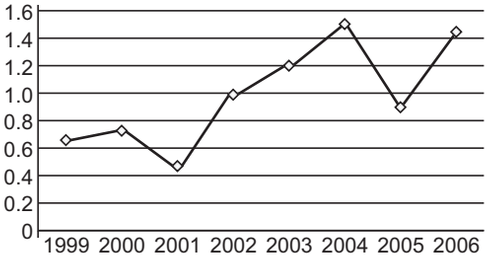
22. Two persons are climbing up on two moving escalators which have 120 steps. The ratio of 1st person's speed to that of 1st escalator is 2 : 3 (steps). The ratio of 2nd person's speed to that of 2nd escalator is 3:5 (steps). Find the total number of steps they both have taken together.
(1) 85 (2) 93
(3) 80 (4) 75

23. A rainy day occurs once in every 10 days. Half of the rainy days produce rainbows. What percent of all the days do not produce rainbow ?
(1) 95% (2) 10%
(3) 50% (4) 5%

24. If $n = 1 + x$, where *x* is the product of 4 consecutive positive integers, then which of the following is/are true ?
I. *n* is odd.
II. *n* is prime.
III. *n* is a perfect square.
(1) I and III only
(2) I and II only
(3) I only
(4) None of the above

25. In a factory, each day the expected number of accidents is related to the number of overtime hour by a linear equation. Suppose that on one day there were 1000 overtime hours logged and 8 accidents reported and on another day there were 400 overtime hours logged and 5 accidents. What is the expected number of accidents when no overtime hours are logged ?
(1) 2 (2) 3
(3) 4 (4) 5

Directions (Q. 26-28) *The following line graph shows the ratio of the wheat production to rice production of a state in India over a period 7 yr from 1998-2006. Answer the questions based on this data.*



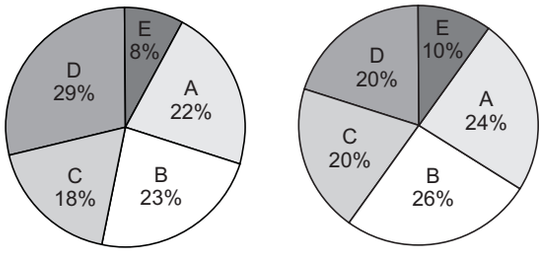
26. In how many years wheat production was more than the rice production ?
(1) 2 (2) 3
(3) 4 (4) 5

27. If the total annual grain requirement (wheat and rice together) of the state is 5 lakh ton and if the rice production during 2001 was 3 lakh ton, how much grain had to be imported from the other state of India to fulfill the requirement ?
(1) 2.00 lakh ton (2) 1.55 lakh ton
(3) 1.05 lakh ton (4) 0.65 lakh ton

28. If the rice production in 2003 was 4 lakh ton what was its wheat production during 2003 ?
(1) 1.2 lakh ton (2) 3.6 lakh ton
(3) 4.8 lakh ton (4) 5.0 lakh ton

Directions (Q. 29-32) *The following pie charts give the percentage distribution of employees in different. Departments in XYZ Company Ltd. during the years 2005 and 2006. Answer the questions using this data.*

2005 (No. of Employees → 18,000) 2006 (No. of Employees → 20,000)



A-Administration; B-Operations; C-Sales and Marketing; D-Finance and Accounts; E-Corporate HQ

29. In which Dept is the variation in strength the maximum in 2006 ?
(1) A (2) B
(3) D (4) E

30. If 300 employees left operations at the end of 2005, how many joined in 2006 ?
(1) 480 (2) 960
(3) 1360 (4) None of these

31. What is the percentage increase in number of employees in sales and marketing ?
(1) 2% (2) 2.34%
(3) 23.45% (4) None of these

32. If the average monthly salary of employees in administration was Rs 12000 in 2005, what was the approximate total salary expense of administration in 2005 ?
(1) Rs 57 lakh (2) Rs 57 crore
(3) Rs 66 crore (4) Rs 66 lakh

Directions (Q. 33-35) : *The following table gives cost data of select stock prices on 3rd Dec 2003 in two markets, BSE of India and NQE of Kya Kya island. Closing stock refers to the price at the close of trading hours and opening stock to the price at the beginning of the day. The currency of Kya Kya is # and the exchange rate is # = Rs 11.*

3rd Dec 2003				
	Closing		Opening	
	BSE (Rs)	NQE (in #)	BSE (Rs)	NQE (in #)
SIFY	247	22	232	21
INFY	115	10.5	105	9.5
Wipro	65	6.5	60	5.5
TCS	444	40	450	40.5

33. Which share and which market showed the highest percentage increase on 3rd Dec 2003 ?
(1) SIFY on BSE (2) INFY on NQE
(3) Wipro on NQE (4) TCS on NQE

34. Arbitrage means buying in one market and selling simultaneously in another market. If Mr. Ghosh Babu were to buy a share at the opening price on one exchange and be able to sell it at the opening price on the other exchange, on which share does he make maximum percentage profit ?
(1) SIFY (2) INFY
(3) Wipro (4) TCS

35. In an M and A deal SIFY is purchased by its parent company SATYAM which purchases 15% of SIFY's equity shares. Total SIFY's equity shares is 1 million. How much does Satyam pay in Rs for the stake if 50% of its purchases were on BSE's operating price and balance on BSE's closing price ?
(1) 36 million (2) 3.5 million
(3) 363 million (4) 217 million

36. In March 2007, Computers Ltd. made a bundled offer of its laptops together with deskjet printers to boost sales, though both the laptop and the printer were also available individually.
What is the price of the printer, if purchased separately ? Decide whether the information given in the two statements is sufficient to solve the problem.
Statement (A) The bundled offer price was Rs 42,600.
Statement (B) The Laptop, without the bundle offer was priced at Rs 39,400.
(1) Any one of the two statements (A) or (B) is sufficient to answer the question
(2) Each of the statements (A) or (B) taken alone, is sufficient to answer the question
(3) Both statements, taken together, are sufficient to answer the question, but neither of them alone is sufficient
(4) Both the statements together are insufficient to answer the question

Directions (Q. 37-40) *The following chart presents the data collected for an online job portal Infinite jobs.com. For each question, there are two statements, each of which could be either true or false. Choose one of the following options as an answer.*

(1) Only [A] is true
(2) Only [B] is true
(3) Both [A] and [B] are true
(4) Neither [A] nor [B] is true

Years	Category	Number of Registrations	Number of Candidates who posted their CVs	Number of Candidates short-listed by Employers	Number of offered Jobs
2004	Technical	61,205	59,981	684	181
	Managerial	19,236	15,389	138	48
2005	Technical	63,298	60,133	637	115
	Managerial	45,292	40,763	399	84

37. **Statement A** The success rate of candidates getting shortlisted based on CVs is higher for Managerial category than for Technical category in 2005.
Statement B The success rate of candidates getting shortlisted based on CV's is better for Managerial category in 2005 than in 2004.

38. **Statement A** In 2004, the number of candidates offered jobs as a proportion of the number of CVs posted was higher for Technical than for Managerial candidates.
Statement B In 2004, among those short-listed by Employers, the Technical category had a higher success rate in securing jobs than the Managerial category.

39. **Statement A** The percentage of drop outs (from Registration state to posting CVs) had decreased from 2004 to 2005 for Managerial category.
Statement B The percentage of drop-outs was higher for Technical than for Managerial category in 2005.

40. **Statement X** The percentage increase in number of Registrations from 2004 to 2005 is higher for Managerial candidates than for Technical condidates.
Statement Y The overall registrations have grown by more than 25%.

■ General English

Directions (Q. 41-46) *Read the following passage and answer within its context.*

The world dismisses curiosity by calling it idle, or mere idle curiosity—even though curious persons are seldom idle. Parents do their best to extinguish curiosity in their children because it makes life difficult to be faced every day with a string of answerable questions about what makes fire hot or why grass grows. Children whose curiosity survives parental discipline are invited to join our university. Within the university, they go on asking their questions and trying to find the answers. In the eyes of a scholar, that is mainly what a university is for. Some of the questions that scholars ask seem to the world to be scarcely worth asking let alone answering. They ask questions too minute and specialized for you and me to understand without years of explanation. If the world inquires of one of them why he wants to know the answer to a particular question he may say, especially if he is a scientist, that the answer will in some obscure way make possible a new machine or weapon or gadget. He talks that way because he knows that the world understands and respects utility. But to you who are now part of the university, he will say that he wants to know the answer simply because he does not know it. The way a mountain climber wants to climb a mountain simply because it is there. Similarly a historian when asked by outsiders why he studies history may come out with argument that he has learnt to repeat on such occasions, something about knowledge of the past making it possible to understand the present and mould the future. But if you really want to know why a historian studies the past, the answer is much simpler : something happened, and he would like to know what. All this does not mean that the answers which scholars find to their questions have no consequences. They may have enormous consequences but these seldom from the reason for asking the question or pursuing the answers. It is true that scholars can be put to work answering questions for the sake of the consequences as thousands are working now, for example, in search of a cure for cancer. But this is not the primary function of the scholar, for the consequences are usually subordinate to the satisfaction of curiosity.

41. Common people consider some of the questions asked by scholars as unimportant
(1) since they are not worth asking or answering
(2) because the question is related to new machines and gadgets
(3) because the common man doesn't understand questions without years of explanations
(4) scholars ask very minute, specialized questions beyond the comprehension of the common man

42. In the statement 'that is mainly what a university is for ' 'that' refers to
(1) parents refusal to answer questions
(2) children's curiosity that survives parents strictures
(3) questions not worth answering
(4) the aim and scope of the university to provide an opportunity to curious minds to find out the answers to their questions

43. According to the passage the general public respects
(1) new inventions
(2) any useful invention
(3) any invention that makes life easier for them
(4) a scientist who invents gadgets and machines for them

44. The writer compares the scientist to
(1) a historian and mountain climber
(2) a historian
(3) a mountain climber
(4) a scholar

45. The primary function of a scholar is different from the search for a cure for cancer because
(1) the answers to the scholar's question have no consequence unlike the results of the research involving a cure for cancer

(2) the answer sought by the scholar is selfish unlike the consequences of cancer research which are for the common weal
(3) the primary function of a scholar is satisfaction of his mental curiosity, while research involving a cure for cancer demands a constant, systematic and planned pursuit by several scholars
(4) several scholars work for a cancer cure while a single scholar works with a selfish motive

46. Idle curiosity means
(1) curiosity is lazy
(2) idle people are curious
(3) curiosity is apt
(4) casual curiosity

47. Find the correct match of definition/meaning with usage for the word : WOULD

	Definition/meaning		Usage
(1)	willingness	(5)	He would get for a walk even it was raining
(2)	obstinacy persistence	(6)	He would do as you say
(3)	determination	(7)	He beat the ox, but it wouln't move
(4)	improbable or real condition	(8)	If you came across a snake what would you do ?

(1) 1-8, 2-5, 3-6, 4-7 (2) 1-6, 2-7, 3-5, 4-8
(3) 1-5, 2-7, 3-8, 4-6 (4) 1-7, 2-5, 3-6, 4-8

48. Choose the most appropriate passive construction of the sentence: 'He is doing his job well'.
(1) His job is done well by him
(2) His job is being done well
(3) He has been doing his job well
(4) His job is being well done

49. There are three underlined words below, followed by their usages. Determine the sentences, in which the use of words is correct or appropriate.
Pray, Prey, Prying
(A) If you pray with faith, they say, it will be answered.
(B) He has fallen a prey to cheats.
(C) Prying into the affairs of others is bad.
(1) A and B (2) A and C
(3) A, B and C (4) B and C

Directions (Q. 50-54) *The following is an excerpt from a recent article by David Ewing Duncan. Read the passage and answer the questions within its context.*

Eye surgeon Virendar Sangwan has perfected a procedure so cutting-edge that most who have tried it have failed. In an operating theatre in the central Indian city of Hyderabad, he surgically implants corneas grown in a petri dish from stem cells by his colleague Geeta Vemuganti in patients with damaged eyes. Together they perform about 80 corneal regeneration procedures a year, making the L.V. Prasad Eye Institute, where they work, one of the most prolific facilities in the world using stem cells to regenerate tissues of any kind.

The Sangwan-Vemuganti team uses stem cells found in the tissues of living adults, not ones derived from embryos. Teams all over the world are working with adult stem cells, trying to coax them to regrow cells in hearts, brains, livers and other organs, but progress is slow. Besides corneas, scientists have had some success regrowing skin cells and bone tissues, but those procedures remain experimental. "A number of programs around the world have tried to perfect this treatment, but they have had bad outcomes," says University of Cincinnati eye surgeon and stem cell specialist Edward Holland. "It is impressive what they are doing at Prasad." In addition to the Hyderabad project, only Holland's program and a half-dozen others in the world conduct operations using corneas grown from stem cells.

The treatment uses stem cells harvested from the limbus, located where the cornea touches the white of the eye. For those with damaged corneas, these cells called "limbic" and "conjunctiva" are harvested from a patients good eye, if he has one, or from a close relative. They are placed in a petri dish and chemically tweaked to grow into the lower layer of a cornea, called epithelium. It is then transplanted into the eye of the patient where in most cases it takes hold and grows. In 56% of the cases at the Prasad institute, patient could still see clearly after 40 months later.

Indians are well known for reverse engineering, meaning they can deduce how drugs are made in order to produce generic versions. But in this case, Sangwan and Vemuganti, a pathologist, developed the technique on their own from reading papers and running experiments in the lab. Sangwan says he had a number of patients with burned eyes who could not be helped with standard corneal transplants from cadavers, so he persuaded Vemuganti to try growing corneas in her lab. "You know how to grow cells, and I know how to do the transplant surgery," Vemuganti recalls him saying. "Why don't we work together?" She smiles and shakes her head. "I had no clue if this was going to work."

Vemuganti's major innovation was developing a platform on which to grow corneas. First she designed a circular glass tube about the size of a stack of coins. Then she overlaid the glass with tissue from a human placenta which is "a good surface to grow corneas on." She says. After that she placed stem cells in four places around a circle, added a growth medium, and watched the corneas begin to grow.

Commercial interests among stem cell companies for the procedure has been scant because of the perceived small volume of patients, says venture capitalist Antoun Nabhan of Bay Capital, who sits on the board of Cellerant, a leading stem cell company in San Carlos, Calif. But corneal stem cell treatment may have wider applications, say ophthalmologist Ivan Schwab of University of California at Davis. "These stem cells are similar to others in the body that make mucous membrane," he says. "These techniques of growing stem cells might one day be used to treat mucous-membrane tissue in the sinuses, bladder, and other organs."

50. According to the article Sangwan-Vemuganti team's cutting edge procedure of implanting cornea grown from stem cells is considered a major advancement by the experts because
(1) they derive stems cells from embryos
(2) their labs are customised to grow stem cells
(3) they regrow cells in hearts, brains, livers with stems cells from tissues of living adults
(4) they derive stem cells from tissues of living adults and grow cells in labs

51. Sangwan-Vemuganti procedure is carried out on
(1) patents requiring any corneal transplant
(2) patients with damaged corneas
(3) patients with damaged eyes of any kind
(4) None of the above

52. The world recognizes this Indian innovation because Indian scientists are normally known
(1) to be good at analysing and finding out a method of how an existing drug is made
(2) as they are good researchers of drugs
(3) as they are good at carrying out experiments to create generic drugs
(4) as they are able to carry out drug trials on large samples

53. The pathologist, Vemuganti, started growing cornea in a petri dish
(1) by following procedures published in research papers
(2) by inventing a totally new procedure
(3) by experimenting with procedures published in journals
(4) by following the instructions of the transplant surgeon

54. In the context of the passage choose the correct set of meanings for the words: 'platform' and 'generic'
(1) lab table; related to genes
(2) method; related to genes
(3) lab experiments; without a brand name
(4) methodology; without a brand name

55. Choose the option which is closest in meaning to the word **SUBTLE**
(1) Innocent (2) Elusive
(3) Dangerous (4) Insidious

56. The following sentence has a missing punctuation mark, choose the right answer.
My mother who is from the village is very superstitious.
(1) Brackets (2) Comma
(3) Semicolon (4) Apostrophe

57. For the pair of sentences below choose the right option
(A) Those are them
(B) Those are they
(1) The first sentence has an error
(2) The second sentence is erroneous
(3) Both sentences are incorrect
(4) Both sentences are correct

Directions (Q. 58-62) *Read the following passage and answer within its context.*

TRIPs agreement provides a comprehensive set of global trade rules for the protection of copyright patents, trademarks, industrial designs, trade secrets, semiconductor lay out designs, and geographical indications, that apply to all the number-countries irrespective of their levels of development, natural and human endowments and history. Every member-country has been asked by the WTO to amend its national patent law to conform to that universal globalized format for legislation relating to pharmaceutical, agrochemical, food, alloys, etc.

Under Article 65, the developed countries have been asked to change their laws within another five years, and the less developed countries within an additional five years. The least developed countries have been asked to make those changes by 2005 AD.

This attempt at global standardisation and uniformity by way of TRIPs agreement is in conflict with the main thrust of the Rio Earth Summit of 1992 that set out the conditions for sustainable development. These two reveal two contrasting types of international approaches and norms.

While the 1992 Earth Summit and the 1993 convention on bio-diversity (CBD) focused on 'diversity' as being fundamental to sustain life and development, TRIPs and WTO are pushing for 'conformity' to international standardized norms on patents, services, labour, investment and what not irrespective of their history, ecology, level of economic development, etc. But despite their diametrically opposed viewpoints, 170 countries signed CBD upholding the need for diversity, and 50 countries signed the TRIPs agreement in 1994 claiming the urgency of uniformity, with a very large element of common names (130) in both.

The convention on bio-diversity (CBD) in its Article 16.5 specifically asserts that intellectual property right must not be in conflict with conservation and sustainable use of bio-diversity, a provision that has been totally ignored by those who composed the TRIPs agreement. While in case of agriculture the higher yield of patented products induces the farmers to switch form a more varied production pattern, the resulting narrowing of genetic base makes the economy and society more vulnerable to plant diseases and epidemics. It is true that the move towards cultivation of a smaller number of higher yielding varieties and the uniform spread of the same variety over a large space predates the present debate on patent, particularly since the introduction of the green revolution technology in the mid-sixties, but there can be no doubt that the latter has brought about a qualitative change in the scenario and has created possibility of a vast quantitative change too in that direction. So far, no attempt has been made to reconcile the two conflicting approaches of CBD and TRIPs. If diversity is so important for sustaining life, how can WTO demand conformity to standardised global formats ?

58. The author points out that intellectual property rights and their administration mechanism
(1) is throttling the interest of global bio-diversity
(2) is working to help sustain golbal bio-diversity
(3) is being sustained by golbal bio-diversity
(4) is what the global bio-diversity needs

59. Which of the following has been said by the author in the passage ?
(1) A high number of countries have signed both CBD and TRIPs, two conflicting treaties
(2) A narrow genetic base, if stuck to for ling, is fraught with danger
(3) Although a nondiscriminatory approach has been followed in the applicability of TRIPs, there has been a confessional attitude in prescribing a timeframe for Transition, as per needs of the respective countries
(4) The author is supportive of international conventions and treaties such as TRIPs, CBD etc.

60. Out of the countries that signed CBD, the percentage of those that signed the TRIPs also, is
(1) 76.5 (2) 74.5
(3) 78.5 (4) 80.2

61. According to the author, a higher-yield seed variety is not always welcome as it also ultimately leads to
(1) diseases among the consumers
(2) diseases among the plants
(3) monopoly of developed countries
(4) monopoly of developed countries

62. As per the TRIPs agreement not much differentiation is made between a developed country such as the USA and an undeveloped country such as Sudan. This is

(1) definitely true (2) probably true
(3) probably false (4) definitely false

63. A single word equivalent for the statement 'Speak falsely with deliberate intent' is
(1) repudiate (2) prevaricate
(3) disparage (4) equivocate

64. Choose the word with the correct spelling
(1) paraphrenalia (2) progarmme
(3) pediatrics (4) beserk

65. Which two sentences in the following convey the same idea ? Choose from the combinations listed below
(i) Wasn't there any checking at the ariport ?
(ii) I want to know if there was any checking at the airport.
(iii) I wonder if there should have been any checking at the airport.
(iv) There should have been checking at the airport.
(1) (ii), (iii) (2) (i), (iv)
(3) (iii), (iv) (4) (ii), (iv)

66. Choose the most appropriate sentence from the following
(1) You should at once report it to the concerned authority
(2) You should report it at once, to the authority concerned
(3) You should report it at once, to the concerned authority
(4) You should at once report it to the authority concerned

67. Find the correct match of grammatical function with usage for the word
AFTER

	Grammatical function		Usage
1.	adjective	5.	You may go after having your lunch.
2.	adverb	6.	It appears to be the after of the disease.
3.	conjunction	7.	Many graduates are hankering after jobs.
4.	preposition	8.	He came soon after.

(1) 1-8, 2-5, 3-6, 4-7 (2) 1-6, 2-5, 3-8, 4-7
(3) 1-5, 2-8, 3-7, 4-6 (4) 1-6, 2-8, 3-5, 4-7

68. 'Neophyte' is the opposite of
(1) student (2) clown
(3) veteran (4) professional

69. Find the maximum number of times that any one of the given words fits the set of sentences
disabled flimsy crippled lame
1. Don't make _____ excuses.
2. Liberalization may have _____ smaller manufacturers.
3. Being a defaulter at the stock exchange makes him a _____ duck.
4. A _____ person may limp.
(1) in all the four sentences
(2) in three sentences
(3) in two sentences
(4) in only one

70. Which of the following does not make a sensible word/phrase when added to the given word ?
FIRE
(1) fly (2) engine
(3) stick (4) escape

71. Arrange the sentence 1, 2, 3, 4 to form a logical sequence between sentences I and II. Choose the alternative where the four combinations make a meaningful sentence.
I. We all value having the freedom
1. Which many of us fail to honour
2. to make the choices we want in our careers
3. but with great freedom comes great responsibility
4. so most companies fall prey to the policies which become rigid
II. and that's probably one reason we find most companies not following what they preach
(1) 1, 3, 4, 2 (2) 2, 3, 1, 4
(3) 1, 4, 2, 3 (4) 3, 2, 1, 4

Directions (Q. 72-74) *Read the edited excerpt of an article by NELSON VINOD MOSES and answer the questions in this context.*

A successful non-resident Indian employed in the United States returns to a backward Indian village and transforms the lives of the villagers. Sounds familiar ? At 31, Ashwin Naik is pacing through the path Shah Rukh Khan traced in his off beat Bollywood movie, Swades. Naik had just quit his cushy job in a genomics firm in the US to join MIT Sloan School of Business. With a month in hand, he headed home a travelled through the remote areas of Bagalkot district in Karnataka. The woeful social conditions he saw moved him. Naik chucked the MBA course and in six months set up Vaatsalya Healthcare, a rural healthcare delivery system.

In February 2005, Vaatsalya's first hospital opened in Hubli. Two more centrs were opened in Gadag and Karwar to offer specialist services of surgeons and facilities such as physiotherapy for children suffering from cerebral palsy. "We introduced paediatric surgery for infants below six months," says Naik. "Else, patients would have to be taken to distant cities of Hubli or Bangalore." Naik plans 100 more units in five states in the next three years. Mere charity by an affluent, middle-class professional ? Far from it, Vaatsalya is one among rapidly spreading 'for profit' social enterprises that serves the poor and brings in profit. Mumbai-based Ziqitza, an imbalance serives company, is another. It never refuses a patient for money, and charges Rs 50 to Rs 200.

Done fleetingly in India and elsewhere till now, entrepreneurial minds with a social conscience are methodically creating such models at a greater pace. There has been a boom in the past two years, "says Varun Sahni, country director of Acumen Fund, a US-based social fund that invests in companies that target low income communities. "Currently, there are about 1,000 in India."

The timing seems perfect. There is a wide market acceptance and funding has been coming in easily. These enterprises work across a swathe of areas including healthcare, education, rural energy, agriculture, arts and crafts, banking and more. 'For profit' entrepreneurs are obsessed with social and environmental impact in addition to the financial returns. Since they are answerable to the investors, they try expanding the business rapidly. SKS Microfinance, for instance, started in 1998 and has now over 900,000 customers, 440 branches and an outstanding loan disbursement of over Rs 452 crore as of August 2007.

72. Identify the appropriate business model of the kind of enterprise described by the author.
(1) Servicing societies at no profit
(2) Profiting from poor people

(3) Setting up enterprises for masses of low-income groups on experimental basis
(4) Setting up enterprises for social causes for profit and expand rapidly

73. Which of the following companies does not illustrate the idea explained by the author ?
(1) SKS Microfinance
(2) Acumen Fund
(3) Ziqitza
(4) Vaatsalya Healthcare

74. According to the author, which of the following options describes 'for profit' entrepreneurs most appropriately ?
(1) NRI's paying back to their motherland
(2) Those affluent, middle-class professionals treating it as charity
(3) Those who work towards getting financial returns on social business by expanding quickly
(4) Those who have sympathetic investors for their business ideas for poor

75. A contextual usage is provided for the world below. Pick the word that is most inappropriate. MALINGER : The young man made it a point to malinger inspite of the assigned work load.
(1) Wander (2) Laze
(3) Evade (4) Argue

76. The following is a scrambled sentence with the segments marked 1, 2, 3 and 4. Choose the alternative with the order segments that best reconstruct the sentence
A. For all the padre's rhetoric about the English as God's Chosen People, the padre had a whole tribe of Anglo-Indian first cousins
B. Padre Rotton was an even more striking case
C. by various Indian wives, all of whom were all that moment engaged in flghting on the rebel side in Avadh, where they took an active part in besieging the British Residency in Lucknow
D. These included James Rotton who could not speak English and the twenty two Muslim sons of his convert cousin, Felix Rotton
(1) A, B, C, D (2) B, A, D, C
(3) A, D, B, C (4) B, D, A, C

77. Choose the sentence in which the given word is used correctly (grammatically and semantically)
ALMOST
(1) As I crossed the road a scooterist almost hit me
(2) Crossing the road a scooterist hit me almost
(3) A scooterist across the road almost hit me
(4) A scooterist almost hit me crosing the road

78. In the following sentence choose the erroneous segment

A	B	C
We took a taxi	so we would be on time	for the meeting

(1) Error in segment A
(2) Error in segment B
(3) Error in segment C
(4) No error

79. Find the ODD one out from the group of words which are related in some way or the other
(1) din (2) cacophony
(3) racket (4) cadence

80. Fill in the blanks with the correct alternative
Caw is to crows as ___ is to cows.
(1) bleat (2) snort
(3) low (4) bellow

General Awareness

81. A new coin called 'Rupiya' was issued for the first time by
(1) Sher Shah Suri
(2) Akbar
(3) Ala-ud-din Khalji
(4) Mohammed Shah Tughlaq

82. Monsoon is caused by
(1) impact of summer temperatures on the seas
(2) movement of clouds
(3) seasonal reversal of winds
(4) rise in temperature

83. In the world of sports, the sobriquet 'Indo-Pak Express' is applied to
(1) Shree Santh and Shoaib Akhtar
(2) Rohan Bopanna and Aisam-Ul-Haq Quareshi
(3) Sania Mirza and Shahar Peer
(4) Rohan Bopanna and Shahar Peer

84. The world's busiest port today
(1) Port of Rotterdam
(2) Port of Shanghai
(3) Port of Singapore
(4) Port of Dubai

85. Which Company owns 'Sugar Free', the leading brand of artificial sweetener in India ?
(1) Cipla (2) Cadila
(3) Glaxo (4) Monsanto

86. Nand Lal Bose, Manjit Bawa, Tyeb Mehta are
(1) classical singers (2) english poets
(3) photographers (4) painters

87. What is Phishing ?
(1) A Govt. plan
(2) A stem cell
(3) Fraudulent way of acquiring PIN and bank passwords using email
(4) None of the above

88. The Indian company that manufactures automobiles for Mitsubishi is
(1) Tata Motors (2) Bajaj Tempo
(3) Mahindra (4) Hindustan Motors

89. Which of these is not a film directed by Mira Nair ?
(1) Earth (2) Monsoon Wedding
(3) Kamasutra (4) Salaam Bombay

90. The major US Toy manufacturer that recently suffered a breakdown in supply chain from China on account of legally unacceptable toxic substances in its products is
(1) Leo Toys (2) Mattel Toys
(3) Universal Toys (4) Funskool Toys

91. Some of the large deals entered into by IT majors in India are listed here as options. They are all billion dollar deals except one. Which one ?
(1) TCS-Nielson (2) Tech Mahindra-BT
(3) Infosys-ABN AMRO (4) IBM-Bharti

92. A suicide car bomb caused havoc in England in July 2007 at the following airport
(1) Glasgow (2) Heathrow
(3) Burmingham (4) Nottingham

93. Pakistan signed Free Trade Agreement (FTA) and a five year investment package with
(1) USA (2) European Union
(3) China (4) Russia

94. Major oil finds are recently reported in
(1) Krishna Godavari Basin
(2) Cauvery Basin
(3) Rajasthan Desert
(4) Lower Assam Valley

95. The propagation of radio signals is greatly influenced by
(1) troposphere (2) ionosphere
(3) exosphere (4) thermosphere

96. According to Dun and Bradsstreet which one of the following is the largest employer among Indian ITES and BPO companies ?
(1) Genpact (2) Accenture Services
(3) Wipro BPO (4) IBM Daksh

97. "In God we trust, the rest have to bring data on the table." This statement was made by the famous industrialist
(1) J.R.D. Tata (2) Aditya Birla
(3) Dhirubhai Ambani (4) Narayan Murthy

98. Following options are a list of forex rates for US Dollar on a day recently. Identify the entry that is reversed
(1) 1 US $ = Rs 39.35 (2) 1 US $ = €1.42
(3) 1 US $ = £0.48 (4) 1 US $ = Y 114

99. Which of the following is not provided in the Constitution ?
(1) Election Commission
(2) Planning Commission
(3) Public Service Commission
(4) Finance Commission

100. When did Governor General's rule end in India ?
(1) 15th August, 1947 (2) 9th August, 1948
(3) 26th January, 1950 (4) 2nd October, 1950

101. The newspaper which was recently voted as the largest selling newspaper in the world is
(1) The Los Angeles Times
(2) The Times of India
(3) The London Times
(4) The New York Times

102. Which of the following countries is the top source of FDI inflows into India at present ?
(1) Singapore (2) USA
(3) UK (4) Mauritius

103. The human cell contains
(1) 44 chromosomes (2) 48 chromosomes
(3) 46 chromosomes (4) 23 chromosomes

104. In 2006, the three largest economies in the world were
(1) USA, Japan and Germany
(2) USA, China and Japan
(3) USA, China and Germany
(4) USA, UK and France

105. The nunner-up in the recent Vice-President election was
(1) Mr. Bhairon Singh Shekhawat
(2) Dr. Najma Heptullah
(3) Mr. Praful Patel
(4) Mr. Hameed Ansari

106. Egg is a rich source of nutrients except
(1) Vitamin C (2) Protein
(3) Vitamin D (4) None of these

107. In banking terminology, CRR mean
(1) Credit Reserve Ratio
(2) Cash Reserve Ratio
(3) Credit Rating Ratio
(4) Cash Rating Ratio

108. "Bollywood" name has been granted as a trademark to which US based Media and Entertainment company by Indian Trademark Registry ?
(1) Viacom (2) Universal Studios
(3) Disney Enterprises (4) Miramax

109. Lakshmi Mittal, the famous NRI industrialist, is partnering with Govt. of India for the following big project
(1) Oil Refinery
(2) Atomic Power Plant
(3) Biotechnology
(4) Mining

110. Which of the following allows one to start and continue to share regularly any of his/her own personal experience, knowledge, opinion or thought with the internet community ?
(1) Portal (2) Email
(3) Blog (4) Chat

111. Zapak Digital Entertainment is an online gaming venture promoted by
(1) UTV
(2) Balaji Telefilm
(3) Zee TV
(4) Anil Dhirubhai Ambani Enterprises

112. In August 2007, Chennai High Court passed a landmark judgment that may have far reaching consequences on pharmaceuticals industry, dismissing a writ petition filed by
(1) Ranbaxy (2) Sandoz
(3) Novartis (4) Glaxo

113. Koneru Humpy plays
(1) Badminton (2) Circket
(3) Chess (4) Weight Lifting

114. Inrecent months the monks of Myanmar marched the streets of Yangon in hundreds
(1) to spread the message of peace
(2) to celebrate Buddha Purnima
(3) to protest against the ruling government
(4) to take the normal route for alms

115. To recover the national loss suffered by small investors in the IPO allotment scam from the National Securities Depository Services Ltd and Central Depository Services Ltd. and eight depository participants, a second interim order was passed by
(1) SEBI (2) RBI
(3) AFMI (4) Supreme Court

116. Indian Broadcasting Service was renamed in 1936 as
(1) Akashwani (2) Nabhowani
(3) Doorwani (4) All India Radio

117. Which of the following is not a principal organ of the UNO ?
(1) General Assembly
(2) Trusteeship Council
(3) Security Council
(4) World Health Organization

118. Who was the composer of the classical composition 'Moonlight Sonata' ?
(1) Ludwig Van Beethoven
(2) Joseph Haydn
(3) Johann
(4) Handel

119. During the year 2006–07, Indian Railways earned a profit of approximately
(1) Rs 20,000 crore (2) Rs 14,000 crore
(3) Rs 18,000 crore (4) Rs 16,500 crore

120. Starbucks, the coffee chain retailer's name is inspired from which book ?
(1) The Scarlet Letter
(2) Mobydick
(3) Uncle Tom's Cabin
(4) None of the above

■ Analytical and Logical Reasoning Section

Directions (Q. 121-123) *Read the following passage and answer the questions.*

An employee has been assigned the task of allotting offices to six of the staff members. The offices are numbered 1-6. The offices are arranged in a row and they are separated from each other by six foot high dividers. Hence voices, sounds and cigarette smoke flow easily from one office to another.

Miss Ruby needs to use the telephone quite often throughout the day. Mr. Minhas and Mr. Brar need adjacent offices as they need to consult each other often while working. Miss Harsha is a senior employee and has to be allotted the office number 5 having the biggest window.

Mr. Dongre requires silence in the offices next to his. Mr. Tanjore, Mr. Minhas and Mr. Dongre are all smokers. Miss Harsha finds tobacco smoke allergic and consequently the offices next to hers are to be occupied by non-smokers.

Unless specifically stated all the employees maintain an atmosphere of silence during office hours.

121. The ideal candidate to occupy the office furthest from Mr. Brar would be
(1) Miss Harsha (2) Mr. Minhas
(3) Mr. Tanjore (4) Mr. Dongre

122. The three employees who are smokers should be seated in the offices
(1) 1, 2 and 4 (2) 2, 3 and 6
(3) 1, 2 and 6 (4) 1, 2, and 3

123. In the event of what occurrence, within a period of one month since the assignment of the offices, would a request for a change in office be put forth by one or more employees
(1) Mr. Dongre quitting smoking
(2) The installation of a noisy teletype machine by Miss Harsha in her office
(3) Miss Ruby needs silence in the office(s) next to her own
(4) Mr. Tanjore taking over the duties formerly taken care of by Miss. Ruby

124. You have three chests in front of you. The first chest is labeled "GOLD", the second is labeled "SILVER" and the third is labeled "GOLD OR SILVER". You have been told that all the labels are on the wrong chests and that one chest contain gold coins, one contains silver coins and one contains bronze coins. How many chests do you need to open to deduce which label goes on which chest ?
(1) 0 (2) 1
(3) 2 (4) Cannot deduce

125. How many minutes before 12 noon is it when it is 27 min past 10 am ?
(1) 39 (2) 93
(3) 49 (4) 94

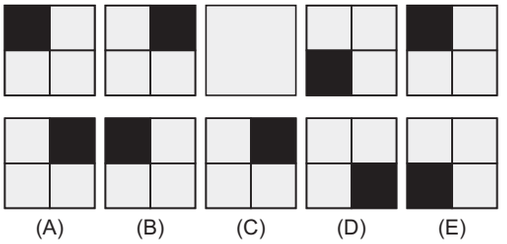
126. Shankar and Jwala are both members of a Youth club, though they are not speaking to each other and refuse to work with each other. Chaya, the club president, is appointing members to the fundraising committee, but she has resolved that she will not appoint anyone without his or her explicit consent. Shankar tells Chaya, "I will not consent to appointment on that committee unless I know whether Jwala is to be a member of it." And Jwala says, "I will not consent to be a member of that committee unless I know whether Shankar will be appointed to it." If all three of these people stick by these resolutions. then
(1) Neither of them can be appointed to the committee
(2) They must either both be appointed or both be left out
(3) The committee may finally have one of them, both of them, or neither of them as members
(4) Either one of them can be appointed, but not both

127. A bank customer had Rs 1000 in his account. He then made 6 withdrawals, totaling Rs 100. He kept a record of these withdrawals, and the balance remaining in the account, as follows

Withdrawals	Balance left
Rs 50	Rs 50
Rs 25	Rs 25
Rs 10	Rs 15
Rs 8	Rs 7
Rs 5	Rs 2
Rs 2	Rs 0
Rs 100	Rs 99

So, why are the totals not exactly right ?
(1) There is a mistake in the total of withdrawals
(2) There is a mistake in the total of the balance
(3) The two totals need not be equal
(4) The bank has cheated the customer

128. Which pattern from the bottom line (A, B C, D or E) is missing from the top line ?



(1) A (2) B
(3) D (4) E

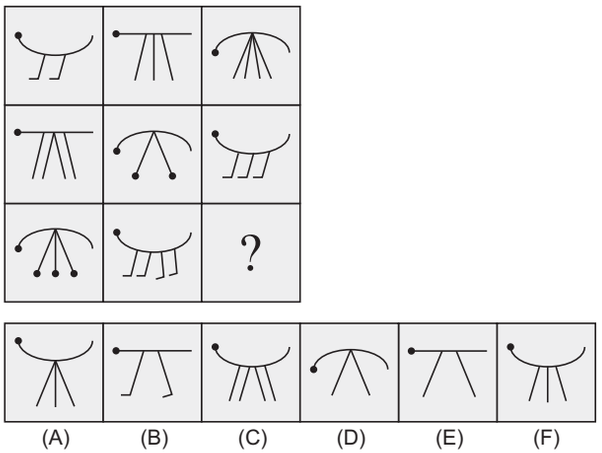
Directions (Q. 129-131) *During their school Silver Jubilee Reunion, four alumni were discussing their starting annual salaries back in 1981. The salaries in question were Rupees 40, 50, 60 and 70 thousand per year. Of course the present MD of a private company earned the most. Arvind earned more than Biswajeet, and the doctor earned more than Dhruv the engineer. Chinmay could not remember what he started on. Biswajeet the lawyer did not start on Rs 50,000, nor did Dhruv.*

129. What is Chinmay's current profession ?
(1) MD (2) Lawyer
(3) Doctor (4) Engineer

130. What was the Lawyer's starting salary ?
(1) 40,000 (2) 50,000
(3) 60,000 (4) 70,000

131. Who received the highest starting salary ?
(1) Arvind (2) Biswajeet
(3) Chinmay (4) Dhruv

132. Which image from the bottom row should replace the question mark ?

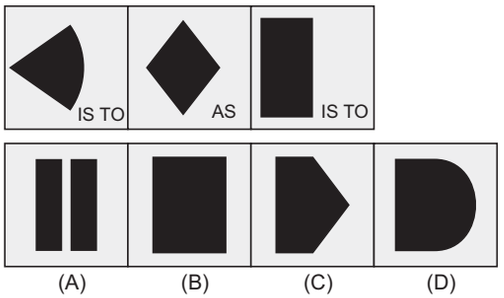


(1) C (2) D
(3) B (4) E

133. A man has a job which requires him to work 8 straight days and rest on the ninth day. If he started work on a Monday, the 12th time he rests will be on what day of the week ?
(1) Sunday (2) Wednesday
(3) Tuesday (4) Friday

134. Babloo and Bunty were excitedly describing the result of the First Annual Running Race at Damapur High School. Snehal, Tanmay and Waman had been three contestants, "Tanmay won the race; Waman was in second place," reported Babloo. Bunty disagreed, "It was Snehal who won. Tanmay came second." In fact, neither Babloo nor Bunty had given a correct version of the result as each had made one true and one false statement.
What was the actual placing of the three contestants ?
(1) Snehal, Waman, Tanmay
(2) Snehal, Tanmay, Waman
(3) Waman, Snehal, Tanmay
(4) Tanmay, Waman, Snehal

135.



(1) A (2) B
(3) C (4) D

136. It is possible to arrange eight of nine numbers 2, 3, 4, 5, 7, 10, 11, 12, 13 in the vacant squares of the 3 by 4 array shown below so that the arithmetic average of the numbers in each row and column is the same integer.

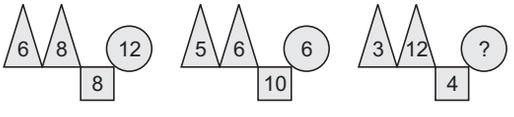
1			15
	9		
		14	

Which of the nine numbers must be left out when completing the array ?
(1) 4 (2) 10
(3) 15 (4) 7

137. At a family reunion were the following people: one grandfather, one grandmother, two fathers, two mothers, four children, three grandchildren, one brother, two sisters, two sons, two daughters, one father-in-law, one mother-in-law, and one daughter-in-law. But not as many people attended as it sounds. How many were there ?
(1) 5 (2) 10
(3) 15 (4) 7

138. A census taker approaches a house and asks the woman who answers the door. "How many children do you have, and what are their ages?"
Woman : "I have three children, the product of their ages is 36, the sum of their ages is equal to the address of the house next door."
The census taker walks next door, comes back and says, "I need more information."
The woman replies, "I have to go, my oldest child is sleeping upstairs."
Census taker : "Thank you, I now have everything I need."
What are the ages of each of the three children ?
(1) 9, 2, 2 (2) 6, 6, 1
(3) 12, 3, 1 (4) 6, 3, 2

139. What number will replace the question mark ?



(1) 40 (2) 18
(3) 19 (4) 24

140. An ingredient in coffee, known as RTC, has been found to inactivate common cold viruses in experiments. In previous experiments, researchers found that inactivated common cold viruses can convert healthy cells into cancer cells. It can be concluded that the use of coffee can cause cancer.
Which one of the following, if true, most seriously weakens the argument ?
(1) Several teams of scientists performed the various experiments, and all of the teams had similar results
(2) The carcinogenic effect of RTC could be neutralized by the other ingredients found in coffee
(3) When RTC kills common cold viruses it weakens the immune system, and it might thus diminish the body's ability to fight other viruses, including viruses linked to cancers
(4) If chemists modify the structure of RTC, RTC can be safely incorporated into medications to prevent the common cold

141. Jaideep was given some money by his mother on his birthday. Jaideep spent all of it in five stores. In each store he spent one rupee more than half of what he had when he came in. How much did he get from his mother ?
(1) Rs 25 (2) Rs 30
(3) Rs 46 (4) Rs 52

Directions (Q. 142-144) *Use the data given below to answer the questions.*

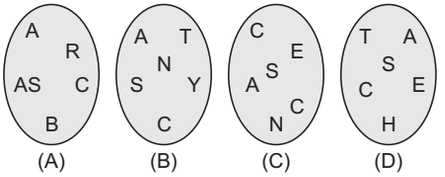
The following are the results of a survey conducted on a small cross section of students from Symbiosis Group of Institutes, to determine the readership of three Magazines. This survey was conducted in Dec. 2006.
— Number of students who read only Business India was 40
— 60 students read only Outlook
— 110 students read only India Today
— 30 students read all three magazines
— 20 read Business India and India Today, but not Outlook
— 50 read Business India and Outlook, but not India Today
— 40 read Outlook and India Today, but not Business India

142. What was the total number of students surveyed ?
(1) 210 (2) 350
(3) 390 (4) None of these

143. How many students did not read Business India ?
(1) 40 (2) 170
(3) 240 (4) None of these

144. When another survey was conducted in May 2007 with the same set of students, their tastes had changed and the findings were different. All of them read India Today. 120 read Outlook, and no one read Business India. Hence, in May 2007, how many students read only India Today ?
(1) 60 (2) 110
(3) 230 (4) None of these

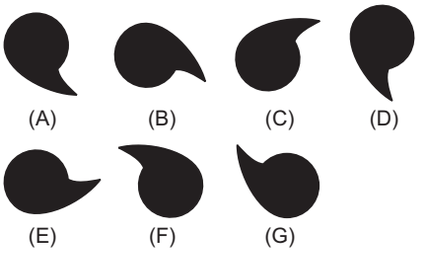
145. Find the circle, which does not include a word using all letters.



(A) (B) (C) (D)

(1) A (2) B
(3) C (4) D

146. From the image below (A-G) find 3 odd ones out.



(1) A, B, C (2) B, C, G
(3) C, G, D (4) D, A, F

147. On the counter are six squares marked 1, 2, 3, 4, 5, 6. Players are invited to place as much money as they wish on any one square. Three dice are then thrown.
- If your number appears on one dice only, you get your money back plus the same amount.
- If two dice show your number, you get your money back plus twice the amount you placed on the square.
- If your number appears on all three dice, you get your money back plus three times the amount.
- If the number is not on any of the dice, the operator gets your money.

For example, suppose that you bet one Rupee on square No. 6. If one dice shows a 6, you get your Rupee back plus another Rupee. If two dice show 6, you get back your Rupee plus two Rupees. If three dice show 6, you get your Rupee back plus three Rupees.
From a player's point of view, the chance of his number showing on one dice is, 1/6, but since there are three dice, the chances must be 3/6 or 1/2, therefore the game is a fair one. Of course this is the way the operator of the game wants everyone to reason, for it is quite fallacious.
What is the probable story ?
(1) Operator gets a profit of 6% on each Rupee bet
(2) Operator suffers a loss of 7.8% on each Rupee bet
(3) Operator gets a profit of 7.8% on each Rupee bet
(4) The player suffers a loss of 6% on each Rupee bet

148. Consider the following two statements to be true if they seem to be at variance from commonly known facts. Then decide which of the given conclusions logically follows from two given statements. Select the correct alternative.
Statements All Lawyers are extrovert.
Some wise men are extrovert.
Conclusions
(ja) All lawyers are wisemen
(jb) All wisemen are lawyers.
(jc) Some extrovert are wisemen.
(jd) All extrovert are lawyers.
(1) Only (ja) follows
(2) Only (jb) and (jc) follow
(3) Only (ja) and (jc) follow
(4) Only (jc) follows

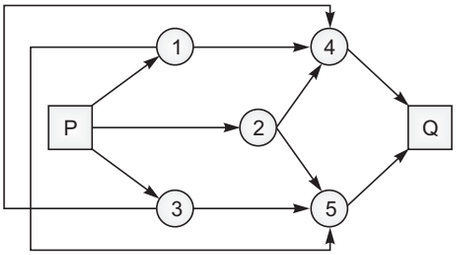
149. Where should the letter 'W' be placed ?

B	C	D	E	J	K	L	P
A	H	M	T	U	V		

(1) Above (2) Below
(3) Both (4) None

(1) 5 (2) 6
(3) 9 (4) 12

150. What is the number of routes from P to Q ?



Answers

1. (4)	**2.** (3)	**3.** (4)	**4.** (1)	**5.** (1)	**6.** (4)	**7.** (3)	**8.** (4)	**9.** (3)	**10.** (3)
11. (4)	**12.** (3)	**13.** (4)	**14.** (*)	**15.** (4)	**16.** (2)	**17.** (3)	**18.** (1)	**19.** (2)	**20.** (3)
21. (4)	**22.** (2)	**23.** (1)	**24.** (1)	**25.** (2)	**26.** (2)	**27.** (4)	**28.** (3)	**29.** (3)	**30.** (3)
31. (3)	**32.** (*)	**33.** (3)	**34.** (4)	**35.** (1)	**36.** (4)	**37.** (1)	**38.** (2)	**39.** (4)	**40.** (3)
41. (1)	**42.** (4)	**43.** (2)	**44.** (1)	**45.** (3)	**46.** (4)	**47.** (2)	**48.** (2)	**49.** (3)	**50.** (4)
51. (2)	**52.** (1)	**53.** (3)	**54.** (4)	**55.** (2)	**56.** (2)	**57.** (3)	**58.** (1)	**59.** (3)	**60.** (1)
61. (2)	**62.** (1)	**63.** (1)	**64.** (3)	**65.** (2)	**66.** (2)	**67.** (4)	**68.** (3)	**69.** (2)	**70.** (1)
71. (2)	**72.** (4)	**73.** (2)	**74.** (3)	**75.** (1)	**76.** (2)	**77.** (1)	**78.** (2)	**79.** (4)	**80.** (4)
81. (1)	**82.** (3)	**83.** (2)	**84.** (4)	**85.** (2)	**86.** (4)	**87.** (3)	**88.** (4)	**89.** (1)	**90.** (2)
91. (3)	**92.** (1)	**93.** (3)	**94.** (1)	**95.** (2)	**96.** (1)	**97.** (4)	**98.** (2)	**99.** (2)	**100.** (3)
101. (2)	**102.** (4)	**103.** (3)	**104.** (1)	**105.** (2)	**106.** (1)	**107.** (2)	**108.** (3)	**109.** (1)	**110.** (3)
111. (4)	**112.** (3)	**113.** (3)	**114.** (3)	**115.** (1)	**116.** (4)	**117.** (4)	**118.** (1)	**119.** (1)	**120.** (2)
121. (4)	**122.** (4)	**123.** (4)	**124.** (2)	**125.** (2)	**126.** (4)	**127.** (3)	**128.** (3)	**129.** (3)	**130.** (3)
131. (1)	**132.** (4)	**133.** (2)	**134.** (1)	**135.** (3)	**136.** (4)	**137.** (4)	**138.** (*)	**139.** (2)	**140.** (2)
141. (2)	**142.** (2)	**143.** (4)	**144.** (3)	**145.** (1)	**146.** (2)	**147.** (3)	**148.** (4)	**149.** (2)	**150.** (2)

NOTE * No option is correct.

Answers with **Explanations**

1. (4) The month having exactly three days less than maximum of 31 will be the month having 28 days. February is the only month that has 28 days.

$$\therefore \quad \text{Probability } (E) = \frac{1}{12}$$

2. (3) Let the principal be x.

$$\text{Simple interest} = \frac{7}{25}x$$

$$\text{SI} = \frac{PRT}{100}$$

$$\frac{7}{25}x = \frac{x \times R \times 4}{100}$$

$$\therefore \quad R = \frac{7 \times 4}{4} = 7\%$$

3. (4) Let the triangle be a right angled triangle and the base and the perpendicular of the triangle be b and h respectively.

$$\therefore \quad \text{Area} = \frac{1}{2} \times b \times h$$

$$\text{Area of flower bed} = \frac{1}{2} \times \frac{b}{2} \times \frac{h}{2} = \frac{bh}{8}$$

$$\therefore \quad \text{Area of the path} = \frac{1}{2}bh - \frac{1}{8}bh = \frac{3}{8}bh$$

$$\therefore \quad \frac{\text{Area of the path}}{\text{Area of the flower bed}} = \frac{\frac{3}{8}bh}{\frac{bh}{8}} = 3:1$$

4. (1) The intersection of a three dimensional figure cannot be three dimensional. It will be two dimensional. So it cannot be a cube.

5. (1) m is the median temperature. By arranging all temperatures in ascendng order 66, 69, 70, 75, 77, 78, 78. Middle number is 75. So, $m = 75$

f is the temperature that most often 78 has occured two times. $f = 78$

a is the averge of the seven temperature

$$= \frac{66 + 78 + 75 + 69 + 78 + 77 + 70}{7}$$

$$= \frac{513}{7} = 73.2$$

$a = 73.2, \quad m = 75, \quad f = 78$

$a < m < f$

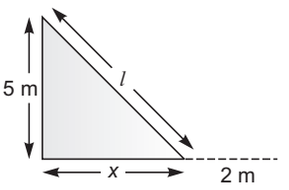
6. (4) After $2\frac{1}{2}$ h $= \frac{5}{2} \times 60$ min $= 150$ min. It is clear from the given table that the speed of the train was 65 km/h.

7. (3) Percentage increase in speed during the three hour period

$$= \frac{70 - 40}{40} \times 100 = 75\%$$

8. (4) $40 + \frac{2.5t}{15} = 40 + \frac{t}{6}$

9. (3) Let the length of the ladder be l metre and distance of the ladder from the wall x metres.



$\therefore \quad l^2 = 5^2 + x^2$

Again, $l = (x + 2)$ m

$\therefore \quad 5^2 + x^2 = (x + 2)^2$

or $25 + x^2 = x^2 + 4 + 4x$

$4x = 21$

$x = \frac{21}{4}$ m

$\therefore \quad l = x + 2 = \frac{21}{4} + 2 = 7.25$ m

10. (3) Data given; in statement I.
Expense in 2005–06 = Rs 1220
Expense in 2006–07 = Rs 1400 crores
and in statement II.
Sale in 2006–07 = Rs 4300
Profit in 2006–07 = Rs 4300 − 1400 crores
= Rs 2900 crores
Profit in 2006–07 increased by 32%

$\therefore$ Profit in 2005–06 $= \frac{2900}{1.32} \approx 2196.97$

$\therefore$ Sale in 2005–06 $= 1220 + 2196.97$
$= 3216.97$

Thus, the percentage increase in sale may be calculated using both the statement.

11. (4) Running in opposite direction they cross each other in 120 s. When they have $12 + 12 = 24$ boggies attatched but now they have $12 + 16 = 28$ boggies attatched.

They will take $\frac{120}{24} \times 28 = 140$ s

ie, 20 s more.

12. (3) $9M = 3P + 6N$

or $3M = P + 2N$

13. (4) Total number of names in the lottery.
$100 \times 3 + 150 \times 2 + 200 \times 1 = 800$
Number of year-III names $= 150 \times 2 = 300$

$\therefore \quad P(E) = \frac{^{300}C_1}{^{800}C_1} = \frac{300}{800} = \frac{3}{8}$

14. (*) Number of letters in the word EXAMINATION = 11
Number of vowels = 6 *ie*. *E, A, I, A, I, O*
Number of consonants = 5 *ie*. *X, M, N, N, T*

$\therefore$ Number of arrangements $= \frac{6!}{2! \times 2!} \times \frac{5!}{2!}$
$= 180 \times 60 = 10800$

15. (4) Area = 50 km^2
Side $= \sqrt{50} = 5\sqrt{2}$ km
$\therefore$ Diagonal length $= \sqrt{2} \cdot 5\sqrt{2} = 10$ km
$= 10 \times 1.6 = 16$ mile

16. (2) The number of children between 6th and 16th children = 9.
$\therefore$ Total number of children = 9 + 9 + 2 = 20

17. (3) Let the number of apples and oranges be x and y respectively.
$\therefore \quad x + y = 40$
Let the price of one apple be a and that of an orange be b.
$\therefore \quad ax + by = 17$ …(i)
Again according to question
$bx + ay = 15$ …(ii)
(i) and (ii), $(a + b)(x + y) = 32$

$a + b = \frac{32}{40} =$ Re 0.80
$= 80$ paise

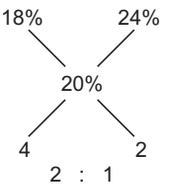
18. (1) $f(-3) = 3 - (-3) = 6$ [$\because -3 < 0$]

19. (2) **Method I**
Let the sum invested in Bank A be x and in Bank B be y.
$\therefore \quad x + y = 9000 \times 24.$

$2\times, 18 + y \times 24 = 1800 \times 100 \times 1$

$6x = 216000 - 18000$
$= 36000$
$x = 6000$

Method II



Amount invested at 6% $= 9000 \times \frac{2}{3} = 6000$

20. (3) Bag-I $\rightarrow 6$
Bag-II $\rightarrow 3$
Bag-III $\rightarrow 1$
} 10 balls

21. (4) The sum of the digits = 35
$\therefore$ The number is not divisible by 3.
The sum of the digits of odd places = 19
and sum of the digits of even places = 16
Difference = 3
$\therefore$ The number is not divisible by 11.

22. (2) 1st person moves only 2 steps in climbing up 5 steps.

$\therefore$ He takes $\frac{2}{5} \times 120 = 48$ steps

While 2nd person takes $\frac{3}{8} \times 120 = 45$ steps

Total steps taken by both = 48 + 45 = 93 steps

23. (1) The rainbow occurs once in 20 days.

$\therefore$ Percentage of days that do not produce rainbow

$$= \frac{19}{20} \times 100 = 95\%$$

24. (1) The product of four consecutive numbers is always an even number.

$\therefore$ $n = 1 + x$ is an odd number

Let the four consecutive numbers be $k-1$, k, $k+1$, $k+2$.

$$\therefore \quad n = 1 + (k-1)k(k+1)(k+2)$$
$$= 1 + (k-1)(k+2)k(k+1)$$
$$= 1 + (k^2 + k - 2)(k^2 + k)$$

Let $k^2 + k = x$

$$\therefore \quad n = 1 + (x-2)(x) = x^2 - 2x + 1$$
$$= (x-1)^2 = (k^2 + k - 1)^2$$

$\therefore$ $n = x + 1$ is definitely a perfect square.

25. (2) Let the number of accidents be n and the number of overtimes logged be a.

$$\therefore \quad n = x + ay$$
$$\therefore \quad 8 = x + 1000y \quad \ldots(i)$$
and $$5 = x + 400y \quad \ldots(ii)$$
$$\Rightarrow \quad 8 - 1000y = 5 - 400y$$
$$3 = 600y$$
$$y = \frac{1}{200}$$

Putting the value of y in Eq. (i),

$$\therefore \quad 8 = x + 1000 \times \frac{1}{200}$$
$$\therefore \quad x = 3$$

For $y = 0$

From the equation $n = x + ay$

$$n = x = 3$$

26. (2) In three years 2003, 2004 and 2006 the wheat production is higher than rice because the ratio of the wheat production to rice production is greater than 1.

27. (4) Ratio of production of wheat to rice

$$0.45 = \frac{45}{100} = \frac{9}{20}$$

$\therefore$ The production of wheat $= \frac{9}{20} \times 3 = 1.35$ lakh ton

$\therefore$ Grain to be imported $= 5 - 3 - 1.35$

$= 0.65$ lakh ton.

28. (3) Ratio in 2003 $= \frac{6}{5}$

Rice production = 4 lakh ton.

$\therefore$ Wheat production $\frac{4}{5} \times 6 = 4.8$ lakh ton

29. (3) It is obvious from the pie chart given

$$= 29 \times 180 - 20 \times 200$$
$$= 5220 - 4000 = 1220$$

30. (3) Number of employees left in $2005 = 23 \times 180 - 300$

$$= 4140 - 300$$
$$= 3840$$

Number of employees in $2006 = 26 \times 200 = 5200$

Number of employees who joined operations in 2006

$$= 5200 - 3840 = 1360$$

31. (3) Required % increase $= \frac{20 \times 200 - 18 \times 180}{18 \times 180} \times 100$

$$= 23.45\%$$

32. (*) Total salary of administrative employees

$$= 12000 \times 18000 \times \frac{22}{100}$$
$$= 12 \times 18 \times 22 \times 10^4$$
$$= 4752 \times 10^4 = 4.752 \text{ crore}$$
$$= 4.7 \text{ crore (approx.)}$$

33. (3) It is quite obvious by a look that Wipro showed highest percentage increase on NQE.

34. (4) Obvious by a look it is TCS.
There is only increases.

35. (1) Total price paid by Satyam

$$= \frac{15 \times 10^6}{200} \times 232 + \frac{15 \times 10^6}{200} \times 247$$
$$= 479 \times \frac{15}{2} \times 10^4 = 479 \times 75 \times 10^3$$
$$= 36 \text{ million (approx.)}$$

36. (4) Both are insufficient.

37. (2) For statement (A)

$$\frac{399}{40763} > \frac{637}{60133} \text{ is not true.}$$

As per (B)

$\frac{399}{40763} > \frac{138}{15333}$ which is true only Y is ture.

38. (4) For statement (A)

$$\frac{181}{59981} > \frac{48}{15389} \text{ not true}$$

For statement (B)

$$\frac{181}{684} > \frac{48}{138} \text{ not true}$$

39. (1) For statement (A),

$$\frac{19236 - 15389}{15389} > \frac{45292 - 40763}{45292}$$

or $$\frac{3847}{15389} > \frac{4529}{45292} \text{ which is true.}$$

For statement (B),

$$\frac{63298 - 60133}{63298} > \frac{45292 - 40763}{45292}$$

or $$\frac{3165}{63298} > \frac{4529}{45292}$$

which is not true.

40. (3) For statement (A),

$$\frac{45292 - 19236}{45292} > \frac{63298 - 61205}{63298} \text{ correct}$$

For statement (B),

$$\left(\frac{26066}{45292} + \frac{2093}{63298}\right) \times 100 > 25 \text{ correct}$$

41. (1) Common people consider the questions asked by curious scholars un-important because they think their answers to be too obvious to ask. This is so because the common people lack deep analytical thinking vis-a-vis learned scholars.

42. (4) Since curiosity of young children may not be satisfied reasonably at home by their parents limitation of knowledge, the university—the vast seat of disseminating variety of disciplines of knowledge is the right place to extinguish thirst for knowledge for which young ones are curious to know.

43. (2) General people respects scholars, especially scientists, who explore curiosity and translate it into new invention which turns out to be useful for general people.

44. (1) A scientist has been compared to a historian, and a mountain climber because each of them does something out of curiosity to know about it without hoping to get benefited in the first place.

45. (3) Even if the answer found by a scholar to a question arising out of his curiosity to know turns out to have immense useful consequence akin to that found by scholars viz. cure for cancer, the primary objective was simply satisfying curiosity to know about unexplored field of knowledge.

46. (4) The literal meaning of 'idle' is 'without purpose'. So the 'idle curiosity' means eagerness to know something without any intended purpose—that is, just in casual manner.

47. (2)
1. 6 (He is willing to do as you say)
2. 7 (The ox is too obstinate to move even if it is beaten)
3. 5 (The determination to go on walk is undaunted by rains)
4. 8 (Coming across a snake presents a real condition which warrants taking a prompt decision as what is to be done to save oneself)

48. (2) The present continuous sentence is changed to 'is being ..... while present perfect sentence is changed to "has been ....." in passive form (4) is not correct because there is a difference between 'done well' and 'well done'. The first case means a job done correctly, while the latter means expression of exclamation for something done well.

49. (3) Pray is to say prayer. Prey is the one, who is victim of a predator. Pry is to peek into something inquisitively.

50. (4) The Sangwan-Vemuganti team uses cells found in the tissues of living adults instead of embryos.

51. (2) Harvested corneas are implanted in patients with damaged cornea.

52. (1) Indians are reputed to deduce drugs to produce generic versions.

53. (3) Sangwan and Vemuganti developed technique on their own experiments in the laboratory.

54. (4) Platform means the material of glass on which cells were grown, and 'generic' means the genra of the chemical instead of its brand name chosen for marketing.

55. (2) Subtle means too delicate to be felt or perceived that is, it is too elusive to be perceived. Incidious is also meaning un-noticed but is used for harmful effects.
So, the difference in shades of a colour may be subtle and so difficult to make distinction as it may be elusive to someone's power of distinguishing.
But, an insidious disease eludes the doctor's diagnosis as it is silently harming the patient.

56. (2) My mother, who is from the village, is very superstitious.

57. (3) Both sentences are incorrect.

58. (1) Intellectual property rights works contrary to the interest of global bio-diversity.

59. (3) The article says that TRIP rules apply to all countries irrespective of levels of developments; but article 65 asks developed country to change existing laws within 5 years but allows underdeveloped country a time frame of 10 years.

60. (1) 130 is 76.5% of 170.

61. (2) High-yield seed variety oftenly leads to plants diseases.

62. (1) The very first paragraph mentioned that TRIPs rules are applicable to all countries irrespective of level of development.

63. (1) Repudiate means disown or deny something.
Prevaricate means speak so to evade the truth
Disparage means speak so to defame someone.
Equivocate means to speak so to carry two differing meanings.

64. (2) Correct spelling of (1) is paraphernalia
(3) is paediatrics
(4) is berserk

65. (2) (1) is a sentence to ascertain whether expected checking did take place at the airport (4) or not.

66. (2) 'at once' is adverb which should come after the verb. 'concerned' should come after authority, as concerned authority means the authority who is concerned *ie*, worried.

67. (4) After is adjective in "after of he disease (6), after is adverb in (8) as it qualifies verb 'came'.
After is conjunction in (5); and it is preposition viz to hanker after.

68. (3) Neophyte means new, untrained, callow, fledgeling all meaning inexperienced which is opposite to veteran, expert.

69. (2) 'Lame' comes in three sentences, viz.;
Lame excuse
Lame duck
Lame person

70. (1) Fire engine — which extinguishes fire
Fire stick — a stick with flame of fire
Fire escape — escape route in case of a building on fire

71. (2) We all value having the freedom (so) to make choices we want in our careers, (remember) that with freedom comes responsibility, but of as fail to do so; therefore, many companies are rigid in treating employees despite 'proclaiming that they give freedom to employees.'

72. (4) Vaatsalya is based on profit with humanitarian approach.

73. (2) All others are engaged in profit-making enterprise with social objective. But Acumen Fund is investment company targetting low-income people.

74. (3) Entrepreneours with empathy for poor are fast expanding now their social business with financial returns.

75. (1) Malinger means evade to do something by feigning illness. One may do so by lying lazy, by arguing also. But, wander is most inappropriate as it means to go off the right course, and move around aimlessly.

76. (2) Padre Rotton case was peculiar in the fact that despite his claim of English people being the God's blessed people *ie*, the most superior race, Padre had a large number of mixed race relatives including James Rotton, Anglo-Indian race, who even did not know the language of English people and had cousin Fetix Rotton who was a Muslim convert and had 22 sons from various Indian wives …... .

77. (1) Almost = just short of, just about to do something; and is adverb—so placed before the verb.

78. (2) It should be "so that…"

79. (4) Cadence is related to musical notes. Din, cacophony, Racket all mean noise.

80. (4) Cows 'bellow'.

For (Q. 121-123)

Miss Harsha occupies office number 5. Since, she finds tobacco smoke allergic, her office cannot be with that of Mr. Tanjore, Mr. Minhas or Mr. Dongre (Smokers). Hence, office number 4 and 6 are to be alloted to Mr. Brar and Miss Ruby.

Now, Mr. Minhas and Mr. Brar need adjacent offices. Hence, Mr. Brar will occupy office number 4 while Mr. Minhas will occupy office number 3. And also, Miss ruby will occupy the office number 6.

Mr. Dongre requires silence. Hence, he will occupy office number 1.

And finally Mr. Tanjore will occupy office number 2. Hence, the arrangement is as

1	2	3	4	5	6
Mr. Dongre	Mr. Tanjore	Mr. Minhas	Mr. Brar	Miss Harsha	Miss Ruby

121. (4) Mr. Dongre is the ideal candidate to occupy the furthest office from Mr. Brar.

122. (4) First three offices are occupied by those three smokers.

123. (4) If Mr. Tanjore needs to use the telephone quite often throughout the day, Mr. Dongre would put forth a request for a change in office because he requires silence in the office next to his.

124. (2) According to the question, we know that all the labels have been given on the wrong chests. So, it is clear that if a chest is labeled with 'GOLD' then it does not have gold coins in it. Similarly, the chest which have been labeled with 'SILVER' and 'GOLD or SILVER' will not have silver and bronze coins in them. So, if we open any of the chest labeled with gold and find silver coins in it, it implies that the chest labled with 'GOLD or SILVER' must have gold coin and the chest labeled with SILVER must have the bronze coin.

125. (2) After 10 : 27, if required 93 min to be 12 noon.

126. (4) Since, Shankar and Jwala do not speak to each other and refuse to work with each other. Either of them can be appointed, but not both.

128. (3) Each time the figure moves at 90° in the clockwise direction.

For (Q. 129-131) On the basis of the data given in the question, we get the chart of those alumni with their income as;

Profession / Alumni	MD Arvind	Lawyer Biswajeet	Engineer Dhruv	Doctor Chinmay
Starting Salary (in Rs)	70000	60000	40000	50000

129. (3) Chinmay is a doctor.

130. (3) Lawyer's (Biswajeet) starting salary is 60000.

131. (1) Arvind received the highest starting salary.

132. (4) In each row, a figure having the head of a straight line (pin) is used. Since, in the third row neither 1st nor 2nd figure is such. Hence, the third figure will be exactly that (having the head of a straight pin).

134. (1) According to Babloo, if 1st statement is true,
Tanmay won the race. That will mean that either Snehal on waman was second. And like this none of Bunty's statement proves right. Hence, the first statement of Babloo is false. Hence, it is true to say that, waman was in second place. And also, Snehal won the race. Hence, Tanmay was in the third place.

135. (3) In the second figure, a figure having one more side is used. That means, the fourth figure will be of five sides.

136. (4)

1	13	3	15
11	9	7	5
12	2	14	4

Here, the arithmetic average of numbers in each row and column is 8.

137. (4)

$A_{(+)}$ $B_{(-)}$ $\leftarrow 1^{st}$ generation
Grand father Grand mother

$C_{(+)}$ $D_{(-)}$ $\leftarrow 2^{nd}$ generation
Father Mother

$E_{(+)}$ $F_{(-)}$ $G_{(-)}$ $\leftrightarrow 3^{rd}$ generation

Now, A was grandfather, B was grandmother, A and C were fathers, D and B were mothers, C, E and F were children, E, F and G were grandchildren, E was the

brother, F and G were sisters, C and E were sons, F and G were daughters, A was father-in-law, B was mother-in-law and D was daughter-in-law.
Hence, there were only seven persons.

138. (*) Cannot be determined.

139. (2) The product of two numbers from the right side is double than the product of the rest two numbers in each case.

$12 \times 8 = 96$ and $(6 \times 8) \times 2 = 96$

$6 \times 10 = 60$ and $(6 \times 5) \times 2 = 60$

$18 \times 4 = 72$ and $(3 \times 12) \times 2 = 72$

142. (2) Total number of students surveyed

$= 40 + 60 + 110 + 30 + 20 + 50 + 40 = 350.$

143. (4) Total number students who did not read Business India

$= 60 + 110 + 40 = 210$

144. (3) Total number of students who read only India Today

$= 350 - 120 = 230$

146. (2) All other figures can be obtained by moving any one of rest four.

148. (4) All lawyers are extroverst
Some wisemen are extrovert NO conclusion.
or some extroverts are wisemen.

Conclusions

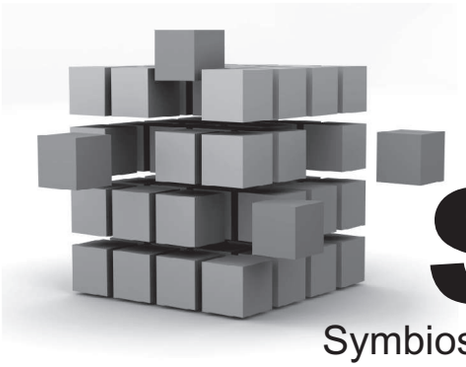
(ja) All lawyers are wisemen → does not follow.

(jb) All wisemen are lawyers → does not follow.

(jc) Some extrovert are wisemen → By converting second statement we can get this conclusion.

(jd) All extrovert are lawyers → does not follow.

150. (2) Possible routes are
P-1-4-Q, P-2-5-Q, P-2-4-Q, P-3-5-Q, P-1-5-Q, P-3-4-Q,



SNAP 2006

Symbiosis National Aptitude Test

Solved Paper

General English

Directions (Q. 1-10) *Select the correct answer from the given choices to fill the blank appropriately.*

1. _____ is to Constitution as prologue is to _____ .
1. Independence 2. Law
3. Preamble 4. Amendment
A. eulogy B. write
C. play D. epilogue
(1) 3C (2) 1D
(3) 4A (4) 4D

2. _____ is to horse as bray is to _____ .
1. Drive 2. Hoof
3. Neigh 4. Saddle
A. relay B. pony
C. wagon D. donkey
(1) 1D (2) 1A
(3) 3C (4) 3D

3. _____ is to distance as kilogram is to _____ .
1. Far 2. Metre
3. Europe 4. Travel
A. heavy B. ounce
C. weight D. noise
(1) 2A (2) 2B
(3) 2C (4) 1A

4. _____ is to never as all is to _____ .
1. Always 2. Usually
3. Seldom 4. Often
A. none B. whole
C. every D. total
(1) 1A (2) 4H
(3) 2B (4) 4A

5. _____ is to rain as dam is to _____.
1. Cloud 2. Fog
3. Water 4. Umbrella
A. electricity B. flood
C. high D. spillway
(1) 3B (2) 1B
(3) 4B (4) 1A

6. _____ is to pacifist as religion is to _____.
1. War 2. Atlantis
3. Object 4. Conscience
A. devout B. sacred
C. atheist D. minister
(1) 4C (2) 4B
(3) 1C (4) 4A

7. _____ is to bird as shedding is to _____.
1. Calling 2. Flying
3. Migrating 4. Moulting
A. barn B. dog
C. hay D. farm
(1) 4A (2) 4B
(3) 2C (4) 2A

8. _____ is to England as Lira is to_____.
1. London 2. Pound
3. King 4. Colony
A. Italy B. Mexico
C. Mandolin D. Money
(1) 1A (2) 1B
(3) 1D (4) 2A

9. _____ is to Canada as Rangoon is to _____.
1. Detroit 2. Florida
3. Toronto 4. Alberta
A. Indonesia B. Burma
C. East Pakistan D. Ceylon
(1) 2B (2) 3B
(3) 3D (4) 3A

10. _____ is to all as part is to _____.
1. Each 2. Righ
3. None 4. Full
A. whole B. separate
C. role D. many
(1) 1A (2) 4B
(3) 1D (4) 4D

Directions (Q. 11-15) *Select two words which are most nearly the same in meaning, or opposite in meaning.*

11. A. comical B. pretentious
C. tragedy D. climatic
(1) AC (2) BC
(3) AB (4) DA

12. A. deliberate B. increasing
C. involuntary D. incorrigible
(1) DC (2) BD
(3) AC (4) CD

13. A. sagacious B. extravagant
C. fastidious D. meticulous
(1) CD (2) DB
(3) BC (4) AC

14. A. dilatory B. querulous
C. compatible D. omnivorous
(1) AC (2) BD
(3) DC (4) CB

15. A. extravagance B. prodigality
C. edification D. acrimony
(1) DC (2) AB
(3) BC (4) CA

Directions (Q. 16-21) *Choose the best answer among the listed alternatives.*

16. One major obligation of the social psychologist is to provide his own discipline, the other social sciences and interested laymen with conceptual tools that will increase the range and the reliability of their understanding to social phenomena. Beyond that, responsible government officials are today turning more frequently to the social scientist for insights into the nature and solution of the problems with which they are confronted.
The above argument assumes that
A. Social psychologists must have a strong background in other sciences as well as their own.
B. The social scientist has an obligation to provide the means by which social phenomena may be understood by others.
C. Social phenomena are little understood by those outside the field of social psychology.
D. A good social psychologist is obligated principally by the need to solve inter-disciplinary problems.
(1) B (2) C
(3) A (4) D

17. Once a company has established an extensive sales network in a foreign market and therefore has achieved substantial sales, it seems that these markets should be treated in a very similar fashion to those in one's own country. It is therefore in those countries where only initial sales and representation have been developed where marketing methods will have to differ from domestic activities.
The above statement assumes that
A. Sales network can be the same in both foreign and domestic markets.
B. Extensive sales networks are preferable to less developed ones.
C. Larger markets abroad are more adaptable to domestic marketing methods.
D. A study of marketing should consider the adaptability of advertising campaigns in different countries.
(1) A (2) B
(3) C (4) D

18. Equality of opportunity has long been prominent as a goal in many countries. In Europe and America there has also been advocacy of more equality of income — the results after taxes of what a person gets for his efforts and the yield of his property. Many western politicians believe this concept of equality should be implemented in developing countries in order to speed economic development.
Which of the following, if true, could weaken the argument above ?
A. In a poor society, total income is so low that if it were distributed equally, no one could save enough to provide resources for investment.
B. The marginal dollars in the hands of people with large incomes provide less utility than those with lower income.
C. High achievement in many societies is due to equality of incomes.
D. Equality of opportunity is not necessarily synonymous with equality of income.
(1) D (2) B
(3) A (4) C

19. In 1985 there were 20 deaths from automobile accidents per 1,000 miles travelled. A total of 20,000 miles were travelled via automobiles in 1985. In the same year, 800 people died in airplane crashes and 400 people were killed in train disasters. A statistician concluded that it was more dangerous to travel by plane, train and automobile, in that order.
Which of the following refutes the statistician's conclusions ?
A. There is no common denominator by which to compare the number of deaths resulting from each mode of travel.
B. One year is insufficient to reach such a conclusion.
C. More people travel by car than any other mode of transport, therefore, the probability of a car accident is greater.
D. The probability of being killed in a train disaster and as a result of a car crash is the same.
(1) A (2) B
(3) C (4) D

20. Product shipments of household appliances are expected to rise to $ 17 billion next year, an average annual increase of 8.0 per cent over the past five years. The real growth rate, after allowing for probable price increases, is expected to be about 4.3 per cent each year, resulting in shipments this year of $ 14 billion in 1987 dollars.
Each of the following, if true, could help to account for this trend except :
A. Increased consumer spending for durable products.
B. Consumer disposable income has increased.
C. The consumer price of electricity has decreased.
D. Individual tax advantages have decreased.
(1) C (2) A
(3) D (4) B

21. Each year's increase or decrease in the trade deficit (merchandise imports greater than exports) is calculated in relation to the previous year's. In 1976, imports of private vehicles were 10 percent higher than in 1975, while imports of vehicles including commercial vans was 15 per cent higher than in 1975. That 15 per cent increase was one and half times the increase recorded in 1975.

If the formation above is accurate, which of the following must be true ?

A. In 1976, the increase, if any, of commercial vehicle imports was smaller than the increase in imports of private vehicles.
B. In 1976, the increase, if any, of commercial vehicle imports was greater than the increase in imports of private vehicles.
C. In 1975, more commercial vehicles were imported than private vehicles.
D. The average number of private vehicles imported in 1975 declined.

(1) A (2) C
(3) B (4) D

Directions (Q. 22-25) *Choose the correct alternative.*

22. Somebody told you, _____ ?
(1) Isn't it (2) Wasn't it
(3) Didn't he (4) Didn't they

23. _____ fast.
(1) News travels (2) News travel
(3) A news travels (4) The news travel

24. She _____ in the morning because she gets up late.
(1) always has to hurry
(2) has always to hurry
(3) has to always hurry
(4) has to hurry always

25. There is _____ .
(1) hair in the soup
(2) hair in soup
(3) a hair in soup
(4) a hair in the soup

■ Quantitative Aptitude

26. There are 6 balls of different colours and 3 boxes of different sizes. Each box can hold all the 6 balls. The balls are put in the boxes so that no box remains empty. The number of ways in which this can be done is
(1) 534 (2) 543
(3) 540 (4) 528

27. If $a^x = b^y = c^z$ and $b^2 = ac$, then y equals
(1) $\frac{xz}{x+z}$ (2) $\frac{xz}{2(x-z)}$
(3) $\frac{xz}{2(z-x)}$ (4) $\frac{2xz}{(x+z)}$

28. If the total numbers of relations from set A to set B is 1024, and $n(A) > 1$, $n(B) > 1$, then
(1) $n(A) = 2$, $n(B) = 5$ (2) $n(A) = 5$, $n(B) = 1$
(3) $n(A) = 10$, $n(B) = 1$ (4) $n(A) = 1$, $n(B) = 10$

29. On a railway there are 10 stations. The number of types of tickets required in order that it may be possible to book a passenger from every station to every other is
(1) $\frac{10!}{2!}$ (2) $10! \times 2!$
(3) $\frac{10!}{8!}$ (4) $\frac{10!}{2!\,8!}$

30. The number of ways in which the letters of the word 'ARGUMENT' can be arranged so that only consonants occur at both the ends is
(1) $3! \times 5!$ (2) 14400
(3) 41000 (4) None of these

31. If U_r denotes the number of one-one functions from $(X_1, X_2, \ldots, X_r)$ to $(Y_1, Y_2, \ldots, Y_r)$ such that $f(x_i) \neq Y_i$ for $i = 1, 2, 3, \ldots, r$, then $U_4 = ?$
(1) 9 (2) 44
(3) 265 (4) None of these

32. If $\frac{a}{b} = \frac{4}{3}$, then $\frac{3a + 2b}{3a - 2b}$ would be equal to
(1) –1 (2) 3
(3) 5 (4) 6

33. The number of ways in which the digits of the number 125453752 can be arranged such that no all 5's come together is
(1) 9! 3! 2! (2) 7! 3! 2!
(3) 6! 2! (4) None of these

34. If $2^{x-1} + 2^{x+1} = 320$, then the value of x is
(1) 4 (2) 5
(3) 6 (4) 7

35. After typist had written a ten letters and had addressed the ten corresponding envelops, a careless mailing clerk inserted the letters in the envelopes at random, one letter per envelop. What is the probability that exactly nine letters were inserted in the proper envelop ?
(1) 3 (2) 1
(3) 0 (4) 9

36. On a 26 questions test, five points were deducated for each wrong answer and eight points were credited for each correct answer. If all the questions were answered how many were correct if the score was zero ?
(1) 13 (2) 6
(3) 9 (4) 10

37. There are n players in an elimination type singles tournament. How many matches must be played (or defaulted) to determine the winner ?
(1) $n + 1$ (2) $n - 1$
(3) $n - 2$ (4) $n + 2$

38. If $-1 \leq x \leq 2$ and $1 \leq y \leq 3$, then least possible value of $(2y - 3x)$ is
(1) 0 (2) – 3
(3) – 4 (4) – 5

39. When n is divided by 4, the remainder is 3. What is the remainder when $2n$ is divided by 4 ?
(1) 1 (2) 2
(3) 3 (4) 6

40. Ten turns of wire are helically wrapped around a cylindrical tube with outside circumference 4" and length 9". The ends of the wire coincide with the ends of the same cylindrical element. Find the length of the wire.

(1) 27" (2) 36"
(3) 41" (4) 43"

41. Multiply 5746320819 by 125
(1) 748,290,103,375 (2) 728,490,301,375
(3) 718,90,102,375 (4) 798,290,102,975

42. A company offered its 350 employees a bonus of Rs 10 to each male and Rs 8.15 to each female. All the females but certain percentage of males refused to accept. The total bonus paid was not dependent upon the number of new employees. What was the total amount paid to the women ?
(1) Rs 2425 (2) Rs 1222.50
(3) Rs 1262.50 (4) Rs 2525

43. Which are two numbers whose difference and whose quotient are each equal to 5 ?
(1) $\frac{5}{4}$ and $\frac{25}{4}$ (2) $\frac{15}{2}$ and $\frac{30}{2}$
(3) 11 and 16 (4) 7 and 12

44. A series of books was published at seven years' interval. When the seventh book was issued, the sum of the publication years was 13,524. When was the first book published ?
(1) 1932 (2) 1942
(3) 1911 (4) 1917

45. In a given period of days, it was observed that when it rained in afternoon, it had been clear in the morning, and when it rained in the morning, it was clear in the afternoon. It rained on 9 days and was clear on 6 afternoons and 7 mornings. How long was the period ?
(1) 21 (2) 17
(3) 13 (4) 11

46. The traffic lights at three different road crossings change after every 48 s, 72 s and 108 s respectively. If they all change simultaneously at 8 : 20 : 00 h, then they will again change simultaneously at
(1) 8 : 27 : 12 h (2) 8 : 27 : 24 h
(3) 8 : 27 : 36 h (4) 8 : 27 : 48 h

47. If $\frac{1}{4}$ of 20 is 6, then $\frac{1}{5}$ of 10 is
(1) 3 (2) 4
(3) $\frac{22}{5}$ (4) $\frac{21}{2}$

48. Which pair from below will not satisfy the equation $187x - 104y = 41$?
(1) $x = 3, y = 5$ (2) $x = 107, y = 192$
(3) $x = 211, y = 379$ (4) $x = 314, y = 565$

49. The simplest reduction to the lowest terms of $\frac{116,690,151}{427,863,887}$ is
(1) $\frac{11}{3}$ (2) $\frac{7}{11}$
(3) $\frac{3}{11}$ (4) None of these

50. If $1.125 \times 10^k = 0.001125$, then the value of k is
(1) – 4 (2) – 3
(3) – 2 (4) –1

51. 3 truck drivers went to roadside cafe. 1st purchased 4 sandwiches, 1 coffee, 10 biscuits and paid Rs 1.69 paise. 2nd purchased 3 sandwiches, 1 coffee, 7 biscuits and paid Rs 1.26. What did 3rd truck driver pay for a sandwich, a cup of coffee, and one biscuit ?
(1) 35 paise (2) 89 paise
(3) 68 paise (4) 40 paise

52. 10 people start a club. If there had been five more in the group, the initial expenses to each would have been Rs 100 less. What was initial cost per person ?
(1) 250 (2) 300
(3) 500 (4) 400

53. A 27 quartz capacity car radiator is filled with 18% alcohol solution. How many quartz be drained and then be replaced by a 90% alcohol solution for resulting solution to contain 42% alcohol ?
(1) 19 quartz (2) 9 quartz
(3) 11 quartz (4) 14 quartz

54. If 70% of adult males of a community have brown eyes, 75% have dark hair, 85% are over 5′ 8″ tall and 90% weight more than 140 pounds, what per cent at least have all four characteristics ?
(1) 60% (2) 45%
(3) 20% (4) 48%

55. Henry started a trip into the country between 8 am and 9 am when the hands of clock were together. He arrived at his destination between 2 pm and 3 pm when the hands of the clock were exactly 180° apart. How long did he travel ?
(1) 6 h (2) 7 h
(3) 9 h (4) 11 h

■ Data Interpretation and Data Sufficiency

Directions (Q. 56-59) *Refer to the following table. Given below are the win-loss histories of four tennis players in three set matches.*

Number of Matches Played

	P	Q	R	S
Won 2-0	23	18	10	16
Won 2-1	13	16	13	14
Lost 0-2	8	13	16	13
Lost 1-2	6	8	20	17

56. If a win carries an award of Rs 1500 and a loss of Rs 1000, what is the largest difference in prize money earned among the four ?
(1) Rs 7000 (2) Rs 14000
(3) Rs 5000 (4) Rs 9000

57. As in the case above what is the lowest average prize money won per match by any player ?
(1) 1135 (2) 1300
(3) 1195 (4) 1160

58. As in the case above what is the highest average prize money won per match by any player ?
(1) 1495 (2) 1360
(3) 1460 (4) 1410

59. Who has lost the maximum per cent of matches played ?
(1) P (2) Q
(3) R (4) S

Directions (Q. 60-67) *Follow the given instructions. Each item has a question followed by two statements*

Mark (1) if the question can be answered with the help of statement A alone.
Mark (2) if the question can be answered with the help of statement B alone.
Mark (3) if the question can be answered with the help of both the statements but not with the help of either statement itself.
Mark (4) if the question can not be answered even with the help of both the statements.

60. The average of three quotations for a particular item is Rs 120. Is the highest quotation less than or equal to Rs 139 ?
A. The lowest quotation is Rs 90.
B. One of the quotations is Rs 125.

61. How many people, read both Economic Times and Financial Express ?
A. Out of 300 readers, 200 read Financial Express, 220 read Economic Times and 50 read Indian Express.
B. Out of total 300 readers, 200 read Financial Express, 220 read Economic Times and 50 read Neither.

62. A bus started from bus stop *P*, developed engine trouble and reached bus stop *Q*, 40 min late. What is the distance between bus stops *P* and *Q* ?
A. The engine trouble developed after travelling 40 km from bus stop *P* and the speed reduced to 1/4th of the original speed.
B. The engine trouble developed after travelling 40 km from bus stop *P* in two hours and the speed reduced 1/4th of the original speed.

63. What is the value of prime number *X* ?
A. $X + Y$ is a two digit number > 50.
B. *Y* is a two digit number.

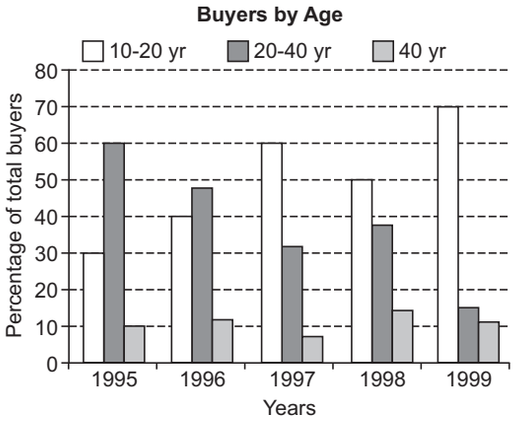
64. What is the time difference in Nairobi and Mumbai ?
A. The departure time at Nairobi is exactly 6 pm local time and the arrival time at Mumbai is next day 3 am local time.
B. The flight time is 6 h.

65. Sujata is the eldest daughter. What is the current age of her father ?
A. Current age of Sujata is 25 yr and current age of her sister Sangeeta is 20 yr.
B. Sangeeta's current age is one-third of father's age five years before.

66. Alka has 2 sisters and 1 brother. How many sisters Sandhya have ?
A. Alka and Sandhya are sisters.
B. Sandhya has one brother.

67. Madhav likes car *X* because its cost is Rs 5 lac cheapest in *C* segment and Makarand likes car *Y* because it costs Rs 7 lac highest in *C* segment. Which vehicle will be selected by Keshav ?
A. Keshav is looking for a car which is average representation of *C* segment.
B. Keshav's budget is around Rs 5.5 lac.

Directions (Q. 68-73) *Questions are based on the following graph of buyers.*



68. By how many has the per cent number of buyers in the age 20-40 yr age group has declined from year 1995 to 1999 ?
(1) 26.6 (2) 73.4
(3) 44 (4) 20

69. If the total number of peoples surveyed is 2000 in each of the five years given, the number of peoples who buy in the age group 10-20 yr has changed from year 1995 to 1999 ?
(1) 800 (2) 600
(3) 1200 (4) 1400

70. Which of the following age groups shows a declining trend in the purchase in the given five years ?
(1) 10-20 yr
(2) 20-40 yr
(3) 40 yr and above
(4) No particular trend in any group

71. By how many has the per cent number of buyers in the 10-20 yr age group has increased from year 1997 to 1999 ?
(1) 13.6 (2) 15
(3) 16.7 (4) 10

72. Which of the age group show no particular trend ?
(1) 10-20 yr
(2) 20-40 yr
(3) 40 yr and above
(4) No particular age group

73. The regain strategy should focus on which age group ?
(1) 10-20 yr
(2) 20-40 yr
(3) 40 yr and above
(4) No particular age group

Directions (Q. 74-81) *Questions are based on the data given in the following table.*

There are Eight major locations in a zoo at Mysore. A survey was conducted to find out the movement of the people within the zoo. Following results were obtained.

		To							
		A	B	C	D	E	F	G	H
From	A		48	46	66	24	4	68	64
	B	20		64	152	64	90	108	86
	C	28	108		88	86	46	130	130
	D	54	92	66		42	82	20	4
	E	70	48	50	110		26	46	22
	F	24	74	92	68	112		120	60
	G	28	32	134	152	6	86		26
	H	50	70	90	108	68	110	86	

74. The least number of peoples go to which location ?
(1) G (2) D
(3) B (4) A

75. What is the total number of peoples visiting the zoo ?
(1) 3868 (2) 5000
(3) 4000 (4) Cannot be determined

76. What percentage of total visiting E comes from D ?
(1) 10 (2) 20
(3) 40 (4) 30

77. What is the difference between the number of peoples going from B to F and G; and D to F and G ?
(1) 48 (2) 96
(3) 72 (4) 84

78. The least number of peoples leave from which place ?
(1) H (2) G
(3) A (4) E

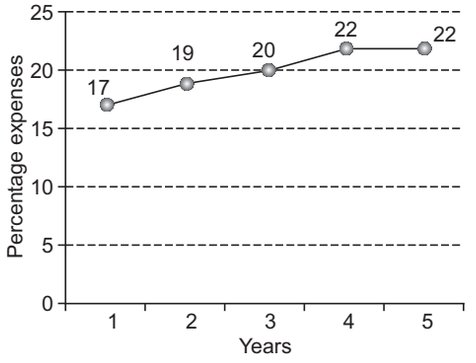
79. The number of peoples leaving F for C is _____ the number of peoples leaving C for F.
(1) greater than
(2) less than
(3) equal to
(4) Cannot say with certainty

80. The number of peoples going to A from B and C is equal to the number of peoples going to from _____ and _____.
(1) E, A and C (2) E, D and C
(3) F, A and C (4) H, E and G

81. The highest number of peoples leave from which place ?
(1) C (2) D
(3) H (4) F

Directions (Q. 82-87) *Questions are based on the following graph.*

Spending by Advertisers and Viewers



Total for all five years = Rs 100 crore

82. If in Year 3 out of money spent by the advertisers and viewers 12% had been spent on contingent expenses, what was the amount spent on contingent expenses (Rs in crore) ?
(1) 2.4 (2) 2.1
(3) 2.5 (4) 2.6

83. The expenses of which year added to one-fourth the expenses of another year equal the expenses of each of two other years ?
(1) 4 (2) 3
(3) 2 (4) 1

84. In how many years from Year 2 has the expenditure risen by the around 15% ?
(1) 1 (2) 3
(3) 2 (4) Insufficient data

85. The difference in the expenditure between which two years equals the difference between the expenditure in Year 2 to Year 1 ?
(1) Year 5 and Year 3 (2) Year 4 and Year 3
(3) Both '1' and '2' (4) Neither '1' nor '2'

86. If in Year 6 the expenses increase at the same rate as they did from Year 3 to Year 5. What will be the expenditure in Year 6 (Rs in crore) ?
(1) 28.2 (2) 24.2
(3) 25.2 (4) 23.2

87. If in Year 4 out of the money spent by the advertisers and viewers, 10% spent on disaster recovery expenses, what was the amount spent on disaster recovery expenses ?
(1) 2.2 (2) 2.6
(3) 2.3 (4) 2.4

Directions (Q. 88-95) *Questions are based on the following table.*

The XYZ manufacturing company produces a product which has a six month demand cycle as below.

Months	Demand in Units	Working Days/Month
January	300	22
February	500	19
March	400	21
April	100	21
May	200	22
June	300	20

If company's policy is to retain safety stock of 20% of monthly demand and each month's safety stock becomes the opening stock for the next month and opening stock for January is 50 unit. Each unit requires 10 labour hours and company works 8 h/day.

88. What is the opening stock for June month ?
(1) 20 (2) 40
(3) 60 (4) 100

89. What is production requirement for March ?
(1) 220 (2) 310
(3) 380 (4) 320

90. In which month production requirement is highest ?
(1) February (2) March
(3) January (4) June

91. In which month opening stock is lowest ?
(1) February (2) March
(3) January (4) May

92. If there are currently 20 workers are employed and how many workers need to be laid off in January to accommodate demand ?
(1) 6 (2) 2
(3) 3 (4) 5

93. If there are currently 20 workers are employed, and number of workers need to be laid off in January (in previous question) to accommodate demand. How many workers need to be hired in February month ?
(1) 18 (2) 16
(3) 14 (4) 22

94. If constant workforce of 20 is retained, how many hours workers sit idle in January month ?
(1) 300 h (2) 420 h
(3) 240 h (4) 200 h

95. If constant workforce of 20 is retained, how many hours workers do overtime in February month ?
(1) 2000 h (2) 3000 h
(3) 2360 h (4) 3360 h

■ Business Awareness

96. The first animal tamed by man of middle stone age was
(1) dog (2) bullock
(3) horse (4) camel

97. The planning commission started functioning on
(1) 1948 (2) 1949
(3) 1950 (4) 1951

98. The age of earth is believed to be
(1) 10 million yr (2) 100 million yr
(3) 1000 million yr (4) 4000 million yr

99. Indian Consitution was prepared in
(1) 2 yr 11 months
(2) 2 yr 18 days
(3) 2 yr 18 months
(4) 2 yr 11 months 18 days

100. In the event of breakdown of Constitution machinery of any state, under which one of the following articles can President Rule by imposed
(1) 352 (2) 356
(3) 360 (4) 370

101. FERA has been replaced by
(1) Competitive Bill (2) Finance Bill
(3) FEMA (4) No replacement

102. The measurement of poverty line is based on criteria of
(1) dwelling houses
(2) caloric consumption
(3) the nature of employment
(4) the level of eduction

103. Largest flower in the world is
(1) Cuscula (2) Nelumbo
(3) Rafflesia (4) Deosera

104. In case of myopia the correct statement is/are
(a) behind the retina
(b) infront of the retina
(c) can be collected by concave lens
(d) can be collected by convex lens
(1) a and d (2) b and d
(3) b and c (4) a, b and d

105. The four most common elements in living organisms in order are
(1) carbon, hydrogen, oxygen, nitrogen
(2) hydrogen, oxygen, carbon, nitrogen
(3) hydrogen, carbon, oxygen, nitrogen
(4) carbon, oxygen, hydrogen, nitrogen

106. The polymer of isoprene is called
(1) natural rubber (2) polyethylene
(3) neoprene (4) monoprene

107. The lightest metal is
(1) K (2) Li
(3) Ca (4) Mg

108. Cinnabar is ore of
(1) Ng (2) Cu
(3) Pb (4) Zn

109. A current of 4 amp from 250 V source flows through an electric iron for 2 h. The energy consumed will be
(1) 1 kWh (2) 4 kWh
(3) 1.5 kWh (4) 2 kWh

110. A coin is dropped at refractive index of 4/3. The real depth of coin is 28 cm. The coin would be apparently appear to be at the depth of
(1) 80 cms (2) 25 cms
(3) 20 cms (4) 21 cms

111. An aircraft takes off at the speed of 100 km/h at an angle of 30° runway. What is the velocity of the aircraft along the runway ?
(1) 86.6 km/hr (2) 76 km/hr
(3) 70 km/hr (4) 50 km/hr

112. The gas primarily responsible for the yellowing of the Taj is
(1) Nitric Oxide (2) Carbon Dioxide
(3) Ozone (4) Sulphur dioxide

113. What annual payment will discharge a debt of Rs 5040 due in 2 yr at 10% per annum compounded annually ?
(1) 2000 (2) 2850
(3) 2575 (4) 2904

114. The base radius of a cylinder is $1\frac{2}{3}$ times its height. The cost of painting is CSA at 20 paise/cm^2 is Rs 92.40. The volume of cylinder is
(1) 80850 cm^3 (2) 80580 cm^3
(3) 80508 cm^3 (4) None of these

115. When a metallic wire is connected to a battery
(1) current is taken to be flowing from negative terminal to positive terminal
(2) electrons flow from negative terminal to positive terminal
(3) electrons flow from positive terminal to negative terminal
(4) current is taken to be flowing from negative to negative and positive to positive terminal

116. In man, the absence of a *Y* chromosome in a fertilised egg causes it to develop into
(1) Male
(2) Female
(3) Inter sex
(4) Dead chromosome

117. Synecology is the study of
(1) Individuals
(2) Relation between plant and animal community
(3) Study of environment
(4) None of the above

118. "No part of the enemy's body may be used as a souvenir", was said by
(1) General Kamikaze
(2) American Pacific Fleet, 1942
(3) Winston Churchill
(4) Stalin

119. Who wrote the book "Selling the Dream" ?
(1) Steve Jobs (2) Guy Kawasaki
(3) Shiv Khera (4) Philip Goldberg

120. The Bombay dabawala have achieved international reputation because of
(1) timely daba delivery
(2) organised team work
(3) six sigma based operation
(4) excellence service for last 50 yr and beyond

■ Logical Reasoning

121. In a beauty contest the order of contestants was to be reported.
I. The editor was asked, "Is the order A-B-C-D-E" The Editor said, "None are in their true position " and not a single one follow her immediate predecessor."
II. The editor was then asked, "Is it D-A-E-C-B ?" The editor replied, "You have two in position and you have two following correctly their immediate predecessors."
III. The editor was asked for the third time, "Is this the order ?" The editor said, "You are right."
Which is this order ?
(1) B-E-C-D-A (2) E-D-A-C-B
(3) C-A-E-D-B (4) A-D-B-E-C

122. On the blackboard, the Mathematics professor wrote a polynomial $f(x)$ with integer coefficients and said, "Today is my son's birthday. When his age A is substituted for x, then $f(A) = A$. You will note also that $f(O) = P$ and that P is prime number greater than A." How old is professor's son ?
(1) 4 (2) 1
(3) 3 (4) 2

123. Select the sequence to be inserted into boxes,
CABAY ☐ ECDY ☐ XGE ☐ WU ☐
(1) ZWVF (2) ZVWF
(3) ZVFW (4) ZWFV

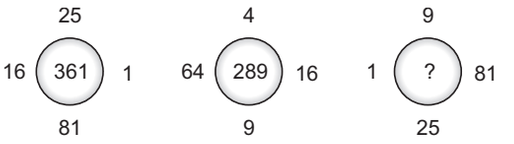
124. Fill in the gaps : SL_Bs_bc_L_dSL_e
(1) AIScD (2) LAScD
(3) ALScd (4) AlSCd

125. Fill in the blanks : 1, 4, 27, 16, ..., 36, 343
(1) 87 (2) 120
(3) 25 (4) 125

126. Which is the odd one ?
(1) ADBEC (2) FIGJH
(3) KNLOM (4) PRTSQ

127. Which is the odd one ?
(1) ZZXAY (2) BBLAM
(3) KKUAV (4) LLSMM

128. Fill in the circle



(1) 269 (2) 260
(3) 324 (4) 429

129. Fill the missing numbers

52	43	34	25
42	33	?	15
45	?	27	18
?	45	36	27

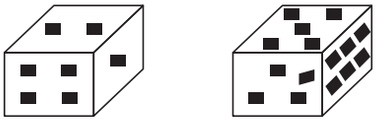
(1) 64, 25, 61 (2) 18, 26, 37
(3) 24, 36, 54 (4) 21, 46, 24

130. Fill the missing number

4C	2B	3A
28A	?	45B
7C	5A	15B

(1) 10A (2) 10C
(3) 13C (4) 7C

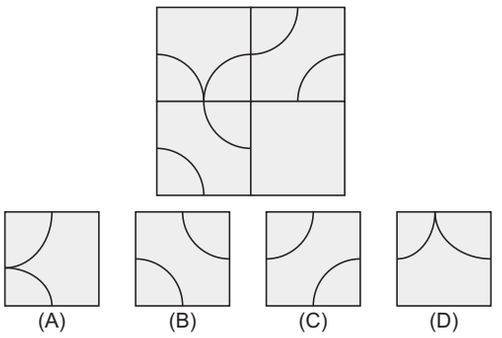
131. If number 4 is placed on the bottom which number will be on top ?



(1) 3 (2) 2
(3) 5 (4) 6

132. If you start your journey on 22nd June 2003, what will be the day of the week after 7 days of your departure ?
(1) Saturday
(2) Monday
(3) Friday
(4) Sunday

133. Fill in the Gap



(1) B (2) C
(3) A (4) D

Directions (Q. 134-138) *Solve the questions if the symbols are as given below.*

*=+ +=/ – = *
/=– →= square ← = cube

134. 65* 89754 – 9856 + 785 / 8647
(1) 1118316.629 (2) 1129316.629
(3) 1119376.629 (4) 1128376.629

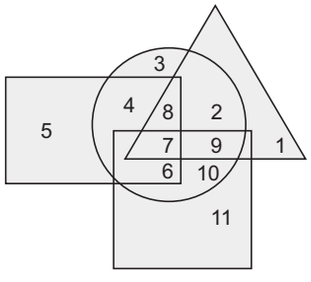
135. 89765 / 897 – 6789 + 89 →
(1) 455839620.7 (2) 45483532.6
(3) 455439719.7 (4) 455440719.7

136. 1234 / 90 * 8978 – 234 ←
(1) 92887128772529 (2) 92874321872529
(3) 98284321772529 (4) 92874581932529

137. 980 – 320 + 56 / 220 →
(1) 28944400 (2) 28934500
(3) 28876700 (4) 27868458

138. 8765* 789654 – 456* 4356 + 660 / 34532
(1) 340056463.6 (2) 360056552.6
(3) 360056463.6 (4) 360090996

Directions (Q. 139-140) *Rectangle, square, circle and triangle represents the regions of wheat, gram, maize and rice cultivation. Based on the below diagram answer the questions.*

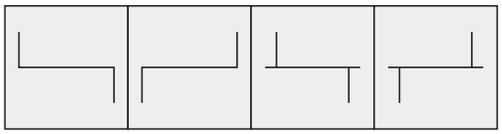


139. Which is area for wheat and maize ?
(1) 8 (2) 6
(3) 5 (4) 4

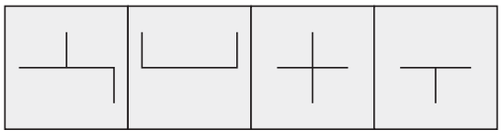
140. Which is the area for maize only ?
(1) 10 (2) 2
(3) 3 (4) 4

Directions (Q. 141-146) *The following questions have two section problem figures and answer figures. Problem figures have four designs in series. Answer figures have the fifth one which should it be ?*

141. Problem figures

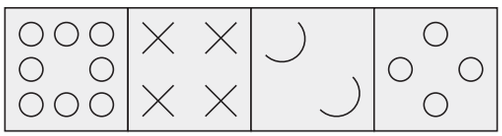


Answer figures

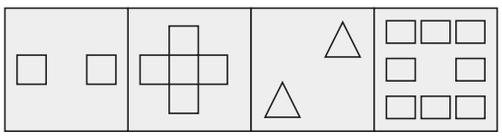


(1) C (2) A
(3) D (4) B

142. Problem figures

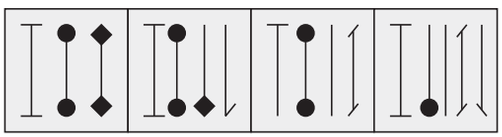


Answer figures

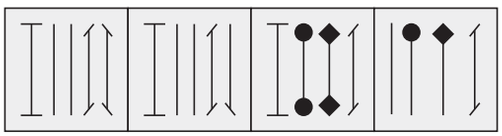


(1) C (2) A
(3) B (4) D

143. Problem figures

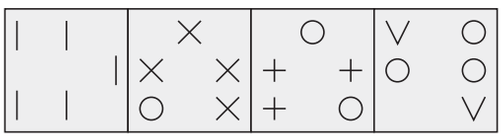


Answer figures

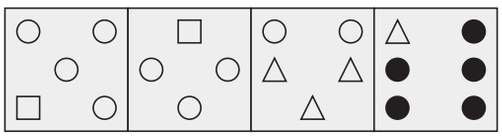


(1) A (2) B
(3) C (4) D

144. Problem figures

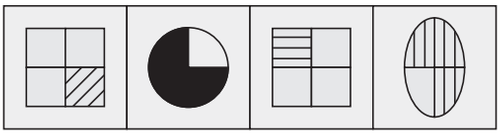


Answer figures

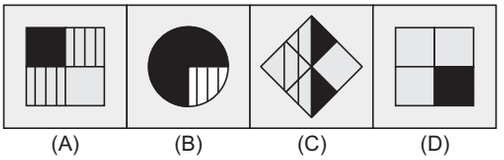


(1) C (2) B
(3) A (4) D

145. Problem figures



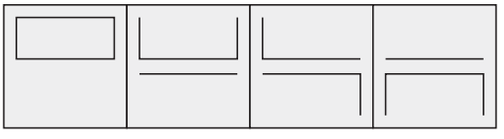
Answer figures



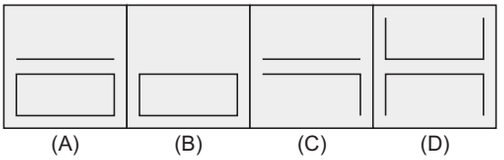
(A) (B) (C) (D)

(1) A (2) B
(3) C (4) D

146. Problem figures



Answer figures



(A) (B) (C) (D)

(1) C (2) D
(3) A (4) B

147. **Statement** Some wise men lies. Whoever lies should be imitated.
Conditions
X. Some wise men deserves to be imitated.
Y. Some fools should not be imitated.
(1) X is valid
(2) Y is valid
(3) X and Y are valid
(4) X and Y are invalid

148. A is a son of C. C is sister of Z, Z is the mother of Q. P is the son of Z. Which one of the following is correct ?
(1) P and A are cousins
(2) P is the maternal uncle of A
(3) Q is the grandfather of A
(4) C and P are sisters

149. If 'MACHINE' is coded as 19-7-9-14-15-20-11, how will you code 'DANCER' ?
(1) 13-7-20-10-11-25
(2) 11-7-20-16-11-24
(3) 10-7-20-13-11-24
(4) 13-7-20-9-11-25

150. In a certain code 'FAVOUR' is written as 'EBUPTS'. How is 'DAGGER' written in that code ?
(1) EBHHFS (2) CBFHDS
(3) CBFFDS (4) EBFHDS

151. If South-East becomes North. North-East becomes West and so on, what will West become ?
(1) North-East (2) North-West
(3) South-East (4) South

152. A and B are standing at the same point. They decide to move exactly in the opposite directions in each other. A goes 5 km East, then goes 7 km North, then goes 5 km West, and finally goes 7 km North. What is the shortest distance between A and B ?
(1) 48 km (2) 38 km
(3) 28 km (4) 118 km

153. X, Y and Z together can do a piece of work in 8 days. While X and Z together can do it in 12 days. Y alone will do the same work in how many days ?
(1) 16 days (2) 20 days
(3) 24 days (4) 28 days

154. If using each letter, meaningful word beginning with B can be formed from 'HACEB', then the fourth letter of that word is your answer. If no such word can be formed then your answer is '5' *ie,* none of these
(1) A (2) C
(3) E (4) H

155. How many meaningful English words can be formed by using any two letters of the word 'NOT' ?
(1) One (2) Two
(3) Three (4) More than three

Direction (Q. 156) *The conversation given below are followed by one question each, to which there are four possible answers. Choose the best answer.*

156. **Host** So here you are at last.
Guest Hope, I am not too late.
What does the guest want to convey ?
(1) He wanted to reach late
(2) He knows it doesn't matter even if he is late
(3) He hopes he is on time
(4) He knows the host doesn't like his coming late

157. Choose the part of the words having similar relationship to each other as the two words Tadpole : Frog.
(1) Frog : Fish
(2) Cat : Feline
(3) Caterpillar : Butterfly
(4) Flower : Tulip

158. What is the synonym of 'Wary' ?
(1) Apprehensive (2) Attentive
(3) Alert (4) Cautious

159. A tank in which fish or water plants are kept is known as
(1) aviary (2) aquarium
(3) nursery (4) apiary

160. This is defect which a boy had even at the time of his birth
(1) genetic (2) genial
(3) congenital (4) biological

Answers

1. (1)	**2.** (4)	**3.** (3)	**4.** (1)	**5.** (3)	**6.** (3)	**7.** (2)	**8.** (4)	**9.** (2)	**10.** (1)
11. (1)	**12.** (3)	**13.** (1)	**14.** (4)	**15.** (2)	**16.** (1)	**17.** (3)	**18.** (1)	**19.** (4)	**20.** (3)
21. (3)	**22.** (1)	**23.** (1)	**24.** (1)	**25.** (4)	**26.** (3)	**27.** (4)	**28.** (1)	**29.** (4)	**30.** (2)
31. (4)	**32.** (2)	**33.** (4)	**34.** (4)	**35.** (3)	**36.** (4)	**37.** (2)	**38.** (3)	**39.** (2)	**40.** (2)
41. (3)	**42.** (2)	**43.** (1)	**44.** (3)	**45.** (4)	**46.** (1)	**47.** (1)	**48.** (4)	**49.** (3)	**50.** (2)
51. (4)	**52.** (2)	**53.** (2)	**54.** (3)	**55.** (1)	**56.** (1)	**57.** (4)	**58.** (2)	**59.** (3)	**60.** (3)
61. (2)	**62.** (2)	**63.** (4)	**64.** (3)	**65.** (3)	**66.** (1)	**67.** (4)	**68.** (2)	**69.** (1)	**70.** (4)
71. (3)	**72.** (4)	**73.** (1)	**74.** (4)	**75.** (4)	**76.** (1)	**77.** (2)	**78.** (3)	**79.** (1)	**80.** (4)
81. (1)	**82.** (1)	**83.** (4)	**84.** (3)	**85.** (3)	**86.** (2)	**87.** (1)	**88.** (2)	**89.** (3)	**90.** (1)
91. (4)	**92.** (2)	**93.** (1)	**94.** (2)	**95.** (3)	**96.** (1)	**97.** (4)	**98.** (4)	**99.** (4)	**100.** (2)
101. (3)	**102.** (2)	**103.** (3)	**104.** (3)	**105.** (1)	**106.** (1)	**107.** (2)	**108.** (3)	**109.** (4)	**110.** (4)
111. (4)	**112.** (4)	**113.** (4)	**114.** (4)	**115.** (3)	**116.** (2)	**117.** (1)	**118.** (3)	**119.** (2)	**120.** (2)
121. (2)	**122.** (4)	**123.** (4)	**124.** (4)	**125.** (4)	**126.** (4)	**127.** (4)	**128.** (3)	**129.** (3)	**130.** (2)
131. (2)	**132.** (4)	**133.** (4)	**134.** (1)	**135.** (3)	**136.** (4)	**137.** (1)	**138.** (3)	**139.** (4)	**140.** (3)
141. (1)	**142.** (4)	**143.** (1)	**144.** (3)	**145.** (4)	**146.** (4)	**147.** (1)	**148.** (1)	**149.** (3)	**150.** (2)
151. (3)	**152.** (3)	**153.** (3)	**154.** (2)	**155.** (3)	**156.** (3)	**157.** (3)	**158.** (4)	**159.** (2)	**160.** (3)

Answers with **Explanations**

1. (1) Preamble is introductory part of Constitution and prologue is that of play.

2. (4) The sound produced by an animal is to be considered. As a horse neighes, a donkey brays.

3. (3) Here, the unit of measurement is to be considered. As distance is measured in metre, weight is measured in kilogram.

4. (1) Here, the relation of antonyms (opposite words) takes place. The word 'Always' is the antonym of the word 'never', whereas the word 'all' is the antonym of 'none'.

5. (3) An Umbrella protects us from rain, whereas a dam protects us from flood.

6. (3) Here, the quality (adjective) and person (noun) relationship is to be considered. A pacifist does not believe in war whereas an atheist does not believe in religion or God.

7. (2) When a bird sheds its feathers, it is called moulting. Moulting is also a natural process in birds. Similarly, when a dog or any animal sheds it skin, it means that they lose the skin naturally.

8. (4) Currency of England is known as Pound whereas Lira is the currency of Italy.

9. (2) Capital of Canada is Toronto whereas Rangoon is the capital of Burma.

10. (1) As each is the one of 'all', part is one of the whole.

11. (1) Here, words comic and tragedy are the antonyms of each other.

12. (3) Deliberate and voluntary are synonyms of each other and mean "Wilfully" something. Involuntary means 'against the will'. Hence, Deliberate and involuntary are the antonyms of each other.

13. (1) Words 'Fastidious' and 'Meticulous' both are synonyms of each other, and mean 'caring for minor details' in doing work.

14. (4) Words 'Querulous' and 'Compatible' both are antonyms of each other. Querulous means complaining and compatible means harmonious.

15. (2) Words 'Extravagance' and 'Prodigality' both are synonyms of each other and mean spending wastefully.

16. (1) The first line of the given information makes us assume that 'the social scientist has an obligation to provide the means by which social phenomena may be understood by others'.

17. (3) The second half of the first sentence of the given information makes us assume that 'larger markets abroad are more adaptable to domestic marketing methods'.

21. (3) The increase in percentage indicates the increase in number also.

22. (1) Here, we'll fill the 'question-tag' of the given sentence. Somebody told you, is not it ?

23. (1) Word 'News' is used as singular and hence, it will take a singular verb also which is travels.

24. (1) This follows the pattern of using the adverbs.

26. (3) There are three ways in which 6 balls can be arranged in three different boxes.
ie, (1, 2, 3), (1, 1, 4) and (2, 2, 2).
But these arrangements can interchange among themselves.
Hence, number of ways of arrangement.

$$= 3!({}^6C_1 \times {}^5C_2 \times {}^3C_3) + \frac{3!}{2}({}^6C_1 \times {}^5C_1 \times {}^4C_4) + {}^6C_2 \times {}^4C_2 \times {}^2C_2$$

$= 6(6 \times 10 \times 1) + 3(6 \times 5 \times 1) + 15 \times 6 \times 1$

$= 360 + 90 + 90 = 540$

27. (4) Let $a^x = b^y = c^z = k$

$\therefore a = k^{1/x}, b = k^{1/y}, c = k^{1/z}$

According to question,

$$b^2 = ac$$

$$\Rightarrow \quad k^{2/y} = k^{1/x} \cdot k^{1/z}$$

$$\Rightarrow \quad \frac{2}{y} = \frac{1}{x} + \frac{1}{z}$$

$$\Rightarrow \quad y = \frac{2xz}{x + z}$$

28. (1) If $n(A) = m$ and $n(B) = n$, then we know from formula, total no. of relations from set A to set B is given by 2^{mn}.

In the question, $2^{mn} = 2^{10}$

$$\Rightarrow \quad mn = 10$$

Clearly, only option (1) satisfy the condition.

29. (4) We require a single ticket for a group of any two stations. Here, number of stations is 10.

$\therefore$ Number of total groups of any two stations $= {}^{10}C_2$

$\therefore$ Required number of tickets $= {}^{10}C_2 = \frac{10!}{2!\,8!}$

30. (2) The given word is ARGUMENT

Number of consonants in 'ARGUMENT' = 5

Hence, required number of ways

$$= {}^5C_2 \times 2 \times 6! = 14400$$

Since, at two end two letters can be chosen from five consonants and then it can also be interchanged hence ${}^5C_2 \times 2$. Moreover remaining 6 letters can be arranged in 6! ways.

31. (4) If set A has m elements and set B has n elements then number of one-one functions from set A to set B is n^m.

Hence, in this case number of one-one functions $= r^r$

$\therefore \quad U_4 = 4^4 = 4 \times 4 \times 4 \times 4 = 256$

32. (2) $\frac{a}{b} = \frac{4}{3} \quad \Rightarrow \quad 3a = 4b.$

$$\Rightarrow \quad \frac{3a + 2b}{3a - 2b} = \frac{4b + 2b}{4b - 2b} = \frac{6b}{2b} = 3$$

33. (4) Given, 125453752

Total number of arrangements $= \frac{9!}{3!\,2!}$

Since, 2 comes twice and 5 comes thrice.

$\therefore$ Total number of arrangements in which 5 comes together $= \frac{7!}{2!}$

Hence, required number of arrangements in which number 5 comes together $= \frac{9!}{12} - \frac{7!}{2} = \frac{9! - 6 \cdot 7!}{12}$

34. (4) $2^{x-1} + 2^{x+1} = 320$

Let $2^x = y$

$$\Rightarrow \quad \frac{y}{2} + 2y = 320$$

or $y + 4y = 640$

or $5y = 640$

$\therefore \quad y = 128$

$$\Rightarrow \quad 2^x = 128 = 2^7$$

$\therefore \quad x = 7$

35. (3) It is never possible to kept one letter in wrong envelop when already 9 letters are kept in right envelop.

36. (4) Let number of correct attempt = x

Then number of wrong attempt = $26 - x$

According to question,

$$8x + (-5)(26 - x) = 0$$

$$\Rightarrow \quad 8x - 130 + 5x = 0$$

$$\Rightarrow \quad 13x = 130$$

$\therefore \quad x = 10$

37. (2) Since, in an elimination type of tournament total number of matches should be one less than the total number of players to determine the winner.

Hence for n player, $n - 1$ matches were played.

38. (3) Given, $-1 \le x \le 2$ and $1 \le y \le 3$

$-3 \le 3x \le 6$ and $2 \le 2y \le 6$

$$\Rightarrow \quad -6 \le -3x \le 3$$

$\therefore$ Least value of $2y - 3x = 2 - 6 = -4$

39. (2) When n is divided by 4, the remainder is 3. Then n will be 7. [Assume quotient is 1]

$\therefore$ Value of $2n = 14$

$\therefore$ Required remainder = 2

40. (2) In wrapping 10 turns, total number of complete round circumference = 9

Circumference of cylinder = 4″

Hence, total length of wire = $4'' \times 9 = 36''$.

41. (3) [100 + 10 + 10 + 5] 5746320819

$= 57463281900 + 57463208190$

$+ 57463208190 + 28731604095$

$= 71890102375$

42. (2) Amount paid to each woman = Rs 8.15

So, by hit and trial method we can reject the options 1, 3 and 4 as they are not divisible by 8.15.

Hence, (2) will be the answer.

43. (1) Let one number be x.

Then other number must be $(x + 5)$.

According to question,

$$\frac{x + 5}{x} = 5$$

$$\Rightarrow \quad x + 5 = 5x \quad \Rightarrow \quad 4x = 5$$

$\therefore \quad x = \frac{5}{4}$

$$x + 5 = \frac{5}{4} + 5 = \frac{25}{4}$$

44. (3) Let first publication year be x, then

$x + (x + 7) + (x + 14) + (x + 21) + \ldots (x + 42)$

$= 13524$

$\Rightarrow \quad 7x + \left[\frac{6}{2}(7 + 42)\right] = 13524$

$\Rightarrow 7x = 13524 - (3 \times 49) = 13524 - 147$

$\Rightarrow \quad 7x = 13377$

$\therefore \quad x = 1911$

45. (4) Given

Rainy afternoon $\rightarrow$ Clear morning

Rainy morning $\rightarrow$ Clear afternoon

Clear morning + Rainy afternoon = 7 …(i)

Rainy morning + Clear afternoon = 6 …(ii)

ie, No rain + Rainy afternoon = 7

and Rainy morning + No rain = 6

Hence, Rain afternoon − Rain morning

$= 7 - 6 = 1$ …(iii)

Rain afternoon + Rain morning = 9

From Eqs. (ii) and (iii)

Rain afternoon = 5

and Rain morning = 4

$\therefore$ No rain = 2

Total number of days = 5 + 4 + 2 = 11 days

46. (1) LCM of 48, 72, 108 = 432 s

= 7 min 12 s

Hence, light will change again after 7 min 12 s *ie*, at 8 : 27 : 12.

47. (1) $\frac{1}{4}$ of $20 = 6 = 5 + 1 \quad \left[\because \frac{1}{4} \times 20 = 5\right]$

$\therefore \quad \frac{1}{5}$ of $10 = \frac{1}{5} \times 10 + 1 = 2 + 1 = 3$

48. (4) $187x - 104y = 41$

Putting the value of x and y from given options we find that $x = 314$ and $y = 565$ does not satisfy the equation. So, we have to apply hit and trial method here to solve the problem.

49. (3) $\frac{116690151}{427863887} = \frac{3}{11}$

Here, common factor is 3896717.

50. (2) Given

$1.125 \times 10^k = 0.001125 = 1.125 \times 10^{-3}$

$\therefore \quad k = -3$

51. (4) If x represents sandwich, y represents coffee and z represents biscuits.

Then, $4x + y + 10z = 169$ …(i)

$3x + y + 7z = 126$ …(ii)

Subtracting Eq. (ii) from Eq. (i), we get

$x + 3z = 43$ …(iii)

From Eqs. (ii) and (iii)

$x + y + z + 2x + 6z = 126$

or $(x + y + z) + 2(x + 3z) = 126$

$(x + y + z) + 2 \times 43 = 126$

$\therefore \quad x + y + z = 126 - 86 = 40$

Hence, cost of one sandwitch, 1 coffee and 1 biscuit = 40 paise.

52. (2) Let initial cost per person be Rs x.

Hence, total expenses = Rs $10x$

Then new cost per person be Rs $(x - 100)$.

Now, total expenses $= 15(x - 100)$

So, $10x = 15(x - 100) = 15x - 1500$

$\Rightarrow \quad 5x = 1500$

$\therefore \quad x = 300$

53. (2)

90		18	
↓	42	↓	[Alligation formula]
24		48	
1	:	2	

Amount of 18% = 2/3

Amount of 90% = 1/3

$\therefore$ Number of quartz which should be replaced

$= \frac{1}{3} \times 27 = 9$

54. (3) Percentage of people who do not have any of four quality

$= (100 - 70)\% + (100 - 75)\% + (100 - 85)\%$

$+ (100 - 90)\%$

$= 30\% + 25\% + 15\% + 10\% = 80\%$

Hence, percentage of people who have all the four quality $= (100 - 80)\% = 20\%$

55. (1) We can easily calculate the time for both cases *ie*, when Henry started his trip and also when he arrived at his destination by using the following formula

Between x and $(x + 1)$ O' clock

Required time $= (5x \pm t) \times \frac{60}{55}$ min past x

where t is the distance between hour hand and minute hand in terms of minute spaces.

We generally use the negative sign when minute hand is behind the hour hand and use the positive sign when hour hand is behind the minute hand.

In case of coincidence, $t = 0$

$\therefore$ Required time when Henry started his trip

$= (5 \times 8 \pm 0) \times \frac{60}{55}$ min past 8

$= 40 \times \frac{12}{11}$ min past 8

$= 43\frac{7}{11}$ min past 8

Required time when Henry arrived at his destination

$$= (5 \times 2 + 30) \times \frac{60}{55} \text{ min past 2}$$

[Here, $t = 30$]

$$= 40 \times \frac{12}{11} \text{ min past 2}$$

$$= 43\frac{7}{11} \text{ min past 2}$$

$\therefore$ Required difference between these two times

$= 6$ h

For (Q. 56-59)

For P

Number of win $= 23 + 13 = 36$

Amount of award $= 36 \times 1500 =$ Rs 54000

Number of loss $= 8 + 6 = 14$

Amount lost $=$ Rs $(14 \times 1000) =$ Rs 14000

Total amount = Rs 68000

Average amount $= \frac{68000}{50} = 1360$

For Q

Number of win $= 18 + 16 = 34$

Amount of win $=$ Rs $(34 \times 1500) =$ Rs 51000

Number of loss $= 13 + 8 = 21$

Amount lost $=$ Rs $(21 \times 1000) =$ Rs 21000

Total amount = Rs 72000

Average amount $=$ Rs $\frac{72000}{55} =$ Rs 1310

For R

Number of win $= 10 + 13 = 23$

Amount of award $= 23 \times 1500 =$ Rs 34500

Number of loss $= 16 + 20 = 36$

Amount lost $= 36 \times 1000 =$ Rs 36000

Total amount = Rs 70500

Average amount $=$ Rs $\frac{70500}{59} \approx 1160$

For S

Number of win $= 16 + 14 = 30$

Amount of award $=$ Rs $(30 \times 1500) =$ Rs 45000

Number of loss $= 13 + 17 = 30$

Amount lost $=$ Rs $(30 \times 1000) =$ Rs 30000

Total amount = Rs 75000

Average amount $=$ Rs $\frac{75000}{60} =$ Rs 1250

Hence, by observing the calculated data carefully.

56. (1) Highest difference in amount = Rs 75000 – 68000 = Rs 7000

57. (4) Lowest average prize money = Rs 1160 [For R]

58. (2) Highest average prize = Rs 1360 [For Q]

59. (3) Out of 59 matches R lost 36. Hence, R lost maximum percentage of match.

60. (3) $\because$ Average money = Rs 120

$\therefore$ The sum of all three quotations = Rs 360

Lowest among three is 90.

[According to statement I]

$\therefore$ Sum of remaining two = Rs $(360 - 90) =$ Rs 270

If one of them is Rs 125.

[According to statement II]

Highest quotations = Rs 145

So, both statements are necessary to give the answer.

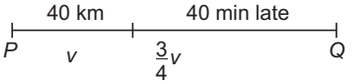
61. (2) By applying the formula

$$n(A \cap B) = n(A) + n(B) - n(A \cup B)$$

we get required number is 170.

Hence, only statement B is sufficient to give answer.

62. (2)



Ratio of speed $= v : \frac{3}{4} v = 4 : 3$

$\therefore$ Ratio of time = 3 : 4

Delay in terms of ratio = 1

$\therefore$ Actual time [If bus travels smoothly]

$= 3 \times 40$ min

$= 120$ min $= 2$ h

From statement B, distance coverd in 2 h = 40 km

$\therefore$ Total distance $= 40 + 40 = 80$ km

63. (4) Both statements are unable to give answer.

64. (3) Total time taken by flight = 6 h

Since, flight reach at 3 am local time of Mumbai.

It means flight take off at 9 pm of local time of Mumbai.

Hence, time difference = 6 h

So, here both statements are simultaneously needed.

65. (3) Sangeeta's current age = 20 yr

$\therefore$ Here, father's age five year before = 60 yr

$\therefore$ Current age of her father $= 60 + 5 = 65$ yr

66. (1) By interpretening only one statement *ie*, A. We are able to say that Sandhya has 2 sisters.

67. (4) We cannot say anything even by using both statements.

68. (2) In 1995 percentage of buyers in the age group 20-40 yr = 60

In 1999 percentage of buyers in the age group 20-40 yr = 16

Percentage of decline $= \frac{(60 - 16) \times 100}{60} = 73.33\%$

69. (1) Total number of peoples = 2000

Percentage increase during 1995-99

$= (70 - 30)\% = 40\%$

Hence, number of peoples increased $= \frac{2000 \times 40}{100}$

$= 800$

70. (4) No particular trend is observed in any group.

71. (3) In 1997, percentage of byers in 10-20 yr group = 60
In 1999, percentage of byers in 10-20 yr group = 70

$$\text{Percentage of increase} = \frac{(70-60)\times 100}{60}$$

$$= \frac{100}{6} = 16.7\%$$

72. (4) By visual interpretation, we can say that there is no particular trend is noticed.

73. (1) 10-20 yr age group increases for Ist three years but fall in 4th year but it regain in 1999 *ie*, in 5th year.

74. (4) By visual interpretation it is clear that least people visit A *ie*, 274 which is least in comparision to other.

75. (4) The given data only clears the number of people visiting different locations. So, here given people may visit one or more locations. Hence, we cannot find the total number of visitors.

76. (1) Total number of peoples visiting E from $D = 42$
Total number of peoples visiting at E from different locations = 402

$$\text{Percentage of people visiting } E \text{ from } D = \frac{42\times 100}{402} \approx 10\%$$

77. (2) Number of people going from B to F and G
$= 90 + 108 = 198$
Number of people going from D to F and G
$= 82 + 20 = 102$
Hence, required difference $= 198 - 102 = 96$

78. (3) The least number of peoples leave from A which is equal to 320 which is compartively low.

79. (1) The number of peoples leaving F for C is 92 which is greater than the number of peoples leaving C for F.

80. (4) The number of peoples going to A from B and C = 20 + 28 = 48
Similarly, after checking all options we find the number of peoples going to H from E and G is also equal to 48.

81. (1) The highest number of peoples leave from C *ie*, 616 which is comparatively highest.

82. (1) Percentate of contigent expense = 12
Percentage of total expense = 20

$$\therefore \text{Amount of contigent expense} = \frac{20\times 12}{100}$$

$$= \frac{24}{10} = 2.4 \text{ crore}$$

83. (4) Expense of 1st year + $\frac{1}{4}$th of expense of 3rd year
= expense of 4th year
= expense of 5th year

84. (3) Expenditure in 4th year = 22
Expenditure in 2nd year = 19
Increasement = (22 – 19) = 3

$$\therefore \text{Per cent increase} = \frac{3\times 100}{19} \approx 15\%$$

85. (3) From the graph it is clear that the difference between the expenditure in Year 2 to Year 1 is equal to 2.
Again, it is also clear, the difference between the expenditure in Year 5 and Year 3 and as well as in Year 4 and Year 3 is also 2.
Hence, both options (1) and (2).

86. (2) Rate of increasement of expenditure $= \frac{2\times 100}{20} = 10$

$$\text{So, expenditure in Year 6} = 22\times\frac{110}{100} = \frac{242}{10}$$

$$= 24.2 \text{ crore}$$

87. (1) Percentage of expense on disaster recovery = 10

$$\therefore \text{ Amount} = \frac{10\times 22}{100} = 2.2 \text{ crore}$$

For (Q. 88-95)

Months	Opening Stock	Demand in Units	Safety Stock 20%	Production = Demand + Safety stock – Opening stock	Working Days
January	50	300	60	300 + (60 – 50) = 310	22
February	60	500	100	500 + (100 – 60) = 540	19
March	100	400	80	400 + (80 – 100) = 380	21
April	80	100	20	100 + (20 – 80) = 40	21
May	20	200	40	200 + (40 – 20) = 220	22
June	40	300	60	300 + (60 – 90) = 320	20

88. (2) Safety stock of month May becomes the opening stock of month June which is 40.

92. (2) ∵ We have, 22 working days in January.
∴ Total number of working hours for a single labour
$= 22\times 8 = 176$ h.
Total man hours required in January to accommodate demand $= 310\times 10 = 3100$ men labours

$$\text{Hence, number of workers} = \frac{3100}{176} = 17.6$$

∵ $17 < 17.6 < 18$
It means 18 workers required.
Hence, number of workers laid off = 20 – 18 = 2

93. (1) In February demand units = 540
Number of labour hours $= 540\times 10 = 5400$
Number of working days in February = 19
Total number of labour hours for a single worker in February $= 19\times 8 = 152$

$$\text{Number of worker} = \frac{5400}{152} \approx 36$$

Hence, number of worker required to be hired
$= 36 - 18 = 18$

94. (2) In January month required labour hours = 3100
Provided labour hours by 20 workers
$= 22 \times 20 \times 8$
$= 3520$ labour hours
Hence, sitting hours $= (3520 - 3100) = 420$ h

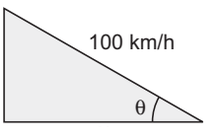
95. (3) In February month required labour hours
$= 5400$ h
Labour hours provided by 20 workers $= 20 \times 8 \times 19$
$= 3040$ h
Overtime hours $= (5400 - 3040)$ h $= 2360$ h

110. (4) $\left[\text{Refractive index} = \dfrac{\text{Real depth}}{\text{Apparent depth}}\right]$

111. (4) $\sin\theta = \dfrac{v}{100}$

$\Rightarrow \quad \sin 30° = \dfrac{v}{100}$

$\therefore \quad v = 50$ km/h



113. (4) Let the annual payment = Rs x
Again, assume p_1 and p_2 are the part of the sum that will be cleared in the first and second year respectively.

Then, $p_1 = \dfrac{x}{\left(1 + \dfrac{10}{100}\right)}$

and $p_2 = \dfrac{x}{\left(1 + \dfrac{10}{100}\right)^2}$

According to question,

$p_1 + p_2 = 5040$

$\dfrac{x}{\frac{11}{10}} + \dfrac{x}{\left(\frac{11}{10}\right)^2} = 5040$

$\Rightarrow \quad \dfrac{10x}{11} + \dfrac{100x}{121} = 5040$

$\Rightarrow \quad \dfrac{110x + 100x}{121} = 5040$

$\Rightarrow \quad x = \dfrac{5040 \times 121}{210} = \text{Rs } 2904$

114. (4) Let height is h.

$\therefore$ Radius $= \dfrac{3}{2}h$

$\therefore$ Curve surface area $= 2\pi rh$

$= 2\pi \times \dfrac{3}{2}h \times h$

$= 3\pi h^2$

Since, cost of painting at 20 paise/cm^2 is Rs 92.40.

$\therefore$ Curve surface area $= \dfrac{92.40}{0.20} = 462$

$\Rightarrow \quad 3\pi h^2 = 462$

$\therefore \quad h^2 = \dfrac{462 \times 7}{3 \times 22}$

$\Rightarrow \quad h = 7$

$\therefore$ Readius $= 10.5$

$\therefore$ Required volume $= \pi r^2 h$

$= \dfrac{22}{7} \times 10.5 \times 10.5 \times 7$

$= 2425.5$ cm^3

121. (2) Here in the statement (2), it is told that the order "D-A-E-C-B" contains two contestants in their correct positions. Hence, in the correct order also two of these five must be in the same positions.
Now, among the given options, we can see that only option (2) has such an order where two are in the same positions as they are in the order "D-A-E-C-B". Hence, the only possible order is (2).

122. (4) $f(x) = 2x - 2$

$\therefore \quad f(p) = 2p - 2$

Putting $p = 2$, we get

$f(2) = 4 - 2 = 2$

123. (4) Here, the sequence will be as
C-AB-A-Y Z -E-CD-Y- W X-G-E F -W-U V.
Here, we can see that after a particular letter, two consecutive letters of alphabet are written in the sequence. By doing so, we get the letters 'ZWFV' to be filled in the boxes.

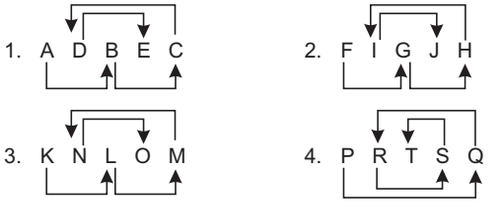
124. (4) Here, the series continues as
SL-AB-sl-bc-SL-CD-sl-de
Hence, we can see that, letters 'AlSCd' are to be filled in the blanks.

125. (4) Here, the series progresses as

1	4	27	16	125	36	343
↓	↓	↓	↓	↓	↓	↓
1^3	2^2	3^3	4^2	5^3	6^2	7^3

Hence, '125' is to be filled in the blank.

126. (4) Here, we shall consider each option one by one.



Hence, option (4) is the odd option.

127. (4) Here, among all the given options, only option (4) contains the two repeating letters while others contain only one such pair. Hence, option (4) is the odd option.

128. (3) Here, the central number is written by adding the square roots of the numbers being out of the circle and then squaring the resultant.

25
16 (361) 1 → $(\sqrt{25} + \sqrt{1} + \sqrt{81} + \sqrt{16})^2 = (5 + 6 + 9 + 4)^2$
$= 19^2 = 361$
81

4
64 (289) 16 → $(\sqrt{4} + \sqrt{16} + \sqrt{9} + \sqrt{64})^2 = (2 + 4 + 3 + 8)^2$
$= (17)^2 = 289$
9

9
1 (324) 81 → $(\sqrt{9} + \sqrt{81} + \sqrt{25} + \sqrt{1})^2 = (3 + 9 + 5 + 1)^2$
$= (18)^2 = 324$
25

Hence, 324 will be filled in the circle.

129. (3) Here, the numbers of each row are written as

52 $\xrightarrow[-9]{}$ 43 $\xrightarrow[-9]{}$ 34 $\xrightarrow[-9]{}$ 25

42 $\xrightarrow[-9]{}$ 33 $\xrightarrow[-9]{}$ 24 $\xrightarrow[-9]{}$ 15

45 $\xrightarrow[-9]{}$ 36 $\xrightarrow[-9]{}$ 27 $\xrightarrow[-9]{}$ 18

54 $\xrightarrow[-9]{}$ 45 $\xrightarrow[-9]{}$ 36 $\xrightarrow[-9]{}$ 27

Hence, "24, 36, 54" are to be filled in the blanks.

130. (2) Here, in each row, A, B and C are written by interchanging their positions among themselves. Also, in each column the upper and the lower is multiplied and then the resultant is written in the middle column. In the middle box of the chart, letter 'C' will be written and the number will be

$$2 \times 5 = 10$$

131. (2) From first figure, we can say that 1 and 2 cannot be opposite to number '4'. Similarly from second figure we can say that 5 and 6 cannot be opposite to number 4. Hence, among the given options, only number '3' is the possible number that can be on the top.

132. (4) It was Sunday on 22nd July 2003 and after 7 days it will be Sunday again.

134. (1) $65 * 89754 - 9856 + 785 / 8647$

$\Rightarrow 65 + 89754 \times 9856 \div 785 - 8647$

$\Rightarrow 65 + \{89754 \times (9856 \div 785)\} - 8647$

$\Rightarrow 65 + \{89754 \times 12.555\} - 8647$

$\Rightarrow 65 + 1126898.6 - 8647$

$\Rightarrow 1126963.6 - 8647 = 1118316.6$

135. (3) $89765 / 897 - 6789 + 89 \rightarrow$

$\Rightarrow (89765 - 897 \times 6789 \div 89)^2$

$\Rightarrow (89765 - 897 \times 76.280899)^2$

$\Rightarrow (89765 - 68423.966)^2$

$\Rightarrow (21341.034)^2 = 455439719.7$

136. (4) $1234 / 90 * 8978 - 234 \rightarrow$

$\Rightarrow (1234 - 90 + 8978 \times 234)^2$

$\Rightarrow (1144 + 2100852)^2$

$\Rightarrow (2101996)^2 = 44183872184016$

137. (1) $980 - 320 + 56 / 220 \rightarrow$

$\Rightarrow (980 \times 320 \div 56 - 220)^2$

$\Rightarrow (980 \times 5.7142 - 220)^2$

$= (5600 - 220)^2$

$= (5380)^2 = 28944400$

138. (3) $8765 * 789654 - 456 * 4356 + 660 / 34532$

$\Rightarrow 8765 + (789654 \times 456) + (4356 \div 660) - 34532$

$\Rightarrow 8765 + 360082224 + 6.6 - 34532$

$\Rightarrow 360090995.6 - 34532 = 360056463.6$

139. (4) Rectangle represents the wheat cultivation area while circle represents the maize cultivation area. Hence, we have to find the number which is only in the rectangle and the circle. Hence, the number is '4'.

140. (3) The circle represents the region for maize. Hence, we have to look for the number which is in the circle only and that number is '3'.

141. (1) Once both upward and downward lines go to the opposite sides of the main line and then it comes more closer in the next figure. The process continues.

142. (4) The fifth figure must contain eight items. Hence, the only possible answer can be 'D'.

143. (1) One line is increasing in second figure and same line is repeated in third figure and hence, the fifth figure must contain five designs.

144. (3) The number of some designs is decreasing one by one while the number of circles is increasing one by one.

146. (4) Every time a line (side) of the upper design comes down and complete the lower design.

147. (1) We can infer that wise men should be imitated.

149. (3)

M	A	C	H	I	N	E
(13)	(1)	(3)	(8)	(9)	(14)	(5)
+6 ↓	↓	↓	↓	↓	↓	↓
19	–7	–9	–14	–15	–20	–11

Hence,

D	A	N	G	E	R
(4)	(1)	(14)	(7)	(5)	(18)
+6 ↓	↓	↓	↓	↓	↓
10	–7	–20	–13	–11	–24

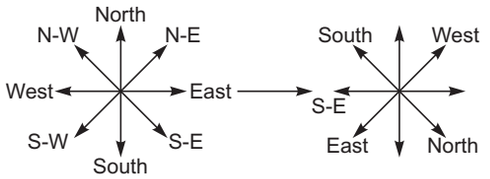
150. (2) F A V O U R → E B U P T S

–1, +1, –1, +1, –1, +1

Hence, D A G G E R → C B F H D S

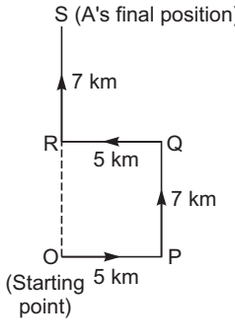
–1, +1, –1, +1, –1, +1

151. (3)



Hence, 'West' becomes 'South-East'.

152. (3)



Here, the shortest distance of A from the starting point is

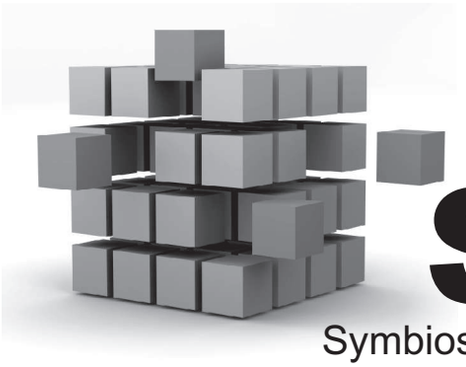
$OS = OR + RS = PQ + RS = (7 + 7)\text{ km} = 14\text{ km}$

The same distance has been covered by B also. Hence, the shortest distance between A and

$= 14 + 14 = 28\text{ km}$

154. (2) The only possible word starting from B is 'BEACH'. Its fourth letter is 'C'.

155. (3) We can make three words 'NO', 'ON' and 'TO'.



SNAP 2005

Symbiosis National Aptitude Test

Solved Paper

1. Milk and water in two vessels *A* and *B* are in the ratio 4 : 3 and 2 : 3 respectively. In what ratio the liquids in both the vessels should be mixed to obtain a new mixture in vessel *C*, containing half milk and half water?
(1) 1 : 1 (2) 7 : 5
(3) 2 : 4 (4) 1 : 3

2. A picture was bought at a certain sum, which was the price paid for its frame. Had the frame cost Rs 100 less and the picture Rs 75 more, the price for the frame would have been only half of that of the picture. What is the price of the frame?
(1) 75 (2) 100
(3) 175 (4) 275

3. What is the closest time between 7 and 8 when the hands of your watch are exactly opposite each other?
(1) 7 h 5 min (2) 7 h 5.5 min
(3) 7 h 6 min (4) 7 h 6.5 min

4. There are 6 tickets to the theater, four of which are for seats in the front row. 3 tickets are selected at random. What is the probability that two of them are for the front row?
(1) 0.6 (2) 0.7
(3) 0.9 (4) 1/3

5. When 75% of a number is added to 75, the result is the same number. The number is
(1) 150 (2) 300
(3) 100 (4) 450

6. Anne travelled 4/7th as many miles on foot as by water and 2/5th as many miles on horse back as by water. If she covered a total of 3036 miles, how many miles did she travel in each manner : water, foot and horseback?
(1) 1540,880,616 (2) 616,880,1540
(3) 1540,880,616 (4) 880,1540,616

7. A family has seven children. Each boy in the family has as many sisters as brothers both. Each girl has twice as many brothers as sisters. How many brother and sisters are there in the family?
(1) 3 Brothers, 4 Sisters (2) 4 Brothers, 4 Sisters
(3) 4 Brothers, 3 Sisters (4) Cannot say

8. You are given 50 white marbles, 50 black marbles and two jars. You need to put 100 marbles in any of these two jars. The jars will, then be shaken and you will be asked to pick one marble from either jar. How would you distribute the marbles in two jars to maximize the possibility of picking a white marble blind folded?
(1) 25 white and 25 black in each
(2) One white in one and rest 99 in the other
(3) 50 white in one and 50 black in the other
(4) All hundred in one

9. The number plate of a bus had peculiarity. The bus number was a perfect square. It was also a perfect square when the plate was turned upside down. The bus company had only five hundred buses numbered from 1 to 500. What was the number?
(1) 169 (2) 36
(3) 196 (4) Cannot say

10. If 5 spiders can catch five flies in 5 min. How many flies can hundred spiders catch in 100 min?
(1) 100 (2) 1000
(3) 500 (4) 2000

11. Recently my brother and I played chess form chocolates. Who ever lost the game gave the other a chocolate. After the last game we counted the chocolates. I had 20 more chocolates than I started with, although he won 7 games. There is no draw. How many games did we play?
(1) 27 (2) 34
(3) 37 (4) 54

12. Study the series carefully 'B 8 4 C R M 9 P D K W F A 2 E J 7 X U Q H L T Y 6 G S'. If it is possible to make a meaningful word with the ninth, the sixteenth, the twenty-fourth and the twenty-seventh letters from the left in the above series, which of the following will be the first letter of that word? If no such word can be made, give 'X' as the answer. If more than one such word can be made, give 'M' as the answer
(1) X (2) M
(3) J (4) Y

13. The value of $5^{1/4} \times (125)^{0.25}$
(1) 5 (2) 25
(3) 50 (4) 10

14. Imagine you have two large pitchers, *A* and *B*. *A* contains 10 L of wine and *B* contains 10 L of water. One litre of water

is removed from B and poured into A. The liquid is mixed very well. Then one litre of the mixture from A is poured into B. Which of the following statements is true?
(1) The water contents in A is more than the wine contents in B
(2) The water contents in A is less than the wine contents in B
(3) The water contents in A is same as the wine contents in B
(4) None of the above

15. Anand, Binoy, Chetan and Dharma together have Rs 47 with them. Anand and Binoy together have Rs 27; Chetan and Anand have Rs 25 and Dharma and Anand have Rs 23. How much money does Binoy have?
(1) Rs 9 (2) Rs 11
(3) Rs 13 (4) Rs 28

16. David gets on the elevator at the 11th floor of a building and rides up at the rate of 57 floors per minute. At the same time Albert gets on an elevator at the 51 it floor of the same building and rides down at the rate of 63 floors per minute. If they continue traveling at these rates, then at which floor will their elevators meet?
(1) 19 (2) 30
(3) 28 (4) 37

17. The average of 5 consecutive numbers is n. If the next two numbers are also included the average will
(1) remain the same (2) increase by 1
(3) increase by 1.4 (4) increase by 2

18. Which is the wrong term in the following sequence?
52, 51, 48, 43, 34, 27, 16
(1) 27 (2) 34
(3) 43 (4) 48

19. P is six times as large as Q. By what per cent is Q less than P?
(1) $16\frac{2}{3}\%$ (2) 60%
(3) $83\frac{1}{3}\%$ (4) 90%

20. When a heap of pebbles is grouped in 32, 40 or 72 it is left with remainders of 10, 18 or 50 respectively. What is the minimum number of pebbles that the heap contains?
(1) 1416 (2) 1418
(3) 1412 (4) 1420

21. Each of the series $S_1 = 2 + 4 + 6 + \ldots\ldots$ and $S_2 = 3 + 6 + 9 + \ldots\ldots$ is continued to 100 terms. Find how many terms are identical?
(1) 34 (2) 33
(3) 32 (4) None of these

22. Ram Singh goes to Pushkar Mela with Rs 10000 to buy exactly 100 animals. He finds that cows are sold at Rs 1000, horses at Rs 300 and chicken at Rs 50. How many chicken should be buy to meet his target of 100 animals?
(1) 92 (2) 94
(3) 90 (4) 88

23. Fill in '+' or '–' sign in between these numbers so that they give the correct answer.

$$1\ 2^3\ 3^3\ 1\ 4^3 = 31$$

(1) + + – – (2) + + + –
(3) – – + + (4) – – – +

24. Symbiosis runs a Corporate Training Programme. At the end of running the first programme its total takings were Rs 38950. There were more than 45 but less than 100 particulars. What was the participant fee the programme?
(1) Rs 410 (2) Rs 450
(3) Rs 500 (4) Rs 510

25. Complete the series 1, 6, 6, 36, 216, ……
(1) 7775 (2) 7776
(3) 7777 (4) 7778

26. Three friends had a dinner at a restaurant. When the bill was received Amita paid 2/3 as much as Veena paid and Veena paid 1/2 as much as Tanya paid. What fraction of the bill did Veena pay?
(1) 1/3 (2) 3/11
(3) 12/31 (4) 5/8

27. 128 players start in the men's singles at a tennis tournament, where this number reduces to half on every succeeding round. How many matches are played totally in the event?
(1) 63 (2) 48
(3) 127 (4) 144

28. A crew can row 10 miles in 5/6th of an hour down-stream and 12 miles upstream in 90 min. Find the current's rale and crew's rate in still water
(1) 12 miles/h, 4 miles/h (2) 10 miles/h, 2 miles/h
(3) 8 miles/h, 4 miles/h (4) 12 miles/h, 2 miles/h

29. If Swamy has two children and he truthfully answers yes to the question "Is at least one of your children a girl?" What is the probability that both his children are girls?
(1) 1/2 (2) 1/3
(3) 1 (4) 0

30. Starting from his house one day, a student walks at a speed of $2\frac{1}{2}$ km/h and reaches his school 6 min late. Next day he increases his speed by 1 km/h and reaches his school 6 min early. How far is the school from the house?
(1) 1 km (2) $1\frac{1}{2}$ km
(3) $1\frac{3}{4}$ km (4) 2 km

31. What year comes next in the sequence 1973, 1979, 1987, 1993, 1997, 1999, ……?
(1) 2001 (2) 2003
(3) 2005 (4) 2007

32. The numbers of students studying Physics, Chemistry and Zoology in a college were in the ratio 4 : 3 : 5 respectively. If the number in these three disciplines increased by 50%, 25% and 10% respectively in the next year, then what was the new respective ratio?
(1) 24 : 15 : 22
(2) 18 : 11 : 13
(3) 24 :13 : 17
(4) Cannot be determined

33. A man earns Rs 20 on the first day and spends Rs 15 on the next day. He again earns Rs 20 on the third day and spends Rs 15 on the fourth day. If he continues to save like this, how soon will he have Rs 60 in hand?
(1) On 17th day (2) On 27th day
(3) On 24th day (4) On 30th day

34. A game of football with 11 players lasts for exactly 90 min. There are four substitutes that alternate equally. If each player plays for the same length of time, what is the duration?
(1) 22.5 min (2) 18 min
(3) 66 min (4) None of these

35. *A*, *B* and *C* start a business each investing Rs 20000. After 5 months *A* withdrew Rs 5000, *B* Rs 4000 and *C* invested Rs 6000 more. At the end of the year a total profit of Rs 69900 was recorded. What is the share of *B*?
(1) Rs 20500 (2) Rs 21200
(3) Rs 28200 (4) Rs 27300

36. The current birth rate per thousand is 32, whereas corresponding death rate is 11 per thousand. The growth rate in terms of population increase in per cent is given by
(1) 0.021% (2) 0.21%
(3) 2.1% (4) 21%

37. The difference between two numbers is 1365. When larger number is divided by the smaller one, the quotient is 6 and the remainder is 15. What is the smaller number?
(1) 240 (2) 360
(3) 270 (4) 295

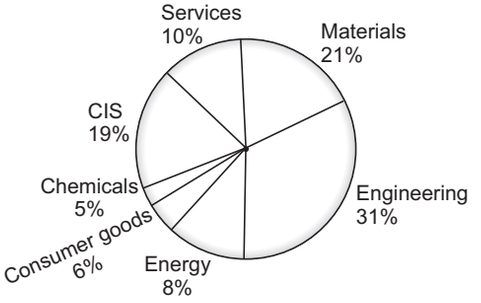
38. Two vertical poles are 40 m apart and the height of one is double that of the other. From the middle point of the line joining their feet, an observer finds the angular elevations of their tops to be complementary. Find their heights.
(1) 14.14 m, 28.28 m (2) 12 m, 22 m
(3) 12.12 m, 24.24 m (4) None of these

39. Meera and Sameera start moving from the same point. Meera goes 4 km West followed by 3 km North. Sameera goes 4 km East and then a right turn and keeps going for 3 km. What is the shortest distance between the two of them?
(1) 10 km (2) 12 km
(3) 8 km (4) 14 km

40. A boy was asked to multiply a certain number by 53. He multiplied it by 35 and got his answer less than the correct one by 1206. Find number to be multiplied.
(1) 68 (2) 67
(3) 77 (4) None of these

Directions (Q. 41-42) *Study the sectorwise sales of a group of companies for the financial year 2004-2005 and answer the following questions.*

Sectorwise Sales 2004-2005



Sectors	(Rs million)	Per cent share
Materials	164430	21.0
Engineering	244830	31.2
Energy	62990	8.0
Consumer goods	47880	6.1
Chemicals	35510	4.5
Communications and information systems	148160	18.9
Services	78950	10.1
Total	782750	

- Sales turnover at US$ 17.4 billion (Rs 78274 crore) representing a year-on-year growth of 30.3% in dollar terms and 27.4% in rupee terms.
- Sales figures shown are net of excise duty.

41. This group of companies operates in multiple sectors. The holding company has decided to closely monitor minimum number of sectors (not exceeding 5) contributing together a major part (at least 80%) of sales of the group. Identify the correct set of sectors.
(1) Engineering, Services, CIS and Consumer Goods
(2) Engineering, Services, CIS, Energy and Consumer Goods
(3) Engineering, Material, Services and CIS
(4) Engineering, Material, CIS and Energy

42. The average profit against the sales across the sectors is 10% while the profit margin of engineering is 12% and CIS, 20%. What is the average profit across the remaining sectors?
(1) 5.87% (2) 7.6%
(3) 4.94% (4) 8.9%

Directions (Q. 43-46) *Each question given below is followed by three statements. Study the question and the statements. Identify which option is necessary to answer the question.*

43. The mean temperature of Monday to Wednesday was 37 degree Celsius and of Tuesday to Thursday was 34 degree Celsius. What was the temperature on Thursday?
I. The temperature on Thursday was 4/5th that of Monday.
II. The mean temperature of Monday and Thursday was 40.5 degree Celsius.
III. The difference between the temperature on Monday and that on Thursday was 9 degree Celsius.
(1) I only (2) Either I or II
(3) Either I or II or III (4) None of these

44. What will be the sum of two numbers?
I. Among the two numbers, the bigger number is greater than the smaller number by 6.
II. 40 % of the smaller number is equal to 30 % of the bigger number.
III. The ratio of half of the bigger number to one third of the smaller number is 2 : 1.
(1) I or II (2) II or III
(3) I or II or III (4) Any two of the three

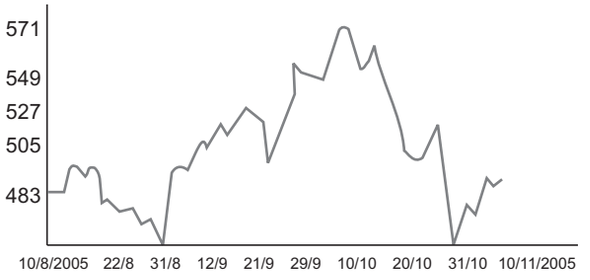
45. A tank is fitted with two taps *A* and *B*. In how much time will the tank be full if both the taps are opened together?
I. *A* is 50% more efficient than *B*
II. *A* alone takes 16 h to fill the tank
III. *B* alone takes 24 h to fill the tank
(1) I and II or III only (2) I, II and III
(3) I and II only (4) I and III only

46. What is the principal sum?
I. The interest amount after 30 months is half the interest amount after 5 yr.
II. The sum amounts to Rs 750 in a five years at simple interest rate.
III. The rate of interest is 8% per annum.
(1) I and III only
(2) II and III only
(3) I and II only
(4) I and III only, or II and III only

Directions (Q. 47-50) *Study the weekly data table and daily data graph for Toya Motors share prices at the National Stock Exchange and answer the following questions.*

Toya Motors NSE Share Prices for 3 Months

10/8	22/8	31/8	12/9	21/9	29/9	10/10	20/10	31/10	10/11
483	475	461	513	516	535	538	500	461	498



47. What is the average share price for 10/8/2005 to 12/9/2005?
(1) 527 (2) 505
(3) 483 (4) 461

48. The drop in the sales price during October 2005 is approximately
(1) 19% (2) 10%
(3) 50% (4) 5%

49. How many times during this 3 month period the share price dropped in the graph?
(1) 6 (2) 17 (3) 15 (4) 10

50. If you purchased 100 shares on 31/8/2005 and sold them on 10/10/2005 what will be your gains?
(1) Rs 7700 (2) Rs 8800
(3) Rs 9900 (4) Rs 6600

Directions (Q. 51-54) *Each question consists of a statement and/or a question followed by two statements I and II. Identify an option which provides sufficient data to answer the given question.*
Mark your answer as
(1) if the data in statement I is sufficient and II is not required;
(2) if the data in statement II is sufficient and I is not required;
(3) if the data in both statements is necessary;
(4) if the data in both statements is not sufficient to answer the question.

51. The total of the present ages of A, B, C and D is 96 yr. What is B's present age?
I. The average age of A, B and D is 20 yr.
II. The average age of C and D is 25 yr.

52. Deepak's marks in Hindi are 15 more than the average marks obtained by him in Hindi, Economics, Sociology and Philosophy. What are his marks in Philosophy?
I. The total marks obtained by him in Hindi and Philosophy together is 112.
II. The difference between the marks obtained by him in Sociology and Economics is 12.

53. What was the cost price of the suitcase purchased by Richard?
I. Richard got 20% concession on the labelled price.
II. Richard sold the suitcase for Rs 2000 with 25% profit on the labelled price.

54. B alone can complete a work in 12 days. How many days will A, B and C together takes to complete the work?
I. A and B together can complete the work in 3 days.
II. B and C together can complete the work in 6 days.

Directions (Q. 55-60) *Study the following two tables of data on Agriculture Import and Export of India during* 1998-2001 *and answer the following questions.*

India's Export of Principal Agricultural Products

Products	1998-1999	Percentage of Agri. Export	1999-2000	Percentage of Agri. Export	2000-2001	Percentage of Agri. Export
1. Tea	538	8.9	412	7.3	433	7.2
2. Coffee	411	6.8	331	5.9	259	4.3
3. Cereals	1495	24.8	724	12.9	744	12.4
4. Tobacco	181	3	233	4.2	191	3.2
5. Spices	388	6.4	408	7.3	354	5.9
6. Cashew	387	6.4	567	10.1	411	6.8
7. Sesame and Niger Seeds	78	1.3	86	1.5	131	2.2
8. Guargum Meal	173	2.9	188	3.4	132	2.2
9. Oil Meals	462	7.7	378	6.7	448	7.5
10. Fruits and Vegetables	184	3	209	3.7	248	4.1
11. Processed Fruit and Vegetables	69	1.1	86	1.5	122	2
12. Meat and Meat Preparations	187	3.1	189	3.4	322	5.4
13. Marine Products	1038	17.2	1183	21.1	1394	23.2

14. Others	446	7.4	614	11	815	13.6
Agri. Exports	6037	100	5608	100	6004	100
Percentage of Agri. to Total Exports	18.2		15.2		13.5	
Total Exports	33218		36822		44560	

India's Agricultural Imports

Commodity	1998-1999	Percentage of Agri. Imports	1999-2000	Percentage of Agri. Imports	2000-2001	Percentage of Agri. Imports
1. Cereal	288	9.9	222	7.8	19	1
2. Pulses	169	5.8	82	2.9	109	5.9
3. Milk and Cream	3	0.1	25	0.9	2	0.1
4. Cashew nuts	230	7.9	276	9.7	211	11.3
5. Nuts and Fruits	159	5.5	136	4.8	175	9.4
6. Sugar	264	9	256	9	7	0.4
7. Oil seeds	2	0.1	4	0.1	2	0.1
8. Veg. oils	1804	61.8	1857	65	1334	71.8
Total Agri. Imports	2919	100	2858	100	1858	100
Percentage of Agri. to Total Imports	6.9		5.8		3.7	
Total Imports	42389		49671		50536	

55. Which was the single largest contributor to the total Agri. exports in 2000-01? If the total Agri. exports were valued at US $ 6 billion, what is its contribution in dollar terms?
(1) Cereal, US $ 1.49 billion
(2) Marine Products, US $ 1.27 billion
(3) Marine Products, US $ 1.39 billion
(4) Cereal, US $1.03 billion

56. Which product has shown strong growth in exports during the 3 yr period?
(1) Meat and Meat Preparations
(2) Fruits and Vegetables
(3) Processed Fruits and Vegetables
(4) None of the above

57. Which was the dominant import commodity in 2000-2001 after the vegetable oil?
(1) Pulses (2) Cereals
(3) Cashew nuts (4) Nuts and Fruits

58. From the import data about Sugar and Cereals it can be said that
(1) India has raised its domestic production of these commodities.
(2) India's demand for these commodities has gone down.
(3) India's gap in production and requirement of these commodities has gone down.
(4) India's population consuming these commodities has gone down.

59. In some circles concerns were expressed that liberalization of imports, resulting from lifting of quantitative restrictions on agri products, would lead to surge of Agri. imports affecting the Indian farmers. What does the data depict?
(1) The concerns were justified because India continued to import Agri. commodities.
(2) The concerns were not justified because the value of Agri. imports in aggregate terms has come down during the period.
(3) The concerns were justified because vegetable oil and pulses formed a major component of the imports.
(4) The concerns were not justified because the total exports in aggregate terms were higher than the total imports in aggregate terms.

60. Over the period under study both percentages of the total Agri. exports to the total exports and that of the total Agri. imports to the total imports show a downward trend. This indicates that
(1) India should not get into export of Agri. products and concentrate on other sectors.
(2) India should find ways and means of increasing imports.
(3) India should restrict its imports and exports only to a limited number of commodities and products.
(4) India should work on strategies to enhance export and reduce imports.

61. Nalini, her brother, her daughter and her son are tennis players and are playing game of doubles. Their positions on the court are as follows
Nalini's brother is directly across the net from her daughter.
Her son is diagonally across the net from the worst player's sibling.
The best player and the worst players are on the same side of the net.
Who is the best players?
(1) Nalini (2) Nalini's brother
(3) Nalini's daughter (4) None of these

62. In certain code language 'PROMOTION' is written as 'QSP89'. How will you write 'DEMOTION'?
(1) DE98 (2) EF98
(3) EF89 (4) EG89

63. Study this matrix.

6	2	5	1
3	1	4	7
4	1	9	5
3	1	2	4

In this game there are two players. The first player can split the matrix vertically into two equal halves and choose one half for further play. The next move on this half is by the other player who will split it only horizontally and choose one half for further play. The game will continue in this manner. At the end, the last number left is the first player's gain.

If you start the game, retain the right half and again right half after your opponent's move, then how should your opponent play to minimize your gain?

(1) Retain upper, retain lower
(2) Retain upper, retain upper
(3) Retain lower, retain upper
(4) Retain lower, retain lower

64. What is the next letter in the series?

U, F, Q, J, M, N

(1) I (2) T
(3) O (4) M

65. Sonal, a mathematician, defines a number as 'connected by 6' if it is divisible by 6, or if the sum of its digits is 6, or if 6 is one of the digits of the number. Other numbers are all 'not connected with 6'. As per this definition, the number of integers, from 1 to 60 (both inclusive) which are not connected with 6 is

(1) 18 (2) 43
(3) 22 (4) 42

66. Leena, Nitin, Arun and Mohan crossed a lake in a canoe that could hold only two persons. The Canoe held two persons on each of three forward trips across the lake and one person on each of two return trips. Leena was unable to paddle when someone else was in the canoe with her. Nitin was unable to paddle when anyone else except Arun was in the canoe with him. Each person paddle continuously for at least one trip. Who paddled twice?

(1) Leena (2) Nitin
(3) Mohan (4) Arun

67. A, B, C, D, E, F and G are the members of a family consisting of 4 adults and 3 children, two of whom, F and G are girls, A and D are brothers and A is doctor. E is an engineer married to one of the brothers and has two children. B is married to D and G is their child. Who is C?

(1) G's father (2) F's father
(3) E's daughter (4) A's son

68. If every alternative letter of English Alphabet from B onwards (including B) is written in lower case (small letters) and the remaining letters are capitalized, then. How will be the first month of the second half of the year be written?

(1) AuGuSt (2) JuLy
(3) jUlY (4) AugUSt

69. A rich merchant had collected many gold coins. He did not want anybody to know about them. One day, his wife asked, "How many gold coins do we have?" After pausing a moment, he replied, "Well! If I divide the coins into two unequal numbers, then 48 times the difference between the two numbers equals the difference between the squares of the two numbers." The wife looked puzzled. Can you help the merchant's wife by finding out how many gold coins the merchant has?

(1) 96 (2) 53
(3) 43 (4) None of these

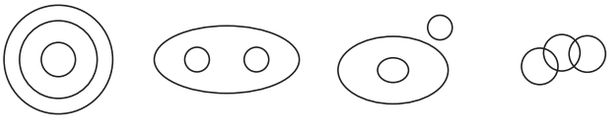
70. An enterprising businessman earns an income of Rs 1 on the first day of his business. On every subsequent day, he earns an income which is just double of that made on the previous day. On the 10th day of business, his income is

(1) Rs 2^9 (2) Rs 2^{10}
(3) Rs 10^2 (4) Rs 10

71. One night three naughty boys stole a basketful of apples from the garden, hid the loot and went to sleep. Before retiring they did some quick counting and found that the fruits were less than a hundred in number. During the night one boy awoke, counted the apples and found that he could divide the apples into three equal parts if he first took one for himself. He, then took one apple, ate it up and took 1/3 of the rest, hid them separately and went back to sleep. Shortly thereafter another boy awoke, counted the apple and he again found that if he took one for himself (he loot could be divided in to three equal parts. He ate up one apple, bagged 1/3 of the remainder, hid them separately and went back to sleep. The third boy also awoke after some time, did the same and went back to sleep. In the morning when all woke up, and counted apples, they found that the remaining apples again totaled I more than could be divided into three equal parts. How many apples did the boys steel?

(1) 67 (2) 79
(3) 85 (4) None of these

Directions (Q. 72-73) *Choose from these four diagrams the one that best illustrates the relationship among three given classes.*



(1) (2) (3) (4)

72. Chilli, Salt, Vegetables.

73. Student of Law, Students of Science, Men.

Directions (Q. 74–78) *Study the following example and answer the questions.*

An electronic device rearranges numbers step-by-step in a particular order according to a set of rules. The device stops when the final result is obtained. In this case the device stops at Step V?

Input :	85	16	36	04	19	97	63	09
Step I	97	85	16	36	04	19	63	09
Step II	97	85	63	16	36	04	19	09
Step III	97	85	63	36	16	04	19	09
Step IV	97	85	63	36	19	16	04	09
Step V	97	85	63	36	19	16	09	04

74. Which of the following will be Step III for the input below?

Input :	09	25	16	30	32	18	17	06
(1)	32	09	25	16	30	18	17	06
(2)	32	30	09	25	16	19	17	06
(3)	32	30	09	25	16	18	17	06
(4)	32	30	25	09	16	18	17	06

75. What is last step for the input below?
Input : 16 09 25 27 06 05
(1) Step II (2) Step III
(3) Step IV (4) None of these

76. What is the output of Step V for the input below?

Input :	25	08	35	11	88	67	23
(1)	88	67	35	25	23	11	08
(2)	88	67	35	25	23	08	11
(3)	08	11	23	25	35	67	88
(4)	None of the above						

77. Which one of the following would be last step for the input below?
Input : 03 31 43 22 11 09
(1) Step II (2) Step III
(3) Step IV (4) None of these

78. If the output of Step IV is as given below, what was the input?

Step IV	92	86	71	69	15	19	06	63	58
(1)	86	92	69	71	15	19	06	63	58
(2)	15	86	19	92	06	69	63	58	71
(3)	15	19	06	63	58	86	92	69	71
(4)	None of the above								

Directions (Q. 79-80) *Read the following information carefully and then answer the questions given below :*

P # Q means P is the father of Q.
P + Q means P is the mother of Q.
P – Q means P is the brother of Q.
P * Q means P is the sister of Q.

79. If A + B # C – D, then A is D's
(1) Sister (2) Grandfather
(3) Grandmother (4) Father

80. Which of the following shows that A is the Aunt of E?
(1) A – B + C # D * E (2) A * B # C * D – E
(3) A # B * C + D – E (4) A + B – C * D # E

Directions (Q. 81-87) *Read the following passage and answer the 7 questions following the same.*

As the climate in the Middle East changed beginning around 7000 B.C.E., conditions emerged that were conducive to a more complex and advanced form of civilization in both Egypt and Mesopotamia. The process began when the swampy valleys of the Nile in Egypt and of the Tigris and Euphrates Rivers in Mesopotamia became driver, producing reverine lands that were both habitable and fertile, and attracting settlers armed with the newly developed techniques of agriculture. This migration was further encouraged by the gradual transformation of the once-hospitable grasslands of these regions into deserts. Human population became increasingly concentrated into pockets of settlement scattered along the banks of the great rivers.

These rivers profoundly shaped the way of life along their banks. In Mesopotamia, the management of water in conditions of upredictable drought, flood and storm became the central economic and social challenge. Villagers began early to build simple earthworks, dikes, canals, and ditches to control the waters and reduce the opposing dangers of drought during the dry season (usually the spring) and flooding at harvest time.

Such efforts required a degree of cooperation among large numbers of people that had not previously existed. The individual village, containing only a dozen or so houses and families, was economically vulnerable; but when several villages, probably under the direction of a council of elders, learned to share their human resources in the building of a coordinated network of water-control systems, the safety, stability, and prosperity of all improved. In this new cooperation, the seeds of the great Mesopotamian civilizations were being sown.

Technological and mathematical invention, too, were stimulated by life along rivers. Such devices as the noria (a primitive waterwheel) and the Archimedean screw (a device for raising water from the low riverbanks to the high ground where it was needed), two forerunners of many more varied and complex machines, were first developed here for use in irrigation systems. Similarly, the earliest methods of measurement and computation and the first developments in geometry were stimulated by the need to keep track of land holdings and boundaries in fields that were periodically inundated.

The rivers served as high roads of the earlist commerce. Traders used boats made of boundles of rushes to transport grains, fruits, nuts, fibers and textiles from one village to another, transforming the rivers into the central spines of nascent commercial kingdoms. Trade expanded surprisingly widely; we have evidence suggesting that, even before the establishment of the first Egyptian dynasty, goods were being exchanged between villagers in Egypt and others as far away as Iran.

Similar developments were occuring at much the same time along the great river valleys in other parts of the world - for example, along the Indus in India and the Hwang Ho in China. The history of early civilization has been shaped to a remarkable degree by the relation of humans and rivers.

81. This passage basically explains
(1) the similarities and differences among several ancient societies
(2) the influence of river settlements on the growth of early civilizations
(3) how climatic changes led to the founding of the earliest recorded cities
(4) the development of primitive technologies in the ancient Middle East

82. According to the passage, the increasing aridity of formally fertile grasslands in Egypt and Mesopotamia caused the settlement patterns in those regions to become
(1) less nomadic (2) less stable
(3) more concentrated (4) more sparse

83. The passage implies that the earliest geometry was practiced primarily by
(1) farm workers
(2) land owners
(3) traders and merchants
(4) mechanical artisans

84. The passage indicates that the social effects of the unpredictability of water supplies in Mesopotamia was
(1) to encourage cooperation in the creation of water management systems
(2) to drive farmers to settle in fertile grasslands far from the uncontrollable rivers
(3) to cause warfare over water rights among rival villages
(4) None of the above

85. The passage refers to the earliest trade routes in the Middle East
(1) between various centrally ruled commercial kingdoms
(2) between linked villages in Egypt with others in Iran
(3) between connected villages that were scattered along the banks of the same river
(4) between the inhabitants of small villages and the dynastic kings who ruled them

86. The passage implies that the emergence of complex civilizations in the Middle East was dependent upon the previous development of
(1) a system of centralized government
(2) symbolic systems for writing and mathematical computation
(3) a method of storing and transferring wealth
(4) basic techniques of agriculture

87. By referring to emerging civilizations in India and China the author wants to emphasize the
(1) relatively advanced position enjoyed by the Middle East in comparison to other regions
(2) rapidity with which social systems developed in the Middle East spread to other places
(3) crucial role played by rivers in the development of human cultures around the world
(4) importance of water transportation in the growth of early trade

Directions (Q. 88-89) *Choose the alternative closest in meaning to the given word.*

88. Askance
(1) side glance (2) quizzical expression
(3) request (4) curious look

89. Virtuoso
(1) skilled performer (2) amateur
(3) good person (4) professional

Directions (Q. 90-92) *Each question below consists of a related pair of words or phrases, followed by four pairs of words or phrases labelel (1) through (4). Select the pair that best expresses the relationship similar to that expressed in the original pair.*

90. Nuance : Subtle
(1) Pun : Sarcastic (2) Fit : Honest
(3) Inquiry : Discreet (4) Hint : Indirect

91. Arena : Conflict
(1) Mirage : Reality (2) Forum : Discussion
(3) Asylum : Pursuit (4) Utopia : Place

92. Hierarchy : Ranked
(1) Equation : Solved
(2) Critique : Biased
(3) Chronology : Sequential
(4) Infinity : Fixed

Directions (Q. 93-96) *Read the following passage and answer the next 4 questions.*

Come with me to Kiebera : the largest shantytown in sub-Saharan Africa. More than 500,000 people live in this vast illegal section of Nairobi, in mud huts on mud streets, with no fresh water or sanitation. Walk down Kiebera's sodden pathways and you'll see a great deal of hunger, poverty and disease. But you'll also find health clinics, beauty salons, grocery stores, bars, restaurants, tailors, clothiers, churches and schools. In the midst of squalor and open sewage, business is booming.

Indeed, Kiebera's underground economy is so vibrant that it has produced its own squatter millionaire, someone I have known for years. From his start a generation ago selling cigarettes and biscuits from the window of his hut, this Kenyan (he asked to remain unnamed) has assembled an empire that includes pharmacies, groceries, bars, beverage-distribution outlets, transportation and manufacturing firms and even real estate.

Families flock to Kiebera for the same reason country folk have always migrated to the city-in search of oppurtunity. In the city they find work but not a place to live. So they build illegally on land they don't own. There are a billion squatters in the world today, almost one in six people on the planet. And their numbers are on the rise. Current projections are that by 2030 there will be two billion squatters and by 2050, three billion, better than one in three people on the planet.

In itself, it is nothing to worry about, for squatting has long had a positive role in urban development. Many urban neighbourhoods in Europe and North America began as squatter outposts. London and Paris boasted huge swaths of mud and stick homes, even during the glory years of the British and French monarchies. Squatters were a significant force in most U.S. cities too. It would no doubt surprise residents paying millions for coop apartments on Manhattens' Upper East and West Sides to know that squatters occupied much of the turf under their buildings until the start of the 20th century..... from an article by Robert Neuwirth.

93. The author argues that Kiebera becoming the shantytown is not unusual because
(1) Kiebera has many poor people who have come to earn but have no land to live on
(2) Researchers have predicted that squatters will continue to grow in numbers
(3) Squatting has long had a positive role in urban development
(4) All of the above

94. The prosperity of Kierbera's underground economy is described by the author through
(1) The description of Kiebera
(2) The description of his friends businesses
(3) The comparison with coop apartments of Manhatten
(4) The history of London and Paris

95. The author puts forward the thesis that
(1) Squatters will continue to rise in numbers in the coming years irrespective of whether they are from poor countries or not
(2) There is nothing wrong in squatting on the land of a stranger
(3) London and Paris too are shantytowns
(4) Even today squatters live under the Manhattan's coop apartments

96. What is the most appropriate title for this passage?
(1) Kiebera-Squatters' Paradise of Nairobi
(2) Squatters of the World
(3) Squatter Cities
(4) Future of Squatters

Directions (Q. 97-99) *Choose the correct alternative to complete the meaning of the given sentence.*

97. The _____ managed to deceive the entire village.
(1) renegade (2) sycophant
(3) charlatan (4) actor

98. She ordered the taxi driver, 'Drive faster, _____ ?'
(1) won't you (2) will you
(3) you must (4) can't you

99. Her written statements failed to be consistent _____ what she had said earlier
(1) on (2) with
(3) in (4) to

100. Choose the antonym nearest in meaning to the word. Facetious
(1) serious (2) uneasy
(3) pleasant (4) cross

101. Which one of the following alternatives is spelt correctly?
(1) extacy (2) ecstasy
(3) ecstacy (4) extasy

102. Choose the correct alternative that best explains the following idiom. Writing on the wall.
(1) graffiti (2) obvious truth
(3) foreboding (4) prediction

103. Choose the correct set of alternatives to fill in the blanks. Although many of the members were ___ about the impending deal, others were ___ about the benefits it would bring.
(1) euphoric _____ confident
(2) optimistic _____ dubious
(3) angry _____ skeptical
(4) confused _____ pleased

104. Select the lettered pair that best expresses a relationship similar to that expressed by the original pair.
BROOK : RIVER
(1) vein artery (2) path highway
(3) yard alley (4) pen paper

105. 'But for cancer I would not have given up smoking'. 'But' in this sentence is
(1) an adverb (2) a preposition
(3) an adjective (4) a verb

106. Here is my list oranges, potatoes, garbage bags and a tooth brush.
After the word 'list' identify which one of the following is required.
(1) colon (2) quotation marks
(3) semicolon (4) None of these

107. Identify the grammatical error in the sentence below by choosing one of the alternatives.
My main reason for learning pharmacy was that my brother was one.
(1) dangling modifier
(2) faulty parallelism
(3) faulty reference by pronoun
(4) the sentence is correct

108. For the following pair a sentences choose the correct option
I. The team quickly took their positions on the field.
II. The team quickly took its position on the field.
(1) the first sentence is wrong
(2) the second sentence is wrong
(3) both are correct
(4) both are wrong

Directions (Q. 109-110) *Choose erroneous underlined segment or option (4) if no error.*

109. He carried his clothes in a black heavy steel trunk.
(1) (2) (3) (4)

110. The corpse had been dead for five days.
(1) (2) (3) (4)

111. Identify the odd one.
(1) Ashoka was one of the greatest kings.
(2) Ashoka was greater than many other kings.
(3) Ashoka was the greatest king.
(4) Very few kings were as great as Ashoka.

112. Identify the incorrect one.
(1) The coach together with his team was praised.
(2) Many a boy is tempted to sing.
(3) The king with all his sons were imprisoned.
(4) Neither James nor his lawyers were there.

113. Identify the sentence that gives the same meaning as the following. He said, "Yes, I'll come and see you."
(1) He accepted that he will come and see me.
(2) He said that he will come and see me.
(3) He agreed that he will come and see me.
(4) He said that he would come and see me.

Directions (Q. 114-117) *Choose one option to fill in the blanks in the following sentences.*

(1) a (2) an
(3) then (4) none

114. I had met him _____ year ago.

115. Can you see _____ moon?

116. He is _____ honourable man.

117. _____ people with little patience rarely succeed.

118. Choose the correct arrangement of the following jumbled sentence of a paragraph to make it coherent. The first sentence is
Barely a year had elapsed before the Pritzker clan began to squabble.
L. Under the plan he has until 2011 to distribute the assets among the heirs.
M. The family was no longer cohesive whole, they wrote, and therefore the business needed the kind of transparency a public corporation might have.
N. A year later the family agreed on a giverning structure for the Pritzker Organisation, requiring Tom to open the books, hold annual meetings of family share holders and issue regular financial reports.
O. In summer 2000, Tom's two brothers and a handful of his cousins sent a letter asking him to restructure the holdings.
(1) NMOL (2) MLON
(3) OMNL (4) ONML

119. "Time to bust some myths about the EPF Organisation. India's only social security fund manager for non-government workers, though what follows is not published on a regular basis. Active membership is just about 5 per cent and only 17 per cent of the members

account for 84 per cent of the balances. That's only Rs 20000 each! Less than 7 per cent have a deposit amount more than 5 lacs!" Choose the statement closest to the idea expressed in this paragraph.
(1) EPF is an efficiently managed organisation about which no one knows correctly.
(2) EPF is an inefficient organisation.
(3) EPF Organisation takes care of future fund requirements of investors adequately.
(4) EPF Organisation does, not have enough funds to take care of secure future.

120. "Two recent World Bank studies on India' rapidly depleting water resources have caused quite a stir. More interesting is how water seems to have become the new focus area for Bank assistance : at $3.2 billion in 2005-08 from a mere $700 million in 1999-04. Within water again, more money is going to rural water, large hydropower projects and water resource management in poorer states."
Choose the statement that summarises the above paragraph best
(1) India's water resources are depleting.
(2) The two World Bank studies on India have caused a stir.
(3) The World Bank assistance to India for developing water resources has increased more than 4 times for 2005-08 as compared to the prior period.
(4) Poorer states of India require water resource management projects such as rural water, large hydropower projects.

121. The International Date Line is located
(1) On the Equator
(2) Along 0 degree Longitude
(3) Along 180 degree Longitude
(4) At Greenwich in UK

122. The Roman Magsaysay award is named after the former President of
(1) Thailand (2) Philippines
(3) Indonesia (4) None of these

123. What level of noise is considered permissible in human habitats?
(1) Upto 50 decibels
(2) Less than 40 decibels
(3) Upto 30 decibels
(4) Less than 20 decibels

124. The Indian city which has a natural harbour and is also one of the largest Indian ports is
(1) Mumbai (2) Kolkata-Haldia
(3) Cochin (4) Vishakhapatnam

125. If saccharine, an artificial sweetener which is 70 times sweeter than sugar, is kept in the open, which one of these insects will it atlract first
(1) Ants (2) Bees
(3) House-flies (4) None of these

126. When a Pepsi bottle is opened, the gas fizzes out because it obeys
(1) Hess's Law (2) Henry's Law
(3) Kohlarsh's Law (4) None of these

127. Television signals from transmission towers are restricted to a limited area because
(1) the signals become weak with distance
(2) there is inteference from other signals
(3) of the Earth's curvature
(4) the atmosphere absorbs the signals

128. Who is the Author of "Food, Nutrition and Poverty in India"?
(1) V K R V Rao (2) T S Eliot
(3) Mark Twain (4) Evelyn Wauh

129. Internet is controlled by
(1) The USA (2) The UK
(3) Switzerland (4) None of these

130. Identify the incorrect statement
(1) A tsunami is a series of waves generated when water in a lake or the sea is rapidly displaced on a massive scale.
(2) A tsunami is an after-effect of an earthquake on the seabed.
(3) A tsunami has smaller amplitude (wave height) offshore and a very long wavelength.
(4) None of the above

131. The largest chunk of revenue for the Union Government comes from
(1) Income Tax (2) Central Excise
(3) Corporation Tax (4) Customs Duty

132. India test-fired three medium range surface-to-air missiles in October 2005. Name the missile
(1) Trishul (2) Nag
(3) Dhanush (4) Akash

133. How many calories are there in one litre of water?
(1) 1000 calories (2) 100 calories
(3) I calorie (4) None of these

134. The city closest to the epicenter of the devastating earthquake that hit Pakistan and India in October 2005 is
(1) Srinagar (2) Balakot
(3) Muzaffarabad (4) Islamabad

135. The CDMA mobile services from BSNL is
(1) Tarang (2) Cell One
(3) Idea (4) Orange

136. The highest ranking country for 2005-2006 in "Global Competitiveness Report" of the World Economic Forum is
(1) Finland (2) USA
(3) Singapore (4) UK

137. Which two independent nations does the 17th Parallel separate?
(1) North and South Korea
(2) North and South Vietnam
(3) East and West Germany
(4) Mexico and Panama

138. In the "One by Six" rule of Income Tax, it is mandatory to file Income Tax Returns if you possess
(1) Television (2) Gold
(3) Credit card (4) Passport

139. The Kyoto Protocol pertains to
(1) banning the hunting of whales
(2) reducing greenhouse gas emissions
(3) securing the release of circus animals
(4) disallowing civilian airports for military use

140. The riots spreading across France in 2005 were caused by
(1) Al Qaeda

(2) French Nationalists
(3) Racial tensions among Paris' immigrants
(4) France's position in European Union

141. One of the UK's leading telecom company and sponsor of the England cricket team is
(1) British Telecom (2) Airtel
(3) AT&T (4) Vodafone

142. The price of any currency in the international market is determined by
(1) The World Bank
(2) The demand for goods and services provided by the country
(3) The amount of gold that country has in reserve
(4) The economic stability of that country

143. The "Whistle Blower Bill" was passed due to the pressure created by the murder of
(1) Madhumita Shukla (2) Satyadev Dubey
(3) Naina Sharma (4) None of these

144. The alleged corruption scam of UN's $64 billion in Iraq disclosed in Nov., 2005 is related to
(1) Iraq war crimes
(2) Iraqi WMD
(3) Saddam Hussein
(4) Food-for-oil programme.

145. The Reserve Bank of India Governor is
(1) Bimal Jain (2) YV Reddy
(3) Deepal Parekh (4) Naresh Chandra

146. Which one is listed in the first place among the largest global corporations in 2005 by the Fortune magazine?
(1) Exxon mobile (2) Toyota motors
(3) General motors (4) Walmart

147. Recently VAT was introduced in India. Choose the correct statement.
(1) VAT has replaced Sale Tax in all states of India.
(2) VAT is charged in place of Sales Tax only in some states of India.
(3) VAT is charged in addition to Sales Tax in all states of India
(4) VAT is charged in addition to Sales Tax in some states of India.

148. Parachute is brand of
(1) Hindustan Lever (2) Proctor and Gamble
(3) Marico (4) Dabur

149. Tamiflu is
(1) A type of influenza originating in Tamil Nadu
(2) A contagious disease of animals
(3) A form of Birdflu
(4) An effective medicine for Birdflu

150. Earlier this year, President Bush revoked a trade embargo against a country that had been declared a terrorist state earlier. This allows US companies to invest in that country for the first time since 1986. Which is the country ?
(1) Egypt (2) Sudan
(3) Nigeria (4) Libya

Answers

1. (2)	**2.** (4)	**3.** (2)	**4.** (1)	**5.** (2)	**6.** (1)	**7.** (3)	**8.** (2)	**9.** (1)	**10.** (4)
11. (2)	**12.** (1)	**13.** (1)	**14.** (3)	**15.** (3)	**16.** (2)	**17.** (2)	**18.** (2)	**19.** (3)	**20.** (2)
21. (2)	**22.** (2)	**23.** (3)	**24.** (1)	**25.** (2)	**26.** (2)	**27.** (3)	**28.** (2)	**29.** (1)	**30.** (3)
31. (2)	**32.** (1)	**33.** (1)	**34.** (1)	**35.** (2)	**36.** (3)	**37.** (3)	**38.** (1)	**39.** (1)	**40.** (2)
41. (3)	**42.** (2)	**43.** (2)	**44.** (4)	**45.** (1)	**46.** (2)	**47.** (3)	**48.** (2)	**49.** (2)	**50.** (1)
51. (4)	**52.** (4)	**53.** (3)	**54.** (3)	**55.** (3)	**56.** (3)	**57.** (3)	**58.** (3)	**59.** (2)	**60.** (4)
61. (2)	**62.** (3)	**63.** (1)	**64.** (1)	**65.** (2)	**66.** (3)	**67.** (4)	**68.** (3)	**69.** (4)	**70.** (1)
71. (2)	**72.** (3)	**73.** (2)	**74.** (4)	**75.** (1)	**76.** (1)	**77.** (4)	**78.** (4)	**79.** (3)	**80.** (2)
81. (2)	**82.** (2)	**83.** (2)	**84.** (1)	**85.** (2)	**86.** (4)	**87.** (3)	**88.** (4)	**89.** (1)	**90.** (1)
91. (2)	**92.** (3)	**93.** (4)	**94.** (2)	**95.** (1)	**96.** (1)	**97.** (3)	**98.** (4)	**99.** (2)	**100.** (1)
101. (2)	**102.** (2)	**103.** (2)	**104.** (2)	**105.** (2)	**106.** (2)	**107.** (3)	**108.** (1)	**109.** (2)	**110.** (2)
111. (3)	**112.** (4)	**113.** (4)	**114.** (1)	**115.** (3)	**116.** (2)	**117.** (4)	**118.** (4)	**119.** (2)	**120.** (4)
121. (3)	**122.** (2)	**123.** (1)	**124.** (1)	**125.** (3)	**126.** (4)	**127.** (3)	**128.** (1)	**129.** (4)	**130.** (4)
131. (2)	**132.** (4)	**133.** (4)	**134.** (3)	**135.** (1)	**136.** (3)	**137.** (2)	**138.** (3)	**139.** (2)	**140.** (3)
141. (4)	**142.** (4)	**143.** (2)	**144.** (4)	**145.** (2)	**146.** (3)	**147.** (2)	**148.** (3)	**149.** (4)	**150.** (4)

Answers with Explanations

1. (2) Let the CP of milk be Re 1 per litre.

Milk in 1 L mixture of $A = \frac{4}{7}$ L

Milk in 1 L mixture of $B = \frac{2}{5}$ L

Milk in 1 L mixture of $C = \frac{1}{2}$ L

$\therefore$ CP of 1 L mixture in $A = \text{Re}\frac{4}{7}$

CP of 1 L mixture in $B = \text{Re}\,\frac{2}{5}$

Mean price = $\text{Re}\,\frac{1}{2}$

By the rule of alligation, we have

CP of 1 L mixture in A CP of 1 L mixture in B

$\left(\frac{4}{7}\right)$ $\left(\frac{2}{5}\right)$

$\left(\frac{1}{2}\right)$

$\left(\frac{1}{2}-\frac{2}{5}=\frac{5-4}{10}=\frac{1}{10}\right)$ $\left(\frac{4}{7}-\frac{1}{2}=\frac{8-7}{14}=\frac{1}{14}\right)$

$\therefore$ Required ratio $= \frac{1}{10}:\frac{1}{14}=7:5$

2. (4) According to question,

$$P+75=(P-100)2$$
$$\Rightarrow \quad P+75=2P-200$$
$$\Rightarrow \quad 2P-P=200+75$$
$$\therefore \quad P=\text{Rs } 275$$

3. (2) The closest time when the hands of the watch will be exactly opposite each other *ie*, in a straight line.

The time will be sometimes after 7 : 05.

The time will be shown by

$$5\times\frac{60}{55}=5\frac{5}{11}\text{ min}$$

4. (1) Probability of selecting two seats in the front row and one seat behind

$$=\frac{{}^4C_2\times{}^2C_1}{{}^6C_3}=\frac{12}{20}=0.6$$

5. (2) Let the number be x.

Now, according to question,

$$x=75\%\text{ of }x+75$$
$$\Rightarrow \quad x=\frac{75}{100}x+75$$
$$\Rightarrow \quad x-\frac{75}{100}x=75$$
$$\Rightarrow \quad \frac{100x-75x}{100}=75$$
$$\Rightarrow \quad 25x=75\times100$$
$$\therefore \quad x=300$$

6. (1) Let the distance travelled by

Anne by water = x miles

According to question,

$$x+\frac{4}{7}x+\frac{2}{5}x=3036$$
$$\Rightarrow \quad \frac{35x+20x+14x}{35}=3036$$
$$\Rightarrow \quad 69x=3036\times35$$
$$\therefore \quad x=1540\text{ miles}$$

Now, $\frac{4}{7}x=\frac{4}{7}\times1540$

$= 880$ miles on foot.

$$\frac{2}{5}x=\frac{2}{5}\times1540$$

$= 616$ miles on horse back.

7. (3) These type of questions can be easily solved through the given options.

Option (3) satisfies both the conditions.

Each boy has equal number of brothers and sisters. Each girl has twice as many brothers as the number of sisters.

8. (2) Probability of picking a white marble blind folded is $\frac{1}{2}$ in each of the given options except option (2).

The probability in option (2) is, $\frac{1}{2}\times1+\frac{1}{2}\times\frac{49}{99}$, which is greater than half.

9. (1) Consider the given options.

In option number (1), 169 is a perfect square of 13. When the number 169 is turned upside down it becomes 961 which is also a perfect square of 31.

So, it satisfies the given condition.

No other option satisfy the given conditions except option number (1).

So, option (1) is the answer.

10. (4) According to question,

5 spiders can catch 5 flies in 5 min.

So, 1 spider can catch 1 fly in 5 min.

$\therefore$ 100 spiders can catch 100 flies in 5 min.

$\therefore$ The number of flies caught by 100 spiders in 100 min is x.

$$\Rightarrow \quad \frac{100\times100}{5}=\frac{100\times x}{100}$$
$$\therefore \quad x=100\times20$$
$$=2000\text{ flies}$$

11. (2) According to question,

My brother won 7 games.

So, he will have 7 chocolates.

Now, if have 20 additional chocolates than my brother.

So, I have $20+7=27$ chocolates.

Thus, I won 27 games.

$\therefore$ Total number of games $= 27+7=34$

12. (1) According to question, the letters are DJYS.

It is not possible to make a meaningful word from these letters. So, 'X' is the the answer.

13. (1) $5^{1/4}\times(125)^{0.25}=5^{1/4}\times(5^3)^{1/4}$

$$=5^{1/4}\times5^{3/4}$$
$$=5^{\left(\frac{1}{4}+\frac{3}{4}\right)}=5^1=5$$

14. (3) In these type of questions, the liquid content in one container *ie*, water is always equal to the liquid content in another container *ie*, wine.

15. (3) According to question,

$$A+B+C+D=47 \quad \ldots(\text{i})$$
$$A+B=27 \quad \ldots(\text{ii})$$
$$C+A=25 \quad \ldots(\text{iii})$$
$$D+A=23 \quad \ldots(\text{iv})$$

Adding Eqs. (ii), (iii) and (iv), we get

$$3A + B + C + D = 75 \quad \ldots(v)$$

Subtracting Eq. (i) from Eq. (v), we get

$$\begin{array}{r} 3A + B + C + D = 75 \\ -A + B + C + D = -47 \\ \hline 2A \qquad\quad = 28 \end{array}$$

$$A = 14$$

Putting the value of A in Eq. (ii), we get

$$A + B = 27$$

$$\Rightarrow \quad 14 + B = 27$$

$$\therefore \quad B = 27 - 14 = 13$$

So, Binoy has Rs 13.

16. (2) Relative velocity of the elevators

$$= 57 + 63 = 120 \text{ floors/min}$$

Distance to be covered

$$= 51 - 11 = 40 \text{ floors}$$

$$\text{Time required} = \frac{40}{120}$$

$\therefore$ Number of floors travelled by David

$$= \frac{40}{120} \times 57 = 19 \text{ floors}$$

$\therefore$ The elevators will meet at,

$$11 + 19 = 30\text{th floor}$$

17. (2) Let the five consecutive numbers be

$x, x+1, x+2, x+3$ and $x+4$

$$\text{The average of these numbers} = \frac{5x+10}{5} = x + 2$$

Now, average of seven consecutive number

$$= \frac{x + (x+1) + (x+2) + (x+3) + (x+4) + (x+5) + (x+6)}{7}$$

$$= \frac{7x + 21}{7} = x + 3$$

So, the average increases by 1.

18. (2) The given series 52, 51, 48, 43, 34, 27, 16

Difference, $52 - 51 = 1$

$51 - 48 = 3$

$48 - 43 = 5$

$43 - 34 = 9 \quad \Rightarrow \quad 43 - 36 = 7$

$34 - 27 = 7 \quad \Rightarrow \quad 36 - 27 = 9$

$27 - 16 = 1$

The decrease in numbers is made by consecutive odd numbers. So, 36 will come in place of 34.

19. (3) $P = 6Q$

So, Q is less than P by $5Q$.

$$\therefore \text{ Required percentage} = \left(\frac{5Q}{P} \times 100\right)\%$$

$$= \left(\frac{5Q}{6Q} \times 100\right)\%$$

$$= 83\frac{1}{3}\%$$

20. (2) LCM of 32, 40 and $72 = 1440$

Remainders, $32 - 10 = 22$

$40 - 18 = 22$

$72 - 50 = 22$

$\therefore$ The minimum number of pebbles that the heap contains $= 1440 - 22 = 1418$

21. (2) $S_1 = 2 + 4 + 6 + \ldots\ldots\ldots + 200$

$S_2 = 3 + 6 + 9 + \ldots\ldots\ldots + 300$

So, in S_2, 6 and the multiples of 6 upto 200 will be identical to the terms in S_1.

Identical terms $= 6 + 12 + 18 \ldots\ldots + 198$

$$\therefore \quad \text{Number of terms} = \frac{198}{6} = 33$$

22. (2) By hit and trial method, in option (2) we find that if he buys 94 chickens, 2 horse and 5 cows, he meets the target of 100 animals and spends Rs 10000 also.

Number of animals,

94 chickens + 1 horse + 5 cows = 100 animals

Money spent,

$$94 \times 50 + 1 \times 300 + 5 \times 1000$$

$$= 4700 + 300 + 5000 = \text{Rs } 10000$$

23. (3) By hit and trial method, we see that option (3) gives the correct answer 31.

$$1 - 2^3 - 3^3 + 1 + 4^3 = 31$$

or $\quad 1 - 8 - 27 + 1 + 64 = 31$

$$\therefore \quad 31 = 31$$

24. (1) By hit and trial method, we see that Rs 38950 is exactly divisible by Rs 410 only *ie*, option (1). So, Rs 410 is the participant fee of the programme.

25. (2) In this series, a number is the product of previous two numbers.

$$1 \times 6 = 6$$

$$6 \times 6 = 36$$

$$6 \times 36 = 216$$

$$\therefore \quad 36 \times 216 = 7776$$

26. (2) Total Bill = Tanya + Veena + Amita

So, $\quad T + T \times \frac{1}{2} + T \times \frac{1}{2} \times \frac{2}{3} = 1$

$$\Rightarrow \quad T + \frac{T}{2} + \frac{T}{3} = 1$$

$$\Rightarrow \quad \frac{6T + 3T + 2T}{6} = 1$$

$$\Rightarrow \quad 11T = 6$$

$$\therefore \quad T = \frac{6}{11}$$

So, Tanya paid $\frac{6}{11}$ part of the bill.

$$\therefore \text{ Part of Bill paid by Veena} = T \times \frac{1}{2}$$

$$= \frac{6}{11} \times \frac{1}{2} = \frac{3}{11} \text{ part}$$

$$\text{and part of Bill paid by Amita} = T \times \frac{1}{2} \times \frac{2}{3}$$

$$= \frac{6}{11} \times \frac{1}{2} \times \frac{2}{3} = \frac{2}{11} \text{ part}$$

27. (3) As, this is a men's singles tournament, so two players will play a match. Therefore, the number of matches will be exactly half of the total number of players in each round.

So, 1st round : 128 players = 64 matches

2nd round : 64 players = 32 matches

3rd round : 32 players = 16 matches

4th round : 16 players = 8 matches

5th round : 8 players = 4 matches

6th round : 4 players = 2 matches

7th round : 2 players = 1 match

∴ Total number of matches played during the whole event

$$= 64 + 32 + 16 + 8 + 4 + 2 = 1$$
$$= 127 \text{ matches}$$

28. (2) Speed downstream = 10 miles in $\frac{5}{6}$th of an hour

$$= 12 \text{ miles/h}$$

Speed upstream = 12 miles in 90 min

$$= 8 \text{ miles/h}$$

$$\text{Rate of current} = \frac{12+8}{2}$$
$$= 10 \text{ miles/h}$$

$$\text{Crew's rate in still water} = \frac{12-8}{2}$$
$$= 2 \text{ miles/h}$$

29. (1) According to question,
We know that one of Swamy's children is a girl.
So, probability of both of Swamy's children being girls $= \frac{1}{2}$

30. (3) Let the distance of the school from the house be x km.

Different in timings = 6 + 6 = 12 min

$$= \frac{12}{60} \text{ h} = \frac{1}{5} \text{ h}$$

$$\therefore \quad \frac{2x}{5} - \frac{2x}{7} = \frac{1}{5}$$

$$\Rightarrow \quad \frac{14x - 10x}{35} = \frac{1}{5}$$

$$\Rightarrow \quad 14x - 10x = 7$$

$$\Rightarrow \quad x = \frac{7}{4} = 1\frac{3}{4} \text{ km}$$

31. (2) The sequence consists of those years that are prime number. So, 2003 is the answer as it is the next prime number.

32. (1) Let the number of students studying Physics, Chemistry and Zoology be $4x$, $3x$ and $5x$ respectively.
Increased number in these three disciplines are (150% of $4x$), (125% of $3x$) and (110% of $5x$)

ie, $\left(\frac{150}{100} \times 4x\right), \left(\frac{125}{100} \times 3x\right)$ and $\left(\frac{110}{100} \times 5x\right)$

ie, $6x, \frac{15}{4}x$ and $\frac{11}{2}x$

$$\therefore \text{ New ratio} = 6x : \frac{15}{4}x : \frac{11}{2}x$$
$$= 24x : 15x : 22x$$
$$= 24 : 15 : 22$$

33. (1) According to question,
a man earns Rs 20 on the first day and spends Rs 15 on the next day.
So, he earns Rs 5 in two days.
∴ he will earn Rs 5×8 = Rs 40 or $2 \times 8 = 16$th day
On the 17th day be will earn Rs 20 more.
So, he will have Rs 40 + Rs 20 = Rs 60 in hand on the 17th day.

34. (1) According to question,
Four substitutes alternate for equal time
The game lasts for 90 min

$$\therefore \text{ Duration per player} = \frac{90}{4} = 22.5 \text{ min}$$

35. (2) Ratio of the capital invested by A, B and C

$$= 20000 \times 5 + 15000 \times 7 : 20000 \times 5 + 16000 \times 7 : 20000 \times 5 + 26000 \times 7$$
$$= 205 : 212 : 282$$

$$\therefore B\text{'s share in total profit} = 69900 \times \frac{212}{699} = \text{Rs } 21200$$

36. (3) Growth rate per thousand = 32 – 11 = 21
Growth rate in terms of population increase in per cent

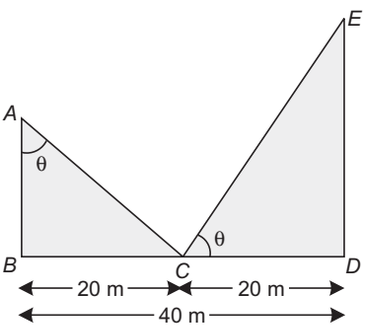
$$= \left(\frac{21}{1000} \times 100\right)\% = 2.1\%$$

37. (3) Since, the difference between two numbers is 1365.
Let the number be x and $(x + 1365)$.

Then, $x + 1365 = 6x + 15$

$$\Rightarrow \quad 6x - x = 1365 - 15$$
$$\Rightarrow \quad 5x = 1350$$
$$\therefore \quad x = 270$$

38. (1)



In ΔABC, $\tan\theta = \frac{20}{AB}$

In ΔCDE, $\tan\theta = \frac{2AB}{20}$

$$\Rightarrow \quad \tan\theta = \frac{2AB}{20}$$

$$\therefore \quad \frac{20}{AB} = \frac{2AB}{20}$$

$$\Rightarrow \quad 2AB^2 = 400$$

$$\Rightarrow \quad AB^2 = 200$$

$\therefore \quad AB = \sqrt{200}$

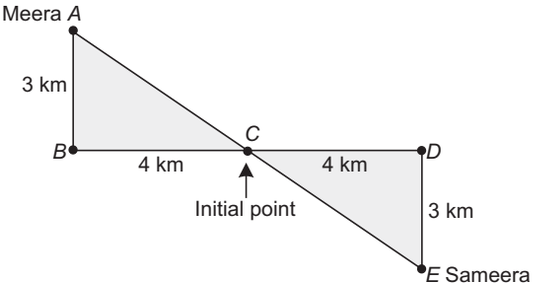
$= 10\sqrt{2} = 14.14$ m

$\therefore \quad 2AB = 2 \times 14.14$

$= 28.28$ m

Height of the two vertical poles are 14.14 m and 28.28 m.

39. (1)



The shortest distance between Meera and Sameera is

$AE = AC + CE$

$= \sqrt{AB^2 + BC^2} + \sqrt{DE^2 + CD^2}$

$= \sqrt{3^2 + 4^2} + \sqrt{3^2 + 4^2}$

$= \sqrt{9 + 16} + \sqrt{9 + 16}$

$= \sqrt{25} + \sqrt{25} = 5 + 5$

$= 10$ km

40. (2) Let the number be x.

According to question,

$53x - 35x = 1206$

$\Rightarrow \quad 18x = 1206$

$\therefore \quad x = 67$

41. (3) Option (1)= 31 + 10 + 19 + 6 = 66%

Option (2) = 31 + 10 + 19 + 8 + 6 = 74%

Option (3) = 31 + 21 + 10 + 19 = 81%

Option (4) = 31 + 21 + 19 + 8 = 79%

42. (2) Total number of sectors = 7

Total profit against the sales across the sectors

= Average profit × 7

= 10 × 7 = 70%

Profit margin of engineering and CIS

= 12 + 20 = 32%

Average profit across the remaining sectors

$= \frac{70 - 32}{5} = \frac{38}{5} = 7.6\%$

43. (2) The temperature on Thursday can be obtained either statement I or II.

From Statement I,

M + Tu + W = 111

Tu + W + Tu = 102

M − Th = 9 …(i)

$\text{Th} = \frac{4}{5}\text{M}$ …(ii)

Putting the value of Th in Eq. (i)

$\text{M} - \frac{4}{5}\text{M} = 9$

$\Rightarrow \quad \frac{5\text{M} - 4\text{M}}{5} = 9$

$\therefore \quad \text{M} = 45$

So, $\text{Th} = \frac{4}{5} \times 45 = 36$

From Statement II,

M − Th = 9 [from Eq. (i)]

M + Th = 81 …(iii)

2M = 90

M = 45

Putting the value of M in Eq. (iii)

M + Th = 81

$\therefore \quad$ Th = 81 − 45 = 36

Statement III is not sufficient to answer the question because it tells us the difference of temperature on Monday and Thursday but it not clear from the given statement as to which of the day's temperature is greater.

44. (4) Let the numbers be x and y.

$x + y = ?$

Statement I Let x is the greater number

$x - y = 6$

Statement II 40% of y = 30% of x

Statement III $\frac{1}{2}x : \frac{1}{3}y = 2 : 1$

Any two statements will give equations which will be sufficient to solve the question.

45. (1) Time taken to fill the tank fitted with taps A and B.

Statement I Tells the relative efficiency of both the taps.

Statement II Time taken by A to fill the tank.

Statement III Time taken by B to fill the tank.

So, statement I and either II or III only can tell the time taken by A and B taps to fill the tank. Therefore, option (1) is the best answer.

46. (2) We have to determine the principal.

Statement I Interest amount after $2\frac{1}{2}$ yr

$= \frac{1}{2}$ (Interest amount after 5 yr)

Statement II $A = P + I$

$A = P + \frac{PRT}{100}$

$750 = P + \frac{P \times R \times 5}{100}$

Statement III Rate % = 8% per annum

The percentage rate given in statement III when put in statement II answers the question.

So, II and III only is the answer.

47. (3) Average share price $= \frac{483 + 475 + 461 + 513}{4} = 483$

48. (2) Price on last day of September = 535

Price on last day of October = 461

$\therefore$ Drop in sales $= \frac{74 \times 100}{535} = 13.83\%$

49. (2) It is evident from the graph itself.

50. (1) Price of 100 shares on 31/8/2005 = 461×100
= Rs 46100
Price of 100 shares on 10/10/2005
= 538×100 = Rs 53800
$\therefore$ Gain = 53800 – 46100 = Rs 7700

51. (4) According to question,
$A + B + C + D = 96$ yr
B's present age = ?
Statement I $A + B + D = 20 \times 3 = 60$ yr
So, $C = 96 - 60 = 36$ yr
Statement II $C + D = 25 \times 2 = 50$ yr
From (I), $C = 36$ yr
$D = 50 - 36 = 14$ yr
From the data given in both statements the age of C, D and $(A + B)$ can only be determined.
So, both statements together are not sufficient to answer the question.

52. (4) According to question,
Deepak's marks in Hindi
$= \left(\frac{\text{Hindi + Economics + Sociology + Philosophy}}{4}\right) + 15$
Deepak's marks in Philosophy = ?
Statement I Hindi + Philosophy = 112
Statement II Sociology – Economics = 12
So, both statements together are not sufficient to determine Deepak's marks in Philosophy.

53. (3) According to question,
CP of suitcase purchased by Richard = ?
Statement I Richard got 20% concession on the Labelled price.
Statement II SP of suitcase = Rs 2000
He sold the suitcase with 25% profit on labelled price.
Labelled price $\Rightarrow 125\% = 2000$
$100\% = \frac{2000 \times 100}{125} = 1600$
Rs 1600 is the labelled price.
So, Richard's cost price after 20% concession
= 80% of 1600
$= \frac{80}{100} \times 1600$ = Rs 1280
So, the data in both the statement's is necessary to answer the question.

54. (3) According to question,
B's 1 day's work $= \frac{1}{12}$
Time taken by $(A + B + C)$ to complete the work = ?
Statement I $(A + B)$'s 1 day's work $= \frac{1}{3}$
A's 1 day's work $= \frac{1}{3} - \frac{1}{12}$
$= \frac{4-1}{12} = \frac{3}{12} = \frac{1}{4}$
Statement II $(B + C)$'s 1 day's work $= \frac{1}{6}$
C's 1 day's work $= \frac{1}{6} - \frac{1}{12} = \frac{2-1}{12} = \frac{1}{12}$
$\therefore$ $(A + B + C)$'s 1 day's work $= \frac{1}{4} + \frac{1}{12} + \frac{1}{12}$
$= \frac{3+1+1}{12} = \frac{5}{12}$
$(A + B + C)$ will complete the work in $\frac{12}{5}$ days.
So, the data in both the statement's is necessary to answer the question.

55. (3) Marine products was the single largest contributor with 23.2%.
23.2% of $ 6 billion = 1.392 billion

56. (3) It is roughly evident from the chart that processed fruits and vegetables have shown strong growth in exports during the 3 yr period.
By calculation,
From 69 to 86, from 1998–99 to 1999–2000
$69 \rightarrow 17$
$100 \rightarrow \frac{17 \times 100}{69} = 24.64\%$
From 86 to 122, from 1999–2000 to 2000–01
$86 \rightarrow 36$
$100 \rightarrow \frac{36 \times 100}{86} = 41.86\%$

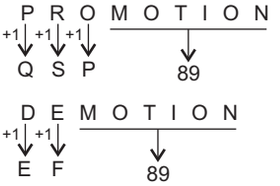
57. (3) Cashow nuts $\rightarrow$ The answer is evident from the table itself.

58. (3) Import of sugar and cereals has decreased yearwise because India's gap in production and requirement of these commodities has gone down.

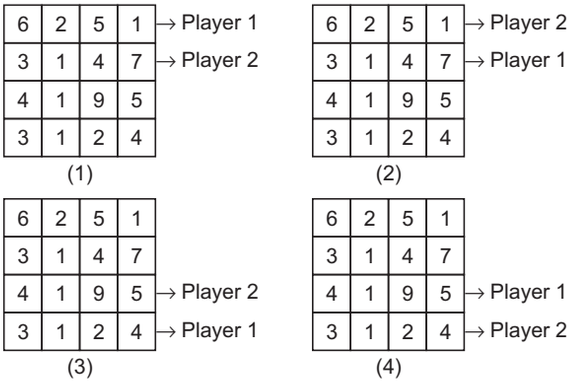
61. (2)

Nalini's brother	Nalini
Nalini's son	Nalini's daughter

62. (3)

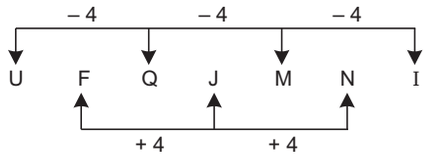


63. (1)



The opponent should retain upper, retain lower to minimize the gain.

64. (1)



So, the next letter is I.

65. (2) Numbers divisible by 6

= 6, 12, 18, 24, 30, 36, 42, 48, 54, 60 = 10

Numbers with 6 as one of the digit

= 6, 16, 26, 36, 46, 56 = 4

Numbers with 6 as the sum of its digits

= 6, 15, 24, 33, 42, 51, 60 = 3

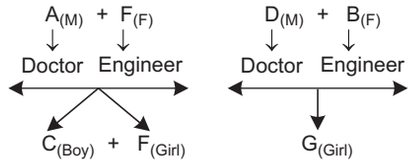
Total of these numbers = 10 + 4 + 3 = 17

∴ Number of integers which are not connected with 6

= 60 – 17 = 43

66. (3) Leena paddled only once on the return trip, when she was alone. Nitin paddled once with Arun and Arun paddled once when he was with somebody else than Nitin. So, only Mohan is the person left. He paddled twice.

67. (4)



G's father is D. F's father is A.

E's daughter is F.

So, C is A's son.

68. (3) The first month of second half of the year is July.

According to question,

Every letter placed at even number

ie, 2, 4, 6, 8, …… in the English alphabet is written in lower case.

j	U	l	Y
↓	↓	↓	↓
10	21	12	25

⇒ jUlY

69. (4) Let the two unequal number of coins be x and y.

Then, according to question,

$$48(x-y) = x^2 - y^2$$

$$48(x-y) = (x+y)(x-y)$$

$$(x-y)(x+y) - 48(x-y) = 0$$

$$(x-y)(x+y-48) = 0$$

But $x - y \neq 0$ (Since, $x \neq y$)

$\therefore$ $x + y - 48 = 0$

$\therefore$ $x + y = 48$

So, the merchant has 48 gold coins.

70. (1) The businessman earns Re 1 on day 1.

Day 1 → Re 1

Day 2 → Rs 2

Day 3 → Rs 2^2

∴ Day 10 → Rs 2^9

71. (2) This question can be solved by hit and trial method.

Solving option (2), we get

$79 - 1 = 78 \Rightarrow 78 \div 3 = 26$

26 apples were taken by the 1st boy.

78 – 26 = 52 apples are left.

$52 - 1 = 51 \Rightarrow 51 \div 3 = 17$

17 apples were taken by the 2nd boy.

51 – 17 = 34 apples are left.

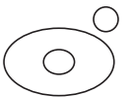
$34 - 1 = 33 \Rightarrow 33 \div 3 = 11$

11 apples were taken by the 3rd boy.

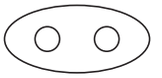
33 – 11 = 22 apples are left.

22 – 1 = 21 which is again divisible by 3.

72. (3) All chilli are vegetables. Salt belongs to a different category.



73. (2) Students of Law and students of Science belong to different category. But both of them are men.



Q. (74–78)

The given numbers are arranged in descending order.

74. (4)

Input :	09	25	16	30	32	18	17	06
Step I :	32	09	25	16	30	18	17	06
Step II :	32	30	09	25	16	18	17	06
Step III :	32	30	25	09	16	18	17	06

75. (1)

Input :	16	09	25	27	06	05
Step I :	27	16	09	25	06	05
Step II :	27	25	16	09	06	05

So, step II is the last step of this Input.

76. (1)

Input :	25	08	35	11	88	67	23
Step I :	88	25	08	35	11	67	23
Step II :	88	67	25	08	35	11	23
Step III :	88	67	35	25	08	11	23
Step IV :	88	67	35	25	23	08	11
Step V :	88	67	35	25	23	11	08

77. (4)

Input :	03	31	43	22	11	09
Step I :	43	03	31	22	11	09
Step II :	43	31	03	22	11	09
Step III :	43	31	22	03	11	09
Step IV :	43	31	22	11	03	09
Step V :	43	31	22	11	09	03

Step V will be the last step for the input below.

78. (4) The previous step or input cannot be determined in these type of questions.

79. (3) A + B ≠ C – D

A is the mother of B.

B is the father of C.

C is the brother of D.

$A_{(F)}$
↓
$B_{(M)}$
↓
$C_{(M)} \leftrightarrow D_{(M/F)}$

So, A is D's Grandmother.

80. (2) By hit and trial method,

A * B # C * D – E

A is the sister of B.

B is the father of C.

C is the sister of D.

D is the brother of E.

$A_{(F)} \longleftrightarrow B_{(M)}$
↓
$C_{(F)} \longleftrightarrow D_{(M)} \longleftrightarrow E_{(M/F)}$

So, A is the aunt of E.

81. (2) The beginning of the second paragraph and the concluding sentence given in the last paragraph of the passage clearly tells us that this passage explains the influence of river settlements on the growth of early civilizations.

82. (2) As the passage tells us that the rivers became the driving force because they produced fertile lands and attracted settlers armed with new techniques of agriculture. The passage also tells us that migration was further encouraged by the transformation of grasslands into desert.
So, option (2) is the answer.

83. (2) This has been mentioned in the last part of fourth paragraph in the given passage.

84. (1) The last part of the second paragraph and the beginning of the third paragraph clearly tells that option (1) is the correct answer.

85. (2) The fifth paragraph of the passage tells us that option (2) is the correct answer.

86. (4) As the passage discusses about the techniques of agriculture and other aspects related to it. So, the emergence of complex civilizations in the Middle East was dependent upon the previous development of basic techniques of agriculture according to passage.

87. (3) The last paragraph of the passage clearly tells that option (3) is the correct answers.

88. (4) Askance means to look at or react to somebody/ something with suspicion or doubt or in a critical way. So, option (1) and (3) cannot be corelated to the meaning. Quizzical expressions shows that a person is slightly surprised or amused. So, option (2) is also not related. Therefore, option (4) is the answer.

89. (1) Virtuoso is a person who is extremely skilful at doing something, especially playing a musical instrument. So he is a skilled performer. Amateur is a person who is not skilled. Profession is a person who does something as a paid job rather than as a hobby.

90. (1) Nuance is a very slight difference in meaning, sound colour or somebody's feeling that is not usually very obvious. Subtle is also not very noticeable or obvious. Pun is the clever or humorous use of a word that has more than one meaning or of words that have different meanings but sound the same. Sarcastic is an expression or way of using words that are opposite of what you mean in order to be unpleasant to somebody.
So, option (1) expresses the same relationship as the original pair.

91. (2) Arena is an area of activity that concerns the public, especially one where there is a lot of opposition or conflict. Forum is a place where discussion takes place and people can exchange opinion or ideas. So, option (2) expresses the same relationship.

92. (3) Hierarchy is a system in which people are ranked into different levels of importance from highest to lowest. Similarly, chronology is the order in which a series of events happen.

93. (4) The third paragraph of the passage refers to option (1) and (2) whereas the fourth paragraph refers to option (3).

94. (2) The first sentence of the second paragraph clearly indicates that option (2) is the correct answers.

95. (1) The third paragraph clearly tells us about the thesis put forward by the author regarding squatters.

96. (1) The major portion of the passage is about the squatters of Kiebera. So, option (2) and (3) cannot be the appropriate title. Only one paragraph talks about the future of squatters but it doesn't talk about their socio-economic condition in future. So, (1) is the most appropriate title for the passage.

97. (3) Charlatan is a person who claims to have knowledge or skills that he doesn't possess. So he deceives others. Renegade is a person who opposes and lives outside a group or society that he used to belong to. Sycophant is a person who praises important or powerful people in a way that is not sincere, to get something from them.

98. (4) Option (2) and (3) are positive options, so they cannot be the answer. The sentence is given in past tense, so option (1) cannot be the answer.

99. (2) 'Consistent with' is the correct use.

100. (1) Facetious means trying to appear amusing and intelligent at a time when other people do not think that it is appropriate and when it would be better to be serious.
So, serious is the correct antonym of the given word.

102. (2) Writing on the wall is an idiom used for an obvious truth.

103. (2) As the sentence beings with although, so it will have words that are contrary to each other. So, (2) is the correct answer because (1) and (3) are not contrary words. In option (4) confused is not exactly contrary to the word pleased.

104. (2) Brook is a small river. Similarly, path is the smaller form of a highway.

105. (2) An adverb modifies the meaning of a verb or adjective or another adverb. An adjective modifies a noun or a pronoun.
So, 'But' in this sentence is a preposition.

106. (2) Quotation mark will be used to list the items together.

107. (3) "For becoming pharmacist" would have been the correct use. "…… my brother was one" is a faulty reference by pronoun.

108. (1) "The team" is a singular term. Hence, the first sentence is wrong and the second sentence is correct.

109. (2) "Heavy black" should be used in place of "black heavy".

110. (2) "Had been dead" is a wrong use for corpse because it implies that it would come alive. The correct use should be "had been lying" or "had been there".

111. (3) Except sentence (3), there is some sort of comparison in all other sentences.

112. (4) As James is a subject, so verb in singular number. The correct sentence is — Neither James nor his lawyers was there.

113. (4) When a sentence is written in indirect speech, the reported speech's verb should be in past tense.

114. (1) "a year" is the correct use, inspite of the first sound of pronunciation of the word 'year' being a vowel.

115. (3) 'The' is used before such common nouns that are names of things and are unique. For Ex.—the sun, the moon, the sky, the earth etc.

116. (2) The word 'honourable' starts with a consonant but its first sound of pronunciation is a vowel. So, an is used.

117. (4) Article is not used before abstract nouns that express qualities, state, feeling, actions.

118. (4) "In summer 2000 ……" is an ideal introductory sentence. "A year later ……" clearly succeeds the matter discussed in sentence '0'. "Plan until 2011 ……" is the concluding sentence.

119. (2) The paragraph clearly expresses the idea that EPF is an inefficient organisation.

120. (4) The above paragraph clearly summarises that more money is going to water resource management projects of poorer states as it is their requirement.

SNAP

Symbiosis National Aptitude Test

Mock Papers

Symbiosis National Aptitude Test

Crack SNAP 1

ENTRANCE PATTERN

The duration for making answers will be 2 hours. Normally the pattern of written test is as follows.

Sections

I. General English
II. Quantitative Aptitude, Data Interpretation & Sufficiency
III. Analytical and Logical Reasoning
IV. General Awareness

Important Instructions

1. *Calculator/Watch calculator will not be allowed in the examination Hall.*
2. *Use only HB Pencils for darkening the ovals.*
3. *Correct method of Marking Answers*

Each questions is followed by four answer choices which are marked a, b, c, d on the answer sheet. Using HB Pencil, darken the oval bearing the correct answer against the serial number of the question.

For example : *If the answer to question number* 2 *is* (4)*, you should mark as shown below*

Q. 2. (1) (2) (3) (4)

General English

Directions (Q. 1-5) *In the following questions, each word in capital letters is followed by four words or phrases. Choose the one which is similiar in meaning to the word.*

1. ACCOUTREMENTS
(1) Relatives (2) Companions
(3) Blemishes (4) Belongings

2. AZURE
(1) Blue (2) Green
(3) Brown (4) Round and big

3. DEFILE
(1) Pollute (2) Disapprove
(3) Delay (4) Reveal

4. HIATUS
(1) Atrocious (2) Gap
(3) Dominance (4) Obscure

5. ZEAL
(1) Initiative (2) Coarseness
(3) Pedagogy (4) Indolence

Directions (Q. 6-10) *In the following questions, words in capital letters are followed by four words. Choose the word that is most nearly opposite in meaning to the given words.*

6. NEBULOUS
(1) Concrete (2) Unkind
(3) Undramatic (4) Uninteresting

7. PENCHANT
(1) Disinclination (2) Lone
(3) Directness (4) Lack of skill

8. PERFUNCTORY
(1) Quick (2) Slow
(3) Careful (4) Loud

9. TRANSIENT
(1) Urgent (2) Youthful
(3) Original (4) Eternal

10. SUBLIME
(1) Base (2) Concise
(3) Partial (4) Insist

Directions (Q. 11-15) *Fill in the blanks with suitable words.*

11. The scholar was so ..... in his field that many of our professors became nervous in his presence.
(1) eminent (2) pathological
(3) petulant (4) amiable

12. If you do not ……, all your monthly expenses would exceed your income.
(1) economise (2) spend
(3) save (4) splurge

13. The cost of this operation has …… our small store of money.
(1) destroyed (2) damaged
(3) depleted (4) affected

14. He had taken the shocking news quietly, neither …… fate nor uttering any word of bitterness.
(1) submitting to (2) railing against
(3) conspiring with (4) dissenting from

15. Criticism that tears down without suggesting areas of improvement is not …… and should be avoided if possible.
(1) constructive (2) mandatory
(3) pertinent (4) sagacious

Directions (Q. 16-18) *The questions given below have an idiomatic expression followed by four options. Choose the option that is closest in meaning to the given idiomatic expressions.*

16. To smell a rat
(1) Signs of plague epidemic
(2) Bad smell
(3) Suspect foul dealings
(4) To be in a bad mood

17. Right-hand man
(1) An honest person
(2) Most efficient assistant
(3) One who cannot use his left hand
(4) A foolish person

18. Bad blood
(1) Infected blood (2) Ill-feeling
(3) Unfaithful (4) Suspicion

Directions (Q. 19-20) *Select the pair of words from the given options that best expresses a relationship SIMILAR to the pair in CAPITAL letters.*

19. Objurgate : Obsecrate
(1) abdicate : abrogate
(2) renegade : relegate
(3) chide : supplicate
(4) obfuscate : obligate

20. Restless : Restive
(1) flammable : inflammable
(2) imminent : eminent
(3) haunted : hunted
(4) oculist : occultist

Directions (Q. 21-22) *Fill in the blanks with the option that has the MOST APPROPRIATE set of words.*

21. Children whose ..... survives parental discipline and who manage to grow up before they blow up are invited to ..... the university faculty.
(1) inquisitiveness, visit
(2) interest, address
(3) curiosity, join
(4) inquiry, join

22. For years, nuclear-power advocates have claimed that nuclear power is the most ..... form of energy available, but in light of a few facts, one begins to ..... this claim.
(1) cheap, support
(2) useful, question
(3) expensive, contest
(4) economical, doubt

Directions (Q. 23-25) *In the following sentences, parts of the sentences are left blank, choose the best answer among the four given option.*

23. The interest generated by the soccer world cup is _(1)_ compared to the way cricket _(2)_ the nation.
(1) milder, fascinates
(2) lukewarm, electrifies
(3) tepid, inspires
(4) milder, gripes

24. The telephone symbolises that awkward _(1)_ in all communication technologies; while it _(2)_ to bring us together it keeps us apart.
(1) paradox, needs (2) irony, intends
(3) paradox, tries (4) irony, wishes

25. The argument that the need for a looser fiscal policy to _(1)_ demand outweights the need to _(2)_ budget deficits is persuasive.
(1) assess, minimise (2) stimulate, control
(3) outtrip, eliminate (4) increase, hide

Direction (Q. 26) *Identify the grammatically INCORRECT option.*

26. (1) The meeting has been preponed by a week.
(2) Either you or Ram is going to look after it.
(3) The argument explains neither what went wrong nor how it should be put right.
(4) Customers want not only good service but also courtesy.

Directions (Q. 27-30) *Choose the word from the following that does NOT match with two given words.*

27. Wonderful, Fabulous
(1) Superb (2) Superlative
(3) Great (4) Splendid

28. Inordinate, Huge
(1) Excessive (2) Considerable
(3) Insatiable (4) Enormous

29. Laconic, Succinct
(1) Terse (2) Brief
(3) Concise (4) Abrupt

30. Obtuse, Unintelligent
(1) Developmentally disabled person
(2) An illogical argument
(3) A careless calculation
(4) An unreasonable demand

Directions (Q. 31-35) *Read the following passage and answer the following questions.*

I met a traveller from an antique land who said : "Two vast and trunkless legs of stone stand in the desert. Near them on the sand, Half sunk, a shattered visage lies, whose frown and wrinkled

lip and sneer of cold command tell that its sculptor well those passions read which yet survive, stamped on those lifeless things, The band that mocked them and heart that fed and on the pedestral these words appear :

'My name is Ozymandias, king of kings :
Lock on my words, yet mighty, and despair!'

Nothing beside remains, Round the decay. Of that colossal wreck, boundless and bare, The lone and level sand stretch far away."

31. From the poem, one can infer that Ozymandias was
(1) A tyrant (2) A megalomaniac
(3) A benign ruler (4) An egoist

32. Which of the following expressions could be considered as the central theme of the poem ?
(1) Pride goes before a fall
(2) Time and tide wait for no men
(3) Ashes to Ashes dust to dust
(4) None of the above

33. The attitude of the traveller towards Ozymandias is one of
(1) Pity (2) Defence
(3) Indifference (4) Derision

34. Why are Ozymandias works no longer there for all to see ?
(1) He never constructed any great monuments or structures
(2) Construction in a desert is never permanent
(3) His works could not stand the vagaries of time
(4) None of the above

35. The term visage implies
(1) image (2) vision
(3) sculptor (4) face

Directions (Q. 36-40) *In these sentence, each sentence has four underlined words or phrases marked with A, B, C and D. Choose one word or phrase that must be changed for the sentence to be correct.*

36. Bangalore is a beautiful city (A) which in (B) people have successfully blended (C) the modern with (D) the traditional.
(1) A (2) B
(3) C (4) D

37. The ways of (A) travelling has changed (B) dramatically since (C) the late nineteenth century (D).
(1) A (2) B
(3) C (4) D

38. Usually (A) the climate in mountainous areas becomes much (B) windy (C) at higher (D) altitudes.
(1) A (2) B
(3) C (4) D

39. Schools in India can be supported (A) either by government budgets (B) and (C) by private foundations (D).
(1) A (2) B
(3) C (4) D

40. No sooner had he (A) come from Mumbai when (B) he was asked (C) to proceed to (D) Delhi.
(1) A (2) B
(3) C (4) D

■ Quantitative Aptitude, Data Interpretation and Sufficiency

41. Dimensions of a cuboid are 25 cm × 20 cm × 4 cm. The volume of a cube is half the volume of the cuboid. Then, total surface of such a cube is
(1) 400 cm^2 (2) 600 cm^2
(3) 800 cm^2 (4) 1200 cm^2

42. A tree 6 m tall casts a 4 m long shadow. At the same time, a flag pole casts a shadow 50 m long. How long is the flag pole ?
(1) 75 m (2) 100 m
(3) 150 m (4) 50 m

43. If P is a number between 0 and 1, which one of the following is true ?
(1) $P > \sqrt{P}$ (2) $\frac{1}{P^2} > \sqrt{P}$
(3) $P < \frac{1}{P}$ (4) $P^3 > P^2$

44. How can the relationship between x and y be best defined, if values of x and y are as follows ?

x	2	3	4	5	6
y	0	2	6	12	20

(1) $y = 2x - 4$
(2) $y = x^2 - 3x + 2$
(3) $y = x^2 - 4x$
(4) $y = x^2 - 4$

45. If a and b are the roots of the equation $x^2 - 6x + 6 = 0$, then the value of $a^2 + b^2$ is
(1) 36 (2) 24
(3) 12 (4) 6

46. There are 6 parallel vertical lines and 7 parallel horizontal lines. These two groups of parallel lines intersect each other. How many parallelograms will be formed ?
(1) 294 (2) 42
(3) 315 (4) None of these

47. What is the missing figure in the expression given below ?
$$\frac{16}{7} \times \frac{16}{7} - \frac{*}{7} \times \frac{9}{7} + \frac{9}{7} \times \frac{9}{7} = 1$$
(1) 1 (2) 7
(3) 4.57 (4) 32

48. If P is a prime number greater than 3, then $(P^2 - 1)$ is always divisible by
(1) 6 but not 12 (2) 24
(3) 12 but not 24 (4) None of these

49. Pintoo gave some cards to Mintoo and himself from a full pack of playing cards and laid the rest aside. Pintoo then said to Mintoo "If you give me a certain number of your cards, I will have 4 times as many cards as you have. If I give you the same number of cards, I will have thrice as many cards you have". How many cards Pintoo has ?

(1) 31 (2) 32
(3) 29 (4) 30

50. A number of points are marked on a plane and are connected pairwise by line segment. If the total number of line segment is 10, how many points are marked on the plane ?
(1) 4 (2) 10
(3) 5 (4) 9

51. A sum of money becomes eight times in 3 years, if the rate is compounded annually. In how much time the same amount at the same compound interest will become sixteen times ?
(1) 6 yr (2) 4 yr
(3) 8 yr (4) 5 yr

52. What is the value of $\frac{P+Q}{P-Q}$, if $\frac{P}{Q}=7$?
(1) $\frac{4}{3}$ (2) $\frac{2}{3}$
(3) $\frac{2}{6}$ (4) $\frac{7}{8}$

53. The dimension of a rectangular room when increased by 4 m are in the ratio of 4 : 3 and when decreased by 4 m are in the ratio of 2 : 1. The dimensions of the room are
(1) 6 m and 4 m (2) 12 m and 8 m
(3) 16 m and 12 m (4) 24 m and 16 m

54. Two trains each of 120 m length, run in opposite directions with a velocity of 40 m/s and 20 m/s. How long will it take for the tail end of the two trains to meet each other during the course of travel ?
(1) 2 s (2) 3 s
(3) 4 s (4) 5 s

55. '*A*' can do a piece of work in 25 days, '*B*' in 20 days. They work together for 5 days and then '*A*' goes away. In how many days will '*B*' finish the remaining work ?
(1) 17 days (2) 11 days
(3) 10 days (4) None of these

56. *P*, *Q* and *R* are the three towns on a river which flows uniformly. *Q* is equidistant from *P* and *R*. I row from *P* to *Q* and back in 10 hours and I can row from *P* to *R* in 4 hours. Compare the speed of my boat in still water that of the river
(1) 4 : 3 (2) 5 : 3
(3) 6 : 5 (4) 7 : 3

57. If $G = H + \sqrt{\frac{4}{L}}$, then L equals
(1) $\frac{4}{(G-H)^2}$ (2) $4(G-H)^2$
(3) $\frac{4}{(G^2-H^2)}$ (4) $4(G^2-H^2)$

58. The average score of boys in an examination of a school is 71 and that of the girls is 73. The average score of the school in that examination is 71.8, find the ratio of the number of boys to the number of girls that appeared in the examination
(1) 4 : 5 (2) 3 : 2
(3) 3 : 5 (4) 5 : 2

59. What is the area of the region in the cartesian plane whose points (x, y) satisfy $|x|+|y|+|x+y| \le 2$?
(1) 2.5 sq unit (2) 3 sq unit
(3) 2 sq unit (4) 4 sq unit

60. In a triangle *ABC*, the length of the sides *AB*, *AC* and *BC* are 3, 5 and 6 cm respectively. If a point *D* on *BC* is drawn such that the line *AD* bisects the angle *A* internally, then what is the length of *BD* ?
(1) 2 cm (2) 2.25 cm
(3) 2.5 cm (4) 3 cm

61. The length of a string between a kite and a point on the ground is 85 m. If the string makes an angle θ with the level ground such that $\tan\theta = 15/8$, then how much high is the kite ?
(1) 78.05 m (2) 75 m
(3) 316 m (4) 226 m

62. A sequence is generated by the rule that the xth term is x^2+1 for each positive integer x. In this sequence, for any value $x > 1$, the value of $(x+1)$th term less than the value of xth term is
(1) $2x^2+1$ (2) x^2+1
(3) $2x+1$ (4) $x+2$

63. The areas of three adjacent faces of a cuboid are a, b and c. If the volume of the cuboid is V, then V^2 is equal to
(1) abc (2) $(ab+bc+ca)$
(3) c/ab (4) None of these

64. A batsman makes a scores of 98 runs in the 19th innings and thus increases his average by 4. What is his average after 19th innings ?
(1) 22 (2) 24
(3) 28 (4) 26

65. Three partners altogether invested Rs 114000 in a business. At the end of the year, one got Rs 337.50, the second Rs 1125.00 and the third Rs 675. What is the percentage of profit ?
(1) 4.8% (2) 3.8%
(3) 1.8% (4) 2.8%

Directions (Q. 66-67) *The table given below shows the thermal and hydel generation from the period* 2004-2008 *in terms of kWh per kW of installed capacity*

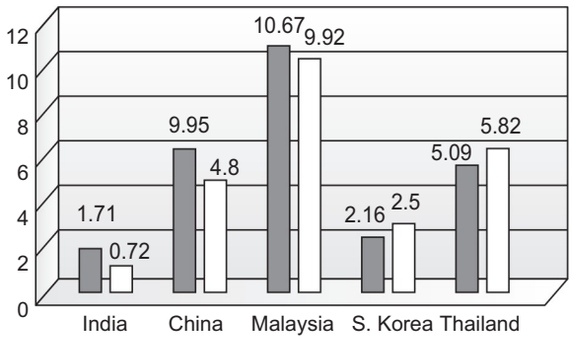
Year	Thermal	Hydel
2004	4000	4240
2005	4200	4010
2006	4020	4160
2007	4050	3700
2008	4040	3930

66. The ratio of thermal and hydel installed capacity over the period 2004–2008 is nearly
(1) 1 : 1 (2) 203 : 200
(3) 20 : 21 (4) 10 : 11

67. The average kWh generated per kW of installed capacity for hydel power generation was approximately
(1) 4100 (2) 3820
(3) 4000 (4) 4050

Directions (Q. 68-71) *Answer these questions from the data presented in the figure below.*

FEI for a country in a year is the ratio of its foreign equity inflows to its GDP. The following figure shows the FEI for select Asian countries for the years 2007 and 2008.



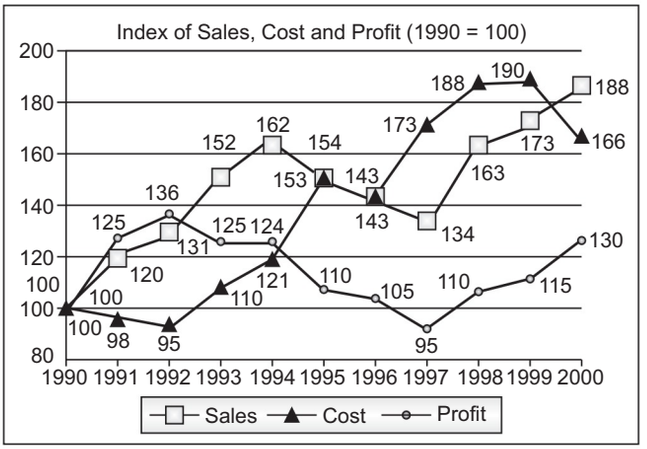
68. The country with the largest change in FEI in 2008 relative to its FEI in 2007 is
(1) India (2) China
(3) Malaysia (4) Thailand

69. Based on the data provided, it can be concluded that
(1) absolute value of foreign equity inflows in 2008 was higher than that in 2007 for both Thailand and South Korea.
(2) absolute value of foreign equity inflows was higher in 2008 for Thailand and lower for China than the corresponding values in 2007.
(3) absolute value of foreign equity flows was lower in 2008 for both India and China than the corresponding values in 2007.
(4) None of the above can be inferred.

70. It is known that China's GDP in 2008 was 7% higher than its value in 2007, while India's GDP grew by 2% during the same period. The GDP of South Korea, on the other hand, fell by 5%. Justify the correct statement from the following
I. Foreign equity inflows to China were higher in 2008 than in 2007.
II. Foreign equity inflows to China were lower in 2008 than in 2007.
III. Foreign equity inflows to South Korea increased in 2008 relative to 2007.
(1) I, II, III (2) I, II
(3) I, III (4) II, III

71. China's foreign equity inflows in 2008 were 10 times than into India. It can be concluded that
(1) China's GDP in 2008 was 40% higher than that of India.
(2) China's GDP in 2008 was 70% higher than that of India.
(3) China's GDP in 2008 was 50% higher than that of India.
(4) None of the above

Directions (Q. 72-75) *Study the following graph to answer these questions.*



Profit = Sales – Cost –Tax Gross Profit = Sales – Cost

72. If the profit was Rs 600 in 1993, then what was the profit in 1990 ?
(1) Rs 441 (2) Rs 395
(3) Rs 480 (4) Rs 545

73. If base of the profit index is changed to 1993 = 100, then what would be the index of profit for the year 2000 ?
(1) 104 (2) 100
(3) 109 (4) None of these

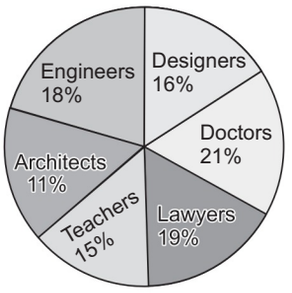
74. If sales were Rs 1200 crore in 1990, then what were the total sales in the period 1990–1995 ?
(1) Rs 8628 crore (2) Rs 9828 crore
(3) Rs 9156 crore (4) Rs 8136 crore

75. What was the difference between the average sales index and average cost index ?
(1) 7.3 (2) 7.7
(3) 7.5 (4) 7.9

Directions (Q. 76-80) *Study the following Pie-chart carefully and answer the questions given below.*

Survey conducted on 10500 people find out Various Professionals in the town and percentage of Female Professionals amongst them

Various Professionals = 10500



Percentage of Female Professionals

Doctors	20%	Architects	40%	Lawyers	40%
Engineers	60%	Teachers	80%	Designers	35%

76. What is the ratio of the male Engineers and Designers to the same occupation female professionals in the town ?
(1) 41 : 44 (2) 44 : 41
(3) 35 : 44 (4) 44 : 35

77. The total number of Lawyers in town is approximately what per cent of the total number of Doctors in the town ?
(1) 95 (2) 98
(3) 90 (4) 85

78. What is the difference between the total number of male and female professionals in the town ?
(1) 1284 (2) 1134
(3) 1054 (4) 1164

79. Female Doctors are what per cent of the female Teachers in the town ?
(1) 42 (2) 28
(3) 15 (4) 35

80. What is the ratio of the number of male Architects to the number of male Teachers in the town ?
(1) 11 : 5 (2) 3 : 2
(3) 5 : 11 (4) 2 : 3

■ Analytical and Logical Reasoning

81. How many meaningful words can be formed with the letters EAML ?
(1) 1 (2) 3
(3) 4 (4) 2

82. If 'light' is called 'morning', 'morning' is called 'dark', 'dark' is called 'night', 'night' is called 'sunshine' and 'sunshine' is called 'dusk', when do we sleep ?
(1) Dusk (2) Dark
(3) Night (4) Sunshine

83. If 'CIGARETTE' is coded as 'GICERAETT', then 'DIRECTION' will be coded in
(1) IRDCTIONE (2) NOIETCRID
(3) RIDTCENOI (4) NORTECDII

84. If 'SMOOTH' is coded as '135579', 'ROUGH' as '975531' and 'HARD' as 9498, then 'SOFT' will be coded as
(1) 1527 (2) 1347
(3) 4998 (4) 8949

85. A man travels 3 km to the West, turns left and goes 3 km, turns right and goes 1 km, again turns right and goes 3 km. How far is he from the starting point ?
(1) 7 km (2) 6 km
(3) 5 km (4) 4 km

86. A's mother is sister of 'B' and has a daughter 'C'. How can 'A' be related to 'B' from among the following ?
(1) Niece (2) Uncle
(3) Daughter (4) Father

87. Showing a lady in the park, Vineet said; "She is the daughter of my grandfather's only son". How is Vineet related to that lady ?
(1) Father (2) Son
(3) Brother (4) Mother

88. In a row at a bus stop, A is the 7th from the left and B is 9th from the right. Both of them interchange their position and so, A becomes 11th from the left. Find the number of peoples in that row.
(1) 18 (2) 19
(3) 20 (4) 21

Directions (Q. 89-91) *The below 3 questions are to be solve on the basis of following informations.*

A five member research group is to be chosen from mathematicians A, B, C and D and the physicists E, F, G and H. It is declared that group must has 3 mathematicians. However,
A refuses to work with D.
B refuses to work with E.
F refuses to work with G.
D refuses to work with F.

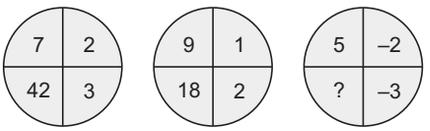
89. If B and C are chosen, which one is true ?
P : A is chosen.
Q : D is chosen.
R : Either F or G is choose.
(1) P only (2) Q only
(3) R only (4) Q and R only

90. If G is rejected, which other member could not work with the group ?
(1) A (2) B
(3) D (4) F

91. If B is chosen, who else would have to be in the group ?
(1) F (2) G
(3) A (4) C

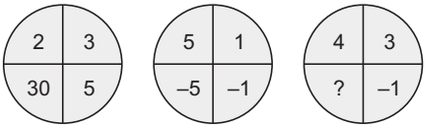
Directions (Q. 92-94) *Choose the exact number for the question mark quadrant.*

92.



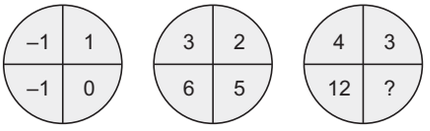
(1) 13 (2) 18
(3) 30 (4) –30

93.



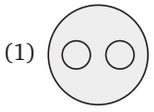
(1) 7 (2) –12
(3) 12 (4) 9

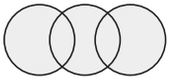
94.

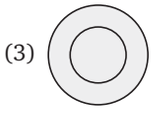


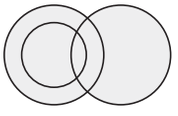
(1) 8 (2) 11
(3) 7 (4) 9

95. Which of the following diagrams shows the last relation between anti-social elements, pick pockets and black marketer ?

(1) 

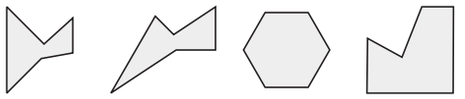
(2) 

(3) 

(4) 

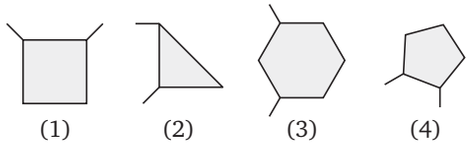
Directions (Q. 96-99) : *In the following four questions, identify the one which is out of place.*

96.



(1) (2) (3) (4)

97.

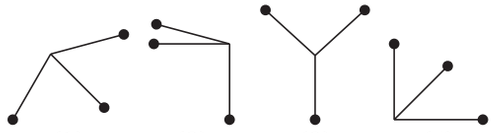


(1) (2) (3) (4)

98.

(1) (2) (3) (4)

99.



(1) (2) (3) (4)

100. Five newly born babies were weighed by doctor. She noticed that child A is lighter than child B. Child C is lighter than D. Child B is lighter than child D, but heavier than child E. Which child is the heaviest one ?

(1) E (2) C
(3) D (4) A

101. Among four friends, Senthill is twice the age of Karthik Gopi is one and a half times younger than Karthik. Bala is six years elder to Karthik, but six years younger to Senthil. Who is the eldest ?

(1) Senthil (2) Karthik
(3) Bala (4) Gopi

Directions (Q. 102-103) *From the sets of number in the four alternatives given below, which is the most similiar to the given set.*

102. Given set : (6, 15, 28)

(1) (50, 59, 71) (2) (46, 56, 66)
(3) (60, 69, 72) (4) (60, 69, 82)

103. Given set : (81, 77, 69)

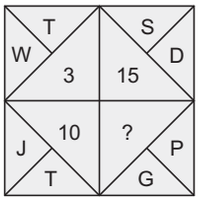
(1) (56, 52, 44) (2) (64, 61, 53)
(3) (92, 88, 79) (4) (75, 71, 60)

104. In a certain coding system, RBM STD BRO PUS means 'the cat is beautiful'. TNH PUS DIM STD means 'the dog is brown'. PUS DIM BRO PUS CUS means 'the dog has the cat'. What is the code for 'has' ?

(1) CUS (2) BRO
(3) DIM (4) STD

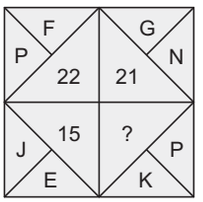
Directions (Q. 105-106) *Find out the missing number in place of ?*

105.



(1) 11 (2) 5
(3) 9 (4) 13

106.



(1) 7 (2) 27
(3) 16 (4) 26

Directions (Q. 107-109) *In an imaginary language, the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9 are substituted by a, b, c, d, e, f, g, h, i and j. If 10 is written as ba, then find the value in the questions given below.*

107. $(cd + ef) \times bc$ is equal to

(1) 684 (2) 816
(3) 916 (4) 1564

108. $(dc \times f) - (bf - d) \times d$ is equal to

(1) *bce* (2) *bcf*
(3) *abe* (4) *abb*

109. $baf \div bf \times d$ is equal to

(1) *dc* (2) *be*
(3) *cb* (4) *df*

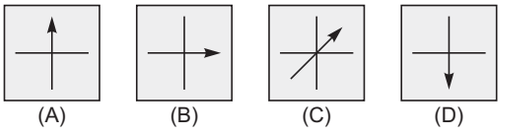
110. Three views of the same cube are shown below :

The figure on the face opposite the triangle is

(1) Pentagon (2) Circle
(3) Question mark (4) Rectangle

Directions (Q. 111-113) *In these questions, select the odd one out.*

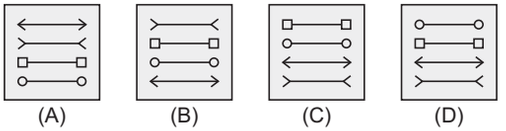
111.



(A) (B) (C) (D)

(1) B (2) A
(3) D (4) C

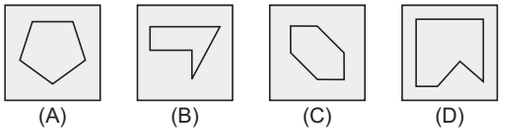
112.



(A) (B) (C) (D)

(1) C (2) D
(3) B (4) A

113.



(A) (B) (C) (D)

(1) C (2) A
(3) B (4) D

Directions (Q. 114-115) *Read the statement and answer the questions.*

A cube is coloured green on all sides. It is cut into 125 smaller cubes of equal size.

114. How many cubes are not coloured on any face ?
(1) 30 (2) 23
(3) 27 (4) 35

115. How many cubes are coloured on three faces ?
(1) 12 (2) 8
(3) 16 (4) 18

Directions (Q. 116-120) *Study the following information carefully to answer these questions.*

Three families X, Y and Z comprising of ten members A, B, C, D, E, F, G, H, J and K. Each family has at least three members with at least one male and one female. They travel in three cars I, II and III in such a way that at least three members travel in each car and all the members of any of the families do not travel together. At least one female and one male travel in each car.

A is wife of J and mother of F and they belong to family X, E is sister of B who is son of C and belongs to family Y. G is father of H and husband of K. D is father of B. A and K travel in Car I only with H. E and B do not travel in the same car. G and C travel in Car II. B does not travel with D or F. J does not travel with F.

116. Which of the following group of people represent one complete family ?
(1) GJF (2) GKF
(3) GKH (4) AJH

117. How is J related to F ?
(1) Father (2) Brother
(3) Son (4) Cannot be determined

118. How is H related to K ?
(1) Daughter (2) Son
(3) Mother (4) Cannot be determined

119. Four members travel in which Car ?
(1) Car I (2) Car II
(3) Car III (4) Car I or II

120. Which of the following combination of family and its members is correct ?
(1) X-A, F, H (2) Y-G, K, H
(3) Z-A, J, F (4) Y-C, D, E, B

■ General Awareness

121. Who of the following wrote the book *"PRECEPTS OF JESUS"* ?
(1) Raja Rammohan Roy
(2) Devendranath Tagore
(3) Ishwarchandra Vidyasagar
(4) Keshab Chandra Sen

122. In the middle of the year 2008 the Parliament of which one of the following countries became the first in the world to enact a climate Act by passing "The Climate Change Accountability Bill".
(1) Australia (2) Canada
(3) Germany (4) Japan

123. Among the following, who was not a co-founder of NAM ?
(1) Gamal Abodel Nasser
(2) Josip Broz Tito
(3) Jawaharlal Nehru
(4) Yasser Arafat

124. Where shall the 16th Asian Games, 2010 be held ?
(1) Kuala Lumpur (2) Guangzhou
(3) Hanoi (4) Ho Chi Minh City

125. Which one of the following countries is the largest producer of petroleum in South America ?
(1) Brazil (2) Venezuela
(3) Columbia (4) Argentina

126. What was the name of the party formed by Subhash Chandra Bose after leaving Indian National Congress ?
(1) Congress Socialist Party
(2) Forward Bloc
(3) Indian National Conference
(4) Swaraj Party

127. The Civil Disobedience Movement started with which of the following ?
(1) No-tax in Barodoli district
(2) Demonstration of Akali Sikhs at Nabha
(3) Demonstration against the Simon Commission
(4) Dandi March

128. Who was the Governor-General of the East India Company when the Revolt of 1857 took place ?
(1) Lord Bentinck (2) Lord Hastings
(3) Lord Dalhousie (4) Lord Canning

129. Which of the following in human body manufacture the antibodies that are responsible for immunity ?
(1) Muscle cells (2) Nerve cells
(3) Red blood cells (4) White blood cells

130. Which of the following enzymes is produced by the salivary glands ?
(1) Pepsin (2) Ptyalin
(3) Rennin (4) Trypsin

131. Which one of the following locations does not have a steel plant of SAIL ?
(1) Bhilai (2) Bokaro
(3) Rourkela (4) Visakhapatnam

132. Which one of the following is not a member of ASEAN ?
(1) Brunei Darussalam (2) East Timor
(3) Laos (4) Myanmar

133. Which one of the following countries produces phalcon airborne warning and control systems (AWACS) ?
(1) India (2) Israel
(3) Russia (4) USA

134. How is international personality Zadie Smith well-known as ?

(1) Environmentalist
(2) Human Rights activist
(3) Litterateur
(4) Musician

135. For which one of the following States, is Guwahati High court not its high court ?
(1) Arunachal Pradesh (2) Meghalaya
(3) Mizoram (4) Sikkim

136. Which of the following has/have velocity equal to that of light ?
(1) X-ray (2) γ-ray
(3) Ultraviolet (4) All of these

137. The Narmada and Tapti rivers flank
(1) Vindhyan Mountains (2) Satpura Ranges
(3) Rajmahal Hills (4) Aravalli Hills

138. Silvassa is the capital of
(1) Daman and Diu
(2) Dadra and Nagar Haveli
(3) Lakshadweep
(4) Arunachal Pradesh

139. Special Economic Zone Act was passed in
(1) May 2005 (2) February 2005
(3) May 2006 (4) February 2006

140. Jaduguda is famous for
(1) Iron ore (2) Manganese
(3) Gold (4) Uranium

141. Which one of the following is not a Natural port in India ?
(1) Kandla (2) Chennai
(3) Paradeep (4) Mumbai

142. What is Jarvik-7 ?
(1) Electronic leg (2) Pace-maker
(3) Artificial heart (4) Artificial eye

143. The first computer was made by
(1) Bill Gates (2) Bill Clinton
(3) Charles Babbage (4) Marconi

144. The temperature of a normal human body is
(1) 98.4°F (2) 98°F
(3) 98.8°F (4) None of these

145. Which of the following is the most common element in the Universe ?
(1) Hydrogen (2) Oxygen
(3) Nitrogen (4) Carbon

146. Shortage of which element causes Goitre disease ?
(1) Nitrogen (2) Calcium
(3) Iodine (4) Phosphorus

147. Miniaturization of present day computers had been possible due to the use of
(1) Transistors
(2) Integrated Circuit Chips
(3) Nano-materials
(4) Super Conductors

148. Who declared "Every blow that is hurled on my back will be a nail in the coffin of the British Empire" ?
(1) Lala Lajpat Rai
(2) Bhagat Singh
(3) Chandrashekhar Azad
(4) Bal Gangadhar Tilak

149. The 'Manu Smriti' mainly deals with
(1) Social order (2) Laws
(3) Economics (4) State-Craft

150. Who sent 'Hieun Tsang' as an envoy to Harsh's Court ?
(1) Tai Tsung (2) Tung-Khan
(3) Ku Yen-Wu (4) None of these

151. Lord Macaulay is associated with
(1) Reforms in Army
(2) Abolition of Sati system
(3) English education
(4) Permanent Settlement

152. Who were directly related to the Poona Pact of 1932 ?
(1) Indian women
(2) Indian Labour class
(3) Indian farmers
(4) Indian depressed classes

153. At present in the Constitution of India 'Right to Property' is a
(1) Fundamental right
(2) Legal right
(3) Moral right
(4) None of the above

154. Who was the Nationalist Leader to be elected as Speaker of the Legislative Assembly in 1925 ?
(1) Motilal Nehru
(2) Vitthal Bhai Patel
(3) Vallabh Bhai Patel
(4) C.R. Das

155. Who was the Ist Lady President of All India Congress ?
(1) Smt. Annie Besant
(2) Smt. Sarojini Naidu
(3) Smt. Nelli Sengupta
(4) Aruna Asaf Ali

156. Who decides whether a bill is a Money bill or not ?
(1) The President
(2) The Speaker
(3) The Finance Secretary
(4) The Finance Minister

157. 'Charkula' is famous folk dance of
(1) Bundelkhand (2) Brij Bhumi
(3) Avadh (4) None of these

158. IRDA regulates
(1) Banking Companies
(2) Insurance Companies
(3) Retail Trade
(4) None of the above

159. The decadal growth of Urban Population (in per cent) was highest in India during
(1) 1961-71 (2) 1971-81
(3) 1981-91 (4) 1991-2001

160. The benefit of which one of the following River valley projects is shared by more than one State ?
(1) Chambal Valley Project
(2) Mayurakshi Project
(3) Sharavati Project
(4) Hirakund Project

(OMR SHEET)

SNAP (Mock Test 1)

(Symbiosis National Aptitude Test)

General English

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
001	① ② ③ ④	009	① ② ③ ④	017	① ② ③ ④	025	① ② ③ ④	033	① ② ③ ④
002	① ② ③ ④	010	① ② ③ ④	018	① ② ③ ④	026	① ② ③ ④	034	① ② ③ ④
003	① ② ③ ④	011	① ② ③ ④	019	① ② ③ ④	027	① ② ③ ④	035	① ② ③ ④
004	① ② ③ ④	012	① ② ③ ④	020	① ② ③ ④	028	① ② ③ ④	036	① ② ③ ④
005	① ② ③ ④	013	① ② ③ ④	021	① ② ③ ④	029	① ② ③ ④	037	① ② ③ ④
006	① ② ③ ④	014	① ② ③ ④	022	① ② ③ ④	030	① ② ③ ④	038	① ② ③ ④
007	① ② ③ ④	015	① ② ③ ④	023	① ② ③ ④	031	① ② ③ ④	039	① ② ③ ④
008	① ② ③ ④	016	① ② ③ ④	024	① ② ③ ④	032	① ② ③ ④	040	① ② ③ ④

Quantitative Aptitude, Data Interpretation and Sufficiency

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
041	① ② ③ ④	049	① ② ③ ④	057	① ② ③ ④	065	① ② ③ ④	073	① ② ③ ④
042	① ② ③ ④	050	① ② ③ ④	058	① ② ③ ④	066	① ② ③ ④	074	① ② ③ ④
043	① ② ③ ④	051	① ② ③ ④	059	① ② ③ ④	067	① ② ③ ④	075	① ② ③ ④
044	① ② ③ ④	052	① ② ③ ④	060	① ② ③ ④	068	① ② ③ ④	076	① ② ③ ④
045	① ② ③ ④	053	① ② ③ ④	061	① ② ③ ④	069	① ② ③ ④	077	① ② ③ ④
046	① ② ③ ④	054	① ② ③ ④	062	① ② ③ ④	070	① ② ③ ④	078	① ② ③ ④
047	① ② ③ ④	055	① ② ③ ④	063	① ② ③ ④	071	① ② ③ ④	079	① ② ③ ④
048	① ② ③ ④	056	① ② ③ ④	064	① ② ③ ④	072	① ② ③ ④	080	① ② ③ ④

Analytical and Logical Reasoning

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
081	① ② ③ ④	089	① ② ③ ④	097	① ② ③ ④	105	① ② ③ ④	113	① ② ③ ④
082	① ② ③ ④	090	① ② ③ ④	098	① ② ③ ④	106	① ② ③ ④	114	① ② ③ ④
083	① ② ③ ④	091	① ② ③ ④	099	① ② ③ ④	107	① ② ③ ④	115	① ② ③ ④
084	① ② ③ ④	092	① ② ③ ④	100	① ② ③ ④	108	① ② ③ ④	116	① ② ③ ④
085	① ② ③ ④	093	① ② ③ ④	101	① ② ③ ④	109	① ② ③ ④	117	① ② ③ ④
086	① ② ③ ④	094	① ② ③ ④	102	① ② ③ ④	110	① ② ③ ④	118	① ② ③ ④
087	① ② ③ ④	095	① ② ③ ④	103	① ② ③ ④	111	① ② ③ ④	119	① ② ③ ④
088	① ② ③ ④	096	① ② ③ ④	104	① ② ③ ④	112	① ② ③ ④	120	① ② ③ ④

General Awarness

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
121	① ② ③ ④	129	① ② ③ ④	137	① ② ③ ④	145	① ② ③ ④	153	① ② ③ ④
122	① ② ③ ④	130	① ② ③ ④	138	① ② ③ ④	146	① ② ③ ④	154	① ② ③ ④
123	① ② ③ ④	131	① ② ③ ④	139	① ② ③ ④	147	① ② ③ ④	155	① ② ③ ④
124	① ② ③ ④	132	① ② ③ ④	140	① ② ③ ④	148	① ② ③ ④	156	① ② ③ ④
125	① ② ③ ④	133	① ② ③ ④	141	① ② ③ ④	149	① ② ③ ④	157	① ② ③ ④
126	① ② ③ ④	134	① ② ③ ④	142	① ② ③ ④	150	① ② ③ ④	158	① ② ③ ④
127	① ② ③ ④	135	① ② ③ ④	143	① ② ③ ④	151	① ② ③ ④	159	① ② ③ ④
128	① ② ③ ④	136	① ② ③ ④	144	① ② ③ ④	152	① ② ③ ④	160	① ② ③ ④

Symbiosis National Aptitude Test

Crack SNAP 2

ENTRANCE PATTERN

The duration for making answers will be 2 hours. Normally the pattern of written test is as follows.

Sections

I. General English
II. Quantitative Aptitude, Data Interpretation & Sufficiency
III. Analytical and Logical Reasoning
IV. General Awareness

Important Instructions

1. *Calculator/Watch calculator will not be allowed in the examination Hall.*
2. *Use only HB Pencils for darkening the ovals.*
3. *Correct method of Marking Answers*

Each questions is followed by four answer choices which are marked a, b, c, d on the answer sheet. Using HB Pencil, darken the oval bearing the correct answer against the serial number of the question.

For example : *If the answer to question number* 2 *is* (4)*, you should mark as shown below*

Q. 2. (1) (2) (3) (4)

General English

1. Which of the following does not make sensible combination with the word 'Red' ?
(1) Carpet (2) Rag
(3) Faced (4) Blood

2. A single word for 'going back on one's promise' is
(1) Prevaricate (2) Retrospect
(3) Renege (4) Repudiate

3. Find the sentences in which following words have been correctly used.
incise, excise, precise
(a) The surgeon incised into the patient's limb.
(b) The diseased gall bladder has to be excised.
(c) He made a very precise guess.
(d) He excised the flower plant from the pot.
(1) a and b (2) b and c
(3) a, c and d (4) a, b and c

4. Find the word with incorrect spelling.
(1) Embarass (2) Harass
(3) Immerse (4) Emerge

5. Identify the odd one out of the following.
(1) Trail (2) Rut
(3) Spoor (4) Groove

Directions (Q. 6-9) *Read the following passage and answer the questions given below.*

Everyone seems to be talking about carbon tax. The idea is to tax environment's polluters which should be commensurate with the carbon emission. It would act as deterrent and result in less carbon emissions. No wonder, people from both sides of the political divide have come out in support of it. Denmark, Finland, Norway and Sweden have already had carbon tax in place since 90's, but the tax has not led to large decline in emission in most of these countries. In case of Norway, emission has actually increased by 43% per capita. An economist might say this is fine, as long as the cost of polluting the environment is exacted which otherwise also might have been done with impunity.

6. What do you understand from carbon tax ?
(1) It is penalty tax for polluting environment by burning organic fossils.
(2) It is tax for producing coal.
(3) It is tax for causing vehicles emitting fumes.
(4) It is tax for using coal.

7. Why do people in the Government as well as in opposition support it ?

(1) Because it was expected to generate revenue.
(2) Because offenders have been polluting environment with impunity.
(3) Because it was expected to act as a measure to make people desist from using materials responsible for environment pollution.
(4) Because the tax was to be levied in proportion to the quantum of emissions.

8. Choose the expression/word which is most inappropriate in meaning for 'commensurate'.
(1) unequal in quantum
(2) in measure with something
(3) unjust
(4) unequal in measurement

9. Which one of the following sentences has used the word 'Impunity' appropriately ?
(1) The VIP's have been granted impunity from punishment.
(2) The vaccine gives children impunity from polio.
(3) The hardened criminal perpetrates crime with impunity.
(4) Vitamin 'C' is supposed to give impunity to people suffering from cold and cough.

10. In how many ways the word 'Address' may be used ?
(a) The particulars/details where one used to live/stay.
(b) Speech made on formal occasions.
(c) Give attention to a problem in order to solve it.
(d) Send a letter to someone.
(1) a and b (2) b and c
(3) c and d (4) a, b and c

11. Choose the odd one out of the followings
(1) Junket (2) Jaunt
(3) Voyage (4) Sojourn

12. Identify the various grammatical inflections of the word 'Aesthete'

1. Noun	A. Aestheticize
2. Adjective	B. Aesthetic
3. Adverb	C. Aesthete
4. Verb	D. Aesthetically

(1) 1-A, 2-B, 3-C, 4-D (2) 1-C, 2-B, 3-D, 4-A
(3) 1-B, 2-C, 3-D, 4-A (4) 1-D, 2-A, 3-B, 4-C

Directions (Q. 13-16) *Read the following passage and answer the questions given below.*

So what if the Kohinoor diamond—the ultimate symbol of India's wealth and grandeur, now resides with the Queen of England ? Now the icon of British luxury rides—Jaguar and Land Rover, passed into Tata's Indian hands. The irony of it all was not lost in either India or the UK—a company from former colonial country took over a company of its coloniser country. In contrast to the high drama that preceded Tata Steel's acquisition of Corus Steel last year, this transaction was a quiet affair.

13. The acquisition of Land Rover is the
(1) maiden take-over of Tata's overseas
(2) second take-over of Tata's overseas
(3) the most controversial take-over
(4) costliest take-over

14. The new-age take-over of foreign businesses
(1) symbolises revanchist turn around of subjugation era
(2) symbolises return of Indian grandeur of possessing Kohinoor diamond
(3) symbolises quiet state of affairs
(4) symbolises passion for luxurious things

15. What is the synonym of 'icon' ?
(1) Idol (2) Image
(3) Symbol (4) Graphics

16. What is the meaning of 'Irony' ?
(1) Resembling iron
(2) Toughness
(3) Expression using words opposite to what is meant
(4) Act of ridiculing

Directions (Q. 17-20) *Read the following passage and answer the questions given below.*

Good business schools boost your self-confidence by providing you with excellent business tools—discounted cash flows, game theory, organisational behaviour, financial accounting and so on—all of which can help you manage and understand diverse work situations.

So when I finished business school, I was filled with the confidence of being able to handle the toughest problems and always emerge a winner. After all, we had cracked every case-study thrown at us. We understood the mistakes made along the way and learnt the best practices of successful corporates around the globe. Thus armed, I returned to India with the adjective of running my own company and a vow : never to be an employee again.

I had decided to start a venture in manufacturing diamond jewellery for export. No sooner than we started, the Government of India banned the export of jewellery containing gold. And that was it!

All the case-studies had not prepared me for such an eventuality. Naturally, the venture failed. Education assumes optimistic outcomes, starting with passing out from college, so we had not been taught how to handle failure.

So there I was, a young 30-year-old, newly married, fresh out of a premier B-school, with the rug pulled from under my feet. There was little capital left, except for a roof over our head, which was courtesy my dad. So what's next ? Should I give up and take up a job, or try again ?

Let me share a framework that worked for me. First, I took a break with my wife, just to get away from the "scene of the crime", so to speak. It allowed me to clear my mind. Then I resisted the pressure to take the first thing that came along by allowing some time to think about why things happened as they did. It also gave me an opportunity to talk with friends and well-wishers who had nothing to do with the business. I could deliberate on what I did wrong, and what I did right, allowing me to answer the key question "am I a failure as an entrepreneur or can I succeed ?" When I resolved that my fundamental approach to being an entrepreneur was valid, I could move ahead with confidence in myself and in my ability to succeed.

It was very important to treat the business failure as a lesson, as a step in the game of life, to be able to persist with my objectives, with a determination that drew from the inner strength of deep introspection.

So when faced with a 'game changing' failure, whether as an entrepreneur or as a manager, one needs to take a break, think through one's objectives and lessons learnt, before moving onto the next phase, with persistence and a renewed determination to succeed.

17. The best recipe to succeed in business is to
(1) earn a management degree from premier business school
(2) acquire the confidence of being able to handle the toughest problems and emerge a winner
(3) learn lessons from failures and move ahead with clear vision of objective and confidence in ability to succeed
(4) consult friends and well-wishers who have nothing to do with business

18. The weakness of management schools is
(1) its failure to train students to handle failures and contingent adverse situations
(2) its failure to educate students on governmental policies controlling business
(3) its failure to prepare students to handle diverse work situations
(4) its failure to train students to take optimistic view of a business

19. Choose the opposite of 'Fill'.
(1) Deplete (2) Evacuate
(3) Vacate (4) Exhaust

20. Choose the synonym of 'Emerge'.
(1) Submerge (2) Peer
(3) Plop (4) Immerse

21. Choose the odd one out.
(1) Catechize (2) Crib
(3) Dressage (4) Preach

22. Choose the incorrect use of phrase/idiom.
(1) He is living in fool's paradise.
(2) He is living in paradise of fools.
(3) He used to talk Turkey.
(4) He made a tongue-in-check remark.

23. Mark the words with incorrect spellings.
(a) Impliment (b) Quietitude
(c) Quintessence (d) Quarrantine
(1) a and b (2) b, c and d
(3) d, a and b (4) c and d

24. Choose correct combination/combination of words with 'Fly'
(a) Fly ash (b) Fly in the face
(c) Fly by night (d) Fly freely
(1) a and b (2) a, b and c
(3) a, b, c and d (4) only d

25. There are some underlined words below followed by their uses in sentences. Find out, the wrong sentence if any
Factious Factitious Fictitious Facetious
(a) Factious quarrels destroyed the party.
(b) He showed factitious affection.
(c) He narrated fictitious story.
(d) He made facetious remark.
(1) only d (2) b and c
(3) only a (4) None of these

26. The following words are expected to be related to one thing. Choose one, if any, which is not related.
(a) Physiognomy (b) Profile
(c) Facet (d) Visage
(1) None (2) only a
(3) only c (4) only d

27. Which of the following means 'a huge waterfall' ?
(1) Cascade (2) Cataract
(3) Catarrah (4) Connecticut

Directions (Q. 28-35) *Read the following passage and answer the questions given below.*

Oedipus was born to a king. An oracle prophesied that this child was doomed to be the killer of his father, and that he would ignorantly marry his mother. The king got his son nailed by his feet on a hill top and abandoned. A passer-by picked injured child up from there, and took him to his king—who was the king of another kingdom, who became foster father of the child—later christened as Oedipus which meant "swollen fact".

When he grew up and learnt of his destiny, he left his home in wandering. He was buffeted into the kingdom of his real father by fate. One day the real father-king was on a journey with his entourage. The bodyguards ordered Oedipus to steer clear of the king's route. The brave Oedipus challenged them and killed the king in the fight that ensued. Thus the first part of oracle's prophecy came true.

It was proclaimed that the throne of the slain king would be offered to a wise man who would solve series of riddles. Oedipus did solve them though he did not know he himself had killed the same king. Thus Oedipus became the king, and his mother became his queen. The second part of prophecy too came true. The new king, Oedipus, ordered an enquiry into the king's killing, which eventually revealed that it was he himself who had killed the predecessor.

The psychological phenomenon of a child harbouring subconcious sexual attachment to the parent of opposite sex has been assigned the name, "Oedipus Complex".

28. One who pronounces divine wish is known as
(1) Sage (2) Oracle
(3) Recluse (4) Hermit

29. What is the one word for 'Cursed by fate to suffer' ?
(1) Doomed (2) Punished
(3) Ordained (4) Destined

30. The situation which a person is assigned by his fate or the divine wish is called
(1) destination (2) destiny
(3) limbo (4) abyss

31. What is the synonym of "Christen" ?
(1) To adopt Christianity
(2) To make the symbol of cross
(3) To baptize
(4) Crucify

32. Choose the most appropriate word which is opposite in meaning of 'Reveal'.
(1) Hide (2) Unveil
(3) Disclose (4) Conceal

33. Choose the synonym of 'Riddle'.
(1) Conundrum (2) Secret
(3) Problem (4) Ridicule

34. What is the synonym of 'harbour' ?
(1) to shelter (2) sea port
(3) to ensconce (4) support

35. Choose the opposite of 'Predecessor'.
(1) Incumbent (2) Successor
(3) Heir (4) Progeny

Directions (Q. 36-40) *In each question below, a sentence is given with a part of it printed in bold type. That part may contain a grammatical error. Each sentence is followed by phrases (1), (2) and (3). Find out which phrase should replace the phrase given in bold to correct the error, if there is any, and to make the sentence grammatically meaningful and correct. If the sentence is correct as it is and no correction is required, mark (4) as the answer.*

36. Many public servants can become **untrusworthy and corruption.**
(1) untrustful and corruption
(2) untrusting and corruptive
(3) untrustworthy and corrupt
(4) No correction required

37. **Mere being equipped by** more programming does not mean we practice moral behaviour.
(1) Mere being equipped with
(2) Merely being equipped with
(3) Merely being equipped for
(4) No correction required

38. His works of art **rank high in the appraisal for** competent critics
(1) are ranking high with the appraisal of
(2) rank high in the appraisal of
(3) rank high by the appraisal of
(4) No correction required

39. Blessed with much intelligence, he **saw very few meanings** in religious rites that are performed as mere rituals.
(1) saw quite a few meanings
(2) looked for very few meanings
(3) saw little meaning
(4) No correction required

40. He **was devoted his time meticulously** for the good of his people.
(1) was meticulous to be devoted his time
(2) devoted his time meticulously
(3) was meticulous in his time to devote
(4) No correction required

■ Quantitative Aptitude, Data Interpretation and Sufficiency

41. There are 10 points in a plane out of which 4 are collinear. Find the number of straight lines formed by joining them.
(1) 30 (2) 45
(3) 6 (4) 40

42. A man's income increases by 10%, then decreases by 10%. Calculate the percentage change in his income.
(1) 1% decrease (2) 2% decrease
(3) 3% increase (4) None of these

43. What sum at compound interest will amount to Rs 650 at the end of the 1st year and Rs 676 at the end of the second year ?
(1) 620 (2) 625
(3) 650 (4) None of these

44. The sum deviation of n variable from mean n will be
(1) 0 (2) 1
(3) n (4) None of these

45. Two dice are thrown simultaneously. What is the probability of obtaining a total score of 7 ?
(1) 1 (2) $\frac{1}{6}$
(3) $\frac{1}{4}$ (4) None of these

46. A coin is tossed, successively three times. Find the probability of getting exactly one head or two heads.
(1) $\frac{3}{4}$ (2) $\frac{1}{2}$
(3) $\frac{3}{7}$ (4) None of these

47. Form a well shuffled deck of 52 cards, a card is drawn at random. Find the probability that the card drawn is red and a king.
(1) $\frac{1}{13}$ (2) $\frac{1}{26}$
(3) $\frac{3}{13}$ (4) None of these

48. A gentleman has 6 friends to invite. In how many ways can he send invitation cards to them if he has 3 servants to carry the cards ?
(1) 216 (2) 729
(3) ${}^6P_3$ (4) None of these

49. In what time will the interest of Rs 10000 be Rs 1576.25 at 5% per annum compound rate of interest ?
(1) 2 yr (2) 3 yr
(3) $2\frac{1}{2}$ yr (4) None of these

50. If Peter can do a piece of work in 12 days. Mohan is found to be 50% more efficient than Peter. If Mohan is given the same piece of work, how long will he take to do it ?
(1) 7 days (2) 12 days
(3) 8 days (4) None of these

51. A can do a work in 32 days and B in 40 days. They worked together for 10 days and then B leaves. In how many days will A finish the remaining work ?
(1) 10 days (2) 14 days
(3) 7 days (4) None of these

52. If $\frac{a}{b}=\frac{4}{3}$, then value of $\frac{a+2b}{b}$ will be
(1) $\frac{5}{3}$ (2) $1\frac{2}{3}$
(3) 3 (4) None of these

53. $\left(\frac{x^a}{x^b}\right)^{(a+b)}\cdot\left(\frac{x^b}{x^c}\right)^{(b+c)}\cdot\left(\frac{x^c}{x^a}\right)^{c+a}$ is equal to
(1) 1 (2) 2
(3) 0 (4) None of these

54. Find the smallest number which when increased by 8 is exactly divisible by 30, 45, 65 and 78.
(1) 1170 (2) 1162
(3) 162 (4) None of these

55. Find the least number of square tiles required to pave the floor of a room 15 m 17 cm in length and 9 m 2 cm breadth.
(1) 14 (2) 89
(3) 814 (4) None of these

56. Find the value of $\cfrac{1}{2+\cfrac{1}{2+\cfrac{1}{2-\frac{1}{2}}}}$.
(1) 19/8 (2) 8/19
(3) 7/19 (4) None of these

57. The average age of 30 children in a class is 9 years. If the teacher's age is included, the average age becomes 10 years. Find the teacher's age.
(1) 30 (2) 35
(3) 40 (4) None of these

58. If $a^x = b$, $b^y = c$ and $c^z = a$, then xyz is equal to
(1) abc (2) $\frac{1}{abc}$
(3) 1 (4) None of these

59. Find a fraction which bears the same ratio to $\frac{1}{27}$ that $\frac{3}{7}$ does to $\frac{5}{9}$.
(1) 1/35 (2) 35
(3) 1/27 (4) 5/9

60. $\frac{(0.623)^3 + (0.377)^3}{(0.623)^2 - (0.623 \times 0.377) + (0.377)^2} = x$, then x will be
(1) 0 (2) 1
(3) 2.25 (4) None of these

61. If 5% of men and 0.25% of women have grey hair. A grey haired person is selected. Assuming that there are equal number of males and females the probability of the person being male is
(1) $\frac{21}{20}$ (2) $\frac{20}{21}$
(3) $\frac{1}{400}$ (4) None of these

62. 3 men and 2 boys can complete a work in 6 days. The same work was done by 5 men and 7 boys in 3 days. If so, how many days will 2 men and 5 boys require to complete the same work ?
(1) 6 days (2) 3 days
(3) 4 days (4) None of these

63. In a 35 lbs mixture, the percentage of quinine is 60. How much pure quinine should be added to the mixture so that the adulteration is 35% of total ?
(1) 14 lbs (2) 5 lbs
(3) 20 lbs (4) None of these

64. A dealer desires to gain 15% after allowing a discount of 5%. How does should be the marked price of the article costing Rs 7.20 ?
(1) Rs 8.72 (2) Rs 8
(3) Rs 7.82 (4) None of these

65. An agent by selling a bicycle gets Rs 4500 but he remits to his principal Rs 4200 only and keep the balance with him as commission, find the rate of commission.
(1) 6% (2) $6\frac{2}{3}\%$
(3) $5\frac{2}{3}\%$ (4) None of these

Directions (Q. 66-70) *Study the table given below and answer the questions.*

Percentage of income generated by the sale of different types of food grains produced by six different farmers out of their total income in a year.

Farmer	Total income	Percentage of income generated				
		Wheat	Rice	Bajra	Jowar	Maize
J	2.55	28	36	12	15	9
K	2.00	27	26	16	13	18
L	1.85	19	32	18	18	13
M	1.75	23	29	21	17	10
N	2.15	14	19	26	23	18
O	2.40	16	14	27	21	22

66. Which farmer has generated the maximum amount of income from the sale of rice ?
(1) K (2) L
(3) M (4) None of these

67. Income generated by farmer M from the sale of bajra is approximately what per cent of the total income generated by all farmers together from the same food grain ?
(1) 26% (2) 8%
(3) 21% (4) 15%

68. What is the respective ratio of income generated from the sale of jowar by farmer N to that by farmer O ?
(1) 989 : 1008 (2) 873 : 967
(3) 967 : 873 (4) 1008 : 989

69. What is the income generated by farmer K from the sale of wheat and rice together ?
(1) Rs 98900 (2) Rs 115000
(3) Rs 106000 (4) Rs 110500

70. What is the total income generated by the sale of maize by all farmers together ?
(1) Rs 154000 (2) Rs 192000
(3) Rs 180600 (4) Rs 167500

Directions (Q. 71-75) *Study the table carefully to answer the following questions.*

Price per kilo (in Rs) at which different farmers sell six different products.

Produce → Farmer ↓	Wheat	Rice	Gram	Sugar	Jowar	Bajra
L	3	8	10.5	4	7	3
M	3.9	7.5	12	2.5	6.5	4
N	4.6	5	9.75	4	4.5	4.5
O	2.5	6.5	9	6.5	5	5
P	6.5	4	8.25	5	5.5	6.5
Q	5.5	9	11	3.5	8	4

71. If farmer O sells 40 kilos of each of the six produces, what is the total money earned by him ?
(1) Rs 1260 (2) Rs 1540
(3) Rs 1360 (4) Rs 1380

72. Which produce has the lowest average selling price ?
(1) Wheat (2) Sugar
(3) Jowar (4) Bajra

73. If each farmer sells 250 kilos fo Wheat and 150 kilos of Gram, which farmer earns the maximum money ?
(1) M (2) P
(3) Q (4) None of these

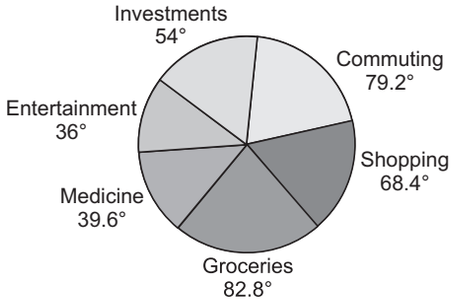
74. Farmer M sells 170 kilos of Jowar and 120 kilos of Bajra. Farmer L sells 120 kilos of Jowar and 170 kilos of Bajra. What is the difference between the total money earned by each farmer from both produces together ?
(1) Rs 235 (2) Rs 195
(3) Rs 260 (4) Rs 315

75. Average selling price of sugar is approximately what per cent of average selling price of Rice ?
(1) 64% (2) 76%
(3) 52% (4) 48%

Directions (Q. 76-80) *Study the following pie-chart carefully to answer these questions.*

Degree wise break-up of expenditure of a family in a month.

Total amount spent in a month is Rs 45,800%.



76. What is the amount spent by the family on Commuting ?
(1) Rs 10,076 (2) Rs 10,534
(3) Rs 6,870 (4) Rs 8,702

77. What is the respective ratio of amount spent by family on Medicine to the amount spent on Groceries ?
(1) 1 : 2 (2) 13 : 21
(3) 3 : 5 (4) 11 : 23

78. What is the total amount spent by the family on Entertainment and Shopping together ?
(1) Rs 9,618 (2) Rs 13,282
(3) Rs 13,740 (4) Rs 11,908

79. Total amount spent by the family on Groceries, Entertainment and Investment together forms approximately what per cent of amount spent on commuting ?
(1) 209 (2) 76
(3) 154 (4) 218

80. Amount spent by the family on Medicine forms what per cent of amount spent on Shopping ? (Rounded off to two digits after decimal)
(1) 43.67 (2) 49.52
(3) 57.89 (4) 61.89

■ Analytical and Logical Reasoning

Directions (Q. 81-83) *Read the following information and answer the questions based on them.*

There are five teams P, Q, R, S and T playing in a tournament where each team plays against every other team. Each match can result in a draw where each team gets '2' points or a team winning where it gets '3' points and the other team gets '1' point.

81. If P has won all the matches and T has lost all the matches and all the remaining three teams get equal points, how many points each of three got ?
(1) 5 (2) 7
(3) 8 (4) 5 or 7

82. What is the sum of the points achieved by all the five teams in the tournaments ?
(1) 43 (2) 40
(3) Data insufficient (4) None of these

83. What is the number of points scored by each team, if all the five teams have an equal score ?
(1) 7 (2) 8
(3) Not possible (4) None of these

Directions (Q. 84-85) *Read the following information and answer the questions based on them.*

The house number in my lane starts with natural numbers *ie*, 1, 2, 3, ... Sum of all the house numbers on the left side of my house is equal to sum of all the house numbers on the right side of my house. There are more than 100 but less than 3000 houses on the right side of my house.

84. What is my house number ?
(1) 204 (2) 6929
(3) 288 (4) 7014

85. How many houses are there in the lane ?
(1) 9885 (2) 288
(3) 9801 (4) Data inademate

Directions (Q. 86-88) : *Refer to the data below and answer the questions that follow.*

There are three farmers A, B and C. All the three kept cows, hens and goats. Each farmer has a different number (either 1 or 2 or 3) of each type of animals. A had more goats, but fewer hens, than B, both of them had more than one cow. The farmer with two cows had three hens.

86. The number of cows and goats owned by 'A' is
(1) 3 (2) 4
(3) 5 (4) 6

87. B own ..... cows ..... hens and ..... goats.
(1) 1 .... 2 .... 3 (2) 3 .... 2 1
(3) 2 .... 3 .... 1 (4) 1 .... 3 2

88. C owns ..... cows, ... hens and ..... goats.
(1) 1 .... 2 ..... 3 (2) 3 .... 2 1
(3) 2 .... 3 .... 1 (4) 1 .... 3 2

89. Choose the correct alternative complete the series EFONIJKJMN....
(1) FE (2) KL
(3) OP (4) GF

Directions (Q. 90-92) *Answer the question on the basis of the information given below.*

There are 12 members in the three generation of a family Rahul has three married children and four grand children. Out of these four grand children three are males and one is a female. Karan, who is from second generation is married to Ravi's only sister. There are four couples in the family. Neha is a female from 3rd generation and has an uncle Aashu. Aakash is Ravati's grand son who has only one male sibling and two cousins. Ravi is married but does not have any children.

90. Who are the children of Aashu if Aadi and Vicky are the two more male members of the family ?
(1) Aakash and Neha
(2) Vicky and Aadi
(3) Aadi and Aakash
(4) Cannot be determined

91. Karan is Aakash's
(1) father (2) uncle
(3) brother-in-law (4) None of these

92. If Kanchan is Revati's daughter then who is Kanchan's husband ?
(1) Ravi
(2) Aashu
(3) Karan
(4) Cannot be determined

Directions (Q. 93-96) *Choose the correct alternative.*

93. In a code language if TEARS is coded as VWXYZ and MAN is coded as 123. Then how could you code RESENTMENT in that language ?
(1) YWZW3V1W3Y (2) YWZ3WV1W3Y
(3) YWZW3V1WY3 (4) YWZW3V1W3V

94. COOLER is coded as HTTQJW, what is the code for SUNDAY ?
(1) XZSIFD (2) XSIZFD
(3) YATJGE (4) WYRHEC

95. If KITCHEN is coded as 5263417 what is CHICKEN coded as ?
(1) 4321753 (2) 3423517
(3) 2435317 (4) 3423571

96. If MONEY is coded as NNODZ and if NAVY is coded as OZWX what is the code for MEAN ?
(1) NDZO (2) NFZM
(3) NDBM (4) None of these

Directions (Q. 97-101) *Read the following information carefully to answer these questions.*

A sample poll of 200 votes revealed the following information concerning three candidates A, B and C of a party who were running for three different offices.
28 in favour of both A and B
98 in favour of A or B but not C
42 in favour of B but not A or C
122 in favour of B or C but not A
64 in favour of C but not A or B
14 in favour of A and C but not B

97. How many voters were in favour of all the three candidates ?
(1) 14 (2) 8
(3) 20 (4) 16

98. How many voters were in favour of A irrespective of B or C ?
(1) 78 (2) 36
(3) 42 (4) None of these

99. How many voters were in favour of B irrespective of A or C ?
(1) 36 (2) 78
(3) 42 (4) 86

100. How many voters where in favour of only one of the candidates ?
(1) 58 (2) 78
(3) 106 (4) 142

101. How many voters were in favour of A and C but not B ?
(1) 16 (2) 14
(3) 36 (4) 42

102. There are twenty people working in an office. The first group of five works between 8:00 am and 2:00 pm. The second group of ten works between 10:00 am and 4:00 pm and the third group of five works between 12:00 noon and 6:00 pm. There are three computers in the office which all the employees frequently use. During which of the following hours the computers are likely to be used most
(1) 10:00 am-12:00 Noon
(2) 12:00 Noon-2:00 pm
(3) 1:00 pm-3:00 pm
(4) None of the above

Directions (Q. 103-104) *From a same point K, Indu and Soumya walked in opposite directions. After walking 5 km, Indu turned right and walked 3 km. She again turned right and walked 5 km. From the starting point Soumaya walked 4 km and turned left.*

103. At this point, if Indu was facing East, Soumya would be facing which direction ?
(1) North (2) South
(3) East (4) North-West

104. How far is Indu from the starting point ?
(1) 1 km (2) 2 km
(3) 3 km (4) 4 km

105. If 2 is subtracted from the middle digit of the following numbers and then the positions of the digits are reversed. Which of the following will be the last digit of the middle number after they are arranged in descending order ?
489, 362, 154, 271, 542
(1) 5 (2) 4
(3) 2 (4) 3

Directions (Q. 106-107) *Determine which sets of sentences are logically related that is the third statement follows logically from the first two.*

106. (A) Many singers are not writers. All poets are singers. Some poets are not writers.
(B) Giants climb beanstalks. Some chicken donot climb beanstalks. Some chickens are not giants.
(C) All explorers live in snowdrifts. Some Penguins live in snowdrifts. Some penguins are explorers.
(D) Amar is taller than Akbar. Anthony is shorter than Amar. Akbar is shorter than Anthony.
(1) A only (2) B only
(3) B and C (4) D only

107. (A) No sun is not white. All moon is sun. All moon is white.
(B) All windows are open. No open space is allocated. All window is closed space.
(C) No German can fire. All Americans bombard. Both Germans and Americans can fight.
(D) Aishwarya is an actress. Some actresses are beautiful. Aishwarya is beautiful.
(1) A only (2) B only
(3) C only (4) D only

108. A bird shooter was asked how many birds he had in the bag ? He replied there were all sparrows but six all pigeons but six and all ducks but six. How many birds he had in the bag in all ?
(1) 9 (2) 18
(3) 27 (4) 36

Directions (Q. 109-110) *Indicate the answer choice to the question mark.*

109. CIRCLE : RICELC :: SQUARE : ?
(1) QSUERA (2) QUSERA
(3) UQSAER (4) UQSERA

110. BUCKET : ACTVBDILDFSU : BONUS : ?
(1) ACMNMOTVRT
(2) ACNPMOTURT
(3) ACNPMOTVRT
(4) ACMNMOTURT

111. **Statement** If betting and match fixing play a vital role in cricket, which makes the game ungentlemanly and ugly, there is absolutely no use of wasting precious time to watch the game of cricket which cheats the innocent ?
Assumptions
I. The contagious disease may spray to other games.
II. Cricketers indulge in betting and match fixing and are exploiting innocent public feelings.
(1) Only I is implicit
(2) Only II is implicit
(3) Neither I nor II is implicit
(4) Both I and II are implicit

Directions (Q. 112-113) *In each question below one statement is followed by a blank space which is followed by a conclusion which can be drawn from either the statement given in the question or from the statement given in the question and an additional statement which may list in the blank space.*

112. **Statement I** The Government has to be ready with the disaster management plan to handle the natural calamities like Tsunami or Hurricane.
Blank space : .....
Conclusion It will be possible to avoid the damage to life and property if the Government is ready with a strategy in place.
(A) **Statement II** One has to know how to handle the natural calamities.
(B) **Statement II** It is not possible for the public to handle the situation arising due to natural calamities without help from the Government.
(C) **Statement II** Planning is most important aspect in handling any unforeseen situation.
(1) Only A
(2) Only B
(3) Only C
(4) No additional statement is required to draw the Conclusion.

113. **Statement I** The Government has decided to award only grades and not marks to the students up to tenth class.
Blank space : .....
Conclusion High scorers are unhappy because there will be no difference between the high scores and the moderately high scores.
(A) **Statement II** There are mixed reactions from different groups of students and parents.
(B) **Statement II** This was necessary since our examination system puts a lot of pressure on students.
(C) **Statement II** The practicle is being followed by some of the foreign universities.
(1) Only A
(2) Only B
(3) Only C
(4) No additional statement is required to draw the conclusion.

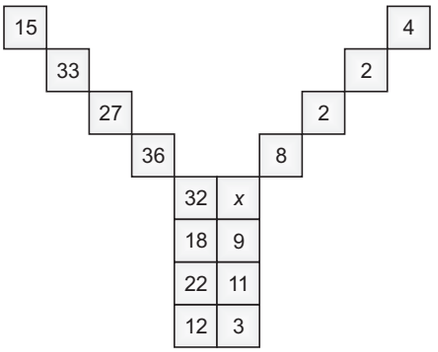
Directions (Q. 114-115) *Arrange the following in a meaningful words.*

114. (A) Fruit (B) Flower (C) Seed
(D) Pollination (E) Bud
(1) A, B, C, D, E (2) D, B, E, C, A
(3) E, B, D, C, A (4) E, B, D, A, C

115. (A) Accident (B) Judge (C) Doctor
(D) Lawyer (E) Police
(1) A, C, D, B, E (2) A, C, E, D, B
(3) A, B, C, D, E (4) A, B, E, D, C

116. If Ist May, 2000 was Monday than what was the day on 31st July of the same year ?
(1) Sunday (2) Monday
(3) Tuesday (4) Wednesday

117. Find the value of x in the given question.



(1) 3 (2) 4
(3) 8 (4) 12

Directions (Q. 118-119) *In each of the following letter series some of the letters are missing which are given, in that order as one of the alternative below it. Choose the correct alternatives.*

118. c...ac...aa...aa...bc...bcc
(1) cabba (2) ccbbb
(3) bbbbb (4) aabba

119. gfe...ig...eii...fei...gf...ii
(1) eifgi (2) figie
(3) ifgie (4) None of these

120. A is the brother of B. C is married to D. If C is the nephew of B. How is A related to D ?
(1) Father-in-law
(2) Mother-in-law
(3) Son-in-law
(4) None of the above

■ General Awareness

121. Which State of India has the largest number of child labourers ?
(1) Andhra Pradesh (2) Bihar
(3) Madhya Pradesh (4) Uttar Pradesh

122. Now-a-days, what is the approximate number of domestic tourists visiting different parts of the country ?
(1) 200 million (2) 250 million
(3) 320 million (4) 380 million

123. India has recently signed a MOU for agricultural cooperation with
(1) China (2) Nigeria
(3) Hungary (4) South Africa

124. According to the regular Budget 2009-10, outlays for Commonwealth Games, 2010 has been stepped up from Rs 2112 crore in Interim Budget 2009-10 to .....
(1) Rs 3100 crore (2) Rs 3112 crore
(3) Rs 3472 crore (4) Rs 4211 crore

125. According to the Union Budget 2009-10, revenue deficit has been projected at ..... of Gross Domestic Product (GDP) compared to 1 per cent in 2008-09 and 4.6 per cent as per provisional accounts of 2008-09.
(1) 4.2 per cent (2) 4.3 per cent
(3) 4.8 per cent (4) 5.2 per cent

126. Government of India has decided to set up two more All India Institute of Medical Sciences (AIIMS) like institutes in near future. These new institutions would be established at
(1) West Bengal and Uttar Pradesh
(2) Haryana and Punjab
(3) Punjab and Tamil Nadu
(4) Tamil Nadu and Andhra Pradesh

127. In UK, India is the second largest investor, second only to
(1) S Korea (2) US
(3) Japan (4) China

128. In 2008, India's trade with EU would be about
(1) $ 111 billion (2) $ 150 billion
(3) $ 170 billion (4) $ 200 billion

129. Identify the most expensive office markets among them, according to a recent survey
(1) Khan market
(2) Bandrakurla complex
(3) Connaught place
(4) Crawford market

130. A JV of Arcelor-mittal is to open a metal cladding unit in India at
(1) Raipur (2) Alwar
(3) Indore (4) Chennai

131. Which company has been awarded the Golden Peacock Global Award for corporate social responsibility ?
(1) Pepsico India (2) Coca-Cola India
(3) BHEL (4) L & T

132. The Direct to Home (DTH) service 'Big TV' has been launched by
(1) Aditya Birla (2) Reliance Com
(3) Star India (4) Sony Entertainment

133. Where is the headquarters of Asian Development Bank ?
(1) Jakarta (2) Manila
(3) Singapore (4) Bangkok

134. Who among the following is the author of the book "My Country, My Life" ?
(1) A.P.J. Abdul Kalam (2) Atal Bihari Vajpayee
(3) L.K. Advani (4) Shashi Tharoor

135. According to Rail Budget, a new rail coach factory will be set up in
(1) Kerala (2) Tamil Nadu
(3) Karnataka (4) Madhya Pradesh

136. Which of the following was the theme of the Olympic Torch ?
(1) Journey of Harmony
(2) Journey of Peace
(3) Green World Clean World
(4) Journey for Hunger-free World

137. The construction of railway line in Kashmir valley is planned to be completed by
(1) 2008 (2) 2009
(3) 2010 (4) 2011

138. Delimitation commission has made a recommendation that next Census should be Panchayat-wise. When is the next census due ?
(1) 2010 (2) 2011
(3) 2015 (4) 2012

139. Who has been appointed the new Prime Minister of Malaysia ?
(1) Najeeb Abdul Razaaq (2) P.M. Sayed
(3) Ahmed Abideen (4) Jainul Badavi

140. Who is the Present Attorney General of India ?
(1) Milon Benerjee (2) Pramod Batra
(3) Divaker Pandey (4) Shambhu Shiva

141. Which of the following countries signed the Tshwane Declaration in October, 2006 ?
(1) China and South Africa
(2) India and South Africa
(3) South Africa and Botswana
(4) Saudi Arabia and South Africa

142. Which one of the following Himalayan passes was reopened around in the middle of the year 2006 to facilitate trade between India and China ?
(1) Chang La (2) Jara La
(3) Nathu La (4) Shipki La

143. Basel II relates to which one of the following ?
(1) International standards for safety in civil aviation
(2) Measures against cyber crimes
(3) Measures against drug abuse by sportspersons
(4) International standards for measuring the adequacy of a bank's capital

144. NASA's Deep Impact Space Mission was employed to take detailed pictures of which comet nucleus ?
(1) Halley's comet (2) Hale-Bopp
(3) Hyakutake (4) Tempel I

145. Where was the G-20 Summit held recently ?
(1) London (2) Rome
(3) New York (4) Paris

146. The Pulitzer Prize is associated with which one of the following ?
(1) Environmental protection
(2) Olympic games
(3) Journalism
(4) Civil Aviation

147. Who among the following served as the Chief Economist of the International Monetary Fund ?
(1) Ashok Lahiri
(2) Sumantra Ghoshal
(3) Saumitra Chaudhari
(4) Raghuram Rajan

148. Where are Shevaroy Hills loated ?
(1) Andhra Pradesh (2) Karnataka
(3) Kerala (4) Tamil Nadu

149. Which one of the following pairs of countries joined the European Union in January, 2007 ?
(1) Bulgaria and Romania
(2) Bulgaria and Belgium
(3) Romania and Slovenia
(4) Hungary and Croatia

150. In the human body, which structure is the appendix attached to ?
(1) The large intestine (2) The small intestine
(3) The gall bladder (4) The stomach

151. Which one among the following rivers is the longest ?
(1) Amazon (2) Amur
(3) Congo (4) Lena

152. Which one of the following rivers originates at Amarkantak ?
(1) Damodar (2) Mahanadi
(3) Narmada (4) Tapti

153. Production of which one of the following is a function of the liver ?
(1) Lipase (2) Urea
(3) Mucus (4) Hydrocloric acid

154. Which one of the following is not a digestive enzyme in the human system ?
(1) Trypsin (2) Gastrin
(3) Ptyalin (4) Pepsin

155. Which of the following types of light are strongly absorbed by plants ?
(1) Violet and Orange (2) Blue and Red
(3) Indigo and Yellow (4) Yellow and Violet

156. White Coal is
(1) Uranium (2) Hydro-electricity
(3) Diamond (4) Ice

157. Who among the following started the newspaper 'Shome Prakash' ?
(1) Dayanand Saraswati
(2) Ishwar Chandra Vidyasagar
(3) Raja Rammohan Roy
(4) Surendranath Benerjee

158. The ruler of which one of the following States was removed from power by the British on the pretext of misgovernance ?
(1) Awadh (2) Jhansi
(3) Nagpur (4) Satara

159. Who among the following Europeans were the last to come to pre-independence India as traders ?
(1) Dutch (2) English
(3) French (4) Portuguese

160. Which one of the following is the correct sequence in the order of decreasing length of the three structure parts given below of small intestine in the human body ?
(1) Jejunum-Duodenum-Ileum
(2) Ileum-Duodenum-Jejunum
(3) Jejunum-Ileum-Duodenum
(4) Ileum-Jejunum-Duodenum

(OMR SHEET)

SNAP (Mock Test 2)

(Symbiosis National Aptitude Test)

General English

Q. No.	Response
001	① ② ③ ④
002	① ② ③ ④
003	① ② ③ ④
004	① ② ③ ④
005	① ② ③ ④
006	① ② ③ ④
007	① ② ③ ④
008	① ② ③ ④

Q. No.	Response
009	① ② ③ ④
010	① ② ③ ④
011	① ② ③ ④
012	① ② ③ ④
013	① ② ③ ④
014	① ② ③ ④
015	① ② ③ ④
016	① ② ③ ④

Q. No.	Response
017	① ② ③ ④
018	① ② ③ ④
019	① ② ③ ④
020	① ② ③ ④
021	① ② ③ ④
022	① ② ③ ④
023	① ② ③ ④
024	① ② ③ ④

Q. No.	Response
025	① ② ③ ④
026	① ② ③ ④
027	① ② ③ ④
028	① ② ③ ④
029	① ② ③ ④
030	① ② ③ ④
031	① ② ③ ④
032	① ② ③ ④

Q. No.	Response
033	① ② ③ ④
034	① ② ③ ④
035	① ② ③ ④
036	① ② ③ ④
037	① ② ③ ④
038	① ② ③ ④
039	① ② ③ ④
040	① ② ③ ④

Quantitative Aptitude, Data Interpretation and Sufficiency

Q. No.	Response
041	① ② ③ ④
042	① ② ③ ④
043	① ② ③ ④
044	① ② ③ ④
045	① ② ③ ④
046	① ② ③ ④
047	① ② ③ ④
048	① ② ③ ④

Q. No.	Response
049	① ② ③ ④
050	① ② ③ ④
051	① ② ③ ④
052	① ② ③ ④
053	① ② ③ ④
054	① ② ③ ④
055	① ② ③ ④
056	① ② ③ ④

Q. No.	Response
057	① ② ③ ④
058	① ② ③ ④
059	① ② ③ ④
060	① ② ③ ④
061	① ② ③ ④
062	① ② ③ ④
063	① ② ③ ④
064	① ② ③ ④

Q. No.	Response
065	① ② ③ ④
066	① ② ③ ④
067	① ② ③ ④
068	① ② ③ ④
069	① ② ③ ④
070	① ② ③ ④
071	① ② ③ ④
072	① ② ③ ④

Q. No.	Response
073	① ② ③ ④
074	① ② ③ ④
075	① ② ③ ④
076	① ② ③ ④
077	① ② ③ ④
078	① ② ③ ④
079	① ② ③ ④
080	① ② ③ ④

Analytical and Logical Reasoning

Q. No.	Response
081	① ② ③ ④
082	① ② ③ ④
083	① ② ③ ④
084	① ② ③ ④
085	① ② ③ ④
086	① ② ③ ④
087	① ② ③ ④
088	① ② ③ ④

Q. No.	Response
089	① ② ③ ④
090	① ② ③ ④
091	① ② ③ ④
092	① ② ③ ④
093	① ② ③ ④
094	① ② ③ ④
095	① ② ③ ④
096	① ② ③ ④

Q. No.	Response
097	① ② ③ ④
098	① ② ③ ④
099	① ② ③ ④
100	① ② ③ ④
101	① ② ③ ④
102	① ② ③ ④
103	① ② ③ ④
104	① ② ③ ④

Q. No.	Response
105	① ② ③ ④
106	① ② ③ ④
107	① ② ③ ④
108	① ② ③ ④
109	① ② ③ ④
110	① ② ③ ④
111	① ② ③ ④
112	① ② ③ ④

Q. No.	Response
113	① ② ③ ④
114	① ② ③ ④
115	① ② ③ ④
116	① ② ③ ④
117	① ② ③ ④
118	① ② ③ ④
119	① ② ③ ④
120	① ② ③ ④

General Awarness

Q. No.	Response
121	① ② ③ ④
122	① ② ③ ④
123	① ② ③ ④
124	① ② ③ ④
125	① ② ③ ④
126	① ② ③ ④
127	① ② ③ ④
128	① ② ③ ④

Q. No.	Response
129	① ② ③ ④
130	① ② ③ ④
131	① ② ③ ④
132	① ② ③ ④
133	① ② ③ ④
134	① ② ③ ④
135	① ② ③ ④
136	① ② ③ ④

Q. No.	Response
137	① ② ③ ④
138	① ② ③ ④
139	① ② ③ ④
140	① ② ③ ④
141	① ② ③ ④
142	① ② ③ ④
143	① ② ③ ④
144	① ② ③ ④

Q. No.	Response
145	① ② ③ ④
146	① ② ③ ④
147	① ② ③ ④
148	① ② ③ ④
149	① ② ③ ④
150	① ② ③ ④
151	① ② ③ ④
152	① ② ③ ④

Q. No.	Response
153	① ② ③ ④
154	① ② ③ ④
155	① ② ③ ④
156	① ② ③ ④
157	① ② ③ ④
158	① ② ③ ④
159	① ② ③ ④
160	① ② ③ ④

Symbiosis National Aptitude Test

Crack SNAP 3

ENTRANCE PATTERN

The duration for making answers will be 2 hours. Normally the pattern of written test is as follows.

Sections

I. General English
II. Quantitative Aptitude, Data Interpretation & Sufficiency
III. Analytical and Logical Reasoning
IV. General Awareness

Important Instructions

1. *Calculator/Watch calculator will not be allowed in the examination Hall.*
2. *Use only HB Pencils for darkening the ovals.*
3. *Correct method of Marking Answers*

Each questions is followed by four answer choices which are marked a, b, c, d on the answer sheet. Using HB Pencil, darken the oval bearing the correct answer against the serial number of the question.

For example : *If the answer to question number* 2 *is* (4)*, you should mark as shown below*

Q. 2. (1) (2) (3) (4)

General English

1. The following words have been used in sentences. Determine the sentences in which uses of words are correct or appropriate.
Red, Green, Blue
A. The arrival of police at the site of the accident proved red herring to the mob assembled there.
B. He has green fingers.
C. He claims descendancy from blue blood.
(1) A and B (2) A and C
(3) A, B and C (4) B and C

2. A single word for "be thin and worn out in memory of someone loved" is
(1) pine (2) lament
(3) emaciate (4) grieve

3. Find the sentence in which the following words have been wrongly used
enabled, unabled, disabled, crippled
A. He is crippled with disease .
B. He is physically disabled child.
C. He has computer enabled work experience.
D. He is mentally unabled child.
(1) A and B (2) A and C
(3) B and D (4) D only

4. Find the incorrectly spelt word among the following.
(1) Heir (2) Heirarchy
(3) Hieroglyphic (4) Hierolatry

5. Identify the odd one out.
(1) Tolerate (2) Brook
(3) Countenance (4) Persevere

Directions (Q. 6-8) *Read the passage and answer the questions given below.*

When television meant black and white pictures, and there was a two-year waiting period for a simple telephone connection, and a computer was an object to wonder at, Arther Clark's house boasted things like privately owned satellite dish antenna. He showed me amazing number of channels brought to his television. He told, "Time will come when every home will have these things. Every one will have a portable telephone, and entire house will be controlled by a computer." I thought to myself, "What planet this guy is living on ?" I remember Arther Clark today everytime I switch on my cable TV, use my mobile phone, or log on to internet. How prophetic was he ? He used to say that we always over-estimated the change that would take place in the next decade.

6. The passage seeks to convey that
(1) change in life-style is more dynamic than we foresee to happen in near-future
(2) the change in life-style is more dynamic than we foresee to happen in distant future
(3) change in life-style takes place due to television, mobile phone, etc
(4) change in life-style has taken place due to computer-controlling of all facilities

7. Arther Clark was a man of
(1) clairvoyant vision
(2) prophetic ability
(3) having all modern gadgets when these were rarely available
(4) having great liking for modern gadgets

8. Why did the author of the passage remember Arther Clark oftenly today ?
(1) Because he was enjoying several modern facilities his house was equipped with
(2) Because modern gadgets were scarcely available in those days when he had met Arther Clark
(3) Because of the prophecy of Arther Clark which has since come true
(4) Because Arther Clark was a respected person

9. Match the given word's various uses with correct applications in sentences.
Hail

A.	to call out	(i)	The govt's decision was hailed.
B.	to praise	(ii)	Hail Hitler! the Germans used to cry of welcome.
C.	to exclaim	(iii)	He hailed a taxi for going to the station.

(1) A–(ii) B–(i) C–(iii) (2) A–(iii) B–(i) C–(ii)
(3) A–(i) B–(ii) C–(iii) (4) A–(i) B–(iii) C–(ii)

10. What is the one word used to describe supposed transmigration of soul from the body of a living person after he dies ?
(1) Re-incarnation (2) Transcension
(3) Metempsychosis (4) Emigration

11. What is the closest meaning of the word "Behest" ?
(1) Diktat (2) Edict
(3) Order (4) Verdict

Directions (Q. 12-15) *Read the following passage and answer the questions given below.*

Within a matter of months, we have moved from a situation of high growth and moderate inflation to slower growth and spiralling inflation. Wholesale prices rose 6.68 per cent for the week ended March 15, its highest rate of increase over 13 months. However, unlike February 2007, the price rise this time is not driven entirely by foodstuffs. Inflation in manufactured goods touched a one-year high of 6.27 per cent for the week ended March 15. Lower interest rates are virtually ruled out when the Reserve Bank makes its next policy announcement in April.

One hopes RBI does not press the panic button and hike interest rates.That would spell disaster for an economy that is faced with declining industrial growth over the last quarter. There has also been a decline in the growth of bank credit. Bank credit for the period April 2007-February 2008 was up 16.7 per cent over the same period last year. However, the loans given out between April 2006 and February 2007 were 22 per cent higher than that disbursed in the corresponding period of the previous year. This is a sure sign that the economy is cooling, not overheating. If inflation is on a march, it is because of high commodity and food prices. Food, particularly cereals, has turned expensive all over the world, because of supply-side constraints rather than a sudden increase in demand. While diversion of cropland to biofuels in the US has led to high wheat and corn prices all over the world, wheat and rice yields have hit a plateau in India. Producers need incentives, including credit, as opposed to a credit squeeze that will hurt both demand and supply without bridging the gap.

Commodity prices could taper off, as China completes its preparations for the Olympics and the US leads the rest of the world into a slowdown. The government should wait and watch. It need not have come down heavily on commodity and farm exports. That could send wrong signals to commodity and farm producers, with the possibility of prices taking a turn for the worse. While the government's concern over inflation is understandable in an election year, its populism should be geared towards improving agriculture output. Turning the screws on credit makes sense when the productive capacities of an economy are close to being fully utilised. India is nowhere near that stage.

It should increasingly seek supply-side solutions by producing goods and services that are consumed by large sections of our population, while promoting demand through fiscal measures such as tax cuts. Inflation points to deficiencies in our production process, rather than an increase in demand. In fact, there is plenty of room for demand to rise in an economy where 22 per cent live below the poverty line. The solution is to free up supply in ever new areas by extending credit, not by denying it.

12. According to the passage the sudden crisis in economy has been caused by
(1) high rate of interests on bank credit
(2) glut in industrial products
(3) slow-down of economy growth
(4) inordinate increase in inflation rate

13. The suggested remedial measures are
A. increase in production of goods
B. increase in services which are needed by people
C. easy access to credits
D. controlling bank credits
(1) A and B (2) B and D
(3) A, B and C (4) A, B, C and D

14. What could send wrong signals to food producers ?
(1) Prices taking a turn for the worse
(2) Turning screw tight on credit policy
(3) Controlling/curbing exports of farm products
(4) Slow-down in production of farm products

15. What, according to this article, is feared to spell disaster for the economy already hit by declining industrial growth ?
(1) Hike in interest rates
(2) Further decline in industrial growth
(3) Decline in growth of bank credits
(4) Very high rate of inflation

Directions (Q. 16-17) *Choose the word which carries the same meaning of the words given under.*

16. Inflation
(1) Deflation (2) Stagflation
(3) Conflagration (4) Spiralation

17. Disaster
(1) Cataclysm (2) Distraction
(3) Disruption (4) Epiphany

Directions (Q. 18-22) *How many options carry the meaning of the given words ?*

18. Brook
(a) Waterfall (b) Tolerate
(c) Mangrove (d) Ravine
(1) a and b (2) a and c
(3) a and d (4) b and c

19. Credit
(a) Credence
(b) Honour
(c) Loan
(d) Acknowledgement of achievement
(1) a and b (2) a, b and c
(3) a, c and d (4) b, c and d

20. Fiscal
(a) Of government treasury or taxes
(b) Of financial matters
(c) Of financial accounting year
(d) Of monetary management
(1) a, b and c (2) a, b and d
(3) b, c and d (4) a, c and d

21. Orient
(a) East (b) To align
(c) Mystique (d) West
(1) a and d (2) b and c
(3) a and b (4) c and d

22. Exact
(a) Precise (b) Approximate
(c) Appropriate (d) Inflict
(1) a and b (2) a, b and c
(3) b and c (4) a and d

Directions (Q. 23-29) *Read the following passage and answer the questions given below.*

Given the dynamic, borderless world, with an extremely discerning customer at every level, product design and development is assuming an ever more important role in the future success of companies. At present, customers demand greater product variety and quickly shift to new, innovative products. Their purchase decisions are based on factors that are affected by product design. Earlier, customers made purchase decisions based primarily on product price and/or quality.

While these factors are still important, customers are looking at other dimensions, too, such as customisability, order-to-delivery time, product safety and ease and cost of maintenance. Especially in India, the business scenario has undergone a 10X change. Five years ago, industrial customers expected delivery once a day; now they demand 20 shipments a day. Earlier, they expected six new products to be developed in a year; today they expect 11.

Environmental concerns have grown to include impacts during production, during the product's operating life, and also at the end of its life (recycle-ability). In addition, customers demand greater protection from defective products, which leads to lower product liability losses.

While these seem like too many aspects to consider, when firms are able to develop distinctive products that meet these customer requirements, they have opportunities to command premium pricing. According to Professor Shoji Shiba, an expert in breakthrough management, to develop successful new products, it is important to have the Big M perspective, where, M refers to manufacturing. The Big M perspective includes societal change, environment change, technology change, regulatory change, supply chain, feedback from customer, after sales service, suppliers and so on.

I believe that Indian companies need to expand their risk-taking appetite in order to come out with new products that can help to expand their markets and bring new customers into their fold.

Therefore, it becomes imperative for the R and D team or the product design team to interact closely with the customer-facing team and the customers themselves. It is also the responsibility of other functions in the organisation to observe and absorb the weak signals from the environment and provide inputs to the product development team. In this sense, an organisation would do well to have a cross-functional team working on a new product design.

Taking inputs from these functions can help define latent customer needs—needs that the customer himself does not know exist. The product design team must focus on developing products to meet these latent needs.

What is most important is creating products based on a deep understanding of the customer. You have to begin with understanding what would be most relevant, useful or appealing to the customer. How can you arrive at an understanding of the "most relevant" when, at times, the customer himself may not know what would meet his need ?

This can be achieved by what Shiba calls the "Jumping into the Fishbowl" methodology. You must jump into the fishbowl and swim with the fish to make observations. It implies that the manager must visit customers where the product or service is being used. He must experience the environment in which the product is used and then chart out the future course with the insights gained. Sometimes the insights gained may lead the organisation to develop a completely new business, or realise a new customer group or market—it may not mean just a new product.

This exploration of new markets is what will lead to sustainable businesses. We need to change our mindsets about product design to include "new features" to a much larger picture of satisfying latent customer needs and finding new customers.

23. What is the opposite word of 'Dynamic' ?
(1) Unmoving (2) Unchanging
(3) Static (4) Sluggish

24. Which synonym best explains the word 'Discerning' ?
(1) Not concerned
(2) Analytical
(3) Deciding suitably
(4) Differing from others

25. Which one is closest in meaning to the word 'Scenario' ?
(1) Scene
(2) Scenery
(3) Site
(4) Setting of various situations

26. Which of the options closely explains the meaning of the word "Perspective" ?

(1) Ability of looking through something clearly and closely
(2) Manner of doing something
(3) Manner or way of thinking about something
(4) Correctness of situation

27. According to the above passage, the success of a new launch in the market depends upon
(1) the correct appreciation of the relevance of the product to the potential customers
(2) exploration of market
(3) finding new customers
(4) design of the product

28. Customers turn to the products available in the market on the basis of
(1) price of the product
(2) quality of the product
(3) innovative/attractive design of the product
(4) after-sale service

29. The manufacturer gets premium price of the product, if
A. design is attractive
B. made to suit individual demand
C. reduced order-to- delivery time
(1) A, B and C (2) B and C
(3) A and B (4) A and D

Directions (Q. 30-33) *Read the following passage and answer the questions given below.*

For the 59-year-old Ehsaan Qureshi and his wife Mehraam, gaming parlours were places where their children frequented. Today, the retired banker spends his evenings, with his better half, at the nearest entertainment centre in a multiplex trying his hand in "bumper cars" and "an electronic racing game." Qureshi is not alone. Gone are the days when grannies and grandpas spent their evenings taking a walk in a park of playing bridge with their cronies.

It is more likely that you would find them gliding on a dance floor or manoeuvering gaming equipment at the nearest multiplex. "I would recommend salsa or ballroom dancing for every couple that wants to pursue an activity together", beams the 54-year-old Mumbai-based Jagjeevan Pathare,who, along with his wife, joined a salsa training class in the evening. "I read about the salsa classes on the notice board of our community centre and asked my wife to accompany me to the initial sessions," he says. The Pathares are now hooked to dancing and having discovered the charm of salsa they are now planning to enrol in ballroom dancing too. Mihir Shah, a professional salsa trainer, admits that there has been a rise in the number of elderly taking up dancing as a serious hobby. He says, "It helps them in socialising and also provides easy exercise. Some enthusiasts even enter ballroom and salsa dance competitions."

India is home to over 80 million elderly, according to a report by Help Age India, expected to increase to 177 million by 2025. Arguably, this is a new market waiting to be tapped. Nandan Piramal, executive vice-chairman, Piramyd Retail, has launched its sixth family entertainment centre—Jammin—that is aimed at senior members in the family. "We are targeting grown-ups and senior members in a family with our entertainment centres. For this, we have kept the whole look and feel of the games a little mature," he adds. Piramal is looking at launching another six family entertainment centres, primarily in west and north India. He says, "We expect to garner 20 percent growth per store and will have professional gaming experts to help players."

The Oxford Institute of Ageing estimates that in India, people over 60 give over 1.3 billion voluntary hours per year, worth around Rs 15.8 billion to the economy. Minoo Shroff, a retired professor, spends close to Rs 500 each weekend, trying her hand at the local bowling alley along with a group of like-minded seniors. "I don't expect my NRI son or my daughter to look after me," she says confidently, adding "I am independent, they are independent." Shroff and her buddies carefully plan their weekends in advance, allocating money to various outdoor activities, from gaming to yoga classes.

Shroff's neighbour in Navi Mumbai, the 60-year-old Uma Paranjpe, has been practicing Tai Chi and now takes classes for the elderly. "It's an alternative to a more dynamic workout. Similar to martial arts, such as karate, the movements of Tai Chi are rooted in a warrior stance. However, like yoga, Tai Chi combines the concentration of breath and movement to encourage the body to increase efficiency and relieve the mind of stress."

Looks like some people never retire; they just change careers.

30. How many of the options are valid to mean 'Frequent' ?
(a) Often (b) Sporadic
(c) Continual (d) Visit regularly
(1) a and b (2) b and c
(3) c and d (4) a and d

31. Which of the options most closely resemble in meaning with 'Manoeuvre ?
(1) Drive (2) Manipulate
(3) Guide (4) Handle skilfully

32. Contemporary elderly persons expect
(1) that their son or daughter should look after them
(2) to join fun and dance clubs
(3) to live peacefully post retirement
(4) to live independent life

33. Now the present-day elderly people want to
(1) spend time peacefully taking evening walk and play cards with peers
(2) enjoy high-energy activities defying age
(3) join dance school or learn martial art
(4) drive sports cars

34. The under-mentioned word carries meaning in which option ?
STANCE
(1) Tack (2) Trait
(3) Posture (4) Gait

35. Which branch of medicine/medical study is related to elderly persons ?
(1) Paediatrics (2) Geriatrics
(3) Orthopaedics (4) Laryngities

36. Choose the odd one out.
(1) Look (2) Behold
(3) Ogle (4) Notice

37. Which one is a wrong pair out of the following pairs given below ?
(1) Alumnus — Alumna
(2) Baker — Baxter
(3) Confidant — Confidante
(4) Master — Miss

38. Which of the verb is wrongly paired with its noun form ?
(1) Expunge — Expunction
(2) Breathe — Breath
(3) Denounce — Denunciation
(4) Assend — Assention

39. Pick up the words with wrong spelling
(a) Beatify (b) Dinner
(c) Diner (d) Emmigrate
(1) a and c (2) a and d
(3) d only (4) c and d

40. Pick out the wrong combination of word with 'white'.
(1) white collar (2) white elephant
(3) white lie (4) white weather

■ Quantitative Aptitude, Data Interpretation and Sufficiency

41. If $f(x) = x + \frac{1}{x}$, then $[f(x)]^3$ is equal to
(1) $f(x^3) + 3f\left(\frac{1}{x}\right)$ (2) $f(x^3) + 2f(x)$
(3) $f(x^3) + f\left(\frac{1}{x}\right)$ (4) None of these

42. If a function $f : R \to R$ is defined as $f(x) = x^2 + 1$, then find f^{-1} (26), where R is the set of rational number.
(1) {2, – 2} (2) {5, 5}
(3) {– 5, 5} (4) None of these

43. Find the capacity of a 24 cm high cone having covered surface area 550 cm^2 ?
(1) 1232 cm^3 (2) 4826 cm^3
(3) 2469 cm^3 (4) None of these

44. The area of lateral faces of a cuboid is 9 cm^2, 25 cm^2 and 16 cm^2. Then its volume will be
(1) 30 cm^3 (2) 60 cm^3
(3) 3600 cm^3 (4) None of these

45. Find n, if ${}^nC_6 : {}^{n-3}C_3 = 33 : 4$.
(1) 12 (2) 11
(3) 10 (4) None of these

46. Find the number of ways in which 20 different beads can be arranged to form a necklace.
(1) 20! (2) 13!
(3) $\frac{1}{2} \cdot 19!$ (4) None of these

47. In how many ways can 5 Indian and 5 Englishmen be seated along a circle so that they can sit alternate to each other ?
(1) 5! (2) 4!
(3) 5! 4! (4) None of these

48. A telegraph pole has 5 arms and each arm is capable of 4 distinct positions including the position of rest. What is the number of signals that can be made ?
(1) 1024 (2) 1023
(3) 512 (4) None of these

49. If $\frac{a}{b} = \frac{4}{5}$ and $\frac{b}{c} = \frac{15}{16}$, then $\frac{c^2 - a^2}{c^2 + a^2}$ would be
(1) $\frac{1}{7}$ (2) $\frac{7}{25}$
(3) $\frac{3}{4}$ (4) None of these

50. If $x + \frac{1}{x} = 2$, then $x^4 + \frac{1}{x^4} = ?$
(1) 2 (2) – 2
(3) 4 (4) None of these

51. Two urns contains 1 white, 6 red and 4 white, 8 red balls. One of the urn is selected at random and a ball is drawn from it. Find the probability of drawing a white ball.
(1) $\frac{5}{14}$ (2) $\frac{3}{17}$
(3) $\frac{1}{14}$ (4) None of these

52. Two balls are drawn at random with replacement from a box containing 10 black and 8 red balls. Find the probability that both balls are red.
(1) $\frac{4}{5}$ (2) $\frac{16}{81}$
(3) $\frac{1}{4}$ (4) None of these

53. A wine dealer has two kinds of mixture of wine. The first mixture contains 40% water and the second 20%. The dealer mixes 90 gallons of the first mixture with 210 gallons of the second. What is the percentage of water in the final mixture ?
(1) 20% (2) 25%
(3) 26% (4) None of these

54. A man can row 5 km/h in still water. If the river is running at 1.5 km/h, it takes him 1 hour to row to a place and back. How far is the place ?
(1) 2.275 km (2) 22.75 km
(3) 2 km (4) None of these

55. A man can row downstream at the rate of 14 km/h and upstream at the rate of 5 km/h. Find man's rate in still water and the rate of current.
(1) 9.5 km/h and 4.5 km/h
(2) 3 km/h and 2 km/h
(3) 1.5 km/h and 1.5 km/h
(4) None of the above

56. If α and β are roots of the equation $2x^2 - 4x + 9 = 0$, then value of $\alpha^3 + \beta^3$ will be
(1) 19 (2) – 19
(3) 17 (4) None of these

57. If 16th term of AP is 24 and 24th term of AP is 16, then 40th term of AP will be
(1) 0 (2) 24
(3) 40 (4) None of these

58. If $\frac{a}{b} = \frac{4}{9}$, then find the value of $\frac{9a^2 + 4b^2}{9a^2 - 4b^2}$.

(1) 13 (2) 1
(3) $-\frac{13}{5}$ (4) None of these

59. If $4^{1+x} + 4^{1-x} = 10$, then x will be
(1) $2, \frac{1}{2}$ (2) $\frac{1}{2}, -\frac{1}{2}$
(3) $-2, \frac{1}{2}$ (4) None of these

60. In how many ways can 10 examination papers be arranged so that the best and the worst paper never come together ?
(1) $8 \cdot 9!$ (2) $8!$
(3) $9!$ (4) None of these

61. If x, y, z are positive real numbers, then $\sqrt{x^{-1}y} \cdot \sqrt{y^{-1}z} \cdot \sqrt{z^{-1} \cdot x}$ is equal to
(1) 1 (2) 0
(3) xyz (4) None of these

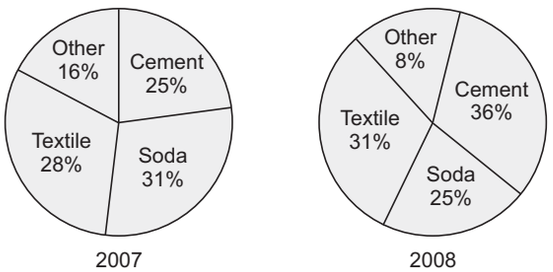
62. A, B and C became partners in a trade. A and B invested Rs 700 and Rs 840 respectively. Out of a total profit of Rs 1520, if C got Rs 640, then what was the investment of C ?
(1) 1120 (2) 120
(3) 1570 (4) None of these

63. The ages of Madhu and Seema are in the ratio of 5 : 2 respectively. After 7 yr the ratio of their ages will be 4 : 3. What is the age of Madhu ?
(1) 10 yr (2) 5 yr
(3) 6 yr (4) 12 yr

64. A job is completed by 10 men in 20 days and by 20 women in 15 days. How many days will it take for 5 men and 10 women to finish that work ?
(1) $17\frac{1}{2}$ (2) $17\frac{1}{7}$
(3) 17 (4) $17\frac{1}{120}$

65. A trader earns 20% profit on selling price. He reduces the price by 10%. Calculate his net profit per cent on cost price.
(1) $12\frac{1}{2}$ (2) 10%
(3) 16% (4) None of these

Directions (Q. 66–70) *The percentage of revenue that comes from the four different product segments for a diversified manufacturing company are shown in the following pie-chart. Total sale in 2007 was 120 crore and in 2008 was 150 crore. Answer the questions on the basis of the pie-chart given below.*



66. Ratio of the sale of other products in the two years will be
(1) 3 : 5 (2) 8 : 5
(3) 1 : 5 (4) None of these

67. What is the difference in the amount of revenue earned from other products during the two years ?
(1) 3.8 crore (2) 4.8 crore
(3) 3.2 crore (4) None of these

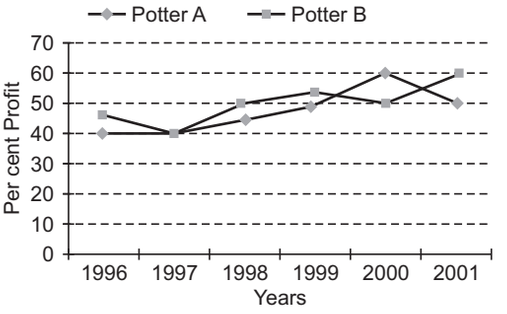
68. What is the ratio of revenue received from other products during the two years ?
(1) 5 : 4 (2) 2 : 3
(3) 1 : 3 (4) None of these

69. Least change in percent revenue occurs in which of the following sector ?
(1) Cement (2) Textile
(3) Soda (4) None of these

70. What is the ratio of revenue earned from other sector to that of cement in 2008 ?
(1) 1 : 9 (2) 2 : 7
(3) 2 : 9 (4) None of these

Directions (Q. 71-75) *Study the graph carefully to answer the following questions.*

Per cent profit earned by two Potters over the years.



$$\text{Per cent Profit} = \frac{\text{Selling Price} - \text{Cost Price}}{\text{Cost Price}} \times 100$$

71. If Potter A sold 3,650 pots in the year 1999 at the price of Rs 12 per pot, what is the cost price of all the pots together ?
(1) Rs 22,600 (2) Rs 28,500
(3) Rs 25,300 (4) Rs 29,200

72. If the amount of profit earned by Potter B in the year 1996 is Rs 40500, what was the selling price of all the pots in that year ?
(1) Rs 1,30,500 (2) Rs 2,06,200
(3) Rs 1,44,500 (4) Rs 1,87,500

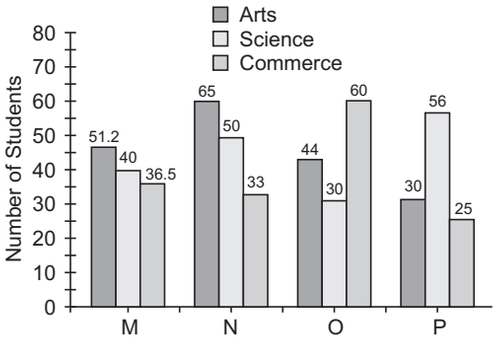
73. If in the year 2000, the total selling price of Potter B is equal to the total cost price of Potter A, what is the respective ratio of their selling prices ?
(1) 7 : 8 (2) 5 : 8
(3) 5 : 7 (4) 7 : 9

74. Which of the following statement(s) is/are definitely true ?
(1) Amount of profit earned by Potter B in the year 1999 and 2000 is the same.
(2) Potter A earned the highest amount of profit in the year 2000.
(3) Amount of profit earned by Potter A and B is the same in the year 1997.
(4) None of the above

75. What is the approximate per cent rise in per cent profit of Potter B in the year 2001 from the previous year ?
(1) 16% (2) 4%
(3) 9% (4) 19%

Directions (Q. 76-80) *Study the following graph carefully and answer the questions given below it.*

Number of Students Studying in Various Colleges from Various Faculties (Number in thousands)



76. What is the difference between the total number of students studying in college M and those studying in college P ?
(1) 16100 (2) 15800
(3) 16300 (4) 16700

77. What is the total number of students studying in all the colleges together ?
(1) 520900 (2) 520700
(3) 610200 (4) 510800

78. What is the respective ratio of the students from the faculty of Science from colleges M and N together to the students from the same faculty from colleges O and P together ?
(1) 43 : 45 (2) 41 : 43
(3) 45 : 43 (4) 43 : 41

79. The number of students from the faculty of Science from college N are approximately, what per cent of the total number of students studying in that college ?
(1) 34 (2) 37
(3) 29 (4) 31

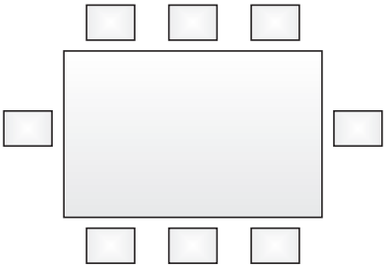
80. What is the average number of students from the faculty of commerce from all the colleges together ?
(1) 36825 (2) 38655
(3) 38625 (4) 36585

■ Analytical and Logical Reasoning

Directions (Q. 81–83) *Study the following information carefully and answer the questions given below.*

A, B, C, D, E, F, G and H are eight friends sitting along the length and the breath of the rectangular table as shown below.

A, H and D are sitting on the same side of the table. A is sitting between H and D. B is sitting adjacent to G, F is sitting opposite E. A is sitting opposite G, F and H and are not sitting on the same side but H is sitting on the right of F.



81. Who is sitting on the immediate right of G ?
(1) B (2) C
(3) B or C (4) Data inadequate

82. Who is sitting opposite C ?
(1) H (2) D
(3) H or D (4) None of these

83. According to the positions of the persons three of the following four pairs are alike and hence form a group. Which of the following does not belong to that group ?
(1) HF (2) DA
(3) ED (4) FB

Directions (Q. 84–86) *These questions are based on the statements given below.*

Madhu and Shivani are good in Dramatics and Computer Science.
Asha and Madhu are good in Computer Science and Physics.
Asha, Pratibha and Namita are good in Physics and History.
Namita and Asha are good in Physics and Mathematics.
Pratibha and Shivani are good in Dramatics and Computer Science each. One person is not good in Physics and History each. Two persons are not good in Mathematics.

84. Who is good in Physics, History and Mathematics, but not in Computer Science ?
(1) Pratibha (2) Asha
(3) Madhu (4) Namita

85. Who is good in History, Physics, Computer Science and Mathematics ?
(1) Namita (2) Asha
(3) Madhu (4) None of these

86. Who is good in Physics, History and Dramatics ?
(1) Pratibha (2) Madhu
(3) Shivani (4) Asha

87. After a get together every person present shakes the hand of every other person. If there were 105 hand shakes in all. How many persons were present in the party ?
(1) 14 (2) 13
(3) 15 (4) 16

88. There are eight poets A, B, C, D, E, F, G and H. A, B, C and D are Medieval Poets whereas E, F, G and H are Modern Poets. Questions on Modern Poets and Medieval Poets are set in the question paper every alternate year. Among the Modern Poets, question setters who like E also like F and those who like G also like H. The question-setter has written a book on F and so he does not like to ask any question on F. Last year there was a question on A. On which poet will be the probable question this year ?
(1) F (2) E
(3) D (4) C

89. A one rupee coin is placed on a plain paper. How many coins of the same size can be placed round it so that each one touches the central and adjacent coins ?
(1) 3 (2) 4
(3) 7 (4) 6

90. 40 # 18 @ 2 Δ 10 • 130 = 0,
then which of the following shows the correct meaning of #, @ Δ and • ?
(1) + − × ÷ (2) − + ÷ ×
(3) × − + ÷ (4) + ÷ × −

Directions (Q. 91–95) *Study the following information carefully and answer the questions given below.*

A, B, C, D, E and F are six persons. They study in different colleges. These colleges are P, Q, R, S, T and U. They belong to different states J, K, L, M, N and O.

(i) A, E and the person who studies in the college 'P' are vegetarians, whereas the persons who belong to states K and N and the person who studies in the college R are non-vegetarians.
(ii) Neither A nor E belongs to state J or L. Neither B nor C studies in the college P or R.
(iii) One who belongs to the state K studies in the college Q, B and the person who studies in the college S are friends.
(iv) A does not study in the college T, the person who belong to the state M does not study in the college U. One who belong to state N studies in the college S.

91. D belongs to which of the following states ?
(1) J (2) L
(3) either J or L (4) eitehr J or K

92. Who among the following persons is studying in the college R ?
(1) D (2) F
(3) B (4) Data inadequate

93. Who among the following persons is associated with the state N ?
(1) B (2) C
(3) A (4) E

94. Which of the following statements is definitely false ?
I. D belongs to state J.
II. F belongs to state L.
III. B belongs to state N.
(1) All I, II and III (2) Only I and II
(3) Only I and III (4) Only III

95. Which of the following statements is definitely true ?
I. A and E study in the colleges U and T respectively.
II. D and F study in the colleges P and R respectively.
III. B and C study in the colleges Q and S respectively.
(1) All I, II and III (2) Only I and II
(3) Only I and III (4) Only II and III

96. In a month of 31 days, the third Wednesday falls on the 15th. Which will be the last day of that month ?
(1) Fifth Thursday (2) Fifth Wednesday
(3) Fourth Sunday (4) Fifth Friday

97. In a party every one gave gift to every one else. If the total number of gifts exchanged in the party was 600. How many persons were there in the party ?
(1) 20 (2) 15
(3) 10 (4) 25

98. In a get together, every person present shakes hand with every other person. If there were 105 handshakes in all, how many persons were present in the party ?
(1) 15 (2) 14
(3) 13 (4) 16

99. In a class of 35 students, Kiran is placed 7th from the bottom whereas Mohan is placed 9th from the top. Sohan is placed exactly in between Kiran and Mohan. What is Kiran's position from Sohan ?
(1) 13th (2) 11th
(3) 10th (4) 9th

100. The letters skipped between adjacent letters is in the order of 1, 2, 3, 4, … . Which alternative follows this rule ?
(1) DEIMR (2) DFINR
(3) DFIMR (4) DFIMS

Directions (Q. 101–102) *Refer the data below and answer the questions that follows.*

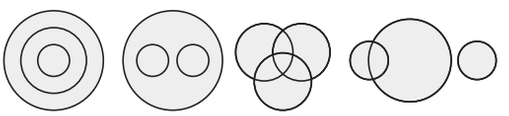
Recardo buys 800 bags of sand. All bags weigh 10 gms except one bag. To determine the bag with lesser weight he uses a two pan balance.

Jason buys 80 bags of sand. All bags except one weighs 100 gms. To determine the bag with lesser weight he uses a spring balance.

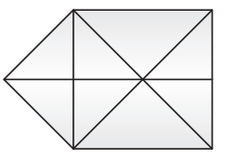
101. What is the minimum number of weighings that Recardo requires to determine the faulty bag ?
(1) 6 (2) 7
(3) 8 (4) 10

102. What is the minimum number of weighings that Jason requires to determine the faulty bag ?
(1) 6 (2) 7
(3) 9 (4) 16

Directions (Q. 103–107) *Choose the venn diagram which best illustrates the three given classes in each question.*


(1) (2) (3) (4)

103. Girl, Athlete, Singer
104. Window, Room, Wall
105. State, Country, City
106. Copper, Paper, Wire
107. Teacher, Graduate, Player
108. Count the number of triangles in the following figure.


(1) 15 (2) 16
(3) 17 (4) 18

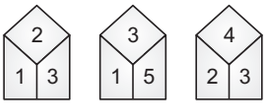
109. Find the missing character in the given questions.

BD_3	CE_5	DF_{15}
EG_2	FH_4	GI_8
HJ_4	IK_6	?

(1) JL_{24} (2) IJ_{18}
(3) JK_{18} (4) JL_{12}

110. Nimmi returned home three days earlier than the time she had told her mother. Nimmi's sister Vimmi reached five days later than the day Nimmi was supposed to return. If Nimmi returned on Thursday on what day did Vimmi return ?
(1) Tuesday (2) Wednesday
(3) Saturday (4) Friday

111. What should be the number opposite 3 ?



(i) (ii) (iii)

(1) 1 (2) 6
(3) 5 (4) 4

112. How many such pairs of digits are there in the number 148958799 each of which has as many digits between them in the number as in the given number sequence ?
(1) Less than 5 (2) 5
(3) 6 (4) More than 7

113. Three of the following five are alike in a certain way and hence form a group. Which is the one that does not belong to that group ?
(1) Paddy (2) Pea
(3) Grain (4) Floor

114. Mr. 'X' started walking towards North form a point P. After walking 20 m he turned right and walked 24 m. Again he turned right and walked 15 m. Again he turned right and walked 20 m. Again he turned left and walked 40 m. Again he turned left and walked 20 m. Again, he turned left and walked 10 m and reached a point Q. Assume that every time he walked in a straight line. Approximetly what is the shortest distance between the two points P and Q ?
(1) 26 m (2) 31.8 m
(3) 34.65 m (4) 48.9 m

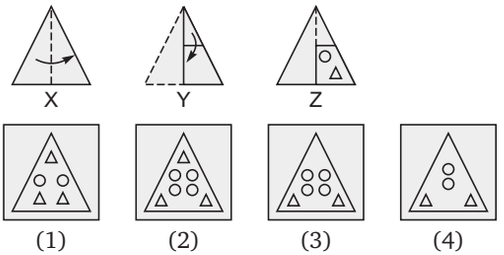
115. As a part of a delicately balanced system the human heart secretes a hormone, a substance that controls the amount of salt in the blood and the volume of blood circulating within the body. Only very small quantities of the hormone are required. The hormone is extremely important in regulating blood pressure and is found in large amounts in the blood of those suffering a heart attack.
If the statements above are true, then it must also be true that.
(1) Any drug that regulates blood pressure will have its effect by influencing the amount of the heart hormone secreted.
(2) It is large quantities of the heart hormone that cause heart attacks to occur.
(3) If a device that is only a mechanical pump is used as an artificial heart, it will not perform all the functions of the human heart.
(4) The effects of a small amount of the heart hormone will be long-lasting in the body.

116. Recent surveys show that many people who seek medical help are under a great deal of stress. Medical research also shows that stress can adversely affect an individual's immune system which is responsible for combating many infections. Thus when a person is under stress, she or he is more likely to be come ill. Which of the following, if true, would most strengthen the conclusion above ?
(1) Most people report that being in a hospital is a stressful situation.
(2) Many businesses that provide health insurance for their employees also provide seminars on Stress Management.
(3) There is a marked increase in the number of illness treated at college infirmaries around the time of examinations.
(4) There is a marked decrease in the number of complaints presented at college infirmaries during vacation times.

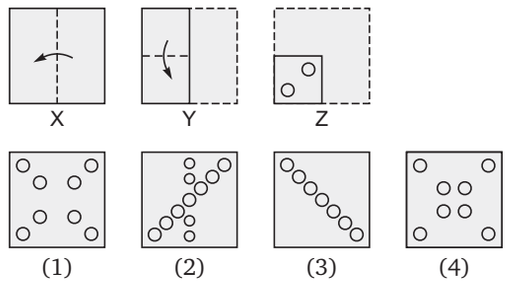
117. Pointing to Kedar, Veena said, his mother's brother is the father of my son Nitin. How is kedar related to Veena ?
(1) Niece (2) Aunt
(3) Nephew (4) None of these

Directions (Q. 118–119) *A paper sheet has been folded in the manner as shown in X, Y and Z respectively and punched. You have to choose from the alternatives how it will look when unfolded ?*

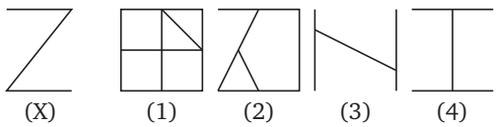
118.



119.



120. In the question given below, figure (X) is in one of the following figures. Find out the correct alternative.



General Awareness

121. Nand Lal Bose, Manjit Bawa, Tyeb Mehta are
(1) Classical singers (2) English poets
(3) Photographers (4) Painters

122. Major oil fields have been reported in
(1) Krishna-Godavari Basin
(2) Cauvery Basin
(3) Rajasthan Desert
(4) Lower Assam Valley

123. When did Governor General's rule end in India ?
(1) 15th August 1947 (2) 9th August 1948
(3) 26th January 1950 (4) 2nd October 1950

124. Who amongst the following is the winner of Ramon Magsaysay Award 2008 ?
(1) Dr. Prakash and Mandakini Amte
(2) Dr. Amartya Sen
(3) Ms. Barkha Dutt
(4) None of the above

125. What is the official language of Argentina ?
(1) Portguese (2) Spanish
(3) Italian (4) German

126. The function of DNA in the body is
(1) to help in the synthesis of proteins
(2) to control the heredity
(3) to assist in the release of energy
(4) None of the above

127. A 'bear' on the Stock Exchange in a speculator
(1) who sells shares which he does not possess
(2) who applies for new issues of shares in the hope that the price will go up
(3) whose holding of the shares in a company is so large that sellng them could affect the market price
(4) who buys shares in the hope that price will go up

128. Who was the composer of the classical composition 'Moonlight Sonata' ?
(1) Ludwig Van Beethoven
(2) Joseph Haydn
(3) Johann Bach
(4) Handel

129. The 'Interest Rate Policy' is a component of
(1) Direct control (2) Fiscal policy
(3) Trade policy (4) Monetary policy

130. Starbucks, the coffee chain retailer's name is inspired from which book
(1) The Scarlet letter (2) Modydick
(3) Uncle Tom's cabin (4) None of these

131. Who has been appointed as chairman and managing director of Rural Electrification Corporation ?
(1) R.P. Singh (2) G.N. Bajpai
(3) P. Uma Shankar (3) None of these

132. Senior Advocate who, on June 15, 2009 was appointed Solicitor General by the Union Government is
(1) Abhishek Solanki
(2) Subrata Dey
(3) Gopal Subramaniam
(4) Manohar Singh

133. The 2009 SEED Awards have been recognised in the field of
(1) Communication (2) Entrepreneurship
(3) Energy (4) None of these

134. As per revised estimates for the year 2008-09, released by CSO, GDP registers growth of
(1) 6.4% (2) 6.5%
(3) 6.7% (4) 6.9%

135. In which year was "Rajiv Gandhi Grameen Vidyutikaran Yojana" launched ?
(1) 2004 (2) 2005
(3) 2006 (4) 2007

136. Which of the following countries has the second-largest rail networks in the world ?
(1) China (2) India
(3) USA (4) Russia

137. Position of India in world telecommunication network is
(1) First (2) Second
(3) Third (4) Fourth

138. What is the budget allocation for Sarva Shiksha Abhiyan ?
(1) Rs 8000 crore (2) Rs 13100 crore
(3) Rs 4554 crore (4) Rs 16534 crore

139. The Rashtriya Swasthya Bima Yojana for every worker of unorganised sector will be launched in
(1) Delhi (2) Haryana
(3) Rajasthan (4) All of these

140. The national programme for the elderly will be started in 2008-09 with a plan outlay of
(1) Rs 500 crore (2) Rs 300 crore
(3) Rs 400 crore (4) Rs 200 crore

141. Centre of DNA fingerprinting is located at
(1) Bengaluru (2) New Delhi
(2) Pune (4) Hyderabad

142. Cryogenic engines are used in
(1) Atomic reactors
(2) Rockets
(3) Defrost refrigerators
(4) Doing research connected with superconductivity

143. Threat of Global Warming is increasing due to increasing concentration of
(1) Nitrous oxide (2) Ozone
(3) Sulphur Dioxide (3) Carbon dioxide

144. The beneficiary states of Sardar Sarovar Project are
(1) Gujarat, M.P., A.P. and Maharashtra
(2) A.P., M.P., Gujarat and U.P.
(3) M.P., Gujarat, Karnataka and Maharashtra
(4) Orissa, M.P., Gujarat and Maharashtra

145. EI Nino is
(1) A sea storm
(2) A warm ocean current
(3) A tropical disturbance
(4) Another name of typhoon

146. The World Trade Organization (WTO) was earlier known as
(1) UNICEF (2) GATT
(3) UNCTAD (4) FAO

147. The outcome of 'devaluation of currency' is
(1) increased export and foreign reserve deficiency
(2) increased export and improvement in balance of payments
(3) increased import and improvement in balance of payments
(4) increased export and import

148. The common currency which has been introduced among 11 European Nations is known as
(1) Euro (2) Euro Pound
(3) Euro Dollar (4) None of these

149. Nuclear reactors used to produce electricity are based on
(1) Nuclear fusion
(2) Nuclear fission
(3) Cold fusion
(4) Superconductivity

150. Which bank has launched a new health cover scheme under the name of 'Arogya Shree' ?
(1) PNB (2) SBI
(3) Andhra Bank (4) OBC

151. ASEAN, the major trade block of the world, consists of the following nations
(1) Indonesia, Malaysia, Thailand, Philippines and Singapore
(2) India, Bangladesh, Myanmar, Sri Lanka and Pakistan
(3) Indonesia, India, Thailand, Malaysia and Singapore
(4) Vietnam, Thailand, Mauritius, Singapore and Philippines

152. The states through which the Cauvery river flows is
(1) Kerala, Karnataka and Tamil Nadu
(2) Andhra Pradesh, Karnataka and Tamil Nadu
(3) Kerala, Andhra Pradesh and Tamil Nadu
(4) Maharashtra, Karnataka and Tamil Nadu

153. In India, National Income is estimated by
(1) Central Statistical Organisation
(2) Planning Commission
(3) Indian Statistical Institute
(4) National Sample Survey Organisation

154. 'Peace Mission 2007' was the anti-terrorism exercise of
(1) NATO
(2) Shanghai Coop. Organisation
(3) EU
(4) BRIC

155. Who among the following became the first Prime Minister of Nepal after it got the status of a democratic republic nation ?
(1) Ram Baran Yadav
(2) Subhash Nembang
(3) Pushpa Kamal Dahal (Prachanda)
(4) G.P. Koirala

156. Which of the following airlines has entered into a deal with BCCI
(1) Air India (2) Spice jet
(3) Kingfisher (4) Jet Airways

157. Which BPO company of Indian has been recently listed on New York Stock Exchange ?
(1) Genpact (2) Progeon
(3) Nipun (4) Convergys

158. ONGC-Mittal has recently won a gas block in
(1) Russia (2) Venezuela
(3) Trinidad and Tobago (4) Libya

159. Which IT company will provide event management system to FIFA ?
(1) Infosys (2) Satyam
(3) Microsoft (4) TCS

160. A third petroleum refinery is being built at 'Ceyhan', which is in
(1) Libya (2) Iran
(3) Turkey (4) UAE

(OMR SHEET)

SNAP (Mock Test 3)

(Symbiosis National Aptitude Test)

General English

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
001	① ② ③ ④	009	① ② ③ ④	017	① ② ③ ④	025	① ② ③ ④	033	① ② ③ ④
002	① ② ③ ④	010	① ② ③ ④	018	① ② ③ ④	026	① ② ③ ④	034	① ② ③ ④
003	① ② ③ ④	011	① ② ③ ④	019	① ② ③ ④	027	① ② ③ ④	035	① ② ③ ④
004	① ② ③ ④	012	① ② ③ ④	020	① ② ③ ④	028	① ② ③ ④	036	① ② ③ ④
005	① ② ③ ④	013	① ② ③ ④	021	① ② ③ ④	029	① ② ③ ④	037	① ② ③ ④
006	① ② ③ ④	014	① ② ③ ④	022	① ② ③ ④	030	① ② ③ ④	038	① ② ③ ④
007	① ② ③ ④	015	① ② ③ ④	023	① ② ③ ④	031	① ② ③ ④	039	① ② ③ ④
008	① ② ③ ④	016	① ② ③ ④	024	① ② ③ ④	032	① ② ③ ④	040	① ② ③ ④

Quantitative Aptitude, Data Interpretation and Sufficiency

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
041	① ② ③ ④	049	① ② ③ ④	057	① ② ③ ④	065	① ② ③ ④	073	① ② ③ ④
042	① ② ③ ④	050	① ② ③ ④	058	① ② ③ ④	066	① ② ③ ④	074	① ② ③ ④
043	① ② ③ ④	051	① ② ③ ④	059	① ② ③ ④	067	① ② ③ ④	075	① ② ③ ④
044	① ② ③ ④	052	① ② ③ ④	060	① ② ③ ④	068	① ② ③ ④	076	① ② ③ ④
045	① ② ③ ④	053	① ② ③ ④	061	① ② ③ ④	069	① ② ③ ④	077	① ② ③ ④
046	① ② ③ ④	054	① ② ③ ④	062	① ② ③ ④	070	① ② ③ ④	078	① ② ③ ④
047	① ② ③ ④	055	① ② ③ ④	063	① ② ③ ④	071	① ② ③ ④	079	① ② ③ ④
048	① ② ③ ④	056	① ② ③ ④	064	① ② ③ ④	072	① ② ③ ④	080	① ② ③ ④

Analytical and Logical Reasoning

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
081	① ② ③ ④	089	① ② ③ ④	097	① ② ③ ④	105	① ② ③ ④	113	① ② ③ ④
082	① ② ③ ④	090	① ② ③ ④	098	① ② ③ ④	106	① ② ③ ④	114	① ② ③ ④
083	① ② ③ ④	091	① ② ③ ④	099	① ② ③ ④	107	① ② ③ ④	115	① ② ③ ④
084	① ② ③ ④	092	① ② ③ ④	100	① ② ③ ④	108	① ② ③ ④	116	① ② ③ ④
085	① ② ③ ④	093	① ② ③ ④	101	① ② ③ ④	109	① ② ③ ④	117	① ② ③ ④
086	① ② ③ ④	094	① ② ③ ④	102	① ② ③ ④	110	① ② ③ ④	118	① ② ③ ④
087	① ② ③ ④	095	① ② ③ ④	103	① ② ③ ④	111	① ② ③ ④	119	① ② ③ ④
088	① ② ③ ④	096	① ② ③ ④	104	① ② ③ ④	112	① ② ③ ④	120	① ② ③ ④

General Awarness

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
121	① ② ③ ④	129	① ② ③ ④	137	① ② ③ ④	145	① ② ③ ④	153	① ② ③ ④
122	① ② ③ ④	130	① ② ③ ④	138	① ② ③ ④	146	① ② ③ ④	154	① ② ③ ④
123	① ② ③ ④	131	① ② ③ ④	139	① ② ③ ④	147	① ② ③ ④	155	① ② ③ ④
124	① ② ③ ④	132	① ② ③ ④	140	① ② ③ ④	148	① ② ③ ④	156	① ② ③ ④
125	① ② ③ ④	133	① ② ③ ④	141	① ② ③ ④	149	① ② ③ ④	157	① ② ③ ④
126	① ② ③ ④	134	① ② ③ ④	142	① ② ③ ④	150	① ② ③ ④	158	① ② ③ ④
127	① ② ③ ④	135	① ② ③ ④	143	① ② ③ ④	151	① ② ③ ④	159	① ② ③ ④
128	① ② ③ ④	136	① ② ③ ④	144	① ② ③ ④	152	① ② ③ ④	160	① ② ③ ④

Symbiosis National Aptitude Test

Crack SNAP 4

ENTRANCE PATTERN

The duration for making answers will be 2 hours. Normally the pattern of written test is as follows.

Sections

I. General English
II. Quantitative Aptitude, Data Interpretation & Sufficiency
III. Analytical and Logical Reasoning
IV. General Awareness

Important Instructions

1. *Calculator/Watch calculator will not be allowed in the examination Hall.*
2. *Use only HB Pencils for darkening the ovals.*
3. *Correct method of Marking Answers*

Each questions is followed by four answer choices which are marked a, b, c, d on the answer sheet. Using HB Pencil, darken the oval bearing the correct answer against the serial number of the question.

For example : *If the answer to question number* 2 *is* (4)*, you should mark as shown below.*

Q. 2. (1) (2) (3) (4)

General English

Directions (Q. 1-8) *Read the passage and answer the questions given below.*

Salman Rushdie has now revealed what many had suspected. He had only pretended to embrace Islam in 1990 in order to defuse the 'Satanic Verses' controversy that was raging at the time. In the preceding year Ayatollah Ruhollah Khomeini, supreme ruler of Iran, had issued a fatwa to kill Rushdie as well as the publishers of 'Satanic Verses'. Hitoshi Igarashi, the Japanese translator of the book, was stabbed to death. Its Italian translator and Norwegian publisher were grievously injured in separate assassination attempts.

There is no doubt that Rushdie's profession of belief was extracted under duress. In a globally interconnected world, however, everyone is bound to have exposure to faiths other than his own, or even non-faith. In this context the best way to protect faith has to be through freely chosen belief rather than coerced conformity, whose negative effects are many. The commitment of those who have freely chosen a faith will be many times that of people who have no option. In a world where no one faith is dominant, such coercion can also invite a backlash. For a peaceful world, religious pluralism is the best option. But more fundamentally religion belongs to the spiritual domain. Whoever brings force and violence into the propagation of a religion is bound to falsify it.

That's why those who drive painters like M F Husain or writers like Taslima Nasreen out of India do a disservice to any religion they claim to be defending. Compared to today's religious fundamentalists India's Mughal rulers had a canny sense of the benefits of freely chosen faith, which is why they didn't try to get the majority of their subjects to follow their faith even when they had the power to do so. Akbar, for example, issued an edict that a Hindu who had been made a Muslim against his will should be allowed to go back to his former religion.

Hindu fundamentalists like to point to Aurangzeb, the exception who proves the rule of Mughal tolerance. Aurangzeb, it must not be forgotten, is a usurper in the Mughal tradition. He killed his elder brother Dara Shikoh, who was in the line of succession for the throne as well as a connoisseur of Hindu scriptures. Aurangzeb also placed his father, Shah Jahan, under life imprisonment. That Aurangzeb presided over the demise of the Mughal empire reinforces the point about the superiority of freely chosen faith. Iran's ayatollahs may have a thing or two to learn from India's Mughal emperors who preceded Aurangzeb.

1. What does the following sentence as contained in the passage convey ?
Rushdie embraced Islam in 1990 in order to defuse 'Satanic Verses' controversy that was raging at that time.
(1) Rushdie became a Muslim only to explode public ire to boost sale of his book on the controversy bandwagon.
(2) Rushdie embraced Islam hoping to blunt the ire of the Muslims over controversial contents written in the book.
(3) Rushdie embraced Islam to end the controversy, which led to the burning of his book, 'Satanic Verses' by the muslim mob.
(4) Rushdie changed his faith to Islam because he wrote the controversial book, 'Satanic Verses'.

2. What does the following sentence in the passage mean ?
There was no doubt that Rushdie's profession of belief was extracted under duress.
(1) The changing belief in religion is the profession of Rushdie.
(2) Rushdie was coerced into changing his religious profession.
(3) No doubt, Rushdie pretended to believe in Islam due to public threat.
(4) No doubt, public forced Rushdie to embrace Islam.

3. In the present era of religious plurality, faith in a religion forced upon someone is fraught with
(1) his exposure to another religion
(2) his turning atheist
(3) many negative effects
(4) forced conformity

4. Superiority of freely chosen religious faith manifests itself in
(1) Emperor Akbar allowing Hindus, who were forcibly converted into Islam, to return to their faith.
(2) the fall of Mughal empire during Aurangzeb regime.
(3) spontaneous and unflinchising commitment of adherents.
(4) plurality of religions.

5. Overzealous vigilantes of a religion do a great disservice to the cause by
(1) driving MF Hussain and his likes out of the country
(2) showing and indulging in acts of intolerance to non-conformists
(3) indulging in hooliganism
(4) denying religion its spiritual domain

6. Choose the word having close resemblance in meaning with the word
CONFORM
(1) Comply (2) Confound
(3) Obey (4) Resemble

7. Choose the synonym of the word
OPTION
(1) Alternative (2) Choice
(3) Selection (4) Freedom

8. What is the antonym of
CONNOISSEUR
(1) Reconnoitre (2) Coiffeuse
(3) Laity (4) Gourmet

Directions (Q. 9-10) *Arrange the following jumbled sentences into meaningful passages.*

9. A. India and Africa wrongly preferred industries at the cost of agriculture.
B. Fertile soils of India and Africa are favourable for cultivation of every important crop.
C. These countries are struggling to fight the menace of food shortage and spiralling prices of foodgrains.
D. India and Africa have the potential of becoming granaries of the world.
E. Both India and African nations must undergo a paradigm shift in thinking about agriculture.
(1) A, B, D, C, E (2) A, B, C, D, E
(3) E, A, C, B, D (4) D, B, A, C, E

10. A. The guest invited to the party got the perk of being punctual.
B. The invitation card mentioned 8 a.m. as the time of the party.
C. Those who reached on time found things were not quite as expected.
D. A few busy-for-nothing people were darting in and out of the hall.
E. Joyous laughter of those who came late made punctual guests feel being stupid.
(1) BCDEA (2) BDACE
(3) ECDBA (4) ECBDA

11. Find out the sentence/sentences below which has used the word "COMMIT" incorrectly.
A. He has committed unpardonable crime.
B. The passengers commit their luggages to the cloak room on railway stations.
C. He is a committed Christian.
D. The case was committed to sessions court.
(1) B only (2) B and C
(3) B and D (4) None of these

12. Match the words with their correct use in sentences.

1.	Rash	(i)	The grains of wheat are—out of husk.
2.	Trash	(ii)	He was censured for—action.
3.	Thrash	(iii)	The police—the thief.
4.	Thresh	(iv)	Litters were thrown in—can.

(1) 1–(i), 2–(iv), 3–(ii), 4–(iii)
(2) 1–(ii), 2–(iv), 3–(iii), 4–(i)
(3) 1–(iv), 2–(i), 3–(ii), 4–(iii)
(4) 1–(ii), 2–(i), 3–(iv), 4–(iii)

Directions (Q. 13-16) *Read the following passage and answer the questions given below.*

Of all the possible economic crisis, inflation is perhaps the one which receives the most public attention. And why not ? If prices keep on rising continuously over several weeks, then every household except the fortunate few has a significantly harder time balancing its monthly budget. Quite inevitably, all the parties outside the government, ranging from the BJP to the CPM,have found the current bout of inflation a convenient stick with which to beat up the government.

The rate of inflation has just touched 7 per cent, which is the highest it has been in the last 40 months. The biggest contributory factor has been the huge increase in prices of minerals. Cement and metals prices have also shot through the roof. Fortunately, these items do not enter household budgets directly, and so do not affect their welfare at least in the short run.

Unfortunately, there has also been a very large increase in the prices of staples such as vegetables, rice, wheat and edible oils.

With several state elections just around the corner and perhaps even the general elections not very far away, the parties constituting the UPA ministry find themselves with a noose around their collective necks. And the noose is tighter because it seems that there is very little that the government can do to bring down prices in a hurry.

Prices rise when there is an excess of overall demand over supply in the economy. If the economy has been growing very rapidly over long periods of time then it makes sense to restrict inflationary pressures by reducing aggregate demand. In such situations, the government or central bank of a country can use a range of monetary and fiscal instruments to curtail aggregate demand. For instance, a higher interest rate regime and other actions to restrict money supply can choke off demand.

Of course, such efforts to cool the economy can have potentially deleterious effects on the economy. In particular, if the government "overshoots" and implements its anti-inflationary policies too enthusiastically, then the economy can actually slow down much more than is desirable. Or the economy may already have started to slow down and then it becomes even more of a danger to restrict demand.

There are now clear signs that the Indian economy and indeed economies all over the world are slowing down. Despite growth elsewhere in the world, the US economy still retains its overwhelming global presence. The subprime crisis, the turmoil in US financial markets and the threat that its economy is slipping into a recession, have cast a shadow all over the world. Not surprisingly, even the normally proactive Reserve Bank has not yet taken any steps to curb the current inflationary pressures.

This leaves the government with only one option—it has to take steps to increase the availability of commodities at reasonable prices. In the short run, this is easier said than done. Obviously, the government does not possess any magic wand that it can wave over the economy and achieve an instantaneous increase in agricultural production.

We are sitting on a huge pile of foreign exchange reserves. Since foreign exchange is no longer a constraint, it is natural to wonder whether it is possible to augment the domestic availability of essential commodities by large-scale imports. To a large extent, this is the thrust of the recent policy measures announced by the government. It has slashed import duties on a large number of commodities such as wheat and edible oils, in the hope that the reduced import duties will bring about a reduction in the final price of imported food items.

There are times when this strategy can work. Unfortunately, the prevailing conditions in the rest of the world mean that this strategy is doomed to failure today. This is because world prices of most essential goods are appreciably higher than domestic prices in India. For instance, the price of wheat in global markets is almost twice that in India. Similarly, edible oil prices are also significantly higher than those in India. So, even the complete absence of import duties will not really make imported food affordable.

And there are reasons to believe that the upward pressure on world prices is not likely to ease up in the near future. The huge increase in crude oil prices have forced western countries to look for alternative sources of energy. European countries have been blending vegetable oils with diesel, while maize and wheat are being used in the US to make bio-ethanol. These alternative sources of demand mean a rather large-scale diversion of the available supply of these commodities away from domestic households. The failure of the wheat crop in Australia—a major supplier to the rest of the world—has exacerbated the global shortage of wheat.

The intense political pressure may force the government to take steps that it should not adopt in a hurry. In particular, the RBI may well step in before very long and raise the structure of interest rates to reduce money supply. This will undoubtedly help to curb inflationary pressures to some extent, although at the cost of a possibly lower rate of growth. Prudence demands that the government wait for a little while longer before it adopts this extreme step. After all, there will be some improvement in the domestic supply of foodgrains once the output from the rabi harvest reaches the markets.

13. The household budget in middle class families is in turmoil because of
(1) inordinate price-rise of steel, cement, and minerals
(2) inordinate price-rise of food items
(3) inordinate price-rise of sundry necessities of common man
(4) inordinate price-rise has adversely affected people's purchasing potential

14. The government can do very little to contain inflation because
(1) economy is growing at a fast pace of around 8 to 9 per cent
(2) harsh economic regime may have negative effects on economy
(3) government cannot further enhance interest rates to cut demand
(4) government do not possess any magic wand to change the situation in a trice

15. The age-old practice of importing scarce commodities to contain price-rise in not likely to work this time because
(1) the government is not willing to reduce import duties on commodities
(2) the prices of essential commodities in international market are higher than in local market
(3) the commodities are scarce in international markets too
(4) the government is unable to meet foreign currencies needed to import such commodities

16. What is the author's prescription to government to combat current price-rise menace ?
(1) Reduce import duties
(2) Import food items
(3) Enhance interest rates on bank deposits to reduce money supply
(4) To wait and watch in the hope of abundant arrival of fresh harvest in the market

Directions (Q. 17-20) *Choose the appropriate conjunction to join the given sentences.*

17. He was reading newspaper.
He was taking his meals too.
(1) Even as he was taking his meals, he was reading newspaper.
(2) He was reading newspaper and taking his meals simultaneously.
(3) He was reading newspaper and taking his meals at the same time.
(4) He was reading newspaper as he was taking his meals.

18. He is a teacher.
He is a musician as well.
(1) He is a teacher and a musician.
(2) He is a teacher and a musician to boot.
(3) He is a teacher and musician.
(4) He is a teacher and a musician as well.

19. He is a student studying regularly.
His regular studies will keep him in advantageous position at the time of exams.
(1) Regular studies will keep a student in advantageous position at the time of exams.
(2) Regular studies will stand a student in good stead at the time of exams.
(3) Regular studies is a must to be in advantageous position at the time of exams.
(4) Regular studies will help a student in exams.

20. Tread carefully.
You may trample on my toes.
(1) Tread carefully unless you want to trample on my toes.
(2) Tread carefully, otherwise you will trample on my toes.
(3) Tread carefully lest you may trample on my toes.
(4) Tread carefully to avoid trampling on my toes.

Directions (Q. 21-25) *Read the passage and answer the questions given below.*

The Report of the Justice Srikrishna Commission on the revision of pay scales of Central Government employees has allayed apprehensions of a severe fiscal burden. By all accounts, additional fiscal stress from the revision is expected to be much lower than what was feared. There are 3.39 million Union Government employees and an unspecified number of defence personnel. The pay revision for them is expected to entail an additional annual burden of Rs 7,975 crore and arrears of Rs 18,060 crore on government finances. However, in combination with other unfunded commitments such as loan waiver and underfunded fertiliser and food subsidies, it is likely to add to the fiscal stress at a time when the need of the hour is austerity.

There is nothing wrong with increasing the pay scales of government employees. Surely, if you pay peanuts, you will only get monkeys. A measure of parity with private sector pay is necessary to attract talent to government service. A World Bank study in 2005 showed that in 1999-2000, public sector employees received higher than formal private sector wages for different categories of employees by 62 to 102 per cent and it was higher than informal private sector employees by 164 to 259 per cent. Notably, governments are bottom-heavy and employees at the bottom of the pyramid receive much higher salaries than comparable employees in the private sector. Unfortunately, the governments do not believe in slimming (and remain ugly) and increases in pay scales do not make them more efficient in service delivery. However, in respect of senior cadres, comparable private sector officials receive multiple times the salary of government officials and during the last eight years, the increase in organised private sector pay has been phenomenal and the differences have been increasing.

Unlike the Fifth, the Sixth Pay Commission does not seem to have put forth much of a slimming plan, probably because the Government never bothered to implement that part of the recommendations anyway. However, it has continued the plan of rationalising the pay scales by reducing the number of scales from 35 to 20 and has prescribed a minimum qualification at the entry level. It may be recalled that the Fifth Pay Commission had reduced the number of scales from 51 to 35. The extent of increase seems to be more for higher scales of pay, partly to reflect the public-private sector differentials in the pay.

What are the fiscal implications of pay revision on state finances ? Although not mandatory, most state governments follow the central lead and implement the Central Pay Commission recommendations for their employees. In fact, following the central pay revision in 1997-98, 20 state governments besides the Union Territories implemented the Central Pay Commission recommendations to create an additional revenue and fiscal deficit to the tune of 1.5 percent of GDP. As a ratio of revenue expenditures, the expenditures on salaries in the states increased from 35 per cent in 1996-97 to 41.3 per cent in 1999-00. With at least 11 state governments going for elections in the next one year and with political parties indulging in competitive populism, there is a strong likelihood that more states will follow the Centre in implementing the pay revision. There are over 8.5 million employees at the state level and in addition, there are aided institutions and some states pay the salaries of private school teachers as well.

Fortunately, even if all the states adopt Central pay scales, it is not likely to create severe fiscal stress in the states for a variety of reasons. First, the burden of pay commission recommendations is not likely to exceed 0.5 per cent of GDP, unlike in the earlier case where it increased by over 1.5 per cent of GDP. Second, the revision of pay scales comes in the background of a much healthier state finances. The last pay revision was undertaken when the states were in a stressed situation. First, the central transfers were declining in the wake of declining tax revenues of the Centre whereas both central tax revenues and transfers to states continue to show remarkable buoyancy now. It may be recalled that the gross tax revenues of the Centre relative to GDP declined from 10.2 per cent in 1991-92 to 8.2 per cent in 2001-02 following the decline in customs and excise duties relative to GDP. In contrast, Central tax revenues have grown at over 20 per cent per annum during the last six years. Second, unlike the last time, when the states own revenues were stagnant, the recent buoyancy in their revenues will provide the needed cushion. The states revenues relative to GDP have increased by over 1 per cent during the last three years mainly due to the VAT-related reforms, better collections from stamp duties and registration fees and state excise duties. Moreover, in the earlier case the states were faced with much higher effective rates of interest on their borrowings whereas, now thanks to the debt swap scheme, lower interest rates and the relief given by the Twelfth Finance Commission, the effective rates of interest are much lower. Above all, the severely stressed fiscal situation in the states during the past decade has led to many states applying a freeze on the employment, particularly at lower levels, and the benefits of that can be reaped now in terms of fewer employees relative to population, although the improved fiscal situation in the last couple of years has led to relaxation on this front.

Thus, the states are in a much better situation to face pay revision today than a decade ago. In fact, except for Sikkim and West Bengal, all the states are signatories to fiscal responsibility legislation and except for Kerala, West Bengal and to some extent Punjab, all the states are well on their way to achieving the fiscal restructuring targets. In fact, the budget estimates for 2007-08 show that in the aggregate the states would have a small revenue surplus and exceed their fiscal deficit reduction target. Of course,

the impact will differ from state to state and some states will have difficult times. Surely, the open and hidden deficits will continue to be a cause of concern, but for the fiscal stress, Pay Commission report can not be blamed this time.

21. The steep hike in central government employees' pay as recommended by 6th Pay Commission is likely to
(1) put unbearable burden on states which have to willy-nilly follow the suit for their employees too.
(2) put states in unviable position to straightaway deny their employees, any such hike in salary.
(3) be implemented in states without a fuss as the increase in burden would be one-third to that of 5th Pay Commission implementation.
(4) be implemented in states which are now getting increased revenue shares from the Centre as well as increase in own revenues.

22. The Central Government is likely to face fiscal stress due to wage revision because of
(1) loan waiver and funding food and other subsidies as other commitments
(2) compulsion to make wage parity of its employees with private sector employees in order to attract talent to government service
(3) the government have to bear the burden of bottom-heavy lethargic non-performing lower rank staff who are already getting much higher salary than their counterpart in private sector
(4) the unwillingness of the government to prune the strength of redundant lower-rank employees

Directions (Q. 23-25) *What does the undermentioned sentences mean in the context of the passage ?*

23. The hike in salary in organised private sector, during the last decade, has been phenomenal.
(1) The hike in salary in organised private sector, during the last decade, has been prodigious.
(2) The hike in salary in organised private sector, during the last decade, has been extraordinary.
(3) The hike in salary in organised private sector, during the last decade, has been perceptible to the senses.
(4) The hike in salary in organised private sector, during the last decade, has been stupendous.

24. However, it has continued the plan of rationalising the pay scales by reducing from 35 to 20.
(1) However, it has continued the plan of justifying the action/decision of reducing pay scales from 35 to 20.
(2) However, it has continued the plan of reorganizing to make reducing pay scales from 35 to 20 more logical.
(3) However, it has continued the plan of making more efficient by dispensing with the pay scales from 35 to 20.
(4) However, it has continued the plan of betterment of pay scales by reducing from 35 to 20.

25. With at least 11 states going in election in next one year, political parties indulge in competitive populism.
(1) With at least 11 states going in election in next one year, political parties indulge in competition to become popular.
(2) With at least 11 states going in election in next one year, political parties indulge in competition to make own leader popular.
(3) With at least 11 states going in election in next one year, political parties indulge in one-up-manship in supporting such cheap issues as are appealing to people's interest.
(4) With at least 11 states going in election in next one year, political parties indulge in competition to make more people join the ranks of their party.

Directions (Q. 26-32) *Read the passage and answer the questions given below.*

One of the biggest social problems today is child nutrition. All the much hyped strides taken by our country notwithstanding, this problem of child malnutrition has not shown any sign of abating. There has been rhetorics, but not determined action to combat this menace which casts its shadow on future potential of human resource.

The problem of child malnutrition begins to affect them in foetal stage. Early marriage of women, and their ignorance about nutritional demands during pregnancy make the matter worse.

The lack of girls' education keeps them unaware of the importance of preventing infections by innoculations, inculcating good hygiene in kids, and balanced diets in warding off malnutrition.

26. What does 'Malnutrition' mean ?
(1) Bad nutrition
(2) Improper balance in diet
(3) Lack of nourishment in body
(4) Unhygienic food

27. The problem of child malnutrition has continued to affect the children on large scale because
(1) the country has taken long strides in economy
(2) economic growth in the country has not addressed this problem
(3) there has been rhetorics only sans efforts
(4) the problem has not yet abated

28. What, according to the passage, is the possible disastrous effect of child malnutrition ?
(1) Country turning into a nation of diseased people
(2) Country spending stupendous amount of resources on medicines
(3) Country having dearth of healthy and intelligent work-force
(4) Country facing problem of disbalanced ratio of child population to grown-up people

29. What, according to the passage, step or steps are needed to combat this menace ?
(1) Potential mothers' inculcating good hygiene in kids
(2) Running innoculation drive *viz.* polio eradication programme
(3) Whole-hearted effort to provide eduction to girl child as par with male-child, and parents giving them equal importance in every respect
(4) Stopping early marriage of girls

30. What is the meaning of the word 'HYPE' ?
(1) Publicity (2) Hoax
(3) Excite (4) Stimulate

31. Which of the options does not mean 'notwithstanding' ?
(1) Nevertheless (2) Not standing with
(3) In spite of (4) Although

32. Which one is the closest in meaning of 'Rhetoric' ?
(1) Repetition of speech (2) High-flown speech
(3) Roundabout speech (4) Insincere speech

Directions (Q. 33-40) *Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are four numbered words or sets of words. Choose the word or set of words for each blank that best fits the meaning of the sentence as a whole.*

33. The notion that a parasite can alter the behaviour of a host organism is not mere fiction; indeed, the phenomenon is not even..... .
(1) observable (2) comprehended
(3) imaginable (4) rare

34. That many of the important laws of science were discovered during experiments designed to.....other phenomena suggests their experimental results are the.....of inevitable natural forces rather than of planning.
(1) alter, adjuncts
(2) verify, essence
(3) illuminate, consequence
(4) analyse, foundation

35. Darwin's method did not really.....the idea of race as an important conceptual category; even the much more central idea of species was little more than a theoretical ..... .
(1) subsume, validation
(2) exclude, practice
(3) require, convenience
(4) reject, fact

36. People frequently denigrate books about recent catastrophes as morally.....attempts to profit from misforture, but in my view our desire for such books, together with the venerable tradition to which they belong, ...them.
(1) fortuitous, fosters
(2) corrupt, generates
(3) despicable, legitimises
(4) inopportune, encourages

37. Although Shakespeare received little formal education, scholars have in recent years.....the view that he was.....the work of classical authors.
(1) questioned, influenced by
(2) supported, oblivious to
(3) substantiated, unimpressed by
(4) undermined, unfamiliar with

38. Borrowing a copyrighted book from a library amounts to a form of theft.....by entrenched custom; the copyright owner's property, the book, is used repeatedly without .....for such use.
(1) perpetrated, permission
(2) provoked, adjustment
(3) anticipated, acknowledgement
(4) sanctioned, compensation

39. Although in eighteenth century England an active cultural life accompanied the beginnings of middle class consumerism, the.....of literacy was.....with the rise of such consumerism in the different areas of the country.
(1) degree, uncorrelated
(2) spread, compatible
(3) promotion, combined
(4) renewal, inconsistent

40. The trainees were given copies of a finished manual to see whether they could themselves begin to.....the inflexible, though tacit, rules for composing more of such instructional materials.
(1) derive (2) design
(3) revise (4) standardise

■ Quantitative Aptitude, Data Interpretation and Sufficiency

41. How many words can be formed by rearranging the letters of the word 'NEERAJ' ?
(1) 790 (2) 1440
(3) 60 (4) None of these

42. If radius of circle become 2 times, then the ratio of area will be
(1) 1 : 4 (2) 4 : 1
(3) 2 : 1 (4) None of these

43. A bucket having radii 28 cm and 7 cm and its height 45 cm. Then its volume will be
(1) 4850 cm^3 (2) 48500 cm^3
(3) 48510 cm^3 (4) None of these

44. If the cost of 2 goats and 3 cows is Rs 550 and cost of 3 goats and 2 cows is Rs 450. The cost of 3 goats and 3 cows will be
(1) 200 (2) 400
(3) 600 (4) None of these

45. If $4^{1+x} + 4^{1-x} = 10$, then the value of x^2 will be
(1) $\frac{1}{4}$ (2) 4
(3) –9 (4) None of these

46. Two radio sets were purchased for Rs 624 and one was sold of 14% profit and the other at 14% loss. If both the SP was same, then calculate profit or loss per cent
(1) 1.96% profit (2) 1.96% loss
(3) 2% loss (4) None of these

47. If $\tan\theta = \sqrt{2} - 1$, then the value of $\sin\theta \cdot \cos\theta$ will be
(1) $\sqrt{2}$ (2) $\frac{1}{\sqrt{2}}$
(3) $\frac{1}{2\sqrt{2}}$ (4) None of these

48. If $\sin\theta - \cos\theta = 1$, then the value of $\sin\theta + \cos\theta$ will be
(1) –1 (2) 1
(3) ±1 (4) None of these

49. A sum was lent out at simple interest and compound interest. After 2 yr, simple interest was Rs 200 and compound interest was Rs 210. Then find the rate of interest
(1) 5% (2) 10%
(3) 15% (4) None of these

50. Given that E and F are event such that $P(E)\,0.4\,P(F) = 0.5$ and $P(E \cap F) = 0.3$, find $P(E/F)$, $P(F/E)$
(1) $\frac{3}{5}, \frac{3}{4}$ (2) $\frac{1}{3}, \frac{1}{4}$
(3) $\frac{2}{5}, \frac{3}{4}$ (4) None of these

51. Puja and Sony lent out Rs 1682 on compund interest. A sum received by Puja in 3 yr is equal to sum received by Sony in 5 yr, then if the share of Puja is x where rate of interest is 5% per annum then x is equal to
(1) 800 (2) 441
(3) 882 (4) None of these

52. If 8 men and 12 boys can do a piece of work in 16 days, in how many days 20 men and 6 boys would finish that work ?
(1) $5\frac{1}{3}$ days (2) $5\frac{1}{2}$ days
(3) $4\frac{1}{3}$ days (4) None of these

53. A trader sells his goods at 10% discount. What price should he mark on an article that costs him Rs 1200, to gain 12.5% ?
(1) 1000 (2) 1200
(3) 1500 (4) None of these

54. How many litres of water flows out of a pipe of cross section area 5 cm^2 in 1 min, if the speed of the water in the pipe is 30 cm/s ?
(1) 9 L (2) 8 L
(3) 5 L (4) None of these

55. If $\frac{x}{y} = \frac{2}{7}$, then find the value of $\frac{7x + 2y}{7x - 3y}$.
(1) 4 (2) – 4
(3) 8 (4) None of these

56. Divide Rs 560 between A and B so that A gets twice the money and Rs 80 more that B gets.
(1) 60, 40 (2) 100, 20
(3) 180, 80 (4) 400, 160

57. If 75 workers working 8 h a day earn Rs 800 in 15 days, then how much will 24 workers earn in 9 days working 10 h a day ?
(1) 92 (2) 292
(3) 192 (4) None of these

58. In a 1500 m race x beats y by 100 m, y beats z by 120 m, by how much metres does x beat z ?
(1) 212 (2) 220
(3) 156 (4) None of these

59. What is the total number of candidates at an examination if 31% fail and the number of those who pass exceeds the number of those who fail by 247 ?
(1) 650 (2) 800
(3) 700 (4) None of these

60. How many bottles of hair oil a sales girl needs to sell at Rs 9.55 per bottle to earn a commission of Rs 95.50. Considering the rate of commission to be 5%.
(1) 20 (2) 200
(3) 250 (4) None of these

61. It is decided that a loan of Rs 10000 will be paid of @ Rs 800 per month in 15 equal instalments. Find out the rate of interest taken on the loan.
(1) 12% (2) 20%
(3) 16% (4) None of these

62. A trader marks his goods with two prices, one for ready money and the other for 6 month's credit. What ratio should the two prices bear to each other if 5% simple interest is taken on 6 months credit.
(1) 40 : 41 (2) 2 : 3
(3) 4 : 1 (4) None of these

63. The true discount on a certain sum of money due 3 yr hence in Rs 100 and the simple interest on the same sum for the same time and at the same rate is Rs 120. Find the sum
(1) 256 (2) 600
(3) 360 (4) None of these

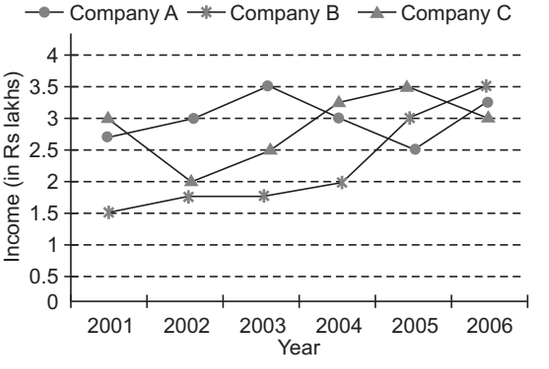
64. If $f(x) = \frac{2x + 3}{x - 2}$, $x \neq 2$, $x \in R$, then $\frac{f(10)}{f\left(\frac{1}{2}\right)}$ is equal to
(1) $\frac{1}{4}$ (2) $-\frac{69}{64}$
(3) 69 (4) None of these

65. If $\log 2^{x+3} = \log 64$, then find x
(1) 3 (2) $\frac{1}{3}$
(3) – 3 (4) None of these

Directions (Q. 66-70) *Study the graph carefully to answer the questions that follow.*

Income (in lakhs) of three companies over the year

$$\text{Per cent profit} = \frac{\text{Income} - \text{Expenditure}}{\text{Expenditure}} \times 100$$



66. If the expenditure of company C in 2003 was Rs 1.75 lakh, what was its per cent profit in that year ? (Rounded off to two digits after decimal)
(1) 38.29 (2) 42.86
(3) 53.41 (4) 58.64

67. What is the average income of company A over the years ?
(1) Rs 275000 (2) Rs 3000000
(3) Rs 300000 (4) Rs 30000

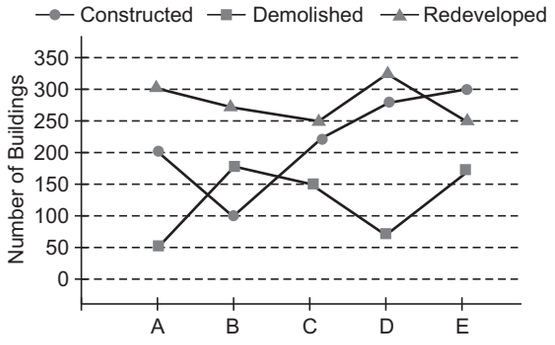
68. What is the approximate per cent increase in income of company B in the year 2006 from the previous year ?
(1) 28 (2) 11
(3) 17 (4) 22

69. If the per cent profit of company A in the year 2002 was 20, what was its expenditure in that year ?
(1) Rs 250000 (2) Rs 275000
(3) Rs 175000 (4) Rs 150000

70. Per cent increase/decrease in income of company C was highest for which year ?
(1) 2004 (2) 2006
(3) 2003 (4) 2002

Directions (Q. 71-75) *Study the following graph carefully and answer the questions given below.*

Number of buildings constructed, demolished and redeveloped across various cities in a year.



71. What is the difference between the total constructions (constructed, demolished and redeveloped) in city B and city E ?
(1) 125 (2) 180
(3) 175 (4) 200

72. What is the average number of buildings demolished across the cities ?
(1) 100 (2) 125
(3) 132 (4) 115

73. The total number of buildings constructed across the cities is approximately what per cent of the total number of buildings redeveloped across the cities ?
(1) 73 (2) 74
(3) 85 (4) 79

74. What is the approximate average number of constructions (constructions, demolished and redeveloped) in the city C ?
(1) 222 (2) 198
(3) 216 (4) 208

75. What is the ratio of the number of buildings demolished in city E to the number of buildings redeveloped in city A ?
(1) 7 : 12 (2) 3 : 5
(3) 2 : 3 (4) 1 : 2

Directions (Q. 76-80) *Study the information carefully to answer the questions that follow.*

In a school of 2,500 students, the students are taught four different languages viz., English, Hindi, Marathi and Sanskrit. The ratio of girls to boys in the school is 3 : 2 respectively. 16% of the boys learn only Marathi. 23% of the girls learn only Marathi and Hindi. 14% of the total number of students learn only English. Boys learning only Marathi and Hindi are 120% of the girls learning the same. One-tenth of the boys learn only Sanskrit. Girls learning only Marathi are 130% of the boys learning the same. 19% of the boys learn only English. Remaining boys learn only Sanskrit, Hindi and English. 32% of the girls learn only Hindi and Sanskrit. Remaining girls learn only Hindi, English and Sanskrit.

76. What is the total number of girls learning English and Sanskrit ?
(1) 787 (2) 867
(3) 947 (4) 897

77. What is the total number of students learning Marathi ?
(1) 1017 (2) 1464
(3) 1024 (4) 1127

78. How many students study at least two languages at a time ?
(1) 1682 (2) 1462
(3) 1670 (4) 1550

79. What is the total number of boys learning Hindi ?
(1) 480 (2) 550
(3) 620 (4) 650

80. How many girls do not learn Hindi ?
(1) 368 (2) 416
(3) 346 (4) 428

■ Analytical and Logical Reasoning

Directions (Q. 81-83) *Refer to the data below and answer the questions that follow.*

A Δ B means A > B; A □ B means A < B; A O B means A = B

81. If P Δ Q, O R and Q Δ S, then which of the following are true ?
I. P O Q II. P Δ R III. R Δ S
(1) II only (2) III only
(3) I and II only (4) II and III only

82. If V O W, W □ X and X Δ Y, then which of the following is true ?
(1) V O Y (2) W O Y
(3) W □ Y (4) V □ X

83. If I O J, J □ K, K Δ L which of the following will necessarily yield positive result ?
I. I – J II. K – J III. K – L
(1) I only (2) II only
(3) III only (4) II and III only

Directions (Q. 84-87) *Each question is followed by two statements I and II.*

Mark (1) If the question can be answered by using one of the statements alone but cannot be answered using the other statement alone.

Mark (2) If the question can be answered by using both statements together but cannot be answered using the other statement alone.

Mark (3) If the question can be answered by using either of the statements alone.

Mark (4) If the question cannot be answered even by both statements together.

84. Which is the heaviest out of iron, aluminium and gold ?
Statement I Gold is very costly.
Statement II Aluminium is lightest of all three.

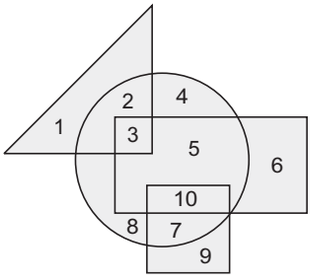
85. Who is the shortest among Aditya, Srinivas and Kalyani ?
Statement I Kalyani is shorter than Aditya but taller than Srinivas.
Statement II Srinivas is shorter than Aditya.

86. Out of Vidya, Vinod and Ashish, who is married ?
Statement I Ashish has one child whose name is Amar.
Statement II Vinod and Vidya are husband and wife.

87. Is Patna a metropolitan city ?
Statement I A metropolitan city must have a population over 20 lakhs.
Statement II This year Patna's population exceeded 20 lakhs.

Directions (Q. 88-90) *Refer to the data below and answer the questions that follow.*

The following figure represents university area. Circle stands for departments area, triangle for hostels region, rectangle for classroom and square for staff quarters.



88. Which number represents the region that has departments, hostels and classrooms ?
(1) 3 (2) 3 and 10
(3) 2, 4, 5 (4) 10

89. Which number shows the part of departments area that has classroom, but not hostels ?
(1) 3 (2) 5
(3) 10 (4) 3, 5 and 10

90. Which number represents staff members that live in department area ?
(1) 7 (2) 9
(3) 10 (4) 7 and 10

91. Once a thief was charged with stealing a vehicle. Five witnesses swore that they had seen him while stealing. But the thief argued that he could prove his innocence by presenting fifty persons who could swear that they did not see him while stealing the vehicle. The thiefs argument is not acceptable because
(1) five witnesses are enough
(2) defence witnesses do not always speak the truth
(3) those who did not see him were not there and therefore, their statement has little merit
(4) a thief will steal the vehicle despite many people watching him

92. If according to a certain code
I. 'tow low sow bow' means Prasad likes personal computer.
II. 'dow pow row low' means computer is expensive gift.
III. 'pow fow gow tow' means Murthy likes expensive hardware.
IV. 'your dow bow fow' means Prasad gave Murthy gift.
What is the code word for 'is' ?.
(1) dow (2) pow
(3) row (4) low

93. Roll numbers of those candidates were not displayed by the university who did not fail. Roll number of Ranu was displayed by the university but roll number of Somu was not displayed. Which of the following statement(s) is/are not valid ?
I. Ranu had failed.
II. Somu had failed.
(1) I only (2) II only
(3) I and II (4) Neither I nor II

Directions (Q. 94-98) *Given an input line the machine arranges the words and numbers step by step in a systematic manner as illustrated below.*

Input : subsidy 76 48 follow 35 next 63 must
Step I : 35 subsidy 76 48 follow next 63 must
Step II : 35 subsidy 48 76 follow next 63 must
Step III : 35 subsidy 48 next 76 follow 63 must
Step IV : 35 subsidy 48 next 63 76 follow must
Step V : 35 subsidy 48 next 63 must 76 follow
Output in Step V is the final output and Step VI is the last step.

As per the rules followed in the above steps, find out in each of the following questions the appropriate step for the given input.

94. If the second step of an input is "21 white direct 72 status front 37 69", how many more steps are required to complete the arrangement ?
(1) Three (2) Four
(3) Five (4) Six

95. What will be the third step for the input "17 85 pearls garland 67 93 restriction judgement" ?
(1) 17 restriction 67 pearls garland 85 93 judgement
(2) 17 restriction 67 pearls garland 85 judgement
(3) 17 restriction 85 pearls 67 garland 93 judgement
(4) 17 restriction 67 pearls 85 garland 93 judgement

96. If the fourth step of an input is "35 wealth 52 task 72 57 jogging playground". What will definitely be the first step ?
(1) task wealth 35 52 57 jogging 72 playground
(2) wealth task 35 52 57 jogging 72 playground
(3) wealth task 35 52 jogging 57 72 playground
(4) Cannot be determined

97. If the second step of an input is "26 shop finance 48 game music 63 37"; what will be the fifth step ?
(1) 26 shop 37 music 48 finance game 63
(2) 26 shop 37 music finance 48 game 63
(3) 26 shop 37 music 48 game finance 63
(4) There is no fifth step

98. How many steps are required to complete the arrangement for the input "56 punish 48 find design 23 lavish 36" ?
(1) Seven (2) Five
(3) Six (4) Eight

99. If ACRIMONY is written as CASJOMOZ in a certain code, how will NAXALITE be written in the same code ?
(1) ANYBILUF (2) ANWBLIUF
(3) PYYBILUF (4) PYWBLIUF

Directions (Q. 100-104) *Refer to the data below and answer the questions that follow.*

Mr. P of ABC Co. is recruiting five persons to do wiring and plumbing on a site. He must have a minimum of two electricians. Nine persons are sent by employment exchange M, N, O are electricians while R, S, T, U, V and W are plumbers. Mr. P knows that U and V fight all the time.

S and T are friends and will work together. O would not work with R.

100. If M, N, O are recruited, the team of plumbers can consist of
(1) S and T only
(2) S and T or U and V
(3) U and W or V and W
(4) S and T or U and W or V and W

101. Mr. P has the greatest number of choices for recruiting plumber if the electricians he chooses are
(1) M, N and O (2) M and N
(3) M and O (4) N and O

102. If R is recruited, the other persons hired must be
(1) M, N, S and T
(2) M, N, O and either U, V or W
(3) M and N together with either S and T or U and W
(4) M and N together with either U and V or V and W

103. Mr. P can put together the rest of his crew in the greatest number of ways if he recruits
(1) S and T (2) O
(3) U (4) W

104. If M is recruited and N is not, which of the following statement must be true ?
I. S and T are recruited.
II. Either U or V is recruited, but not both.
(1) I only
(2) II only
(3) Either I or II but not both
(4) Neither I nor II

105. 'Teacher' is related to 'Students' in the same way as 'Coach' is related to
(1) Team (2) Player
(3) Sport (4) Play

106. Pointing to Raju Ayesha says, "he is the son of my mother's brother's wife". How is Ayesha related to Raju ?
(1) Sister (2) Niece
(3) Cousin sister (4) Data inadequate

107. Tom walked 10 miles from point P towards the East. He, then took right turn and walked 5 mile and taking another right turn walked again for another 5 mile. In which direction is point P from where Tom is standing now ?
(1) West (2) North-West
(3) North-East (4) One of these

108. How many such digits are there in the number 724513 each of which is as for away from the beginning in the number as when they are arranged in ascending order starting from left ?
(1) More than 3 (2) One
(3) Two (4) None of these

109. If each of the odd digits in the number 54638 is decreased by '1' and each of the even digits is increased by '1' which of the following will be the sum of the digits of the new number ?
(1) 26 (2) 29
(3) 25 (4) None of these

110. Five friends M, K, T, P and N are sitting around a circle facing towards the centre K is sitting between M and P, N is to the immediate right of M. Who is to the immediate right of T ?
(1) K (2) P
(3) N (4) Data inadequate

Directions (Q. 111-112) *In each of the following questions a matrix of certain characters is given these characters follow a certain trend, row wise or columnwise. Find out this tread and choose the missing character according.*

111.

18	24	32
12	14	16
3	?	4
72	112	128

(1) 2 (2) 3
(3) 4 (4) 5

112.

3	6	8
5	8	4
4	7	?

(1) 6 (2) 7
(3) 8 (4) 9

113. Not only do major diseases afflict the elderly in particular of which these are more—they are generally very expensive diseases to treat which one of the following options. If true most strengthens this statement ?
(1) Studies reveal that 60% of elderly people suffer from cataract
(2) A dentists major income comes from fixing artificial teeth for elderly patients
(3) Insurance companies do not cover parkin. Son's disease because of the expenses involved
(4) Private nursing homes have reported that cancer accerents for 50% of prolonged hospitalisation of the elderly

Directions (Q. 114-118) *Study the following information carefully and answer the questions given below.*

P, Q, R, S, T and U are six teachers. They teach different subjects. Those subjects are Hindi, English, Mathematics, Sanskrit, General studies and Science. Also all the teachers teach in different schools. The schools are I, J, K, L, M and N. The name of the subjects and schools are not necessarily in the same order as the names of the teachers.

P teaches in the school L. Hindi is taught in the school N. Q teaches Science in the school I. R does not teach in the school J or K. General studies is not taught by P, S, T or U. S does not teach in the school N or K. Neither English nor Mathematics is taught in the school J.

114. Mathematics is taught by
(1) P (2) T
(3) U (4) Data inadequate

115. R teaches in the school
(1) J (2) K
(3) M (4) None of these

116. Who among the following teaches in the school N ?
(1) T (2) U
(3) S (4) Either T or U

117. In which school does S teach ?
(1) J (2) L
(3) M (4) I

118. Which of the following is/are definitely false ?
I. U teaches in the school N.
II. T teaches Hindi.
III. J teaches English.
(1) Only III (2) Only II and III
(3) Any one of the three (4) All I, II and III

119. Ram's age was square of number last year and it will be cube of a number next year. How long must he wait before his age be again the cube of a number ?
(1) 10 yr (2) 39 yr
(3) 38 yr (4) 64 yr

120. An office has a many four legged chairs and as many four legged tables as workers, and as many three legged stools as four legged almirahs. If the number of stools be one more than the number of workers and the total number of legs be 585. The number of workers in the office is
(1) 17 (2) 34
(3) 16 (4) Cannot be determined

■ General Awareness

121. India opted for 'Mixed Economy' in
(1) Framing of the Constitution
(2) Second Five Year Plan
(3) Industrial Policy of 1948
(4) None of the above

122. Which industry in India was worst affected because of the partition ?
(1) Paper and Iron
(2) Jute and Cotton
(3) Cotton and Sugar
(4) Engineering and Cement

123. In ancient peninsular India, who of the following assumed the title 'Vatapikonda' ?
(1) Mayurasharman (2) Narasimhavarman
(3) Pulakeshin II (4) Vikramaditya II

124. 'Indian cotton merchant; banker; Congressman and a close associate of Mahatma Gandhi'. This description fits
(1) Jamnalal Bajaj
(2) G.D. Birla
(3) V.S. Srinivasa Shastri
(4) M.R. Jayakar

125. The Gutenberg discontinuity is between
(1) the crust and the mantle
(2) the upper and lower mantles
(3) the outer core and the lower mantle
(4) the inner and outer cores

126. Who was the first speaker of Independent India's Lok Sabha ?
(1) Hukam Singh (2) Bali Ram Bhagat
(3) Rabi Ray (4) G.V. Mavalankar

127. Which of the following awards is instituted by the UNESCO ?
(1) Nehru award for International understanding
(2) Kalinga prize
(3) Arjuna award
(4) Nobel prize

128. Greenhouse effect is caused by
(1) nitrogen
(2) carbon dioxide
(3) carbon monoxide
(4) nitrogen dioxide

129. Which of the following is not in the infrastructure sector ?
(1) Power generation
(2) Construction of roads
(3) Food production
(4) Expansion of airports

130. Green Revolution in India has so far been most successful in case of
(1) Sugarcane (2) Coarse grains
(3) Wheat (4) Rice

131. Which city is referred to as the 'Silicon Valley of India' ?
(1) Mumbai (2) Chennai
(3) Hyderabad (4) Bangaluru

132. The name of the first cloned sheep was
(1) Molly (2) Dolly
(3) Jolly (4) Roly

133. In India, the First Municipal Corporation was set up in which place among the following ?
(1) Calcutta (2) Madras
(3) Bombay (4) Delhi

134. Which one of the following Constitutional Amendments states that the total number of Ministers including the Prime Minister in the Council of Ministers shall not exceed fifteen percent of the total number of members of the House of the People ?
(1) 90th (2) 91st
(3) 92nd (4) 93rd

135. The first conqueror of Mt. Everest alongwith Tenzing Norgya, Sir Edmund Hillary, has died. He belonged to
(1) New Zealand (2) US
(3) Australia (4) UK

136. Service Tax was introduced in India for the first time in the year
(1) 1990-91 (2) 1991-92
(3) 1994-95 (4) 1980-81

137. Which of the following has now become the third largest economy of the world ?
(1) Brazil (2) India
(3) China (4) Japan

138. Which country is the largest investor in India ?
(1) USA (2) UK
(3) China (4) Japan

139. Which one among the following has the highest energy ?
(1) Blue light (2) Green light
(3) Red light (4) Yellow light

140. Which of the following rivers in popularly called 'Dakshin Ganga' ?
(1) Cauvery (2) Krishna
(3) Godavari (4) Tungabhadra

141. Which group has recently launched its mobile services into the European market ?
(1) Tata group (2) Bharti group
(3) Idea cellular (4) Hutch

142. Where is Rail-Wheel factory situated ?
(1) Mysore (2) Hubli
(3) Bidar (4) Yelahanka

143. Navratna status has been conferred on
(1) HAL (2) BEL
(3) PFC (4) All of these

144. G-8 group of nations consists of …… nations of the world.
(1) developing (2) most industrialised
(3) poor nations (4) populous

145. The baseline 'Because life insurance is also about other people's lives' belongs to
(1) TATA-AIG (2) HDFC
(3) ICICI Pru. Life (4) Birla Sunlife

146. Which is not a product of Bajaj ?
(1) Blaze (2) Platina
(3) Sapphire (4) Brave

147. Which country owns Silk Airlines ?
(1) Indonesia (2) Singapore
(3) Bangladesh (4) Bhutan

148. The Chairman of the National Development Council (NDC) is the
(1) Minister of Information Technology
(2) Finance Minister
(3) Prime Minister
(4) Speaker of Lok Sabha

149. BILT means
(1) Ballarpur Industries Ltd
(2) Bombay International Ltd
(3) British International Technologies
(4) None of the above

150. The brand D.Dams is associated with
(1) Retail chain (2) Diamonds
(3) Gold (4) Textiles

151. "Bravia" is a new LCD TV from
(1) Sony (2) Samsung
(3) Philips (4) None of these

152. The name Hang Sang is associated with
(1) Stock exchange
(2) Textile industry
(3) Fast food
(4) IT Software solutions

153. Blue Chip companies are the ones who have
(1) Maximum sales
(2) Maximum Branches
(3) Minimum fluctuation or minimum risk in shares
(4) None of the above

154. True value is to Maruti, so is advantage to
(1) Ford (2) Tata
(3) Mahindra (4) Hyundai

155. Hritik Roshan does not endorse the following tires
(1) Coke (2) John Player
(3) Parle-Hide and Seek (4) Pulsar

156. The name Adlabs is associated with
(1) Lintas (2) O and M
(3) Reliance Industries (4) Nexus

157. "Har Jeb Mein Rang" is a punchline associated with
(1) Asian Paints (2) Berger Paints
(3) Jenson Nicholson (4) None of these

158. Kishore Biyani is the promoter of
(1) Manager (2) Reliance Fresh
(3) Amartex (4) Future Group

159. "Centurion Bank of Punjab" has recently been merged with
(1) ICICI (2) SBI
(3) HDFC (4) IDBI

160. French Car Maker "Renault" is setting up a new plant near
(1) Chennai (2) Kochi
(3) Vijaywada (4) Hyderabad

(OMR SHEET)

SNAP (Mock Test 4)

(Symbiosis National Aptitude Test)

General English

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
001	① ② ③ ④	009	① ② ③ ④	017	① ② ③ ④	025	① ② ③ ④	033	① ② ③ ④
002	① ② ③ ④	010	① ② ③ ④	018	① ② ③ ④	026	① ② ③ ④	034	① ② ③ ④
003	① ② ③ ④	011	① ② ③ ④	019	① ② ③ ④	027	① ② ③ ④	035	① ② ③ ④
004	① ② ③ ④	012	① ② ③ ④	020	① ② ③ ④	028	① ② ③ ④	036	① ② ③ ④
005	① ② ③ ④	013	① ② ③ ④	021	① ② ③ ④	029	① ② ③ ④	037	① ② ③ ④
006	① ② ③ ④	014	① ② ③ ④	022	① ② ③ ④	030	① ② ③ ④	038	① ② ③ ④
007	① ② ③ ④	015	① ② ③ ④	023	① ② ③ ④	031	① ② ③ ④	039	① ② ③ ④
008	① ② ③ ④	016	① ② ③ ④	024	① ② ③ ④	032	① ② ③ ④	040	① ② ③ ④

Quantitative Aptitude, Data Interpretation and Sufficiency

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
041	① ② ③ ④	049	① ② ③ ④	057	① ② ③ ④	065	① ② ③ ④	073	① ② ③ ④
042	① ② ③ ④	050	① ② ③ ④	058	① ② ③ ④	066	① ② ③ ④	074	① ② ③ ④
043	① ② ③ ④	051	① ② ③ ④	059	① ② ③ ④	067	① ② ③ ④	075	① ② ③ ④
044	① ② ③ ④	052	① ② ③ ④	060	① ② ③ ④	068	① ② ③ ④	076	① ② ③ ④
045	① ② ③ ④	053	① ② ③ ④	061	① ② ③ ④	069	① ② ③ ④	077	① ② ③ ④
046	① ② ③ ④	054	① ② ③ ④	062	① ② ③ ④	070	① ② ③ ④	078	① ② ③ ④
047	① ② ③ ④	055	① ② ③ ④	063	① ② ③ ④	071	① ② ③ ④	079	① ② ③ ④
048	① ② ③ ④	056	① ② ③ ④	064	① ② ③ ④	072	① ② ③ ④	080	① ② ③ ④

Analytical and Logical Reasoning

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
081	① ② ③ ④	089	① ② ③ ④	097	① ② ③ ④	105	① ② ③ ④	113	① ② ③ ④
082	① ② ③ ④	090	① ② ③ ④	098	① ② ③ ④	106	① ② ③ ④	114	① ② ③ ④
083	① ② ③ ④	091	① ② ③ ④	099	① ② ③ ④	107	① ② ③ ④	115	① ② ③ ④
084	① ② ③ ④	092	① ② ③ ④	100	① ② ③ ④	108	① ② ③ ④	116	① ② ③ ④
085	① ② ③ ④	093	① ② ③ ④	101	① ② ③ ④	109	① ② ③ ④	117	① ② ③ ④
086	① ② ③ ④	094	① ② ③ ④	102	① ② ③ ④	110	① ② ③ ④	118	① ② ③ ④
087	① ② ③ ④	095	① ② ③ ④	103	① ② ③ ④	111	① ② ③ ④	119	① ② ③ ④
088	① ② ③ ④	096	① ② ③ ④	104	① ② ③ ④	112	① ② ③ ④	120	① ② ③ ④

General Awarness

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
121	① ② ③ ④	129	① ② ③ ④	137	① ② ③ ④	145	① ② ③ ④	153	① ② ③ ④
122	① ② ③ ④	130	① ② ③ ④	138	① ② ③ ④	146	① ② ③ ④	154	① ② ③ ④
123	① ② ③ ④	131	① ② ③ ④	139	① ② ③ ④	147	① ② ③ ④	155	① ② ③ ④
124	① ② ③ ④	132	① ② ③ ④	140	① ② ③ ④	148	① ② ③ ④	156	① ② ③ ④
125	① ② ③ ④	133	① ② ③ ④	141	① ② ③ ④	149	① ② ③ ④	157	① ② ③ ④
126	① ② ③ ④	134	① ② ③ ④	142	① ② ③ ④	150	① ② ③ ④	158	① ② ③ ④
127	① ② ③ ④	135	① ② ③ ④	143	① ② ③ ④	151	① ② ③ ④	159	① ② ③ ④
128	① ② ③ ④	136	① ② ③ ④	144	① ② ③ ④	152	① ② ③ ④	160	① ② ③ ④

Symbiosis National Aptitude Test

Crack SNAP 5

ENTRANCE PATTERN

The duration for making answers will be 2 hours. Normally the pattern of written test is as follows.

Sections

I. General English
II. Quantitative Aptitude, Data Interpretation & Sufficiency
III. Analytical and Logical Reasoning
IV. General Awareness

Important Instructions

1. *Calculator/Watch calculator will not be allowed in the examination Hall.*
2. *Use only HB Pencils for darkening the ovals.*
3. *Correct method of Marking Answers*

Each questions is followed by four answer choices which are marked a, b, c, d on the answer sheet. Using HB Pencil, darken the oval bearing the correct answer against the serial number of the question.

For example : *If the answer to question number* 2 *is* (4)*, you should mark as shown below*

Q. 2. (1) (2) (3) (4)

■ General English

Directions (Q. 1-5) *Read the passage and answer the questions that follow.*

There is a good deal of ambivalence in the current debates on energy security. Can energy security be left entirely to markets or is it too strategic a commodity that warrants state intervention ? Is overseas oil equity a commercial investment decision that does not necessarily enhance energy security ? Should we, in India, compete with China for access to overseas energy assets or is our objective better achieved through a cooperative, or better still, collaborative approach ? Should we go for vertical integration of our national oil companies so that they acquire the scale to compete in the global market place or should they remain focused on their core competencies ?

Should we diversify our sources of supply ? Should we prioritise fuel use on the basis of our developmental goals or should we let market decide who gets which fuel ? Should we go all out to develop renewables even if they require huge subsidies and incentives ? Should we focus all our efforts on gas pipelines from the neighbourhood ? Can nuclear energy bridge the ever-widening gap between electricity demand and supply ? Should we pass on the burden of higher crude prices to consumers or should we avoid inflation at all costs ?

Most of these dilemmas are unavoidable, not just for India, but for virtually every energy importing country. After all, isn't energy too strategic and vital a commodity to escape the ominous cloud of geopolitics ? Commercial energy—its availability and affordability—is not only essential for the well-being of the citizens, but is often linked to the legitimacy of political regimes. Not even the most ardent champion of markets can afford to recommend total reliance on markets to deliver energy.

The United States, the largest oil consumer in the world, professes supreme faith in markets. Yet, throughout its history, it has frequently, and with impunity, wielded both the carrot and stick to protect its own energy interests. Many believe the war on Iraq is one such manoeuvre. The EU, for its part, may have networked and synchronised its gas and electricity markets, but has been unable to do so for the energy policies of its members.

Even as other EU members are fretting over the momentary disruption of Russian gas supplies to Ukraine, the German chancellor went ahead and signed energy supply agreements with Russia. Now the others are breaking ranks as well. Asian countries are no saints either. Japan's quest for overseas oil assets seems to have inspired China which has now excelled in this game, snapping up assets whatever the price or location.

As for India, we seem to be clutching at every straw to ensure that we do not drown under the weight of our own impressive, but energy-intensive growth. With crude prices hovering around $90 a barrel, with oil consumption galloping out of control—spurred by a phenomenal rise in vehicles on Indian roads—with domestic reserves dwindling and no significant new discoveries on the horizon yet, the desperation to clutch at every straw is understandable. Yet, it is time we prioritised the various options available to us instead of frittering away our energy and effort on multiple and diffuse initiatives. A SWOT analysis is the starting point. Acknowledging and assessing our weaknesses is as critical as identifying and building on our strengths.

The Planning Commission's integrated energy policy (IEP) does precisely that. It is a comprehensive document that examines the country's strengths and weaknesses and charts the way forward. After outlining and discussing various energy scenarios, IEP settles on a coaldominant energy paradigm as the most likely and feasible one. We may or may not agree with the approach suggested by the document, but at least it is decisive and purposeful and makes a good starting point.

Deliberate, focused, purposeful and implementable energy policy is imperative if we are to deal effectively with our energy vulnerability. If we conclude that energy is too strategic a commodity to be left to markets, we would do well to define clearly the areas that are best left to market forces and those that call for state intervention. Then comes the task of defining the nature, context and scope of state intervention.

An ambivalent approach or even inordinate delay in making up our minds can do more harm than good. Similarly, if overseas oil assets are deemed necessary or desirable, it might be worthwhile to identify the assets that might interest us and undertake a credible political risk analysis. India's going out strategy should be informed by long-term vision backed by political, diplomatic, commercial and strategic support. This will improve our success rate enormously. Presently, our staggering oil import bill dwarfs and pre-empts investments in renewable technologies. But if only we can conquer our myopic obsession with the present and factor in long-term benefits, we might reverse the trend sooner rather than later.

While there are many energy conundrums that are not within our power to resolve, we must resolutely address those that are. After all, in a country of billion-plus population, the stakes of indecision are very high.

1. The article suggests that
(1) energy is too strategic to be subjected to state control
(2) energy policy needs strategic state intervention depending upon identifying nature and scope of intervention
(3) energy is the subject which must be put under state control in order to prioritise the use of this scarce resource
(4) energy problem should be left to take its own course

2. Which one is true ?
(1) USA advocates free market policy in the matter of energy resources.
(2) USA goes all out to secure its stranglehold or oil rich countries whatever be its political cost.
(3) USA advocates free oil market but does not practise itself.
(4) USA does not believe in free market policy.

3. It is suggested that
(1) India should encourage growth in motor vehicles in the country for people's transport needs.
(2) India should discourage growth in moter vehicles in the country for transport needs.
(3) India should find means to increase oil production which is ever dwindling.
(4) India should judiciously use oil instead of wasting it unnecessarily by adopting various initiatives.

4. The article suggests that
(1) India should not rely on alternative source of energy
(2) India is spending hugely on development of renewable source of energy for ensuring future demands
(3) Investments in developing renewable sources of energy for ensuring gains in long run will be a correct option
(4) India should assess and acknowledge its weaknesses

5. Which of the given options is the closest in meaning of monoeuvre ?
(1) Strategy (2) Manipulation
(3) Tactic (4) Planning

Given sentences carries an underlined word. Which of the options is the opposite in meaning of it ?

6. The flying crow dropped the meat in its beak, and a chasing eagle dived towards the meat that had fallen on the ground
(1) plummeted (2) swooped
(3) plunged (4) soared

7. The workers did their job perfunctorily.
(1) Routinely (2) Meticulously
(3) Strenuously (4) Hesitatingly

Choose the most appropriate synonym of the underlined word in the given sentences out of the given options.

8. He made overtures to befriend his new neighbour
(1) advances (2) attempts
(3) tenders (4) moves

9. Doctor's prognosis of my disease was ominous
(1) diagnosis (2) forecast
(3) prescription (4) identification

10. The party spokesperson was quick to **rejoin**.
(1) retort (2) riposte
(3) clarify (4) retaliate

Directions (Q. 11-14) *Read the passage and answer the questions that follow.*

Here's something for critics of globalisation to chew on. Enrolment in English-medium schools in India has gone up sharply in just two years, from 4.3 per cent of total enrolment in 2003-04 to 6.3 per cent in 2005-06. The surge in English enrolment propels it to third place after Hindi and Marathi, in terms of total number of Indian children being instructed in that language. This shows that despite what parochially-minded politicians would have us believe about the allegedly elitist trappings of English language teaching, there's actually a grass-roots movement favouring it.

Increasing enrolment in English language schools in India is in line with international trends, where English is establishing itself as the language of global commerce. The world's top business schools and universities, for example, are increasingly moving towards adopting English as the language of instruction.

According to premier linguistic expert David Crystal there is no historical precedent for today's situation, where there's a language that's spoken in every country of the world : English.

Within India states like Tamil Nadu, Andhra Pradesh and Maharashtra are leading the surge towards English, with the number of students enrolled in English-medium schools actually doubling within two years in AP. There has been no comparable increase in Hindi-speaking states like UP, Bihar and MP. That's not because people living in the southern states are less proud of their culture than those in the Hindi heartland. That's caused, rather, by the wide availability of globalisationdriven jobs, such as in the IT or BPO sectors, in the southern states. Since candidates knowing English are at an advantage when applying for those jobs, it's natural that there should be a demand for English inschools. The economic figures for states such as UP or Bihar suggest, on the other hand, that they are missing out on a lot of job opportunities.

Rapidly rising enrolment in English-medium schools despite official discouragement, suggests that more room ought to be made for English in school curricula. Apart from being a link language to the world, English is also an indispensable link language between Indian states. Studies have shown that knowledge of English can break down caste and gender inequalities while allowing the hitherto underprivileged to break into whitecollar professions. It's the highway not just to jobs but political modernity as well. In Maharashtra, NCP ministers have been urging the reservation of 80 per cent jobs for locals. That's not only unconstitutional, it shows the harm that can be done by the forces of cultural chauvinism. English, which breaks down barriers between states, can act as an effective antidote.

11. The demand of NCP ministers in Maharashtra for reserving 80% jobs for locals is a reflection of
(1) their love for Marathi language
(2) their belief that Maharashtrians are having edge over others so far as knowledge of English is concerned
(3) their parochial and populist mentality
(4) their belief that this would provide locals opportunities in global job market

12. The students enrollment in schools is ever-increasing where medium of teaching is English, because
(1) English has come to be recognized as link language between states with various local languages
(2) People have lost interest in own languages and developed liking for English language
(3) Knowledge of English as medium of communications has opened opportunities in job markets on global scale
(4) English language knowing is a sign of modernity

13. Students in southern states viz Tamil Nadu, Andhra Pradesh, etc. are leading the surge in enrollments in English-medium schools as compared to that in Hindi belt because
(1) students in Hindi belt have more affinity towards own language
(2) the availablity of globalization driven jobs in IT, BPO sectors in southern states has spurred students to get educated through English medium
(3) students in Hindi belt are getting more jobs in their states with knowledge of local language
(4) people in Hindi belt donot like that English knowledge break the caste-based society

14. The parochial political leaders' antagonism of English as medium of instruction did not find favour with people at large because
(1) people have lost faith in leaders
(2) knowledge of English has elitist trappings
(3) people donot want to miss out opportunities requiring skill in English language
(4) knowledge of English engenders unity in diversity amongst people of different cultures and languages

Directions (Q. 15-18) *Choose appropriate word or words to fill in the blanks in the sentences from the given options.*

15. The …… of the bollywood attended the film award ceremony.
(1) literati (2) glitterati
(3) litterateurs (4) pedant

16. …… sale of the property yielded less return.
(1) Immediate (2) Unpublicised
(3) Distress (4) Publicized

17. His mood swings from …… to …… at the slightest provocation.
(1) elation, exasperation
(2) happy, fury
(3) jovial, erascible
(4) arrogance, haughty

18. The boy …… at the story-book with …… attention.
(1) looked, engrossed (2) stared, rapt
(3) pored, rapt (4) saw, serious

Directions (Q. 19-23) *Unscramble the following sentences in meaningful and logical sequence*

19. 1. Bihar's development commissioner S. Vijay Raghavan gave a power-point presentation on investment potential in Bihar.
2. Stressing on public-private ventures in agro-processing sector, the Chief Minister pointed out that Bihar had a large consumer market which makes it ideal for investment.
3. Mittal described the change in Bihar as impressive
4. Bihar C.M. and Bharti Chief Mittal identified agriculture and tourism as key sector for private investment in the state.
5. The two met at an interactive session organized by Bihar State Council of the CII.
(1) 5-1-4-2-3 (2) 1-5-2-3-4
(3) 3-2-4-1-5 (4) 1-3-5-4-2

20. 1. Turkey has the chance to bhaze the trail for the Muslim world.
2. Islamic governments must respect individual freedom and equality of all citizens.
3. The west does not know quite what to think of Turkey's proposed ban on wearing headscarf on university campus.
4. This had pervers initial effect on devout women staying away from higher education.
5. Turkey is to continue its integration into European civilization by adopting liberal values.
(1) 3-4-1-2-5 (2) 1-2-5-4-3
(3) 3-1-4-2-5 (4) 5-4-1-2-3

21. 1. Norms and standards of various teacher-training courses were revised and notified.
2. It was decided to consolidate these regulations under one notification for clarity.
3. Regulations laying down the procedure for grant of recognition to teachers' programmes were promulgated by the National Council for Teachers Education (NCTE) on 13.1.2006.
4. Regulations listed in para 12 of these regulations were superseded in exercise of the powers conferred under section 32(C) of the NCTE Act, 1993.
5. The NCTE consequently makes the following regulations.

(1) 4-5-3-2-1 (2) 4-2-1-5-3
(3) 1-2-4-5-3 (4) 5-4-2-3-1

22. 1. Reservations for SC/ST/OBC categories in the appointments shall be made as per government norms.
2. Management of the institution shall abide by statutory regulations on PF, Gratuity, etc for its employees.
3. All the appointments are to be made on full-time basis.
4. Appointments shall be made on the basis of recommendation of the selection committee constituted as per the policy of UGC/affiliating university.
5. The academic staff of the institution shall be paid salary as per prescribed pay scales.

(1) 4-3-1-5-2 (2) 1-3-4-2-5
(3) 3-4-1-5-2 (4) 1-2-3-4-5

23. 1. General Body Meeting of a society may be called anytime if one-third of its members so desired by giving 30 days advance notice to the chairman.
2. Annual General Body Meeting is called by the chairman once every year to discuss society's affairs.
3. Managing Committee of the Society meets quarterly to discuss pre-circulated agenda.
4. The chairman may call managing committee meeting anytime on written request by at least one-third members of this committee.
5. Chairman will have the right to nominate against any vacancy arising out of long absence of any of its office bearers.

(1) 1-2-3-4-5 (2) 2-3-1-4-5
(3) 2-1-3-4-5 (4) 3-4-2-1-5

Directions (Q. 24-25) *Transform the following sentences in shorter ones using one word for a number of words.*

24. The government wants to suppress the facts from the people if these are not welcome.
(1) The government must suppress uncomfortable facts.
(2) The government keeps unwelcome facts out of people's notice.
(3) The government is unwilling to suppress facts.
(4) The government is adept in suppressio veri.

25. The students who do regular studies all through the year find themselves in comfortable position at the time of final examination.
(1) Regular students feel comfortable in final examination.
(2) Regular studies help students in final exams.
(3) Regular studies stand students in good stead at the time of exams.
(4) Regular studies throughout the year is a must.

Directions (Q. 26-30) *Read the passage and answer the questions.*

When the Communist states of Eastern Europe collapsed in the early-1990s, there were anxious debates in Marxist circles over what had gone wrong. The common sense explanation was pretty simple : the dictatorship of the party was tyrannical, opaque, excessively bureaucratic and plain inefficient. The system violated human nature. For those who felt history was on their side, the collapse lent itself to a curious explanation. The socialist bloc, they argued, had crumbled because it had deviated from the true path.

Dogmatists and ideologues hate admitting they were ever wrong. Disaster is never attributed to a had idea but to human foibles. The remedy is never to abandon flawed beliefs but to cling to it more tenaciously.

As Budget Day approaches, the dogmatists have mounted a campaign to elevate a failed idea into a national catastrophe. There is pressure on the government to widen the scope of the National Rural Employment Guarantee Act, now operational in 330 districts, nationally. Last year, the NREG Programme devoured Rs 12,000 crore of taxpayers' money; this year, its drum-beaters want double that amount.

There is nothing wrong per se in the idea that all citizens should be able to get a guaranteed 100 days of paid work. It has more self-respect than queuing up for a dole. A spurt in rural employment can contribute to greater liquidity and increased consumption which, in turn, is good for the economy.

The problem arises in how work is defined. By getting people to dig a great big pit one day and filling it up the next, a lot of man hours will be generated. The job cards will be dutifully ticked. However, manual work that does not have a productive outcome—in this case, creating assets—is a national drain. It's also a crime when you consider the opportunity costs.

The NREGA deems that 60% of any approved project must comprise labour costs. The other 40% can include material and skilled labour. It's an absurd stipulation that virtually rules out the meaningful use of bricks and mortar. It has meant that a huge amount of manpower has been expended in the creation of non-tangible assets such as *kuccha* roads which will be washed away during the rains. True, the first year has also witnessed the cleaning of many village ponds and the creation of other water bodies. But you can't keep doing this year after year.

Secondly, as those with a stake in India would have predicted, an incompetent state machinery, riddled with corruption, is incapable of implementing a programme of this magnitude with sincerity. A draft report by the Comptroller and Auditor General reveals that only 3.2% of the registered households could avail of 100 days 'guaranteed' work. The average employment under NREGP was just 18 days. Rahul Gandhi says that only 5% of the money reaches the beneficiaries. Yet, the Centre persists with the fiction that nearly all the 2.73 crore households seeking work have got it.

A report on the NREGA website says that in the six chosen districts of Karnataka only 13 households were provided 50 mandays employment in seven months. This 'achievement' cost Rs 1.47 crore in wages, Rs 1.65 crore on material and Rs 2.84 crore on 'contingencies'!

To correct the follies, the fanatics want a dedicated babudom to run NREGP—a remedy worse than the ailment.

A normal world is impatient with dogmatic adventurism. In the Indian Wonderland where nothing succeeds like failure, ideological loonies have acquired the right to blow up public money. They were clever : they just gifted the copy-right of a bad idea to Sonia Gandhi.

26. The mentioned title of the article may be replaced with
(1) Unproductive Scheme of Employment Guaranty
(2) Symbol of Beaurocratic Inefficiency
(3) Marxist Dogmatism
(4) Squandering Public Money

27. The communism failed in Eastern Europe in 1990's because
(1) it was contrary to human nature
(2) it had deviated from true path
(3) communist Party's dictatorship was tyrannical
(4) the policy itself was flawed and inefficient

28. The demand to double the budget for employment guaranty scheme is based on
(1) the fact that a sum of Rs 12000 crore has been spent this year on this scheme
(2) there is no dearth of people seeking employment under this scheme
(3) it is an economically productive and assets generating sheme
(4) popular demand of the public

29. The failure of this scheme is attributed to
(1) the fact that only 3.2% households could get the work this year
(2) the fact that all households were given work
(3) the fact that the average employment under this scheme was only for 18 days instead of proposed 100 days in a year
(4) incompetent state machinery which is entrusted with implementation of the scheme

30. This scheme was a remedy worse than the ailment because
(1) it costed 1.47 crore in wage, Rs 1.65 crore on material purchasing, and Rs 2.85 crore on contingencies
(2) it squandered public money on wasteful expenditures
(3) it costed exchequer heavily
(4) the officials rightfully blew up public money

Directions (Q. 31-36) *Read the passage and answer the questions.*

In Vedanta, which is the crest jewel of Hindu philosopy the highest ideal is Self-realisation. This involves joining the spark of divinity within us, the Atman, with the divinity that pervades the entire universe, the Brahmn.

There are many methodologies for achieving this, classed under the generic term yoga, a word from the same root as the English word yoke, and involves joining the atman and the Brahmn or, in Semitic terms, God Immanent and God Transcendent.

There are four main paths of yoga—jnana yoga, the way of wisdom; bhakti yoga, the way of devotion; karma yoga, the way of dedicated work and raja yoga, the way of psycho-spiritual practices.

While there are hundreds of different paths to the divine, the Rig Veda says : "The truth is one, the wise call it by many names". The Upanishads in particular are the pre-eminent structures in Hinduism because they contain illuminating expressions of multifaceted truths by realised souls, the rishis. These verses are full of ecstatic glimpses into the divine, about Self-realisation.

One short piece, the Wisdom of Shandilya, from the Chandogya Upanishad brings out the essence of Self-realisation : "This universe comes forth from Brahmn and will return to Brahmn. Verily, all is Brahmn".

It goes on to say : "A person is what his deep desire is. It is our deepest desire in this life that shapes the life to come. So let us direct our deepest desires to realise the Self".

The Self, that can be realised by the pure in heart, is life, light, space, gives rise to all works, desires, odours and all tastes, is beyond words, is joy abiding. This is the Self dwelling in my heart.

"Smaller than a grain of rice, smaller than a grain of barley, smaller than a mustard seed, smaller than a grain of millet, smaller even than the kernel of a grain of a millet is the Self. This is the Self dwelling in my heart, greater than the earth, greater than the sky, greater than all the worlds.

"This Self that gives rise to all works, desires, odours, and all tastes, pervades the universe, is beyond words, is joy abiding, is ever present in my heart, is Brahmn indeed. To him I shall attain when my ego dies. So said Shandilya, so said Shandilya".

As the Self resides in all beings, there is really no scope in Vedanta for discrimination on the basis of birthplace, religion, caste, class, gender or economic status. True, such discrimination based on an inequitable caste hierarchy has been part of Hindu society for centuries, but all Hindu reformers and spiritual leaders have roundly condemned such restrictive practice. Whether it was Raja Rammohun Roy in Bengal. Ranade and Bhandarkar in Maharashtra, or Swami Dayananda Saraswati in Punjab, they all sought to remove the restrictive aspects of Hindu society to return to the principles of Vedanta.

Vivekananda and his brother monks Ramana Maharshi and Aurobindo had no room in their soaring philosophies for the ridiculous taboos based on caste discrimination. Indeed, in the global society towards which we are hurtling astride the irreversible arrow of time, what is needed is an inclusive philosophy that cuts across all the artificial barriers that divide human beings and prevent them from becoming what the Vedanta calls the world as family — "Vasudhaiva Kutumbakam".

31. "Vedanta is the crest jewel of Hindu philosopy". Does it mean ?
(1) Vedanta is the jewel of Hinduism
(2) Vedanta is the most important Hindu philosophy
(3) Vedanta is most attractive philosophy of Hinduism
(4) Vedanta is the quintessence of Hindu philosophy

32. According to this article, the word 'yoga' is a means to achieve
(1) transcendental meditation
(2) spiritual state of atman
(3) hitching one's immanent atman to the universe
(4) self realisation of one's inner strength

33. The self is to …… as the heart is to ……
(1) desire, realise the goal of life
(2) atman, Brahman
(3) a tiny object, vast expanse
(4) spirituality, desire

34. The Vedanta philosphy is contradictory to discrimination on the basis of caste, creed, birth, or gender because

(1) it is contrary to self realisation
(2) hinders achieving goal of divinity
(3) it is inequitous practice
(4) the self is immanent in all beings irrespective of their social milieu

35. Noted social reformers have done their best to emancipate society from the yoke of discrimination because
(1) it was riduculous taboo
(2) it was not in tune with the philosophy of the world being a single family
(3) the discrimination in society was a hinderance to self realisation
(4) it was contrary to the principle of equality

36. The movement of society to the desired goal of global society is taking place
(1) despite the social taboo of discrimination
(2) slowly but surely
(3) sitting on the horse of fast running time
(4) not as fast as it should have

Directions (Q. 37-41) *Read the passage and answer the questions.*

A businessman approached the Wise One, "All my decisions seem wrong, I never make enough money… you must help me".

The Wise One simply gave him a tiny locked treasure chest cham, and said, "Take this with you everywhere, and shake it three times whenever you enter a new space or meet someone for the first time". Somewhat sceptical, he looked at the tiny charm. "Oh well", he thought, "maybe the magic object inside will bring me luck; no harm in trying it out".

He carried the little charm everywhere, and in the space that he shook it thrice, waited, and paid attention, he noticed an opportunity here, a danger or pitfall there. A year went by and his luck had totally changed.

He went back to the Wise One and excitedly asked for the key. At first she refused, but he persisted, and she reluctantly opened the tiny little treasure chest. The man reached out and almost snatched it from her. He looked inside.

"Why… there's nothing inside!" he exclaimed in shock. "But there is on the outside", she smiled.

The power of talismans to open us to our own capacities needs to be better understood, and this lesson is best communicated through story, as in this beautiful traditional one—but modern tales too have had much the same point to make to a rather different audience, as in the story of Dumbo, the Flying Elephant.

Born into a circus, a baby elephant is rejected by the others because of his unusually big ears. Jeered at by children, rejected by the other performers, poor Dumbo is demoted to appearing as a clumsy, bumbling creature in an act with the clowns.

Only little Timothy Mouse befriends him, and devises a to make lonely little Dumbo a star. He gets a crafty crow to inform Dumbo that magic flying feathers are what really make crows able to fly, and if Dumbo had just one such magic feather, then he would be able to fly too.

Plucking a feather out of a friend's tail and giving it to Dumbo, he says, "This is a magic flying feather for you". And wonder of wonders—Dumbo flies on his first try, and because of this, is made part of a more daring performance, where he has to leap from a high platform as part of the clown's fire rescue act.

All goes well till one day, about to dive off the high platform in the middle of the act, Dumbo somehow loses the magic feather. Timothy Mouse, terrified, immediately yells and convinces him, "it isn't really magic. You can fly on your own". And Dumbo, flapping his enormous ears madly, discovers in that moment that it was not the feather at all which had the power of flight, but Dumbo himself.

The feather was only a bridge to put him in touch with a gift that was his all along. In our lives, we are in need of magic feathers in the form of talismans or beliefs that can help us reach new heights. Our lesson is to utilise them for as long as we need them, but recognising them for the temporary gift they are, we are challenged to transcend the need for an outer reminder of what has been ours all along.

37. On being given a magic box of treasure, the businessman felt sceptical—that is
(1) happy (2) lucky
(3) satisfied (4) suspicious

38. The Wise One who gave the businessman a magic box was
(1) a fairy (2) magician
(3) oracle (4) wise man

39. What do you think a talisman is ?
(1) It is a device to make one capable of achieving what is desired
(2) It is an inanimate object with magical power
(3) It is a device which makes one get money
(4) The circus elephant's device to help it fly in the air

40. The circus elephant was the butt of ridicule because
(1) it was capable of flying
(2) a little mouse, 'Timothy' was its friend
(3) it was performing with clowns in the circus
(4) it had had large-size ears

41. The little elephant, Dumbo, was able to fly in a daring circus act because
(1) a crow gave him a magical flying feather
(2) it could fly on its own without crow's feather
(3) it could fly with its flapping big ears
(4) it thought it could do so

■ Quantitative Aptitude, Data Interpretation and Sufficiency

42. Munna Bhai deals in second hand cars. His associate "Circuit Kumar" specializes in identifying cars with a good resale value. Munna Bhai bought 2 cars : an Esteem for Rs 270000, and a Zen for Rs 180000. After one year Munna Bhai wanted to sell both the cars to his favourite customer Lucky Singh. The Esteem bought by Munna Bhai was manufactured 5 years back and Zen bought by Munna Bhai was manufactured 4 years back. Munna Bhai also spent Rs 27000 on repair of the Esteem and Rs 18000 on repair of the Zen. According to the income tax guidelines, Munna Bhai could charge depreciation proportional to the cost of the asset (in this case the assets were cars). Thus, the value of any car could decrease by a maximum of $9\frac{1}{11}\%$ each year. Munna Bhai also wanted to earn a profit $11\frac{1}{9}\%$ on the

sale of both the cars. Find the approximate best price (combined) at which he should sell both the cars ? (Assume that the profit percentage is calculated on the Book value of the assets)

NOTE Best price is lowest price at which Lucky Singh would sell.

(1) 488889 (2) 500000
(3) 480000 (4) 540000

43. I (N) represents the integer value of a given number. If $N = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \ldots + \frac{1}{\sqrt{10000}}$. What is the value of $I(N)$?

(1) 197 (2) 196
(3) 296 (4) 297

44. 10 lines are drawn in a plane. Which of the following numbers cannot be the number of points of intersections that can be generated (by these 10 lines) ?

(1) 0 (2) 44
(3) 42 (4) None of these

45. In an Antakshari tournament there are 3 categories of teams. They are named as Deewane, Parwane and Mastane. There are 3 teams from the Deewane category, coded as D_1, D_2, D_3. Each team from Deewane category plays exactly 4 games of Antakshari. Similarly there are 5 and 2 teams respectively in Parwane and Mastane category. These teams are coded as P_1, P_2, P_3, P_4 and P_5; and M_1 and M_2 respectively. The rules of the tournament allow any team from any category to play a game of Antakshari against any other team from either their own category or a different category. The number of teams that participate in each game are two. Further, each of the 5 teams in the Parwane category played exactly 3 games. Which of the following could be the exact number of games played by each team of Mastane category ?

(1) 1 (2) 5
(3) 3 (4) None of these

46. There are 'n' baskets in the Reliance Fresh outlet at Bhopal. Each basket contains a different number of oranges. One basket was counted at random by the store manager and there were 100 oranges found in that basket. Further, his deputy who was a fresh MBA, noted that if the bakset with 100 oranges were taken away, then the average number of oranges per basket drops by 1%. Find the largest possible number of oranges in any basket if the initial average of oranges per basket was 80.

(1) 1680 (2) 1800
(3) 1755 (4) 1825

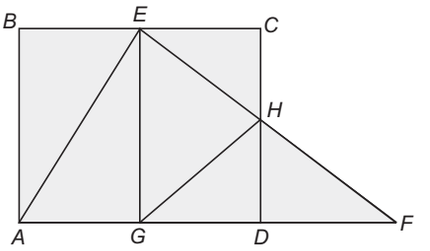
47. Let $n(>1)$ be a positive integer. Then the largest integer 'm' such that $[n^m]$ divides $(1 + n + n^2 + \ldots + n^{127})$ is (given $[n^m]$ represents the least integer greater than n^m)

(1) 127 (2) 63
(3) 64 (4) 32

48. The value of $\log_3 2 \cdot \log_4 3 \cdot \ldots \cdot \log_{10} 9 \cdot \log_{11} 10$ is

(1) $\log_{11} 2$ (2) $\log_{10} 3$
(3) $\log_4 5$ (4) $\log_2 11$

49. E and G are the mid points of sides BC and AD respectively of rectangle $ABCD$. AEF is a right angle and $AF = \frac{4}{3} AD$. What is the ratio of the areas of triangle ECH and rectangle $ABCD$?



(1) 3 : 22 (2) 1 : 6
(3) 1 : 5 (4) 3 : 20

50. If $g(x) = 1 - x^2$ and $f[g(x)] = \frac{x^2 - 1}{x^2}$, when $x \neq 0$, then $f\left(\frac{1}{2}\right) = ?$

(1) 3 (2) 3/4
(3) 1 (4) None of these

51. Chantu and Bantu bet on reaching the top of a $30\sqrt{5}$ m high building first. Chantu has to reach by pole to a height equal to the height of the building while Bantu along a spiral staircase of the same height. The radius of the spiral is 7 m and there are 5 complete spirals in the staircase. If the bet ended in a tie, what was the ratio of the speeds of Chantu and Bantu respectively ?

(1) $4\sqrt{5} : 23$ (2) $5\sqrt{3} : \sqrt{23}$
(3) $3\sqrt{5} : 23$ (4) $5\sqrt{3} : 23$

52. If $n = 3(1 + 2 + 3 + 4 + 5) + 4(2 + 3 + 4 + 5 + 6) + 5(3 + 4 + 5 + 6 + 7) + 6(4 + 5 + 6 + 7 + 8) +$ upto 18 terms, what is the value of n ?

(1) 14375 (2) 10545
(3) 14325 (4) 10525

53. Find the term independent of x in the expansion of $\left(\frac{1}{2} x^{\frac{1}{3}} - x^{\frac{-1}{5}}\right)^8$.

(1) 7 (2) 8
(3) 15 (4) None of these

54. A woman can complete a certain job in 8 days. She works for a day and then is replaced by a woman who can do the same job in 7 days. Then next day, this woman is also replaced by a woman who can do this job in 6 days and so on. On which day is the job completed ?

(1) Fourth (2) Seventh
(3) Sixth (4) Fifth

55. The number of integer solutions of the equation $y^2 = 1 + x + x^2$ are

(1) 1 (2) 2
(3) 3 (4) 4

56. A group of masons, all of equal capacities, were hired to work for Rs 10000. From this group one mason started the work, second day one more mason joined him and like this every day one more mason joined the group. If every mason worked at half of their own efficiency then the work was completed in exactly 15 days. Find the maximum amount earned by any mason in the group

(1) Rs 750 (2) Rs 1250
(3) Rs 1500 (4) Rs 1300

57. Ram Lal professes to sell 2 kg of Kajoo at a loss of 20%, Now to avoid the loss, he uses two weighing stones, each of which is marked 1 kg, but weighs less. By doing so he gains $6\frac{2}{3}\%$ on selling evey 2 kg of Kajoo. If it is given that one of the stones weights only 750 g, then how much does the second stone weight ?

(1) 900 g (2) 850 g
(3) 750 g (4) $583\frac{1}{3}$ g

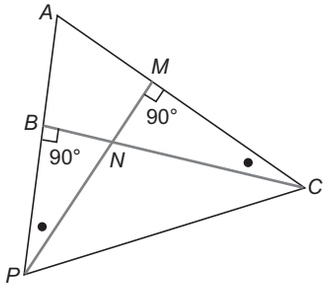
58. Amitabh, Aishwarya, Abhishek, Nana and Madhuri are among 7 film celebrities invited at TISS to speak on different subjects in psychology. Abhishek says he must speak after Aishwarya and Nana says he must speak before Amitabh, who in turn requested the college management to schedule his speech before Aishwarya. How many different orders of speakers are possible [given the above constrainsts] ?

(1) 5040 (2) 1260
(3) 210 (4) 420

59. If $P = \sum_{i=1}^{x} x_i$. Further it is known that $x_1 = x_2 = x_3 = x_i = \ldots = x_x = x$. where x is a natural number which of the following is a possible value of P ?

(1) 3473569949 (2) 5775794629
(3) 5773416289 (4) 32296449

60. M is the mid point of the hypotenuse AC of a right angled triangle ABC. The perpendicular MP to AC meet AB produced at P and intersects BC at N. $MN = 3$ cm and $PN = 9$ cm. Find the length of the hypotenuse AC.



(1) 10 cm (2) 15 cm
(3) 12 cm (4) 18 cm

61. ABC is a triangle $\angle A = 30°$, $\angle B = 60°$ and $AB = 10$ cm. Find the length of the shorter trisector of $\angle C$.

(1) $2.5\sqrt{3}$ cm (2) $3\sqrt{2}$ cm
(3) $2\sqrt{5}$ cm (4) 4 cm

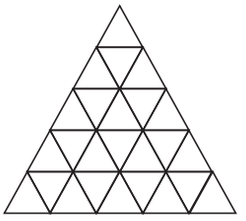
62. Two boys with one bicycle between them set out from A in the direction of B, one by bicycle and the other on foot. At a certain distance from A the one riding the bicycle left it by the road and continued towards B on foot. The one who had started out on foot reached the bicycle and rode the rest of the distance. Both reached B at the same time. On the return trip from B to A, they did as before, but this time the first cyclist rode 1 km more than the first time and so his comrade arrived in 21 min after he did. Find the speed of the faster boy on foot if they both did 20 km/h cycling and on foot, the first takes 3 min less to cover each kilometre then the second

(1) 3 km/h (2) 4 km/h
(3) 5 km/h (4) 6 km/h

63. The percentages (by weight) of alcohol in three solutions form a geometric progression. If we mixture the first, second and third solutions in the weight ratio of 2 : 3 : 4, we obtain a solution containing 32% alcohol. If we mixture them in the weight ratio 3 : 2 : 1, we obtain a solution containing 22% alcohol. What is the percentage of alcohol in each solution ?

(1) 6%, 12%, 24% (2) 9%, 18%, 36%
(3) 8%, 12%, 18% (4) 12%, 24%, 48%

64. What is the total number of triangles in the following figure ?



(1) 26 (2) 36
(3) 60 (4) 48

65. Four of my friends love collecting books. Last year, they purchased a total of 70 books. When I saw their purchases, I noted that each of them had purchased a different number of books and the number of books purchased by each was an integral multiple of purchases of every other friend who purchased less books than him. What is the least number of books purchased by the friend who had purchased the most books ?

(1) 56 (2) 64
(3) 24 (4) 48

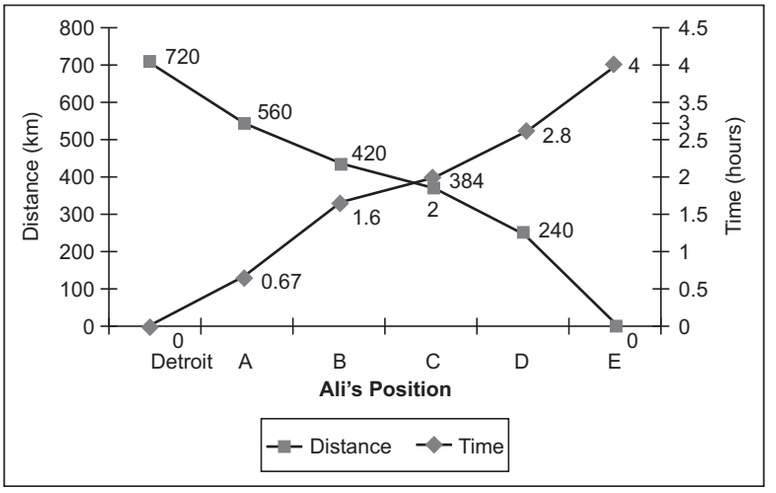
66. Cost of 15 sets, each set consisting of a ball pen, a gel pen, and a fountain pen is Rs 165. Shobhna was given Rs 65 by her father so that she could buy 5 ball pens, 6 gel pens, and 7 fountain pens. However, if Shobhna wanted to buy 5 fountain pens, 6 gel pens and 7 ball pens, how much more money would she require ?

(1) Rs 2 (2) Rs 4
(3) Rs 5 (4) Rs 10

Directions (Q. 67-70) *Onthe basis of the information given below.*

Ali and Sonali live in Detroit and Cincinnati respectively and during weekends one of them drives down to the other's city so as to spend time together. However at one of the weekends, there was some confusion resulting in both of them starting at the same time (6 pm on Friday).

The line chart below shows time and distance (km) between Ali and Sonali when Ali reached town A, B, C, etc on his way.



Traffic faced by Ali was quite dense initially and he could drive at uniform 60 km/h only till 8 pm. From 8 pm, Ali uniformly raised his speed to 180 km/h for next 40 min and kept constant at that speed for another 40 min. Then till 10 pm, he uniformaly reduced his speed to 120 km/h to reach town E.

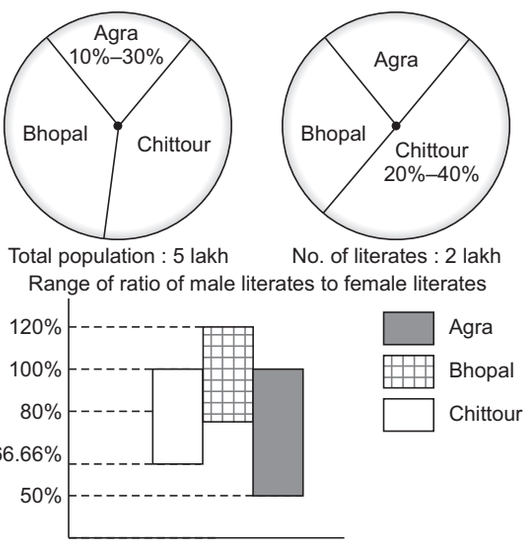
67. What is the distance, by road, of town D from Cincinnati ?
(1) 196 km (2) 224 km
(3) 256 km (4) 496 km

68. Between 6 pm and 10 pm, average speed of Ali and average speed of Sonali were in the ratio
(1) 8 : 7 (2) 7 : 5
(3) 5 : 3 (4) 3 : 2

69. During the above period, what maximum speed did Sonali assume ?
(1) 90 km/h (2) 120 km/h
(3) 180 km/h (4) Can't be determined

70. Ali's speed bore maximum ratio to Sonali's average in the corresponding interval speed at the same time when Ali was driving between
(1) Towns A and B (2) Towns B and C
(3) Towns C and D (4) Towns D and E

Directions (Q. 71-74) *These questions on the basis of following information.*



$$\text{Literacy rate of city} = \frac{\text{Number of literates in the city}}{\text{Total population of the city}}$$

Minimum population of at least one of Bhopal and Chttour is 30% of total population. Population of none of the cities is less than 10% of total population.

Minimum number of literates in each of the cities, Agra and Bhopal, is not less than 30% of total literates.

71. If number of literates in Chittour is 80000, then minimum rate of literacy in Agra could be
(1) 33.33 % (2) 66.67%
(3) 40 % (4) Indeterminate

72. If number of literates in Agra is 100000, then the minimum possible rate of literacy in Bhopal is appoximately
(1) 17% (2) 12%
(3) 20% (4) 23%

73. Number of literate males in which of the cities can be the highest of the three cities ?
(1) Agra or Bhopal or Chittour
(2) Bhopal
(3) Agra or Bhopal
(4) Agra

74. If the population distribution is Bhopal 30%, Chittour 60%, the then maximum number of literate females in Agra city is approximately
(1) 30000 (2) Data inconsistent
(3) 20000 (4) 25000

Directions (Q. 75-76) *On the basis of following information.*

A geneticist carried out a survey in Hamburg among newborns. It was revealed in her survey that at least 65% of newborns had their eyes dissimilar to their mother's, at least 75% newborns had their nose dissimilar to their mother's and at least 85% newborns had their fingers dissimilar to their mother's.

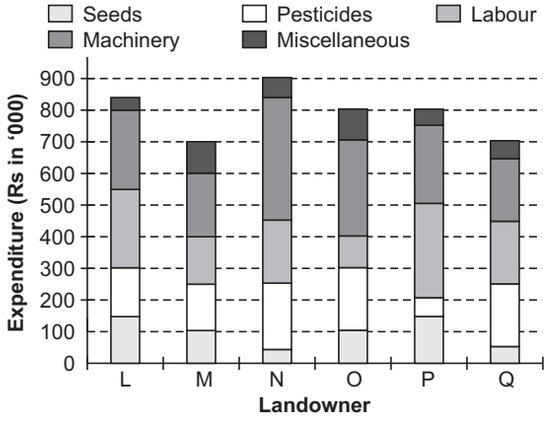
NOTE Throughout this passage and in questions figures for newborn refer to live births only.

75. Assuming, 1200 live births took place in January in Hamburg which of the following statements does not disagree with survey results ?

(1) 200 newborns had their eyes, nose as well as fingers similar to their mother's
(2) 950 newborns had their eyes, and nose dissimilar to their mother's but not fingers
(3) 300 newborns had their eyes, nose as well as fingers dissimilar to their parents
(4) 450 newborns had all of eyes, nose and fingers dissimilar to their mother's

76. Assuming survey results to be true and considering data of Q. 75, if every newborn is dissimilar to its mother in either its nose, or eyes or fingers and exactly 15% newborns are dissimilar to their mother in respect of just one of these at least how many new borns are dissmilar to their mother in respect of all of these ?
(1) 120 (2) Indeterminate
(3) 360 (4) 480

Directions (Q. 77-81) *Study the graph carefully to answer the following questions. Expenditure (in Rs '000') incurred by six Landowners in a year on the cultivation of their lands.*



77. What is the amount of expenditure incurred by P on machinery ?
(1) Rs 100000 (2) Rs 150000
(3) Rs 200000 (4) Rs 2500000

78. The amount of expenditure incurred by L on pesticides is what per cent of the amount of expenditure incurred by Q on pesticides ?
(1) 75 (2) 60
(3) 50 (4) 85

79. What is the total amount of expenditure incurred on labour by all landowners together ?
(1) Rs 12000 (2) Rs 120000
(3) Rs 1200000 (4) Rs 1200

80. Expenditure on seeds by Q is approximately what percent of its total expenditure ?
(1) 12 (2) 8
(3) 3 (4) 7

81. Miscellaneous expenditure incurred by all landowners together is approximately what per cent of the total expenditure incurred on all items together by all landowners ?
(1) 6 (2) 10
(3) 15 (4) 20

■ Analytical and Logical Reasoning

Directions (Q. 82-83) *In an institute there are five identical rooms having different items in it. Every identical looking room has a name indicating its contents. The details of the contents and the name of each room are as given below.*

Contents	Name of Rooms
Two Printers	Pashupatti
Two Computers	Chandrachud
Two Bags	Bagbahadur
One Printer and One Computer	Purnachandra
One Bag and One Computer	Biswachakra

One day somebody in the institute changed the name plate of every room such that no room contains the name correctly explaining contents.

82. If supervisor of the institute opened a room named Biswachakra and found that one item was a bag. Which of the following would **difinitely be correct ?**
(1) The other item will be a bag
(2) The other item will not be bag
(3) The other room named bagbahadur will contain a bag
(4) The other item is a computer

83. If it is known that room named Purna chandra does not contain either any printer or any computer, the room named Pashupatti does not contain any printer and room named Chandrachud contains one computer and one bag which of the following will **definitely be true** if only one of the remaining rooms is opened ?
(1) It will have at least one computer and printer
(2) It will have two printers
(3) It will have at least one computer
(4) It will have at least one printer

Directions (Q. 84-87) *Study the following information corefully and answer the questions that follow.*

Mr. Bashu recently redecorated his house by coordinating orange and three other colours for the walls, carpets and curtains of four different rooms from the information below. Determine the colours of the carpet, walls and curtains for each of the room and answer the following questions.

(i) Yellow was the only colour used in all the four rooms. It was used at least once for walls, carpets and curtains.
(ii) Three different colours were used in each room but only the dining-room and the bed-room were decorated in the same three colours.

(iii) The same colour was chosen for the curtains in the bed-room, the carpet in the living-room and the walls in the dining-room. That colour was not used at all in the study-room.
(iv) The only room with both green and grey in its colour scheme had carpet of the same colour as in the dining-room.
(v) Grey was the only colour used exactly twice both times for curtains.
(vi) The study-room walls were painted the same colour as the living-room walls.

84. Which of the following rooms had orange curtains and green walls ?
(1) Dining-room (2) Living-room
(3) Bed-room (4) Study-room

85. Which of the two rooms had green carpets ?
(1) Dining-room and bed-room
(2) Study-room and living-room
(3) Living-room and dining-room
(4) Study-room and dining-room

86. Which room did not use grey colour at all ?
(1) Dining-room (2) Study-room
(3) Cannot say (4) Living-room

87. The dining-room has …… curtains.
(1) green (2) yellow
(3) orange (4) grey

Directions (Q. 88-92) *It being given that + dinotes greater than – dinotes 'Les than'. × denotes not 'greater than' : ÷ denotes not less than □ denotes 'not' equal to and Δ denotes equal to*

88. $a \square b \div$ implies that
(1) $b - c + a$ (2) $b \times c \Delta a$
(3) $c \square a + b$ (4) $a \div b - c$

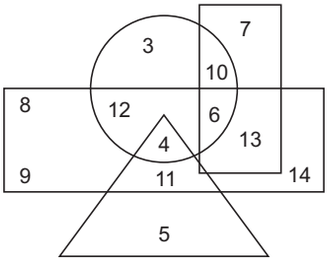
89. $a \Delta b \square c$ implies that
(1) $c - a \times b$ (2) $b + a + c$
(3) $b \times c - a$ (4) $c + b \div a$

90. $c \square b \div a$ implies
(1) $a - c \Delta b$ (2) $a \Delta c + b$
(3) $c \Delta a - b$ (4) $b + c - a$

91. $a + b + c$ does not imply
(1) $b - a + c$ (2) $b - a - c$
(3) $c - b - a$ (4) $c - a + b$

92. $a \div b \times c$ does not imply
(1) $b - c \Delta a$ (2) $c + b \times a$
(3) $a \Delta b - c$ (4) $c \times a \Delta b$

Directions (Q. 93-95) *The following questions are based on the diagram given below.*



Represents
(i) Rectangle → mates
(ii) Triangle → educated
(iii) Circle → urban
(iv) Square → civil servants

93. Who among the following is only a civil servant but not a male nor urban oriented and uneducated ?
(1) 7 (2) 8
(3) 9 (4) 11

94. Who among the following is a male, urban oriented and also a civil servant but not educated ?
(1) 13 (2) 12
(3) 6 (4) 7

95. Who among the following is a female, urban resident and also a civil servant ?
(1) 6 (2) 7
(3) 10 (4) 12

96. P is father of G, A is sister of C. D is uncle of A, C is brother of G and G is father of B, who is uncle of B ?
(1) D (2) P
(3) C (4) None of these

97. If the order of letters of each of the following words is reversed then which of the following will be a meaningful word thus formed ? If more than one such word can be formed mark 'S' as the answer and if no such word can be formed mark 'X' as the answer
NAIL, PAIL, RAIL, MADAM, REST
(1) RAIL (2) MADAM
(3) S (4) X

98. If REQUEST is written as S2R52TU, how will ACID be written ?
(1) BDJE (2) B3J4
(3) IC94 (4) None of these

Directions (Q. 99-103) *Study the following information corefully and answer the question given below.*

Gautam passes through seven lanes to reach his school. He finds that 'Yellow Lane' is between his house and 'Kama Lane'. The third lane from his school is 'Apple Lane' and 'Peacock lane' is immediately before the 'Park Lane'. He passes 'Ash Lane' at the end 'Kama Lane' is between 'Yellow lane' and 'Peacock Lane'. The sixth lane from his house is 'Rao Lane'.

99. How many lanes are there between 'Kama Lane' and 'Rao Lane' ?
(1) One (2) Two
(3) Three (4) Four

100. After passing the 'Park Lane', how many lanes does Gautam cross to reach the school ?
(1) Four (2) Three
(3) Two (4) One

101. After passing the 'Yellow Lane', how many lanes does Gautam cross to reach the school ?
(1) Four (2) Six
(3) Two (4) One

102. Which lane is between 'Park Lane' and 'Rao Lane' ?
(1) Yellow Lane (2) Kama Lane
(3) Apple Lane (4) Peacock Lane

103. If the house of Gautam, each lane and his school are equidistant and he takes two minutes to pass one lane then how much time will he take to reach school from his house ?
(1) 18 min (2) 16 min
(3) 14 min (4) 12 min

104. Shailendra takes casual leave on the first working day of every month. The office remains closed on every saturday and sunday. If the first working day of a 30 day month is Tuesday, then his next. Casual leave will be on which of the following days ?
(1) Wednesday (2) Thursday
(3) Friday (4) None of these

105. If it is possible to make a nunber which is perfect square of a two digit odd number with the second. The sixth and ninth digits of the number 187642539, which of the following is the digit in the unit's place of that two digit odd number ?
(1) 1
(2) 7
(3) 9
(4) No such number can be made

106. Sitting at a square table are a smuggler, a mafia boss, a bootlegger and a contract killer. Only two of these men, mama and kalia are genuine criminals. The other two are CID officers posing as criminals. Mama is sitting opposite the mafia boses the junior CID officer is sitting to the left of the smuggler and the senior CID officer is sitting opposite the bootlegger. The junior officer as
(1) smuggler (2) mafia boss
(3) bootlegger (4) contract killer

Directions (Q. 107-109) *Following questions are based on the five three digit numbers given below.*

518 725 849 387 634

107. If the positions of the first and the third digits in each of the numbers are interchanged. Which of the following will be the lowest after the rearrangement ?
(1) 518 (2) 725
(3) 849 (4) None of these

108. If the positions of the first and the second digits in each of the numbers are interchanged, which of the following will be the third from the top when they are arranged in descending order after the rearrangement ?
(1) 518 (2) 725
(3) 849 (4) None of these

109. If the positions of the second and the third digits in each number are interchanged which of the following will be the difference between the highest and the lowest number after the rearrangement ?
(1) 518 (2) 725
(3) 849 (4) 516

Directions (Q. 110-112) *Refer to the following data and answer the questions that follow.*

There is an empty three shelf bookcase. T, U, V, W, X, Y and Z are seven objects to be placed either on the bottom, middle or top shelf of the bookcase with the following conditions.
(i) At least two objects should be on the top shelf
(ii) Three should be no more than four subjects on any shelf.
(iii) T and U should be on different shelves
(iv) V is to be either on one shelf or two shelves above the shelf where X is placed.
(v) W is to be either on one shelf or two shelves above the shelf that X is placed.

110. If V and W are the only two objects on one of the shelves and four objects are on the middle shelf which of the following must be true ?
(1) Y is on the top shelf
(2) U is on the bottom shelf
(3) T is on the middle shelf
(4) Z is on the middle shelf

111. If all the seven objects are on two shelves. Which of the following must be true ?
(1) T, V and W are on the top shelf
(2) No more than three objects are on the middle shelf
(3) At least three objects are on the top shelf
(4) U is on the top shelf

112. If V and T are the only objects on one of the shelves, which of the following must be true ?
(1) U is either on the middle or top shelf
(2) X is on the bottom shelf
(3) W is on the top shelf
(4) if Y is on the bottom shelf, Z is on the middle shelf

Directions (Q. 113-114) *In each of the following question, a matrix of certain characters is given. These characters follow a certain trend row wise or column wise. Find out this trend and choose the missing character.*

113.

6	6	8
5	7	5
4	3	?
120	126	320

(1) 12 (2) 8
(3) 4 (4) 16

114.

7B	5C	6B
3C	9B	19A
15A	17A	?

(1) 14B (2) 12C
(3) 10C (4) 16C

Directions (Q. 115-116) *The following questions are based on the information given below.*

The following are the data on 500 students who appeared for examination in Geography, Economics and Commerce.

Passed in all three subjects	191
Passed in Geography only	72
Passed in Economics only	61
Passed in Commerce only	59
Failed in Geography	199
Failed in Economics	206
Failed in Commerce	201
Failed in all the three subjects	70

115. How many have passed in exactly two subjects ?
(1) 152 (2) 144
(3) 47 (4) 56

116. How many have passed in at least one subject ?
(1) 430 (2) 390
(3) 427 (4) 327

Directions (Q. 117-119) *A, B, C, D and E are five persons holding certain amount of money each (all different) when B, C, D and E exchanged their amounts amongst themselves so that no one had their original amount, it is observed that*

I. B possesses the highest amount amongst all persons.
II. D possesses the lowest amount amongst all persons, which is Rs 20 lessthan what A has.
III. E and C possess Rs 50 and Rs 70 respectively.

When A, C, D and E exchanged their initial amounts amongst themselves so that no one had their original amount it is observed that

I. A has the highest amount amongst all persons which is Rs 40 more than what B has
II. C has Rs 30 less than what A has
III. E has Rs 10 less than what D has

117. The initial amount possessed by A was
(1) Rs 80
(2) Cannot be determined
(3) 40
(4) 60

118. The initial amount possessed by B was
(1) Rs 60
(2) Cannot be determined
(3) Rs 40
(4) Rs 50

119. The initial amount possessed by C was
(1) Rs 80
(2) Cannot be determined
(3) Rs 50
(4) Rs 70

120. Should it be made compulsory for all the private sector organisation to be reserve quota for socially backward classes ?
Arguments
I. No, the private sector should not be governed by the government rules.
II. Yes, private sector organisation should also contribute in upliftment of socially backward classes.
(1) Only I is strong
(2) Only II is strong
(3) Both I and II are strong
(4) None of the above

121. Should the women be advised not to travel alone at night inview of the increasing incidence of rape and sixual abuse ?
Arguments
I. No, instead the government should take measures to control such incidences.
II. Yes, it is difficult even for the police department to control such cases.
(1) Only I is strong
(2) Only II is strong
(3) Both I and II are strong
(4) None of the above

■ General Awareness

122. Which right in India is a constitutional right but not a fundamental right ?
(1) Right to Freedom of Religion
(2) Cultural and Educational Rights
(3) Right against Exploitation
(4) Right to Property

123. The book 'The Shape of the Beast' is written by
(1) Khushwant Singh (2) Kuldeep Nayeir
(3) Arundhati Roy (4) Shabana Azami

124. Who is the Governor of the Reserve Bank of India ?
(1) Y.V. Reddy (2) D. Subbarao
(3) Montek Singh (4) None of these

125. Who has become the India's first women basketball player to play in a foreign league ?
(1) Dola Banerjee (2) Gethu Annajore
(3) Divya Singh (4) Jhoolan Goswami

126. Who has been appointed as the chairman of Prasar Bharti ?
(1) K.V. Kamath (2) Rahul Bos
(3) Nrimal Pandey (4) Arun Bhatnagar

127. What name has Michael Jackson given to his forth coming tour of concerts termed by him as his final tour ?
(1) Final Curtain Calls
(2) That's Folks
(3) The Final cerentdown
(4) The Final Stage is Beckoning

128. Which of the following aircraft retired from IAF service in March 2008 ?
(1) Mirage F-2000 (2) MIG 21
(3) MIG 23-BDI (4) MIG 27

129. Which of the following is the largest power distribution company in India ?
(1) Torrent power (2) Tata power
(3) BSES (4) Reliance power

130. Who tops the above ranking of the world's most attractive tourist destinations ?
(1) Switzerland (2) South Africa
(3) Singapore (4) Mauritius

131. Who has recently been appointed the first ever Director General (administration) of the Indian Airforce ?
(1) Air Vice-Marshal T. Kumaran
(2) Air Marshal Naresh Verma
(3) Air Marshal K. S. Gahlot
(4) Air Commodore V.K. Puri

132. Which of the following insurance companies has tied up with India post to collect the insurance premium through post offices ?
(1) LIC India (2) Birla Sunlife
(3) HDFC Standard (4) ICICI Prudential

133. Bill Gates has replaced Worren Buffettas as the richest man in the world in the Forbes 2009 list of the world's Billionaries. Who among these is at No. 3 ?
(1) Callos Slim Helu (2) Lawrence Ellison
(3) Karl Abrecht (4) Mukesh Ambani

134. Olympic Games 2012 will be held in
(1) London (2) Paris
(3) Washington (4) Tokyo

135. Who has been appointed as the next chief of the India Air Force (IAF) ?
(1) S. Subramanium
(2) Virendra Rana
(3) Yashwant Singh
(4) Pradeep Vasant Naik

136. In which one of the following states has India's largest private sector sea part been commissioned recently ?
(1) Andhra Pradesh (2) Karnataka
(3) Kerala (4) Tamil Nadu

137. Which of the following companies is the latest to get the license for operating NLD and ILD services in India ?
(1) AT and T (2) Singtel
(3) Orange Buginess (4) Verizon

138. The company in the question above has a joint venture with
(1) Bharti Enterprises (2) Lee la group
(3) Both (1) and (2) (4) None of these

139. BMW pushed Mercedez Benz to the second spot in terms of numbers sold in the Indian luxary car market. Which of the third company currently operating in this segment in India ?
(1) Audi (2) GM Cadila
(3) Toyota Lexus (4) Honda Acura

140. Which is the third European country to ligalise euthansia after the Hether Lands and Belgium ?
(1) France (2) Spain
(3) Norway (4) Luxemburg

141. Recently, the Earth Hour was celebrated across the globe on
(1) March 22 (2) March 30
(3) March 25 (4) March 28

142. The winner of the Magistral Ciudael De Linaares chess tournament 2009 is
(1) Viswanathan Anand
(2) Levon Aronias
(3) Alexander Grischuk
(4) Veselin Topalov

143. Which country has become the first nation ever to win all five titles in the open era at the All-England Championship ?
(1) China (2) Malaysia
(3) Indonesia (4) Denmark

144. Mark H. Mc Cormack Award is given to a sportsperson of which sport ?
(1) Athletics (2) Football
(3) Hockey (4) Golf

145. As per RBI guidelines, what is the period of default after which a lender can recall back the entire dues in a home loan extended to a borrower ?
(1) Two irregular instalments paying
(2) Default of two instalments
(3) Default of three instalments
(4) Default of six instalments

146. Which of these noted Indian origin writer and actor has been appointed chancellor of the University of Sussex in UK ?
(1) Amit Chaudhary
(2) Sanjeev Bhaskar
(3) Sayyed Jaffrey
(4) None of the above

147. Who has been appointed the Director of the All India Institute of Medical Sciences (AIIMS) for the next 5 years ?
(1) Dr. Rakesh Mittal (2) Dr. Girish Ahuja
(3) Dr. Krishna Aiyar (4) Dr. KC Deka

148. Rain water harvesting has become compulsory in which of the following cities ?
(1) Bengaluru (2) Chennai
(3) Mumbai (4) Chandigarh

149. India's first Independent maintenance Repair and overhaul (MRO) unit of aircrafts is to the set up at
(1) Bangaluru (2) Ahmedabad
(3) Delhi (4) Nagpur

150. …… is the world's biggest producer of solar power.
(1) Russia (2) Germany
(3) USA (4) UK

151. The world's biggest pharmaceutical company is
(1) Novartis (2) Glaxo
(3) Pfizer (4) Daiche

152. Which of these statements is/are correct ?
A. HDFC Bank has pipped ICICI Bank interms of number of branchess.
B. ICICI Bank is the largest private sector bank in terms of assests and market capitalization.
(1) Only A (2) Only B
(3) Both A and B (4) Neither A nor B

153. Italian Motorbike giant Dukati has entered into a multi-million dollar agreement with which Indian IT company ?
(1) Educamp (2) Wipro
(3) TCS (4) HCL

154. National Highway No. 3 runs from
(1) Delhi to Chennai
(2) Amritsar to Kolkata
(3) Delhi to Ahemdabad
(4) Agra to Mumbai

155. According to 'WHO' which disease will be the world's top killer by 2010 ?
(1) AIDS (2) Cancer
(3) Cardiac (4) Hepatities

156. Which country recently launched Yaogan '5' a remote sensing satellite ?
(1) Japan (2) China
(3) South Korea (4) North Korea

(OMR SHEET)

SNAP (Mock Test 5)

(Symbiosis National Aptitude Test)

General English

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
001	① ② ③ ④	009	① ② ③ ④	017	① ② ③ ④	025	① ② ③ ④	033	① ② ③ ④
002	① ② ③ ④	010	① ② ③ ④	018	① ② ③ ④	026	① ② ③ ④	034	① ② ③ ④
003	① ② ③ ④	011	① ② ③ ④	019	① ② ③ ④	027	① ② ③ ④	035	① ② ③ ④
004	① ② ③ ④	012	① ② ③ ④	020	① ② ③ ④	028	① ② ③ ④	036	① ② ③ ④
005	① ② ③ ④	013	① ② ③ ④	021	① ② ③ ④	029	① ② ③ ④	037	① ② ③ ④
006	① ② ③ ④	014	① ② ③ ④	022	① ② ③ ④	030	① ② ③ ④	038	① ② ③ ④
007	① ② ③ ④	015	① ② ③ ④	023	① ② ③ ④	031	① ② ③ ④	039	① ② ③ ④
008	① ② ③ ④	016	① ② ③ ④	024	① ② ③ ④	032	① ② ③ ④	040	① ② ③ ④

Quantitative Aptitude, Data Interpretation and Sufficiency

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
041	① ② ③ ④	049	① ② ③ ④	057	① ② ③ ④	065	① ② ③ ④	073	① ② ③ ④
042	① ② ③ ④	050	① ② ③ ④	058	① ② ③ ④	066	① ② ③ ④	074	① ② ③ ④
043	① ② ③ ④	051	① ② ③ ④	059	① ② ③ ④	067	① ② ③ ④	075	① ② ③ ④
044	① ② ③ ④	052	① ② ③ ④	060	① ② ③ ④	068	① ② ③ ④	076	① ② ③ ④
045	① ② ③ ④	053	① ② ③ ④	061	① ② ③ ④	069	① ② ③ ④	077	① ② ③ ④
046	① ② ③ ④	054	① ② ③ ④	062	① ② ③ ④	070	① ② ③ ④	078	① ② ③ ④
047	① ② ③ ④	055	① ② ③ ④	063	① ② ③ ④	071	① ② ③ ④	079	① ② ③ ④
048	① ② ③ ④	056	① ② ③ ④	064	① ② ③ ④	072	① ② ③ ④	080	① ② ③ ④

Analytical and Logical Reasoning

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
081	① ② ③ ④	089	① ② ③ ④	097	① ② ③ ④	105	① ② ③ ④	113	① ② ③ ④
082	① ② ③ ④	090	① ② ③ ④	098	① ② ③ ④	106	① ② ③ ④	114	① ② ③ ④
083	① ② ③ ④	091	① ② ③ ④	099	① ② ③ ④	107	① ② ③ ④	115	① ② ③ ④
084	① ② ③ ④	092	① ② ③ ④	100	① ② ③ ④	108	① ② ③ ④	116	① ② ③ ④
085	① ② ③ ④	093	① ② ③ ④	101	① ② ③ ④	109	① ② ③ ④	117	① ② ③ ④
086	① ② ③ ④	094	① ② ③ ④	102	① ② ③ ④	110	① ② ③ ④	118	① ② ③ ④
087	① ② ③ ④	095	① ② ③ ④	103	① ② ③ ④	111	① ② ③ ④	119	① ② ③ ④
088	① ② ③ ④	096	① ② ③ ④	104	① ② ③ ④	112	① ② ③ ④	120	① ② ③ ④

General Awarness

Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response	Q. No.	Response
121	① ② ③ ④	129	① ② ③ ④	137	① ② ③ ④	145	① ② ③ ④	153	① ② ③ ④
122	① ② ③ ④	130	① ② ③ ④	138	① ② ③ ④	146	① ② ③ ④	154	① ② ③ ④
123	① ② ③ ④	131	① ② ③ ④	139	① ② ③ ④	147	① ② ③ ④	155	① ② ③ ④
124	① ② ③ ④	132	① ② ③ ④	140	① ② ③ ④	148	① ② ③ ④	156	① ② ③ ④
125	① ② ③ ④	133	① ② ③ ④	141	① ② ③ ④	149	① ② ③ ④		
126	① ② ③ ④	134	① ② ③ ④	142	① ② ③ ④	150	① ② ③ ④		
127	① ② ③ ④	135	① ② ③ ④	143	① ② ③ ④	151	① ② ③ ④		
128	① ② ③ ④	136	① ② ③ ④	144	① ② ③ ④	152	① ② ③ ④		

SNAP

Symbiosis National Aptitude Test

Answers

Answers

Mock Test 1

1.	(4)	**2.**	(1)	**3.**	(1)	**4.**	(2)	**5.**	(1)	**6.**	(1)	**7.**	(1)	**8.**	(3)	**9.**	(4)	**10.**	(1)
11.	(1)	**12.**	(1)	**13.**	(3)	**14.**	(2)	**15.**	(1)	**16.**	(3)	**17.**	(2)	**18.**	(2)	**19.**	(3)	**20.**	(1)
21.	(1)	**22.**	(4)	**23.**	(2)	**24.**	(3)	**25.**	(2)	**26.**	(2)	**27.**	(2)	**28.**	(3)	**29.**	(4)	**30.**	(2)
31.	(2)	**32.**	(3)	**33.**	(4)	**34.**	(3)	**35.**	(4)	**36.**	(2)	**37.**	(2)	**38.**	(2)	**39.**	(3)	**40.**	(2)
41.	(2)	**42.**	(1)	**43.**	(3)	**44.**	(2)	**45.**	(2)	**46.**	(3)	**47.**	(4)	**48.**	(2)	**49.**	(1)	**50.**	(3)
51.	(2)	**52.**	(1)	**53.**	(2)	**54.**	(3)	**55.**	(2)	**56.**	(3)	**57.**	(1)	**58.**	(2)	**59.**	(3)	**60.**	(2)
61.	(2)	**62.**	(3)	**63.**	(1)	**64.**	(4)	**65.**	(3)	**66.**	(2)	**67.**	(3)	**68.**	(1)	**69.**	(4)	**70.**	(4)
71.	(3)	**72.**	(4)	**73.**	(1)	**74.**	(2)	**75.**	(3)	**76.**	(2)	**77.**	(3)	**78.**	(2)	**79.**	(4)	**80.**	(1)
81.	(2)	**82.**	(4)	**83.**	(3)	**84.**	(4)	**85.**	(4)	**86.**	(1)	**87.**	(3)	**88.**	(2)	**89.**	(3)	**90.**	(3)
91.	(2)	**92.**	(3)	**93.**	(2)	**94.**	(3)	**95.**	(1)	**96.**	(3)	**97.**	(3)	**98.**	(2)	**99.**	(1)	**100.**	(3)
101.	(1)	**102.**	(4)	**103.**	(1)	**104.**	(1)	**105.**	(3)	**106.**	(2)	**107.**	(2)	**108.**	(1)	**109.**	(3)	**110.**	(3)
111.	(3)	**112.**	(1)	**113.**	(3)	**114.**	(3)	**115.**	(2)	**116.**	(3)	**117.**	(1)	**118.**	(2)	**119.**	(3)	**120.**	(4)
121.	(1)	**122.**	(2)	**123.**	(4)	**124.**	(2)	**125.**	(2)	**126.**	(2)	**127.**	(4)	**128.**	(4)	**129.**	(4)	**130.**	(2)
131.	(4)	**132.**	(2)	**133.**	(2)	**134.**	(3)	**135.**	(4)	**136.**	(2)	**137.**	(2)	**138.**	(2)	**139.**	(1)	**140.**	(4)
141.	(3)	**142.**	(3)	**143.**	(3)	**144.**	(1)	**145.**	(3)	**146.**	(3)	**147.**	(2)	**148.**	(1)	**149.**	(1)	**150.**	(1)
151.	(3)	**152.**	(4)	**153.**	(2)	**154.**	(4)	**155.**	(1)	**156.**	(2)	**157.**	(2)	**158.**	(2)	**159.**	(3)	**160.**	(1)

Mock Test 2

1.	(4)	**2.**	(3)	**3.**	(4)	**4.**	(1)	**5.**	(1)	**6.**	(1)	**7.**	(3)	**8.**	(4)	**9.**	(3)	**10.**	(4)
11.	(4)	**12.**	(2)	**13.**	(2)	**14.**	(1)	**15.**	(3)	**16.**	(3)	**17.**	(3)	**18.**	(1)	**19.**	(1)	**20.**	(2)
21.	(2)	**22.**	(2)	**23.**	(3)	**24.**	(2)	**25.**	(4)	**26.**	(3)	**27.**	(2)	**28.**	(2)	**29.**	(1)	**30.**	(2)
31.	(3)	**32.**	(4)	**33.**	(1)	**34.**	(1)	**35.**	(2)	**36.**	(3)	**37.**	(2)	**38.**	(2)	**39.**	(3)	**40.**	(2)
41.	(4)	**42.**	(1)	**43.**	(2)	**44.**	(1)	**45.**	(2)	**46.**	(1)	**47.**	(2)	**48.**	(2)	**49.**	(2)	**50.**	(3)
51.	(2)	**52.**	(1)	**53.**	(1)	**54.**	(2)	**55.**	(3)	**56.**	(2)	**57.**	(3)	**58.**	(3)	**59.**	(1)	**60.**	(2)
61.	(2)	**62.**	(1)	**63.**	(2)	**64.**	(1)	**65.**	(2)	**66.**	(4)	**67.**	(4)	**68.**	(1)	**69.**	(3)	**70.**	(2)
71.	(4)	**72.**	(1)	**73.**	(3)	**74.**	(1)	**75.**	(1)	**76.**	(1)	**77.**	(4)	**78.**	(2)	**79.**	(4)	**80.**	(3)
81.	(3)	**82.**	(2)	**83.**	(2)	**84.**	(2)	**85.**	(3)	**86.**	(3)	**87.**	(3)	**88.**	(1)	**89.**	(4)	**90.**	(4)
91.	(2)	**92.**	(3)	**93.**	(4)	**94.**	(1)	**95.**	(2)	**96.**	(3)	**97.**	(2)	**98.**	(1)	**99.**	(4)	**100.**	(4)
101.	(2)	**102.**	(2)	**103.**	(1)	**104.**	(3)	**105.**	(4)	**106.**	(2)	**107.**	(1)	**108.**	(1)	**109.**	(4)	**110.**	(3)
111.	(2)	**112.**	(4)	**113.**	(1)	**114.**	(3)	**115.**	(2)	**116.**	(2)	**117.**	(2)	**118.**	(2)	**119.**	(3)	**120.**	(1)
121.	(4)	**122.**	(4)	**123.**	(4)	**124.**	(3)	**125.**	(3)	**126.**	(1)	**127.**	(2)	**128.**	(1)	**129.**	(2)	**130.**	(3)
131.	(2)	**132.**	(2)	**133.**	(2)	**134.**	(3)	**135.**	(1)	**136.**	(1)	**137.**	(2)	**138.**	(2)	**139.**	(1)	**140.**	(1)
141.	(2)	**142.**	(3)	**143.**	(4)	**144.**	(4)	**145.**	(1)	**146.**	(3)	**147.**	(4)	**148.**	(4)	**149.**	(1)	**150.**	(1)
151.	(1)	**152.**	(3)	**153.**	(2)	**154.**	(2)	**155.**	(2)	**156.**	(2)	**157.**	(2)	**158.**	(1)	**159.**	(3)	**160.**	(4)

Mock Test 3

1.	(3)	**2.**	(1)	**3.**	(4)	**4.**	(2)	**5.**	(4)	**6.**	(2)	**7.**	(2)	**8.**	(3)	**9.**	(2)	**10.**	(3)
11.	(3)	**12.**	(4)	**13.**	(3)	**14.**	(3)	**15.**	(1)	**16.**	(2)	**17.**	(1)	**18.**	(1)	**19.**	(3)	**20.**	(1)
21.	(3)	**22.**	(4)	**23.**	(3)	**24.**	(2)	**25.**	(4)	**26.**	(1)	**27.**	(1)	**28.**	(3)	**29.**	(1)	**30.**	(4)
31.	(4)	**32.**	(4)	**33.**	(2)	**34.**	(3)	**35.**	(2)	**36.**	(4)	**37.**	(4)	**38.**	(4)	**39.**	(3)	**40.**	(4)
41.	(1)	**42.**	(3)	**43.**	(1)	**44.**	(2)	**45.**	(2)	**46.**	(3)	**47.**	(3)	**48.**	(2)	**49.**	(2)	**50.**	(1)
51.	(1)	**52.**	(2)	**53.**	(3)	**54.**	(1)	**55.**	(1)	**56.**	(2)	**57.**	(1)	**58.**	(3)	**59.**	(2)	**60.**	(1)
61.	(1)	**62.**	(1)	**63.**	(2)	**64.**	(2)	**65.**	(1)	**66.**	(2)	**67.**	(2)	**68.**	(1)	**69.**	(2)	**70.**	(3)
71.	(4)	**72.**	(1)	**73.**	(2)	**74.**	(4)	**75.**	(3)	**76.**	(4)	**77.**	(2)	**78.**	(3)	**79.**	(1)	**80.**	(3)
81.	(3)	**82.**	(3)	**83.**	(4)	**84.**	(4)	**85.**	(2)	**86.**	(1)	**87.**	(3)	**88.**	(2)	**89.**	(4)	**90.**	(4)
91.	(3)	**92.**	(4)	**93.**	(2)	**94.**	(4)	**95.**	(3)	**96.**	(4)	**97.**	(4)	**98.**	(1)	**99.**	(3)	**100.**	(3)
101.	(2)	**102.**	(2)	**103.**	(3)	**104.**	(2)	**105.**	(1)	**106.**	(4)	**107.**	(3)	**108.**	(2)	**109.**	(1)	**110.**	(4)
111.	(2)	**112.**	(4)	**113.**	(4)	**114.**	(3)	**115.**	(2)	**116.**	(3)	**117.**	(3)	**118.**	(1)	**119.**	(1)	**120.**	(2)
121.	(4)	**122.**	(1)	**123.**	(3)	**124.**	(1)	**125.**	(2)	**126.**	(1)	**127.**	(1)	**128.**	(1)	**129.**	(4)	**130.**	(2)
131.	(3)	**132.**	(3)	**133.**	(2)	**134.**	(3)	**135.**	(2)	**136.**	(2)	**137.**	(3)	**138.**	(2)	**139.**	(4)	**140.**	(3)
141.	(4)	**142.**	(2)	**143.**	(4)	**144.**	(1)	**145.**	(2)	**146.**	(2)	**147.**	(2)	**148.**	(1)	**149.**	(3)	**150.**	(1)
151.	(1)	**152.**	(1)	**153.**	(1)	**154.**	(1)	**155.**	(3)	**156.**	(4)	**157.**	(1)	**158.**	(3)	**159.**	(2)	**160.**	(3)

Mock Test 4

1.	(2)	**2.**	(3)	**3.**	(3)	**4.**	(3)	**5.**	(2)	**6.**	(1)	**7.**	(2)	**8.**	(3)	**9.**	(4)	**10.**	(1)
11.	(4)	**12.**	(2)	**13.**	(4)	**14.**	(2)	**15.**	(2)	**16.**	(4)	**17.**	(1)	**18.**	(3)	**19.**	(2)	**20.**	(3)
21.	(4)	**22.**	(3)	**23.**	(1)	**24.**	(2)	**25.**	(3)	**26.**	(3)	**27.**	(4)	**28.**	(3)	**29.**	(3)	**30.**	(1)
31.	(2)	**32.**	(2)	**33.**	(4)	**34.**	(4)	**35.**	(3)	**36.**	(3)	**37.**	(1)	**38.**	(4)	**39.**	(1)	**40.**	(1)
41.	(4)	**42.**	(1)	**43.**	(3)	**44.**	(3)	**45.**	(1)	**46.**	(2)	**47.**	(3)	**48.**	(3)	**49.**	(2)	**50.**	(1)
51.	(3)	**52.**	(1)	**53.**	(3)	**54.**	(1)	**55.**	(2)	**56.**	(4)	**57.**	(3)	**58.**	(1)	**59.**	(1)	**60.**	(2)
61.	(3)	**62.**	(1)	**63.**	(2)	**64.**	(2)	**65.**	(1)	**66.**	(2)	**67.**	(3)	**68.**	(3)	**69.**	(1)	**70.**	(4)
71.	(3)	**72.**	(2)	**73.**	(4)	**74.**	(4)	**75.**	(1)	**76.**	(3)	**77.**	(4)	**78.**	(1)	**79.**	(2)	**80.**	(1)
81.	(4)	**82.**	(4)	**83.**	(4)	**84.**	(4)	**85.**	(1)	**86.**	(1)	**87.**	(2)	**88.**	(1)	**89.**	(2)	**90.**	(4)
91.	(3)	**92.**	(3)	**93.**	(2)	**94.**	(3)	**95.**	(4)	**96.**	(4)	**97.**	(1)	**98.**	(3)	**99.**	(1)	**100.**	(1)
101.	(2)	**102.**	(2)	**103.**	(4)	**104.**	(4)	**105.**	(2)	**106.**	(4)	**107.**	(2)	**108.**	(2)	**109.**	(4)	**110.**	(2)
111.	(3)	**112.**	(1)	**113.**	(4)	**114.**	(4)	**115.**	(3)	**116.**	(4)	**117.**	(1)	**118.**	(1)	**119.**	(3)	**120.**	(2)
121.	(3)	**122.**	(2)	**123.**	(2)	**124.**	(2)	**125.**	(3)	**126.**	(4)	**127.**	(2)	**128.**	(2)	**129.**	(3)	**130.**	(3)
131.	(4)	**132.**	(2)	**133.**	(2)	**134.**	(2)	**135.**	(1)	**136.**	(3)	**137.**	(2)	**138.**	(2)	**139.**	(1)	**140.**	(3)
141.	(2)	**142.**	(4)	**143.**	(2)	**144.**	(2)	**145.**	(4)	**146.**	(1)	**147.**	(2)	**148.**	(3)	**149.**	(1)	**150.**	(2)
151.	(1)	**152.**	(1)	**153.**	(4)	**154.**	(4)	**155.**	(4)	**156.**	(3)	**157.**	(4)	**158.**	(4)	**159.**	(3)	**160.**	(1)

Mock Test 5

1.	(3)	**2.**	(2)	**3.**	(4)	**4.**	(3)	**5.**	(2)	**6.**	(4)	**7.**	(2)	**8.**	(3)	**9.**	(2)	**10.**	(1)
11.	(3)	**12.**	(3)	**13.**	(2)	**14.**	(3)	**15.**	(2)	**16.**	(3)	**17.**	(1)	**18.**	(3)	**19.**	(1)	**20.**	(1)
21.	(2)	**22.**	(3)	**23.**	(2)	**24.**	(4)	**25.**	(3)	**26.**	(1)	**27.**	(4)	**28.**	(4)	**29.**	(3)	**30.**	(2)
31.	(4)	**32.**	(3)	**33.**	(3)	**34.**	(4)	**35.**	(2)	**36.**	(3)	**37.**	(4)	**38.**	(1)	**39.**	(2)	**40.**	(4)
41.	(3)	**42.**	(2)	**43.**	(1)	**44.**	(4)	**45.**	(4)	**46.**	(1)	**47.**	(3)	**48.**	(1)	**49.**	(4)	**50.**	(4)
51.	(3)	**52.**	(3)	**53.**	(4)	**54.**	(3)	**55.**	(4)	**56.**	(2)	**57.**	(3)	**58.**	(3)	**59.**	(3)	**60.**	(3)
61.	(1)	**62.**	(3)	**63.**	(4)	**64.**	(4)	**65.**	(4)	**66.**	(1)	**67.**	(4)	**68.**	(2)	**69.**	(4)	**70.**	(4)
71.	(3)	**72.**	(1)	**73.**	(1)	**74.**	(2)	**75.**	(4)	**76.**	(4)	**77.**	(2)	**78.**	(1)	**79.**	(3)	**80.**	(4)
81.	(3)	**82.**	(1)	**83.**	(4)	**84.**	(3)	**85.**	(4)	**86.**	(1)	**87.**	(2)	**88.**	(3)	**89.**	(1)	**90.**	(2)
91.	(3)	**92.**	(3)	**93.**	(1)	**94.**	(3)	**95.**	(3)	**96.**	(3)	**97.**	(4)	**98.**	(4)	**99.**	(3)	**100.**	(2)
101.	(2)	**102.**	(3)	**103.**	(2)	**104.**	(2)	**105.**	(2)	**106.**	(2)	**107.**	(4)	**108.**	(4)	**109.**	(4)	**110.**	(4)
111.	(3)	**112.**	(2)	**113.**	(2)	**114.**	(4)	**115.**	(3)	**116.**	(1)	**117.**	(4)	**118.**	(3)	**119.**	(1)	**120.**	(2)
121.	(4)	**122.**	(4)	**123.**	(3)	**124.**	(2)	**125.**	(2)	**126.**	(4)	**127.**	(1)	**128.**	(3)	**129.**	(1)	**130.**	(1)
131.	(2)	**132.**	(4)	**133.**	(3)	**134.**	(1)	**135.**	(4)	**136.**	(1)	**137.**	(2)	**138.**	(3)	**139.**	(1)	**140.**	(4)
141.	(4)	**142.**	(3)	**143.**	(1)	**144.**	(4)	**145.**	(2)	**146.**	(2)	**147.**	(4)	**148.**	(1)	**149.**	(4)	**150.**	(2)
151.	(3)	**152.**	(3)	**153.**	(3)	**154.**	(4)	**155.**	(2)	**156.**	(2)								

SNAP

Symbiosis National Aptitude Test

Answers *with* Explanations

Answers with **Explanations**

■ Mock Test 1

1. (4) Accoutrements means extra items necessary for a dress etc. Belongings means movable things. Blemish is a spot or scar that spoils beauty.

2. (1) Azure means bright blue in colour like the sky.

3. (1) Defile means to make dirty or to corrupt. Disapprove means to reject.

4. (2) Hiatus means a break or gap in a series of something. Atrocious means cruel or bad. Obscure is not distinct and Dominance stands for overwhelming influence.

5. (1) Zeal means enthusiasm and one can take initiative only when one is enthusiastic about something. Indolence is laziness; Coarseness is roughness or unrefinement in behaviour or composition. Pedagogy is concerned with teaching.

6. (1) Nebulous is without a definite shape and concrete is solid and definite.

7. (1) Penchant means a tendency or liking or inclination for something.

8. (3) Perfunctory means done carelessly as a duty only.

9. (4) Transient means temporary. Eternal is permanent or without end.

10. (1) Sublime is lofty and noble and base means low and mean.

11. (1) Eminent means famous in one's profession. Petulant means peevish or bad-tempered. Amiable means friendly.

12. (1) Economise means to control one's expenditure. Splurge means to spend money freely.

13. (3) When a stock of something is made thinner it is called depleted.

14. (2) To rail is to curse heavily or complain about something. Dissent means to disagree.

15. (1) Mandatory = Necessary by rules, Pertinent = relevant, Sagacious = showing good decision.

18. (2) It means a feeling of animosity.

19. (3) The given two words have the relationship of antonym. Only chide and supplicate have the relationship of antonym.

20. (1) The given two words have the relationship of synonym. Only flammable and inflammable have the relationship of synonym.

26. (2) If the words that follow neither or either is singular, then the verb needs to be singular. If one or both of the words is plural, then the verb needs to be plural. Therefore, the verb should have been 'are' instead of 'is'.

27. (2) Superlative cannot give a sense of comparison and it means the highest degree of something or highest quality among many things.

28. (3) Insatiable means that cannot be satiated or placated. Always wanting more of something.

29. (4) All the words except (4) means brief in size or content. Abrupt means sudden.

30. (2) Obtuse is intellectually dull and illogical in speaking or understanding.

36. (2) 'where' will be used in place of 'which'.

37. (2) 'have changed' will be used in place of 'has changed'.

38. (2) Comparative; so 'more' will be used.

39. (3) 'or' will be used with 'either' in place of 'and'

40. (2) No sooner-than is the correct use.

41. (2) Volume of a cuboid $= 25 \times 20 \times 4 = 2000\ \text{cm}^3$

So, volume of a cube $= \dfrac{2000}{2}$

$= 1000\ \text{cm}^3$ or $(\text{Side})^3$

$= 1000\ \text{cm}^3$

$\therefore$ Side $= 10$ cm

$\therefore$ Total surface area of the cube $= 6 \cdot (10)^2$

$= 6 \times 100 = 600\ \text{cm}^2$

42. (1) According to the question, $6 : 4 = h : 50$

$\Rightarrow \quad h = \dfrac{50 \times 6}{4} = 75\ \text{m}$

43. (3) Clearly, if $0 < P < 1$ say 0.5, then $0.5 < \dfrac{1}{0.5}$

$\therefore \quad P < \dfrac{1}{P}$

44. (2) Putting $x = 2, 3, 4, 5, 6$ in the equation

$y = x^2 - 3x + 2$

We get $y = 0, 2, 6, 12, 20$.

45. (2) Sum of roots $= a + b = 6$

Product of roots $= a \times b = 6$

$a^2 + b^2 = (a + b)^2 - 2ab$

$= (6)^2 - 2 \cdot 6$

$= 36 - 12 = 24$

46. (3) To form a parallelogram, we need 2 parallel vertical and 2 horizontal lines.

$\therefore$ Total number of parallelogram

$= {}^6C_2 \times {}^7C_2$

$= 15 \times 21 = 315$

47. (4) $\dfrac{256}{49} - \dfrac{* \times 9}{49} + \dfrac{81}{49} = 1$

$\Rightarrow \quad 256 - * \times 9 + 81 = 49$

$\Rightarrow \quad - * \times 9 = 49 - 256 - 81$

$\Rightarrow \quad - * \times 9 = - 288$

$\therefore$ * will be replaced by $\dfrac{288}{9} = 32$

48. (2) Let $P = 5, 7, 11, 13, \ldots$

for $P = 5, (P^2 - 1) = 24$

for $P = 7, (P^2 - 1) = 48$

for $P = 11, (P^2 - 1) = 120$

for $P = 13, (P^2 - 1) = 168$

Clearly, all above numbers are divisible by 24.

49. (1) $(p + n) = 4(m - n) \Rightarrow p - 4m = -5n$ ...(i)

and $(p - n) = 3(m + n) \Rightarrow p - 3m = 4n$ ...(ii)

Solving Eqs. (i) and (ii), we get

$m = 9n$ and $p = 31n$

Put $n = 1$, we get $p = 31$

50. (3) Let there be in points marked on a plane.

$\therefore$ Total number of line segments $= {}^nC_2 = 10$

$$\Rightarrow \frac{n(n-1)}{2} = 10$$

$$\Rightarrow n^2 - n - 20 = 0$$

$$\Rightarrow (n-5)(n+4) = 0$$

$$\Rightarrow n = 5$$

51. (2) Let the sum of money be Rs x.

$$8x = x\left(1 + \frac{r}{100}\right)^3$$

$$\Rightarrow \left(1 + \frac{r}{100}\right)^3 = (2)^3$$

$$\Rightarrow 1 + \frac{r}{100} = 2$$

Again, let the sum become 16 times in n years, then

$$16x = x\left(1 + \frac{r}{100}\right)^n$$

$\Rightarrow 16 = 2^n \Rightarrow 2^4 = 2^n \Rightarrow n = 4$ yr

52. (1) $\frac{P+Q}{P-Q} = \frac{P/Q + 1}{P/Q - 1} = \frac{7+1}{7-1} = \frac{8}{6} = \frac{4}{3}$

53. (2) $\frac{L+4}{B+4} = \frac{4}{3} \Rightarrow 3L + 12 = 4B + 16$

or $3L - 4B = 4$ ...(i)

$\frac{L-4}{B-4} = \frac{2}{1} \Rightarrow L - 4 = 2B - 8$

or $L - 2B = -4$ ...(ii)

Solving Eqs. (i) and (ii), we get

$L = 12$ m and $B = 8$ m

54. (3) Relative speed $= (40 + 20) = 60$ m/s

Distance $= (120 + 120) = 240$ m

$\therefore$ Time taken by trains to cross each other $= \frac{240}{60} = 4$ s

55. (2) $(A + B)$'s 5' day's work $= 5\left(\frac{1}{25} + \frac{1}{20}\right) = \frac{45}{100} = \frac{9}{20}$

Remaining $\left(1 - \frac{9}{20}\right) = \frac{11}{20}$ of work will be finished by B in 11 days.

56. (3) $\frac{d}{v_1 + v_2} = 4$ and $\frac{d}{v_1 + v_2} = 6$

$$\Rightarrow \frac{v_1 + v_2}{v_1 - v_2} = \frac{6}{4} = \frac{2v_1}{2v_2} = \frac{10}{2} = \frac{v_1}{v_2} = 5:1$$

Required ratio $= (5 + 1) : 5 = 6 : 5$

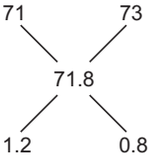
57. (1) $G = H + \sqrt{\frac{4}{L}}$

$$\Rightarrow (G - H) = \sqrt{\frac{4}{L}}$$

$$\Rightarrow (G - H)^2 = \frac{4}{L}$$

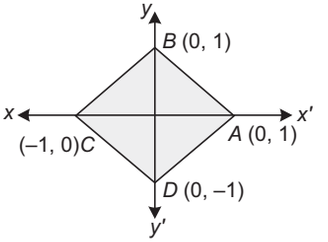
$$\Rightarrow L = \frac{4}{(G-H)^2}$$

58. (2)



Required ratio $= 1.2 : 0.8 = 3 : 2$

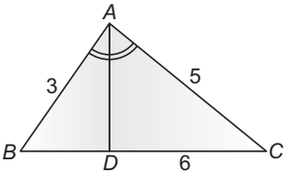
59. (3) The region given by the inequality is shown by the graph



Area of one triangle $= \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$ sq unit

Total area $= 4 \times \frac{1}{2} = 2$ sq unit

60. (2)



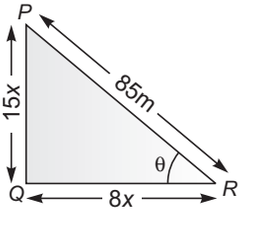
$\therefore \frac{BD}{DC} = \frac{3}{5}$

(Since, AD bisects angle A internally)

$\therefore BD : DC = 3 : 5$

Now, $BD = \frac{3}{(3+5)} \times 6 = \frac{18}{8} = \frac{9}{4} = 2.25$ cm

61. (2) Given, let $PQ = 15x$ and $QR = 8x$



$(PR)^2 = (PQ)^2 + (QR)^2$

$$(85)^2 = (15x)^2 + (8x)^2$$

$$(85)^2 = 225x^2 + 64x^2$$

$$\Rightarrow \quad x^2 = 25 \quad \text{or} \quad x = 5$$

$\therefore$ Height of the kite $= 15 \times 5 = 75$ m

62. (3) $(x + 1)$th term $- x$th term

$$= (x + 1)^2 + 1 - (x^2 + 1)$$

$$= x^2 + 2x + 1 + 1 - x^2 - 1$$

$$= 2x + 1$$

63. (1) The area of three faces are a, b and c.

$\therefore$ Sides are $\sqrt{a}$, $\sqrt{b}$ and $\sqrt{c}$.

Hence, Volume, $V = \sqrt{abc}$

and $(V)^2 = abc$

64. (4) Let the average of 19 innings be x, then

$$\frac{18x + 98}{19} = x + 4$$

$$\Rightarrow \quad 18x + 98 = 19x + 76$$

$$\Rightarrow \quad x = 22$$

Thus, average after 19th innings

$$= x + 4 = 22 + 4 = 26$$

65. (3) Total profit $= 337.50 + 1125.00 + 675$

$$= \text{Rs } 2137.50$$

$\therefore$ Required percentage $= \frac{2137.50}{114000} \times 100 = 1.8\%$

66. (2) Required ratio $= 20310 : 30040 = 203 : 200$

67. (3) Required average

$$= \frac{4240 + 4010 + 4160 + 3700 + 3930}{5}$$

$$= \frac{20040}{5} = 4008 \approx 4000$$

68. (1) Visual question : See the gap between the two bars for each country.

69. (4) We can say nothing about absolute values in the absence of data.

70. (4) Though China's GDP has increased, EFI has fallen, South Korea's inflow increased.

71. (3) We have to compare 4.80/10 and 0.72/1.

72. (4) In the year 1993, cost $= 110$

profit $=$ Rs 600

In 1990, profit was Rs 100

If profit in 1993 was Rs 600

Then, profit in 1990 $= \frac{600}{110} \times 100 =$ Rs 545

73. (1) Index of profit for the year, 2000

$$= \frac{100}{125} \times 130 = \text{Rs } 104$$

74. (2) Total sales during the period 1990 – 1995

$$= 100 + 120 + 131 + 152 + 162 + 154 = 819$$

If sales of Rs 100 becomes Rs 1200 crore, then sale of Rs 819 becomes Rs 9828 crore.

75. (3) Average sales index

$$= \frac{100 + 120 + 131 + 152 + 162 + 154}{11} + \frac{143 + 134 + 163 + 173 + 188}{11}$$

$$= \frac{1620}{121} = 147.27$$

Average cost index

$$= \frac{100 + 98 + 95 + 110 + 121 + 153}{11} + \frac{143 + 173 + 188 + 190 + 166}{11}$$

$$= \frac{1537}{11} = 139.73$$

Required difference $= 147.27 - 139.73 = 7.54$

$$= 7.5$$

76. (2) Male Engineers + Male Designers

$= 40\%$ of $(18\%$ of $10500) + 65\%$ of $(16\%$ of $10500)$

Female Engineers + Female Designers

$= 60\%$ of $(18\%$ of $10500) + 35\%$ of $(16\%$ of $10500)$

$\therefore$ Required ratio

$= (40 \times 18) + (65 \times 16) : (60 \times 18 + 35 \times 16)$

$= (720 + 1040 : 1080 + 560)$

$= 1760 : 1640 = 44 : 41$

77. (3) Required percentage $= \frac{19}{21} \times 100 = 90\%$

78. (2) Percentage of Female Professionals

$= [20\%$ of $21\% + 60\%$ of $18\% + 40\%$ of 11%

$+ 80\%$ of $15\% + 40\%$ of $19\% + 35$ of $16\%]$

$= \frac{1}{100}[420 + 1080 + 440 + 1200 + 760 + 560]\%$

$= \frac{4460}{100}\% = 44.6\%$

$\therefore$ Percentage of Male Professionals

$$= 100\% - 44.6\% = 55.4\%$$

$\therefore$ Required Difference $= (55.4 - 44.6)\%$ of 10500

$$= 10.8\% \text{ of } 10500$$

$$= 10.8 \times 105 = 1134$$

79. (4) Required percentage

$$\frac{20\% \text{ of } 21}{89\% \text{ of } 15} \times 100\% = \frac{20 \times 21}{80 \times 15} \times 100\%$$

$$\frac{420}{12} = 35\%$$

80. (1) Required ratio $= \frac{60 \times 11}{20 \times 15} = 11 : 5$

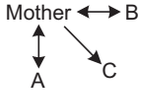
81. (2) Three meaningful words Meal, Lame and Male can be formed.

82. (4) We sleep at night. Night is called sunshine. Hence, we sleep in sunshine.

83. (3) 'CIG' is coded as 'GIC', 'ARE' is coded as 'ERA' and 'TTE' is coded as 'ETT'. Therefore, 'DIRECTION' is coded as 'RIDTCENOI'.

84. (4) The antonym of Smooth is Rough. So, the given number is written in reverse order in the code of Rough. Similarly, the code of Soft will be 8949.

86. (1) Clearly, A is niece of B.



87. (3) The only son of Vineet's grandfather is Vineet's father. The lady is the daughter of Vineet's father. Thus, Vineet is brother of that lady.

88. (2) Total number of peoples in the row

$$= (9 + 11) - 1 = 19$$

89. (3) If A is chosen, then mathematician chosen are ABC and physicists chosen are F and H or G and H. If D is chosen mathematician chosen are B, C, and D and Physicist chosen are GH.
From the above two cases it is clear that whether A is chosen or D is chosen either F or G is chosen.

90. (3) If G is rejected then, the member selected would be ABC-FH.
Hence, if G is rejected D cannot be in the group.

91. (2) Keeping in view all the conditions given, B is chosen in two ways.

Mathematician	Physicists
(i) ABC	FH or GH
(ii) BCD	GH

From (i) Two physicists are FH and GH.
From (ii) Only one way of choosing Physicist exists.
Hence, If 'B' is chosen, then' G' will also be chosen.

92. (3) $2 \times 3 = 6$ and $7 \times 6 = 42$, $1 \times 2 = 2$ and $9 \times 2 = 18$
$\therefore \quad -2 \times -3 = 6$ and $5 \times 6 = 30$
$\therefore$ 30 will comes in place of '?'.

93. (2) $2 \times 5 = 10$ and $3 \times 10 = 30$,
$-1 \times 5 = -5$ and $-5 \times 1 = 5$
$\therefore -4 \times 1 = -4$ and $3 \times -4 = -12$
$\therefore -12$ will comes in place of '?'.

94. (3) $1 + (-1) = 0$, $3 + 2 = 5$ and $4 + 3 = 7$
$\therefore$ 7 will comes in place of '?'.

95. (1) Both pickpockets and black marketer are anti-social elements.

96. (3) All the figures except (3) are concave. Figure (3) is convex.

97. (3) In all other figures lines are attached to the adjacent vertices.

98. (2) Construction of all other figures are clockwise.

99. (1) Among all, other figures has one angle as a right angle.

101. (1) Age wise order of all the four persons in ascending order is given below
Gopi, Karthik, Bala, Senthill
Hence, Senthill is eldest.

102. (4) $6 + \underline{9} = 15$, $15 + \underline{13} = 28$
$60 + 9 = 69$, $69 + \underline{13} = 82$

103. (1) $81 - \underline{4} = 77$, $77 - \underline{8} = 69$
$56 - 4 = 52$, $52 - 8 = 44$.

104. (1) The code for 'has' is CUS.

105. (3) $W - T = 23 - 20 = 3$
$S - D = 19 - 4 = 15$
$T - J = 20 - 10 = 10$
$P - G = 16 - 7 = 9$

106. (2) $P + F = 16 + 6 = 22$
$G + N = 7 + 14 = 21$
$J + E = 10 + 5 = 15$
$K + P = 11 + 16 = 27$

107. (2) $(cd + ef) \times bc = (23 + 45) \times 12 = 816$

108. (1) $(dc \times f) - (bf - d) \times d = (32 \times 5) - (15 - 3) \times 3$
$= 160 - 12 \times 3 = 124 = bce$

109. (3) $baf \div bf \times d = 105 \div 15 \times 3$
$= 7 \times 3 = 21 = cb$

110. (3) The symbols which are adjacent to triangle will not be on the opposite face.
Hence, the remaining symbol *ie*, question mark will be opposite to triangle.

111. (3) All other figure have one arrow and one line while figure (C) has two lines and one arrow.

112. (1) In all other figures all the four figures are identical.

113. (3) Except (3) all other figures are five sided. Figure (3) is hexagon.

114. (3) The inner 3×3 cubes will not have any coloured face So, 27 cubes will not be coloured.

115. (2) 8 corners are there in a cube.
All the corner cube will have 3 faces coloured.

116. (3) GKH represents one complete family.

117. (1) J is father of F.

118. (2) H is son of K.

119. (3) Car III

120. (4) Y → C, D, B, E.

Mock Test 2

1. (4) Red carpet, Red rag, and Red faced are phrases meaning accord welcome to a distinguished guest by spreading red carpet for him to walk on; a red coloured cloth-piece used to enrage a bull by a matador in the game of bull-fighting; and having a face red by anger or shame.

2. (3) Prevaricate = to speak elusively so as to conceal fact
Retrospect = to consider/think about past events
Repudiate = to refuse to accept

3. (4) incise = cut into in surgery
excise = cut to remove in surgery
precise = exact, accurate

4. (1) Correct spelling is Embarrass

5. (1) Rest three mean a mark made by something which others follow to reach someone or something viz, police followed the trail of blood on ground left by an injured culprit; mark left by a passing vehicle's tyres on the ground; and the mark left by a fleeing animal's scent or pug.

6. (1) Carbon tax means penal tax for producing carbon dioxide gas which pollutes environment.

7. (3) Read sentences 3 and 4 of the passage.

8. (4) Commensure $\Rightarrow$ com = with, mensura = measure

9. (3) In rest of the sentences, 'immunity' is the correct word which means protection from something harmful, whereas 'impurity' means disregard of any fear/danger of getting punishment for doing something wrong.

10. (4) (a), (b) and (c) all mean journey.

11. (4) Junket = journey made on public cost for pleasure
Jaunt = short distance journey for pleasure
Voyage = journey by sea-route
but, Sojourn = a temporary stay in a distant place

13. (2) First one was take-over of steel industry, Corus.

14. (1) revanchist = of revenge, so, it is like taking revenge on UK for colonising India in the past.

15. (3) Icon means symbol of particular field
whereas Idol = image of god which is worshipped

17. (3) See last-but-one para.

18. (1) The last line of 4th para.

19. (1) deplete = remove a part of stored quantity
evacuate = a place occupied
vacate = a room/house occupied
exhaust = remove something completely by consuming

20. (2) emerge = come out of something viz water, house
submerge = go under something viz water
immerse = to drown something under water
peer = to become visible
plop = sound produced by falling on a liquid surface

21. (2) Rest are the words of teaching methods.

22. (2) Wrong use of phrase, 'fool's paradise'.

23. (3) Corrects spellings are Implement, Quetude, Qurantine

24. (2) Fly ash = ask remanant of burnt coal which floats in air
Fly in the face = be visible/experienced what is unexpected
Fly by right = untrust worthy (business, financial venture)

25. (4) Factious = related to faction; Factitious = falsely made; Fictitious = of imagined, fiction; Facetious = inappropriate

26. (3) Physiognomy = face indicating race viz. mongoloid physiognomy
Profile = face as seen from one side of it
Visage = face with particular expression viz. Happy visage
Facet = one side of many-sided object viz. many faceted gem

29. (1) destined = pre determined destination in life
ordained = as ordered by divine wish

31. (3) to give child name in religious ceremony

32. (4) Reveal = make known something hitherto unknown to general people
Disclose = make known a secret
Unveil = to keep something away from other's view
Conceal = to keep a matter/news from getting known to others.

35. (2) Incumbent = holder of a post at present
Predecessor = holder of a post before the incumbent
Successor = holder of a post after the incumbent has left

41. (4) Since out of 10 points 4 are collinear.

$\therefore$ Number of straight lines formed $= {}^{10}C_2 - {}^{4}C_2 + 1$

$= 45 - 6 + 1$

$= 40$

42. (1) Decrease percent $= \frac{P^2}{100} = \frac{10^2}{100} = 1\%$

NOTE Income will always decrease in such a case.

43. (2)

P	I
650	26

$\therefore \quad r = \frac{26 \times 100}{5650 \times 1} = 4\%$

$\therefore$ Rate = 4%

$\therefore$ Amount at the end of one year = 100

A	P
104	100
650	y

$\therefore \quad y = \frac{650 \times 100}{104} = 625$

44. (1) $\bar{X} = a + \frac{\Sigma dx}{n} \quad \bar{X} = \bar{X} + \frac{\Sigma dx}{n}$

$0 = \frac{\Sigma dx}{n} \qquad \therefore \ \Sigma dx = 0$

45. (2) Here, $n(S) = 6 \times 6 = 36$

$n(E) = \{(1, 6); (6, 1); (2, 5); (5, 2); (4, 3); (3, 4)\} = 6$

$\therefore \quad P(E) = \frac{6}{36} = \frac{1}{6}$

46. (1) Here, $n(E) = 6$ and $n(S) = 8$

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{6}{8} = \frac{3}{4}$

47. (2) Here, $n(S) = 52 \qquad n(E) = 2$

$\therefore \quad P$ (Setting a red king) $= \frac{2}{52} = \frac{1}{26}$

48. (2) Since, any of the servant can carry invitation card of any of the 6 friends.

$\therefore$ Required number $= 3^6 = 729$

49. (2) $$I = P\left[\left(1 + \frac{r}{100}\right)^n - 1\right]$$

$$1576.25 = 10000\left[\left(1 + \frac{5}{100}\right)^n - 1\right]$$

or $$\left(1 + \frac{5}{100}\right)^n = 1 + \frac{1576.25}{10000}$$

$$\left(\frac{21}{20}\right)^n = \frac{11576.25}{10000}$$

or $$\left(\frac{21}{20}\right)^n = \frac{9261}{8000} = \left(\frac{21}{20}\right)^3$$

$$n = 3 \text{ yr}$$

50. (3) Peter's 1 day's work $= \frac{1}{12}$

Mohan's 1 day work $= \frac{1}{12} + 50\%$ of $\frac{1}{12}$

$= \frac{1}{12} + \frac{50}{100} \times \frac{1}{12}$

$= \frac{1}{12} + \frac{1}{24} = \frac{2+1}{24} = \frac{1}{8}$

$\therefore$ Mohan will take 8 days.

51. (2) Since, A's 1 day work $= \frac{1}{32}$ and B's is $\frac{1}{40}$.

$\therefore \quad (A + B)$' s 1 day work $= \frac{1}{32} + \frac{1}{40} = \frac{9}{160}$

$\Rightarrow (A + B)$'s 10 days work $= \frac{9 \times 10}{160} = \frac{9}{16}$

Remaining work $= 1 - \frac{9}{16} = \frac{7}{16}$

$\therefore A$ will finish in $\frac{\frac{7}{16}}{\frac{1}{32}} = \frac{7}{16} \times 32 = 14$ days

52. (1) By putting the value of a and b

$$\frac{a + 2b}{b} = \frac{2+3}{3} = \frac{5}{3}$$

53. (1) $\left(\frac{x^a}{x^b}\right)^{a+b} \cdot \left(\frac{x^b}{x^c}\right)^{b+c} \cdot \left(\frac{x^c}{x^a}\right)^{c+a}$

$= x^{a^2 - b^2} \cdot x^{b^2 - c^2} \cdot x^{c^2 - a^2}$

$= x^0 = 1$

54. (2) The required number = (LCM of 30, 45, 65, 78) $-$ 8}

$= (5 \times 3 \times 13 \times 2 \times 3) - 8$

$= 1170 - 8 = 1162$

55. (3) HCF of 1517 and 902 is 41.

$\therefore$ Number of tiles $= \frac{1517 \times 902}{41 \times 41} = 814$

56. (2) The value of $\cfrac{1}{2 + \cfrac{1}{2 + \cfrac{1}{2 - \frac{1}{2}}}}$ is

$= \cfrac{1}{2 + \cfrac{1}{2 + \frac{2}{3}}} = \cfrac{1}{2 + \frac{1}{\frac{8}{3}}} = \frac{1}{\frac{19}{8}} = \frac{8}{19}$

57. (3) Average age of 30 children = 9 yr

Total age of children $= 9 \times 30 = 270$

Average age of 30 children and 1 teacher = 10 yr

Sum of ages $= 10 \times 31 = 310$

$\therefore$ Teacher's age $= 310 - 270 = 40$ yr

58. (3) $c^z = a$

$(b^y)^z = a \quad$ or $\quad (a^x)^{yz} = a^1$

$a^{xyz} = a^1$

$\therefore \quad xyz = 1$

59. (1) Let the number be x.

$$x : \frac{1}{27} = \frac{3}{7} : \frac{5}{9}$$

or $\quad 27x : 1 = 27 : 35$

$\therefore \quad x = \frac{1 \times 27}{27 \times 35} = \frac{1}{35}$

60. (2) $\frac{a^3 + b^3}{a^2 - ab + b^2} = (a + b)$

where $\quad a = 0.623, \quad b = 0.377$

$\therefore \quad a + b = (0.623 + 0.377) = 1$

61. (2) $P(E_1) = P(E_2) = \frac{1}{2}$

$\therefore \quad P(E/E_1) = \frac{5}{100} \; P(E/E_2) = \frac{0.25}{100} = \frac{1}{400}$

$\therefore$ Required probability

$$= \frac{P(E_1) \cdot P(E/E_1)}{P(E_1) \cdot P(E/E_1) + P(E_2) \cdot P(E/E_2)}$$

$$= \frac{\frac{1}{2} \times \frac{5}{100}}{\frac{1}{2} \times \frac{5}{100} + \frac{1}{2} \times \frac{1}{400}}$$

$$= \frac{5}{5 + \frac{1}{4}} = \frac{20}{21}$$

62. (1) 3 men and 2 boys can complete a work in 6 days.

$\therefore$ 5 men and 7 boys can complete the same work in 3 days

$\therefore$ 2 men and 5 boys can complete $\frac{1}{2}$ of work in 3 days.

$\therefore$ 2 men and 5 boys can complete the work in 6 days.

63. (2) Total amount of mixture = 35 lbs

Amount of quinine $= \frac{35 \times 60}{100} = 21$ lbs

Amount of adulteration $= \frac{35 \times 40}{100} = 14$ lbs

$\therefore \quad 21 + x : 65 :: 14 : 35$

$\therefore \quad 21 + x = \frac{65 \times 14}{35} = 26$

$\therefore \quad x = 26 - 21 = 5$ lbs

64. (1)

CP	SP	MP
100	115	y
	95	100

$\therefore$ An article having cost Rs 7.20.

$$\text{MP} = \frac{115 \times 100 \times 7.20}{100 \times 95}$$

$$= \frac{165.60}{19} = \text{Rs } 8.72$$

65. (2) Amount obtained by sale = Rs 4500

Amount sent for owner = Rs 4200

Commission = Rs 300

$\therefore \quad R\% = \frac{100 \times 300}{4500} = 6\frac{2}{3}\%$

66. (4) Farmer J generated maximum amount of income and sale of rice has the largest share.

67. (4) Total income from bajra

$$= \frac{2.55\times 12 + 2\times 16 + 1.85\times 18 + 1.75\times 21 + 2.15\times 26 + 2.4\times 27}{100}$$

$$= \frac{30 + 32 + 34 + 35 + 57 + 65}{100} \approx 2.52 \text{ lakh}$$

Percentage of M's income $= \frac{1.75\times 21}{100\times 2.52}\times 100 \approx 15\%$

68. (1) Ratio $= \frac{2.15\times 23}{2.40\times 21} = \frac{989}{1008}$

69. (3) Income $= \frac{27 + 26}{100}\times 2 = 1.06$ lakh = Rs 106000

70. (2) Total income

$$= \frac{9\times 2.59 + 18\times 2 + 13\times 1.85 + 10\times 1.75 + 18\times 2.15 + 22\times 2.40}{100}$$

$$= \frac{22.95 + 36 + 24.05 + 17.5 + 38.7 + 52.8}{100}$$

$$= \frac{192}{100} = 1.92 \text{ lakh} = \text{Rs } 192000$$

71. (4) Average price per kilo of the six produces

$$= \frac{2.5 + 6.5 + 9 + 6.5 + 5 + 5}{6} = \frac{34.5}{6}$$

Total money earned by Farmer O

$$= \frac{34.5}{6}\times 240 = \text{Rs } 1380$$

72. (1) Lowest average selling price = Lowest sum of selling price by each of the six farmers. From the given data we can conclude that wheat has the lowest average selling price.

73. (3) Quantity of Wheat : Quantity of Gram

$$250 : 150 = 5 : 3$$

Now, each farmers earning from the sale of Wheat and Gram.

L = $3\times 5 + 10.5\times 3 = 15 + 31.5 = $ Rs 46.50

M = $3.9\times 5 + 12\times 3 = 19.5 + 36 = $ Rs 55.50

N = $4.6\times 5 + 9.75\times 3 = 23.0 + 29.25 = $ Rs 52.25

O = $2.5\times 5 + 9\times 3 = 12.5 + 27 = $ Rs 39.50

P = $6.5\times 5 + 8.25\times 3 = 32.5 + 24.75 = $ Rs 57.25

Q = $5.5\times 5 + 11\times 3 = 27.5 + 33 = $ Rs 60.50

∴ It is clear that farmer Q earns Rs 60.50 *ie,* the maximum money.

74. (1) Required difference

$= (170\times 6.5 + 120\times 4) - (120\times 7 + 170\times 3)$

$= 1585 - 1350 = $ Rs 235

75. (1) $\frac{\text{Average SP of Sugar}}{\text{Average SP of Rice}}\times 100 = \frac{25.5/6}{40/6}\times 100 = 64\%$

76. (1) Amount spent by the family on commuting

$$= \frac{79.2}{360}\times 45800 = \text{Rs } 10076$$

77. (4) Required ratio $= \frac{39.6}{82.8} = \frac{11}{23} = 11 : 23$

78. (2) Total amount spent on Entertainment and Shopping

$$= \frac{36 + 68.4}{360}\times 45800 = \text{Rs } 13282$$

79. (4) Required percentage $= \frac{82.8 + 36 + 54}{79.2}\times 400$

$= 218.2 = 218\%$

80. (3) Required percentage $= \frac{39.6}{68.4}\times 100 = 57.89\%$

81. (3) P has won all matches and T none. Each Q, R and S scores 3 points against T and 1 point against P. Of the two matches that Q, R and S play against each other, they have to win one match and lose one in this case the points will be 3 + 1 = 4 or all the matches will end in draw in this case the points will be 2 + 2 = 4. Hence, the total points = 3 + 1 + 4 = 8

Alternatively

P has won all and T none

∴ P got 12 and T got 4 points in all 10 matches that were played, therefore out of $10\times 4 = 40$ points, three remaining teams got $(40 - 60) = 24$ points. Hence, each gets $24/3 = 8$ points

82. (2) They all play in all 4 + 3 + 2 + 1 = 10 matches

If a match is not a draw, winning team gets 3 points and loser team gets 1.

Hence, total points scored for a match = 4 and if the match ends in a draw, each of the team gets 2 points. In both the cases total points scored remain the same. Hence, total points scored in the tournament $= 10\times 4 = 40$

83. (2) It is given that each team plays 4 matches.

Hence, for the scores of all the teams to be equal. Every team has to win two games and lose two or all the four games should end in a draw. In both the cases the points scored will be 8

$\because \quad 2\times 4 = 8 = 3\times 2 + 1\times 2$

84. (2) Suppose there are n houses in the left side and the total number of houses in the lane is m.

Then my house number will be $(n + 1)$, now

$$\frac{n(n+1)}{2} = \frac{m(m+1)}{2} - \left[\frac{(n+1)+(n+2)}{2}\right]$$

$$= \frac{m - n - 1}{2}(m + n + 2)$$

There are many pairs of n and m that satisfy the equation but $n = 6929$ and $m = 9801$ is the only pair from which we shall get more than 100 but less than 3000 houses on the right side of my house.

For (Q. 86-88)

Neither A nor B had the single cow, so C must have had it. Therefore, C cannot, has a single hen or goat. Since, A has fewer hens than B, A must has only 1 hen. By the same logic since A have more goats than B, B has only 1 goat. Since the farmer with 2 cows and 3 hens can be neither C nor A, it must be B.

Therefore, A has 3 cows, 1 hen and 2 goats, C has 1 cow, 2 hens and 2 goats.

89. (4) Two series alternated

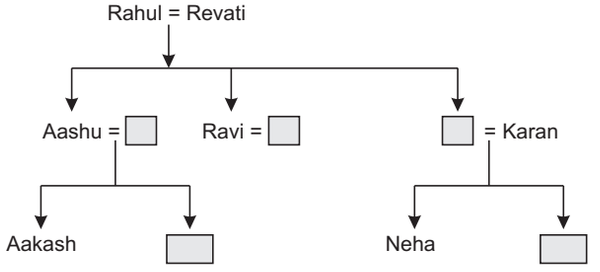
(i) EF (+ 2) IJ (+ 2) MN

(ii) ON (– 2) KJ (– 2) GF (reverse)

Hence option (4).

For (Q. 90-92)

Rahul has three children and four grand children which means Rahul is from the first generation. So is Revati and they are married. Ravi and Karan are from the second generation so is Aashu. Aakash and Neha are from the third generation and are cousins as Aakash has a male sibling. Also Aashu is Neha's uncle and Ravi has no child. So, Neha is Karan's daughter and Aakash is Aashu's son. Hence, the following family tree can be shown.



90. (4) Among Vicky and Aadi one can be the son of Aashu. Therefore, the answer cannot be determined from the given information.

91. (2) Karan is Aakash's uncle. Hence option (2).

92. (3) Kanchan being Revati's daughter is the only sister of Ravi who is married to Karan. Hence, Karan is Kanchan's husband. Hence option (3).

93. (4) This question is of direct substitution method.
RESENTMENT → YWZW3V1W3V

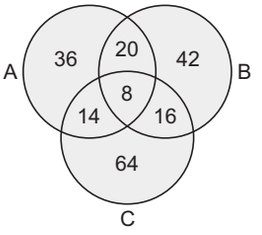
94. (1) Every letter is substituted by the fifth letter succeding in the English Alphabet
COOLER → HTTQJW (+ 5)
SUNDAY → XZSIFD (+ 5)

95. (2) Every letter is equated to the corresponding digit just as is given
KITCHEN → 5263417
CHICKEN → 3423517
C = 3, H = 4, I = 2, K = 5, E = 1, N = 7

96. (3) M + 1 = N N + 1 = O
O – 1 = N A – 1 = Z
N + 1 = O V + 1 = W
E – 1 = D Y – 1 = X
Y + 1 = Z
Thus, M + 1 = N, E – 1 = D,
A + 1 = B, N – 1 = M

For (Q. 97-101)

On the basis of information given in question following venn diagram can be prepared.



97. (2) 8 votes were in favour of all the three candidates A, B and C.

98. (1) The number of voters in favour of A irrespective of B or C = 36 + 20 + 8 + 14 = 78

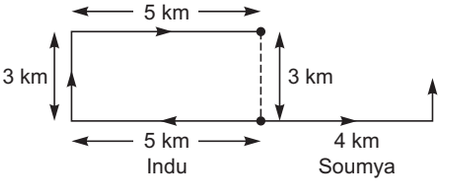
99. (4) Number of voters in favour of B irrespective of A and C = 42 + 20 + 16 + 8 = 86

100. (4) The number of voters in favour of only one of the candidates = 36 + 42 + 64 = 142

101. (2) Number of voters in favour of A and C but not B = 14

102. (2) Clearly, the computers would be used most when all the three groups are working simultaneously and this happens during the period 12 noon to 2 pm.

For (Q. 103-104)



103. (1) North

104. (3) 3 km

105. (4) As same number is subtracted from them it will not change their relative positives when they are arranged in descending order. So, only reversing them and arranging as directed we get 984, 451, 263, 245, 172.

107. (1) All moon is sun + [conversion of number sun is not white =] All suns are white A + A = A = All moon is white.

108. (1) There were 9 birds in all – 3 sparrows + 3 pigeons + 3 ducks.

109. (4) The word is divided into two-halves of three letters each and then each half is written in reverse order.

110. (3) Each letter in the word is written as letters preceding and following it in the alphabet.

111. (2) Statement I is not implicit. The statement has no relationship to other games. Statement II is implicit because the speaker talks about the consequence only alter assuming this.

113. (1) Statement I mentions only the decison of the government. It does not mention the reaction of any group of people. Hence, it is clear that statement I needs additional statement. Now look at the given options. According to the given options only one additional statement is needed to get the conclusion. Among the given three statements (A) is the additional statement that brings us to the given conclusion.

114. (3) A bud develops into flower. After that pollination take place which helps flower develop into a fruit and formation of seeds takes place when a fruit ripens.

115. (2) An accident is followed by visit to a doctor which is followed by registration of a case by the police. This is followed by lawyer's plea and the matter comes to a halt after a verdict by the Judge.

116. (2) Total odd days = 30 (May) + 30 (June) + 31 (July)
= 91
When 91 is divided by 7 it leaves no remainder. Hence, zero odd days implies that it was Monday.

117. (2) The top left hand number is obtained by adding the bottom two numbers. The top right hand number is the result of dividing the bottom two numbers.

Thus, $12 + 3 = 15$, $12 \div 3 = 4$

$22 + 11 = 33$, $22 \div 11 = 2$

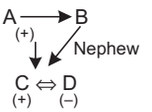
$18 + 9 = 27$, $18 \div 9 = 2$

So, $32 + x = 36$ and $32 \div x = 8$

or $x = 4$

118. (2) The series is ccacc/aabaa/bbcbb/cc.

119. (3) The series is gfeii/gfeii/gfeii/gfeii. Thus, the pattern gfeii is repeated.

120. (1) 

Mock Test 3

1. (3) Red herring means distraction of attention.
Green fingers means ability to grow plants, vegetables abundantly.
Blue blood means nobility, aristocrat family.

3. (4) It should be disabled/challenged.

4. (2) Correct word is Hierarchy.

5. (4) It means, to continue in one's efforts despite failures/difficulty.

6. (2) As mentioned in last sentence of paragraph.

7. (2) Prophesy means make prediction of future like a saint does.
Clairvoyant means one who has super-natural ability to see something what ordinary people cannot see.

8. (3) We can arrive at the answer from the last part of para 1.

10. (3) Re-incarnation means coming into physical form again of God.
Transcension means the ascending beyond physical feeling.
Emigration means going to another country for living.

11. (3) Diktat means order imposed on someone unfairly.
Edict means official order.
At somebody's behest means because somebody has ordered or requested it.
Verdict means a decision that is made by a jury in court.

12. (4) See 1st line of the paragraph.

13. (3) See last para.

14. (3) See para number 3.

15. (1) 2nd sentence of 2nd para.

16. (2) Stagflation = stagnant inflation
Deflation = opposite of inflation
Conflagration = raging fire causing great destruction

17. (1) Cataclysm = violent disaster (destruction)
Distraction = diversion of attention
Disruption = hinderance, breakage in process
Epiphany = revelation of God's existence/ manifestation.

20. (1) Fiscal = connected with government or public money, especially taxes and so matters related to it.

21. (3) Countries lying in the East of the world, orient something in right direction.

22. (4) Exact amount and exact punishment on culprit.

25. (4) Present economic scenario in the country is not Good.
Scene means the thing presented before eyes viz gory scene.
Scenery means picturesque presentation of view.

26. (1) Perspective ⇒ per = through + spect (ive) = to see.

30. (4) He goes there frequently/He makes frequent visit there and He frequents hill stations for vacations.

31. (4) He manoeuvred the car through crowded traffic.

34. (3) Stance = posture of body taken deliberately
He took a tough stance.
Tack = manner of dealing with something
Trait = distinguishing quality in a person
Gait = manner of walking

35. (2) Paediatrics — concerns children
Orthopaedics — concerns bones
Laryngitis means inflammation of larynx

36. (4) Behold = to hold eyes on something
Ogle = to look with eyes wide open
Notice = to give attention

37. (4) These are masculine–feminine pairs. But,
Master = title of a child and
Miss = of an unmarried woman

38. (4) Correct word is Ascension.

39. (3) Correct word is Emigrate.

41. (1) $[f(x)]^3 = \left(x + \frac{1}{x}\right)^3$

$= x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right)$

$= f(x^3) + 3f\left(\frac{1}{x}\right)$

42. (3) Let $f^{-1}(26) = x$

$\therefore \quad 26 = f(x)$

$\Rightarrow \quad 26 = x^2 + 1$

$\Rightarrow \quad x = \pm 5$

$\therefore \quad f^{-1}(26) = \{-5, 5\}$

43. (1) $\pi rl = 550$

$\frac{22}{7} \times r\sqrt{24^2 + r^2} = 550$

$\Rightarrow \quad r\sqrt{24^2 + r^2} = \frac{550 \times 7}{22}$

$\Rightarrow \quad r^2(24^2 + r^2) = 625 \times 49$

$\therefore \quad r^4 + 576r^2 - 625 \times 49 = 0$

$\therefore \quad r^4 + 625r^2 - 49r^2 - 625 \times 49 = 0$

$r^2(r^2 + 625) - 49(r^2 + 625) = 0$

$(r^2 + 625)(r^2 - 49) = 0$

$r^2 = 49$

$\Rightarrow \quad r = 7 \text{ cm}$

$v = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{27}{7} \times 7 \times 7 \times 24$

$= 1232 \text{ cm}^3$

44. (2) V^2 = Product of area of faces

$V^2 = 9 \times 25 \times 16$

$V = \sqrt{9 \times 25 \times 16} = 3 \times 5 \times 4 = 60 \text{ cm}^3$

45. (2) $\frac{n!}{6!(n-6)!} \times \frac{3!(n-3-3)!}{(n-3)!} = \frac{33}{4}$

$\therefore \quad \frac{n!}{(n-3)!} \cdot \frac{3!}{6!} = \frac{33}{4}$

or $\quad n(n-1)(n-2) = 6 \times 5 \times 33$

$n(n-1)(n-2) = 11 \times 10 \times 9$

$\therefore \quad n = 11$

46. (3) $\because$ Since necklace is in circular form and anti- clockwise and clockwise arrangement are not different.

$\therefore$ Required number $= \frac{1}{2}(n-1)!$

$= \frac{1}{2}(20-1)! = \frac{1}{2} \cdot 19!$

47. (3) 5 Indian can be seated along a circle

$= (5-1)! = 4!$ ways

Now, 5 Englishmen can seat at 5 places in 5! ways.

$\therefore$ Required number $= 4!\,5!$

48. (2) Each of the 5 arms can have any one of the 4 positions in 4 ways. But all the 5 arms will be in rest position in $1 \times 1 \times 1 \times 1 \times 1 = 1$ way.

$\therefore$ Total number of signals $= 4^5 - 1 = 1023$

49. (2) $\frac{a}{b} = \frac{4}{5}$ and $\frac{b}{c} = \frac{15}{16}$

$\frac{a}{b} \times \frac{b}{c} = \frac{4}{5} \times \frac{15}{16} = \frac{3}{4}$

$\frac{a}{c} = \frac{3}{4} \Rightarrow \frac{c}{a} = \frac{4}{3}$

$\frac{c^2}{a^2} = \frac{16}{9}$

$\therefore \quad \frac{c^2 + a^2}{c^2 - a^2} = \frac{16+9}{16-9} = \frac{25}{7}$

$\frac{c^2 - a^2}{c^2 + a^2} = \frac{7}{25}$

50. (1) $x + \frac{1}{x} = 2$

$\Rightarrow \quad x^2 + \frac{1}{x^2} = 4 - 2$

$x^4 + \frac{1}{x^4} = (2)^2 - 2 = 2$

51. (1) $P(E_1) = \frac{1}{7}, \; P(E_2) = \frac{1}{2}$

$\therefore \; P(A) = \left(\frac{1}{2} \times \frac{1}{7} + \frac{1}{2} \times \frac{4}{7}\right) = \frac{5}{14}$

52. (2) Number of red balls = 8

Total ball = 10 + 8 = 18

$\therefore$ Required probability $= \frac{4}{9} \times \frac{4}{9} = \frac{16}{81}$

53. (3) Total mixture = 90 + 210 = 300

Amount of water in 1st gallon $= \frac{90 \times 40}{100} = 36$

Amount of water in 2nd gallon $= \frac{210 \times 20}{100} = 42$

Amount of water in final mixture = 36 + 42 = 78

Percentage $= \frac{100 \times 78}{300} = 26\%$ water

54. (1) Man's rate downstream = 5 + 1.5 = 6.5 km/ h

Man's rate upstream = (5 – 1.5) = 3.5 km/h

Let distance be x km.

Total time taken to row x km and back $= \left(\frac{x}{6.5} + \frac{x}{3.5}\right)$

$\therefore \quad \frac{x}{6.5} + \frac{x}{3.5} = 1 \quad \text{or} \quad \frac{2x}{13} + \frac{2x}{7} = 1$

$\therefore \quad x = 2.275 \text{ km}$

55. (1) Rate in still water $= \frac{1}{2}(14 + 5) = 9.5$ km/h

Rate of current $= \frac{1}{2}(14 - 5) = 4.5$ km/h

56. (2) $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

$= \left(\frac{4}{2}\right)^3 - 3 \times \frac{9}{2} \times \frac{4}{2}$

$= 8 - 27 = -19$

57. (1) If pth term is q and qth term is p.

Then, $(p + q)$th term will be zero.

$\therefore$ 16th + 24th = 40th term will be zero.

58. (3) $\frac{a}{b} = \frac{4}{9} \quad \therefore \quad \frac{3a}{2b} = \frac{3 \times 4}{2 \times 9} = \frac{12}{18} = \frac{2}{3}$

$\therefore \quad \frac{(3a)^2}{(2b)^2} = \frac{(2)^2}{(3)^2}$

$\therefore \quad \frac{9a^2}{4b^2} = \frac{4}{9}$

$\Rightarrow \quad \frac{9a^2 + 4b^2}{9a^2 - 4b^2} = \frac{4+9}{4-9} = -\frac{13}{5}$

59. (2) $4^{1+x} + 4^{1-x} = 10$

$4 \times 4^x + \frac{4}{4^x} = 10$

Let $\quad 4^x = a$

$4a + \frac{4}{a} = 10$

$2a + \frac{2}{a} = 5$

$$2a^2 - 5a + 2 = 0$$
$$(2a - 1)(a - 2) = 0$$
$$a = \frac{1}{2}\ a = 2$$
$$4^x = 2^{-1}, \quad 4^x = 2^1$$
$$2^{2x} = 2^{-1}, \quad 2^{2x} = 2^1$$
$$\therefore \quad x = -\frac{1}{2}, \quad x = \frac{1}{2}$$

60. (1) Ten papers can be arranged in 10! ways.
When the best and worst paper come together considering two as one paper, we have only 9 paper.
∴ 9 papers can be arranged in 9! ways.
∴ Number of arrangements when best and worst papers come together is 10! − 9! 2! = 8.9!

61. (1) $\sqrt{x^{-1}y} \cdot \sqrt{y^{-1}z} \cdot \sqrt{z^{-1}x}$

$$= (x^{-1}y)^{1/2} \cdot (y^{-1}z)^{1/2} \cdot (z^{-1}x)^{1/2}$$
$$= x^{\frac{-1}{2}+\frac{1}{2}} \cdot y^{\frac{-1}{2}+\frac{1}{2}} \cdot z^{\frac{-1}{2}+\frac{1}{2}}$$
$$= x^0 \cdot y^0 \cdot z^0 = 1 \cdot 1 \cdot 1 = 1$$

62. (1) Total investment of A and B = 700 + 840 = Rs 1540

Total profit = 1520

Profit	Investment
880	1540
640	?

$$\therefore \quad ? = \frac{640 \times 1540}{880} = \text{Rs } 1120$$

63. (2) Let the ages of Madhu and Seema be $5x$ and $2x$ years respectively.
Now, according to the question

$$\frac{5x + 7}{2x + 7} = \frac{4}{3}$$
$$\text{or} \quad 15x + 21 = 8x + 28$$
$$\text{or} \quad 15x - 8x = 28 - 21$$
$$\text{or} \quad 7x = 7$$
$$\text{or} \quad x = 1$$
$$\therefore \quad \text{Age of Madhu} = 5x = 5 \times 1 = 5 \text{ yr}$$

64. (2) According to question,

$$10M \times 20D = 20W \times 15D$$
$$\text{or} \quad 200M = 300W$$
$$\text{or} \quad 2M = 3W$$
$$\therefore \quad M = \frac{3}{2}W$$

Now, $5M + 10W = 5 \times \frac{3}{2}W + 10W$

$$= \frac{15}{2}W + 10W$$
$$= 17.5W$$

Number of days taken by them to complete the work

$$\Rightarrow \quad 20W = 15D$$
$$\therefore \quad 17.5W = \frac{20 \times 15}{17.5} = 17\frac{1}{7} \text{ days}$$

65. (1) Let SP = Rs 100
CP = Rs 80
New, SP = Rs 90

$$\text{Profit} = \frac{(90 - 80) \times 100}{80} = \frac{10 \times 100}{80} = 12\frac{1}{2}\%$$

66. (2) $$\text{Required ratio} = \frac{120 \times \frac{16}{100}}{150 \times \frac{8}{100}} = \frac{8}{5} = 8 : 5$$

67. (2) In 2004 sale $= 120 \times \frac{16}{100} = 19.2$ crore

In 2005 sale $= 150 \times \frac{8}{100} = 24$ crore

Difference = 24 − 19.2 = 4.8 crore

68. (1) $$\text{Required ratio} = \frac{24}{19.2} = \frac{5}{4} = 5 : 4$$

69. (2) Textile changes by the least percentage *ie*, 3%.

70. (3) $$\text{Required ratio} = \frac{8\%}{36\%} = \frac{2}{9} = 2 : 9$$

71. (4) Total selling price of Potter A in the year 1999
= Rs (3650 × 12) = Rs 43800
Percent profit in the year 1999 = 50% (From the graph)
Let the cost price of all pots together be Rs x.
From the formula given in the question we have,

$$50 = \frac{43800 - x}{x} \times 100$$
$$\text{or} \quad 50x = 4380000 - 100x$$
$$\therefore \quad x = \frac{4380000}{150} = 29200$$

∴ Cost Price = Rs 29200

72. (1) Amount of profit earned by the Potter B in the year 1999 = Rs 40500.
Percent profit earned by the Potter B in the year 1999 = 45 percent (from the graph)
Now, from the formula

$$45 = \frac{40500}{\text{Cost Price}} \times 100$$
$$\therefore \quad \text{Cost Price} = \frac{40500 \times 100}{45} = \text{Rs } 90000$$

Selling Price = Cost Price + Profit
= Rs 90000 + Rs 40500 = 130500

73. (2) In the year 2000,
The total Selling Price of Potter B
= Total Price of Potter A
Percent profit earned by the Potter B and Potter A are 55% and 60% respectively.

Now,	Cost Price (in Rs)	Selling Price (in Rs)
Potter A	100	(100 + 55) = 155
Potter B	155	$155\left(\frac{100 + 60}{100}\right)$
		$= \frac{155 \times 160}{100} = 248$

$$\therefore \quad \text{Required ratio} = \frac{155}{248} = 5 : 8$$

75. (3) Per cent profit of Potter B in the year 2001 = 60
Per cent profit of Potter B in the year 2000 = 55
Per cent rise in per cent profit $= \frac{60-55}{55} \times 100$
$= \frac{5}{55} \times 100 = \frac{100}{11}$
$= 9\frac{1}{11} = 9\%$

76. (4) Required difference
$= [(51.2 + 40 + 36.5) - (30 + 56 + 25)]$ thousand
$= (127.7 - 111)$ thousand
$= 16.7$ thousand
$= 16700$

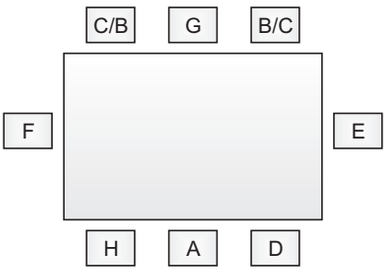
77. (2) Total number of students
$= (51.2 + 40 + 36.5 + 65 + 50 + 33 + 44 + 30 + 60 + 30 + 56 + 25)$ thousand
$= (127.7 + 148 + 134 + 111)$ thousand
$= 520.7$ thousand
$= 520700$

78. (3) Required ratio $= \frac{(40+50)}{(30+56)} = \frac{90}{86} = 45:43$

79. (1) Required percentage $= \frac{50 \times 100}{148}\% = 33.78\%$
$= 34\%$ (appox.)

80. (3) Required average number
$= \left(\frac{36.5 + 33 + 60 + 25}{4}\right)$ thousand
$= \left(\frac{154.5}{4}\right)$ thousand
$= 38625$

For (Q. 81-83)



For (Q. 84-86)

Persons	Dramatics	Computer Science	Physics	History	Mathe matics
Madhu	Yes	Yes	Yes	No	Yes
Shivani	Yes	Yes	X	Yes	No
Asha	No	Yes	Yes	Yes	Yes
Pratibha	Yes	No	Yes	Yes	No
Namita	No	No	Yes	Yes	Yes

87. (3) Suppose, the total number of persons in the group was x. Then, we have
$\frac{x(x-1)}{2} = 105$
or $x(x-1) = 210$
$\therefore \quad x = 15$

89. (4) Number of coins $= \frac{2 \times 2\pi r}{2r} = 2\pi = 6$

90. (4) $40 \# 18 @ 2 \Delta 10 \bullet 130$
$40 + 18 \div 2 \times 10 - 130$
$40 + 90 - 130 = 0$

For (Q. 91-95)

S. No.	Persons	College	State
1.	A	U	O
2.	E	T	M
3.	D/F	P	J/L
4.	B	Q	K
5.	C	S	N
6.	F/D	R	L/J

96. (4) Third Wednesday falls on 15th of the month.
Fourth Wednesday will fall on 22nd of the month.
Fifth Wednesday will fall on 29th of the month.
Fifth Friday will fall on 31st of the month.

97. (4) $n(n-1) = 600$
$\therefore \quad n = 25$

98. (1) Here, $\frac{x(x-1)}{2} = 105$
$\therefore \quad x = 15$

99. (3) Mohan's position = 9th from the top
Kiran's position = $(35 - 7 + 1) = 29$th from the top
Sohan's position $= \frac{9 + 29}{2} = 19$th from the top
Hence, Kiran's position is 10th from Sohan's position.

100. (3) D F I M R
E GH JKL NOPQ

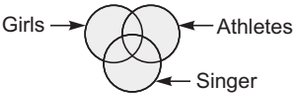
101. (2) 800 bags can be divided as group of

$\rightarrow$ 26∃	26∃	266
$\rightarrow$ 89	89	29
$\rightarrow$ 30	30	29
$\rightarrow$ 10	10	10
$\rightarrow$ 3	3	4
$\rightarrow$ 111		$\rightarrow$ 2 2
		$\rightarrow$ 1 1

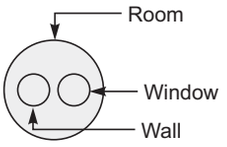
Thus, 7 weightings are required.

102. (2) The 80 bags can be distributed in 2 groups of
* 40 40
* 20 20
* 10 10
* 5 5
* 221
At the last step 3 weightings are required as a spring balance is used.
$\therefore$ Minimum number of weightings = 7

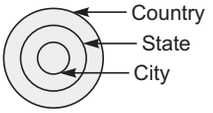
103. (3) Some girls can be athletes. Some athletes can be singers. Some girls can be singers.



104. (2) Both wall and windows are parts of a room. But wall and window are entirely different

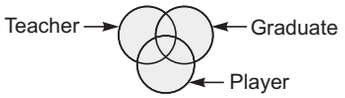


105. (1) A city lies within a state which lies within a country

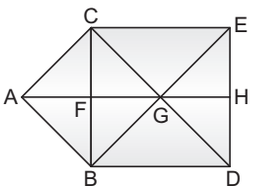


106. (4) Some wires are made of copper, but paper is entirly different.

107. (3) All three items are partly related to each other.



108. (2)



First count triangles are
AFC, AFB, BGF, CGF, CGE, BGD, DHG, ABC, ACG, CGB, ABG, GDE, BCD, CEB, EDC, EDB.
Thus, there are 16 triangles.

109. (1) The number in the third column in each row is obtained by multiplying the numbers in the first and second column in the same row. So, in the first row, $3 \times 5 = 15$ and in second row, $2 \times 4 = 8$

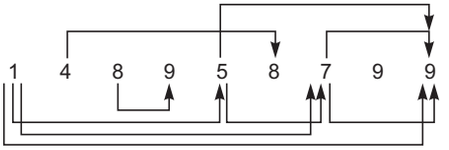
$\therefore$ Number to be filled in blank spaces $= 4 \times 6 = 24$

JL_{24}

110. (4) Nimmi returned home on Thursday. Nimmi was supposed to return three days later *ie*, on Sunday. Vimmi returned five days later from Sunday *ie*, Friday.

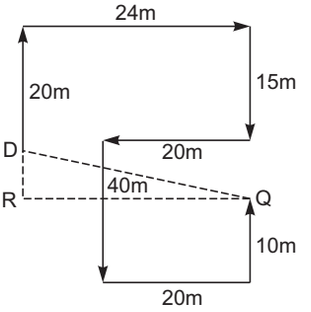
111. (2) Clearly, the number adjacent 3 are 1, 2, 5 and 4. So, the number opposite 3 is 6.

112. (4)



113. (4) Others are found in natural form.

114. (3)



Here, $PR = 40 + 15 - (10 + 20) = 25$ m

$QR = 24$ m

$\therefore \quad PQ = \sqrt{PR^2 + QR^2} = \sqrt{25^2 + 24^2}$

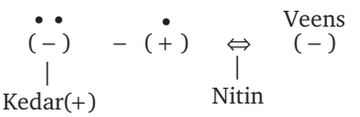
$= \sqrt{625 + 576} = \sqrt{1201}$

$= 34.65$ m

115. (2) 1 is not necessarily true because it is possible that the drug does not influence the amount of the hormone secreted but cerenters its ill-effect independently 2 is true it does not seem to be a mere co-incidence as the passage projects the role of the harmone as Nital.

116. (3) We need a situation where stress loads to illness but in 1. It is the other way round. 4 may happen simply because there are lesser number of students present during vacation times 2 may streng, then the argument bus is mere of a conclusion from the given situation. 3 strengthens the conclusion examinations are times of stress and therefore increase the number of illness cases.

117. (3)



Hence, Kedar is Veena's nephew.

■ Mock Test 4

1. (2) See the second sentence of the passage.

2. (3) 'Profess' has been used in the passage to mean "pretend by speaking false".

3. (3) It is contained in 2nd para.

4. (3) See 2nd para.

5. (2) Opening sentence in 3rd para says so.

7. (2) Alternative is another (alter) possibility instead of one.

8. (3) Laity = ignorant person.
Connoisseur = person knowing intricate art.

11. (4) Commit = 1. to do wrong act 2. give to someone for safe keeping, 3. having allegiance 4. send/refer to higher authority.

12. (2) Rash = hasty; Trash = litters; Thrash = to beat severely; Thresh = beat grains to remove husk from.

13. (4) See 1st para.

14. (2) 1st line of 5th para says so.

15. (2) See 4th para.

16. (4) See last para.

18. (3) Only first article 'a' is used to show both the qualities in one person.

19. (2) Stand someone in good stead is a phrase.

20. (3) Lest in order to prevent something from happening.

21. (4) See para 5.

22. (3) See para 2.

23. (1) Phenomenal = Prodigious.

24. (2) Rationalise = to make something justified, logical, or efficient.

25. (3) Populism = action appealing to the masses/people.

27. (4) See para 1.

28. (3) See last line of para 1.

29. (3) See last para.

30. (1) Hype = intensive publicity to create undue public interest.

41. (4) Here given letters are 6 in which 2 are similar.

$\therefore$ Required number $= \frac{6!}{2!} = \frac{720}{2} = 360.$

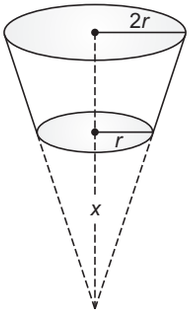
42. (1) Since, if radius becomes n time, area becomes n^2 times.

$\therefore$ Ratio of area will be $\frac{1}{4}$.

43. (3) $\frac{45+x}{x} = \frac{28}{7}$

$x = 15$

Total height of cone = 45 + 60



$\therefore$ Volume $= \frac{1}{3}\pi R^2 H - \frac{1}{3}\pi r^2 h$

$= \frac{1}{3}\times\frac{22}{7}\times 28\times 28\times 60 - \frac{1}{3}\times\frac{22}{7}\times 7\times 7\times 15$

$= \frac{1}{3}\times\frac{22}{7}\times 7\times 7\times 15(4\times 4\times 4 - 1)$

$= 48510\ \text{cm}^3$

IInd Method

Volume $= \frac{\pi h}{3}(R^2 + r^2 + rR)$

$= \frac{22}{7\times 3}\times 14[784 + 49 + 28\times 7]$

$= \frac{22\times 15}{7}\times 1029$

$= 330\times 147 = 48510\ \text{cm}^3$

44. (3) $2x + 3y = 550$

$3x + 2y = 450$

$5x + 5y = 1000$

$x + y = 200$

Cost of 3 goats and 3 cows $= 3(x + y) = 3\times 200$

$= 600$

45. (1) $4^{1+x} + 4^{1-x} = 10$

$4\times 4^x + \frac{4}{4^x} = 10$

Let $4^x = a$

or $4a + \frac{4}{a} = 10$

or $2a^2 + 2 = 5a$

or $2a^2 - 5a + 2 = 0$

$\therefore$ $a = 2$ or $\frac{1}{2}$

$4^x = 2^1$ or $4^x = 2^{-1}$

$2^{2x} = 2^1$ or $2^{2x} = 2^{-1}$

$x = \frac{1}{2}$ $x = \frac{-1}{2}$

$\therefore$ $x^2 = \frac{1}{4}$

46. (2) Loss $P\% = \frac{P^2}{100} = \frac{(14)^2}{100} = 1.96\%$

47. (3) $\tan\theta = \sqrt{2} - 1$

$h = \sqrt{(\sqrt{2}-1)^2 + (1)^2}$

$= \sqrt{2 + 1 - 2\sqrt{2} + 1}$

$= \sqrt{2\sqrt{2}(\sqrt{2}-1)}$

or $\tan\theta = \frac{l}{b} = \frac{\sqrt{2}-1}{1}$

$\therefore$ $h = \sqrt{(\sqrt{2}-1)^2 + 1^2}$

$= \sqrt{2 + 1 - 2\sqrt{2} + 1}$

$= \sqrt{4 - 2\sqrt{2}}$

$= \sqrt{2\sqrt{2}(\sqrt{2}-1)}$

$\therefore \sin\theta\cdot\cos\theta = \frac{p}{h}\times\frac{b}{h} = \frac{pb}{h^2}$

$= \frac{\sqrt{2}-1}{2\sqrt{2}(\sqrt{2}-1)} = \frac{1}{2\sqrt{2}}$

48. (3) $\sin\theta - \cos\theta = 1$

$\sin\theta + \cos\theta = x$

$\sin^2\theta + \cos^2\theta + 2\sin\theta\cdot\cos\theta$

$- 2\sin\theta\cdot\cos\theta + \sin^2\theta + \cos^2\theta = 1 + x^2$

$\therefore$ $x^2 = 2 - 1$

$x^2 = 1$

$x = \pm 1$

IInd Method

$\because$ $\sin\theta - \cos\theta = 1$

$\therefore$ $(\sin\theta - \cos\theta)^2 = 1^2$

$\sin^2\theta + \cos^2\theta - 2\sin\theta\cdot\cos\theta = 1$

$1 - 1 = 2\sin\theta\cos\theta = 0$

$\because (\sin\theta + \cos\theta)^2 = (\sin\theta - \cos\theta)^2 + 4\sin\theta \cdot \cos\theta$

$\sin\theta + \cos\theta = 1 + 2 \times 0 = 1 = \pm 1$

49. (2) $SI = \dfrac{P \times R \times T}{100}$

$200 = \dfrac{P \times R \times 2}{100}$ ...(i)

$CI = P\left[\left(1 + \dfrac{r}{100}\right)^n - 1\right]$

$210 = P\left[\left(\dfrac{100 + r}{100}\right)^2 - 1\right]$ ...(ii)

Dividing Eq. (ii) from Eq. (i), we get

$$\frac{210}{200} = \frac{\dfrac{P[10000 + R^2 + 200R - 10000]}{10000}}{P \times \dfrac{R \times 2}{100}}$$

$$\frac{21}{20} = \frac{R(R + 200)}{200R}$$

$210 = R + 200$

$R = 10\%$

IInd Method

SI = 200 Rs

CI = 210 Rs

100

100 10

$\text{Rate} = \dfrac{10}{100} \times 100 = 10\%$

50. (1) $P\left(\dfrac{E}{F}\right) = \dfrac{0.3}{0.5} = \dfrac{3}{5} \quad P\left(\dfrac{F}{E}\right) = \dfrac{0.3}{0.4} = \dfrac{3}{4}$

51. (3) Let share of Puja = x

Sony = $1682 - x$

$$x\left(1 + \frac{5}{100}\right)^3 = (1682 - x)\left(1 + \frac{5}{100}\right)^5$$

$$x = (1682 - x) \times \frac{441}{400}$$

$441x + 400x = 1682 \times 441$

$$x = \frac{1682 \times 441}{841} = 882$$

IInd Method

$$P_1 : P_2 = \left(\frac{100}{100 + 5}\right)^3 : \left(\frac{100}{100 + 5}\right)^5$$

$$P_1 : P_2 = 1 : \frac{400}{441} = 441 : 400$$

841 unit = 1682

1 unit = 2

$\therefore \quad 441 = 441 \times 2 = 882$

52. (1) In the first case either 8 men or 12 boys are taken when in the second case 20 men and 6 boys are taken. So first of all, we shall find that 20 men and 6 boys are equivalent to how many men or how many boys.

Given 8 men = 12 boys

$$20 \text{ men} = \frac{12}{8} \times 20 = 30 \text{ boys}$$

12 boys can do the work in 16 days

36 boys can do the work in $\dfrac{16 \times 12}{36} = 5\dfrac{1}{3}$ days

53. (3) CP = 1200 and gain = 12.5%

$\therefore \quad SP = \dfrac{112.5}{100} \times 1200 = 1350$

Let MP be Rs x

$$x\left(1 - \frac{10}{100}\right) = 1350$$

$$x = 1500$$

$$M \times 0.9 = C \times 1.125$$

$$\frac{C}{M} = \frac{900}{1125}$$

$$\therefore \quad MP = \frac{1125}{900} \times 1200 = 1500$$

54. (1) Volume of water flowing in 1 s = 5×30

$= 150 \text{ cm}^3$

Volume of water flowing in 1 min = 150×60

$= 9000 \text{ cm}^3$

$= 9 \text{ L}$

55. (2) Given, $\dfrac{x}{y} = \dfrac{2}{7} \Rightarrow 7x = 2y$

$$\therefore \quad \frac{7x + 2y}{7x - 3y} = \frac{2y + 2y}{2y - 3y} = \frac{4y}{-y} = -4$$

56. (4) Let B receives Rs x.

$\therefore$ Share of $A = 2x + 80$

Now, $560 - 80 = 480$

$\therefore \quad 2x : x = 2 : 1$

$\therefore$ Share of $A = 480 \times \dfrac{2}{3} = 320 + 80 = 400$

Share of $B = 480 \times \dfrac{1}{3} = 160$

57. (3)

Labour	Hour	Day	Earning
75	8	15	800
24	10	9	x

$$\therefore \quad x = \frac{24 \times 10 \times 9 \times 800}{75 \times 8 \times 15} = 192$$

58. (1) $x : y :: 1500 : 100$ or $\dfrac{x}{y} = \dfrac{1500}{1400}$

$y : z :: 1500 : 1380$ or $\dfrac{y}{z} = \dfrac{1500}{1380}$

$$\therefore \frac{x}{y} \times \frac{y}{z} = \frac{x}{z} = \frac{1500}{1400} \times \frac{1500}{1380} = \frac{1500}{1288}$$

$\therefore$ He will defeated by $1500 - 1288 = 212$ m

59. (1) Percentage of candidates who fail in the examination = 31%

$\therefore$ Percentage of candidates who pass in the examination

$= (100 - 31)\% = 69\%$

According to question,
Number of candidates who pass exceeds the number of those who fail by 247.

$\therefore \quad 69\% - 31\% = 247$

$\Rightarrow \quad 38\% = 247$

So, the total number of candidates.

$\Rightarrow \quad 38\% = 247$

$\therefore \quad 100\% = \frac{247 \times 100}{38} = 650$

60. (2) Let SP = 100

$\therefore$ Commission = 5

100 5

x 95.50

$\therefore \quad x = \frac{100 \times 95.50}{5} = 1910$

$\therefore$ Number of bottle $= \frac{1910 \times 100}{955} = 200$

61. (3) Number of instalment = 15

Monthly payment = 800

$\therefore$ Total paid amount $= 800 \times 15 = 12000$

Interest $= 12000 - 10000 = 2000$

$\therefore$ Rate $= \frac{100 \times 2000 \times 1 \times 4}{10000 \times 5} = 16\%$

62. (1) Let cash value Rs 100

Credit value = 100 + Interest

Interest $= 100 \times \frac{61}{2} \times \frac{5}{100} = 2\frac{1}{2}$

Credit value $= 100 + 2\frac{1}{2} = 102\frac{1}{2}$

Now, Cash value : Credit value $= 100 : 102\frac{1}{2}$

$= 200 : 205$

$= 40 : 41$

63. (2) Sum due $= \frac{SI \times TD}{SI - TD} = \frac{120 \times 100}{20} =$ Rs 600

64. (2) $\frac{f(10)}{f\left(\frac{1}{2}\right)} = \frac{\frac{2 \times 10 + 3}{10 - 2}}{\frac{2 \times \frac{1}{2} + 3}{\frac{1}{2} - 2}} = -\frac{69}{64}$

65. (1) If $\log 2^{x+3} = \log 64$

$\therefore \quad \log 2^{x+3} = \log 2^6$

$x + 3 = 6$

$\therefore \quad x = 6 - 3 = 3$

66. (2) Percentage profit of company C in 2003

$= \frac{2.5 - 1.75}{1.75} \times 100$

$= \frac{0.75 \times 100}{1.75} = \frac{7500}{175} = \frac{300}{7}$

$= 42.857 = 42.86$

67. (3) Required average

$= \frac{1}{6}(2.75 + 3 + 3.5 + 3 + 2.5 + 3.25)$ lakh

$= \frac{18}{6}$ lakh = Rs 300000 = 3 lakh

68. (3) Percentage increase in income of company B in year 2006

$= \frac{3.5 - 3}{3} \times 100 = \frac{50}{3} = 17$

69. (1) Income of A in 2002 = Rs 300000

Suppose expenditure of A in 2002 = Rs x

Then, $x\left\{\frac{120}{100}\right\} = 300000$

Expenditure = x

$= \frac{300000 \times 100}{120} =$ Rs 250000

70. (4) Percentage change in income of company C was the highest for year 2002. We see that from 2001 to 2002 the percentage change is 33% (from 3 to 2). In other years percentage change is less than 33%.

71. (3) Total construction in B = 100 + 175 + 275 = 550

Total construction in E = 175 + 250 + 300 = 725

$\therefore$ Required difference = 175

72. (2) Required average $= \frac{50 + 175 + 150 + 75 + 175}{5}$

$= \frac{625}{5} = 125$

73. (4) Required percentage $= \frac{1100}{1400} \times 100 = 79$

74. (4) Required approximate average $= \frac{150 + 225 + 250}{3}$

$= \frac{625}{3} = 208$

75. (1) Required ratio = 175 : 300 = 7 : 12

For (Q. 76-80)

Total number of students in the school = 2500

Total number of girls $= \frac{2500}{3+2} \times 3 = 1500$

Total number of boys $= \frac{2500}{3+2} \times 2 = 1000$

Number of boys learning only Marathi

= 16% of the boys

$= \frac{16}{100} \times 1000 = 160$

Number of girls learning only Marathi and Hindi

= 23% of the girls

$= \frac{23}{100} \times 1500 = 345$

Number of students learning only English

= 14% of the total number of students

$= \frac{2500 \times 14}{100} = 350$

Number of boys learning only Marathi and Hindi
= 120% of the number of girls learning only Marathi and Hindi
$= \frac{120 \times 345}{100} = 414$
Number of boys learning only Sanskrit
$= \frac{1}{10} \times 1000 = 100$
Number of girls learning only Marathi
= 130% of the number of boys learning only Marathi
$= \frac{130 \times 160}{100} = 208$
Number of boys learning only English
$= 19\% \text{ of } 1000 = \frac{19}{100} \times 1000 = 190$
Number of boys learning only Sanskrit, Hindi and English
= 1000 – (160 + 414 + 100 + 190) = 136
Number of girls learning only Hindi and Sanskrit
$= 32\% \text{ of } 1500 = \frac{32}{100} \times 1500 = 480$
Number of girls learning only Hindi, English and Sanskrit
= 1500 – (345 + 208 + 480 + 160) = 307

76. (3) Total number of girls learning English and Sanskrit
= 480 + 307 + 160 = 947

77. (4) Total number of student learning Marathi
= 160 + 414 + 345 + 208 = 1127

78. (1) Number of students studying two languages at a time
= 414 + 136 + 345 + 480 + 307 = 1682

79. (2) Total number of boys learning Hindi
= 414 + 136 = 550

80. (1) Number of girls who do not learn Hindi
= (208 + 160) = 368

81. (4) P Δ Q means P > Q, Q O R means Q = R and Q Δ S means Q > S
∴ We get P > Q = R > S
∴ P O Q means P = is not true.
P Δ R means P > R true R Δ S means R > S true.
Hence, option (4) is correct.

82. (4) V O W means V = W, W □ X means W < X,
W Δ Y means W > Y
∴ V = W < X and Y < X
∴ V < X *ie*, V □ X is true.
Hence, option (4) is correct.

83. (4) As I O J means
I = J ⇒ I – J = O
J □ K means J < K ⇒ K – J > O
K Δ L means K > L ⇒ K – L > O
Hence, option (4) is correct.

84. (4)
(i) This statement is irrelevent as it is not discussing the weights of the three items.
(ii) This also does not gives any information about which is th eheaviest. We can only say that either of iron or gold is the heaviest.

85. (1)
(i) Kalyani < Aditya and Kalyani > Srinivas. Thus, Srinivas will be shorter than Aditya. Hence, Srinivas is the shortest.
(ii) Srinivas < Aditya but we cannot say whether Kalyani is shorter than Srinivas or not. Hence, we cannot come to the answer.

86. (1)
(i) We cannot assume that Ashish is married. He may have adopted the child whose name is Amar.
(ii) Vinod and Vidya are married as they are husband and wife.

87. (2)
(i) This statement sets a condition about when a city can be called metropolitan. But we do not know anything about Patna's population.
(ii) This statement gives Patna's population but does not given as the condition about when a city can be classified metropolitan.
Combining both (i) and (ii) we can say that Patna is a metropolitan city.

92. (3) From first two sentences the common word computer ↔ low
From second and third sentences the common word expensive ↔ pow
From first and fourth sentences the common word Prasad ↔ bow
From second and fourth sentences the common word Prasad ↔ bow
From second and fourth sentences the common word gift ↔ dow
∴ is ↔ row.

93. (2) I. Ranu had failed is 'true' is valid.
II. Somu had failed is 'not true' is not valid.

For (Q. 94-98)
Numbers and words are arranged alternately starting with numbers. All the numbers are arranged in ascending order whereas all the words are arranged in a pattern opposite to dictionary. (The words coming last in the dictionary are arranged first).

94. (3) **Step II :** 21 white direct 72 status front 37 69.
Step III : 21 white 37 direct 72 status front 69.
Step IV : 21 white 37 status direct 72 front 69.
Step V : 21 white 37 status 69 direct 72 front.
Step VI : 21 white 37 status 69 front direct 72.
Step VII : 21 white 37 status 69 front 72 direct.
Step VII is the final step.

95. (4) **Input :** 17 85 pearls garland 67 93 restriction judgement.
Step I : 17 restriction 85 pearls garland 69 93 judgement.
Step II : 17 restriction 67 85 pearls garland 93 judgement.
Step III : 17 restriction 67 pearls 85 garland 93 judgement.

96. (4) We cannot determine the previous steps definitely in such an arrangement.

97. (1) **Step II :** 26 shop finance 48 game music 63 37.
Step III : 26 shop 37 finance 48 game music 63.
Step IV : 26 shop 37 music finance 48 game 63.
Step V : 26 shop 37 music 48 finance game 63.

98. (3) **Input :** 56 punish 48 find design 23 lavish 36.
Step I : 23 56 punish 48 find design lavish 36.
Step II : 23 punish 56 48 find design lavish 36.
Step III : 23 punish 36 56 48 find design lavish.
Step IV : 23 punish 36 lavish 56 48 find design.
Step V : 23 punish 36 lavish 48 56 find design.
Step VI : 23 punish 36 lavish 48 find 56 design.
Step VI is the final step.

99. (1) A C R I M O N Y
+1 +1 +1 +1
C A S J O M O Z
Similarly,
N A X A L I T E
+1 +1 +1 +1
A N Y B I L U F

100. (1) (3), (4) and (2) are obviously not true.

101. (2) Since by omitting O, R can be included.

102. (2) Cannot be chosen and 2 more plumbers are to be chosen.

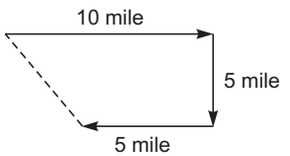
103. (4) Since no condition is attached to W.

104. (4) Since neither I nor II is compulsory.

105. (2) 'Teachers' train 'Students'. Similarly 'Coach' trains players.

106. (4) Mother's brother's wife means maternal uncle's wife *ie*, Raju is the only son of Ayesha's maternal uncle's wife. Therefore, Raju is the cousin brother of Ayesha. The sex of Ayesha is not given.

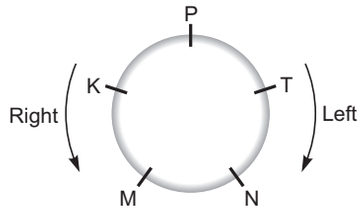
107. (2) It is clear from the diagram that point P is in North.



108. (2) Given number is 724513 number when arranged in ascending order = 123457, clearly there is only one number 2 which is as far away from the beginning in the number as when they are arranged in ascending order starting from left.

109. (4) 5 4 6 3 8
↓ ↓ ↓ ↓ ↓
4 5 7 2 9 = 27

110. (2) Sitting arrangement is given below.



P is to the immediate right of T.

111. (3) In column (1) and column (3), we have $12 \times 2 \div 72 = 3$ and $16 \times 2 \div 128 = 4$, similarly in column (2) we have $14 \times 2 \div 112 = 4$

112. (1) In column (1) and column (2) we have $(3 + 5) \div 2 = 4$ and $(6 + 8) \div 2 = 7$, similarly in column (3) we have
$(8 + 4) \div 2 = 6$

For (Q. 114-118)

School	Teacher	Subject
I	Q	Science
J	S	Sanskrit
K	T/U	English/Mathematics
L	P	Mathematics or English
M	R	General Science
N	U/T	Hindi

119. (3) Ram's present age is 26 yr $[\because 25 = 5^2 \text{and } 27 = 3^3]$
When Ram will be 64 yr old than his age will be the cube of 4.
Hence, required time = 64 – 26 = 38 yr.

120. (2) Through options = $34(4 + 4 + 2) + (34 + 1)(3 + 4)$
$= 34 \times 10 + 35 \times 7$
$= 340 + 245 = 585$

■ Mock Test 5

1. (3) The matter needs to be a mixed bag of free enterprise as well as state control whereever it is in the public interest.

2. (2) It is evident from USA policy on Iran, Afganistan, etc.

3. (4) India needs more and more energy in proportion to increasing population, dwindling oil stock is also a reality.

4. (3) India can't cope up with ever-growing demand of energy with just fossil fuel.

5. (2) Strategy and Tactic are related to plans to gain advantage. Manoeuvre and manipulation mean clever handling of something.

6. (4) Plummet and Plunge mean to fall down headlong. Swoop means descend down fast. Soar means to ascend upward.

7. (2) Perfunctory means with serious attention. Meticulously means giving great care to perfection.

8. (3) Tenders means "indicative actions to show one's intention".

9. (2) Prognosis means forecast about the future course of progress of the disease.

10. (1) Rejoin means retort an accusation.

15. (2) It means glittering/glamorous people of film world.

16. (3) It means panic leading to desperation.

17. (1) Elation and exasperation are indicative of someone's mood.

21. (2) Earlier regulations are superseded, if found unsuitable (4) which were done for consolidation under one notification (2), for revising norms (1) as were mentioned (5) pertaining to training programmes (3).

24. (4) It means "expert in concealing facts/truth."

25. (3) It means help someone by putting him in advantageous position.

26. (1) The scheme was flawed, because the money spent on it did not generate wealth for future.

30. (2) The given expenditure explains squandering of funds in the name of the scheme which least benefitted targeted poor people.

31. (4) The crest jewel does not import literary meaning here and hence is not something like ornament adorning the Hindu philosophy. It is indeed the pithy part *ie*, quintessence, core of the philosophy.

32. (3) 'Yoga' is the same in origin as yoke—it means to harness, attach—here attach self to the Brahm *ie*, universe.

33. (3) The article says that Self is as small as a tiny grain which is situated in a vast space—larger than the whole universe—of heart.

34. (4) God is imminent—present inside—in every creature, and therefore, God can't be discriminated against God.

35. (2) The philosophy of Vedanta teaches us the lesson of "Vashudhaiva Kutumbkam". Hence discriminating one from another in a family is a wrong practice and society needs to get rid of it.

36. (3) Time is the fastest moving vehicle of change. The society too is moving fast towards the goal of global family with the passage of time.

37. (4) It is the synonym of given word 'sceptical'.

38. (1) A woman, as pronoun 'she' has been used in the passage for 'Wise One'.

42. (2) Combined cost of both the cars

$= 270000 + 27000 + 180000 + 18000$

$= 495000.$

Depreciation for 1 year $= 9\frac{1}{11}\%$

Cost of both the cars after deducting depreciation

$= 495000 - 45000 = \text{Rs } 450000$

For a profit of $11\frac{1}{9}\%$ the selling price combined must be $= 450000 \times \frac{10}{9} = \text{Rs } 500000.$

43. (1) $\frac{1}{\sqrt{k}} = \frac{2}{\sqrt{k}+\sqrt{k}} < \frac{2}{\sqrt{k}+\sqrt{k-1}}$

$\therefore \quad \frac{1}{\sqrt{k}} < \frac{1}{\sqrt{k}+\sqrt{k-1}}$

On multiplying and dividing RHS by $\sqrt{k}-\sqrt{k-1}$, we have

$$\frac{1}{\sqrt{k}} < 2(\sqrt{k}-\sqrt{k-1})$$

$\therefore$ For $k = 2$

$$\frac{1}{\sqrt{2}} < 2(\sqrt{2}-\sqrt{1})$$

Similarly,

$$\frac{1}{\sqrt{3}} < 2(\sqrt{3}-\sqrt{2})$$

$$\frac{1}{\sqrt{4}} < 2(\sqrt{4}-\sqrt{3})$$

$$\frac{1}{\sqrt{10000}} < 2(\sqrt{10000}-\sqrt{9999})$$

Adding all of the above, we get

$$N = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} + \ldots \frac{1}{\sqrt{10000}} < 2(\sqrt{10000}-1)$$

$\therefore \quad N < 198$

Similarly, we have

$$\frac{1}{\sqrt{k}} = \frac{2}{\sqrt{k}+\sqrt{k}} > \frac{2}{\sqrt{k+1}+\sqrt{k}}$$

$$= 2(\sqrt{k}+1)-\sqrt{k})$$

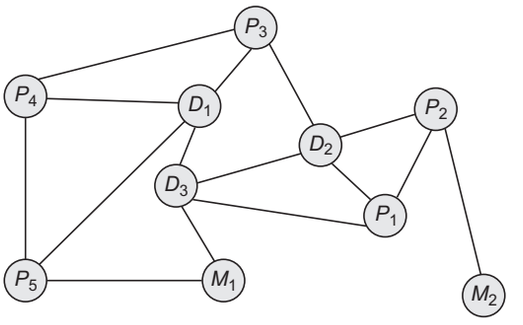
Hence, $N > 2(\sqrt{10001}-\sqrt{2})$

$\therefore \quad N > 197$

Therefore, the value of $I(N) = 197$

44. (4) All the above points possible of intersection are possible. 44 points of intersection are possible if their are 2 lines parallel out of 10 lines, thus the points of intersection ${}^{10}C_2 - {}^{2}C_2 = 44$, zero points of intersection are possible with all lines parallel to each other.
Similarly, with 3 out of 10 lines parallel and none of the lines parallel we can have 42 and 45 points of intersection respectively.

45. (4) Assume the teams to be nodes of a graph as shown in the figure below



Thus, a match between 2 teams is represented by a line drawn between the 2 nodes, *eg*, line drawn from node (D_1) to node (P_3) represents the game between team D_1 and P_3. From the above graph, we can easily see that total of games played by all the teams must be $2\times$ [the number of lines connecting the nodes]. Thus, we conclude that the total no. of games played by all the teams must be an even number.
In this question, the total games played by Deewane, Parwane, Mastane is $(3\times 4 + 5\times 3 + (2)x)$ where 'x' represents the number of games exactly played by each team from category Mastane. We can very clearly see that the above total can never be even for any value of 'x'.

46. (1) $80n - 100 = 79.2(n-1)$

$0.8n = 20.8$

$n = 26$

Thus, the total number of oranges are $26 \times 80 = 2080$. One of the baskets contains 100 and the other 24 baskets could contain $1, 2, 3, \ldots, 24$ oranges respectively. So, largest number of oranges.

$$= 2080 - 100\,\frac{24}{2}(1 + 24) = 1680$$

47. (3) The least integer greater than $[n^m]$ is $n^m + 1$. Thus, the question reduces to identifying for which values of m(largest) shall $n^m + 1$ divide $[1 + n + n^2 + \ldots + n^{127}]$.

$$1 + n + n^2 + \ldots + n^{127} = \frac{1(n^{128} - 1)}{(n-1)}$$

[By applying the formula for sum of GP]

$$= \frac{(n^{64} - 1)(n^{64} + 1)}{(n-1)}.$$

Thus, the greatest value of $m = 64$.

48. (1) $\log_3 2 = \dfrac{\log_{11} 2}{\log_{11} 3}, \quad \log_4 3 = \dfrac{\log_{11} 3}{\log_{11} 4}$

[By basic properties]

Thus, on substituting the value of $\log_3 2 \cdot \log_4 3$ and so we get

$\log_3 2 \cdot \log_4 3 \cdot \ldots \cdot \log_{10} 9 \cdot \log_{11} 10$

$$= \frac{\log_{11} 2}{\log_{11} 3} \times \frac{\log_{11} 3}{\log_{11} 4} \times \ldots \times \log_{11} 10$$

$$= \log_{11} 2$$

Hence, option (1) is correct.

49. (4) From similar triangles EGF and HDF,

$$\frac{EG}{HD} = \frac{GF}{DF} = \frac{\left(\frac{4}{3} - \frac{1}{2}\right) AD}{\left(\frac{4}{3} - 1\right) AD} = \frac{\frac{5}{6}}{\frac{1}{3}} = \frac{5}{2}$$

$$HD = \frac{2}{5} EG \quad \Rightarrow \quad CH = \frac{3}{5} EG = \frac{3}{5} AB$$

Area of triangle $ECH = \dfrac{1}{2} \times EC \times CH$

$$= \frac{1}{2} \times \left(\frac{1}{2} BC\right) \times \frac{3}{5} AB$$

Area of rectangle $ABCD = AB \times BC$

The ratio of their areas is

$$\frac{1}{2} \times \frac{1}{2} \times \frac{3}{5} = \frac{3}{20} = 3 : 20$$

50. (4) $g(x) = 1 - x^2$

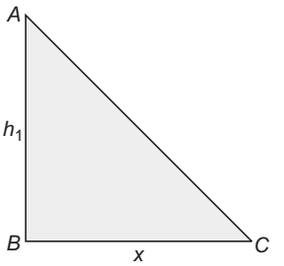
$$f(1 - x^2) = \frac{x^2 - 1}{x^2}$$

$$f(x) = \frac{-x}{1 - x}$$

$$\Rightarrow \qquad f\left(\frac{1}{2}\right) = \frac{-\frac{1}{2}}{1 - \frac{1}{2}} = -1$$

51. (3) Since, time for both is same so ratio of speed is the same as ratio of distance travelled.

Distance of Chantu = $30\sqrt{5}$ m. Distance travelled by Bantu is equal the length of one spiral × the number of spirals. Length of each spiral is calculated below.



Let l be the length of 1 spiral staircase, then

In ΔABC

$$h_1 = \frac{30\sqrt{5}}{5} = 6\sqrt{5}$$

x = circumference of circle = $2\pi r$

$$= 2 \times \frac{22}{7} \times 7 = 44$$

Using Pythagoras theorem

$$l = \sqrt{(h_1)^2 + x^2} = \sqrt{(6\sqrt{5})^2 + (44)^2} = 46 \text{ m}$$

Ratio of speeds of Chantu and Bantu

$$= 30\sqrt{5} : 5 \times 46$$

$$= 3\sqrt{5} : 23$$

52. (3) We can observe that the nth term of the series is $(n + 2)$

$[n + (n + 1) + (n + 2) + (n + 3) + (n + 4)]$

$$= (n + 2)[5(n + 2)] = 5(n + 2)^2$$

[It may be noted that the terms in the bracket are in AP and their sum is middle term x number of terms.]

$\therefore$ Sum of the first 18 terms

$$= 5[3^2 + 4^2 + \ldots + 20^2]$$

$$= 5[(1^2 + 2^2 + \ldots + 20^2) - (1^2 + 2^2)]$$

$$= 5\left[\frac{20 \times 21 \times 41}{6} - 5\right]$$

$$= 5[2870 - 5] = 14325$$

53. (4) In the expansion of $(x - a)^n$, the $(r + 1)$ th term

$$T_{r+1} = (-1)^r \, {}^nC_r \, x^{n-1} a^r$$

For $\left(\dfrac{1}{2} x^{\frac{1}{3}} - x^{\frac{-1}{5}}\right)^8$

$$T_{r+1} = (-1)^r \, {}^8C_r \left(\frac{1}{2}\right)^{8-r} = x^{\frac{8-r}{3} - \frac{r}{5}}$$

For term independent of x, power of x has to be zero.

$\dfrac{8-r}{3} - \dfrac{r}{5} = 0 \Rightarrow r = 5$; 6th term is independent of x

$$\therefore \quad T_6 = (-1)^5\, {}^8C_5 \left(\frac{1}{2}\right)^{8-5}$$

$$= -\frac{8\times 7\times 6}{3\times 2\times 1}\times\frac{1}{8} = -7$$

54. (3) The first woman completes $\frac{1}{8}$th of the job in a day.

Thus, on the first day $\frac{1}{8}$th of the work is done.

On the second day $\left(\frac{1}{8}+\frac{1}{7}\right)$, *ie*, $\frac{15}{56}$th of the work is done.

On the third day $\left(\frac{1}{8}+\frac{1}{7}+\frac{1}{6}\right)$, *ie*, $\frac{73}{168}$ th of the work is done.

On the fourth day $\left(\frac{1}{8}+\frac{1}{7}+\frac{1}{6}+\frac{1}{5}\right)$, *ie*, $\frac{533}{840}$ th of the work is done.

On the fifth day $\left(\frac{1}{8}+\frac{1}{7}+\ldots+\frac{1}{4}\right)$, *ie*, $\frac{743}{840}$th of the work is done.

On the sixth day $\left(\frac{1}{8}+\frac{1}{7}+\ldots+\frac{1}{3}\right)$, *ie*, $\frac{341}{280}$th of the work is done.

Now, as $\frac{341}{280} > 1$, the work is completed.

Thus, the job is completed on the 6th day.

55. (4) If $x > 0$, then $x^2 < x^2 + 1 + x < x^2 + 2x + 1$ or $(x+1)^2$, therefore $x^2 + x + 1$ lies between 2 consecutive square integers and hence, cannot be a perfect square. If $x = 0$, $y^2 = 1 + 0 + 0 = 1$ is a square number. Thus the solutions in this case are (0, 1) and (0, –1). Again, if $x < -1$, then $x^2 > x^2 + x + 1 > x^2 + 2x + 1$ and hence, there exists no solution. For $x = -1$, we have

$$y^2 = 1 - 1 + (-1)^2 = 1$$

$$\therefore \quad y \pm 1$$

Thus, 2 more solutions exist (1, 1), (1, –1). Therefore, there are 4 solutions in all.

56. (2) Here, each mason works at the same rate finally. Therefore, only the number of days worked by any person is relevant for finding the wage share of that person.

So, maximum possible share is of the person who started *ie*,

$$= \frac{15}{15+14+\ldots+2+1}\times 10000$$

$$= \frac{15\times 2}{15\times 16}\times 10000 = \text{Rs } 1250$$

57. (3) Let the CP of 2 kg of Kajoo = Rs 100

SP = Rs 80 (for 20% loss claimed by Ramlal)

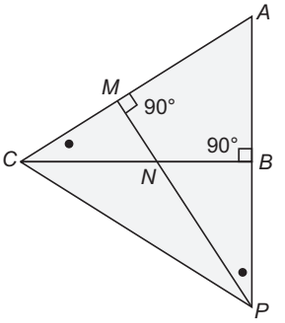
For a gain of $6\frac{2}{3}$% the SP of 2 kg of Kajoo must be 80×2 = Rs 160 and therefore the cost price must be Rs 150. This can happen if and only if the other stone weighed 750 g as well.

58. (3) The relative order of the speakers is Nana, Amitabh. Aishwarya, Abhishek. For other speakers there is no constraint. Hence, the number of possible arrangement of speakers shall be $\frac{7!}{4!} = 210$.

59. (3) The above definition tells us that P must be square of a natural number x. Therefore, the problem boils down to identifying which of these number might be a square, since 'none of these' is not one of the options. Hence one of the options must be a square. Option (1) cannot be a square, since they are not in the form $9k$ or $3k + 1$. Similarly, option (2) cannot be the answer as it is not in the form $8k + 1$.

60. (3) ΔMNC and ΔMAP are similar.

For PC subtends right angles at B, M and hence B, M, C, P are concyclic and hence



$$\angle MCN = \angle MPB = \angle MPA$$

and $\quad \angle NMC = 90^\circ = \angle AMP$

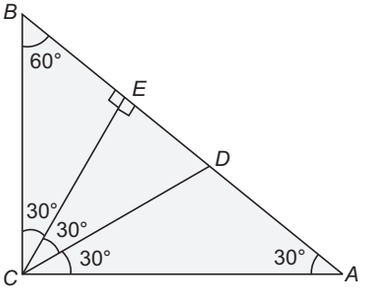
$$\therefore \quad \frac{MN}{MC} = \frac{MA}{MP}$$

[Since M is the mid point of AC, $MC = MA$]

$$\Rightarrow \quad MC \cdot MA = MC^2 = MN \cdot MP = 3(3+9) = 36$$

$$\Rightarrow \quad MC = MA = 6 \quad \text{and} \quad AC = 12$$

61. (1) In the figure CE and CD are trisectors of $\angle C$, $\angle CED = 90^\circ$ and hence $CE < CD$. (In ΔCED, CD is the hypotenuse).



Thus, it is required to find the length of CE.

$AB = 10$ cm, $\angle B = 60^\circ$, $\angle A = 30^\circ$

$\Rightarrow$ $BC = 5$ cm and $AC = 5\sqrt{3}$ cm

Again in ΔBCE,

$\angle CEB = 90^\circ$, $BC = 5$ cm, $CE = 5\cos 30^\circ$

$$\Rightarrow \quad CE = \frac{5}{2}\sqrt{3}\ \text{cm} \qquad \left(BE = \frac{5}{2}\right)$$

[CD can also be calculated from the right angled ΔECD, $\angle CED = 90°$, $\angle ECD = 30°$, $CE = \frac{5}{2}\sqrt{3}$ cm, $CD = 5$ cm $= DA$. Thus, $BE = ED = \frac{5}{2}$ cm, $DA = 5$ cm, $CE = \frac{5}{2}\sqrt{3}$ cm, $CD = 5$ cm and clearly $\frac{5}{2}\sqrt{3} < 5$ and the shorter trisector has a length of $\frac{5}{2}\sqrt{3}$ cm]

62. (3) Let S be the distance between A and B. v km/h for the rate on foot of the first boy and w km/h be the rate on foot of the second boy. Let a km be the distance that the first boy cycled from A to B. (He thus left his cycle a km from A and covered the rest of the distance on foot).

$$\frac{a}{20} + \frac{s-a}{v} = \frac{a}{w} + \frac{s-a}{20} \quad \ldots(i)$$

For the return from B to A, we have

$$\frac{a+1}{20} + \frac{s-a-1}{v} = \frac{a+1}{w} + \frac{s-a-1}{20} = -\frac{7}{20} \quad \ldots(ii)$$

Since, the first boy's spends $\frac{1}{v}$ hours/km and second boy spent $\frac{1}{w}$ hours/km, we get

$$\frac{1}{w} - \frac{1}{v} = \frac{1}{20}$$

On subtracting Eq. (i) from Eq. (ii), we get

$$\frac{1}{w} + \frac{1}{v} = \frac{9}{20}$$

Thus, we get $v = 5$ km/h and $w = 4$ km/h.

63. (4) Just substitute the values from the options and the answer can be obtained very easily.

64. (4) $25 + 13 + 6 + 3 + 1 = 48$
The above value represents triangles of different sizes that can be verified by counting.

65. (4) Let the number of books be z, yz, xyz, $wxyz$.
Then, $z + yz + xyz + wxyz = 70$ …(i)
$\Rightarrow \quad z(1 + y + xy + wxy) = 70 = 1 \times 2 \times 5 \times 7$
If $z = 1$, $y + xy + wxy = 69$
$\Rightarrow \quad y\,(1 + x + wx) = 3 \times 23$
Since, $y \neq 1$ (number with each friend different) or $y \neq 23$ hence the sum cannot be equal to 70, $y = 3$
Then, $x + wx = 22$
$\Rightarrow \quad x(1 + w) = 2 \times 11$
Since, x cannot be 11, $x = 2$, $w = 10$
Thus in Eq. (i), we have
$1 + (3 \times 1) + (2 \times 3 \times 1) + (10 \times 2 \times 3 \times 1) = 70$
If $z = 2$,
$y + xy + wxy = (70 \div 2) - 1 = 34$
or $y(1 + x + wx) = 2 \times 17$
Since, $y \neq 17$ hence sum would exceed 70, $y = 2$
$\therefore \quad x + wx = 16 \quad \Rightarrow \quad x(1 + w) = 16$
x has to be 2 or 4 $\Rightarrow w = 7$ or 3
$\therefore$ In Eq. (i), we get
$2 + (2 \times 2) + (2 \times 2 \times 2) + (7 \times 2 \times 2 \times 2) = 70$
or $2 + (2 \times 2) + (4 \times 2 \times 2) + (3 \times 4 \times 2 \times 2) = 70$
(z cannot assume higher values; 3, 4 do not divide 70 while $z = 5$ gives a higher sum).
Thus, minimum number of books purchased by the one purchasing most books $= 3 \times 4 \times 2 \times 2 = 48$

66. (1) $B + G + F = 165 \div 11$ …(i)
It is also given that
$5B + 6G + 7F = 65$ …(ii)
She required $7B + 6G + 5F$
we have,
$12B + 12G + 12F = 12 \times 11 = 132$ …(iii)
Subtracting Eq. (ii) from Eq. (iii), we get
$7B + 6G + 5F = 67$.
So she requires Rs 2 more.

For (Q. 67-70)

Ali's	Time	Dist. Moved	Ali's		Sonali
Movement		(Ali + Sonali)	Speed	Dist. Moved	Dist. Moved
Detroit–A	0–0.67	160	60	40	120
A–B	0.67–1.6	140	60	56	84
B–C	1.6–2.0	36	60	24	12
C–D	2.0–2.8	144	9–2.67 hr –120	80	40
D–E	2.8–4.0	240	2.8–3.33 hr –180	96	44
			3.33–4.0 hr–150	100	

67. (4) Total distance moved by Ali from detroit to Town D $= 40 + 56 + 24 + (80 + 24) = 224$ km
$\therefore$ Distance from Cincinnati to Town D
$= 720 - 224 = 496$ km

68. (2) Distance moved by Ali in the given time period
$= 40 + 56 + 24 + (80 + 24) + (96 + 100)$
$= 420$ km
Distance moved by Sonali in the same period
$= 720 - 420 = 300$ km
Required ratio $= \frac{420}{300} = 7 : 5$

69. (4) Only average speed of Sonali during different periods can be calculated and not the maximum speed.

70. (4) Between D and E, distance moved by Ali = 196 and distance moved by Sonali at the same time = 44 km hence the required ratio is the maximum.

71. (3) Minimum literates in Agra = 30% of 2 lakh = 60000
For literacy rate of Agra to be minimum, its population must be highest. It will be 30% of 5 lakh, *ie*, 1.5 lakh.
Required literacy rate $= \frac{60000}{150000} \times 100 = 40\%$.

72. (1) Share of Agra in total literates = 50%
Minimum possible number of literates at Bhopal = 30% of 2 lakh = 60000. For minimum rate of literacy, its population must be the highest possible. Agra city cannot have less than 100000 population (otherwise literacy rate will be > 100%). So, minimum possible population of Agra city is 20% of total population

(100000). Similarly, population of Chittour can't be less than 50000 (10% of 5 lakh). Minimum population of (Chittour + Agra) = 150000. So, maximum population of Bhopal can be 350000.

Required literacy rate $= \frac{60000}{350000} \times 100 \cong 17\%$

73. (1)

	Agra	Bhopal	Chittour
Maximum number of literates	100000	100000	80000
Maximum number of male literates	50000	54545	40000
Minimum number of literates	60000	60000	40000
Minimum number of male literates	24000	26666	13333

Thus, any city can have highest number of literates (when corresponding numbers for other two cities are minimum).

74. (2) Population of Agra = 50000
But minimum number of literates in Agra = 60000 which is not possible.

For (Q. 75-76)

$$65 + 75 + 85 = 225$$

Thus, at least 25% had all three (eyes, nose and fingers) dissimilar to their mother's. For a population of 1200, at least 300 had eyes, nose as well as fingers dissimilar to their mother's. Also, at least $85 \times 12 = 1020$ newborns had at least one of eyes, nose and fingers dissimilar.

75. (4) Choice (1) and (2) are in disagreement with above conclusions. Choice (3) also cannot be said to agree since survey is not about dissimilarity with parents. Then, choice (4) is the correct answer.

76. (4) 100 newborns have at least 225 similarities, out of which 15 have only 15 dissimilarities. So, 85 newborns have at least 210 dissimilarities $210 = 85 \times 2 + 40$. So, at least 40% have all the 3 dissimilarities with their mothers.

77. (2) Amount of expenditure incurred by P on Machinery
= Rs 150 thousand = Rs 150000

78. (1) Amount of expenditure incurred by L on Pesticides
= Rs 150 thousand = Rs 150000
Amount of expenditure incurred by Q on Pesticides
= Rs 200 thousand = Rs 200000
$\therefore$ Required percent $= \frac{150000}{200000} \times 100 = 75\%$

79. (3) Total amount of expenditure incurred on Labour by all Landowners together
= Rs (250 + 150 + 200 + 100 + 300 + 200) thousand
= Rs 1200000

80. (4) Expenditure on Seeds by Q = Rs 50000
Total expenditure by Q = Rs 700000
$\therefore$ Required percent $= \frac{50000}{700000} \times 100 = 7\%$

81. (3) Miscellaneous expenditure incurred by all Landowners together
= Rs (150 + 100 + 50 + 100 + 200 + 50) thousand
= Rs 650000
Total expenditure incurred on all items together by all landowners
= Rs (850 + 700 + 900 + 800 + 750 + 700) thousand
= Rs 4700000
$\therefore$ Required percent $= \frac{650000}{4700000} \times 100 = 15\%$

82. (1) The only other room except Biswachakra having a bag is Bagbahadur. Bagbahadur has two bags. So if one item is a bag the other item will also be a bag.

83. (4) If room Purnachandra does not contain either any printer or any computer, it must contain two bags. Room Chandrachud contains one computer and one bag. So, if room Pashupatti does not contain any printer, it must contain two computers. So, rooms having two printers and the room having one computer and one printer is left. In this case it will have at least one printer if any one of the remaining two rooms is opened.

For (Q. 84-87)

Rooms	Walls	Carpets	Curtains
Bed-room	Green	Yellow	Orange
Living-room	Yellow	Orange	Grey
Dining-room	Orange	Green	Yellow
Study-room	Yellow	Green	Grey

88. (3) $a \# b \geq c$ or $c \# a > b$

89. (1) $a = b \# c$ or $c < a \leq b$

90. (2) $c \neq b \geq a$ or $a = c > b$

91. (3) $a > b > c$ or $c < b < a$

92. (3) $a \geq b \leq c$ or $a = b < c$

96. (3) P $\xleftarrow{\text{Father}}$ G $\xrightarrow{\text{Brother}}$ C $\xrightarrow{\text{Sister}}$ A $\xrightarrow{\text{Uncle}}$ D
$\uparrow$ Father
B

So, C is uncle of B because G is B's father and C is G's brother.

97. (4) When order of the letter of the words are reversed they become as follows.

1. LIAN 2. LIAP 3. LIAR
4. MADAM 5. TSER

Among the above five LIAR and MADAM are meaningful words.

98. (4)

R	E	Q	U	E	S	T
+1		+1			+1	+1
S	2	R	5	2	T	U

The logic used here is very simple. Vowels have been coded as the xth vowel from the left end of English Alphabet whereas the consonants have been coded as the next letter.

Therefore,

A	C	I	D
↓	↓ + 1	↓	↓ + 1
1	D	3	E

For (Q. 99-103)

HOUSE

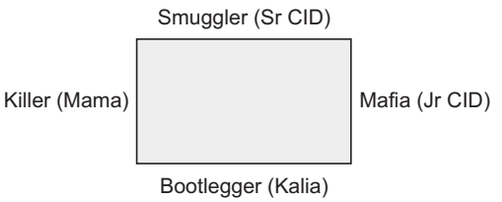
1. Yellow Lane
2. Kama Lane
3. Peacock Lane
4. Park Lane
5. Apple Lane
6. Rao Lane
7. Ash Lane

SCHOOL

104. (2) If first working day of a month is Tuesday then the last working day of the ongoing Month (30 days) will be Wednesday. Therefore, the first working day in the coming month will be Thursday.

105. (2) The specified digits are 8, 2 and 9. Now, we know a perfect square number does not have 8 and 2 at unit's place. Therefore, we can make only two three digit number from it *ie*, 829 and 289. Among these two numbers 289 is a perfect square number *ie*, square of 17. Thus, unit digits is 7 and ten's digit is 1.

106. (2)



107. (4) After interchanging the positions of the first and the third digits in each number, we get a new set of numbers as given below.
815 527 948 783 436
Hence, the lowest number is 436 which is the new form of 634.

108. (4) After interchanging the positions of the first and the second digits in each number we get a new set of numbers as given below.
158 275 489 837 364
Hence, the third number from the top is 364 which is the new form of 634.

109. (4) After interchanging the positions of the second and the third digits in each number we get a new set of numbers as given below
581 752 894 378 643
Hence, the difference between the highest and the lowest number

$$894 - 378 = 516$$

110. (4) Z is on the middle shelf.

111. (3) At least three objects are on the top shelf.

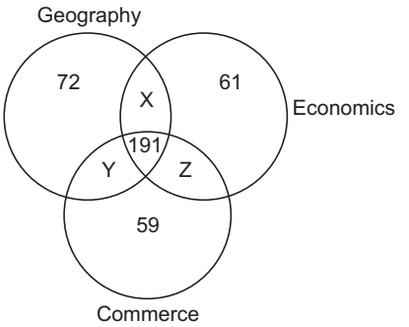
112. (2) X is on the bottom shelf.

113. (2) The trend is

$$6 \times 5 \times 4 = 120$$
$$6 \times 7 \times 3 = 126$$
$$8 \times 5 \times 8 = 320$$

114. (4) In each column, the sum of the top and bottom digits is 22 and in each column three letters A, B and C must be present. Hence, option (4) is the correct answer.

For (Q. 115-116)



116. (1) As given above, 70 students failed in all three subjects. Hence,
Number of students passed in at least one subject

$$= 500 - 70 = 430$$

Students who failed in one subject only means students who passed in two subjects only. Number of students who passed in two subjects only

$$= X + Y + Z$$
$$X + Y + Z = 450 - (72 + 61 + 59 + 191)$$
$$430 - 383 = 47$$

For (Q. 117-119)

Persons	After Ist Exchange	After 2nd Exchange	Initial Amount
A	60	80	60
B	80	40	40
C	70	50	80
D	40	70	50
E	50	60	70

120. (2) I is not the right way of arguing. Hence, I is not strong. Upliftment of socially backward classes is necessary. Hence, II is strong.

121. (4) Neither I nor II is strong to argue as the best Arguments.